```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     2 ************************************************************************
                                                     3 *
                                                     4 *        PLO-Extension Facility function tests
                                                     5 *
                                                     6 *        James Wekel October 2026
                                                     7 ************************************************************************


                                                     9 ************************************************************************
                                                    10 *
                                                    11 *        basic instruction tests
                                                    12 *
                                                    13 ************************************************************
                                                    14 *  This program tests proper functioning of the PLO-Extension
                                                    15 *  facility functions. Specification exceptions and access-register
                                                    16 *  mode are not tested.
                                                    17 *
                                                    18 *  PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch
                                                    19 *  obvious coding errors.  None of the tests are thorough.  They are
                                                    20 *  NOT designed to test all aspects of any of the instructions.
                                                    21 *
                                                    22 ************************************************************************
                                                    23 *
                                                    24 *  Example Hercules Testcase:
                                                    25 *
                                                    26 *    *Testcase plo-01-extension
                                                    27 *    *
                                                    28 *    *   PLO-Extension Functions Facility tests
                                                    29 *    *
                                                    30 *    *   # ---------------------------------------------------------
                                                    31 *    *   #  This tests only the basic function of the instruction.
                                                    32 *    *   #  Exceptions are NOT tested.
                                                    33 *    *   # ---------------------------------------------------------
                                                    34 *    *
                                                    35 *    mainsize    2
                                                    36 *    numcpu      1
                                                    37 *    sysclear
                                                    38 *    archlvl     z/Arch
                                                    39 *
                                                    40 *    loadcore    "$(testpath)/plo-01-extension.core" 0x0
                                                    41 *
                                                    42 *    diag8cmd    enable    # (needed for messages to Hercules console)
                                                    43 *    runtest     5         #
                                                    44 *    diag8cmd    disable   # (reset back to default)
                                                    45 *
                                                    46 *    *Done
                                                    47 *
                                                    48 ************************************************************************
```

```
 LOC         OBJECT CODE       ADDR1     ADDR2     STMT
                                                      50 ************************************************************************
                                                      51 *        FCHECK Macro - Is a Facility Bit set?
                                                      52 *
                                                      53 *        If the facility bit is NOT set, an message is issued and
                                                      54 *        the test is skipped.
                                                      55 *
                                                      56 *        Fcheck uses R0, R1 and R2
                                                      57 *
                                                      58 * eg.    FCHECK 134,'vector-packed-decimal'
                                                      59 ************************************************************************
                                                      60          MACRO
                                                      61          FCHECK &BITNO,&NOTSETMSG
                                                      62 .*                      &BITNO : facility bit number to check
                                                      63 .*                      &NOTSETMSG : 'facility name'
                                                      64          LCLA  &FBBYTE             Facility bit in Byte
                                                      65          LCLA  &FBBIT              Facility bit within Byte
                                                      66
                                                      67          LCLA  &L(8)
                                                      68 &L(1)    SetA  128,64,32,16,8,4,2,1  bit positions within byte
                                                      69
                                                      70 &FBBYTE  SETA  &BITNO/8
                                                      71 &FBBIT   SETA  &L((&BITNO-(&FBBYTE*8))+1)
                                                      72 .*       MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'
                                                      73
                                                      74          B     X&SYSNDX
                                                      75 *                                  Fcheck data area
                                                      76 *                                  skip messgae
                                                      77 SKT&SYSNDX DC  C'          Skipping tests: '
                                                      78          DC    C&NOTSETMSG
                                                      79          DC    C' facility (bit &BITNO) is not installed.'
                                                      80 SKL&SYSNDX EQU *-SKT&SYSNDX
                                                      81 *                                  facility bits
                                                      82          DS    FD                  gap
                                                      83 FB&SYSNDX DS   4FD
                                                      84          DS    FD                  gap
                                                      85 *
                                                      86 X&SYSNDX EQU *
                                                      87          LA    R0,((X&SYSNDX-FB&SYSNDX)/8)-1
                                                      88          STFLE FB&SYSNDX           get facility bits
                                                      89
                                                      90          XGR   R0,R0
                                                      91          IC    R0,FB&SYSNDX+&FBBYTE     get fbit byte
                                                      92          N     R0,=F'&FBBIT'            is bit set?
                                                      93          BNZ   XC&SYSNDX
                                                      94 *
                                                      95 * facility bit not set, issue message and exit
                                                      96 *
                                                      97          LA    R0,SKL&SYSNDX       message length
                                                      98          LA    R1,SKT&SYSNDX       message address
                                                      99          BAL   R2,MSG
                                                     100
                                                     101          B     EOJ
                                                     102 XC&SYSNDX EQU *
                                                     103          MEND
                                                     104
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     106 * ================================================================================
                                                     107 * LOW CORE
                                                     108 * ================================================================================
                                                     109 *
                               00000000  00070997    110 PLOETST  START 0
00000000                       00000000              111          USING PLOETST,R0          Low core addressability
                                                     112
                               00000140  00000000    113 SVOLDPSW EQU   PLOETST+X'140'      z/Arch Supervisor call old PSW


00000000                       00000000  000001A0    115          ORG   PLOETST+X'1A0'      z/Architecure RESTART PSW
000001A0  00000001 80000000                         116          DC    X'0000000180000000'
000001A8  00000000 00000200                         117          DC    AD(BEGIN)

000001B0                       000001B0  000001D0    119          ORG   PLOETST+X'1D0'      z/Architecure PROGRAM CHECK PSW
000001D0  00020001 80000000                         120          DC    X'0002000180000000'
000001D8  00000000 0000DEAD                         121          DC    AD(X'DEAD')


000001E0                       000001E0  00000200    123          ORG   PLOETST+X'200'      Start of actual test program...
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                     125 ************************************************************************
                                                     126 *               The actual "PLOETST" program itself...
                                                     127 ************************************************************************
                                                     128 *
                                                     129 *  Architecture Mode: z/Arch
                                                     130 *  Register Usage:
                                                     131 *
                                                     132 *   R0-R4    (work)
                                                     133 *   R5       Testing control table - base current entry
                                                     134 *   R6-R7    (work)
                                                     135 *   R8       First base register
                                                     136 *   R9       Second base register
                                                     137 *   R10-R13  (work)
                                                     138 *   R14      Subroutine call return
                                                     139 *   R15      Secondary Subroutine call return or work
                                                     140 *
                                                     141 ************************************************************************

00000200                       00000200              143          USING  BEGIN,R8           FIRST Base Register
00000200                       00001200              144          USING  BEGIN+4096,R9      SECOND Base Register

00000200  0580                                       146 BEGIN    BALR   R8,0               Initalize FIRST base register
00000202  0680                                       147          BCTR   R8,0               Initalize FIRST base register
00000204  0680                                       148          BCTR   R8,0               Initalize FIRST base register

00000206  4190 8800                    00000800      150          LA     R9,2048(,R8)       Initalize SECOND base register
0000020A  4190 9800                    00000800      151          LA     R9,2048(,R9)       Initalize SECOND base register
                                                     152
0000020E  B600 8664                    00000864      153          STCTL  R0,R0,CTLR0        Store CR0 to enable AFP
00000212  9604 8665                    00000865      154          OI     CTLR0+1,X'04'      Turn on AFP bit
00000216  9602 8665                    00000865      155          OI     CTLR0+1,X'02'      Turn on Vector bit
0000021A  B700 8664                    00000864      156          LCTL   R0,R0,CTLR0        Reload updated CR0
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     158 ************************************************************************
                                                     159 * Is PLO-extension facility installed  (bit 87)
                                                     160 ************************************************************************
                                                     161
                                                     162          FCHECK 87,'PLO-extension'
0000021E  47F0 80A0                       000002A0   163+          B     X0001
                                                     164+*                                 Fcheck data area
                                                     165+*                                 skip messgae
00000222  40404040 40404040                          166+SKT0001  DC    C'          Skipping tests: '
0000023C  D7D3D660 85A7A385                          167+         DC    C'PLO-extension'
00000249  40868183 899389A3                          168+         DC    C' facility (bit 87) is not installed.'
                            0000004B  00000001       169+SKL0001  EQU   *-SKT0001
                                                     170+*                                 facility bits
00000270  00000000 00000000                          171+         DS    FD                 gap
00000278  00000000 00000000                          172+FB0001   DS    4FD
00000298  00000000 00000000                          173+         DS    FD                 gap
                                                     174+*
                            000002A0  00000001       175+X0001    EQU   *
000002A0  4100 0004                       00000004   176+          LA    R0,((X0001-FB0001)/8)-1
000002A4  B2B0 8078                       00000278   177+          STFLE FB0001             get facility bits
000002A8  B982 0000                                  178+          XGR   R0,R0
000002AC  4300 8082                       00000282   179+          IC    R0,FB0001+10             get fbit byte
000002B0  5400 85D4                       000007D4   180+          N     R0,=F'1'                 is bit set?
000002B4  4770 80C8                       000002C8   181+          BNZ   XC0001
                                                     182+*
                                                     183+* facility bit not set, issue message and exit
                                                     184+*
000002B8  4100 004B                       0000004B   185+          LA    R0,SKL0001         message length
000002BC  4110 8022                       00000222   186+          LA    R1,SKT0001         message address
000002C0  4520 84F0                       000006F0   187+          BAL   R2,MSG
000002C4  47F0 85B8                       000007B8   188+          B     EOJ
                            000002C8  00000001       189+XC0001   EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     191 ************************************************************************
                                                     192 *    Do tests in the PLOETST table
                                                     193 ************************************************************************
                                                     194
000002C8  58C0 85D8                       000007D8  195          L     R12,=A(PLOTESTS)    get table of test addresses
                                                     196
                                 000002CC  00000001  197 NEXTTST  EQU   *
000002CC  5850 C000                       00000000  198          L     R5,0(0,R12)         get test address
000002D0  1255                                      199          LTR   R5,R5                have a test?
000002D2  4780 80E4                       000002E4  200          BZ    ENDTEST                 done?
                                                     201
000002D6                         00000000           202          USING PLOETEST,R5
                                                     203
000002D6  58B0 5000                       00000000  204          L     R11,TSUB            get address of test routine
000002DA  05BB                                      205          BALR  R11,R11             do test
                                                     206 *
000002DC  41C0 C004                       00000004  207          LA    R12,4(0,R12)        next test address
000002E0  47F0 80CC                       000002CC  208          B     NEXTTST


                                                     210 ************************************************************************
                                                     211 * end of testing; set ending psw
                                                     212 ************************************************************************
                                 000002E4  00000001  213 ENDTEST  EQU   *
000002E4  5810 8E00                       00001000  214          L     R1,FAILED           did a test fail?
000002E8  1211                                      215          LTR   R1,R1
000002EA  4780 85B8                       000007B8  216          BZ    EOJ                 No, exit
000002EE  47F0 85D0                       000007D0  217          B     FAILTEST            Yes, exit with BAD PSW
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    219 ************************************************************************
                                                    220 * CCMSG:       cc was not as expected
                                                    221 *              R5  = address of test control table
                                                    222 *              R15 = return address
                                                    223 ************************************************************************
                             000002F2  00000001     224 CCMSG    EQU   *
                                                    225 *
                                                    226 ** check for ignore cc as PLO did not update CC
                                                    227 *
000002F2  9563 500A                    0000000A     228          CLI   CC,99               99 -->  cc not updated
000002F6  478F 0000                    00000000     229          BE    0(R15)
                                                    230
000002FA  5800 85D4                    000007D4     231          L     R0,=F'1'            set GLOBAL failed test indicator
000002FE  5000 8E00                    00001000     232          ST    R0,FAILED
                                                    233 *
                                                    234 * extract CC extracted PSW
                                                    235 *
00000302  5810 5010                    00000010     236          L     R1,CCPSW
00000306  8810 000C                    0000000C     237          SRL   R1,12
0000030A  5410 85DC                    000007DC     238          N     R1,=XL4'3'
0000030E  4210 500C                    0000000C     239          STC   R1,CCFOUND          save cc
                                                    240 *
                                                    241 * FILL IN MESSAGE
                                                    242 *
00000312  4820 5004                    00000004     243          LH    R2,TNUM                 get test number and convert
00000316  4E20 8FA1                    000011A1     244          CVD   R2,DECNUM
0000031A  D211 8F8B 8F75       0000118B  00001175     245          MVC   PRT3,EDIT
00000320  DE11 8F8B 8FA1       0000118B  000011A1     246          ED    PRT3,DECNUM
00000326  D202 8E91 8F98       00001091  00001198     247          MVC   CCPRTNUM(3),PRT3+13 fill in message with test #
                                                    248
0000032C  B982 0022                                 249          XGR   R2,R2               get FUN as U8
00000330  4320 5007                    00000007     250          IC    R2,FUN
00000334  4E20 8FA1                    000011A1     251          CVD   R2,DECNUM           and convert
00000338  D211 8F8B 8F75       0000118B  00001175     252          MVC   PRT3,EDIT
0000033E  DE11 8F8B 8FA1       0000118B  000011A1     253          ED    PRT3,DECNUM
00000344  D201 8EBD 8F99       000010BD  00001199     254          MVC   CCPRTFUN(2),PRT3+14 fill in message with CC field
                                                    255
0000034A  B982 0022                                 256          XGR   R2,R2               get CC as U8
0000034E  4320 500A                    0000000A     257          IC    R2,CC
00000352  4E20 8FA1                    000011A1     258          CVD   R2,DECNUM           and convert
00000356  D211 8F8B 8F75       0000118B  00001175     259          MVC   PRT3,EDIT
0000035C  DE11 8F8B 8FA1       0000118B  000011A1     260          ED    PRT3,DECNUM
00000362  D200 8ECE 8F9A       000010CE  0000119A     261          MVC   CCPRTEXP(1),PRT3+15 fill in message with CC field
                                                    262
00000368  B982 0022                                 263          XGR   R2,R2               get CCFOUND as U8
0000036C  4320 500C                    0000000C     264          IC    R2,CCFOUND
00000370  4E20 8FA1                    000011A1     265          CVD   R2,DECNUM           and convert
00000374  D211 8F8B 8F75       0000118B  00001175     266          MVC   PRT3,EDIT
0000037A  DE11 8F8B 8FA1       0000118B  000011A1     267          ED    PRT3,DECNUM
00000380  D200 8EDE 8F9A       000010DE  0000119A     268          MVC   CCPRTGOT(1),PRT3+15 fill in message with ccfound
                                                    269 *
                                                    270 *        Use Hercules Diagnose for Message to console
                                                    271 *
00000386  4100 005F                    0000005F     272          LA    R0,CCPRTLNG         message length
0000038A  4110 8E81                    00001081     273          LA    R1,CCPRTLINE        message address
                                                    274
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0000038E  EBF5 81A0 0024                 000003A0   275          STMG  R15,R5,FCCSAV       save regs r15 to R5
                                                    276
00000394  4520 84F0                      000006F0   277          BAL   R2,MSG
                                                    278
00000398  EBF5 81A0 0004                 000003A0   279          LMG   R15,R5,FCCSAV       restore regs
0000039E  07FF                                      280          BR    R15                 Return to caller

000003A0  00000000 00000000                         282 FCCSAV   DC     16D'0'             save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     284 ************************************************************************
                                                     285 * URMSG:       UR was not as expected
                                                     286 *              R5  = address of test control table
                                                     287 *              R15 = return address
                                                     288 *
                                                     289 *              note: UR 99 - UR was not updated
                                                     290 ************************************************************************
                               00000420  00000001    291 URMSG    EQU   *
                                                     292 *
                                                     293 ** check for ignore UR as PLO did not update UR
                                                     294 *
00000420  9563 5008                    00000008     295          CLI   UR,99               99 -->  UR not updated
00000424  478F 0000                    00000000     296          BE    0(R15)
                                                     297
00000428  5800 85D4                    000007D4     298          L     R0,=F'1'            set GLOBAL failed test indicator
0000042C  5000 8E00                    00001000     299          ST    R0,FAILED
                                                     300 *
                                                     301 * FILL IN MESSAGE
                                                     302 *
00000430  4820 5004                    00000004     303          LH    R2,TNUM             get test number and convert
00000434  4E20 8FA1                    000011A1     304          CVD   R2,DECNUM
00000438  D211 8F8B 8F75       0000118B  00001175     305          MVC   PRT3,EDIT
0000043E  DE11 8F8B 8FA1       0000118B  000011A1     306          ED    PRT3,DECNUM
00000444  D202 8EF0 8F98       000010F0  00001198     307          MVC   URPRTNUM(3),PRT3+13 fill in message with test #
                                                     308
0000044A  B982 0022                                  309          XGR   R2,R2               get FUN as U8
0000044E  4320 5007                    00000007     310          IC    R2,FUN
00000452  4E20 8FA1                    000011A1     311          CVD   R2,DECNUM            and convert
00000456  D211 8F8B 8F75       0000118B  00001175     312          MVC   PRT3,EDIT
0000045C  DE11 8F8B 8FA1       0000118B  000011A1     313          ED    PRT3,DECNUM
00000462  D201 8F1C 8F99       0000111C  00001199     314          MVC   URPRTFUN(2),PRT3+14 fill in message with Fun field
                                                     315
00000468  B982 0022                                  316          XGR   R2,R2               get UR as U8
0000046C  4320 5008                    00000008     317          IC    R2,UR
00000470  4E20 8FA1                    000011A1     318          CVD   R2,DECNUM            and convert
00000474  D211 8F8B 8F75       0000118B  00001175     319          MVC   PRT3,EDIT
0000047A  DE11 8F8B 8FA1       0000118B  000011A1     320          ED    PRT3,DECNUM
00000480  D200 8F2D 8F9A       0000112D  0000119A     321          MVC   URPRTEXP(1),PRT3+15 fill in message with UR field
                                                     322
00000486  B982 0022                                  323          XGR   R2,R2               get URFOUND as U8
0000048A  4320 5009                    00000009     324          IC    R2,URFOUND
0000048E  4E20 8FA1                    000011A1     325          CVD   R2,DECNUM            and convert
00000492  D211 8F8B 8F75       0000118B  00001175     326          MVC   PRT3,EDIT
00000498  DE11 8F8B 8FA1       0000118B  000011A1     327          ED    PRT3,DECNUM
0000049E  D200 8F3D 8F9A       0000113D  0000119A     328          MVC   URPRTGOT(1),PRT3+15 fill in message with urfound
                                                     329 *
                                                     330 *        Use Hercules Diagnose for Message to console
                                                     331 *
000004A4  4100 005F                    0000005F     332          LA    R0,URPRTLNG         message length
000004A8  4110 8EE0                    000010E0     333          LA    R1,URPRTLINE        messagfe address
                                                     334
000004AC  EBF5 82C0 0024               000004C0     335          STMG  R15,R5,URSAV        save regs r15 to R5
                                                     336
000004B2  4520 84F0                    000006F0     337          BAL   R2,MSG
                                                     338
000004B6  EBF5 82C0 0004               000004C0     339          LMG   R15,R5,URSAV       restore regs
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000004BC  07FF                                340          BR    R15                Return to caller
000004C0  00000000 00000000                   342 URSAV    DC     16D'0'            save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     344 ************************************************************************
                                                     345 * Fail Message to console
                                                     346 *              R5  = address of test control table
                                                     347 *              R15 = return address
                                                     348 ************************************************************************
                                 00000540  00000001  349 FAILMSG  EQU   *
00000540  5800 85D4                        000007D4  350          L     R0,=F'1'            set GLOBAL failed test indicator
00000544  5000 8E00                        00001000  351          ST    R0,FAILED
                                                     352
00000548  4820 5004                        00000004  353          LH    R2,TNUM             get test number and convert
0000054C  4E20 8FA1                        000011A1  354          CVD   R2,DECNUM
00000550  D211 8F8B 8F75       0000118B    00001175  355          MVC   PRT3,EDIT
00000556  DE11 8F8B 8FA1       0000118B    000011A1  356          ED    PRT3,DECNUM
0000055C  D202 8E18 8F98       00001018    00001198  357          MVC   FAILTNUM(3),PRT3+13 fill in message with test #
                                                     358
00000562  D263 8E1D 85E4       0000101D    000007E4  359          MVC   FAILTXT(L'FAILTXT),=CL128' '  set to blanks
00000568  B982 0022                                  360          XGR   R2,R2
0000056C  5820 5018                        00000018  361          L     R2,FAILTXTA
00000570  D263 8E1D 2000       0000101D    00000000  362          MVC   FAILTXT(L'FAILTXT),0(R2)      load message text
                                                     363
                                                     364 *
                                                     365 *        Use Hercules Diagnose for Message to console
                                                     366 *
00000576  4100 0079                        00000079  367          LA    R0,FAILLNG          message length
0000057A  4110 8E08                        00001008  368          LA    R1,FAILLINE         messagfe address
                                                     369
0000057E  EBF5 8390 0024                   00000590  370          STMG  R15,R5,FLSAV        save regs r15 to R5
                                                     371
00000584  4520 84F0                        000006F0  372          BAL   R2,MSG
                                                     373
00000588  EBF5 8390 0004                   00000590  374          LMG   R15,R5,FLSAV        restore regs
0000058E  07FF                                       375          BR    R15                 Return to caller

00000590  00000000 00000000                          377 FLSAV    DC     16D'0'             save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     379 ***************************************************************************
                                                     380 * Test: Not Available Message
                                                     381 *              R5  = address of test control table
                                                     382 *              R15 = return address
                                                     383 ***************************************************************************
                               00000610  00000001    384 NAMSG    EQU   *
00000610  5800 85D4                      000007D4    385          L     R0,=F'1'            set GLOBAL failed test indicator
00000614  5000 8E00                      00001000    386          ST    R0,FAILED
                                                     387
00000618                       00000000              388          USING PLOAVAL,R5
                                                     389
00000618  4820 5004                      00000004    390          LH    R2,ATNUM            get test number and convert
0000061C  4E20 8FA1                      000011A1    391          CVD   R2,DECNUM
00000620  D211 8F8B 8F75       0000118B  00001175    392          MVC   PRT3,EDIT
00000626  DE11 8F8B 8FA1       0000118B  000011A1    393          ED    PRT3,DECNUM
0000062C  D202 8F4F 8F98       0000114F  00001198    394          MVC   NAPRTNUM(3),PRT3+13 fill in message with test #
                                                     395
00000632  B982 0022                                  396          XGR   R2,R2               get FUN as U8
00000636  4320 5007                      00000007    397          IC    R2,AFUN
0000063A  4E20 8FA1                      000011A1    398          CVD   R2,DECNUM           and convert
0000063E  D211 8F8B 8F75       0000118B  00001175    399          MVC   PRT3,EDIT
00000644  DE11 8F8B 8FA1       0000118B  000011A1    400          ED    PRT3,DECNUM
0000064A  D201 8F61 8F99       00001161  00001199    401          MVC   NAPRTFUN(2),PRT3+14 fill in message with Fun field
                                                     402
00000650                                             403          DROP R5
                                                     404 *
                                                     405 *        Use Hercules Diagnose for Message to console
                                                     406 *
00000650  4100 0036                      00000036    407          LA    R0,NAPRTLNG         message length
00000654  4110 8F3F                      0000113F    408          LA    R1,NAPRTLINE        messagfe address
                                                     409
00000658  EBF5 8470 0024                 00000670    410          STMG  R15,R5,NACCSAV      save regs r15 to R5
                                                     411
0000065E  4520 84F0                      000006F0    412          BAL   R2,MSG
                                                     413
00000662  EBF5 8470 0004                 00000670    414          LMG   R15,R5,NACCSAV      restore regs
00000668  07FF                                       415          BR    R15                 Return to caller

00000670  00000000 00000000                          417 NACCSAV  DC     16D'0'             save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    419 ************************************************************************
                                                    420 *        Issue HERCULES MESSAGE
                                                    421 *              R0 = length of message
                                                    422 *              R1 = message address
                                                    423 *              R2 = return address
                                                    424 ************************************************************************

000006F0  4900 85E0                    000007E0    426 MSG      CH    R0,=H'0'            Do we even HAVE a message?
000006F4  07D2                                     427          BNHR  R2                  No, ignore

000006F6  9002 852C                    0000072C    429          STM   R0,R2,MSGSAVE       Save registers

000006FA  4900 85E2                    000007E2    431          CH    R0,=AL2(L'MSGMSG)   Message length within limits?
000006FE  47D0 8506                    00000706    432          BNH   MSGOK               Yes, continue
00000702  4100 005F                    0000005F    433          LA    R0,L'MSGMSG         No, set to maximum

00000706  1820                                     435 MSGOK    LR    R2,R0               Copy length to work register
00000708  0620                                     436          BCTR  R2,0                Minus-1 for execute
0000070A  4420 8538                    00000738    437          EX    R2,MSGMVC           Copy message to O/P buffer

0000070E  4120 200A                    0000000A    439          LA    R2,1+L'MSGCMD(,R2)  Calculate true command length
00000712  4110 853E                    0000073E    440          LA    R1,MSGCMD           Point to true command

00000716  83120008                                 442          DC    X'83',X'12',X'0008' Issue Hercules Diagnose X'008'
0000071A  4780 8526                    00000726    443          BZ    MSGRET              Return if successful
                                                   444
0000071E  1222                                     445          LTR   R2,R2               Is Diag8 Ry (R2) 0?
00000720  4780 8526                    00000726    446          BZ    MSGRET                an error occurred but coninue
                                                   447
00000724  0000                                     448          DC    H'0'                CRASH for debugging purposes

00000726  9802 852C                    0000072C    450 MSGRET   LM    R0,R2,MSGSAVE       Restore registers
0000072A  07F2                                     451          BR    R2                  Return to caller




0000072C  00000000 00000000                        453 MSGSAVE  DC    3F'0'               Registers save area
                                                   454
00000738  D200 8547 1000      00000747 00000000    455 MSGMVC   MVC   MSGMSG(0),0(R1)     Executed instruction


0000073E  D4E2C7D5 D6C8405C                        457 MSGCMD   DC    C'MSGNOH * '        *** HERCULES MESSAGE COMMAND ***
00000747  40404040 40404040                        458 MSGMSG   DC    CL95' '             The message text to be displayed
                                                   459
000007A8                                           460          DS    0D
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     462 ***************************************************************************
                                                     463 *        Normal completion or Abnormal termination PSWs
                                                     464 ***************************************************************************


000007A8  00020001 80000000                          466 EOJPSW   DC    0D'0',X'0002000180000000',AD(0)
000007B8  B2B2 85A8                        000007A8  468 EOJ      LPSWE EOJPSW               Normal completion


000007C0  00020001 80000000                          470 FAILPSW  DC    0D'0',X'0002000180000000',AD(X'BAD')
000007D0  B2B2 85C0                        000007C0  472 FAILTEST LPSWE FAILPSW              Abnormal termination


                                                     474 ***************************************************************************
                                                     475 *        Working Storage
                                                     476 ***************************************************************************


000007D4                                             478          LTORG ,                   Literals pool
000007D4  00000001                                   479                =F'1'
000007D8  00070800                                   480                =A(PLOTESTS)
000007DC  00000003                                   481                =XL4'3'
000007E0  0000                                       482                =H'0'
000007E2  005F                                       483                =AL2(L'MSGMSG)
000007E4  40404040 40404040                          484                =CL128' '
                                                     485
00000864  00000000                                   486 CTLR0    DS    F
00000868  00000000                                   487          DS    F
                                                     488
                                                     489 * Constants
                                                     490
                          00000400  00000001         491 K        EQU   1024                One KB
                          00001000  00000001         492 PAGE     EQU   (4*K)               Size of one page
                          00010000  00000001         493 K64      EQU   (64*K)              64 KB
                          00100000  00000001         494 MB       EQU   (K*K)               1 MB
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     496 *=====================================================================
                                                     497 *
                                                     498 *  NOTE: start data on an address that is easy to display
                                                     499 *        within Hercules
                                                     500 *
                                                     501 *=====================================================================
                                                     502
0000086C                       0000086C  00001000    503          ORG   PLOETST+X'1000'
00001000  00000000                                   504 FAILED   DC    F'0'                some test failed?
00001004  00000000                                   505 TESTING  DC    F'0'                current test number


                                                     507 ************************************************************************
                                                     508 *        TEST failed : instruction data
                                                     509 ************************************************************************
                                                     510 *
                                                     511 *        failed message and associated editting
                                                     512 *
00001008  40404040 40404040                          513 FAILLINE DC    C'         Test # '
00001018  A7A7A7                                     514 FAILTNUM DC    C'xxx'
0000101B  7A40                                       515          DC    C': '
                               00000015  00000001    516 FAILHLNG EQU   *-FAILLINE          fail line header length
                                                     517
0000101D  40404040 40404040                          518 FAILTXT  DC    CL100' '             Test fail message
                               00000079  00000001    519 FAILLNG  EQU   *-FAILLINE
                                                     520


                                                     522 ************************************************************************
                                                     523 *        TEST failed : CC message
                                                     524 ************************************************************************
                                                     525 *
                                                     526 *        failed message and associated editting
                                                     527 *
00001081  40404040 40404040                          528 CCPRTLINE DC   C'         Test # '
00001091  A7A7A7                                     529 CCPRTNUM DC    C'xxx'
00001094  7A40A699 96958740                          530          DC    c': wrong cc for PLO instruction'
000010B2  404D86A4 9583A389                          531          DC    c' (function '
000010BD  A7A7                                       532 CCPRTFUN DC    CL2'xx'
000010BF  5D                                         533          DC    c')'
000010C0  4085A797 8583A385                          534          DC    C' expected: cc='
000010CE  A7                                         535 CCPRTEXP DC    C'x'
000010CF  6B                                         536          DC    C','
000010D0  40998583 8589A585                          537          DC    C' received: cc='
000010DE  A7                                         538 CCPRTGOT DC    C'x'
000010DF  4B                                         539          DC    C'.'
                               0000005F  00000001    540 CCPRTLNG   EQU   *-CCPRTLINE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    542 ************************************************************************
                                                    543 *        TEST failed : unequal results (UR) field message
                                                    544 ************************************************************************
                                                    545 *
                                                    546 *        failed message and associated editting
                                                    547 *
000010E0  40404040 40404040                         548 URPRTLINE DC   C'         Test # '
000010F0  A7A7A7                                    549 URPRTNUM DC    C'xxx'
000010F3  7A40A699 96958740                         550           DC    c': wrong UR for PLO instruction'
00001111  404D86A4 9583A389                         551           DC    c' (function '
0000111C  A7A7                                      552 URPRTFUN DC    CL2'xx'
0000111E  5D                                        553           DC    c')'
0000111F  4085A797 8583A385                         554           DC    C' expected: UR='
0000112D  A7                                        555 URPRTEXP DC    C'x'
0000112E  6B                                        556           DC    C','
0000112F  40998583 8589A585                         557           DC    C' received: UR='
0000113D  A7                                        558 URPRTGOT DC    C'x'
0000113E  4B                                        559           DC    C'.'
                            0000005F  00000001      560 URPRTLNG   EQU   *-URPRTLINE



                                                    562 ************************************************************************
                                                    563 *        TEST failed : PLO function is Not Available
                                                    564 ************************************************************************
                                                    565 *
                                                    566 *        failed message and associated editting
                                                    567 *
0000113F  40404040 40404040                         568 NAPRTLINE DC   C'         Test # '
0000114F  A7A7A7                                    569 NAPRTNUM DC    C'xxx'
00001152  7A40D7D3 D64086A4                         570           DC    C': PLO function '
00001161  A7A7                                      571 NAPRTFUN DC    CL2'xx'
00001163  4089A240 D5D6E340                         572           DC    C' is NOT installed.'
                                                    573
                            00000036  00000001      574 NAPRTLNG EQU   *-NAPRTLINE
                                                    575



                                                    577 ************************************************************************
                                                    578 *        TEST failed : message working storge
                                                    579 ************************************************************************
00001175  40212020 20202020                         580 EDIT      DC    XL18'402120202020202020202020202020202020'
                                                    581
00001187  7E7E7E6E                                  582           DC    C'===>'
0000118B  40404040 40404040                         583 PRT3      DC    CL18' '
0000119D  4C7E7E7E                                  584           DC    C'<==='
000011A1  00000000 00000000                         585 DECNUM    DS    CL16
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     587 ********************************************************************************
                                                     588 *        PLOETEST DSECT
                                                     589 ********************************************************************************


                                                     591 PLOETEST DSECT ,
00000000  00000000                                   592 TSUB     DC    A(0)               pointer  to test
00000004  0000                                       593 TNUM     DC    H'00'              Test Number
00000006  00                                         594          DC    XL1'00'
00000007  00                                         595 FUN      DC    HL1'00'            Function
00000008  00                                         596 UR       DC    HL1'00'            expected unequal results (UR)
00000009  00                                         597 URFOUND  DS    HL1'00'            extracted UR from PL
                                                     598
0000000A  00                                         599 CC       DC    HL1'00'            cc
0000000B  00                                         600 CCMASK   DC    HL1'00'            not expected CC mask
                                                     601
0000000C  00                                         602 CCFOUND  DS    X                  extracted cc
00000010  00000000 00000000                          603 CCPSW    DS    2F                 extract PSW after test (has CC)
                                                     604
00000018  00000000                                   605 FAILTXTA DC    A(0)               address of test fail message
0000001C  00000000                                   606 OPSRCA   DC    A(0)               address of operand source
                           00000020  00000001        607 OPREPA   EQU   *                  address of operand replacement
00000020  00000000                                   608 OPDSTA   DC    A(0)               address of operand destination
00000024  00000000                                   609 PLA      DC    A(0)               address of parameter list
00000028  00000000                                   610 RANDXA   DC    A(0)               address of random source data


                                                     612 ********************************************************************************
                                                     613 *        PLOEAVL DSECT
                                                     614 ********************************************************************************


                                                     616 PLOAVAL  DSECT ,
00000000  00000000                                   617 ATSUB    DC    A(0)               pointer  to test
00000004  0000                                       618 ATNUM    DC    H'00'              Test Number
00000006  00                                         619          DC    XL1'00'
00000007  00                                         620 AFUN     DC    HL1'00'            Function
00000008                                             621          DS    0F
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                      623 ********************************************************************************
                                                      624 *     Macros to help build test tables
                                                      625 ********************************************************************************


                                                      627 *-------------------------------------------------------------------------
                                                      628 * TSTPLO : test whether a PLO function is available/installed.
                                                      629 *          If a functionm is installed, CC=0. Not installed, CC=3.
                                                      630 *          - &fun: the function number
                                                      631 *-------------------------------------------------------------------------
                                                      632          MACRO
                                                      633          TSTPLO  &FUN
                                                      634          GBLA  &TNUM
                                                      635 &TNUM    SETA  &TNUM+1
                                                      636 *
                                                      637 * test if PLO function is available
                                                      638 *
                                                      639          USING *,R5                base for test data and test routine
                                                      640          USING PLOAVAL,R5
                                                      641
                                                      642 T&TNUM   DC    A(CODE&TNUM)           address of test routine
                                                      643          DC    H'&TNUM'              test number
                                                      644          DC    X'00'
                                                      645          DC    HL1'&FUN'             Function Number
                                                      646 *
                                                      647 CODE&TNUM DS    0F
                                                      648          LA    R1,(256+&FUN)         T bit 55  (+256)
                                                      649          LR    R0,R1
                                                      650          PLO   0,0,0,0
                                                      651          BZ    0(R11)                Function is installed, return
                                                      652
                                                      653          BAL   R15,NAMSG             Function not available message
                                                      654          BR    R11                   return
                                                      655          DROP  R5
                                                      656 *
                                                      657          MEND
                                                      658
                                                      659 *-------------------------------------------------------------------------
                                                      660 * TSTLBL : insert a label with a test number suffix
                                                      661 *          - &name: the label name
                                                      662 *-------------------------------------------------------------------------
                                                      663          MACRO
                                                      664          TSTLBL  &NAME
                                                      665          GBLA  &TNUM
                                                      666          LCLC  &X
                                                      667 &X       SETC  '&NAME&TNUM'
                                                      668 &X       DS  0C
                                                      669          MEND
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   671 *-------------------------------------------------------------------
                                                   672 * TSTEND : end a test
                                                   673 *-------------------------------------------------------------------
                                                   674          MACRO
                                                   675          TSTEND  &NAME
                                                   676          GBLA  &TNUM
                                                   677          DS    0H
                                                   678          DC    CL32'End of test: &TNUM'
                                                   679          MEND
                                                   680
                                                   681 *-------------------------------------------------------------------
                                                   682 * TSTCODE : The code for the test follows the label
                                                   683 *-------------------------------------------------------------------
                                                   684          MACRO
                                                   685          TSTCODE  &NAME
                                                   686          GBLA  &TNUM
                                                   687          LCLC  &X
                                                   688 &X       SETC  'CODE&TNUM'
                                                   689 &X       DS    0F
                                                   690          MEND
                                                   691
                                                   692 *-------------------------------------------------------------------
                                                   693 * TSTDATA: The data for the test follows.
                                                   694 *          Data is aligned to the 1K boundary after the TSTHDR
                                                   695 *-------------------------------------------------------------------
                                                   696          MACRO
                                                   697          TSTDATA  &NAME
                                                   698          GBLA  &THDRNUM
                                                   699          ORG   PLOETST+(200*K)+(&THDRNUM*4*K)+K
                                                   700          MEND
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   702 *-------------------------------------------------------------------
                                                   703 * TSTHDR: The start of a PLOE Test
                                                   704 *         Test is aligned to 4K boundary starting at 200K
                                                   705 *              1 K - test constants plus code
                                                   706 *              3 K - test data area
                                                   707 *
                                                   708 *        Note: PLOETEST DSECT needs to be in-sync with TSTHDR
                                                   709 *-------------------------------------------------------------------
                                                   710          MACRO
                                                   711          TSTHDR  &CC,&UR,&FUN
                                                   712 .*                        &CC    - expected CC
                                                   713 .*                        &UR    - expected unequal results
                                                   714 .*                        &FUN   - function
                                                   715
                                                   716          LCLA  &XCC(4)  &CC has mask values for FAILED condition codes
                                                   717 &XCC(1)  SETA  7                   CC != 0
                                                   718 &XCC(2)  SETA  11                  CC != 1
                                                   719 &XCC(3)  SETA  13                  CC != 2
                                                   720 &XCC(4)  SETA  14                  CC != 3
                                                   721
                                                   722          GBLA  &TNUM
                                                   723 &TNUM    SETA  &TNUM+1
                                                   724
                                                   725          GBLA  &THDRNUM
                                                   726 &THDRNUM SETA  &THDRNUM+1
                                                   727
                                                   728          ORG   PLOETST+(200*K)+(&THDRNUM*4*K)
                                                   729          DC    CL32'START of test: &TNUM'
                                                   730
                                                   731          USING *,R5                base for test data and test routine
                                                   732          USING PLOETEST,R5
                                                   733
                                                   734 T&TNUM   DC    A(CODE&TNUM)        address of test code
                                                   735          DC    H'&TNUM'            test number
                                                   736          DC    X'00'
                                                   737          DC    HL1'&FUN'           Function
                                                   738          DC    HL1'&UR'            expected unequal results (UR)
                                                   739          DC    HL1'00'             found UR
                                                   740
                                                   741          DC    HL1'&CC'            cc
                                                   742          AIF   ('&CC' EQ '99').NOCC
                                                   743          DC    HL1'&XCC(&CC+1)'    not expected CC mask
                                                   744          AGO   .CCCONT
                                                   745 .NOCC    ANOP
                                                   746          DC    HL1'00'             cc not checked
                                                   747 .CCCONT  ANOP
                                                   748
                                                   749          DS    X                   extracted cc
                                                   750          DS    2F                  extract PSW after test (has CC)
                                                   751          DC    A(FMSG&TNUM)        address of test fail message
                                                   752          DC    A(OPSRC&TNUM)       address of operand source
                                                   753          DC    A(OPDST&TNUM)       address of operand destination
                                                   754          DC    A(PL&TNUM)          address of parameter list
                                                   755          DC    A(RANDX+&TNUM)      address of random source data
                                                   756 .*
                                                   757          MEND
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

                                                      759 *-------------------------------------------------------------------
                                                      760 * PTTABLE: generate table of pointers to individual tests
                                                      761 *-------------------------------------------------------------------
                                                      762
                                                      763          MACRO
                                                      764          PTTABLE
                                                      765          GBLA  &TNUM
                                                      766          LCLA  &CUR
                                                      767 &CUR     SETA  1
                                                      768 .*
                                                      769 TTABLE   DS    0F
                                                      770 .LOOP    ANOP
                                                      771 .*
                                                      772          DC    A(T&CUR)
                                                      773 .*
                                                      774 &CUR     SETA  &CUR+1
                                                      775          AIF   (&CUR LE &TNUM).LOOP
                                                      776 *
                                                      777          DC    A(0)              END OF TABLE
                                                      778          DC    A(0)
                                                      779 .*
                                                      780          MEND
                                                      781
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     783 ********************************************************************************
                                                     784 *        PLO-Ext tests
                                                     785 ********************************************************************************
                               00000000  00070997    786 PLOETST  CSECT
                                                     787          PRINT DATA
                                                     788
                                                     789 * ------------------------------------------------------------------------
                                                     790 * Random Bytes for Tests
                                                     791 * ------------------------------------------------------------------------
000011B1  7A1FC308 9DE246B1                          792 RANDX    DC    XL16'7A 1F C3 08 9D E2 46 B1 5C F7 20 8B 64 DA 03 91'
000011B9  5CF7208B 64DA0391
000011C1  BE550CA7 3ED87219                          793          DC    XL16'BE 55 0C A7 3E D8 72 19 F4 6B 80 2D C9 14 EF 5A'
000011C9  F46B802D C914EF5A
000011D1  31AC77E0 4B960FD3                          794          DC    XL16'31 AC 77 E0 4B 96 0F D3 68 B9 25 FE 52 8D 17 C6'
000011D9  68B925FE 528D17C6
000011E1  EB40A16E 09D5832A                          795          DC    XL16'EB 40 A1 6E 09 D5 83 2A F8 5F BC 12 74 C1 9A 36'
000011E9  F85FBC12 74C19A36
000011F1  0DE759B2 43FC861B                          796          DC    XL16'0D E7 59 B2 43 FC 86 1B D0 6C 28 A5 93 4E F1 7D'
000011F9  D06C28A5 934EF17D
00001201  C81165DE 3B90FA27                          797          DC    XL16'C8 11 65 DE 3B 90 FA 27 54 AD 02 79 E4 B7 4C 88'
00001209  54AD0279 E4B74C88
00001211  16D970A3 5ECB34F5                          798          DC    XL16'16 D9 70 A3 5E CB 34 F5 81 0A BD 62 E9 47 9F 23'
00001219  810ABD62 E9479F23
00001221  D65801AF 75CA3DE1                          799          DC    XL16'D6 58 01 AF 75 CA 3D E1 92 6F 18 B4 50 FD 87 2C'
00001229  926F18B4 50FD872C
00001231  A844DB0E 69C235F0                          800          DC    XL16'A8 44 DB 0E 69 C2 35 F0 9B 56 ED 21 7F B8 04 D4'
00001239  9B56ED21 7FB804D4
00001241  63AA1DE6 4895CF30                          801          DC    XL16'63 AA 1D E6 48 95 CF 30 F9 82 5B 13 DC 76 A0 2F'
00001249  F9825B13 DC76A02F
00001251  8E41F3B5 0B67CE24                          802          DC    XL16'8E 41 F3 B5 0B 67 CE 24 99 DA 53 E8 1A 7C C7 32'
00001259  99DA53E8 1A7CC732
00001261  F68549B0 1ED26A94                          803          DC    XL16'F6 85 49 B0 1E D2 6A 94 3F EB 07 5D A9 70 C4 26'
00001269  3FEB075D A970C426
00001271  DD1084F2 38AB61C5                          804          DC    XL16'DD 10 84 F2 38 AB 61 C5 0C 97 4A E3 7B B6 29 D1'
00001279  0C974AE3 7BB629D1
00001281  5F8A15CC 73F800A6                          805          DC    XL16'5F 8A 15 CC 73 F8 00 A6 42 9C E5 37 BD 68 02 D7'
00001289  429CE537 BD6802D7
00001291  904DEA22 7EC059F1                          806          DC    XL16'90 4D EA 22 7E C0 59 F1 0F A4 6D DB 35 88 B3 1C'
00001299  0FA46DDB 3588B31C
000012A1  E0712BC9 469EF460                          807          DC    XL16'E0 71 2B C9 46 9E F4 60 AD 18 D5 83 3A BF 54 07'
000012A9  AD18D583 3ABF5407
000012B1  A73CF108 6DB249DE                          808          DC    XL16'A7 3C F1 08 6D B2 49 DE 15 83 5A C6 2F 91 E4 70'
000012B9  15835AC6 2F91E470
000012C1  D30B58AC E9347F12                          809          DC    XL16'D3 0B 58 AC E9 34 7F 12 C1 66 9D 42 F8 25 B7 5E'
000012C9  C1669D42 F825B75E
000012D1  81DA3704 9AE26BC5                          810          DC    XL16'81 DA 37 04 9A E2 6B C5 18 F3 4D 76 A0 2C BF 53'
000012D9  18F34D76 A02CBF53
000012E1  E7901D68 C83FA572                          811          DC    XL16'E7 90 1D 68 C8 3F A5 72 0E DB 46 B1 59 F6 23 8C'
000012E9  0EDB46B1 59F6238C
000012F1  35CE7A11 E0945FB8                          812          DC    XL16'35 CE 7A 11 E0 94 5F B8 27 D4 63 0A F9 41 AD 86'
000012F9  27D4630A F941AD86
00001301  C31A75EC 509F28D7                          813          DC    XL16'C3 1A 75 EC 50 9F 28 D7 62 B4 0D E8 39 87 FA 14'
00001309  62B40DE8 3987FA14
00001311  6FC03295 4ABE07D1                          814          DC    XL16'6F C0 32 95 4A BE 07 D1 7C E5 21 89 56 F2 A9 3D'
00001319  7CE52189 56F2A93D
00001321  B648DF10 649C2AF5                          815          DC    XL16'B6 48 DF 10 64 9C 2A F5 73 C7 0F 82 ED 31 A4 5B'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001329  73C70F82 ED31A45B
00001331  D926904F B36AE10C                          816          DC    XL16'D9 26 90 4F B3 6A E1 0C 78 C2 55 AF 13 F7 84 3E'
00001339  78C255AF 13F7843E
00001341  69D5209B 47EA7106                          817          DC    XL16'69 D5 20 9B 47 EA 71 06 BD 52 F0 38 8E C9 16 A2'
00001349  BD52F038 8EC916A2
00001351  5DE62B97 40BC037E                          818          DC    XL16'5D E6 2B 97 40 BC 03 7E D0 19 A8 65 F4 33 8A C4'
00001359  D019A865 F4338AC4
00001361  1F74EB57 920AD843                          819          DC    XL16'1F 74 EB 57 92 0A D8 43 B5 6C F1 2E 80 CD 39 A6'
00001369  B56CF12E 80CD39A6
00001371  E3519E24 7BC80DB0                          820          DC    XL16'E3 51 9E 24 7B C8 0D B0 67 F5 32 8F DA 14 46 AC'
00001379  67F5328F DA1446AC
00001381  79E90593 2DC658B1                          821          DC    XL16'79 E9 05 93 2D C6 58 B1 7F 30 ED 84 12 A5 6E D2'
00001389  7F30ED84 12A56ED2
00001391  4CF8219A 63BE07D6                          822          DC    XL16'4C F8 21 9A 63 BE 07 D6 50 C3 88 1B F0 35 A9 72'
00001399  50C3881B F035A972
000013A1  DB448D17 E269B72C                          823          DC    XL16'DB 44 8D 17 E2 69 B7 2C 95 0F C1 5A F6 38 AE 73'
000013A9  950FC15A F638AE73
000013B1  A73DF218 9CE641BB                          824          DC    XL16'A7 3D F2 18 9C E6 41 BB 72 0F D5 84 2A C1 67 9E'
000013B9  720FD584 2AC1679E
000013C1  54D823FA 814CB716                          825          DC    XL16'54 D8 23 FA 81 4C B7 16 ED 90 3B 65 C2 A9 7F 08'
000013C9  ED903B65 C2A97F08
000013D1  CE31956A F41B80D7                          826          DC    XL16'CE 31 95 6A F4 1B 80 D7 2D A3 5E C8 74 19 BF 60'
000013D9  2DA35EC8 7419BF60
000013E1  12E54A93 DC2778B1                          827          DC    XL16'12 E5 4A 93 DC 27 78 B1 0D F6 3C A8 55 EA 82 1F'
000013E9  0DF63CA8 55EA821F
000013F1  69B407FD 38C7912E                          828          DC    XL16'69 B4 07 FD 38 C7 91 2E AB 54 E0 1C 73 8D F9 46'
000013F9  AB54E01C 738DF946
00001401  D12B865F C418A7E3                          829          DC    XL16'D1 2B 86 5F C4 18 A7 E3 79 0A BD 62 F0 35 9C 4E'
00001409  790ABD62 F0359C4E
00001411  83DA24B8 6F11E75A                          830          DC    XL16'83 DA 24 B8 6F 11 E7 5A C1 9D 30 FC 47 A2 68 BE'
00001419  C19D30FC 47A268BE
00001421  1594CB73 2FE801AD                          831          DC    XL16'15 94 CB 73 2F E8 01 AD 56 39 F2 8C D7 40 B5 1A'
00001429  5639F28C D740B51A
00001431  EF620CB9 45D37E28                          832          DC    XL16'EF 62 0C B9 45 D3 7E 28 A4 F1 9A 16 C0 5B 87 3D'
00001439  A4F19A16 C05B873D
00001441  71E620DF 8B34CA59                          833          DC    XL16'71 E6 20 DF 8B 34 CA 59 13 AC F8 42 9F 65 0E B7'
00001449  13ACF842 9F650EB7
00001451  2C98E174 06D5BA3F                          834          DC    XL16'2C 98 E1 74 06 D5 BA 3F C7 50 8E 19 F4 63 AD 21'
00001459  C7508E19 F463AD21
00001461  DB457AC0 16E9923B                          835          DC    XL16'DB 45 7A C0 16 E9 92 3B F5 6C A1 08 7F D4 2E B8'
00001469  F56CA108 7FD42EB8
00001471  58F31D84 CA6730BE                          836          DC    XL16'58 F3 1D 84 CA 67 30 BE 09 D2 E7 4F 91 25 AC 6B'
00001479  09D2E74F 9125AC6B
00001481  C51AF07D 3289E45C                          837          DC    XL16'C5 1A F0 7D 32 89 E4 5C B6 03 DA 78 2F 95 41 CE'
00001489  B603DA78 2F9541CE
00001491  8A27BD60 F914C35E                          838          DC    XL16'8A 27 BD 60 F9 14 C3 5E A0 73 1F E8 46 D1 9C 25'
00001499  A0731FE8 46D19C25
000014A1  B35D02FC 8739A610                          839          DC    XL16'B3 5D 02 FC 87 39 A6 10 ED 74 CB 58 21 9A F7 4E'
000014A9  ED74CB58 219AF74E
000014B1  64E13A8D B057C926                          840          DC    XL16'64 E1 3A 8D B0 57 C9 26 FA 13 7C D6 40 AF 95 0B'
000014B9  FA137CD6 40AF950B
000014C1  D84CF165 2E9B07B4                          841          DC    XL16'D8 4C F1 65 2E 9B 07 B4 73 CA 18 E5 5A 80 3F D2'
000014C9  73CA18E5 5A803FD2
000014D1  29BE54C7 0DF861A3                          842          DC    XL16'29 BE 54 C7 0D F8 61 A3 9E 35 DC 42 17 EB 76 90'
000014D9  9E35DC42 17EB7690
000014E1  F508B16D 24C8935F                          843          DC    XL16'F5 08 B1 6D 24 C8 93 5F EA 30 7B A4 D9 12 CE 67'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000014E9  EA307BA4 D912CE67
000014F1  813EF758 AC04D16A                          844          DC    XL16'81 3E F7 58 AC 04 D1 6A B5 29 E0 7C 13 9F 42 DB'
000014F9  B529E07C 139F42DB
00001501  6FC215E9 8457B038                          845          DC    XL16'6F C2 15 E9 84 57 B0 38 FA A1 2D 96 0B C5 73 E4'
00001509  FAA12D96 0BC573E4
00001511  378FD04B A91E65FC                          846          DC    XL16'37 8F D0 4B A9 1E 65 FC 22 B7 58 DE 90 3C E1 74'
00001519  22B758DE 903CE174
00001521  CB06A579 2FD843EF                          847          DC    XL16'CB 06 A5 79 2F D8 43 EF 14 B2 6D 80 5C FA 31 97'
00001529  14B26D80 5CFA3197
00001531  04ED7B2A C6913FD4                          848          DC    XL16'04 ED 7B 2A C6 91 3F D4 68 A8 15 BE 50 E3 7C 29'
00001539  68A815BE 50E37C29
00001541  F15A0EB3 87DC2469                          849          DC    XL16'F1 5A 0E B3 87 DC 24 69 CE 10 A7 4D F8 32 95 6B'
00001549  CE10A74D F832956B
00001551  D741BC78 1FE2530A                          850          DC    XL16'D7 41 BC 78 1F E2 53 0A AD 64 F9 35 8E C0 27 B6'
00001559  AD64F935 8EC027B6
00001561  5BE812CF 70A43D99                          851          DC    XL16'5B E8 12 CF 70 A4 3D 99 26 DF 81 4A B7 05 EC 63'
00001569  26DF814A B705EC63
00001571  9820D57F BA4613C1                          852          DC    XL16'98 20 D5 7F BA 46 13 C1 6E F4 39 8B 02 AD 75 E7'
00001579  6EF4398B 02AD75E7
00001581  1CC952FE 8437B068                          853          DC    XL16'1C C9 52 FE 84 37 B0 68 D3 19 A6 4F E1 72 0D BC'
00001589  D319A64F E1720DBC
00001591  65F2289D C744EB16                          854          DC    XL16'65 F2 28 9D C7 44 EB 16 80 5A B3 0F D8 71 CE 24'
00001599  805AB30F D871CE24
000015A1  AF3964E5 12C07B98                          855          DC    XL16'AF 39 64 E5 12 C0 7B 98 2D F6 41 AC 57 0E D3 89'
000015A9  2DF641AC 570ED389
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    857 * ------------------------------------------------------------------------
                                                    858 * Test PLO_Extension functions are available/installed
                                                    859 * ------------------------------------------------------------------------
                                                    860          TSTPLO 24
                                                    861+*
                                                    862+* test if PLO function is available
                                                    863+*
000015B1                      000015B1              864+         USING *,R5                base for test data and test routine
000015B1                      00000000              865+         USING PLOAVAL,R5
000015B4  000015BC                                  866+T1       DC    A(CODE1)               address of test routine
000015B8  0001                                      867+         DC    H'1'              test number
000015BA  00                                        868+         DC    X'00'
000015BB  18                                        869+         DC    HL1'24'           Function Number
                                                    870+*
000015BC                                            871+CODE1    DS    0F
000015BC  4110 0118                     00000118    872+         LA    R1,(256+24)       T bit 55  (+256)
000015C0  1801                                      873+         LR    R0,R1
000015C2  EE00 0000 0000      00000000  00000000    874+         PLO   0,0,0,0
000015C8  478B 0000                     00000000    875+         BZ    0(R11)            Function is installed, return
000015CC  45F0 8410                     00000610    876+         BAL   R15,NAMSG         Function not available message
000015D0  07FB                                      877+         BR    R11               return
000015D2                                            878+         DROP  R5
                                                    879+*
                                                    880          TSTPLO 25
                                                    881+*
                                                    882+* test if PLO function is available
                                                    883+*
000015D2                      000015D2              884+         USING *,R5                base for test data and test routine
000015D2                      00000000              885+         USING PLOAVAL,R5
000015D4  000015DC                                  886+T2       DC    A(CODE2)               address of test routine
000015D8  0002                                      887+         DC    H'2'              test number
000015DA  00                                        888+         DC    X'00'
000015DB  19                                        889+         DC    HL1'25'           Function Number
                                                    890+*
000015DC                                            891+CODE2    DS    0F
000015DC  4110 0119                     00000119    892+         LA    R1,(256+25)       T bit 55  (+256)
000015E0  1801                                      893+         LR    R0,R1
000015E2  EE00 0000 0000      00000000  00000000    894+         PLO   0,0,0,0
000015E8  478B 0000                     00000000    895+         BZ    0(R11)            Function is installed, return
000015EC  45F0 8410                     00000610    896+         BAL   R15,NAMSG         Function not available message
000015F0  07FB                                      897+         BR    R11               return
000015F2                                            898+         DROP  R5
                                                    899+*
                                                    900          TSTPLO 26
                                                    901+*
                                                    902+* test if PLO function is available
                                                    903+*
000015F2                      000015F2              904+         USING *,R5                base for test data and test routine
000015F2                      00000000              905+         USING PLOAVAL,R5
000015F4  000015FC                                  906+T3       DC    A(CODE3)               address of test routine
000015F8  0003                                      907+         DC    H'3'              test number
000015FA  00                                        908+         DC    X'00'
000015FB  1A                                        909+         DC    HL1'26'           Function Number
                                                    910+*
000015FC                                            911+CODE3    DS    0F
000015FC  4110 011A                     0000011A    912+         LA    R1,(256+26)       T bit 55  (+256)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00001600  1801                                      913+          LR    R0,R1
00001602  EE00 0000 0000      00000000  00000000    914+          PLO   0,0,0,0
00001608  478B 0000                     00000000    915+          BZ    0(R11)              Function is installed, return
0000160C  45F0 8410                     00000610    916+          BAL   R15,NAMSG           Function not available message
00001610  07FB                                      917+          BR    R11                 return
00001612                                            918+          DROP  R5
                                                    919+*
                                                    920           TSTPLO 27
                                                    921+*
                                                    922+* test if PLO function is available
                                                    923+*
00001612                  00001612                  924+          USING *,R5                base for test data and test routine
00001612                  00000000                  925+          USING PLOAVAL,R5
00001614  0000161C                                  926+T4        DC    A(CODE4)               address of test routine
00001618  0004                                      927+          DC    H'4'                test number
0000161A  00                                        928+          DC    X'00'
0000161B  1B                                        929+          DC    HL1'27'             Function Number
                                                    930+*
0000161C                                            931+CODE4     DS    0F
0000161C  4110 011B                     0000011B    932+          LA    R1,(256+27)         T bit 55  (+256)
00001620  1801                                      933+          LR    R0,R1
00001622  EE00 0000 0000      00000000  00000000    934+          PLO   0,0,0,0
00001628  478B 0000                     00000000    935+          BZ    0(R11)              Function is installed, return
0000162C  45F0 8410                     00000610    936+          BAL   R15,NAMSG           Function not available message
00001630  07FB                                      937+          BR    R11                 return
00001632                                            938+          DROP  R5
                                                    939+*
                                                    940           TSTPLO 28
                                                    941+*
                                                    942+* test if PLO function is available
                                                    943+*
00001632                  00001632                  944+          USING *,R5                base for test data and test routine
00001632                  00000000                  945+          USING PLOAVAL,R5
00001634  0000163C                                  946+T5        DC    A(CODE5)               address of test routine
00001638  0005                                      947+          DC    H'5'                test number
0000163A  00                                        948+          DC    X'00'
0000163B  1C                                        949+          DC    HL1'28'             Function Number
                                                    950+*
0000163C                                            951+CODE5     DS    0F
0000163C  4110 011C                     0000011C    952+          LA    R1,(256+28)         T bit 55  (+256)
00001640  1801                                      953+          LR    R0,R1
00001642  EE00 0000 0000      00000000  00000000    954+          PLO   0,0,0,0
00001648  478B 0000                     00000000    955+          BZ    0(R11)              Function is installed, return
0000164C  45F0 8410                     00000610    956+          BAL   R15,NAMSG           Function not available message
00001650  07FB                                      957+          BR    R11                 return
00001652                                            958+          DROP  R5
                                                    959+*
                                                    960           TSTPLO 29
                                                    961+*
                                                    962+* test if PLO function is available
                                                    963+*
00001652                  00001652                  964+          USING *,R5                base for test data and test routine
00001652                  00000000                  965+          USING PLOAVAL,R5
00001654  0000165C                                  966+T6        DC    A(CODE6)               address of test routine
00001658  0006                                      967+          DC    H'6'                test number
0000165A  00                                        968+          DC    X'00'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0000165B  1D                                        969+          DC    HL1'29'             Function Number
                                                    970+*
0000165C                                            971+CODE6     DS    0F
0000165C  4110 011D                       0000011D  972+          LA    R1,(256+29)         T bit 55  (+256)
00001660  1801                                      973+          LR    R0,R1
00001662  EE00 0000 0000      00000000    00000000  974+          PLO   0,0,0,0
00001668  478B 0000                       00000000  975+          BZ    0(R11)              Function is installed, return
0000166C  45F0 8410                       00000610  976+          BAL   R15,NAMSG           Function not available message
00001670  07FB                                      977+          BR    R11                 return
00001672                                            978+          DROP  R5
                                                    979+*
                                                    980           TSTPLO 30
                                                    981+*
                                                    982+* test if PLO function is available
                                                    983+*
00001672                      00001672              984+          USING *,R5                base for test data and test routine
00001672                      00000000              985+          USING PLOAVAL,R5
00001674  0000167C                                  986+T7        DC    A(CODE7)               address of test routine
00001678  0007                                      987+          DC    H'7'                test number
0000167A  00                                        988+          DC    X'00'
0000167B  1E                                        989+          DC    HL1'30'             Function Number
                                                    990+*
0000167C                                            991+CODE7     DS    0F
0000167C  4110 011E                       0000011E  992+          LA    R1,(256+30)         T bit 55  (+256)
00001680  1801                                      993+          LR    R0,R1
00001682  EE00 0000 0000      00000000    00000000  994+          PLO   0,0,0,0
00001688  478B 0000                       00000000  995+          BZ    0(R11)              Function is installed, return
0000168C  45F0 8410                       00000610  996+          BAL   R15,NAMSG           Function not available message
00001690  07FB                                      997+          BR    R11                 return
00001692                                            998+          DROP  R5
                                                    999+*
                                                   1000           TSTPLO 31
                                                   1001+*
                                                   1002+* test if PLO function is available
                                                   1003+*
00001692                      00001692             1004+          USING *,R5                base for test data and test routine
00001692                      00000000             1005+          USING PLOAVAL,R5
00001694  0000169C                                 1006+T8        DC    A(CODE8)               address of test routine
00001698  0008                                     1007+          DC    H'8'                test number
0000169A  00                                       1008+          DC    X'00'
0000169B  1F                                       1009+          DC    HL1'31'             Function Number
                                                   1010+*
0000169C                                           1011+CODE8     DS    0F
0000169C  4110 011F                       0000011F 1012+          LA    R1,(256+31)         T bit 55  (+256)
000016A0  1801                                     1013+          LR    R0,R1
000016A2  EE00 0000 0000      00000000    00000000 1014+          PLO   0,0,0,0
000016A8  478B 0000                       00000000 1015+          BZ    0(R11)              Function is installed, return
000016AC  45F0 8410                       00000610 1016+          BAL   R15,NAMSG           Function not available message
000016B0  07FB                                     1017+          BR    R11                 return
000016B2                                           1018+          DROP  R5
                                                   1019+*
                                                   1020           TSTPLO 32
                                                   1021+*
                                                   1022+* test if PLO function is available
                                                   1023+*
000016B2                      000016B2             1024+          USING *,R5                base for test data and test routine
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000016B2                       00000000            1025+          USING PLOAVAL,R5
000016B4  000016BC                                 1026+T9        DC    A(CODE9)               address of test routine
000016B8  0009                                     1027+          DC    H'9'                   test number
000016BA  00                                       1028+          DC    X'00'
000016BB  20                                       1029+          DC    HL1'32'                Function Number
                                                   1030+*
000016BC                                           1031+CODE9     DS    0F
000016BC  4110 0120                      00000120  1032+          LA    R1,(256+32)            T bit 55  (+256)
000016C0  1801                                     1033+          LR    R0,R1
000016C2  EE00 0000 0000       00000000  00000000  1034+          PLO   0,0,0,0
000016C8  478B 0000                      00000000  1035+          BZ    0(R11)                 Function is installed, return
000016CC  45F0 8410                      00000610  1036+          BAL   R15,NAMSG              Function not available message
000016D0  07FB                                     1037+          BR    R11                    return
000016D2                                           1038+          DROP  R5
                                                   1039+*
                                                   1040           TSTPLO 33
                                                   1041+*
                                                   1042+* test if PLO function is available
                                                   1043+*
000016D2                       000016D2            1044+          USING *,R5                   base for test data and test routine
000016D2                       00000000            1045+          USING PLOAVAL,R5
000016D4  000016DC                                 1046+T10       DC    A(CODE10)              address of test routine
000016D8  000A                                     1047+          DC    H'10'                  test number
000016DA  00                                       1048+          DC    X'00'
000016DB  21                                       1049+          DC    HL1'33'                Function Number
                                                   1050+*
000016DC                                           1051+CODE10    DS    0F
000016DC  4110 0121                      00000121  1052+          LA    R1,(256+33)            T bit 55  (+256)
000016E0  1801                                     1053+          LR    R0,R1
000016E2  EE00 0000 0000       00000000  00000000  1054+          PLO   0,0,0,0
000016E8  478B 0000                      00000000  1055+          BZ    0(R11)                 Function is installed, return
000016EC  45F0 8410                      00000610  1056+          BAL   R15,NAMSG              Function not available message
000016F0  07FB                                     1057+          BR    R11                    return
000016F2                                           1058+          DROP  R5
                                                   1059+*
                                                   1060           TSTPLO 34
                                                   1061+*
                                                   1062+* test if PLO function is available
                                                   1063+*
000016F2                       000016F2            1064+          USING *,R5                   base for test data and test routine
000016F2                       00000000            1065+          USING PLOAVAL,R5
000016F4  000016FC                                 1066+T11       DC    A(CODE11)              address of test routine
000016F8  000B                                     1067+          DC    H'11'                  test number
000016FA  00                                       1068+          DC    X'00'
000016FB  22                                       1069+          DC    HL1'34'                Function Number
                                                   1070+*
000016FC                                           1071+CODE11    DS    0F
000016FC  4110 0122                      00000122  1072+          LA    R1,(256+34)            T bit 55  (+256)
00001700  1801                                     1073+          LR    R0,R1
00001702  EE00 0000 0000       00000000  00000000  1074+          PLO   0,0,0,0
00001708  478B 0000                      00000000  1075+          BZ    0(R11)                 Function is installed, return
0000170C  45F0 8410                      00000610  1076+          BAL   R15,NAMSG              Function not available message
00001710  07FB                                     1077+          BR    R11                    return
00001712                                           1078+          DROP  R5
                                                   1079+*
                                                   1080           TSTPLO 35
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   1081+*
                                                   1082+* test if PLO function is available
                                                   1083+*
00001712                     00001712              1084+         USING *,R5                base for test data and test routine
00001712                     00000000              1085+         USING PLOAVAL,R5
00001714  0000171C                                 1086+T12      DC    A(CODE12)              address of test routine
00001718  000C                                     1087+         DC    H'12'             test number
0000171A  00                                       1088+         DC    X'00'
0000171B  23                                       1089+         DC    HL1'35'           Function Number
                                                   1090+*
0000171C                                           1091+CODE12   DS    0F
0000171C  4110 0123                    00000123    1092+         LA    R1,(256+35)       T bit 55  (+256)
00001720  1801                                     1093+         LR    R0,R1
00001722  EE00 0000 0000     00000000  00000000    1094+         PLO   0,0,0,0
00001728  478B 0000                    00000000    1095+         BZ    0(R11)            Function is installed, return
0000172C  45F0 8410                    00000610    1096+         BAL   R15,NAMSG         Function not available message
00001730  07FB                                     1097+         BR    R11               return
00001732                                           1098+         DROP  R5
                                                   1099+*
                                                   1100          TSTPLO 36
                                                   1101+*
                                                   1102+* test if PLO function is available
                                                   1103+*
00001732                     00001732              1104+         USING *,R5                base for test data and test routine
00001732                     00000000              1105+         USING PLOAVAL,R5
00001734  0000173C                                 1106+T13      DC    A(CODE13)              address of test routine
00001738  000D                                     1107+         DC    H'13'             test number
0000173A  00                                       1108+         DC    X'00'
0000173B  24                                       1109+         DC    HL1'36'           Function Number
                                                   1110+*
0000173C                                           1111+CODE13   DS    0F
0000173C  4110 0124                    00000124    1112+         LA    R1,(256+36)       T bit 55  (+256)
00001740  1801                                     1113+         LR    R0,R1
00001742  EE00 0000 0000     00000000  00000000    1114+         PLO   0,0,0,0
00001748  478B 0000                    00000000    1115+         BZ    0(R11)            Function is installed, return
0000174C  45F0 8410                    00000610    1116+         BAL   R15,NAMSG         Function not available message
00001750  07FB                                     1117+         BR    R11               return
00001752                                           1118+         DROP  R5
                                                   1119+*
                                                   1120          TSTPLO 37
                                                   1121+*
                                                   1122+* test if PLO function is available
                                                   1123+*
00001752                     00001752              1124+         USING *,R5                base for test data and test routine
00001752                     00000000              1125+         USING PLOAVAL,R5
00001754  0000175C                                 1126+T14      DC    A(CODE14)              address of test routine
00001758  000E                                     1127+         DC    H'14'             test number
0000175A  00                                       1128+         DC    X'00'
0000175B  25                                       1129+         DC    HL1'37'           Function Number
                                                   1130+*
0000175C                                           1131+CODE14   DS    0F
0000175C  4110 0125                    00000125    1132+         LA    R1,(256+37)       T bit 55  (+256)
00001760  1801                                     1133+         LR    R0,R1
00001762  EE00 0000 0000     00000000  00000000    1134+         PLO   0,0,0,0
00001768  478B 0000                    00000000    1135+         BZ    0(R11)            Function is installed, return
0000176C  45F0 8410                    00000610    1136+         BAL   R15,NAMSG         Function not available message
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00001770  07FB                                    1137+          BR    R11                 return
00001772                                          1138+          DROP  R5
                                                  1139+*
                                                  1140           TSTPLO 38
                                                  1141+*
                                                  1142+* test if PLO function is available
                                                  1143+*
00001772                      00001772            1144+          USING *,R5                base for test data and test routine
00001772                      00000000            1145+          USING PLOAVAL,R5
00001774  0000177C                                1146+T15       DC    A(CODE15)              address of test routine
00001778  000F                                    1147+          DC    H'15'               test number
0000177A  00                                      1148+          DC    X'00'
0000177B  26                                      1149+          DC    HL1'38'             Function Number
                                                  1150+*
0000177C                                          1151+CODE15    DS    0F
0000177C  4110 0126                     00000126  1152+          LA    R1,(256+38)         T bit 55  (+256)
00001780  1801                                    1153+          LR    R0,R1
00001782  EE00 0000 0000      00000000  00000000  1154+          PLO   0,0,0,0
00001788  478B 0000                     00000000  1155+          BZ    0(R11)              Function is installed, return
0000178C  45F0 8410                     00000610  1156+          BAL   R15,NAMSG           Function not available message
00001790  07FB                                    1157+          BR    R11                 return
00001792                                          1158+          DROP  R5
                                                  1159+*
                                                  1160           TSTPLO 39
                                                  1161+*
                                                  1162+* test if PLO function is available
                                                  1163+*
00001792                      00001792            1164+          USING *,R5                base for test data and test routine
00001792                      00000000            1165+          USING PLOAVAL,R5
00001794  0000179C                                1166+T16       DC    A(CODE16)              address of test routine
00001798  0010                                    1167+          DC    H'16'               test number
0000179A  00                                      1168+          DC    X'00'
0000179B  27                                      1169+          DC    HL1'39'             Function Number
                                                  1170+*
0000179C                                          1171+CODE16    DS    0F
0000179C  4110 0127                     00000127  1172+          LA    R1,(256+39)         T bit 55  (+256)
000017A0  1801                                    1173+          LR    R0,R1
000017A2  EE00 0000 0000      00000000  00000000  1174+          PLO   0,0,0,0
000017A8  478B 0000                     00000000  1175+          BZ    0(R11)              Function is installed, return
000017AC  45F0 8410                     00000610  1176+          BAL   R15,NAMSG           Function not available message
000017B0  07FB                                    1177+          BR    R11                 return
000017B2                                          1178+          DROP  R5
                                                  1179+*
                                                  1180           TSTPLO 40
                                                  1181+*
                                                  1182+* test if PLO function is available
                                                  1183+*
000017B2                      000017B2            1184+          USING *,R5                base for test data and test routine
000017B2                      00000000            1185+          USING PLOAVAL,R5
000017B4  000017BC                                1186+T17       DC    A(CODE17)              address of test routine
000017B8  0011                                    1187+          DC    H'17'               test number
000017BA  00                                      1188+          DC    X'00'
000017BB  28                                      1189+          DC    HL1'40'             Function Number
                                                  1190+*
000017BC                                          1191+CODE17    DS    0F
000017BC  4110 0128                     00000128  1192+          LA    R1,(256+40)         T bit 55  (+256)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000017C0  1801                                     1193+          LR    R0,R1
000017C2  EE00 0000 0000      00000000  00000000   1194+          PLO   0,0,0,0
000017C8  478B 0000                     00000000   1195+          BZ    0(R11)              Function is installed, return
000017CC  45F0 8410                     00000610   1196+          BAL   R15,NAMSG           Function not available message
000017D0  07FB                                     1197+          BR    R11                 return
000017D2                                           1198+          DROP  R5
                                                   1199+*
                                                   1200           TSTPLO 41
                                                   1201+*
                                                   1202+* test if PLO function is available
                                                   1203+*
000017D2                        000017D2           1204+          USING *,R5                base for test data and test routine
000017D2                        00000000           1205+          USING PLOAVAL,R5
000017D4  000017DC                                 1206+T18       DC    A(CODE18)              address of test routine
000017D8  0012                                     1207+          DC    H'18'               test number
000017DA  00                                       1208+          DC    X'00'
000017DB  29                                       1209+          DC    HL1'41'             Function Number
                                                   1210+*
000017DC                                           1211+CODE18    DS    0F
000017DC  4110 0129                     00000129   1212+          LA    R1,(256+41)         T bit 55  (+256)
000017E0  1801                                     1213+          LR    R0,R1
000017E2  EE00 0000 0000      00000000  00000000   1214+          PLO   0,0,0,0
000017E8  478B 0000                     00000000   1215+          BZ    0(R11)              Function is installed, return
000017EC  45F0 8410                     00000610   1216+          BAL   R15,NAMSG           Function not available message
000017F0  07FB                                     1217+          BR    R11                 return
000017F2                                           1218+          DROP  R5
                                                   1219+*
                                                   1220           TSTPLO 42
                                                   1221+*
                                                   1222+* test if PLO function is available
                                                   1223+*
000017F2                        000017F2           1224+          USING *,R5                base for test data and test routine
000017F2                        00000000           1225+          USING PLOAVAL,R5
000017F4  000017FC                                 1226+T19       DC    A(CODE19)              address of test routine
000017F8  0013                                     1227+          DC    H'19'               test number
000017FA  00                                       1228+          DC    X'00'
000017FB  2A                                       1229+          DC    HL1'42'             Function Number
                                                   1230+*
000017FC                                           1231+CODE19    DS    0F
000017FC  4110 012A                     0000012A   1232+          LA    R1,(256+42)         T bit 55  (+256)
00001800  1801                                     1233+          LR    R0,R1
00001802  EE00 0000 0000      00000000  00000000   1234+          PLO   0,0,0,0
00001808  478B 0000                     00000000   1235+          BZ    0(R11)              Function is installed, return
0000180C  45F0 8410                     00000610   1236+          BAL   R15,NAMSG           Function not available message
00001810  07FB                                     1237+          BR    R11                 return
00001812                                           1238+          DROP  R5
                                                   1239+*
                                                   1240           TSTPLO 43
                                                   1241+*
                                                   1242+* test if PLO function is available
                                                   1243+*
00001812                        00001812           1244+          USING *,R5                base for test data and test routine
00001812                        00000000           1245+          USING PLOAVAL,R5
00001814  0000181C                                 1246+T20       DC    A(CODE20)              address of test routine
00001818  0014                                     1247+          DC    H'20'               test number
0000181A  00                                       1248+          DC    X'00'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0000181B  2B                                       1249+          DC    HL1'43'              Function Number
                                                   1250+*
0000181C                                           1251+CODE20    DS    0F
0000181C  4110 012B                      0000012B  1252+          LA    R1,(256+43)          T bit 55  (+256)
00001820  1801                                     1253+          LR    R0,R1
00001822  EE00 0000 0000      00000000   00000000  1254+          PLO   0,0,0,0
00001828  478B 0000                      00000000  1255+          BZ    0(R11)               Function is installed, return
0000182C  45F0 8410                      00000610  1256+          BAL   R15,NAMSG            Function not available message
00001830  07FB                                     1257+          BR    R11                  return
00001832                                           1258+          DROP  R5
                                                   1259+*
                                                   1260           TSTPLO 44
                                                   1261+*
                                                   1262+* test if PLO function is available
                                                   1263+*
00001832                  00001832                 1264+          USING *,R5                 base for test data and test routine
00001832                  00000000                 1265+          USING PLOAVAL,R5
00001834  0000183C                                 1266+T21       DC    A(CODE21)               address of test routine
00001838  0015                                     1267+          DC    H'21'                test number
0000183A  00                                       1268+          DC    X'00'
0000183B  2C                                       1269+          DC    HL1'44'              Function Number
                                                   1270+*
0000183C                                           1271+CODE21    DS    0F
0000183C  4110 012C                      0000012C  1272+          LA    R1,(256+44)          T bit 55  (+256)
00001840  1801                                     1273+          LR    R0,R1
00001842  EE00 0000 0000      00000000   00000000  1274+          PLO   0,0,0,0
00001848  478B 0000                      00000000  1275+          BZ    0(R11)               Function is installed, return
0000184C  45F0 8410                      00000610  1276+          BAL   R15,NAMSG            Function not available message
00001850  07FB                                     1277+          BR    R11                  return
00001852                                           1278+          DROP  R5
                                                   1279+*
                                                   1280           TSTPLO 45
                                                   1281+*
                                                   1282+* test if PLO function is available
                                                   1283+*
00001852                  00001852                 1284+          USING *,R5                 base for test data and test routine
00001852                  00000000                 1285+          USING PLOAVAL,R5
00001854  0000185C                                 1286+T22       DC    A(CODE22)               address of test routine
00001858  0016                                     1287+          DC    H'22'                test number
0000185A  00                                       1288+          DC    X'00'
0000185B  2D                                       1289+          DC    HL1'45'              Function Number
                                                   1290+*
0000185C                                           1291+CODE22    DS    0F
0000185C  4110 012D                      0000012D  1292+          LA    R1,(256+45)          T bit 55  (+256)
00001860  1801                                     1293+          LR    R0,R1
00001862  EE00 0000 0000      00000000   00000000  1294+          PLO   0,0,0,0
00001868  478B 0000                      00000000  1295+          BZ    0(R11)               Function is installed, return
0000186C  45F0 8410                      00000610  1296+          BAL   R15,NAMSG            Function not available message
00001870  07FB                                     1297+          BR    R11                  return
00001872                                           1298+          DROP  R5
                                                   1299+*
                                                   1300           TSTPLO 46
                                                   1301+*
                                                   1302+* test if PLO function is available
                                                   1303+*
00001872                  00001872                 1304+          USING *,R5                 base for test data and test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001872                     00000000              1305+         USING PLOAVAL,R5
00001874  0000187C                                 1306+T23      DC    A(CODE23)              address of test routine
00001878  0017                                     1307+         DC    H'23'                  test number
0000187A  00                                       1308+         DC    X'00'
0000187B  2E                                       1309+         DC    HL1'46'                Function Number
                                                   1310+*
0000187C                                           1311+CODE23   DS    0F
0000187C  4110 012E                      0000012E  1312+         LA    R1,(256+46)            T bit 55  (+256)
00001880  1801                                     1313+         LR    R0,R1
00001882  EE00 0000 0000      00000000   00000000  1314+         PLO   0,0,0,0
00001888  478B 0000                      00000000  1315+         BZ    0(R11)                 Function is installed, return
0000188C  45F0 8410                      00000610  1316+         BAL   R15,NAMSG              Function not available message
00001890  07FB                                     1317+         BR    R11                    return
00001892                                           1318+         DROP  R5
                                                   1319+*
                                                   1320          TSTPLO 47
                                                   1321+*
                                                   1322+* test if PLO function is available
                                                   1323+*
00001892                     00001892              1324+         USING *,R5                   base for test data and test routine
00001892                     00000000              1325+         USING PLOAVAL,R5
00001894  0000189C                                 1326+T24      DC    A(CODE24)              address of test routine
00001898  0018                                     1327+         DC    H'24'                  test number
0000189A  00                                       1328+         DC    X'00'
0000189B  2F                                       1329+         DC    HL1'47'                Function Number
                                                   1330+*
0000189C                                           1331+CODE24   DS    0F
0000189C  4110 012F                      0000012F  1332+         LA    R1,(256+47)            T bit 55  (+256)
000018A0  1801                                     1333+         LR    R0,R1
000018A2  EE00 0000 0000      00000000   00000000  1334+         PLO   0,0,0,0
000018A8  478B 0000                      00000000  1335+         BZ    0(R11)                 Function is installed, return
000018AC  45F0 8410                      00000610  1336+         BAL   R15,NAMSG              Function not available message
000018B0  07FB                                     1337+         BR    R11                    return
000018B2                                           1338+         DROP  R5
                                                   1339+*
                                                   1340          TSTPLO 48
                                                   1341+*
                                                   1342+* test if PLO function is available
                                                   1343+*
000018B2                     000018B2              1344+         USING *,R5                   base for test data and test routine
000018B2                     00000000              1345+         USING PLOAVAL,R5
000018B4  000018BC                                 1346+T25      DC    A(CODE25)              address of test routine
000018B8  0019                                     1347+         DC    H'25'                  test number
000018BA  00                                       1348+         DC    X'00'
000018BB  30                                       1349+         DC    HL1'48'                Function Number
                                                   1350+*
000018BC                                           1351+CODE25   DS    0F
000018BC  4110 0130                      00000130  1352+         LA    R1,(256+48)            T bit 55  (+256)
000018C0  1801                                     1353+         LR    R0,R1
000018C2  EE00 0000 0000      00000000   00000000  1354+         PLO   0,0,0,0
000018C8  478B 0000                      00000000  1355+         BZ    0(R11)                 Function is installed, return
000018CC  45F0 8410                      00000610  1356+         BAL   R15,NAMSG              Function not available message
000018D0  07FB                                     1357+         BR    R11                    return
000018D2                                           1358+         DROP  R5
                                                   1359+*
                                                   1360          TSTPLO 49
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   1361+*
                                                   1362+* test if PLO function is available
                                                   1363+*
000018D2                     000018D2              1364+         USING *,R5                base for test data and test routine
000018D2                     00000000              1365+         USING PLOAVAL,R5
000018D4  000018DC                                 1366+T26      DC    A(CODE26)              address of test routine
000018D8  001A                                     1367+         DC    H'26'             test number
000018DA  00                                       1368+         DC    X'00'
000018DB  31                                       1369+         DC    HL1'49'           Function Number
                                                   1370+*
000018DC                                           1371+CODE26   DS    0F
000018DC  4110 0131                    00000131    1372+         LA    R1,(256+49)       T bit 55  (+256)
000018E0  1801                                     1373+         LR    R0,R1
000018E2  EE00 0000 0000     00000000  00000000    1374+         PLO   0,0,0,0
000018E8  478B 0000                    00000000    1375+         BZ    0(R11)            Function is installed, return
000018EC  45F0 8410                    00000610    1376+         BAL   R15,NAMSG         Function not available message
000018F0  07FB                                     1377+         BR    R11               return
000018F2                                           1378+         DROP  R5
                                                   1379+*
                                                   1380          TSTPLO 50
                                                   1381+*
                                                   1382+* test if PLO function is available
                                                   1383+*
000018F2                     000018F2              1384+         USING *,R5                base for test data and test routine
000018F2                     00000000              1385+         USING PLOAVAL,R5
000018F4  000018FC                                 1386+T27      DC    A(CODE27)              address of test routine
000018F8  001B                                     1387+         DC    H'27'             test number
000018FA  00                                       1388+         DC    X'00'
000018FB  32                                       1389+         DC    HL1'50'           Function Number
                                                   1390+*
000018FC                                           1391+CODE27   DS    0F
000018FC  4110 0132                    00000132    1392+         LA    R1,(256+50)       T bit 55  (+256)
00001900  1801                                     1393+         LR    R0,R1
00001902  EE00 0000 0000     00000000  00000000    1394+         PLO   0,0,0,0
00001908  478B 0000                    00000000    1395+         BZ    0(R11)            Function is installed, return
0000190C  45F0 8410                    00000610    1396+         BAL   R15,NAMSG         Function not available message
00001910  07FB                                     1397+         BR    R11               return
00001912                                           1398+         DROP  R5
                                                   1399+*
                                                   1400          TSTPLO 51
                                                   1401+*
                                                   1402+* test if PLO function is available
                                                   1403+*
00001912                     00001912              1404+         USING *,R5                base for test data and test routine
00001912                     00000000              1405+         USING PLOAVAL,R5
00001914  0000191C                                 1406+T28      DC    A(CODE28)              address of test routine
00001918  001C                                     1407+         DC    H'28'             test number
0000191A  00                                       1408+         DC    X'00'
0000191B  33                                       1409+         DC    HL1'51'           Function Number
                                                   1410+*
0000191C                                           1411+CODE28   DS    0F
0000191C  4110 0133                    00000133    1412+         LA    R1,(256+51)       T bit 55  (+256)
00001920  1801                                     1413+         LR    R0,R1
00001922  EE00 0000 0000     00000000  00000000    1414+         PLO   0,0,0,0
00001928  478B 0000                    00000000    1415+         BZ    0(R11)            Function is installed, return
0000192C  45F0 8410                    00000610    1416+         BAL   R15,NAMSG         Function not available message
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00001930  07FB                                    1417+         BR    R11                 return
00001932                                          1418+         DROP  R5
                                                  1419+*
                                                  1420          TSTPLO 52
                                                  1421+*
                                                  1422+* test if PLO function is available
                                                  1423+*
00001932                    00001932              1424+         USING *,R5                base for test data and test routine
00001932                    00000000              1425+         USING PLOAVAL,R5
00001934  0000193C                                1426+T29      DC    A(CODE29)              address of test routine
00001938  001D                                    1427+         DC    H'29'               test number
0000193A  00                                      1428+         DC    X'00'
0000193B  34                                      1429+         DC    HL1'52'             Function Number
                                                  1430+*
0000193C                                          1431+CODE29   DS    0F
0000193C  4110 0134                     00000134  1432+         LA    R1,(256+52)         T bit 55  (+256)
00001940  1801                                    1433+         LR    R0,R1
00001942  EE00 0000 0000    00000000    00000000  1434+         PLO   0,0,0,0
00001948  478B 0000                     00000000  1435+         BZ    0(R11)              Function is installed, return
0000194C  45F0 8410                     00000610  1436+         BAL   R15,NAMSG           Function not available message
00001950  07FB                                    1437+         BR    R11                 return
00001952                                          1438+         DROP  R5
                                                  1439+*
                                                  1440          TSTPLO 53
                                                  1441+*
                                                  1442+* test if PLO function is available
                                                  1443+*
00001952                    00001952              1444+         USING *,R5                base for test data and test routine
00001952                    00000000              1445+         USING PLOAVAL,R5
00001954  0000195C                                1446+T30      DC    A(CODE30)              address of test routine
00001958  001E                                    1447+         DC    H'30'               test number
0000195A  00                                      1448+         DC    X'00'
0000195B  35                                      1449+         DC    HL1'53'             Function Number
                                                  1450+*
0000195C                                          1451+CODE30   DS    0F
0000195C  4110 0135                     00000135  1452+         LA    R1,(256+53)         T bit 55  (+256)
00001960  1801                                    1453+         LR    R0,R1
00001962  EE00 0000 0000    00000000    00000000  1454+         PLO   0,0,0,0
00001968  478B 0000                     00000000  1455+         BZ    0(R11)              Function is installed, return
0000196C  45F0 8410                     00000610  1456+         BAL   R15,NAMSG           Function not available message
00001970  07FB                                    1457+         BR    R11                 return
00001972                                          1458+         DROP  R5
                                                  1459+*
                                                  1460          TSTPLO 54
                                                  1461+*
                                                  1462+* test if PLO function is available
                                                  1463+*
00001972                    00001972              1464+         USING *,R5                base for test data and test routine
00001972                    00000000              1465+         USING PLOAVAL,R5
00001974  0000197C                                1466+T31      DC    A(CODE31)              address of test routine
00001978  001F                                    1467+         DC    H'31'               test number
0000197A  00                                      1468+         DC    X'00'
0000197B  36                                      1469+         DC    HL1'54'             Function Number
                                                  1470+*
0000197C                                          1471+CODE31   DS    0F
0000197C  4110 0136                     00000136  1472+         LA    R1,(256+54)         T bit 55  (+256)
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00001980  1801                                    1473+          LR    R0,R1
00001982  EE00 0000 0000      00000000  00000000  1474+          PLO   0,0,0,0
00001988  478B 0000                     00000000  1475+          BZ    0(R11)             Function is installed, return
0000198C  45F0 8410                     00000610  1476+          BAL   R15,NAMSG          Function not available message
00001990  07FB                                    1477+          BR    R11                return
00001992                                          1478+          DROP  R5
                                                  1479+*
                                                  1480           TSTPLO 55
                                                  1481+*
                                                  1482+* test if PLO function is available
                                                  1483+*
00001992                  00001992                1484+          USING *,R5               base for test data and test routine
00001992                  00000000                1485+          USING PLOAVAL,R5
00001994  0000199C                               1486+T32       DC    A(CODE32)             address of test routine
00001998  0020                                    1487+          DC    H'32'              test number
0000199A  00                                      1488+          DC    X'00'
0000199B  37                                      1489+          DC    HL1'55'            Function Number
                                                  1490+*
0000199C                                          1491+CODE32    DS    0F
0000199C  4110 0137                     00000137  1492+          LA    R1,(256+55)        T bit 55  (+256)
000019A0  1801                                    1493+          LR    R0,R1
000019A2  EE00 0000 0000      00000000  00000000  1494+          PLO   0,0,0,0
000019A8  478B 0000                     00000000  1495+          BZ    0(R11)             Function is installed, return
000019AC  45F0 8410                     00000610  1496+          BAL   R15,NAMSG          Function not available message
000019B0  07FB                                    1497+          BR    R11                return
000019B2                                          1498+          DROP  R5
                                                  1499+*
                                                  1500           TSTPLO 56
                                                  1501+*
                                                  1502+* test if PLO function is available
                                                  1503+*
000019B2                  000019B2                1504+          USING *,R5               base for test data and test routine
000019B2                  00000000                1505+          USING PLOAVAL,R5
000019B4  000019BC                                1506+T33       DC    A(CODE33)             address of test routine
000019B8  0021                                    1507+          DC    H'33'              test number
000019BA  00                                      1508+          DC    X'00'
000019BB  38                                      1509+          DC    HL1'56'            Function Number
                                                  1510+*
000019BC                                          1511+CODE33    DS    0F
000019BC  4110 0138                     00000138  1512+          LA    R1,(256+56)        T bit 55  (+256)
000019C0  1801                                    1513+          LR    R0,R1
000019C2  EE00 0000 0000      00000000  00000000  1514+          PLO   0,0,0,0
000019C8  478B 0000                     00000000  1515+          BZ    0(R11)             Function is installed, return
000019CC  45F0 8410                     00000610  1516+          BAL   R15,NAMSG          Function not available message
000019D0  07FB                                    1517+          BR    R11                return
000019D2                                          1518+          DROP  R5
                                                  1519+*
                                                  1520           TSTPLO 57
                                                  1521+*
                                                  1522+* test if PLO function is available
                                                  1523+*
000019D2                  000019D2                1524+          USING *,R5               base for test data and test routine
000019D2                  00000000                1525+          USING PLOAVAL,R5
000019D4  000019DC                                1526+T34       DC    A(CODE34)             address of test routine
000019D8  0022                                    1527+          DC    H'34'              test number
000019DA  00                                      1528+          DC    X'00'
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000019DB  39                                       1529+          DC    HL1'57'              Function Number
                                                   1530+*
000019DC                                           1531+CODE34    DS    0F
000019DC  4110 0139                      00000139  1532+          LA    R1,(256+57)          T bit 55  (+256)
000019E0  1801                                     1533+          LR    R0,R1
000019E2  EE00 0000 0000      00000000   00000000  1534+          PLO   0,0,0,0
000019E8  478B 0000                      00000000  1535+          BZ    0(R11)               Function is installed, return
000019EC  45F0 8410                      00000610  1536+          BAL   R15,NAMSG            Function not available message
000019F0  07FB                                     1537+          BR    R11                  return
000019F2                                           1538+          DROP  R5
                                                   1539+*
                                                   1540           TSTPLO 58
                                                   1541+*
                                                   1542+* test if PLO function is available
                                                   1543+*
000019F2                      000019F2             1544+          USING *,R5                 base for test data and test routine
000019F2                      00000000             1545+          USING PLOAVAL,R5
000019F4  000019FC                                 1546+T35       DC    A(CODE35)               address of test routine
000019F8  0023                                     1547+          DC    H'35'                test number
000019FA  00                                       1548+          DC    X'00'
000019FB  3A                                       1549+          DC    HL1'58'              Function Number
                                                   1550+*
000019FC                                           1551+CODE35    DS    0F
000019FC  4110 013A                      0000013A  1552+          LA    R1,(256+58)          T bit 55  (+256)
00001A00  1801                                     1553+          LR    R0,R1
00001A02  EE00 0000 0000      00000000   00000000  1554+          PLO   0,0,0,0
00001A08  478B 0000                      00000000  1555+          BZ    0(R11)               Function is installed, return
00001A0C  45F0 8410                      00000610  1556+          BAL   R15,NAMSG            Function not available message
00001A10  07FB                                     1557+          BR    R11                  return
00001A12                                           1558+          DROP  R5
                                                   1559+*
                                                   1560           TSTPLO 59
                                                   1561+*
                                                   1562+* test if PLO function is available
                                                   1563+*
00001A12                      00001A12             1564+          USING *,R5                 base for test data and test routine
00001A12                      00000000             1565+          USING PLOAVAL,R5
00001A14  00001A1C                                 1566+T36       DC    A(CODE36)               address of test routine
00001A18  0024                                     1567+          DC    H'36'                test number
00001A1A  00                                       1568+          DC    X'00'
00001A1B  3B                                       1569+          DC    HL1'59'              Function Number
                                                   1570+*
00001A1C                                           1571+CODE36    DS    0F
00001A1C  4110 013B                      0000013B  1572+          LA    R1,(256+59)          T bit 55  (+256)
00001A20  1801                                     1573+          LR    R0,R1
00001A22  EE00 0000 0000      00000000   00000000  1574+          PLO   0,0,0,0
00001A28  478B 0000                      00000000  1575+          BZ    0(R11)               Function is installed, return
00001A2C  45F0 8410                      00000610  1576+          BAL   R15,NAMSG            Function not available message
00001A30  07FB                                     1577+          BR    R11                  return
00001A32                                           1578+          DROP  R5
                                                   1579+*
                                                   1580           TSTPLO 60
                                                   1581+*
                                                   1582+* test if PLO function is available
                                                   1583+*
00001A32                      00001A32             1584+          USING *,R5                 base for test data and test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00001A32                     00000000              1585+         USING PLOAVAL,R5
00001A34  00001A3C                                 1586+T37      DC    A(CODE37)              address of test routine
00001A38  0025                                     1587+         DC    H'37'                  test number
00001A3A  00                                       1588+         DC    X'00'
00001A3B  3C                                       1589+         DC    HL1'60'                Function Number
                                                   1590+*
00001A3C                                           1591+CODE37   DS    0F
00001A3C  4110 013C                      0000013C  1592+         LA    R1,(256+60)            T bit 55  (+256)
00001A40  1801                                     1593+         LR    R0,R1
00001A42  EE00 0000 0000     00000000  00000000  1594+         PLO   0,0,0,0
00001A48  478B 0000                      00000000  1595+         BZ    0(R11)                 Function is installed, return
00001A4C  45F0 8410                      00000610  1596+         BAL   R15,NAMSG              Function not available message
00001A50  07FB                                     1597+         BR    R11                    return
00001A52                                           1598+         DROP  R5
                                                   1599+*
                                                   1600          TSTPLO 61
                                                   1601+*
                                                   1602+* test if PLO function is available
                                                   1603+*
00001A52                     00001A52              1604+         USING *,R5                   base for test data and test routine
00001A52                     00000000              1605+         USING PLOAVAL,R5
00001A54  00001A5C                                 1606+T38      DC    A(CODE38)              address of test routine
00001A58  0026                                     1607+         DC    H'38'                  test number
00001A5A  00                                       1608+         DC    X'00'
00001A5B  3D                                       1609+         DC    HL1'61'                Function Number
                                                   1610+*
00001A5C                                           1611+CODE38   DS    0F
00001A5C  4110 013D                      0000013D  1612+         LA    R1,(256+61)            T bit 55  (+256)
00001A60  1801                                     1613+         LR    R0,R1
00001A62  EE00 0000 0000     00000000  00000000  1614+         PLO   0,0,0,0
00001A68  478B 0000                      00000000  1615+         BZ    0(R11)                 Function is installed, return
00001A6C  45F0 8410                      00000610  1616+         BAL   R15,NAMSG              Function not available message
00001A70  07FB                                     1617+         BR    R11                    return
00001A72                                           1618+         DROP  R5
                                                   1619+*
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    1621 *=======================================================================
                                                    1622 *       Quadruple Load / Store Tests
                                                    1623 *=======================================================================
                                                    1624
                                                    1625 * ---------------------------------------------------------------------
                                                    1626 * Test Quadruple Load(QLO: 48) / Store(QSTO: 61) instructions
                                                    1627 * ---------------------------------------------------------------------
                                                    1628          TSTHDR 99,99,61
00001A72                          00001A72  00033000  1629+         ORG   PLOETST+(200*K)+(1*4*K)
00033000  E2E3C1D9 E3409686                         1630+         DC    CL32'START of test: 39'
00033008  40A385A2 A37A40F3
00033010  F9404040 40404040
00033018  40404040 40404040
00033020                          00033020          1631+         USING *,R5                base for test data and test routine
00033020                          00000000          1632+         USING PLOETEST,R5
00033020  0003304C                                  1633+T39      DC    A(CODE39)           address of test code
00033024  0027                                      1634+         DC    H'39'               test number
00033026  00                                        1635+         DC    X'00'
00033027  3D                                        1636+         DC    HL1'61'             Function
00033028  63                                        1637+         DC    HL1'99'             expected unequal results (UR)
00033029  00                                        1638+         DC    HL1'00'             found UR
0003302A  63                                        1639+         DC    HL1'99'             cc
0003302B  00                                        1640+         DC    HL1'00'             cc not checked
0003302C  00                                        1641+         DS    X                   extracted cc
00033030  00000000 00000000                         1642+         DS    2F                  extract PSW after test (has CC)
00033038  00033600                                  1643+         DC    A(FMSG39)           address of test fail message
0003303C  00033480                                  1644+         DC    A(OPSRC39)          address of operand source
00033040  00033400                                  1645+         DC    A(OPDST39)          address of operand destination
00033044  00033500                                  1646+         DC    A(PL39)             address of parameter list
00033048  000011D8                                  1647+         DC    A(RANDX+39)         address of random source data
                                                    1648
                                                    1649          TSTCODE
0003304C                                            1650+CODE39   DS    0F
                                                    1651 *                                  load random source
0003304C  B982 0011                                 1652          XGR   R1,R1
00033050  5810 501C                       0000001C  1653          L     R1,OPSRCA
00033054  B982 0022                                 1654          XGR   R2,R2
00033058  5820 5028                       00000028  1655          L     R2,RANDXA
0003305C  D27F 1000 2000      00000000    00000000  1656          MVC   0(128,R1),0(R2)
                                                    1657 *
                                                    1658 * Quadruple Load(QLO: 48)
                                                    1659 *
00033062  B982 0033                                 1660          XGR   R3,R3
00033066  5830 5024                       00000024  1661          L     R3,PLA
0003306A  E313 0000 0024                  00000000  1662          STG   R1,0(R3)            load OP1 address in PL
                                                    1663
00033070  4111 0020                       00000020  1664          LA    R1,32(R1)
00033074  E313 0008 0024                  00000008  1665          STG   R1,8(R3)            load OP2 address in PL
                                                    1666
0003307A  4111 0020                       00000020  1667          LA    R1,32(R1)
0003307E  E313 0010 0024                  00000010  1668          STG   R1,16(R3)           load OP3 address in PL
                                                    1669
00033084  4111 0020                       00000020  1670          LA    R1,32(R1)
00033088  E313 0018 0024                  00000018  1671          STG   R1,24(R3)           load OP4 address in PL
                                                    1672
0003308E  B982 0000                                 1673          XGR   R0,R0
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00033092  4100 0030                     00000030  1674          LA    R0,48
00033096  EE10 0000 3000      00000000  00000000  1675          PLO   R1,0,0,0(R3)        Quadruple Load(QLO: 48)
                                                   1676 *
                                                   1677 * Quadruple Store(QSTO: 61)
                                                   1678 *
0003309C  B982 0011                               1679          XGR   R1,R1
000330A0  5810 5020                     00000020  1680          L     R1,OPDSTA           Destination addresses
                                                   1681
000330A4  B982 0033                               1682          XGR   R3,R3
000330A8  5830 5024                     00000024  1683          L     R3,PLA
000330AC  E313 0000 0024                00000000  1684          STG   R1,0(R3)            load OP1 address in PL
                                                   1685
000330B2  4111 0020                     00000020  1686          LA    R1,32(R1)
000330B6  E313 0008 0024                00000008  1687          STG   R1,8(R3)            load OP2 address in PL
                                                   1688
000330BC  4111 0020                     00000020  1689          LA    R1,32(R1)
000330C0  E313 0010 0024                00000010  1690          STG   R1,16(R3)           load OP3 address in PL
                                                   1691
000330C6  4111 0020                     00000020  1692          LA    R1,32(R1)
000330CA  E313 0018 0024                00000018  1693          STG   R1,24(R3)           load OP4 address in PL
                                                   1694
000330D0  B982 0000                               1695          XGR   R0,R0
000330D4  4100 003D                     0000003D  1696          LA    R0,61
000330D8  EE10 0000 3000      00000000  00000000  1697          PLO   R1,0,0,0(R3)        Quadruple Store(QSTO: 61)
                                                   1698 *
                                                   1699 * compare results
                                                   1700 *
000330DE  B982 0011                               1701          XGR   R1,R1
000330E2  5810 501C                     0000001C  1702          L     R1,OPSRCA           source
000330E6  B982 0033                               1703          XGR   R3,R3
000330EA  5830 5020                     00000020  1704          L     R3,OPDSTA           Destination
000330EE  D51F 1000 3000      00000000  00000000  1705          CLC   0(32,R1),0(R3)
000330F4  4770 50F6                     00033116  1706          BNE   F4861               OP1 Different, error
000330F8  D51F 1020 3020      00000020  00000020  1707          CLC   32(32,R1),32(R3)
000330FE  4770 50F6                     00033116  1708          BNE   F4861               OP2 Different, error
00033102  D51F 1040 3040      00000040  00000040  1709          CLC   64(32,R1),64(R3)
00033108  4770 50F6                     00033116  1710          BNE   F4861               OP3 Different, error
0003310C  D51F 1060 3060      00000060  00000060  1711          CLC   96(32,R1),96(R3)
00033112  478B 0000                     00000000  1712          BE    0(R11)              OP4 same, return
                                                   1713
                                        00033116  00000001  1714 F4861    EQU   *
00033116  45F0 8340                     00000540  1715          BAL   R15,FAILMSG
0003311A  07FB                                    1716          BR    R11
                                                   1717
                                                   1718          TSTDATA
0003311C                      0003311C  00033400  1719+         ORG   PLOETST+(200*K)+(1*4*K)+K
                                                   1720          TSTLBL OPDST              Operand Destination (Results)
00033400                                          1721+OPDST39  DS    0C
00033400  00000000 00000000                       1722          DC    XL128'0'
00033408  00000000 00000000
00033410  00000000 00000000
00033418  00000000 00000000
00033420  00000000 00000000
00033428  00000000 00000000
00033430  00000000 00000000
00033438  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00033440  00000000 00000000
00033448  00000000 00000000
00033450  00000000 00000000
00033458  00000000 00000000
00033460  00000000 00000000
00033468  00000000 00000000
00033470  00000000 00000000
00033478  00000000 00000000
                                          1723
                                          1724           TSTLBL OPSRC              Operand Source
00033480                                  1725+OPSRC39   DS    0C
00033480  00000000 00000000               1726           DC    XL128'0'
00033488  00000000 00000000
00033490  00000000 00000000
00033498  00000000 00000000
000334A0  00000000 00000000
000334A8  00000000 00000000
000334B0  00000000 00000000
000334B8  00000000 00000000
000334C0  00000000 00000000
000334C8  00000000 00000000
000334D0  00000000 00000000
000334D8  00000000 00000000
000334E0  00000000 00000000
000334E8  00000000 00000000
000334F0  00000000 00000000
000334F8  00000000 00000000
                                          1727
                                          1728           TSTLBL PL                 Parameter List
00033500                                  1729+PL39      DS    0C
00033500  00000000 00000000               1730           DC    XL256'00'
00033508  00000000 00000000
00033510  00000000 00000000
00033518  00000000 00000000
00033520  00000000 00000000
00033528  00000000 00000000
00033530  00000000 00000000
00033538  00000000 00000000
00033540  00000000 00000000
00033548  00000000 00000000
00033550  00000000 00000000
00033558  00000000 00000000
00033560  00000000 00000000
00033568  00000000 00000000
00033570  00000000 00000000
00033578  00000000 00000000
00033580  00000000 00000000
00033588  00000000 00000000
00033590  00000000 00000000
00033598  00000000 00000000
000335A0  00000000 00000000
000335A8  00000000 00000000
000335B0  00000000 00000000
000335B8  00000000 00000000
000335C0  00000000 00000000
000335C8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000335D0  00000000 00000000
000335D8  00000000 00000000
000335E0  00000000 00000000
000335E8  00000000 00000000
000335F0  00000000 00000000
000335F8  00000000 00000000
                                        1731
                                        1732          TSTLBL FMSG               fail message
00033600                                1733+FMSG39   DS    0C
00033600  86818993 858440D7             1734          DC    C'failed PLO-Extension '
00033608  D3D660C5 A7A38595
00033610  A2899695 40
00033615  D8A48184 99A49793             1735          DC    CL96'Quadruple Load(QLO: 48) / Store(QSTO: 61)        '
0003361D  8540D396 81844DD8
00033625  D3D67A40 F4F85D40
0003362D  6140E2A3 9699854D
00033635  D8E2E3D6 7A40F6F1
0003363D  5D404040 40404040
00033645  40404040 40404040
0003364D  40404040 40404040
00033655  40404040 40404040
0003365D  40404040 40404040
00033665  40404040 40404040
0003366D  40404040 40404040
                                        1736
                                        1737          TSTEND
00033676                                1738+         DS    0H
00033676  C5958440 968640A3             1739+         DC    CL32'End of test: 39'
0003367E  85A2A37A 40F3F940
00033686  40404040 40404040
0003368E  40404040 40404040
                                        1740
                                        1741 * ------------------------------------------------------------------
                                        1742 * Test Quadruple Load(QLX: 47) / Store(QSTX: 60) instructions
                                        1743 * ------------------------------------------------------------------
                                        1744          TSTHDR 99,99,60
00033696                  00033696  00034000  1745+         ORG   PLOETST+(200*K)+(2*4*K)
00034000  E2E3C1D9 E3409686             1746+         DC    CL32'START of test: 40'
00034008  40A385A2 A37A40F4
00034010  F0404040 40404040
00034018  40404040 40404040
00034020                  00034020          1747+         USING *,R5                base for test data and test routine
00034020                  00000000          1748+         USING PLOETEST,R5
00034020  0003404C                          1749+T40      DC    A(CODE40)           address of test code
00034024  0028                              1750+         DC    H'40'               test number
00034026  00                                1751+         DC    X'00'
00034027  3C                                1752+         DC    HL1'60'             Function
00034028  63                                1753+         DC    HL1'99'             expected unequal results (UR)
00034029  00                                1754+         DC    HL1'00'             found UR
0003402A  63                                1755+         DC    HL1'99'             cc
0003402B  00                                1756+         DC    HL1'00'             cc not checked
0003402C  00                                1757+         DS    X                   extracted cc
00034030  00000000 00000000                 1758+         DS    2F                  extract PSW after test (has CC)
00034038  00034600                          1759+         DC    A(FMSG40)           address of test fail message
0003403C  00034480                          1760+         DC    A(OPSRC40)          address of operand source
00034040  00034400                          1761+         DC    A(OPDST40)          address of operand destination
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00034044  00034500                                1762+          DC    A(PL40)             address of parameter list
00034048  000011D9                                1763+          DC    A(RANDX+40)         address of random source data
                                                  1764
                                                  1765           TSTCODE
0003404C                                          1766+CODE40    DS    0F
                                                  1767 *                                   load random source
0003404C  B982 0011                               1768           XGR   R1,R1
00034050  5810 501C                0000001C       1769           L     R1,OPSRCA
00034054  B982 0022                               1770           XGR   R2,R2
00034058  5820 5028                00000028       1771           L     R2,RANDXA
0003405C  D27F 1000 2000      00000000  00000000  1772           MVC   0(128,R1),0(R2)
                                                  1773 *
                                                  1774 * Quadruple Load(QLX: 47)
                                                  1775 *
00034062  B982 0033                               1776           XGR   R3,R3
00034066  5830 5024                00000024       1777           L     R3,PLA
0003406A  E313 0000 0024           00000000       1778           STG   R1,0(R3)           load OP1 address in PL
                                                  1779
00034070  4111 0020                00000020       1780           LA    R1,32(R1)
00034074  E313 0008 0024           00000008       1781           STG   R1,8(R3)           load OP2 address in PL
                                                  1782
0003407A  4111 0020                00000020       1783           LA    R1,32(R1)
0003407E  E313 0010 0024           00000010       1784           STG   R1,16(R3)          load OP3 address in PL
                                                  1785
00034084  4111 0020                00000020       1786           LA    R1,32(R1)
00034088  E313 0018 0024           00000018       1787           STG   R1,24(R3)          load OP4 address in PL
                                                  1788
0003408E  B982 0000                               1789           XGR   R0,R0
00034092  4100 002F                0000002F       1790           LA    R0,47
00034096  EE10 0000 3000      00000000  00000000  1791           PLO   R1,0,0,0(R3)       Quadruple Load(QLX: 47)
                                                  1792 *
                                                  1793 * Quadruple Store(QSTX: 60)
                                                  1794 *
0003409C  B982 0011                               1795           XGR   R1,R1
000340A0  5810 5020                00000020       1796           L     R1,OPDSTA          Destination addresses
                                                  1797
000340A4  B982 0033                               1798           XGR   R3,R3
000340A8  5830 5024                00000024       1799           L     R3,PLA
000340AC  E313 0000 0024           00000000       1800           STG   R1,0(R3)           load OP1 address in PL
                                                  1801
000340B2  4111 0020                00000020       1802           LA    R1,32(R1)
000340B6  E313 0008 0024           00000008       1803           STG   R1,8(R3)           load OP2 address in PL
                                                  1804
000340BC  4111 0020                00000020       1805           LA    R1,32(R1)
000340C0  E313 0010 0024           00000010       1806           STG   R1,16(R3)          load OP3 address in PL
                                                  1807
000340C6  4111 0020                00000020       1808           LA    R1,32(R1)
000340CA  E313 0018 0024           00000018       1809           STG   R1,24(R3)          load OP4 address in PL
                                                  1810
000340D0  B982 0000                               1811           XGR   R0,R0
000340D4  4100 003C                0000003C       1812           LA    R0,60
000340D8  EE10 0000 3000      00000000  00000000  1813           PLO   R1,0,0,0(R3)       Quadruple Store(QSTX: 60)
                                                  1814 *
                                                  1815 * compare results
                                                  1816 *
000340DE  B982 0011                               1817           XGR   R1,R1
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000340E2  5810 501C                      0000001C  1818           L     R1,OPSRCA           source
000340E6  B982 0033                               1819           XGR   R3,R3
000340EA  5830 5020                      00000020  1820           L     R3,OPDSTA           Destination
000340EE  D50F 1000 3000       00000000  00000000  1821           CLC   0(16,R1),0(R3)
000340F4  4770 50F6                      00034116  1822           BNE   F4760               OP1 Different, error
000340F8  D50F 1020 3020       00000020  00000020  1823           CLC   32(16,R1),32(R3)
000340FE  4770 50F6                      00034116  1824           BNE   F4760               OP2 Different, error
00034102  D50F 1040 3040       00000040  00000040  1825           CLC   64(16,R1),64(R3)
00034108  4770 50F6                      00034116  1826           BNE   F4760               OP3 Different, error
0003410C  D50F 1060 3060       00000060  00000060  1827           CLC   96(16,R1),96(R3)
00034112  478B 0000                      00000000  1828           BE    0(R11)              OP4 same, return
                                                   1829
                                00034116  00000001  1830 F4760     EQU   *
00034116  45F0 8340                      00000540  1831           BAL   R15,FAILMSG
0003411A  07FB                                     1832           BR    R11
                                                   1833
                                                   1834           TSTDATA
0003411C                        0003411C  00034400  1835+          ORG   PLOETST+(200*K)+(2*4*K)+K
                                                   1836           TSTLBL OPDST              Operand Destination (Results)
00034400                                           1837+OPDST40   DS    0C
00034400  00000000 00000000                        1838           DC    XL128'0'
00034408  00000000 00000000
00034410  00000000 00000000
00034418  00000000 00000000
00034420  00000000 00000000
00034428  00000000 00000000
00034430  00000000 00000000
00034438  00000000 00000000
00034440  00000000 00000000
00034448  00000000 00000000
00034450  00000000 00000000
00034458  00000000 00000000
00034460  00000000 00000000
00034468  00000000 00000000
00034470  00000000 00000000
00034478  00000000 00000000
                                                   1839
                                                   1840           TSTLBL OPSRC              Operand Source
00034480                                           1841+OPSRC40   DS    0C
00034480  00000000 00000000                        1842           DC    XL128'0'
00034488  00000000 00000000
00034490  00000000 00000000
00034498  00000000 00000000
000344A0  00000000 00000000
000344A8  00000000 00000000
000344B0  00000000 00000000
000344B8  00000000 00000000
000344C0  00000000 00000000
000344C8  00000000 00000000
000344D0  00000000 00000000
000344D8  00000000 00000000
000344E0  00000000 00000000
000344E8  00000000 00000000
000344F0  00000000 00000000
000344F8  00000000 00000000
                                                   1843
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   1844           TSTLBL PL                  Parameter List
00034500                                           1845+PL40      DS     0C
00034500  00000000 00000000                        1846           DC     XL256'00'
00034508  00000000 00000000
00034510  00000000 00000000
00034518  00000000 00000000
00034520  00000000 00000000
00034528  00000000 00000000
00034530  00000000 00000000
00034538  00000000 00000000
00034540  00000000 00000000
00034548  00000000 00000000
00034550  00000000 00000000
00034558  00000000 00000000
00034560  00000000 00000000
00034568  00000000 00000000
00034570  00000000 00000000
00034578  00000000 00000000
00034580  00000000 00000000
00034588  00000000 00000000
00034590  00000000 00000000
00034598  00000000 00000000
000345A0  00000000 00000000
000345A8  00000000 00000000
000345B0  00000000 00000000
000345B8  00000000 00000000
000345C0  00000000 00000000
000345C8  00000000 00000000
000345D0  00000000 00000000
000345D8  00000000 00000000
000345E0  00000000 00000000
000345E8  00000000 00000000
000345F0  00000000 00000000
000345F8  00000000 00000000
                                                   1847
                                                   1848           TSTLBL FMSG                fail message
00034600                                           1849+FMSG40    DS     0C
00034600  86818993 858440D7                        1850           DC     C'failed PLO-Extension '
00034608  D3D660C5 A7A38595
00034610  A2899695 40
00034615  D8A48184 99A49793                        1851           DC     CL96'Quadruple Load(QLX: 47) / Store(QSTX: 60)'
0003461D  8540D396 81844DD8
00034625  D3E77A40 F4F75D40
0003462D  6140E2A3 9699854D
00034635  D8E2E3E7 7A40F6F0
0003463D  5D404040 40404040
00034645  40404040 40404040
0003464D  40404040 40404040
00034655  40404040 40404040
0003465D  40404040 40404040
00034665  40404040 40404040
0003466D  40404040 40404040
                                                   1852
                                                   1853           TSTEND
00034676                                           1854+          DS     0H
00034676  C5958440 968640A3                        1855+          DC     CL32'End of test: 40'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003467E  85A2A37A 40F4F040
00034686  40404040 40404040
0003468E  40404040 40404040
                                          1856
                                          1857 * ------------------------------------------------------------------------
                                          1858 * Test Quadruple Load(QLG: 46) / Store(QSTG: 59) instructions
                                          1859 * ------------------------------------------------------------------------
                                          1860          TSTHDR 99,99,99
00034696                    00034696  00035000  1861+         ORG   PLOETST+(200*K)+(3*4*K)
00035000  E2E3C1D9 E3409686               1862+         DC    CL32'START of test: 41'
00035008  40A385A2 A37A40F4
00035010  F1404040 40404040
00035018  40404040 40404040
00035020                    00035020          1863+         USING *,R5                base for test data and test routine
00035020                    00000000          1864+         USING PLOETEST,R5
00035020  0003504C                            1865+T41      DC    A(CODE41)           address of test code
00035024  0029                                1866+         DC    H'41'               test number
00035026  00                                  1867+         DC    X'00'
00035027  63                                  1868+         DC    HL1'99'             Function
00035028  63                                  1869+         DC    HL1'99'             expected unequal results (UR)
00035029  00                                  1870+         DC    HL1'00'             found UR
0003502A  63                                  1871+         DC    HL1'99'             cc
0003502B  00                                  1872+         DC    HL1'00'             cc not checked
0003502C  00                                  1873+         DS    X                   extracted cc
00035030  00000000 00000000                   1874+         DS    2F                  extract PSW after test (has CC)
00035038  00035600                            1875+         DC    A(FMSG41)           address of test fail message
0003503C  00035480                            1876+         DC    A(OPSRC41)          address of operand source
00035040  00035400                            1877+         DC    A(OPDST41)          address of operand destination
00035044  00035500                            1878+         DC    A(PL41)             address of parameter list
00035048  000011DA                            1879+         DC    A(RANDX+41)         address of random source data
                                              1880
                                              1881          TSTCODE
0003504C                                      1882+CODE41   DS    0F
                                              1883 *                                  load random source
0003504C  B982 0011                           1884          XGR   R1,R1
00035050  5810 501C                 0000001C  1885          L     R1,OPSRCA
00035054  B982 0022                           1886          XGR   R2,R2
00035058  5820 5028                 00000028  1887          L     R2,RANDXA
0003505C  D27F 1000 2000  00000000  00000000  1888          MVC   0(128,R1),0(R2)
                                              1889 *
                                              1890 * Quadruple Load(QLX: 46)
                                              1891 *
00035062  B982 0033                           1892          XGR   R3,R3
00035066  5830 5024                 00000024  1893          L     R3,PLA
0003506A  E313 0000 0024            00000000  1894          STG   R1,0(R3)            load OP1 address in PL
                                              1895
00035070  4111 0020                 00000020  1896          LA    R1,32(R1)
00035074  E313 0008 0024            00000008  1897          STG   R1,8(R3)            load OP2 address in PL
                                              1898
0003507A  4111 0020                 00000020  1899          LA    R1,32(R1)
0003507E  E313 0010 0024            00000010  1900          STG   R1,16(R3)           load OP3 address in PL
                                              1901
00035084  4111 0020                 00000020  1902          LA    R1,32(R1)
00035088  E313 0018 0024            00000018  1903          STG   R1,24(R3)           load OP4 address in PL
                                              1904
0003508E  B982 0000                           1905          XGR   R0,R0
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00035092  4100 002E                     0000002E  1906          LA    R0,46
00035096  EE10 0000 3000      00000000  00000000  1907          PLO   R1,0,0,0(R3)        Quadruple Load(QLG: 46)
                                                   1908 *
                                                   1909 * Quadruple Store(QSTX: 59)
                                                   1910 *
0003509C  B982 0011                               1911          XGR   R1,R1
000350A0  5810 5020                     00000020  1912          L     R1,OPDSTA           Destination addresses
                                                   1913
000350A4  B982 0033                               1914          XGR   R3,R3
000350A8  5830 5024                     00000024  1915          L     R3,PLA
000350AC  E313 0000 0024                00000000  1916          STG   R1,0(R3)            load OP1 address in PL
                                                   1917
000350B2  4111 0020                     00000020  1918          LA    R1,32(R1)
000350B6  E313 0008 0024                00000008  1919          STG   R1,8(R3)            load OP2 address in PL
                                                   1920
000350BC  4111 0020                     00000020  1921          LA    R1,32(R1)
000350C0  E313 0010 0024                00000010  1922          STG   R1,16(R3)           load OP3 address in PL
                                                   1923
000350C6  4111 0020                     00000020  1924          LA    R1,32(R1)
000350CA  E313 0018 0024                00000018  1925          STG   R1,24(R3)           load OP4 address in PL
                                                   1926
000350D0  B982 0000                               1927          XGR   R0,R0
000350D4  4100 003B                     0000003B  1928          LA    R0,59
000350D8  EE10 0000 3000      00000000  00000000  1929          PLO   R1,0,0,0(R3)        Quadruple Store(QSTG: 59)
                                                   1930 *
                                                   1931 * compare results
                                                   1932 *
000350DE  B982 0011                               1933          XGR   R1,R1
000350E2  5810 501C                     0000001C  1934          L     R1,OPSRCA           source
000350E6  B982 0033                               1935          XGR   R3,R3
000350EA  5830 5020                     00000020  1936          L     R3,OPDSTA           Destination
000350EE  D507 1000 3000      00000000  00000000  1937          CLC   0(8,R1),0(R3)
000350F4  4770 50F6                     00035116  1938          BNE   F4659               OP1 Different, error
000350F8  D507 1020 3020      00000020  00000020  1939          CLC   32(8,R1),32(R3)
000350FE  4770 50F6                     00035116  1940          BNE   F4659               OP2 Different, error
00035102  D507 1040 3040      00000040  00000040  1941          CLC   64(8,R1),64(R3)
00035108  4770 50F6                     00035116  1942          BNE   F4659               OP3 Different, error
0003510C  D507 1060 3060      00000060  00000060  1943          CLC   96(8,R1),96(R3)
00035112  478B 0000                     00000000  1944          BE    0(R11)              OP4 same, return
                                                   1945
                                        00035116  00000001  1946 F4659    EQU   *
00035116  45F0 8340                     00000540  1947          BAL   R15,FAILMSG
0003511A  07FB                                    1948          BR    R11
                                                   1949
                                                   1950          TSTDATA
0003511C                      0003511C  00035400  1951+         ORG   PLOETST+(200*K)+(3*4*K)+K
                                                   1952          TSTLBL OPDST              Operand Destination (Results)
00035400                                          1953+OPDST41  DS    0C
00035400  00000000 00000000                       1954          DC    XL128'0'
00035408  00000000 00000000
00035410  00000000 00000000
00035418  00000000 00000000
00035420  00000000 00000000
00035428  00000000 00000000
00035430  00000000 00000000
00035438  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00035440  00000000 00000000
00035448  00000000 00000000
00035450  00000000 00000000
00035458  00000000 00000000
00035460  00000000 00000000
00035468  00000000 00000000
00035470  00000000 00000000
00035478  00000000 00000000
                                          1955
                                          1956           TSTLBL OPSRC              Operand Source
00035480                                  1957+OPSRC41   DS    0C
00035480  00000000 00000000               1958           DC    XL128'0'
00035488  00000000 00000000
00035490  00000000 00000000
00035498  00000000 00000000
000354A0  00000000 00000000
000354A8  00000000 00000000
000354B0  00000000 00000000
000354B8  00000000 00000000
000354C0  00000000 00000000
000354C8  00000000 00000000
000354D0  00000000 00000000
000354D8  00000000 00000000
000354E0  00000000 00000000
000354E8  00000000 00000000
000354F0  00000000 00000000
000354F8  00000000 00000000
                                          1959
                                          1960           TSTLBL PL                 Parameter List
00035500                                  1961+PL41      DS    0C
00035500  00000000 00000000               1962           DC    XL256'00'
00035508  00000000 00000000
00035510  00000000 00000000
00035518  00000000 00000000
00035520  00000000 00000000
00035528  00000000 00000000
00035530  00000000 00000000
00035538  00000000 00000000
00035540  00000000 00000000
00035548  00000000 00000000
00035550  00000000 00000000
00035558  00000000 00000000
00035560  00000000 00000000
00035568  00000000 00000000
00035570  00000000 00000000
00035578  00000000 00000000
00035580  00000000 00000000
00035588  00000000 00000000
00035590  00000000 00000000
00035598  00000000 00000000
000355A0  00000000 00000000
000355A8  00000000 00000000
000355B0  00000000 00000000
000355B8  00000000 00000000
000355C0  00000000 00000000
000355C8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000355D0  00000000 00000000
000355D8  00000000 00000000
000355E0  00000000 00000000
000355E8  00000000 00000000
000355F0  00000000 00000000
000355F8  00000000 00000000
                                          1963
                                          1964          TSTLBL FMSG                fail message
00035600                                  1965+FMSG41   DS    0C
00035600  86818993 858440D7               1966          DC    C'failed PLO-Extension '
00035608  D3D660C5 A7A38595
00035610  A2899695 40
00035615  D8A48184 99A49793               1967          DC    CL96'Quadruple Load(QLG: 46) / Store(QSTG: 59)'
0003561D  8540D396 81844DD8
00035625  D3C77A40 F4F65D40
0003562D  6140E2A3 9699854D
00035635  D8E2E3C7 7A40F5F9
0003563D  5D404040 40404040
00035645  40404040 40404040
0003564D  40404040 40404040
00035655  40404040 40404040
0003565D  40404040 40404040
00035665  40404040 40404040
0003566D  40404040 40404040
                                          1968
                                          1969          TSTEND
00035676                                  1970+         DS    0H
00035676  C5958440 968640A3               1971+         DC    CL32'End of test: 41'
0003567E  85A2A37A 40F4F140
00035686  40404040 40404040
0003568E  40404040 40404040
                                          1972
                                          1973 * ------------------------------------------------------------------
                                          1974 * Test Quadruple Load(QL: 45) / Store(QST: 58) instructions
                                          1975 * ------------------------------------------------------------------
                                          1976          TSTHDR 99,99,99
00035696                    00035696  00036000        1977+         ORG   PLOETST+(200*K)+(4*4*K)
00036000  E2E3C1D9 E3409686               1978+         DC    CL32'START of test: 42'
00036008  40A385A2 A37A40F4
00036010  F2404040 40404040
00036018  40404040 40404040
00036020                    00036020                  1979+         USING *,R5                 base for test data and test routine
00036020                    00000000                  1980+         USING PLOETEST,R5
00036020  0003604C                                    1981+T42      DC    A(CODE42)            address of test code
00036024  002A                                        1982+         DC    H'42'                test number
00036026  00                                          1983+         DC    X'00'
00036027  63                                          1984+         DC    HL1'99'              Function
00036028  63                                          1985+         DC    HL1'99'              expected unequal results (UR)
00036029  00                                          1986+         DC    HL1'00'              found UR
0003602A  63                                          1987+         DC    HL1'99'              cc
0003602B  00                                          1988+         DC    HL1'00'              cc not checked
0003602C  00                                          1989+         DS    X                    extracted cc
00036030  00000000 00000000                           1990+         DS    2F                   extract PSW after test (has CC)
00036038  00036600                                    1991+         DC    A(FMSG42)            address of test fail message
0003603C  00036480                                    1992+         DC    A(OPSRC42)           address of operand source
00036040  00036400                                    1993+         DC    A(OPDST42)           address of operand destination
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00036044  00036500                                1994+          DC    A(PL42)             address of parameter list
00036048  000011DB                                1995+          DC    A(RANDX+42)         address of random source data
                                                  1996
                                                  1997           TSTCODE
0003604C                                          1998+CODE42    DS    0F
                                                  1999 *                                   load random source
0003604C  B982 0011                               2000           XGR   R1,R1
00036050  5810 501C                    0000001C   2001           L     R1,OPSRCA
00036054  B982 0022                               2002           XGR   R2,R2
00036058  5820 5028                    00000028   2003           L     R2,RANDXA
0003605C  D27F 1000 2000      00000000 00000000   2004           MVC   0(128,R1),0(R2)
                                                  2005 *
                                                  2006 * Quadruple Load(QLX: 45`
                                                  2007 *
00036062  B982 0033                               2008           XGR   R3,R3
00036066  5830 5024                    00000024   2009           L     R3,PLA
0003606A  E313 0000 0024               00000000   2010           STG   R1,0(R3)           load OP1 address in PL
                                                  2011
00036070  4111 0020                    00000020   2012           LA    R1,32(R1)
00036074  E313 0008 0024               00000008   2013           STG   R1,8(R3)           load OP2 address in PL
                                                  2014
0003607A  4111 0020                    00000020   2015           LA    R1,32(R1)
0003607E  E313 0010 0024               00000010   2016           STG   R1,16(R3)          load OP3 address in PL
                                                  2017
00036084  4111 0020                    00000020   2018           LA    R1,32(R1)
00036088  E313 0018 0024               00000018   2019           STG   R1,24(R3)          load OP4 address in PL
                                                  2020
0003608E  B982 0000                               2021           XGR   R0,R0
00036092  4100 002D                    0000002D   2022           LA    R0,45
00036096  EE10 0000 3000      00000000 00000000   2023           PLO   R1,0,0,0(R3)       Quadruple Load(QL: 45)
                                                  2024 *
                                                  2025 * Quadruple Store(QST: 58)
                                                  2026 *
0003609C  B982 0011                               2027           XGR   R1,R1
000360A0  5810 5020                    00000020   2028           L     R1,OPDSTA          Destination addresses
                                                  2029
000360A4  B982 0033                               2030           XGR   R3,R3
000360A8  5830 5024                    00000024   2031           L     R3,PLA
000360AC  E313 0000 0024               00000000   2032           STG   R1,0(R3)           load OP1 address in PL
                                                  2033
000360B2  4111 0020                    00000020   2034           LA    R1,32(R1)
000360B6  E313 0008 0024               00000008   2035           STG   R1,8(R3)           load OP2 address in PL
                                                  2036
000360BC  4111 0020                    00000020   2037           LA    R1,32(R1)
000360C0  E313 0010 0024               00000010   2038           STG   R1,16(R3)          load OP3 address in PL
                                                  2039
000360C6  4111 0020                    00000020   2040           LA    R1,32(R1)
000360CA  E313 0018 0024               00000018   2041           STG   R1,24(R3)          load OP4 address in PL
                                                  2042
000360D0  B982 0000                               2043           XGR   R0,R0
000360D4  4100 003A                    0000003A   2044           LA    R0,58
000360D8  EE10 0000 3000      00000000 00000000   2045           PLO   R1,0,0,0(R3)       Quadruple Store(QST: 58)
                                                  2046 *
                                                  2047 * compare results
                                                  2048 *
000360DE  B982 0011                               2049           XGR   R1,R1
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000360E2  5810 501C                      0000001C  2050           L     R1,OPSRCA           source
000360E6  B982 0033                               2051           XGR   R3,R3
000360EA  5830 5020                      00000020  2052           L     R3,OPDSTA           Destination
000360EE  D503 1000 3000       00000000  00000000  2053           CLC   0(4,R1),0(R3)
000360F4  4770 50F6                      00036116  2054           BNE   F4558               OP1 Different, error
000360F8  D503 1020 3020       00000020  00000020  2055           CLC   32(4,R1),32(R3)
000360FE  4770 50F6                      00036116  2056           BNE   F4558               OP2 Different, error
00036102  D503 1040 3040       00000040  00000040  2057           CLC   64(4,R1),64(R3)
00036108  4770 50F6                      00036116  2058           BNE   F4558               OP3 Different, error
0003610C  D503 1060 3060       00000060  00000060  2059           CLC   96(4,R1),96(R3)
00036112  478B 0000                      00000000  2060           BE    0(R11)              OP4 same, return
                                                   2061
                                00036116  00000001  2062 F4558     EQU   *
00036116  45F0 8340                      00000540  2063           BAL   R15,FAILMSG
0003611A  07FB                                     2064           BR    R11
                                                   2065
                                                   2066           TSTDATA
0003611C                        0003611C  00036400  2067+          ORG   PLOETST+(200*K)+(4*4*K)+K
                                                   2068           TSTLBL OPDST              Operand Destination (Results)
00036400                                           2069+OPDST42   DS    0C
00036400  00000000 00000000                        2070           DC    XL128'0'
00036408  00000000 00000000
00036410  00000000 00000000
00036418  00000000 00000000
00036420  00000000 00000000
00036428  00000000 00000000
00036430  00000000 00000000
00036438  00000000 00000000
00036440  00000000 00000000
00036448  00000000 00000000
00036450  00000000 00000000
00036458  00000000 00000000
00036460  00000000 00000000
00036468  00000000 00000000
00036470  00000000 00000000
00036478  00000000 00000000
                                                   2071
                                                   2072           TSTLBL OPSRC              Operand Source
00036480                                           2073+OPSRC42   DS    0C
00036480  00000000 00000000                        2074           DC    XL128'0'
00036488  00000000 00000000
00036490  00000000 00000000
00036498  00000000 00000000
000364A0  00000000 00000000
000364A8  00000000 00000000
000364B0  00000000 00000000
000364B8  00000000 00000000
000364C0  00000000 00000000
000364C8  00000000 00000000
000364D0  00000000 00000000
000364D8  00000000 00000000
000364E0  00000000 00000000
000364E8  00000000 00000000
000364F0  00000000 00000000
000364F8  00000000 00000000
                                                   2075
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    2076           TSTLBL PL                  Parameter List
00036500                                            2077+PL42      DS    0C
00036500  00000000 00000000                         2078           DC    XL256'00'
00036508  00000000 00000000
00036510  00000000 00000000
00036518  00000000 00000000
00036520  00000000 00000000
00036528  00000000 00000000
00036530  00000000 00000000
00036538  00000000 00000000
00036540  00000000 00000000
00036548  00000000 00000000
00036550  00000000 00000000
00036558  00000000 00000000
00036560  00000000 00000000
00036568  00000000 00000000
00036570  00000000 00000000
00036578  00000000 00000000
00036580  00000000 00000000
00036588  00000000 00000000
00036590  00000000 00000000
00036598  00000000 00000000
000365A0  00000000 00000000
000365A8  00000000 00000000
000365B0  00000000 00000000
000365B8  00000000 00000000
000365C0  00000000 00000000
000365C8  00000000 00000000
000365D0  00000000 00000000
000365D8  00000000 00000000
000365E0  00000000 00000000
000365E8  00000000 00000000
000365F0  00000000 00000000
000365F8  00000000 00000000
                                                    2079
                                                    2080           TSTLBL FMSG                fail message
00036600                                            2081+FMSG42    DS    0C
00036600  86818993 858440D7                         2082           DC    C'failed PLO-Extension '
00036608  D3D660C5 A7A38595
00036610  A2899695 40
00036615  D8A48184 99A49793                         2083           DC    CL96'Quadruple Load(QL: 45) / Store(QST: 58)'
0003661D  8540D396 81844DD8
00036625  D37A40F4 F55D4061
0003662D  40E2A396 99854DD8
00036635  E2E37A40 F5F85D40
0003663D  40404040 40404040
00036645  40404040 40404040
0003664D  40404040 40404040
00036655  40404040 40404040
0003665D  40404040 40404040
00036665  40404040 40404040
0003666D  40404040 40404040
                                                    2084
                                                    2085           TSTEND
00036676                                            2086+          DS    0H
00036676  C5958440 968640A3                         2087+          DC    CL32'End of test: 42'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003667E  85A2A37A 40F4F240
00036686  40404040 40404040
0003668E  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2089 *=====================================================================
                                                    2090 *       Triple Load / Store Tests
                                                    2091 *=====================================================================
                                                    2092
                                                    2093 * ---------------------------------------------------------------------
                                                    2094 * Test Triple Load(TLO: 44) / Store(TSTO: 57) instructions
                                                    2095 * ---------------------------------------------------------------------
                                                    2096          TSTHDR 99,99,99
00036696                          00036696  00037000  2097+         ORG   PLOETST+(200*K)+(5*4*K)
00037000  E2E3C1D9 E3409686                          2098+         DC    CL32'START of test: 43'
00037008  40A385A2 A37A40F4
00037010  F3404040 40404040
00037018  40404040 40404040
00037020                          00037020            2099+         USING *,R5                base for test data and test routine
00037020                          00000000            2100+         USING PLOETEST,R5
00037020  0003704C                                    2101+T43      DC    A(CODE43)           address of test code
00037024  002B                                        2102+         DC    H'43'               test number
00037026  00                                          2103+         DC    X'00'
00037027  63                                          2104+         DC    HL1'99'             Function
00037028  63                                          2105+         DC    HL1'99'             expected unequal results (UR)
00037029  00                                          2106+         DC    HL1'00'             found UR
0003702A  63                                          2107+         DC    HL1'99'             cc
0003702B  00                                          2108+         DC    HL1'00'             cc not checked
0003702C  00                                          2109+         DS    X                   extracted cc
00037030  00000000 00000000                           2110+         DS    2F                  extract PSW after test (has CC)
00037038  00037600                                    2111+         DC    A(FMSG43)           address of test fail message
0003703C  00037480                                    2112+         DC    A(OPSRC43)          address of operand source
00037040  00037400                                    2113+         DC    A(OPDST43)          address of operand destination
00037044  00037500                                    2114+         DC    A(PL43)             address of parameter list
00037048  000011DC                                    2115+         DC    A(RANDX+43)         address of random source data
                                                    2116
                                                    2117          TSTCODE
0003704C                                            2118+CODE43   DS    0F
                                                    2119 *                                    load random source
0003704C  B982 0011                                 2120          XGR   R1,R1
00037050  5810 501C                     0000001C    2121          L     R1,OPSRCA
00037054  B982 0022                                 2122          XGR   R2,R2
00037058  5820 5028                     00000028    2123          L     R2,RANDXA
0003705C  D27F 1000 2000      00000000  00000000    2124          MVC   0(128,R1),0(R2)
                                                    2125 *
                                                    2126 * Triple Load(TLO: 44)
                                                    2127 *
00037062  B982 0033                                 2128          XGR   R3,R3
00037066  5830 5024                     00000024    2129          L     R3,PLA
0003706A  E313 0000 0024                00000000    2130          STG   R1,0(R3)            load OP1 address in PL
                                                    2131
00037070  4111 0020                     00000020    2132          LA    R1,32(R1)
00037074  E313 0008 0024                00000008    2133          STG   R1,8(R3)            load OP2 address in PL
                                                    2134
0003707A  4111 0020                     00000020    2135          LA    R1,32(R1)
0003707E  E313 0010 0024                00000010    2136          STG   R1,16(R3)           load OP3 address in PL
                                                    2137
00037084  B982 0000                                 2138          XGR   R0,R0
00037088  4100 002C                     0000002C    2139          LA    R0,44
0003708C  EE10 0000 3000      00000000  00000000    2140          PLO   R1,0,0,0(R3)        Triple Load(TLO: 44)
                                                    2141 *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   2142 * Triple Store(TSTO: 57)
                                                   2143 *
00037092  B982 0011                                2144          XGR   R1,R1
00037096  5810 5020                 00000020       2145          L     R1,OPDSTA           Destination addresses
                                                   2146
0003709A  B982 0033                                2147          XGR   R3,R3
0003709E  5830 5024                 00000024       2148          L     R3,PLA
000370A2  E313 0000 0024            00000000       2149          STG   R1,0(R3)            load OP1 address in PL
                                                   2150
000370A8  4111 0020                 00000020       2151          LA    R1,32(R1)
000370AC  E313 0008 0024            00000008       2152          STG   R1,8(R3)            load OP2 address in PL
                                                   2153
000370B2  4111 0020                 00000020       2154          LA    R1,32(R1)
000370B6  E313 0010 0024            00000010       2155          STG   R1,16(R3)           load OP3 address in PL
                                                   2156
000370BC  B982 0000                                2157          XGR   R0,R0
000370C0  4100 0039                 00000039       2158          LA    R0,57
000370C4  EE10 0000 3000  00000000  00000000       2159          PLO   R1,0,0,0(R3)        Triple Store(TSTO: 57)
                                                   2160 *
                                                   2161 * compare results
                                                   2162 *
000370CA  B982 0011                                2163          XGR   R1,R1
000370CE  5810 501C                 0000001C       2164          L     R1,OPSRCA           source
000370D2  B982 0033                                2165          XGR   R3,R3
000370D6  5830 5020                 00000020       2166          L     R3,OPDSTA           Destination
000370DA  D51F 1000 3000  00000000  00000000       2167          CLC   0(32,R1),0(R3)
000370E0  4770 50D8                 000370F8       2168          BNE   F4457               OP1 Different, error
000370E4  D51F 1020 3020  00000020  00000020       2169          CLC   32(32,R1),32(R3)
000370EA  4770 50D8                 000370F8       2170          BNE   F4457               OP2 Different, error
000370EE  D51F 1040 3040  00000040  00000040       2171          CLC   64(32,R1),64(R3)
000370F4  478B 0000                 00000000       2172          BE    0(R11)              OP3 same, return
                                                   2173
                          000370F8  00000001       2174 F4457    EQU   *
000370F8  45F0 8340                 00000540       2175          BAL   R15,FAILMSG
000370FC  07FB                                     2176          BR    R11
                                                   2177
                                                   2178          TSTDATA
000370FE                  000370FE  00037400       2179+         ORG   PLOETST+(200*K)+(5*4*K)+K
                                                   2180          TSTLBL OPDST              Operand Destination (Results)
00037400                                           2181+OPDST43  DS    0C
00037400  00000000 00000000                        2182          DC    XL128'0'
00037408  00000000 00000000
00037410  00000000 00000000
00037418  00000000 00000000
00037420  00000000 00000000
00037428  00000000 00000000
00037430  00000000 00000000
00037438  00000000 00000000
00037440  00000000 00000000
00037448  00000000 00000000
00037450  00000000 00000000
00037458  00000000 00000000
00037460  00000000 00000000
00037468  00000000 00000000
00037470  00000000 00000000
00037478  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2183
                                                    2184          TSTLBL OPSRC              Operand Source
00037480                                            2185+OPSRC43  DS    0C
00037480  00000000 00000000                         2186          DC    XL128'0'
00037488  00000000 00000000
00037490  00000000 00000000
00037498  00000000 00000000
000374A0  00000000 00000000
000374A8  00000000 00000000
000374B0  00000000 00000000
000374B8  00000000 00000000
000374C0  00000000 00000000
000374C8  00000000 00000000
000374D0  00000000 00000000
000374D8  00000000 00000000
000374E0  00000000 00000000
000374E8  00000000 00000000
000374F0  00000000 00000000
000374F8  00000000 00000000
                                                    2187
                                                    2188          TSTLBL PL                 Parameter List
00037500                                            2189+PL43     DS    0C
00037500  00000000 00000000                         2190          DC    XL256'00'
00037508  00000000 00000000
00037510  00000000 00000000
00037518  00000000 00000000
00037520  00000000 00000000
00037528  00000000 00000000
00037530  00000000 00000000
00037538  00000000 00000000
00037540  00000000 00000000
00037548  00000000 00000000
00037550  00000000 00000000
00037558  00000000 00000000
00037560  00000000 00000000
00037568  00000000 00000000
00037570  00000000 00000000
00037578  00000000 00000000
00037580  00000000 00000000
00037588  00000000 00000000
00037590  00000000 00000000
00037598  00000000 00000000
000375A0  00000000 00000000
000375A8  00000000 00000000
000375B0  00000000 00000000
000375B8  00000000 00000000
000375C0  00000000 00000000
000375C8  00000000 00000000
000375D0  00000000 00000000
000375D8  00000000 00000000
000375E0  00000000 00000000
000375E8  00000000 00000000
000375F0  00000000 00000000
000375F8  00000000 00000000
                                                    2191
                                                    2192          TSTLBL FMSG               fail message
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00037600                                           2193+FMSG43   DS    0C
00037600  86818993 858440D7                        2194          DC    C'failed PLO-Extension '
00037608  D3D660C5 A7A38595
00037610  A2899695 40
00037615  E3998997 938540D3                        2195          DC    CL96'Triple Load(TLO: 44) / Store(TSTO: 57)'
0003761D  9681844D E3D3D67A
00037625  40F4F45D 406140E2
0003762D  A3969985 4DE3E2E3
00037635  D67A40F5 F75D4040
0003763D  40404040 40404040
00037645  40404040 40404040
0003764D  40404040 40404040
00037655  40404040 40404040
0003765D  40404040 40404040
00037665  40404040 40404040
0003766D  40404040 40404040
                                                   2196
                                                   2197          TSTEND
00037676                                           2198+         DS    0H
00037676  C5958440 968640A3                        2199+         DC    CL32'End of test: 43'
0003767E  85A2A37A 40F4F340
00037686  40404040 40404040
0003768E  40404040 40404040
                                                   2200
                                                   2201 * ------------------------------------------------------------------
                                                   2202 * Test Triple Load(TLX: 43) / Store(TSTX: 56) instructions
                                                   2203 * ------------------------------------------------------------------
                                                   2204          TSTHDR 99,99,99
00037696                        00037696  00038000 2205+         ORG   PLOETST+(200*K)+(6*4*K)
00038000  E2E3C1D9 E3409686                        2206+         DC    CL32'START of test: 44'
00038008  40A385A2 A37A40F4
00038010  F4404040 40404040
00038018  40404040 40404040
00038020                        00038020          2207+         USING *,R5                  base for test data and test routine
00038020                        00000000          2208+         USING PLOETEST,R5
00038020  0003804C                                 2209+T44      DC    A(CODE44)             address of test code
00038024  002C                                     2210+         DC    H'44'                 test number
00038026  00                                       2211+         DC    X'00'
00038027  63                                       2212+         DC    HL1'99'               Function
00038028  63                                       2213+         DC    HL1'99'               expected unequal results (UR)
00038029  00                                       2214+         DC    HL1'00'               found UR
0003802A  63                                       2215+         DC    HL1'99'               cc
0003802B  00                                       2216+         DC    HL1'00'               cc not checked
0003802C  00                                       2217+         DS    X                     extracted cc
00038030  00000000 00000000                        2218+         DS    2F                    extract PSW after test (has CC)
00038038  00038600                                 2219+         DC    A(FMSG44)             address of test fail message
0003803C  00038480                                 2220+         DC    A(OPSRC44)            address of operand source
00038040  00038400                                 2221+         DC    A(OPDST44)            address of operand destination
00038044  00038500                                 2222+         DC    A(PL44)               address of parameter list
00038048  000011DD                                 2223+         DC    A(RANDX+44)           address of random source data
                                                   2224
                                                   2225          TSTCODE
0003804C                                           2226+CODE44   DS    0F
                                                   2227 *                                    load random source
0003804C  B982 0011                                2228          XGR   R1,R1
00038050  5810 501C                     0000001C   2229          L     R1,OPSRCA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00038054  B982 0022                       2230          XGR   R2,R2
00038058  5820 5028                 00000028  2231          L     R2,RANDXA
0003805C  D27F 1000 2000      00000000  00000000  2232          MVC   0(128,R1),0(R2)
                                              2233 *
                                              2234 * Triple Load(TLX: 43)
                                              2235 *
00038062  B982 0033                       2236          XGR   R3,R3
00038066  5830 5024                 00000024  2237          L     R3,PLA
0003806A  E313 0000 0024            00000000  2238          STG   R1,0(R3)            load OP1 address in PL
                                              2239
00038070  4111 0020                 00000020  2240          LA    R1,32(R1)
00038074  E313 0008 0024            00000008  2241          STG   R1,8(R3)            load OP2 address in PL
                                              2242
0003807A  4111 0020                 00000020  2243          LA    R1,32(R1)
0003807E  E313 0010 0024            00000010  2244          STG   R1,16(R3)           load OP3 address in PL
                                              2245
00038084  B982 0000                       2246          XGR   R0,R0
00038088  4100 002B                 0000002B  2247          LA    R0,43
0003808C  EE10 0000 3000      00000000  00000000  2248          PLO   R1,0,0,0(R3)        Triple Load(TLX: 43)
                                              2249 *
                                              2250 * Triple Store(TSTX: 56)
                                              2251 *
00038092  B982 0011                       2252          XGR   R1,R1
00038096  5810 5020                 00000020  2253          L     R1,OPDSTA           Destination addresses
                                              2254
0003809A  B982 0033                       2255          XGR   R3,R3
0003809E  5830 5024                 00000024  2256          L     R3,PLA
000380A2  E313 0000 0024            00000000  2257          STG   R1,0(R3)            load OP1 address in PL
                                              2258
000380A8  4111 0020                 00000020  2259          LA    R1,32(R1)
000380AC  E313 0008 0024            00000008  2260          STG   R1,8(R3)            load OP2 address in PL
                                              2261
000380B2  4111 0020                 00000020  2262          LA    R1,32(R1)
000380B6  E313 0010 0024            00000010  2263          STG   R1,16(R3)           load OP3 address in PL
                                              2264
000380BC  B982 0000                       2265          XGR   R0,R0
000380C0  4100 0038                 00000038  2266          LA    R0,56
000380C4  EE10 0000 3000      00000000  00000000  2267          PLO   R1,0,0,0(R3)        Triple Store(TSTX: 56)
                                              2268 *
                                              2269 * compare results
                                              2270 *
000380CA  B982 0011                       2271          XGR   R1,R1
000380CE  5810 501C                 0000001C  2272          L     R1,OPSRCA           source
000380D2  B982 0033                       2273          XGR   R3,R3
000380D6  5830 5020                 00000020  2274          L     R3,OPDSTA           Destination
000380DA  D50F 1000 3000      00000000  00000000  2275          CLC   0(16,R1),0(R3)
000380E0  4770 50D8                 000380F8  2276          BNE   F4356               OP1 Different, error
000380E4  D50F 1020 3020      00000020  00000020  2277          CLC   32(16,R1),32(R3)
000380EA  4770 50D8                 000380F8  2278          BNE   F4356               OP2 Different, error
000380EE  D50F 1040 3040      00000040  00000040  2279          CLC   64(16,R1),64(R3)
000380F4  478B 0000                 00000000  2280          BE    0(R11)              OP3 same, return
                                              2281
                                  000380F8  00000001  2282 F4356     EQU   *
000380F8  45F0 8340                 00000540  2283          BAL   R15,FAILMSG
000380FC  07FB                              2284          BR    R11
                                              2285
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2286           TSTDATA
000380FE                      000380FE  00038400  2287+          ORG   PLOETST+(200*K)+(6*4*K)+K
                                                    2288           TSTLBL OPDST              Operand Destination (Results)
00038400                                            2289+OPDST44   DS    0C
00038400  00000000 00000000                         2290           DC    XL128'0'
00038408  00000000 00000000
00038410  00000000 00000000
00038418  00000000 00000000
00038420  00000000 00000000
00038428  00000000 00000000
00038430  00000000 00000000
00038438  00000000 00000000
00038440  00000000 00000000
00038448  00000000 00000000
00038450  00000000 00000000
00038458  00000000 00000000
00038460  00000000 00000000
00038468  00000000 00000000
00038470  00000000 00000000
00038478  00000000 00000000
                                                    2291
                                                    2292           TSTLBL OPSRC              Operand Source
00038480                                            2293+OPSRC44   DS    0C
00038480  00000000 00000000                         2294           DC    XL128'0'
00038488  00000000 00000000
00038490  00000000 00000000
00038498  00000000 00000000
000384A0  00000000 00000000
000384A8  00000000 00000000
000384B0  00000000 00000000
000384B8  00000000 00000000
000384C0  00000000 00000000
000384C8  00000000 00000000
000384D0  00000000 00000000
000384D8  00000000 00000000
000384E0  00000000 00000000
000384E8  00000000 00000000
000384F0  00000000 00000000
000384F8  00000000 00000000
                                                    2295
                                                    2296           TSTLBL PL                 Parameter List
00038500                                            2297+PL44      DS    0C
00038500  00000000 00000000                         2298           DC    XL256'00'
00038508  00000000 00000000
00038510  00000000 00000000
00038518  00000000 00000000
00038520  00000000 00000000
00038528  00000000 00000000
00038530  00000000 00000000
00038538  00000000 00000000
00038540  00000000 00000000
00038548  00000000 00000000
00038550  00000000 00000000
00038558  00000000 00000000
00038560  00000000 00000000
00038568  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00038570  00000000 00000000
00038578  00000000 00000000
00038580  00000000 00000000
00038588  00000000 00000000
00038590  00000000 00000000
00038598  00000000 00000000
000385A0  00000000 00000000
000385A8  00000000 00000000
000385B0  00000000 00000000
000385B8  00000000 00000000
000385C0  00000000 00000000
000385C8  00000000 00000000
000385D0  00000000 00000000
000385D8  00000000 00000000
000385E0  00000000 00000000
000385E8  00000000 00000000
000385F0  00000000 00000000
000385F8  00000000 00000000
                                          2299
                                          2300          TSTLBL FMSG               fail message
00038600                                  2301+FMSG44   DS    0C
00038600  86818993 858440D7               2302          DC    C'failed PLO-Extension '
00038608  D3D660C5 A7A38595
00038610  A2899695 40
00038615  E3998997 938540D3               2303          DC    CL96'Triple Load(TLX: 43) / Store(TSTX: 56)'
0003861D  9681844D E3D3E77A
00038625  40F4F35D 406140E2
0003862D  A3969985 4DE3E2E3
00038635  E77A40F5 F65D4040
0003863D  40404040 40404040
00038645  40404040 40404040
0003864D  40404040 40404040
00038655  40404040 40404040
0003865D  40404040 40404040
00038665  40404040 40404040
0003866D  40404040 40404040
                                          2304
                                          2305          TSTEND
00038676                                  2306+         DS    0H
00038676  C5958440 968640A3               2307+         DC    CL32'End of test: 44'
0003867E  85A2A37A 40F4F440
00038686  40404040 40404040
0003868E  40404040 40404040
                                          2308
                                          2309 * ------------------------------------------------------------------
                                          2310 * Test Triple Load(TLG: 42) / Store(TSTG: 55) instructions
                                          2311 * ------------------------------------------------------------------
                                          2312          TSTHDR 99,99,99
00038696                      00038696  00039000  2313+         ORG   PLOETST+(200*K)+(7*4*K)
00039000  E2E3C1D9 E3409686               2314+         DC    CL32'START of test: 45'
00039008  40A385A2 A37A40F4
00039010  F5404040 40404040
00039018  40404040 40404040
00039020                      00039020            2315+         USING *,R5                base for test data and test routine
00039020                      00000000            2316+         USING PLOETEST,R5
00039020  0003904C                        2317+T45      DC    A(CODE45)           address of test code
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00039024  002D                                     2318+          DC    H'45'              test number
00039026  00                                       2319+          DC    X'00'
00039027  63                                       2320+          DC    HL1'99'            Function
00039028  63                                       2321+          DC    HL1'99'            expected unequal results (UR)
00039029  00                                       2322+          DC    HL1'00'            found UR
0003902A  63                                       2323+          DC    HL1'99'            cc
0003902B  00                                       2324+          DC    HL1'00'            cc not checked
0003902C  00                                       2325+          DS    X                  extracted cc
00039030  00000000 00000000                        2326+          DS    2F                 extract PSW after test (has CC)
00039038  00039600                                 2327+          DC    A(FMSG45)          address of test fail message
0003903C  00039480                                 2328+          DC    A(OPSRC45)         address of operand source
00039040  00039400                                 2329+          DC    A(OPDST45)         address of operand destination
00039044  00039500                                 2330+          DC    A(PL45)            address of parameter list
00039048  000011DE                                 2331+          DC    A(RANDX+45)        address of random source data
                                                   2332
                                                   2333           TSTCODE
0003904C                                           2334+CODE45    DS    0F
                                                   2335 *                                  load random source
0003904C  B982 0011                                2336           XGR   R1,R1
00039050  5810 501C                     0000001C   2337           L     R1,OPSRCA
00039054  B982 0022                                2338           XGR   R2,R2
00039058  5820 5028                     00000028   2339           L     R2,RANDXA
0003905C  D27F 1000 2000      00000000  00000000   2340           MVC   0(128,R1),0(R2)
                                                   2341 *
                                                   2342 * Triple Load(TLG: 42)
                                                   2343 *
00039062  B982 0033                                2344           XGR   R3,R3
00039066  5830 5024                     00000024   2345           L     R3,PLA
0003906A  E313 0000 0024                00000000   2346           STG   R1,0(R3)           load OP1 address in PL
                                                   2347
00039070  4111 0020                     00000020   2348           LA    R1,32(R1)
00039074  E313 0008 0024                00000008   2349           STG   R1,8(R3)           load OP2 address in PL
                                                   2350
0003907A  4111 0020                     00000020   2351           LA    R1,32(R1)
0003907E  E313 0010 0024                00000010   2352           STG   R1,16(R3)          load OP3 address in PL
                                                   2353
00039084  B982 0000                                2354           XGR   R0,R0
00039088  4100 002A                     0000002A   2355           LA    R0,42
0003908C  EE10 0000 3000      00000000  00000000   2356           PLO   R1,0,0,0(R3)       Triple Load(TLG: 42)
                                                   2357 *
                                                   2358 * Triple Store(TSTG: 55)
                                                   2359 *
00039092  B982 0011                                2360           XGR   R1,R1
00039096  5810 5020                     00000020   2361           L     R1,OPDSTA          Destination addresses
                                                   2362
0003909A  B982 0033                                2363           XGR   R3,R3
0003909E  5830 5024                     00000024   2364           L     R3,PLA
000390A2  E313 0000 0024                00000000   2365           STG   R1,0(R3)           load OP1 address in PL
                                                   2366
000390A8  4111 0020                     00000020   2367           LA    R1,32(R1)
000390AC  E313 0008 0024                00000008   2368           STG   R1,8(R3)           load OP2 address in PL
                                                   2369
000390B2  4111 0020                     00000020   2370           LA    R1,32(R1)
000390B6  E313 0010 0024                00000010   2371           STG   R1,16(R3)          load OP3 address in PL
                                                   2372
000390BC  B982 0000                                2373           XGR   R0,R0
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000390C0  4100 0037                      00000037  2374          LA    R0,55
000390C4  EE10 0000 3000      00000000   00000000  2375          PLO   R1,0,0,0(R3)        Triple Store(TSTG: 55)
                                                   2376 *
                                                   2377 * compare results
                                                   2378 *
000390CA  B982 0011                                2379          XGR   R1,R1
000390CE  5810 501C                      0000001C  2380          L     R1,OPSRCA           source
000390D2  B982 0033                                2381          XGR   R3,R3
000390D6  5830 5020                      00000020  2382          L     R3,OPDSTA           Destination
000390DA  D507 1000 3000      00000000   00000000  2383          CLC   0(8,R1),0(R3)
000390E0  4770 50D8                      000390F8  2384          BNE   F4255               OP1 Different, error
000390E4  D507 1020 3020      00000020   00000020  2385          CLC   32(8,R1),32(R3)
000390EA  4770 50D8                      000390F8  2386          BNE   F4255               OP2 Different, error
000390EE  D507 1040 3040      00000040   00000040  2387          CLC   64(8,R1),64(R3)
000390F4  478B 0000                      00000000  2388          BE    0(R11)              OP3 same, return
                                                   2389
                              000390F8   00000001  2390 F4255    EQU   *
000390F8  45F0 8340                      00000540  2391          BAL   R15,FAILMSG
000390FC  07FB                                     2392          BR    R11
                                                   2393
                                                   2394          TSTDATA
000390FE                      000390FE   00039400  2395+         ORG   PLOETST+(200*K)+(7*4*K)+K
                                                   2396          TSTLBL OPDST              Operand Destination (Results)
00039400                                           2397+OPDST45  DS    0C
00039400  00000000 00000000                        2398          DC    XL128'0'
00039408  00000000 00000000
00039410  00000000 00000000
00039418  00000000 00000000
00039420  00000000 00000000
00039428  00000000 00000000
00039430  00000000 00000000
00039438  00000000 00000000
00039440  00000000 00000000
00039448  00000000 00000000
00039450  00000000 00000000
00039458  00000000 00000000
00039460  00000000 00000000
00039468  00000000 00000000
00039470  00000000 00000000
00039478  00000000 00000000
                                                   2399
                                                   2400          TSTLBL OPSRC              Operand Source
00039480                                           2401+OPSRC45  DS    0C
00039480  00000000 00000000                        2402          DC    XL128'0'
00039488  00000000 00000000
00039490  00000000 00000000
00039498  00000000 00000000
000394A0  00000000 00000000
000394A8  00000000 00000000
000394B0  00000000 00000000
000394B8  00000000 00000000
000394C0  00000000 00000000
000394C8  00000000 00000000
000394D0  00000000 00000000
000394D8  00000000 00000000
000394E0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000394E8  00000000 00000000
000394F0  00000000 00000000
000394F8  00000000 00000000
                                          2403
                                          2404           TSTLBL PL                  Parameter List
00039500                                  2405+PL45      DS    0C
00039500  00000000 00000000               2406           DC    XL256'00'
00039508  00000000 00000000
00039510  00000000 00000000
00039518  00000000 00000000
00039520  00000000 00000000
00039528  00000000 00000000
00039530  00000000 00000000
00039538  00000000 00000000
00039540  00000000 00000000
00039548  00000000 00000000
00039550  00000000 00000000
00039558  00000000 00000000
00039560  00000000 00000000
00039568  00000000 00000000
00039570  00000000 00000000
00039578  00000000 00000000
00039580  00000000 00000000
00039588  00000000 00000000
00039590  00000000 00000000
00039598  00000000 00000000
000395A0  00000000 00000000
000395A8  00000000 00000000
000395B0  00000000 00000000
000395B8  00000000 00000000
000395C0  00000000 00000000
000395C8  00000000 00000000
000395D0  00000000 00000000
000395D8  00000000 00000000
000395E0  00000000 00000000
000395E8  00000000 00000000
000395F0  00000000 00000000
000395F8  00000000 00000000
                                          2407
                                          2408           TSTLBL FMSG                fail message
00039600                                  2409+FMSG45    DS    0C
00039600  86818993 858440D7               2410           DC    C'failed PLO-Extension '
00039608  D3D660C5 A7A38595
00039610  A2899695 40
00039615  E3998997 938540D3               2411           DC    CL96'Triple Load(TLG: 42) / Store(TSTG: 55)'
0003961D  9681844D E3D3C77A
00039625  40F4F25D 406140E2
0003962D  A3969985 4DE3E2E3
00039635  C77A40F5 F55D4040
0003963D  40404040 40404040
00039645  40404040 40404040
0003964D  40404040 40404040
00039655  40404040 40404040
0003965D  40404040 40404040
00039665  40404040 40404040
0003966D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2412
                                                    2413          TSTEND
00039676                                            2414+         DS    0H
00039676  C5958440 968640A3                       2415+         DC    CL32'End of test: 45'
0003967E  85A2A37A 40F4F540
00039686  40404040 40404040
0003968E  40404040 40404040
                                                    2416
                                                    2417 * ------------------------------------------------------------------
                                                    2418 * Test Triple Load(TL: 41) / Store(TST: 54) instructions
                                                    2419 * ------------------------------------------------------------------
                                                    2420          TSTHDR 99,99,99
00039696                      00039696  0003A000    2421+         ORG   PLOETST+(200*K)+(8*4*K)
0003A000  E2E3C1D9 E3409686                       2422+         DC    CL32'START of test: 46'
0003A008  40A385A2 A37A40F4
0003A010  F6404040 40404040
0003A018  40404040 40404040
0003A020                      0003A020            2423+         USING *,R5                base for test data and test routine
0003A020                      00000000            2424+         USING PLOETEST,R5
0003A020  0003A04C                                2425+T46      DC    A(CODE46)           address of test code
0003A024  002E                                    2426+         DC    H'46'               test number
0003A026  00                                      2427+         DC    X'00'
0003A027  63                                      2428+         DC    HL1'99'             Function
0003A028  63                                      2429+         DC    HL1'99'             expected unequal results (UR)
0003A029  00                                      2430+         DC    HL1'00'             found UR
0003A02A  63                                      2431+         DC    HL1'99'             cc
0003A02B  00                                      2432+         DC    HL1'00'             cc not checked
0003A02C  00                                      2433+         DS    X                   extracted cc
0003A030  00000000 00000000                       2434+         DS    2F                  extract PSW after test (has CC)
0003A038  0003A600                                2435+         DC    A(FMSG46)           address of test fail message
0003A03C  0003A480                                2436+         DC    A(OPSRC46)          address of operand source
0003A040  0003A400                                2437+         DC    A(OPDST46)          address of operand destination
0003A044  0003A500                                2438+         DC    A(PL46)             address of parameter list
0003A048  000011DF                                2439+         DC    A(RANDX+46)         address of random source data
                                                    2440
                                                    2441          TSTCODE
0003A04C                                          2442+CODE46   DS    0F
                                                    2443 *                                  load random source
0003A04C  B982 0011                               2444          XGR   R1,R1
0003A050  5810 501C                   0000001C    2445          L     R1,OPSRCA
0003A054  B982 0022                               2446          XGR   R2,R2
0003A058  5820 5028                   00000028    2447          L     R2,RANDXA
0003A05C  D27F 1000 2000      00000000  00000000  2448          MVC   0(128,R1),0(R2)
                                                    2449 *
                                                    2450 * Triple Load(TL: 41)
                                                    2451 *
0003A062  B982 0033                               2452          XGR   R3,R3
0003A066  5830 5024                   00000024    2453          L     R3,PLA
0003A06A  E313 0000 0024              00000000    2454          STG   R1,0(R3)            load OP1 address in PL
                                                    2455
0003A070  4111 0020                   00000020    2456          LA    R1,32(R1)
0003A074  E313 0008 0024              00000008    2457          STG   R1,8(R3)            load OP2 address in PL
                                                    2458
0003A07A  4111 0020                   00000020    2459          LA    R1,32(R1)
0003A07E  E313 0010 0024              00000010    2460          STG   R1,16(R3)           load OP3 address in PL
                                                    2461
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003A084  B982 0000                            2462          XGR   R0,R0
0003A088  4100 0029                 00000029   2463          LA    R0,41
0003A08C  EE10 0000 3000      00000000  00000000  2464          PLO   R1,0,0,0(R3)        Triple Load(TL: 41)
                                                2465 *
                                                2466 * Triple Store(TST: 54)
                                                2467 *
0003A092  B982 0011                            2468          XGR   R1,R1
0003A096  5810 5020                 00000020   2469          L     R1,OPDSTA           Destination addresses
                                                2470
0003A09A  B982 0033                            2471          XGR   R3,R3
0003A09E  5830 5024                 00000024   2472          L     R3,PLA
0003A0A2  E313 0000 0024            00000000   2473          STG   R1,0(R3)            load OP1 address in PL
                                                2474
0003A0A8  4111 0020                 00000020   2475          LA    R1,32(R1)
0003A0AC  E313 0008 0024            00000008   2476          STG   R1,8(R3)            load OP2 address in PL
                                                2477
0003A0B2  4111 0020                 00000020   2478          LA    R1,32(R1)
0003A0B6  E313 0010 0024            00000010   2479          STG   R1,16(R3)           load OP3 address in PL
                                                2480
0003A0BC  B982 0000                            2481          XGR   R0,R0
0003A0C0  4100 0036                 00000036   2482          LA    R0,54
0003A0C4  EE10 0000 3000      00000000  00000000  2483          PLO   R1,0,0,0(R3)        Triple Store(TST: 54)
                                                2484 *
                                                2485 * compare results
                                                2486 *
0003A0CA  B982 0011                            2487          XGR   R1,R1
0003A0CE  5810 501C                 0000001C   2488          L     R1,OPSRCA           source
0003A0D2  B982 0033                            2489          XGR   R3,R3
0003A0D6  5830 5020                 00000020   2490          L     R3,OPDSTA           Destination
0003A0DA  D503 1000 3000      00000000  00000000  2491          CLC   0(4,R1),0(R3)
0003A0E0  4770 50D8                 0003A0F8   2492          BNE   F4154               OP1 Different, error
0003A0E4  D503 1020 3020      00000020  00000020  2493          CLC   32(4,R1),32(R3)
0003A0EA  4770 50D8                 0003A0F8   2494          BNE   F4154               OP2 Different, error
0003A0EE  D503 1040 3040      00000040  00000040  2495          CLC   64(4,R1),64(R3)
0003A0F4  478B 0000                 00000000   2496          BE    0(R11)              OP3 same, return
                                                2497
                                    0003A0F8   00000001  2498 F4154    EQU   *
0003A0F8  45F0 8340                 00000540   2499          BAL   R15,FAILMSG
0003A0FC  07FB                                 2500          BR    R11
                                                2501
                                                2502          TSTDATA
0003A0FE                            0003A0FE   0003A400  2503+         ORG   PLOETST+(200*K)+(8*4*K)+K
                                                2504          TSTLBL OPDST              Operand Destination (Results)
0003A400                                       2505+OPDST46  DS    0C
0003A400  00000000 00000000                    2506          DC    XL128'0'
0003A408  00000000 00000000
0003A410  00000000 00000000
0003A418  00000000 00000000
0003A420  00000000 00000000
0003A428  00000000 00000000
0003A430  00000000 00000000
0003A438  00000000 00000000
0003A440  00000000 00000000
0003A448  00000000 00000000
0003A450  00000000 00000000
0003A458  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0003A460  00000000 00000000
0003A468  00000000 00000000
0003A470  00000000 00000000
0003A478  00000000 00000000
                                          2507
                                          2508          TSTLBL OPSRC              Operand Source
0003A480                                  2509+OPSRC46  DS    0C
0003A480  00000000 00000000               2510          DC    XL128'0'
0003A488  00000000 00000000
0003A490  00000000 00000000
0003A498  00000000 00000000
0003A4A0  00000000 00000000
0003A4A8  00000000 00000000
0003A4B0  00000000 00000000
0003A4B8  00000000 00000000
0003A4C0  00000000 00000000
0003A4C8  00000000 00000000
0003A4D0  00000000 00000000
0003A4D8  00000000 00000000
0003A4E0  00000000 00000000
0003A4E8  00000000 00000000
0003A4F0  00000000 00000000
0003A4F8  00000000 00000000
                                          2511
                                          2512          TSTLBL PL                 Parameter List
0003A500                                  2513+PL46     DS    0C
0003A500  00000000 00000000               2514          DC    XL256'00'
0003A508  00000000 00000000
0003A510  00000000 00000000
0003A518  00000000 00000000
0003A520  00000000 00000000
0003A528  00000000 00000000
0003A530  00000000 00000000
0003A538  00000000 00000000
0003A540  00000000 00000000
0003A548  00000000 00000000
0003A550  00000000 00000000
0003A558  00000000 00000000
0003A560  00000000 00000000
0003A568  00000000 00000000
0003A570  00000000 00000000
0003A578  00000000 00000000
0003A580  00000000 00000000
0003A588  00000000 00000000
0003A590  00000000 00000000
0003A598  00000000 00000000
0003A5A0  00000000 00000000
0003A5A8  00000000 00000000
0003A5B0  00000000 00000000
0003A5B8  00000000 00000000
0003A5C0  00000000 00000000
0003A5C8  00000000 00000000
0003A5D0  00000000 00000000
0003A5D8  00000000 00000000
0003A5E0  00000000 00000000
0003A5E8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0003A5F0  00000000 00000000
0003A5F8  00000000 00000000
                                                   2515
                                                   2516          TSTLBL FMSG                fail message
0003A600                                           2517+FMSG46   DS     0C
0003A600  86818993 858440D7                        2518          DC     C'failed PLO-Extension '
0003A608  D3D660C5 A7A38595
0003A610  A2899695 40
0003A615  E3998997 938540D3                        2519          DC     CL96'Triple Load(TL: 41) / Store(TST: 54)'
0003A61D  9681844D E3D37A40
0003A625  F4F15D40 6140E2A3
0003A62D  9699854D E3E2E37A
0003A635  40F5F45D 40404040
0003A63D  40404040 40404040
0003A645  40404040 40404040
0003A64D  40404040 40404040
0003A655  40404040 40404040
0003A65D  40404040 40404040
0003A665  40404040 40404040
0003A66D  40404040 40404040
                                                   2520
                                                   2521          TSTEND
0003A676                                           2522+         DS     0H
0003A676  C5958440 968640A3                        2523+         DC     CL32'End of test: 46'
0003A67E  85A2A37A 40F4F640
0003A686  40404040 40404040
0003A68E  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2525 *=======================================================================
                                                    2526 *       Double Load / Store Tests
                                                    2527 *=======================================================================
                                                    2528
                                                    2529 * ---------------------------------------------------------------------
                                                    2530 * Test Double Load(DLO: 40) / Store(DSTO: 53) instructions
                                                    2531 * ---------------------------------------------------------------------
                                                    2532          TSTHDR 99,99,99
0003A696                          0003A696  0003B000 2533+         ORG   PLOETST+(200*K)+(9*4*K)
0003B000  E2E3C1D9 E3409686                         2534+         DC    CL32'START of test: 47'
0003B008  40A385A2 A37A40F4
0003B010  F7404040 40404040
0003B018  40404040 40404040
0003B020                          0003B020          2535+         USING *,R5                base for test data and test routine
0003B020                          00000000          2536+         USING PLOETEST,R5
0003B020  0003B04C                                  2537+T47      DC    A(CODE47)           address of test code
0003B024  002F                                      2538+         DC    H'47'               test number
0003B026  00                                        2539+         DC    X'00'
0003B027  63                                        2540+         DC    HL1'99'             Function
0003B028  63                                        2541+         DC    HL1'99'             expected unequal results (UR)
0003B029  00                                        2542+         DC    HL1'00'             found UR
0003B02A  63                                        2543+         DC    HL1'99'             cc
0003B02B  00                                        2544+         DC    HL1'00'             cc not checked
0003B02C  00                                        2545+         DS    X                   extracted cc
0003B030  00000000 00000000                         2546+         DS    2F                  extract PSW after test (has CC)
0003B038  0003B600                                  2547+         DC    A(FMSG47)           address of test fail message
0003B03C  0003B480                                  2548+         DC    A(OPSRC47)          address of operand source
0003B040  0003B400                                  2549+         DC    A(OPDST47)          address of operand destination
0003B044  0003B500                                  2550+         DC    A(PL47)             address of parameter list
0003B048  000011E0                                  2551+         DC    A(RANDX+47)         address of random source data
                                                    2552
                                                    2553          TSTCODE
0003B04C                                            2554+CODE47   DS    0F
                                                    2555 *                                  load random source
0003B04C  B982 0011                                 2556          XGR   R1,R1
0003B050  5810 501C                     0000001C    2557          L     R1,OPSRCA
0003B054  B982 0022                                 2558          XGR   R2,R2
0003B058  5820 5028                     00000028    2559          L     R2,RANDXA
0003B05C  D27F 1000 2000      00000000  00000000    2560          MVC   0(128,R1),0(R2)
                                                    2561 *
                                                    2562 * Double Load(DLO: 40)
                                                    2563 *
0003B062  B982 0033                                 2564          XGR   R3,R3
0003B066  5830 5024                     00000024    2565          L     R3,PLA
0003B06A  E313 0000 0024                00000000    2566          STG   R1,0(R3)            load OP1 address in PL
                                                    2567
0003B070  4111 0020                     00000020    2568          LA    R1,32(R1)
0003B074  E313 0008 0024                00000008    2569          STG   R1,8(R3)            load OP2 address in PL
                                                    2570
0003B07A  B982 0000                                 2571          XGR   R0,R0
0003B07E  4100 0028                     00000028    2572          LA    R0,40
0003B082  EE10 0000 3000      00000000  00000000    2573          PLO   R1,0,0,0(R3)        Double Load(DLO: 40)
                                                    2574 *
                                                    2575 * Double Store(DSTO: 53)
                                                    2576 *
0003B088  B982 0011                                 2577          XGR   R1,R1
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003B08C  5810 5020                     00000020  2578           L     R1,OPDSTA           Destination addresses
                                                  2579
0003B090  B982 0033                               2580           XGR   R3,R3
0003B094  5830 5024                     00000024  2581           L     R3,PLA
0003B098  E313 0000 0024                00000000  2582           STG   R1,0(R3)            load OP1 address in PL
                                                  2583
0003B09E  4111 0020                     00000020  2584           LA    R1,32(R1)
0003B0A2  E313 0008 0024                00000008  2585           STG   R1,8(R3)            load OP2 address in PL
                                                  2586
0003B0A8  B982 0000                               2587           XGR   R0,R0
0003B0AC  4100 0035                     00000035  2588           LA    R0,53
0003B0B0  EE10 0000 3000  00000000      00000000  2589           PLO   R1,0,0,0(R3)        Double Store(DSTO: 53)
                                                  2590 *
                                                  2591 * compare results
                                                  2592 *
0003B0B6  B982 0011                               2593           XGR   R1,R1
0003B0BA  5810 501C                     0000001C  2594           L     R1,OPSRCA           source
0003B0BE  B982 0033                               2595           XGR   R3,R3
0003B0C2  5830 5020                     00000020  2596           L     R3,OPDSTA           Destination
0003B0C6  D51F 1000 3000  00000000      00000000  2597           CLC   0(32,R1),0(R3)
0003B0CC  4770 50BA                     0003B0DA  2598           BNE   F4053               OP1 Different, error
0003B0D0  D51F 1020 3020  00000020      00000020  2599           CLC   32(32,R1),32(R3)
0003B0D6  478B 0000                     00000000  2600           BE    0(R11)              OP2 same, return
                                                  2601
                          0003B0DA      00000001  2602 F4053     EQU   *
0003B0DA  45F0 8340                     00000540  2603           BAL   R15,FAILMSG
0003B0DE  07FB                                    2604           BR    R11
                                                  2605
                                                  2606           TSTDATA
0003B0E0                  0003B0E0      0003B400  2607+          ORG   PLOETST+(200*K)+(9*4*K)+K
                                                  2608           TSTLBL OPDST              Operand Destination (Results)
0003B400                                          2609+OPDST47   DS    0C
0003B400  00000000 00000000                       2610           DC    XL128'0'
0003B408  00000000 00000000
0003B410  00000000 00000000
0003B418  00000000 00000000
0003B420  00000000 00000000
0003B428  00000000 00000000
0003B430  00000000 00000000
0003B438  00000000 00000000
0003B440  00000000 00000000
0003B448  00000000 00000000
0003B450  00000000 00000000
0003B458  00000000 00000000
0003B460  00000000 00000000
0003B468  00000000 00000000
0003B470  00000000 00000000
0003B478  00000000 00000000
                                                  2611
                                                  2612           TSTLBL OPSRC              Operand Source
0003B480                                          2613+OPSRC47   DS    0C
0003B480  00000000 00000000                       2614           DC    XL128'0'
0003B488  00000000 00000000
0003B490  00000000 00000000
0003B498  00000000 00000000
0003B4A0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0003B4A8  00000000 00000000
0003B4B0  00000000 00000000
0003B4B8  00000000 00000000
0003B4C0  00000000 00000000
0003B4C8  00000000 00000000
0003B4D0  00000000 00000000
0003B4D8  00000000 00000000
0003B4E0  00000000 00000000
0003B4E8  00000000 00000000
0003B4F0  00000000 00000000
0003B4F8  00000000 00000000
                                        2615
                                        2616          TSTLBL PL                 Parameter List
0003B500                                2617+PL47     DS    0C
0003B500  00000000 00000000             2618          DC    XL256'00'
0003B508  00000000 00000000
0003B510  00000000 00000000
0003B518  00000000 00000000
0003B520  00000000 00000000
0003B528  00000000 00000000
0003B530  00000000 00000000
0003B538  00000000 00000000
0003B540  00000000 00000000
0003B548  00000000 00000000
0003B550  00000000 00000000
0003B558  00000000 00000000
0003B560  00000000 00000000
0003B568  00000000 00000000
0003B570  00000000 00000000
0003B578  00000000 00000000
0003B580  00000000 00000000
0003B588  00000000 00000000
0003B590  00000000 00000000
0003B598  00000000 00000000
0003B5A0  00000000 00000000
0003B5A8  00000000 00000000
0003B5B0  00000000 00000000
0003B5B8  00000000 00000000
0003B5C0  00000000 00000000
0003B5C8  00000000 00000000
0003B5D0  00000000 00000000
0003B5D8  00000000 00000000
0003B5E0  00000000 00000000
0003B5E8  00000000 00000000
0003B5F0  00000000 00000000
0003B5F8  00000000 00000000
                                        2619
                                        2620          TSTLBL FMSG               fail message
0003B600                                2621+FMSG47   DS    0C
0003B600  86818993 858440D7             2622          DC    C'failed PLO-Extension '
0003B608  D3D660C5 A7A38595
0003B610  A2899695 40
0003B615  C496A482 938540D3             2623          DC    CL96'Double Load(DLO: 40) / Store(DSTO: 53)'
0003B61D  9681844D C4D3D67A
0003B625  40F4F05D 406140E2
0003B62D  A3969985 4DC4E2E3
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003B635  D67A40F5 F35D4040
0003B63D  40404040 40404040
0003B645  40404040 40404040
0003B64D  40404040 40404040
0003B655  40404040 40404040
0003B65D  40404040 40404040
0003B665  40404040 40404040
0003B66D  40404040 40404040
                                       2624
                                       2625          TSTEND
0003B676                               2626+         DS    0H
0003B676  C5958440 968640A3            2627+         DC    CL32'End of test: 47'
0003B67E  85A2A37A 40F4F740
0003B686  40404040 40404040
0003B68E  40404040 40404040
                                       2628
                                       2629 * ------------------------------------------------------------------
                                       2630 * Test Double Load(DLX: 39) / Store(DSTX: 52) instructions
                                       2631 * ------------------------------------------------------------------
                                       2632          TSTHDR 99,99,99
0003B696                  0003B696  0003C000  2633+         ORG   PLOETST+(200*K)+(10*4*K)
0003C000  E2E3C1D9 E3409686            2634+         DC    CL32'START of test: 48'
0003C008  40A385A2 A37A40F4
0003C010  F8404040 40404040
0003C018  40404040 40404040
0003C020                  0003C020     2635+         USING *,R5                base for test data and test routine
0003C020                  00000000     2636+         USING PLOETEST,R5
0003C020  0003C04C                     2637+T48      DC    A(CODE48)           address of test code
0003C024  0030                         2638+         DC    H'48'               test number
0003C026  00                           2639+         DC    X'00'
0003C027  63                           2640+         DC    HL1'99'             Function
0003C028  63                           2641+         DC    HL1'99'             expected unequal results (UR)
0003C029  00                           2642+         DC    HL1'00'             found UR
0003C02A  63                           2643+         DC    HL1'99'             cc
0003C02B  00                           2644+         DC    HL1'00'             cc not checked
0003C02C  00                           2645+         DS    X                   extracted cc
0003C030  00000000 00000000            2646+         DS    2F                  extract PSW after test (has CC)
0003C038  0003C600                     2647+         DC    A(FMSG48)           address of test fail message
0003C03C  0003C480                     2648+         DC    A(OPSRC48)          address of operand source
0003C040  0003C400                     2649+         DC    A(OPDST48)          address of operand destination
0003C044  0003C500                     2650+         DC    A(PL48)             address of parameter list
0003C048  000011E1                     2651+         DC    A(RANDX+48)         address of random source data
                                       2652
                                       2653          TSTCODE
0003C04C                               2654+CODE48   DS    0F
                                       2655 *                                  load random source
0003C04C  B982 0011                    2656          XGR   R1,R1
0003C050  5810 501C            0000001C  2657          L     R1,OPSRCA
0003C054  B982 0022                    2658          XGR   R2,R2
0003C058  5820 5028            00000028  2659          L     R2,RANDXA
0003C05C  D27F 1000 2000  00000000 00000000  2660          MVC   0(128,R1),0(R2)
                                       2661 *
                                       2662 * Double Load(DLX: 39)
                                       2663 *
0003C062  B982 0033                    2664          XGR   R3,R3
0003C066  5830 5024            00000024  2665          L     R3,PLA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003C06A  E313 0000 0024                 00000000  2666          STG   R1,0(R3)            load OP1 address in PL
                                                   2667
0003C070  4111 0020                      00000020  2668          LA    R1,32(R1)
0003C074  E313 0008 0024                 00000008  2669          STG   R1,8(R3)            load OP2 address in PL
                                                   2670
0003C07A  4111 0020                      00000020  2671          LA    R1,32(R1)
0003C07E  E313 0010 0024                 00000010  2672          STG   R1,16(R3)           load OP3 address in PL
                                                   2673
0003C084  B982 0000                                2674          XGR   R0,R0
0003C088  4100 002B                      0000002B  2675          LA    R0,43
0003C08C  EE10 0000 3000      00000000   00000000  2676          PLO   R1,0,0,0(R3)        Double Load(DLX: 39)
                                                   2677 *
                                                   2678 * Double Store(DSTX: 52)
                                                   2679 *
0003C092  B982 0011                                2680          XGR   R1,R1
0003C096  5810 5020                      00000020  2681          L     R1,OPDSTA           Destination addresses
                                                   2682
0003C09A  B982 0033                                2683          XGR   R3,R3
0003C09E  5830 5024                      00000024  2684          L     R3,PLA
0003C0A2  E313 0000 0024                 00000000  2685          STG   R1,0(R3)            load OP1 address in PL
                                                   2686
0003C0A8  4111 0020                      00000020  2687          LA    R1,32(R1)
0003C0AC  E313 0008 0024                 00000008  2688          STG   R1,8(R3)            load OP2 address in PL
                                                   2689
0003C0B2  4111 0020                      00000020  2690          LA    R1,32(R1)
0003C0B6  E313 0010 0024                 00000010  2691          STG   R1,16(R3)           load OP3 address in PL
                                                   2692
0003C0BC  B982 0000                                2693          XGR   R0,R0
0003C0C0  4100 0034                      00000034  2694          LA    R0,52
0003C0C4  EE10 0000 3000      00000000   00000000  2695          PLO   R1,0,0,0(R3)        Double Store(DSTX: 52)
                                                   2696 *
                                                   2697 * compare results
                                                   2698 *
0003C0CA  B982 0011                                2699          XGR   R1,R1
0003C0CE  5810 501C                      0000001C  2700          L     R1,OPSRCA           source
0003C0D2  B982 0033                                2701          XGR   R3,R3
0003C0D6  5830 5020                      00000020  2702          L     R3,OPDSTA           Destination
0003C0DA  D50F 1000 3000      00000000   00000000  2703          CLC   0(16,R1),0(R3)
0003C0E0  4770 50CE                      0003C0EE  2704          BNE   F4352               OP1 Different, error
0003C0E4  D50F 1020 3020      00000020   00000020  2705          CLC   32(16,R1),32(R3)
0003C0EA  478B 0000                      00000000  2706          BE    0(R11)              OP2 same, return
                                                   2707
                                         0003C0EE  00000001  2708 F4352     EQU   *
0003C0EE  45F0 8340                      00000540  2709          BAL   R15,FAILMSG
0003C0F2  07FB                                     2710          BR    R11
                                                   2711
                                                   2712          TSTDATA
0003C0F4                                 0003C0F4  0003C400  2713+          ORG   PLOETST+(200*K)+(10*4*K)+K
                                                   2714          TSTLBL OPDST              Operand Destination (Results)
0003C400                                           2715+OPDST48  DS    0C
0003C400  00000000 00000000                        2716          DC    XL128'0'
0003C408  00000000 00000000
0003C410  00000000 00000000
0003C418  00000000 00000000
0003C420  00000000 00000000
0003C428  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003C430  00000000 00000000
0003C438  00000000 00000000
0003C440  00000000 00000000
0003C448  00000000 00000000
0003C450  00000000 00000000
0003C458  00000000 00000000
0003C460  00000000 00000000
0003C468  00000000 00000000
0003C470  00000000 00000000
0003C478  00000000 00000000
                                          2717
                                          2718           TSTLBL OPSRC              Operand Source
0003C480                                  2719+OPSRC48   DS    0C
0003C480  00000000 00000000               2720           DC    XL128'0'
0003C488  00000000 00000000
0003C490  00000000 00000000
0003C498  00000000 00000000
0003C4A0  00000000 00000000
0003C4A8  00000000 00000000
0003C4B0  00000000 00000000
0003C4B8  00000000 00000000
0003C4C0  00000000 00000000
0003C4C8  00000000 00000000
0003C4D0  00000000 00000000
0003C4D8  00000000 00000000
0003C4E0  00000000 00000000
0003C4E8  00000000 00000000
0003C4F0  00000000 00000000
0003C4F8  00000000 00000000
                                          2721
                                          2722           TSTLBL PL                 Parameter List
0003C500                                  2723+PL48      DS    0C
0003C500  00000000 00000000               2724           DC    XL256'00'
0003C508  00000000 00000000
0003C510  00000000 00000000
0003C518  00000000 00000000
0003C520  00000000 00000000
0003C528  00000000 00000000
0003C530  00000000 00000000
0003C538  00000000 00000000
0003C540  00000000 00000000
0003C548  00000000 00000000
0003C550  00000000 00000000
0003C558  00000000 00000000
0003C560  00000000 00000000
0003C568  00000000 00000000
0003C570  00000000 00000000
0003C578  00000000 00000000
0003C580  00000000 00000000
0003C588  00000000 00000000
0003C590  00000000 00000000
0003C598  00000000 00000000
0003C5A0  00000000 00000000
0003C5A8  00000000 00000000
0003C5B0  00000000 00000000
0003C5B8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003C5C0  00000000 00000000
0003C5C8  00000000 00000000
0003C5D0  00000000 00000000
0003C5D8  00000000 00000000
0003C5E0  00000000 00000000
0003C5E8  00000000 00000000
0003C5F0  00000000 00000000
0003C5F8  00000000 00000000
                                          2725
                                          2726           TSTLBL FMSG               fail message
0003C600                                  2727+FMSG48    DS     0C
0003C600  86818993 858440D7               2728           DC     C'failed PLO-Extension '
0003C608  D3D660C5 A7A38595
0003C610  A2899695 40
0003C615  C496A482 938540D3               2729           DC     CL96'Double Load(DLX: 39) / Store(DSTX: 52)'
0003C61D  9681844D C4D3E77A
0003C625  40F3F95D 406140E2
0003C62D  A3969985 4DC4E2E3
0003C635  E77A40F5 F25D4040
0003C63D  40404040 40404040
0003C645  40404040 40404040
0003C64D  40404040 40404040
0003C655  40404040 40404040
0003C65D  40404040 40404040
0003C665  40404040 40404040
0003C66D  40404040 40404040
                                          2730
                                          2731           TSTEND
0003C676                                  2732+          DS     0H
0003C676  C5958440 968640A3               2733+          DC     CL32'End of test: 48'
0003C67E  85A2A37A 40F4F840
0003C686  40404040 40404040
0003C68E  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2735 *=====================================================================
                                                    2736 *       Double Store Tests
                                                    2737 *=====================================================================
                                                    2738
                                                    2739 * ---------------------------------------------------------------------
                                                    2740 * Test Double Store(DSTG: 51) instruction
                                                    2741 * ---------------------------------------------------------------------
                                                    2742          TSTHDR 99,99,51
0003C696                          0003C696  0003D000  2743+         ORG   PLOETST+(200*K)+(11*4*K)
0003D000  E2E3C1D9 E3409686                         2744+         DC    CL32'START of test: 49'
0003D008  40A385A2 A37A40F4
0003D010  F9404040 40404040
0003D018  40404040 40404040
0003D020                          0003D020          2745+         USING *,R5                base for test data and test routine
0003D020                          00000000          2746+         USING PLOETEST,R5
0003D020  0003D04C                                  2747+T49      DC    A(CODE49)           address of test code
0003D024  0031                                      2748+         DC    H'49'               test number
0003D026  00                                        2749+         DC    X'00'
0003D027  33                                        2750+         DC    HL1'51'             Function
0003D028  63                                        2751+         DC    HL1'99'             expected unequal results (UR)
0003D029  00                                        2752+         DC    HL1'00'             found UR
0003D02A  63                                        2753+         DC    HL1'99'             cc
0003D02B  00                                        2754+         DC    HL1'00'             cc not checked
0003D02C  00                                        2755+         DS    X                   extracted cc
0003D030  00000000 00000000                         2756+         DS    2F                  extract PSW after test (has CC)
0003D038  0003D600                                  2757+         DC    A(FMSG49)           address of test fail message
0003D03C  0003D480                                  2758+         DC    A(OPSRC49)          address of operand source
0003D040  0003D400                                  2759+         DC    A(OPDST49)          address of operand destination
0003D044  0003D500                                  2760+         DC    A(PL49)             address of parameter list
0003D048  000011E2                                  2761+         DC    A(RANDX+49)         address of random source data
                                                    2762
                                                    2763          TSTCODE
0003D04C                                            2764+CODE49   DS    0F
                                                    2765 *                                  load random source
0003D04C  B982 0011                                 2766          XGR   R1,R1
0003D050  5810 501C                     0000001C    2767          L     R1,OPSRCA
                                                    2768
0003D054  B982 0022                                 2769          XGR   R2,R2
0003D058  5820 5028                     00000028    2770          L     R2,RANDXA
                                                    2771
0003D05C  D27F 1000 2000      00000000  00000000    2772          MVC   0(128,R1),0(R2)
                                                    2773 *
                                                    2774 * load of op1-data, op2-data
                                                    2775 *
0003D062  B982 0033                                 2776          XGR   R3,R3
0003D066  5830 5024                     00000024    2777          L     R3,PLA
                                                    2778
0003D06A  E321 0000 0004                00000000    2779          LG    R2,0(R1)            Get OP1 from src
0003D070  E323 0040 0024                00000040    2780          STG   R2,64(R3)           load OP1-Data in PL
                                                    2781
0003D076  E321 0008 0004                00000008    2782          LG    R2,8(R1)            Get OP2 from src
0003D07C  E323 0048 0024                00000048    2783          STG   R2,72(R3)           load OP2-Data in PL
                                                    2784 *
                                                    2785 * Double Store(DSTG: 51)
                                                    2786 *
0003D082  B982 0011                                 2787          XGR   R1,R1
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003D086  5810 5020                     00000020  2788           L     R1,OPDSTA           Destination addresses
                                                  2789
0003D08A  B982 0033                               2790           XGR   R3,R3
0003D08E  5830 5024                     00000024  2791           L     R3,PLA
0003D092  E313 0000 0024                00000000  2792           STG   R1,0(R3)            load OP1 address in PL
                                                  2793
0003D098  4111 0020                     00000020  2794           LA    R1,32(R1)
0003D09C  E313 0008 0024                00000008  2795           STG   R1,8(R3)            load OP2 address in PL
                                                  2796
0003D0A2  B982 0000                               2797           XGR   R0,R0
0003D0A6  4100 0033                     00000033  2798           LA    R0,51
0003D0AA  EE10 0000 3000      00000000  00000000  2799           PLO   R1,0,0,0(R3)        Double Store(DSTG: 51)
                                                  2800 *
                                                  2801 * compare results
                                                  2802 *
0003D0B0  B982 0011                               2803           XGR   R1,R1
0003D0B4  5810 501C                     0000001C  2804           L     R1,OPSRCA           source
0003D0B8  B982 0033                               2805           XGR   R3,R3
0003D0BC  5830 5020                     00000020  2806           L     R3,OPDSTA           Destination
0003D0C0  D507 1000 3000      00000000  00000000  2807           CLC   0(8,R1),0(R3)
0003D0C6  4770 50B4                     0003D0D4  2808           BNE   FXX51               OP1 Different, error
0003D0CA  D507 1008 3020      00000008  00000020  2809           CLC   8(8,R1),32(R3)
0003D0D0  478B 0000                     00000000  2810           BE    0(R11)              OP2 same, return
                                                  2811
                              0003D0D4  00000001  2812 FXX51     EQU   *
0003D0D4  45F0 8340                     00000540  2813           BAL   R15,FAILMSG
0003D0D8  07FB                                    2814           BR    R11
                                                  2815
                                                  2816           TSTDATA
0003D0DA                    0003D0DA  0003D400  2817+          ORG   PLOETST+(200*K)+(11*4*K)+K
                                                  2818           TSTLBL OPDST              Operand Destination (Results)
0003D400                                          2819+OPDST49   DS    0C
0003D400  00000000 00000000                       2820           DC    XL128'0'
0003D408  00000000 00000000
0003D410  00000000 00000000
0003D418  00000000 00000000
0003D420  00000000 00000000
0003D428  00000000 00000000
0003D430  00000000 00000000
0003D438  00000000 00000000
0003D440  00000000 00000000
0003D448  00000000 00000000
0003D450  00000000 00000000
0003D458  00000000 00000000
0003D460  00000000 00000000
0003D468  00000000 00000000
0003D470  00000000 00000000
0003D478  00000000 00000000
                                                  2821
                                                  2822           TSTLBL OPSRC              Operand Source
0003D480                                          2823+OPSRC49   DS    0C
0003D480  00000000 00000000                       2824           DC    XL128'0'
0003D488  00000000 00000000
0003D490  00000000 00000000
0003D498  00000000 00000000
0003D4A0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003D4A8  00000000 00000000
0003D4B0  00000000 00000000
0003D4B8  00000000 00000000
0003D4C0  00000000 00000000
0003D4C8  00000000 00000000
0003D4D0  00000000 00000000
0003D4D8  00000000 00000000
0003D4E0  00000000 00000000
0003D4E8  00000000 00000000
0003D4F0  00000000 00000000
0003D4F8  00000000 00000000
                                        2825
                                        2826           TSTLBL PL                  Parameter List
0003D500                                2827+PL49      DS    0C
0003D500  00000000 00000000             2828           DC    XL256'00'
0003D508  00000000 00000000
0003D510  00000000 00000000
0003D518  00000000 00000000
0003D520  00000000 00000000
0003D528  00000000 00000000
0003D530  00000000 00000000
0003D538  00000000 00000000
0003D540  00000000 00000000
0003D548  00000000 00000000
0003D550  00000000 00000000
0003D558  00000000 00000000
0003D560  00000000 00000000
0003D568  00000000 00000000
0003D570  00000000 00000000
0003D578  00000000 00000000
0003D580  00000000 00000000
0003D588  00000000 00000000
0003D590  00000000 00000000
0003D598  00000000 00000000
0003D5A0  00000000 00000000
0003D5A8  00000000 00000000
0003D5B0  00000000 00000000
0003D5B8  00000000 00000000
0003D5C0  00000000 00000000
0003D5C8  00000000 00000000
0003D5D0  00000000 00000000
0003D5D8  00000000 00000000
0003D5E0  00000000 00000000
0003D5E8  00000000 00000000
0003D5F0  00000000 00000000
0003D5F8  00000000 00000000
                                        2829
                                        2830           TSTLBL FMSG                fail message
0003D600                                2831+FMSG49    DS    0C
0003D600  86818993 858440D7             2832           DC    C'failed PLO-Extension '
0003D608  D3D660C5 A7A38595
0003D610  A2899695 40
0003D615  C496A482 938540E2             2833           DC    CL96'Double Store(DSTX: 51)'
0003D61D  A3969985 4DC4E2E3
0003D625  E77A40F5 F15D4040
0003D62D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003D635  40404040 40404040
0003D63D  40404040 40404040
0003D645  40404040 40404040
0003D64D  40404040 40404040
0003D655  40404040 40404040
0003D65D  40404040 40404040
0003D665  40404040 40404040
0003D66D  40404040 40404040
                                          2834
                                          2835          TSTEND
0003D676                                  2836+         DS    0H
0003D676  C5958440 968640A3               2837+         DC    CL32'End of test: 49'
0003D67E  85A2A37A 40F4F940
0003D686  40404040 40404040
0003D68E  40404040 40404040
                                          2838
                                          2839 * ------------------------------------------------------------------
                                          2840 * Test Double Store(DSTG: 50) instruction
                                          2841 * ------------------------------------------------------------------
                                          2842          TSTHDR 99,99,50
0003D696                  0003D696  0003E000  2843+         ORG   PLOETST+(200*K)+(12*4*K)
0003E000  E2E3C1D9 E3409686               2844+         DC    CL32'START of test: 50'
0003E008  40A385A2 A37A40F5
0003E010  F0404040 40404040
0003E018  40404040 40404040
0003E020                  0003E020        2845+         USING *,R5                base for test data and test routine
0003E020                  00000000        2846+         USING PLOETEST,R5
0003E020  0003E04C                        2847+T50      DC    A(CODE50)           address of test code
0003E024  0032                            2848+         DC    H'50'               test number
0003E026  00                              2849+         DC    X'00'
0003E027  32                              2850+         DC    HL1'50'             Function
0003E028  63                              2851+         DC    HL1'99'             expected unequal results (UR)
0003E029  00                              2852+         DC    HL1'00'             found UR
0003E02A  63                              2853+         DC    HL1'99'             cc
0003E02B  00                              2854+         DC    HL1'00'             cc not checked
0003E02C  00                              2855+         DS    X                   extracted cc
0003E030  00000000 00000000               2856+         DS    2F                  extract PSW after test (has CC)
0003E038  0003E600                        2857+         DC    A(FMSG50)           address of test fail message
0003E03C  0003E480                        2858+         DC    A(OPSRC50)          address of operand source
0003E040  0003E400                        2859+         DC    A(OPDST50)          address of operand destination
0003E044  0003E500                        2860+         DC    A(PL50)             address of parameter list
0003E048  000011E3                        2861+         DC    A(RANDX+50)         address of random source data
                                          2862
                                          2863          TSTCODE
0003E04C                                  2864+CODE50   DS    0F
                                          2865 *                                  load random source
0003E04C  B982 0011                       2866          XGR   R1,R1
0003E050  5810 501C             0000001C  2867          L     R1,OPSRCA
0003E054  B982 0022                       2868          XGR   R2,R2
0003E058  5820 5028             00000028  2869          L     R2,RANDXA
0003E05C  D27F 1000 2000  00000000  00000000  2870          MVC   0(128,R1),0(R2)
                                          2871 *
                                          2872 * load of op1-data, op2-data
                                          2873 *
0003E062  B982 0033                       2874          XGR   R3,R3
0003E066  5830 5024             00000024  2875          L     R3,PLA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2876
0003E06A  5821 0000                     00000000  2877          L     R2,0(R1)            Get OP1 from src
0003E06E  5023 0040                     00000040  2878          ST    R2,64(R3)           load OP1-Data in PL
                                                    2879
0003E072  5821 0008                     00000008  2880          L     R2,8(R1)            Get OP2 from src
0003E076  5023 0048                     00000048  2881          ST    R2,72(R3)           load OP2-Data in PL
                                                    2882 *
                                                    2883 * Double Store(DSTG: 50
                                                    2884 *
0003E07A  B982 0011                               2885          XGR   R1,R1
0003E07E  5810 5020                     00000020  2886          L     R1,OPDSTA           Destination addresses
                                                    2887
0003E082  B982 0033                               2888          XGR   R3,R3
0003E086  5830 5024                     00000024  2889          L     R3,PLA
0003E08A  E313 0000 0024                00000000  2890          STG   R1,0(R3)            load OP1 address in PL
                                                    2891
0003E090  4111 0020                     00000020  2892          LA    R1,32(R1)
0003E094  E313 0008 0024                00000008  2893          STG   R1,8(R3)            load OP2 address in PL
                                                    2894
0003E09A  B982 0000                               2895          XGR   R0,R0
0003E09E  4100 0032                     00000032  2896          LA    R0,50
0003E0A2  EE10 0000 3000      00000000  00000000  2897          PLO   R1,0,0,0(R3)        Double Store(DSTG: 50)
                                                    2898 *
                                                    2899 * compare results
                                                    2900 *
0003E0A8  B982 0011                               2901          XGR   R1,R1
0003E0AC  5810 501C                     0000001C  2902          L     R1,OPSRCA           source
0003E0B0  B982 0033                               2903          XGR   R3,R3
0003E0B4  5830 5020                     00000020  2904          L     R3,OPDSTA           Destination
0003E0B8  D503 1000 3000      00000000  00000000  2905          CLC   0(4,R1),0(R3)
0003E0BE  4770 50AC                     0003E0CC  2906          BNE   FXX50               OP1 Different, error
0003E0C2  D503 1008 3020      00000008  00000020  2907          CLC   8(4,R1),32(R3)
0003E0C8  478B 0000                     00000000  2908          BE    0(R11)              OP2 same, return
                                                    2909
                              0003E0CC  00000001  2910 FXX50    EQU   *
0003E0CC  45F0 8340                     00000540  2911          BAL   R15,FAILMSG
0003E0D0  07FB                                    2912          BR    R11
                                                    2913
                                                    2914          TSTDATA
0003E0D2                      0003E0D2  0003E400  2915+         ORG   PLOETST+(200*K)+(12*4*K)+K
                                                    2916          TSTLBL OPDST              Operand Destination (Results)
0003E400                                          2917+OPDST50  DS    0C
0003E400  00000000 00000000               2918          DC    XL128'0'
0003E408  00000000 00000000
0003E410  00000000 00000000
0003E418  00000000 00000000
0003E420  00000000 00000000
0003E428  00000000 00000000
0003E430  00000000 00000000
0003E438  00000000 00000000
0003E440  00000000 00000000
0003E448  00000000 00000000
0003E450  00000000 00000000
0003E458  00000000 00000000
0003E460  00000000 00000000
0003E468  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0003E470  00000000 00000000
0003E478  00000000 00000000
                                          2919
                                          2920           TSTLBL OPSRC              Operand Source
0003E480                                  2921+OPSRC50   DS     0C
0003E480  00000000 00000000               2922           DC     XL128'0'
0003E488  00000000 00000000
0003E490  00000000 00000000
0003E498  00000000 00000000
0003E4A0  00000000 00000000
0003E4A8  00000000 00000000
0003E4B0  00000000 00000000
0003E4B8  00000000 00000000
0003E4C0  00000000 00000000
0003E4C8  00000000 00000000
0003E4D0  00000000 00000000
0003E4D8  00000000 00000000
0003E4E0  00000000 00000000
0003E4E8  00000000 00000000
0003E4F0  00000000 00000000
0003E4F8  00000000 00000000
                                          2923
                                          2924           TSTLBL PL                 Parameter List
0003E500                                  2925+PL50      DS     0C
0003E500  00000000 00000000               2926           DC     XL256'00'
0003E508  00000000 00000000
0003E510  00000000 00000000
0003E518  00000000 00000000
0003E520  00000000 00000000
0003E528  00000000 00000000
0003E530  00000000 00000000
0003E538  00000000 00000000
0003E540  00000000 00000000
0003E548  00000000 00000000
0003E550  00000000 00000000
0003E558  00000000 00000000
0003E560  00000000 00000000
0003E568  00000000 00000000
0003E570  00000000 00000000
0003E578  00000000 00000000
0003E580  00000000 00000000
0003E588  00000000 00000000
0003E590  00000000 00000000
0003E598  00000000 00000000
0003E5A0  00000000 00000000
0003E5A8  00000000 00000000
0003E5B0  00000000 00000000
0003E5B8  00000000 00000000
0003E5C0  00000000 00000000
0003E5C8  00000000 00000000
0003E5D0  00000000 00000000
0003E5D8  00000000 00000000
0003E5E0  00000000 00000000
0003E5E8  00000000 00000000
0003E5F0  00000000 00000000
0003E5F8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     2927
                                                     2928          TSTLBL FMSG               fail message
0003E600                                             2929+FMSG50   DS    0C
0003E600  86818993 858440D7                          2930          DC    C'failed PLO-Extension '
0003E608  D3D660C5 A7A38595
0003E610  A2899695 40
0003E615  C496A482 938540E2                          2931          DC    CL96'Double Store(DSTG: 50)'
0003E61D  A3969985 4DC4E2E3
0003E625  C77A40F5 F05D4040
0003E62D  40404040 40404040
0003E635  40404040 40404040
0003E63D  40404040 40404040
0003E645  40404040 40404040
0003E64D  40404040 40404040
0003E655  40404040 40404040
0003E65D  40404040 40404040
0003E665  40404040 40404040
0003E66D  40404040 40404040
                                                     2932
                                                     2933          TSTEND
0003E676                                             2934+         DS    0H
0003E676  C5958440 968640A3                          2935+         DC    CL32'End of test: 50'
0003E67E  85A2A37A 40F5F040
0003E686  40404040 40404040
0003E68E  40404040 40404040
                                                     2936
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2938 *=======================================================================
                                                    2939 *       Load / Store Test
                                                    2940 *=======================================================================
                                                    2941
                                                    2942 * ---------------------------------------------------------------------
                                                    2943 * Test Load(LO: 38) / Store(STO: 49) instructions
                                                    2944 * ---------------------------------------------------------------------
                                                    2945          TSTHDR 99,99,99
0003E696                          0003E696  0003F000 2946+         ORG   PLOETST+(200*K)+(13*4*K)
0003F000  E2E3C1D9 E3409686                         2947+         DC    CL32'START of test: 51'
0003F008  40A385A2 A37A40F5
0003F010  F1404040 40404040
0003F018  40404040 40404040
0003F020                          0003F020          2948+         USING *,R5                base for test data and test routine
0003F020                          00000000          2949+         USING PLOETEST,R5
0003F020  0003F04C                                  2950+T51      DC    A(CODE51)           address of test code
0003F024  0033                                      2951+         DC    H'51'               test number
0003F026  00                                        2952+         DC    X'00'
0003F027  63                                        2953+         DC    HL1'99'             Function
0003F028  63                                        2954+         DC    HL1'99'             expected unequal results (UR)
0003F029  00                                        2955+         DC    HL1'00'             found UR
0003F02A  63                                        2956+         DC    HL1'99'             cc
0003F02B  00                                        2957+         DC    HL1'00'             cc not checked
0003F02C  00                                        2958+         DS    X                   extracted cc
0003F030  00000000 00000000                         2959+         DS    2F                  extract PSW after test (has CC)
0003F038  0003F600                                  2960+         DC    A(FMSG51)           address of test fail message
0003F03C  0003F480                                  2961+         DC    A(OPSRC51)          address of operand source
0003F040  0003F400                                  2962+         DC    A(OPDST51)          address of operand destination
0003F044  0003F500                                  2963+         DC    A(PL51)             address of parameter list
0003F048  000011E4                                  2964+         DC    A(RANDX+51)         address of random source data
                                                    2965
                                                    2966          TSTCODE
0003F04C                                            2967+CODE51   DS    0F
                                                    2968 *                                  load random source
0003F04C  B982 0011                                 2969          XGR   R1,R1
0003F050  5810 501C                     0000001C    2970          L     R1,OPSRCA
0003F054  B982 0022                                 2971          XGR   R2,R2
0003F058  5820 5028                     00000028    2972          L     R2,RANDXA
0003F05C  D27F 1000 2000      00000000  00000000    2973          MVC   0(128,R1),0(R2)
                                                    2974 *
                                                    2975 * Load(LO: 38)
                                                    2976 *
0003F062  B982 0033                                 2977          XGR   R3,R3
0003F066  5830 5024                     00000024    2978          L     R3,PLA
0003F06A  E313 0000 0024                00000000    2979          STG   R1,0(R3)            load OP1 address in PL
                                                    2980
0003F070  B982 0000                                 2981          XGR   R0,R0
0003F074  4100 0026                     00000026    2982          LA    R0,38
0003F078  EE10 0000 3000      00000000  00000000    2983          PLO   R1,0,0,0(R3)        Load(LO: 38)
                                                    2984 *
                                                    2985 * Store(STO: 49)
                                                    2986 *
0003F07E  B982 0011                                 2987          XGR   R1,R1
0003F082  5810 5020                     00000020    2988          L     R1,OPDSTA           Destination addresses
                                                    2989
0003F086  B982 0033                                 2990          XGR   R3,R3
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0003F08A  5830 5024                       00000024  2991          L     R3,PLA
0003F08E  E313 0000 0024                  00000000  2992          STG   R1,0(R3)            load OP1 address in PL
                                                    2993
0003F094  B982 0000                               2994          XGR   R0,R0
0003F098  4100 0031                       00000031  2995          LA    R0,49
0003F09C  EE10 0000 3000  00000000        00000000  2996          PLO   R1,0,0,0(R3)        Store(STO: 49)
                                                    2997 *
                                                    2998 * compare results
                                                    2999 *
0003F0A2  B982 0011                               3000          XGR   R1,R1
0003F0A6  5810 501C                       0000001C  3001          L     R1,OPSRCA           source
0003F0AA  B982 0033                               3002          XGR   R3,R3
0003F0AE  5830 5020                       00000020  3003          L     R3,OPDSTA           Destination
0003F0B2  D51F 1000 3000  00000000        00000000  3004          CLC   0(32,R1),0(R3)
0003F0B8  478B 0000                       00000000  3005          BE    0(R11)              OP1 same, return
                                                    3006
0003F0BC  45F0 8340                       00000540  3007          BAL   R15,FAILMSG
0003F0C0  07FB                                    3008          BR    R11
                                                    3009
                                                    3010          TSTDATA
0003F0C2                  0003F0C2        0003F400  3011+         ORG   PLOETST+(200*K)+(13*4*K)+K
                                                    3012          TSTLBL OPDST              Operand Destination (Results)
0003F400                                            3013+OPDST51  DS    0C
0003F400  00000000 00000000                         3014          DC    XL128'0'
0003F408  00000000 00000000
0003F410  00000000 00000000
0003F418  00000000 00000000
0003F420  00000000 00000000
0003F428  00000000 00000000
0003F430  00000000 00000000
0003F438  00000000 00000000
0003F440  00000000 00000000
0003F448  00000000 00000000
0003F450  00000000 00000000
0003F458  00000000 00000000
0003F460  00000000 00000000
0003F468  00000000 00000000
0003F470  00000000 00000000
0003F478  00000000 00000000
                                                    3015
                                                    3016          TSTLBL OPSRC              Operand Source
0003F480                                            3017+OPSRC51  DS    0C
0003F480  00000000 00000000                         3018          DC    XL128'0'
0003F488  00000000 00000000
0003F490  00000000 00000000
0003F498  00000000 00000000
0003F4A0  00000000 00000000
0003F4A8  00000000 00000000
0003F4B0  00000000 00000000
0003F4B8  00000000 00000000
0003F4C0  00000000 00000000
0003F4C8  00000000 00000000
0003F4D0  00000000 00000000
0003F4D8  00000000 00000000
0003F4E0  00000000 00000000
0003F4E8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003F4F0  00000000 00000000
0003F4F8  00000000 00000000
                                              3019
                                              3020           TSTLBL PL                  Parameter List
0003F500                                      3021+PL51      DS     0C
0003F500  00000000 00000000                   3022           DC     XL256'00'
0003F508  00000000 00000000
0003F510  00000000 00000000
0003F518  00000000 00000000
0003F520  00000000 00000000
0003F528  00000000 00000000
0003F530  00000000 00000000
0003F538  00000000 00000000
0003F540  00000000 00000000
0003F548  00000000 00000000
0003F550  00000000 00000000
0003F558  00000000 00000000
0003F560  00000000 00000000
0003F568  00000000 00000000
0003F570  00000000 00000000
0003F578  00000000 00000000
0003F580  00000000 00000000
0003F588  00000000 00000000
0003F590  00000000 00000000
0003F598  00000000 00000000
0003F5A0  00000000 00000000
0003F5A8  00000000 00000000
0003F5B0  00000000 00000000
0003F5B8  00000000 00000000
0003F5C0  00000000 00000000
0003F5C8  00000000 00000000
0003F5D0  00000000 00000000
0003F5D8  00000000 00000000
0003F5E0  00000000 00000000
0003F5E8  00000000 00000000
0003F5F0  00000000 00000000
0003F5F8  00000000 00000000
                                              3023
                                              3024           TSTLBL FMSG                fail message
0003F600                                      3025+FMSG51    DS     0C
0003F600  86818993 858440D7                   3026           DC     C'failed PLO-Extension '
0003F608  D3D660C5 A7A38595
0003F610  A2899695 40
0003F615  D3968184 4DD3D67A                   3027           DC     CL96'Load(LO: 38) / Store(STO: 49)'
0003F61D  40F3F85D 406140E2
0003F625  A3969985 4DE2E3D6
0003F62D  7A40F4F9 5D404040
0003F635  40404040 40404040
0003F63D  40404040 40404040
0003F645  40404040 40404040
0003F64D  40404040 40404040
0003F655  40404040 40404040
0003F65D  40404040 40404040
0003F665  40404040 40404040
0003F66D  40404040 40404040
                                              3028
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

                                                     3029           TSTEND
0003F676                                             3030+          DS    0H
0003F676  C5958440 968640A3                          3031+          DC    CL32'End of test: 51'
0003F67E  85A2A37A 40F5F140
0003F686  40404040 40404040
0003F68E  40404040 40404040
                                                     3032
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   3034 *=====================================================================
                                                   3035 *       Quadruple Compare and Swap Tests
                                                   3036 *=====================================================================
                                                   3037
                                                   3038 * -------------------------------------------------------------------
                                                   3039 * Test Quadruple Compare and Swap (QCSO: 37) instruction : 1st NE
                                                   3040 * -------------------------------------------------------------------
                                                   3041          TSTHDR 2,1,37             1st Operand NOT Equal (cc=2, UR=1)
0003F696                          0003F696  00040000 3042+         ORG   PLOETST+(200*K)+(14*4*K)
00040000  E2E3C1D9 E3409686                        3043+         DC    CL32'START of test: 52'
00040008  40A385A2 A37A40F5
00040010  F2404040 40404040
00040018  40404040 40404040
00040020                          00040020         3044+         USING *,R5                base for test data and test routine
00040020                          00000000         3045+         USING PLOETEST,R5
00040020  0004004C                                 3046+T52      DC    A(CODE52)           address of test code
00040024  0034                                     3047+         DC    H'52'               test number
00040026  00                                       3048+         DC    X'00'
00040027  25                                       3049+         DC    HL1'37'             Function
00040028  01                                       3050+         DC    HL1'1'              expected unequal results (UR)
00040029  00                                       3051+         DC    HL1'00'             found UR
0004002A  02                                       3052+         DC    HL1'2'              cc
0004002B  0D                                       3053+         DC    HL1'13'             not expected CC mask
0004002C  00                                       3054+         DS    X                   extracted cc
00040030  00000000 00000000                        3055+         DS    2F                  extract PSW after test (has CC)
00040038  00040760                                 3056+         DC    A(FMSG52)           address of test fail message
0004003C  00040500                                 3057+         DC    A(OPSRC52)          address of operand source
00040040  00040400                                 3058+         DC    A(OPDST52)          address of operand destination
00040044  00040600                                 3059+         DC    A(PL52)             address of parameter list
00040048  000011E5                                 3060+         DC    A(RANDX+52)         address of random source data
                                                   3061
                                                   3062          TSTCODE
0004004C                                           3063+CODE52   DS    0F
                                                   3064 *                                  load random source
0004004C  B982 0022                                3065          XGR   R2,R2
00040050  5820 5028                     00000028   3066          L     R2,RANDXA
                                                   3067 *
                                                   3068 * Quadruple Compare and Swap (QCSO: 37)
                                                   3069 * Build Parambter List
                                                   3070 *
00040054  B982 0011                                3071          XGR   R1,R1
00040058  5810 501C                     0000001C   3072          L     R1,OPSRCA
                                                   3073
0004005C  B982 0033                                3074          XGR   R3,R3
00040060  5830 5024                     00000024   3075          L     R3,PLA
                                                   3076
00040064  B982 0044                                3077          XGR   R4,R4
00040068  5840 5020                     00000020   3078          L     R4,OPREPA
                                                   3079
0004006C  D27F 1000 2000        00000000  00000000 3080          MVC   0(128,R1),0(R2)     load Op1-Op4 data in source
00040072  D27F 3040 1000        00000040  00000000 3081          MVC   64(128,R3),0(R1)    copy to PL OP1-Op4 Compare
00040078  D27F 4000 2080        00000000  00000080 3082          MVC   0(128,R4),128(R2)   load initial Op1-Op4 data in ???
0004007E  D27F 30C0 4000        000000C0  00000000 3083          MVC   192(128,R3),0(R4)   copy to PL Op1-Op4 Replacement
                                                   3084 ****
                                                   3085 * force a failure by changing the first operand
                                                   3086 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00040084  D207 1000 4000      00000000  00000000  3087           MVC   0(8,R1),0(R4)
                                                   3088 ****
                                                   3089
0004008A  E313 0000 0024                00000000  3090           STG   R1,0(R3)           load OP1 address in PL
                                                   3091
00040090  4111 0020                     00000020  3092           LA    R1,32(R1)
00040094  E313 0008 0024                00000008  3093           STG   R1,8(R3)           load OP2 address in PL
                                                   3094
0004009A  4111 0020                     00000020  3095           LA    R1,32(R1)
0004009E  E313 0010 0024                00000010  3096           STG   R1,16(R3)          load OP3 address in PL
                                                   3097
000400A4  4111 0020                     00000020  3098           LA    R1,32(R1)
000400A8  E313 0018 0024                00000018  3099           STG   R1,24(R3)          load OP4 address in PL
                                                   3100
000400AE  B982 0000                               3101           XGR   R0,R0
000400B2  4100 0025                     00000025  3102           LA    R0,37
000400B6  EE00 0000 3000      00000000  00000000  3103           PLO   0,0,0,0(R3)        Quadruple C/S (QCSO: 37)
                                                   3104
000400BC  B98D 0020                               3105           EPSW  R2,R0              extract psw
000400C0  5020 5010                     00000010  3106           ST    R2,CCPSW             to save CC
000400C4  E310 500B 0076                0000000B  3107           LB    R1,CCMASK          (failure CC mask)
000400CA  8910 0004                     00000004  3108           SLL   R1,4               (shift to BC instr CC position)
000400CE  4410 510C                     0004012C  3109           EX    R1,F371TSTCC         fail if...
                                                   3110
                                        000400D2  00000001  3111 F371CKUR EQU   *
000400D2  D200 5009 3030      00000009  00000030  3112           MVC   URFOUND,48(R3)     Extract UR
000400D8  D500 5008 3030      00000008  00000030  3113           CLC   UR,48(R3)
000400DE  4780 50C6                     000400E6  3114           BE    F371CKRES
000400E2  45F0 8220                     00000420  3115           BAL   R15,URMSG
                                                   3116 *
                                                   3117 * compare results
                                                   3118 *
                                        000400E6  00000001  3119 F371CKRES EQU   *
                                                   3120 *                         reset adresses
000400E6  B982 0011                               3121           XGR   R1,R1
000400EA  5810 501C                     0000001C  3122           L     R1,OPSRCA          source
000400EE  B982 0033                               3123           XGR   R3,R3
000400F2  5830 5024                     00000024  3124           L     R3,PLA             PL
000400F6  B982 0044                               3125           XGR   R4,R4
000400FA  5840 5020                     00000020  3126           L     R4,OPREPA
                                                   3127
000400FE  D51F 1000 3040      00000000  00000040  3128           CLC   0(32,R1),64(R3)
00040104  4770 5106                     00040126  3129           BNE   FXX137             OP1C Different, error
00040108  D51F 1020 3060      00000020  00000060  3130           CLC   32(32,R1),96(R3)
0004010E  4770 5106                     00040126  3131           BNE   FXX137             OP2C Different, error
00040112  D51F 1040 3080      00000040  00000080  3132           CLC   64(32,R1),128(R3)
00040118  4770 5106                     00040126  3133           BNE   FXX137             OP3C Different, error
0004011C  D51F 1060 30A0      00000060  000000A0  3134           CLC   96(32,R1),160(R3)
00040122  478B 0000                     00000000  3135           BE    0(R11)             OP4Csame, return
                                                   3136
                                        00040126  00000001  3137 FXX137    EQU   *
00040126  45F0 8340                     00000540  3138           BAL   R15,FAILMSG
0004012A  07FB                                    3139           BR    R11
                                                   3140
0004012C  4700 5110                     00040130  3141 F371TSTCC BC    0,F371CCMSG        fail if unexpected condition code
                                        00040130  00000001  3142 F371CCMSG EQU   *
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00040130  45F0 80F2                       000002F2  3143           BAL   R15,CCMSG
00040134  47F0 50B2                       000400D2  3144           B     F371CKUR
                                                    3145
                                                    3146           TSTDATA
00040138                      00040138    00040400  3147+          ORG   PLOETST+(200*K)+(14*4*K)+K
                                                    3148           TSTLBL OPDST              Operand Destination
00040400                                            3149+OPDST52   DS    0C
00040400  00000000 00000000                         3150           DC    XL256'0'
00040408  00000000 00000000
00040410  00000000 00000000
00040418  00000000 00000000
00040420  00000000 00000000
00040428  00000000 00000000
00040430  00000000 00000000
00040438  00000000 00000000
00040440  00000000 00000000
00040448  00000000 00000000
00040450  00000000 00000000
00040458  00000000 00000000
00040460  00000000 00000000
00040468  00000000 00000000
00040470  00000000 00000000
00040478  00000000 00000000
00040480  00000000 00000000
00040488  00000000 00000000
00040490  00000000 00000000
00040498  00000000 00000000
000404A0  00000000 00000000
000404A8  00000000 00000000
000404B0  00000000 00000000
000404B8  00000000 00000000
000404C0  00000000 00000000
000404C8  00000000 00000000
000404D0  00000000 00000000
000404D8  00000000 00000000
000404E0  00000000 00000000
000404E8  00000000 00000000
000404F0  00000000 00000000
000404F8  00000000 00000000
                                                    3151
                                                    3152           TSTLBL OPSRC              Operand Source OP1-OP4
00040500                                            3153+OPSRC52   DS    0C
00040500  00000000 00000000                         3154           DC    XL256'0'
00040508  00000000 00000000
00040510  00000000 00000000
00040518  00000000 00000000
00040520  00000000 00000000
00040528  00000000 00000000
00040530  00000000 00000000
00040538  00000000 00000000
00040540  00000000 00000000
00040548  00000000 00000000
00040550  00000000 00000000
00040558  00000000 00000000
00040560  00000000 00000000
00040568  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00040570  00000000 00000000
00040578  00000000 00000000
00040580  00000000 00000000
00040588  00000000 00000000
00040590  00000000 00000000
00040598  00000000 00000000
000405A0  00000000 00000000
000405A8  00000000 00000000
000405B0  00000000 00000000
000405B8  00000000 00000000
000405C0  00000000 00000000
000405C8  00000000 00000000
000405D0  00000000 00000000
000405D8  00000000 00000000
000405E0  00000000 00000000
000405E8  00000000 00000000
000405F0  00000000 00000000
000405F8  00000000 00000000
                                          3155
                                          3156           TSTLBL PL                 Parameter List
00040600                                  3157+PL52      DS    0C
00040600  00000000 00000000               3158           DC    XL352'00'
00040608  00000000 00000000
00040610  00000000 00000000
00040618  00000000 00000000
00040620  00000000 00000000
00040628  00000000 00000000
00040630  00000000 00000000
00040638  00000000 00000000
00040640  00000000 00000000
00040648  00000000 00000000
00040650  00000000 00000000
00040658  00000000 00000000
00040660  00000000 00000000
00040668  00000000 00000000
00040670  00000000 00000000
00040678  00000000 00000000
00040680  00000000 00000000
00040688  00000000 00000000
00040690  00000000 00000000
00040698  00000000 00000000
000406A0  00000000 00000000
000406A8  00000000 00000000
000406B0  00000000 00000000
000406B8  00000000 00000000
000406C0  00000000 00000000
000406C8  00000000 00000000
000406D0  00000000 00000000
000406D8  00000000 00000000
000406E0  00000000 00000000
000406E8  00000000 00000000
000406F0  00000000 00000000
000406F8  00000000 00000000
00040700  00000000 00000000
00040708  00000000 00000000
00040710  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00040718  00000000 00000000
00040720  00000000 00000000
00040728  00000000 00000000
00040730  00000000 00000000
00040738  00000000 00000000
00040740  00000000 00000000
00040748  00000000 00000000
00040750  00000000 00000000
00040758  00000000 00000000
                                        3159
                                        3160          TSTLBL FMSG               fail message
00040760                                3161+FMSG52   DS    0C
00040760  86818993 858440D7             3162          DC    C'failed PLO-Extension '
00040768  D3D660C5 A7A38595
00040770  A2899695 40
00040775  D8A48184 99A49793             3163          DC    CL96'Quadruple Compare and Swap (QCSO: 37): 1st NE'
0004077D  8540C396 94978199
00040785  85408195 8440E2A6
0004078D  8197404D D8C3E2D6
00040795  7A40F3F7 5D7A40F1
0004079D  A2A340D5 C5404040
000407A5  40404040 40404040
000407AD  40404040 40404040
000407B5  40404040 40404040
000407BD  40404040 40404040
000407C5  40404040 40404040
000407CD  40404040 40404040
                                        3164
                                        3165          TSTEND
000407D6                                3166+         DS    0H
000407D6  C5958440 968640A3             3167+         DC    CL32'End of test: 52'
000407DE  85A2A37A 40F5F240
000407E6  40404040 40404040
000407EE  40404040 40404040
                                        3168
                                        3169 * ------------------------------------------------------------------
                                        3170 * Test Quadruple Compare and Swap (QCSO: 37) instruction : 2nd NE
                                        3171 * ------------------------------------------------------------------
                                        3172          TSTHDR 2,2,37             2nd Operand NOT Equal (cc=2, UR=2)
000407F6                  000407F6  00041000  3173+         ORG   PLOETST+(200*K)+(15*4*K)
00041000  E2E3C1D9 E3409686             3174+         DC    CL32'START of test: 53'
00041008  40A385A2 A37A40F5
00041010  F3404040 40404040
00041018  40404040 40404040
00041020                  00041020          3175+         USING *,R5                base for test data and test routine
00041020                  00000000          3176+         USING PLOETEST,R5
00041020  0004104C                        3177+T53      DC    A(CODE53)           address of test code
00041024  0035                            3178+         DC    H'53'               test number
00041026  00                              3179+         DC    X'00'
00041027  25                              3180+         DC    HL1'37'             Function
00041028  02                              3181+         DC    HL1'2'              expected unequal results (UR)
00041029  00                              3182+         DC    HL1'00'             found UR
0004102A  02                              3183+         DC    HL1'2'              cc
0004102B  0D                              3184+         DC    HL1'13'             not expected CC mask
0004102C  00                              3185+         DS    X                   extracted cc
00041030  00000000 00000000               3186+         DS    2F                  extract PSW after test (has CC)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00041038  00041760                                3187+          DC    A(FMSG53)           address of test fail message
0004103C  00041500                                3188+          DC    A(OPSRC53)          address of operand source
00041040  00041400                                3189+          DC    A(OPDST53)          address of operand destination
00041044  00041600                                3190+          DC    A(PL53)             address of parameter list
00041048  000011E6                                3191+          DC    A(RANDX+53)         address of random source data
                                                  3192
                                                  3193           TSTCODE
0004104C                                          3194+CODE53    DS    0F
                                                  3195 *                                   load random source
0004104C  B982 0022                               3196           XGR   R2,R2
00041050  5820 5028                    00000028   3197           L     R2,RANDXA
                                                  3198 *
                                                  3199 * Quadruple Compare and Swap (QCSO: 37)
                                                  3200 * Build Parambter List
                                                  3201 *
00041054  B982 0011                               3202           XGR   R1,R1
00041058  5810 501C                    0000001C   3203           L     R1,OPSRCA
                                                  3204
0004105C  B982 0033                               3205           XGR   R3,R3
00041060  5830 5024                    00000024   3206           L     R3,PLA
                                                  3207
00041064  B982 0044                               3208           XGR   R4,R4
00041068  5840 5020                    00000020   3209           L     R4,OPREPA
                                                  3210
0004106C  D27F 1000 2000      00000000  00000000  3211           MVC   0(128,R1),0(R2)     load Op1-Op4 data in source
00041072  D27F 3040 1000      00000040  00000000  3212           MVC   64(128,R3),0(R1)    copy to PL OP1-Op4 Compare
00041078  D27F 4000 2080      00000000  00000080  3213           MVC   0(128,R4),128(R2)   load initial Op1-Op4 data in ???
0004107E  D27F 30C0 4000      000000C0  00000000  3214           MVC   192(128,R3),0(R4)   copy to PL Op1-Op4 Replacement
                                                  3215 ****
                                                  3216 * force a failure by changing the 2nd operand
                                                  3217 ****
00041084  D207 1020 4000      00000020  00000000  3218           MVC   32(8,R1),0(R4)
                                                  3219 ****
                                                  3220
0004108A  E313 0000 0024                  00000000  3221           STG   R1,0(R3)            load OP1 address in PL
                                                  3222
00041090  4111 0020                       00000020  3223           LA    R1,32(R1)
00041094  E313 0008 0024                  00000008  3224           STG   R1,8(R3)            load OP2 address in PL
                                                  3225
0004109A  4111 0020                       00000020  3226           LA    R1,32(R1)
0004109E  E313 0010 0024                  00000010  3227           STG   R1,16(R3)           load OP3 address in PL
                                                  3228
000410A4  4111 0020                       00000020  3229           LA    R1,32(R1)
000410A8  E313 0018 0024                  00000018  3230           STG   R1,24(R3)           load OP4 address in PL
                                                  3231
000410AE  B982 0000                               3232           XGR   R0,R0
000410B2  4100 0025                       00000025  3233           LA    R0,37
000410B6  EE00 0000 3000      00000000  00000000  3234           PLO   0,0,0,0(R3)         Quadruple C/S (QCSO: 37)
                                                  3235
000410BC  B98D 0020                               3236           EPSW  R2,R0               extract psw
000410C0  5020 5010                       00000010  3237           ST    R2,CCPSW              to save CC
000410C4  E310 500B 0076                  0000000B  3238           LB    R1,CCMASK           (failure CC mask)
000410CA  8910 0004                       00000004  3239           SLL   R1,4                (shift to BC instr CC position)
000410CE  4410 510C                       0004112C  3240           EX    R1,F372TSTCC            fail if...
                                                  3241
                          000410D2        00000001  3242 F372CKUR  EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000410D2  D200 5009 3030      00000009  00000030  3243          MVC   URFOUND,48(R3)      Extract UR
000410D8  D500 5008 3030      00000008  00000030  3244          CLC   UR,48(R3)
000410DE  4780 50C6                     000410E6  3245          BE    F372CKRES
000410E2  45F0 8220                     00000420  3246          BAL   R15,URMSG
                                                  3247 *
                                                  3248 * compare results
                                                  3249 *
                                00410E6   00000001  3250 F372CKRES EQU   *
                                                  3251 *                                reset adresses
000410E6  B982 0011                               3252          XGR   R1,R1
000410EA  5810 501C                     0000001C  3253          L     R1,OPSRCA           source
000410EE  B982 0033                               3254          XGR   R3,R3
000410F2  5830 5024                     00000024  3255          L     R3,PLA              PL
000410F6  B982 0044                               3256          XGR   R4,R4
000410FA  5840 5020                     00000020  3257          L     R4,OPREPA
                                                  3258
000410FE  D51F 1000 3040      00000000  00000040  3259          CLC   0(32,R1),64(R3)
00041104  4770 5106                     00041126  3260          BNE   FXX237              OP1C Different, error
00041108  D51F 1020 3060      00000020  00000060  3261          CLC   32(32,R1),96(R3)
0004110E  4770 5106                     00041126  3262          BNE   FXX237              OP2C Different, error
00041112  D51F 1040 3080      00000040  00000080  3263          CLC   64(32,R1),128(R3)
00041118  4770 5106                     00041126  3264          BNE   FXX237              OP3C Different, error
0004111C  D51F 1060 30A0      00000060  000000A0  3265          CLC   96(32,R1),160(R3)
00041122  478B 0000                     00000000  3266          BE    0(R11)              OP4Csame, return
                                                  3267
                                00041126  00000001  3268 FXX237   EQU   *
00041126  45F0 8340                     00000540  3269          BAL   R15,FAILMSG
0004112A  07FB                                    3270          BR    R11
                                                  3271
0004112C  4700 5110                     00041130  3272 F372TSTCC BC    0,F372CCMSG         fail if unexpected condition code
                                00041130  00000001  3273 F372CCMSG EQU   *
00041130  45F0 80F2                     000002F2  3274          BAL   R15,CCMSG
00041134  47F0 50B2                     000410D2  3275          B     F372CKUR
                                                  3276
                                                  3277          TSTDATA
00041138                        00041138  00041400  3278+         ORG   PLOETST+(200*K)+(15*4*K)+K
                                                  3279          TSTLBL OPDST              Operand Destination
00041400                                          3280+OPDST53  DS    0C
00041400  00000000 00000000               3281          DC    XL256'0'
00041408  00000000 00000000
00041410  00000000 00000000
00041418  00000000 00000000
00041420  00000000 00000000
00041428  00000000 00000000
00041430  00000000 00000000
00041438  00000000 00000000
00041440  00000000 00000000
00041448  00000000 00000000
00041450  00000000 00000000
00041458  00000000 00000000
00041460  00000000 00000000
00041468  00000000 00000000
00041470  00000000 00000000
00041478  00000000 00000000
00041480  00000000 00000000
00041488  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00041490  00000000 00000000
00041498  00000000 00000000
000414A0  00000000 00000000
000414A8  00000000 00000000
000414B0  00000000 00000000
000414B8  00000000 00000000
000414C0  00000000 00000000
000414C8  00000000 00000000
000414D0  00000000 00000000
000414D8  00000000 00000000
000414E0  00000000 00000000
000414E8  00000000 00000000
000414F0  00000000 00000000
000414F8  00000000 00000000
                                          3282
                                          3283          TSTLBL OPSRC              Operand Source OP1-OP4
00041500                                  3284+OPSRC53  DS     0C
00041500  00000000 00000000               3285          DC     XL256'0'
00041508  00000000 00000000
00041510  00000000 00000000
00041518  00000000 00000000
00041520  00000000 00000000
00041528  00000000 00000000
00041530  00000000 00000000
00041538  00000000 00000000
00041540  00000000 00000000
00041548  00000000 00000000
00041550  00000000 00000000
00041558  00000000 00000000
00041560  00000000 00000000
00041568  00000000 00000000
00041570  00000000 00000000
00041578  00000000 00000000
00041580  00000000 00000000
00041588  00000000 00000000
00041590  00000000 00000000
00041598  00000000 00000000
000415A0  00000000 00000000
000415A8  00000000 00000000
000415B0  00000000 00000000
000415B8  00000000 00000000
000415C0  00000000 00000000
000415C8  00000000 00000000
000415D0  00000000 00000000
000415D8  00000000 00000000
000415E0  00000000 00000000
000415E8  00000000 00000000
000415F0  00000000 00000000
000415F8  00000000 00000000
                                          3286
                                          3287          TSTLBL PL                 Parameter List
00041600                                  3288+PL53     DS     0C
00041600  00000000 00000000               3289          DC     XL352'00'
00041608  00000000 00000000
00041610  00000000 00000000
00041618  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00041620  00000000 00000000
00041628  00000000 00000000
00041630  00000000 00000000
00041638  00000000 00000000
00041640  00000000 00000000
00041648  00000000 00000000
00041650  00000000 00000000
00041658  00000000 00000000
00041660  00000000 00000000
00041668  00000000 00000000
00041670  00000000 00000000
00041678  00000000 00000000
00041680  00000000 00000000
00041688  00000000 00000000
00041690  00000000 00000000
00041698  00000000 00000000
000416A0  00000000 00000000
000416A8  00000000 00000000
000416B0  00000000 00000000
000416B8  00000000 00000000
000416C0  00000000 00000000
000416C8  00000000 00000000
000416D0  00000000 00000000
000416D8  00000000 00000000
000416E0  00000000 00000000
000416E8  00000000 00000000
000416F0  00000000 00000000
000416F8  00000000 00000000
00041700  00000000 00000000
00041708  00000000 00000000
00041710  00000000 00000000
00041718  00000000 00000000
00041720  00000000 00000000
00041728  00000000 00000000
00041730  00000000 00000000
00041738  00000000 00000000
00041740  00000000 00000000
00041748  00000000 00000000
00041750  00000000 00000000
00041758  00000000 00000000
                                                3290
                                                3291          TSTLBL FMSG               fail message
00041760                                        3292+FMSG53   DS     0C
00041760  86818993 858440D7                     3293          DC     C'failed PLO-Extension '
00041768  D3D660C5 A7A38595
00041770  A2899695 40
00041775  D8A48184 99A49793                     3294          DC     CL96'Quadruple Compare and Swap (QCSO: 37): 2nd NE'
0004177D  8540C396 94978199
00041785  85408195 8440E2A6
0004178D  8197404D D8C3E2D6
00041795  7A40F3F7 5D7A40F2
0004179D  958440D5 C5404040
000417A5  40404040 40404040
000417AD  40404040 40404040
000417B5  40404040 40404040
000417BD  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000417C5  40404040 40404040
000417CD  40404040 40404040
                                          3295
                                          3296          TSTEND
000417D6                                  3297+         DS    0H
000417D6  C5958440 968640A3               3298+         DC    CL32'End of test: 53'
000417DE  85A2A37A 40F5F340
000417E6  40404040 40404040
000417EE  40404040 40404040
                                          3299
                                          3300 * ------------------------------------------------------------------
                                          3301 * Test Quadruple Compare and Swap (QCSO: 37) instruction : 3rd NE
                                          3302 * ------------------------------------------------------------------
                                          3303          TSTHDR 2,3,37             3rd Operand NOT Equal (cc=2, UR=3)
000417F6                        000417F6  00042000      3304+         ORG   PLOETST+(200*K)+(16*4*K)
00042000  E2E3C1D9 E3409686               3305+         DC    CL32'START of test: 54'
00042008  40A385A2 A37A40F5
00042010  F4404040 40404040
00042018  40404040 40404040
00042020                        00042020              3306+         USING *,R5                base for test data and test routine
00042020                        00000000              3307+         USING PLOETEST,R5
00042020  0004204C                                    3308+T54      DC    A(CODE54)           address of test code
00042024  0036                                        3309+         DC    H'54'               test number
00042026  00                                          3310+         DC    X'00'
00042027  25                                          3311+         DC    HL1'37'             Function
00042028  03                                          3312+         DC    HL1'3'              expected unequal results (UR)
00042029  00                                          3313+         DC    HL1'00'             found UR
0004202A  02                                          3314+         DC    HL1'2'              cc
0004202B  0D                                          3315+         DC    HL1'13'             not expected CC mask
0004202C  00                                          3316+         DS    X                   extracted cc
00042030  00000000 00000000                           3317+         DS    2F                  extract PSW after test (has CC)
00042038  00042760                                    3318+         DC    A(FMSG54)           address of test fail message
0004203C  00042500                                    3319+         DC    A(OPSRC54)          address of operand source
00042040  00042400                                    3320+         DC    A(OPDST54)          address of operand destination
00042044  00042600                                    3321+         DC    A(PL54)             address of parameter list
00042048  000011E7                                    3322+         DC    A(RANDX+54)         address of random source data
                                                      3323
                                                      3324          TSTCODE
0004204C                                              3325+CODE54   DS    0F
                                                      3326 *                                  load random source
0004204C  B982 0022                                   3327          XGR   R2,R2
00042050  5820 5028                     00000028      3328          L     R2,RANDXA
                                                      3329 *
                                                      3330 * Quadruple Compare and Swap (QCSO: 37)
                                                      3331 * Build Parambter List
                                                      3332 *
00042054  B982 0011                                   3333          XGR   R1,R1
00042058  5810 501C                     0000001C      3334          L     R1,OPSRCA
                                                      3335
0004205C  B982 0033                                   3336          XGR   R3,R3
00042060  5830 5024                     00000024      3337          L     R3,PLA
                                                      3338
00042064  B982 0044                                   3339          XGR   R4,R4
00042068  5840 5020                     00000020      3340          L     R4,OPREPA
                                                      3341
0004206C  D27F 1000 2000      00000000  00000000      3342          MVC   0(128,R1),0(R2)     load Op1-Op4 data in source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00042072  D27F 3040 1000      00000040  00000000  3343          MVC   64(128,R3),0(R1)    copy to PL OP1-Op4 Compare
00042078  D27F 4000 2080      00000000  00000080  3344          MVC   0(128,R4),128(R2)   load initial Op1-Op4 data in ???
0004207E  D27F 30C0 4000      000000C0  00000000  3345          MVC   192(128,R3),0(R4)   copy to PL Op1-Op4 Replacement
                                                  3346 ****
                                                  3347 * force a failure by changing the 3nd operand
                                                  3348 ****
00042084  D207 1040 4000      00000040  00000000  3349          MVC   64(8,R1),0(R4)
                                                  3350 ****
                                                  3351
0004208A  E313 0000 0024                00000000  3352          STG   R1,0(R3)            load OP1 address in PL
                                                  3353
00042090  4111 0020                     00000020  3354          LA    R1,32(R1)
00042094  E313 0008 0024                00000008  3355          STG   R1,8(R3)            load OP2 address in PL
                                                  3356
0004209A  4111 0020                     00000020  3357          LA    R1,32(R1)
0004209E  E313 0010 0024                00000010  3358          STG   R1,16(R3)           load OP3 address in PL
                                                  3359
000420A4  4111 0020                     00000020  3360          LA    R1,32(R1)
000420A8  E313 0018 0024                00000018  3361          STG   R1,24(R3)           load OP4 address in PL
                                                  3362
000420AE  B982 0000                               3363          XGR   R0,R0
000420B2  4100 0025                     00000025  3364          LA    R0,37
000420B6  EE00 0000 3000      00000000  00000000  3365          PLO   0,0,0,0(R3)         Quadruple C/S (QCSO: 37)
                                                  3366
000420BC  B98D 0020                               3367          EPSW  R2,R0               extract psw
000420C0  5020 5010                     00000010  3368          ST    R2,CCPSW              to save CC
000420C4  E310 500B 0076                0000000B  3369          LB    R1,CCMASK           (failure CC mask)
000420CA  8910 0004                     00000004  3370          SLL   R1,4                (shift to BC instr CC position)
000420CE  4410 510C                     0004212C  3371          EX    R1,F373TSTCC            fail if...
                                                  3372
                                 000420D2 00000001  3373 F373CKUR  EQU   *
000420D2  D200 5009 3030      00000009  00000030  3374          MVC   URFOUND,48(R3)      Extract UR
000420D8  D500 5008 3030      00000008  00000030  3375          CLC   UR,48(R3)
000420DE  4780 50C6                     000420E6  3376          BE    F373CKRES
000420E2  45F0 8220                     00000420  3377          BAL   R15,URMSG
                                                  3378 *
                                                  3379 * compare results
                                                  3380 *
                                 000420E6 00000001  3381 F373CKRES EQU   *
                                                  3382 *                                  reset adresses
000420E6  B982 0011                               3383          XGR   R1,R1
000420EA  5810 501C                     0000001C  3384          L     R1,OPSRCA           source
000420EE  B982 0033                               3385          XGR   R3,R3
000420F2  5830 5024                     00000024  3386          L     R3,PLA              PL
000420F6  B982 0044                               3387          XGR   R4,R4
000420FA  5840 5020                     00000020  3388          L     R4,OPREPA
                                                  3389
000420FE  D51F 1000 3040      00000000  00000040  3390          CLC   0(32,R1),64(R3)
00042104  4770 5106                     00042126  3391          BNE   F373XX              OP1C Different, error
00042108  D51F 1020 3060      00000020  00000060  3392          CLC   32(32,R1),96(R3)
0004210E  4770 5106                     00042126  3393          BNE   F373XX              OP2C Different, error
00042112  D51F 1040 3080      00000040  00000080  3394          CLC   64(32,R1),128(R3)
00042118  4770 5106                     00042126  3395          BNE   F373XX              OP3C Different, error
0004211C  D51F 1060 30A0      00000060  000000A0  3396          CLC   96(32,R1),160(R3)
00042122  478B 0000                     00000000  3397          BE    0(R11)              OP4Csame, return
                                                  3398
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                               00042126  00000001  3399 F373XX   EQU   *
00042126  45F0 8340                      00000540  3400          BAL   R15,FAILMSG
0004212A  07FB                           00000000  3401          BR    R11
                                                   3402
0004212C  4700 5110                      00042130  3403 F373TSTCC BC    0,F373CCMSG        fail if unexpected condition code
                               00042130  00000001  3404 F373CCMSG EQU   *
00042130  45F0 80F2                      000002F2  3405          BAL   R15,CCMSG
00042134  47F0 50B2                      000420D2  3406          B     F373CKUR
                                                   3407
                                                   3408          TSTDATA
00042138                       00042138  00042400  3409+         ORG   PLOETST+(200*K)+(16*4*K)+K
                                                   3410          TSTLBL OPDST              Operand Destination
00042400                                           3411+OPDST54  DS    0C
00042400  00000000 00000000                        3412          DC    XL256'0'
00042408  00000000 00000000
00042410  00000000 00000000
00042418  00000000 00000000
00042420  00000000 00000000
00042428  00000000 00000000
00042430  00000000 00000000
00042438  00000000 00000000
00042440  00000000 00000000
00042448  00000000 00000000
00042450  00000000 00000000
00042458  00000000 00000000
00042460  00000000 00000000
00042468  00000000 00000000
00042470  00000000 00000000
00042478  00000000 00000000
00042480  00000000 00000000
00042488  00000000 00000000
00042490  00000000 00000000
00042498  00000000 00000000
000424A0  00000000 00000000
000424A8  00000000 00000000
000424B0  00000000 00000000
000424B8  00000000 00000000
000424C0  00000000 00000000
000424C8  00000000 00000000
000424D0  00000000 00000000
000424D8  00000000 00000000
000424E0  00000000 00000000
000424E8  00000000 00000000
000424F0  00000000 00000000
000424F8  00000000 00000000
                                                   3413
                                                   3414          TSTLBL OPSRC              Operand Source OP1-OP4
00042500                                           3415+OPSRC54  DS    0C
00042500  00000000 00000000                        3416          DC    XL256'0'
00042508  00000000 00000000
00042510  00000000 00000000
00042518  00000000 00000000
00042520  00000000 00000000
00042528  00000000 00000000
00042530  00000000 00000000
00042538  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00042540  00000000 00000000
00042548  00000000 00000000
00042550  00000000 00000000
00042558  00000000 00000000
00042560  00000000 00000000
00042568  00000000 00000000
00042570  00000000 00000000
00042578  00000000 00000000
00042580  00000000 00000000
00042588  00000000 00000000
00042590  00000000 00000000
00042598  00000000 00000000
000425A0  00000000 00000000
000425A8  00000000 00000000
000425B0  00000000 00000000
000425B8  00000000 00000000
000425C0  00000000 00000000
000425C8  00000000 00000000
000425D0  00000000 00000000
000425D8  00000000 00000000
000425E0  00000000 00000000
000425E8  00000000 00000000
000425F0  00000000 00000000
000425F8  00000000 00000000
                                          3417
                                          3418          TSTLBL PL                 Parameter List
00042600                                  3419+PL54     DS    0C
00042600  00000000 00000000               3420          DC    XL352'00'
00042608  00000000 00000000
00042610  00000000 00000000
00042618  00000000 00000000
00042620  00000000 00000000
00042628  00000000 00000000
00042630  00000000 00000000
00042638  00000000 00000000
00042640  00000000 00000000
00042648  00000000 00000000
00042650  00000000 00000000
00042658  00000000 00000000
00042660  00000000 00000000
00042668  00000000 00000000
00042670  00000000 00000000
00042678  00000000 00000000
00042680  00000000 00000000
00042688  00000000 00000000
00042690  00000000 00000000
00042698  00000000 00000000
000426A0  00000000 00000000
000426A8  00000000 00000000
000426B0  00000000 00000000
000426B8  00000000 00000000
000426C0  00000000 00000000
000426C8  00000000 00000000
000426D0  00000000 00000000
000426D8  00000000 00000000
000426E0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000426E8  00000000 00000000
000426F0  00000000 00000000
000426F8  00000000 00000000
00042700  00000000 00000000
00042708  00000000 00000000
00042710  00000000 00000000
00042718  00000000 00000000
00042720  00000000 00000000
00042728  00000000 00000000
00042730  00000000 00000000
00042738  00000000 00000000
00042740  00000000 00000000
00042748  00000000 00000000
00042750  00000000 00000000
00042758  00000000 00000000
                                        3421
                                        3422          TSTLBL FMSG               fail message
00042760                                3423+FMSG54   DS    0C
00042760  86818993 858440D7             3424          DC    C'failed PLO-Extension '
00042768  D3D660C5 A7A38595
00042770  A2899695 40
00042775  D8A48184 99A49793             3425          DC    CL96'Quadruple Compare and Swap (QCSO: 37): 3rd NE'
0004277D  8540C396 94978199
00042785  85408195 8440E2A6
0004278D  8197404D D8C3E2D6
00042795  7A40F3F7 5D7A40F3
0004279D  998440D5 C5404040
000427A5  40404040 40404040
000427AD  40404040 40404040
000427B5  40404040 40404040
000427BD  40404040 40404040
000427C5  40404040 40404040
000427CD  40404040 40404040
                                        3426
                                        3427          TSTEND
000427D6                                3428+         DS    0H
000427D6  C5958440 968640A3             3429+         DC    CL32'End of test: 54'
000427DE  85A2A37A 40F5F440
000427E6  40404040 40404040
000427EE  40404040 40404040
                                        3430
                                        3431 * ------------------------------------------------------------------
                                        3432 * Test Quadruple Compare and Swap (QCSO: 37) instruction : 4th NE
                                        3433 * ------------------------------------------------------------------
                                        3434          TSTHDR 2,4,37             4th Operand NOT Equal (cc=2, UR=4)
000427F6                  000427F6  00043000  3435+         ORG   PLOETST+(200*K)+(17*4*K)
00043000  E2E3C1D9 E3409686             3436+         DC    CL32'START of test: 55'
00043008  40A385A2 A37A40F5
00043010  F5404040 40404040
00043018  40404040 40404040
00043020                  00043020          3437+         USING *,R5                base for test data and test routine
00043020                  00000000          3438+         USING PLOETEST,R5
00043020  0004304C                        3439+T55      DC    A(CODE55)           address of test code
00043024  0037                            3440+         DC    H'55'               test number
00043026  00                              3441+         DC    X'00'
00043027  25                              3442+         DC    HL1'37'             Function
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00043028  04                                       3443+          DC    HL1'4'              expected unequal results (UR)
00043029  00                                       3444+          DC    HL1'00'             found UR
0004302A  02                                       3445+          DC    HL1'2'              cc
0004302B  0D                                       3446+          DC    HL1'13'             not expected CC mask
0004302C  00                                       3447+          DS    X                   extracted cc
00043030  00000000 00000000                        3448+          DS    2F                  extract PSW after test (has CC)
00043038  00043760                                 3449+          DC    A(FMSG55)           address of test fail message
0004303C  00043500                                 3450+          DC    A(OPSRC55)          address of operand source
00043040  00043400                                 3451+          DC    A(OPDST55)          address of operand destination
00043044  00043600                                 3452+          DC    A(PL55)             address of parameter list
00043048  000011E8                                 3453+          DC    A(RANDX+55)         address of random source data
                                                   3454
                                                   3455           TSTCODE
0004304C                                           3456+CODE55    DS    0F
                                                   3457 *                                   load random source
0004304C  B982 0022                                3458           XGR   R2,R2
00043050  5820 5028                     00000028   3459           L     R2,RANDXA
                                                   3460 *
                                                   3461 * Quadruple Compare and Swap (QCSO: 37)
                                                   3462 * Build Parambter List
                                                   3463 *
00043054  B982 0011                                3464           XGR   R1,R1
00043058  5810 501C                     0000001C   3465           L     R1,OPSRCA
                                                   3466
0004305C  B982 0033                                3467           XGR   R3,R3
00043060  5830 5024                     00000024   3468           L     R3,PLA
                                                   3469
00043064  B982 0044                                3470           XGR   R4,R4
00043068  5840 5020                     00000020   3471           L     R4,OPREPA
                                                   3472
0004306C  D27F 1000 2000       00000000  00000000  3473           MVC   0(128,R1),0(R2)     load Op1-Op4 data in source
00043072  D27F 3040 1000       00000040  00000000  3474           MVC   64(128,R3),0(R1)    copy to PL OP1-Op4 Compare
00043078  D27F 4000 2080       00000000  00000080  3475           MVC   0(128,R4),128(R2)   load initial Op1-Op4 data in ???
0004307E  D27F 30C0 4000       000000C0  00000000  3476           MVC   192(128,R3),0(R4)   copy to PL Op1-Op4 Replacement
                                                   3477 ****
                                                   3478 * force a failure by changing the 4th operand
                                                   3479 ****
00043084  D207 1060 4000       00000060  00000000  3480           MVC   96(8,R1),0(R4)
                                                   3481 ****
                                                   3482
0004308A  E313 0000 0024                00000000   3483           STG   R1,0(R3)            load OP1 address in PL
                                                   3484
00043090  4111 0020                     00000020   3485           LA    R1,32(R1)
00043094  E313 0008 0024                00000008   3486           STG   R1,8(R3)            load OP2 address in PL
                                                   3487
0004309A  4111 0020                     00000020   3488           LA    R1,32(R1)
0004309E  E313 0010 0024                00000010   3489           STG   R1,16(R3)           load OP3 address in PL
                                                   3490
000430A4  4111 0020                     00000020   3491           LA    R1,32(R1)
000430A8  E313 0018 0024                00000018   3492           STG   R1,24(R3)           load OP4 address in PL
                                                   3493
000430AE  B982 0000                                3494           XGR   R0,R0
000430B2  4100 0025                     00000025   3495           LA    R0,37
000430B6  EE00 0000 3000       00000000  00000000  3496           PLO   0,0,0,0(R3)         Quadruple C/S (QCSO: 37)
                                                   3497
000430BC  B98D 0020                                3498           EPSW  R2,R0               extract psw
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000430C0  5020 5010                    00000010  3499          ST    R2,CCPSW              to save CC
000430C4  E310 500B 0076               0000000B  3500          LB    R1,CCMASK             (failure CC mask)
000430CA  8910 0004                    00000004  3501          SLL   R1,4                  (shift to BC instr CC position)
000430CE  4410 510C                    0004312C  3502          EX    R1,F374TSTCC            fail if...
                                                 3503
                          000430D2     00000001  3504 F374CKUR  EQU   *
000430D2  D200 5009 3030  00000009     00000030  3505          MVC   URFOUND,48(R3)      Extract UR
000430D8  D500 5008 3030  00000008     00000030  3506          CLC   UR,48(R3)
000430DE  4780 50C6                    000430E6  3507          BE    F374CKRES
000430E2  45F0 8220                    00000420  3508          BAL   R15,URMSG
                                                 3509 *
                                                 3510 * compare results
                                                 3511 *
                          000430E6     00000001  3512 F374CKRES EQU   *
                                                 3513 *                                  reset adresses
000430E6  B982 0011                              3514          XGR   R1,R1
000430EA  5810 501C                    0000001C  3515          L     R1,OPSRCA           source
000430EE  B982 0033                              3516          XGR   R3,R3
000430F2  5830 5024                    00000024  3517          L     R3,PLA              PL
000430F6  B982 0044                              3518          XGR   R4,R4
000430FA  5840 5020                    00000020  3519          L     R4,OPREPA
                                                 3520
000430FE  D51F 1000 3040  00000000     00000040  3521          CLC   0(32,R1),64(R3)
00043104  4770 5106                    00043126  3522          BNE   F374XX              OP1C Different, error
00043108  D51F 1020 3060  00000020     00000060  3523          CLC   32(32,R1),96(R3)
0004310E  4770 5106                    00043126  3524          BNE   F374XX              OP2C Different, error
00043112  D51F 1040 3080  00000040     00000080  3525          CLC   64(32,R1),128(R3)
00043118  4770 5106                    00043126  3526          BNE   F374XX               OP3C Different, error
0004311C  D51F 1060 30A0  00000060     000000A0  3527          CLC   96(32,R1),160(R3)
00043122  478B 0000                    00000000  3528          BE    0(R11)              OP4Csame, return
                                                 3529
                          00043126     00000001  3530 F374XX   EQU   *
00043126  45F0 8340                    00000540  3531          BAL   R15,FAILMSG
0004312A  07FB                                   3532          BR    R11
                                                 3533
0004312C  4700 5110                    00043130  3534 F374TSTCC BC    0,F374CCMSG        fail if unexpected condition code
                          00043130     00000001  3535 F374CCMSG EQU   *
00043130  45F0 80F2                    000002F2  3536          BAL   R15,CCMSG
00043134  47F0 50B2                    000430D2  3537          B     F374CKUR
                                                 3538
                                                 3539          TSTDATA
00043138                  00043138     00043400  3540+         ORG   PLOETST+(200*K)+(17*4*K)+K
                                                 3541          TSTLBL OPDST              Operand Destination
00043400                                         3542+OPDST55  DS    0C
00043400  00000000 00000000                      3543          DC    XL256'0'
00043408  00000000 00000000
00043410  00000000 00000000
00043418  00000000 00000000
00043420  00000000 00000000
00043428  00000000 00000000
00043430  00000000 00000000
00043438  00000000 00000000
00043440  00000000 00000000
00043448  00000000 00000000
00043450  00000000 00000000
00043458  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00043460  00000000 00000000
00043468  00000000 00000000
00043470  00000000 00000000
00043478  00000000 00000000
00043480  00000000 00000000
00043488  00000000 00000000
00043490  00000000 00000000
00043498  00000000 00000000
000434A0  00000000 00000000
000434A8  00000000 00000000
000434B0  00000000 00000000
000434B8  00000000 00000000
000434C0  00000000 00000000
000434C8  00000000 00000000
000434D0  00000000 00000000
000434D8  00000000 00000000
000434E0  00000000 00000000
000434E8  00000000 00000000
000434F0  00000000 00000000
000434F8  00000000 00000000
                                          3544
                                          3545           TSTLBL OPSRC              Operand Source OP1-OP4
00043500                                  3546+OPSRC55   DS    0C
00043500  00000000 00000000               3547           DC    XL256'0'
00043508  00000000 00000000
00043510  00000000 00000000
00043518  00000000 00000000
00043520  00000000 00000000
00043528  00000000 00000000
00043530  00000000 00000000
00043538  00000000 00000000
00043540  00000000 00000000
00043548  00000000 00000000
00043550  00000000 00000000
00043558  00000000 00000000
00043560  00000000 00000000
00043568  00000000 00000000
00043570  00000000 00000000
00043578  00000000 00000000
00043580  00000000 00000000
00043588  00000000 00000000
00043590  00000000 00000000
00043598  00000000 00000000
000435A0  00000000 00000000
000435A8  00000000 00000000
000435B0  00000000 00000000
000435B8  00000000 00000000
000435C0  00000000 00000000
000435C8  00000000 00000000
000435D0  00000000 00000000
000435D8  00000000 00000000
000435E0  00000000 00000000
000435E8  00000000 00000000
000435F0  00000000 00000000
000435F8  00000000 00000000
                                                3548
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   3549           TSTLBL PL                  Parameter List
00043600                                           3550+PL55      DS     0C
00043600  00000000 00000000                        3551           DC     XL352'00'
00043608  00000000 00000000
00043610  00000000 00000000
00043618  00000000 00000000
00043620  00000000 00000000
00043628  00000000 00000000
00043630  00000000 00000000
00043638  00000000 00000000
00043640  00000000 00000000
00043648  00000000 00000000
00043650  00000000 00000000
00043658  00000000 00000000
00043660  00000000 00000000
00043668  00000000 00000000
00043670  00000000 00000000
00043678  00000000 00000000
00043680  00000000 00000000
00043688  00000000 00000000
00043690  00000000 00000000
00043698  00000000 00000000
000436A0  00000000 00000000
000436A8  00000000 00000000
000436B0  00000000 00000000
000436B8  00000000 00000000
000436C0  00000000 00000000
000436C8  00000000 00000000
000436D0  00000000 00000000
000436D8  00000000 00000000
000436E0  00000000 00000000
000436E8  00000000 00000000
000436F0  00000000 00000000
000436F8  00000000 00000000
00043700  00000000 00000000
00043708  00000000 00000000
00043710  00000000 00000000
00043718  00000000 00000000
00043720  00000000 00000000
00043728  00000000 00000000
00043730  00000000 00000000
00043738  00000000 00000000
00043740  00000000 00000000
00043748  00000000 00000000
00043750  00000000 00000000
00043758  00000000 00000000
                                                   3552
                                                   3553           TSTLBL FMSG                fail message
00043760                                           3554+FMSG55    DS     0C
00043760  86818993 858440D7                        3555           DC     C'failed PLO-Extension '
00043768  D3D660C5 A7A38595
00043770  A2899695 40
00043775  D8A48184 99A49793                        3556           DC     CL96'Quadruple Compare and Swap (QCSO: 37): 4th NE'
0004377D  8540C396 94978199
00043785  85408195 8440E2A6
0004378D  8197404D D8C3E2D6
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00043795  7A40F3F7 5D7A40F4
0004379D  A38840D5 C5404040
000437A5  40404040 40404040
000437AD  40404040 40404040
000437B5  40404040 40404040
000437BD  40404040 40404040
000437C5  40404040 40404040
000437CD  40404040 40404040
                                       3557
                                       3558          TSTEND
000437D6                               3559+         DS    0H
000437D6  C5958440 968640A3            3560+         DC    CL32'End of test: 55'
000437DE  85A2A37A 40F5F540
000437E6  40404040 40404040
000437EE  40404040 40404040
                                       3561
                                       3562
                                       3563 * ------------------------------------------------------------------
                                       3564 * Test Quadruple Compare and Swap (QCSO: 37) instruction : All EQ
                                       3565 * ------------------------------------------------------------------
                                       3566          TSTHDR 0,99,37              All Equal (cc=0, UR= unchanged)
000437F6                  000437F6  00044000  3567+         ORG   PLOETST+(200*K)+(18*4*K)
00044000  E2E3C1D9 E3409686            3568+         DC    CL32'START of test: 56'
00044008  40A385A2 A37A40F5
00044010  F6404040 40404040
00044018  40404040 40404040
00044020                  00044020             3569+         USING *,R5                base for test data and test routine
00044020                  00000000             3570+         USING PLOETEST,R5
00044020  0004404C                             3571+T56      DC    A(CODE56)           address of test code
00044024  0038                                 3572+         DC    H'56'               test number
00044026  00                                   3573+         DC    X'00'
00044027  25                                   3574+         DC    HL1'37'             Function
00044028  63                                   3575+         DC    HL1'99'             expected unequal results (UR)
00044029  00                                   3576+         DC    HL1'00'             found UR
0004402A  00                                   3577+         DC    HL1'0'              cc
0004402B  07                                   3578+         DC    HL1'7'              not expected CC mask
0004402C  00                                   3579+         DS    X                   extracted cc
00044030  00000000 00000000                    3580+         DS    2F                  extract PSW after test (has CC)
00044038  00044760                             3581+         DC    A(FMSG56)           address of test fail message
0004403C  00044500                             3582+         DC    A(OPSRC56)          address of operand source
00044040  00044400                             3583+         DC    A(OPDST56)          address of operand destination
00044044  00044600                             3584+         DC    A(PL56)             address of parameter list
00044048  000011E9                             3585+         DC    A(RANDX+56)         address of random source data
                                               3586
                                               3587          TSTCODE
0004404C                                       3588+CODE56   DS    0F
                                               3589 *                                  load random source
0004404C  B982 0022                            3590          XGR   R2,R2
00044050  5820 5028                 00000028   3591          L     R2,RANDXA
                                               3592 *
                                               3593 * Quadruple Compare and Swap (QCSO: 37)
                                               3594 * Build Parambter List
                                               3595 *
00044054  B982 0011                            3596          XGR   R1,R1
00044058  5810 501C                 0000001C   3597          L     R1,OPSRCA
                                               3598
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004405C  B982 0033                              3599           XGR   R3,R3
00044060  5830 5024                   00000024   3600           L     R3,PLA
                                                 3601
00044064  B982 0044                              3602           XGR   R4,R4
00044068  5840 5020                   00000020   3603           L     R4,OPREPA
                                                 3604
0004406C  D27F 1000 2000      00000000  00000000 3605           MVC   0(128,R1),0(R2)     load Op1-Op4 data in source
00044072  D27F 3040 1000      00000040  00000000 3606           MVC   64(128,R3),0(R1)    copy to PL OP1-Op4 Compare
00044078  D27F 4000 2080      00000000  00000080 3607           MVC   0(128,R4),128(R2)   load initial Op1R-Op4R data
0004407E  D27F 30C0 4000      000000C0  00000000 3608           MVC   192(128,R3),0(R4)   copy to PL Op1-Op4 Replacement
                                                 3609 ****
                                                 3610 * Do NOT force a failure by changing an operand
                                                 3611 ****
                                                 3612 *          MVC   96(8,R1),0(R4)
                                                 3613 ****
                                                 3614
00044084  E313 0000 0024                00000000 3615           STG   R1,0(R3)            load OP1 address in PL
                                                 3616
0004408A  4111 0020                   00000020   3617           LA    R1,32(R1)
0004408E  E313 0008 0024                00000008 3618           STG   R1,8(R3)            load OP2 address in PL
                                                 3619
00044094  4111 0020                   00000020   3620           LA    R1,32(R1)
00044098  E313 0010 0024                00000010 3621           STG   R1,16(R3)           load OP3 address in PL
                                                 3622
0004409E  4111 0020                   00000020   3623           LA    R1,32(R1)
000440A2  E313 0018 0024                00000018 3624           STG   R1,24(R3)           load OP4 address in PL
                                                 3625
000440A8  B982 0000                              3626           XGR   R0,R0
000440AC  4100 0025                   00000025   3627           LA    R0,37
000440B0  EE00 0000 3000      00000000  00000000 3628           PLO   0,0,0,0(R3)         Quadruple C/S (QCSO: 37)
                                                 3629
000440B6  B98D 0020                              3630           EPSW  R2,R0               extract psw
000440BA  5020 5010                   00000010   3631           ST    R2,CCPSW              to save CC
000440BE  E310 500B 0076              0000000B   3632           LB    R1,CCMASK           (failure CC mask)
000440C4  8910 0004                   00000004   3633           SLL   R1,4               (shift to BC instr CC position)
000440C8  4410 5106                   00044126   3634           EX    R1,F375TSTCC            fail if...
                                                 3635
                                000440CC  00000001 3636 F375CKUR  EQU   *
000440CC  D200 5009 3030      00000009  00000030 3637           MVC   URFOUND,48(R3)      Extract UR
000440D2  D500 5008 3030      00000008  00000030 3638           CLC   UR,48(R3)
000440D8  4780 50C0                   000440E0   3639           BE    F375CKRES
000440DC  45F0 8220                   00000420   3640           BAL   R15,URMSG
                                                 3641 *
                                                 3642 * compare results
                                                 3643 *
                                000440E0  00000001 3644 F375CKRES EQU   *
                                                 3645 *                                  reset adresses
000440E0  B982 0011                              3646           XGR   R1,R1
000440E4  5810 501C                   0000001C   3647           L     R1,OPSRCA           source
000440E8  B982 0033                              3648           XGR   R3,R3
000440EC  5830 5024                   00000024   3649           L     R3,PLA              PL
000440F0  B982 0044                              3650           XGR   R4,R4
000440F4  5840 5020                   00000020   3651           L     R4,OPREPA
                                                 3652
                                                 3653 * All equal; replacement values copied to operand source
                                                 3654
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000440F8  D51F 1000 30C0      00000000  000000C0  3655           CLC   0(32,R1),192(R3)
000440FE  4770 5100                     00044120  3656           BNE   F375XX                OP1R Different, error
00044102  D51F 1020 30E0      00000020  000000E0  3657           CLC   32(32,R1),224(R3)
00044108  4770 5100                     00044120  3658           BNE   F375XX                OP2R Different, error
0004410C  D51F 1040 3100      00000040  00000100  3659           CLC   64(32,R1),256(R3)
00044112  4770 5100                     00044120  3660           BNE   F375XX                OP3R Different, error
00044116  D51F 1060 3120      00000060  00000120  3661           CLC   96(32,R1),288(R3)
0004411C  478B 0000                     00000000  3662           BE    0(R11)                OP4R same, return
                                                  3663
                              00044120  00000001  3664 F375XX    EQU   *
00044120  45F0 8340                     00000540  3665           BAL   R15,FAILMSG
00044124  07FB                                    3666           BR    R11
                                                  3667
00044126  4700 510A                     0004412A  3668 F375TSTCC BC     0,F375CCMSG         fail if unexpected condition code
                              0004412A  00000001  3669 F375CCMSG EQU   *
0004412A  45F0 80F2                     000002F2  3670           BAL   R15,CCMSG
0004412E  47F0 50AC                     000440CC  3671           B     F375CKUR
                                                  3672
                                                  3673           TSTDATA
00044132                      00044132  00044400  3674+          ORG   PLOETST+(200*K)+(18*4*K)+K
                                                  3675           TSTLBL OPDST              Operand Destination
00044400                                          3676+OPDST56   DS    0C
00044400  00000000 00000000                       3677           DC    XL256'0'
00044408  00000000 00000000
00044410  00000000 00000000
00044418  00000000 00000000
00044420  00000000 00000000
00044428  00000000 00000000
00044430  00000000 00000000
00044438  00000000 00000000
00044440  00000000 00000000
00044448  00000000 00000000
00044450  00000000 00000000
00044458  00000000 00000000
00044460  00000000 00000000
00044468  00000000 00000000
00044470  00000000 00000000
00044478  00000000 00000000
00044480  00000000 00000000
00044488  00000000 00000000
00044490  00000000 00000000
00044498  00000000 00000000
000444A0  00000000 00000000
000444A8  00000000 00000000
000444B0  00000000 00000000
000444B8  00000000 00000000
000444C0  00000000 00000000
000444C8  00000000 00000000
000444D0  00000000 00000000
000444D8  00000000 00000000
000444E0  00000000 00000000
000444E8  00000000 00000000
000444F0  00000000 00000000
000444F8  00000000 00000000
                                                  3678
                                                  3679           TSTLBL OPSRC              Operand Source OP1-OP4
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00044500                                            3680+OPSRC56  DS    0C
00044500  00000000 00000000                         3681          DC    XL256'0'
00044508  00000000 00000000
00044510  00000000 00000000
00044518  00000000 00000000
00044520  00000000 00000000
00044528  00000000 00000000
00044530  00000000 00000000
00044538  00000000 00000000
00044540  00000000 00000000
00044548  00000000 00000000
00044550  00000000 00000000
00044558  00000000 00000000
00044560  00000000 00000000
00044568  00000000 00000000
00044570  00000000 00000000
00044578  00000000 00000000
00044580  00000000 00000000
00044588  00000000 00000000
00044590  00000000 00000000
00044598  00000000 00000000
000445A0  00000000 00000000
000445A8  00000000 00000000
000445B0  00000000 00000000
000445B8  00000000 00000000
000445C0  00000000 00000000
000445C8  00000000 00000000
000445D0  00000000 00000000
000445D8  00000000 00000000
000445E0  00000000 00000000
000445E8  00000000 00000000
000445F0  00000000 00000000
000445F8  00000000 00000000
                                                    3682
                                                    3683          TSTLBL PL                 Parameter List
00044600                                            3684+PL56     DS    0C
00044600  00000000 00000000                         3685          DC    XL352'00'
00044608  00000000 00000000
00044610  00000000 00000000
00044618  00000000 00000000
00044620  00000000 00000000
00044628  00000000 00000000
00044630  00000000 00000000
00044638  00000000 00000000
00044640  00000000 00000000
00044648  00000000 00000000
00044650  00000000 00000000
00044658  00000000 00000000
00044660  00000000 00000000
00044668  00000000 00000000
00044670  00000000 00000000
00044678  00000000 00000000
00044680  00000000 00000000
00044688  00000000 00000000
00044690  00000000 00000000
00044698  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000446A0  00000000 00000000
000446A8  00000000 00000000
000446B0  00000000 00000000
000446B8  00000000 00000000
000446C0  00000000 00000000
000446C8  00000000 00000000
000446D0  00000000 00000000
000446D8  00000000 00000000
000446E0  00000000 00000000
000446E8  00000000 00000000
000446F0  00000000 00000000
000446F8  00000000 00000000
00044700  00000000 00000000
00044708  00000000 00000000
00044710  00000000 00000000
00044718  00000000 00000000
00044720  00000000 00000000
00044728  00000000 00000000
00044730  00000000 00000000
00044738  00000000 00000000
00044740  00000000 00000000
00044748  00000000 00000000
00044750  00000000 00000000
00044758  00000000 00000000
                                          3686
                                          3687          TSTLBL FMSG               fail message
00044760                                  3688+FMSG56   DS    0C
00044760  86818993 858440D7               3689          DC    C'failed PLO-Extension '
00044768  D3D660C5 A7A38595
00044770  A2899695 40
00044775  D8A48184 99A49793               3690          DC    CL96'Quadruple Compare and Swap (QCSO: 37): All EQ'
0004477D  8540C396 94978199
00044785  85408195 8440E2A6
0004478D  8197404D D8C3E2D6
00044795  7A40F3F7 5D7A40C1
0004479D  939340C5 D8404040
000447A5  40404040 40404040
000447AD  40404040 40404040
000447B5  40404040 40404040
000447BD  40404040 40404040
000447C5  40404040 40404040
000447CD  40404040 40404040
                                          3691
                                          3692          TSTEND
000447D6                                  3693+         DS    0H
000447D6  C5958440 968640A3               3694+         DC    CL32'End of test: 56'
000447DE  85A2A37A 40F5F640
000447E6  40404040 40404040
000447EE  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   3696 *============================================================================
                                                   3697 *       Quadruple Compare and Swap Tests
                                                   3698 *============================================================================
                                                   3699
                                                   3700 * ----------------------------------------------------------------------
                                                   3701 * Test Quadruple Compare and Swap (QCSX: 36) instruction : 1st NE
                                                   3702 * ----------------------------------------------------------------------
                                                   3703          TSTHDR 2,1,36             1st Operand NOT Equal (cc=2, UR=1)
000447F6                          000447F6  00045000 3704+         ORG   PLOETST+(200*K)+(19*4*K)
00045000  E2E3C1D9 E3409686                        3705+         DC    CL32'START of test: 57'
00045008  40A385A2 A37A40F5
00045010  F7404040 40404040
00045018  40404040 40404040
00045020                          00045020          3706+         USING *,R5                base for test data and test routine
00045020                          00000000          3707+         USING PLOETEST,R5
00045020  0004504C                                 3708+T57      DC    A(CODE57)           address of test code
00045024  0039                                     3709+         DC    H'57'               test number
00045026  00                                       3710+         DC    X'00'
00045027  24                                       3711+         DC    HL1'36'             Function
00045028  01                                       3712+         DC    HL1'1'              expected unequal results (UR)
00045029  00                                       3713+         DC    HL1'00'             found UR
0004502A  02                                       3714+         DC    HL1'2'              cc
0004502B  0D                                       3715+         DC    HL1'13'             not expected CC mask
0004502C  00                                       3716+         DS    X                   extracted cc
00045030  00000000 00000000                        3717+         DS    2F                  extract PSW after test (has CC)
00045038  00045760                                 3718+         DC    A(FMSG57)           address of test fail message
0004503C  00045500                                 3719+         DC    A(OPSRC57)          address of operand source
00045040  00045400                                 3720+         DC    A(OPDST57)          address of operand destination
00045044  00045600                                 3721+         DC    A(PL57)             address of parameter list
00045048  000011EA                                 3722+         DC    A(RANDX+57)         address of random source data
                                                   3723
                                                   3724          TSTCODE
0004504C                                           3725+CODE57   DS    0F
                                                   3726 *                                  load random source
0004504C  B982 0022                                3727          XGR   R2,R2
00045050  5820 5028                    00000028    3728          L     R2,RANDXA
                                                   3729 *
                                                   3730 * Quadruple Compare and Swap (QCSX: 36)
                                                   3731 * Build Parambter List
                                                   3732 *
00045054  B982 0011                                3733          XGR   R1,R1
00045058  5810 501C                    0000001C    3734          L     R1,OPSRCA
                                                   3735
0004505C  B982 0033                                3736          XGR   R3,R3
00045060  5830 5024                    00000024    3737          L     R3,PLA
                                                   3738
00045064  B982 0044                                3739          XGR   R4,R4
00045068  5840 5020                    00000020    3740          L     R4,OPREPA
                                                   3741
0004506C  D23F 1000 2000      00000000  00000000  3742          MVC   0(64,R1),0(R2)      load Op1-Op4 data in source
00045072  D23F 3040 1000      00000040  00000000  3743          MVC   64(64,R3),0(R1)     copy to PL OP1-Op4 Compare
00045078  D23F 4000 2080      00000000  00000080  3744          MVC   0(64,R4),128(R2)    load initial Op1-Op4 data in ???
0004507E  D23F 30C0 4000      000000C0  00000000  3745          MVC   192(64,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                   3746 ****
                                                   3747 * force a failure by changing the first operand
                                                   3748 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00045084  D207 1000 4000      00000000  00000000  3749          MVC   0(8,R1),0(R4)
                                                   3750 ****
                                                   3751
0004508A  E313 0000 0024                00000000  3752          STG   R1,0(R3)           load OP1 address in PL
                                                   3753
00045090  4111 0010                     00000010  3754          LA    R1,16(R1)
00045094  E313 0008 0024                00000008  3755          STG   R1,8(R3)           load OP2 address in PL
                                                   3756
0004509A  4111 0010                     00000010  3757          LA    R1,16(R1)
0004509E  E313 0010 0024                00000010  3758          STG   R1,16(R3)          load OP3 address in PL
                                                   3759
000450A4  4111 0010                     00000010  3760          LA    R1,16(R1)
000450A8  E313 0018 0024                00000018  3761          STG   R1,24(R3)          load OP4 address in PL
                                                   3762
000450AE  B982 0000                               3763          XGR   R0,R0
000450B2  4100 0024                     00000024  3764          LA    R0,36
000450B6  EE00 0000 3000      00000000  00000000  3765          PLO   0,0,0,0(R3)        Quadruple C/S (QCSX: 36)
                                                   3766
000450BC  B98D 0020                               3767          EPSW  R2,R0              extract psw
000450C0  5020 5010                     00000010  3768          ST    R2,CCPSW             to save CC
000450C4  E310 500B 0076                0000000B  3769          LB    R1,CCMASK          (failure CC mask)
000450CA  8910 0004                     00000004  3770          SLL   R1,4               (shift to BC instr CC position)
000450CE  4410 510C                     0004512C  3771          EX    R1,F361TSTCC         fail if...
                                                   3772
                                000450D2  00000001  3773 F361CKUR EQU   *
000450D2  D200 5009 3030      00000009  00000030  3774          MVC   URFOUND,48(R3)     Extract UR
000450D8  D500 5008 3030      00000008  00000030  3775          CLC   UR,48(R3)
000450DE  4780 50C6                     000450E6  3776          BE    F361CKRES
000450E2  45F0 8220                     00000420  3777          BAL   R15,URMSG
                                                   3778 *
                                                   3779 * compare results
                                                   3780 *
                                000450E6  00000001  3781 F361CKRES EQU   *
                                                   3782 *                           reset adresses
000450E6  B982 0011                               3783          XGR   R1,R1
000450EA  5810 501C                     0000001C  3784          L     R1,OPSRCA          source
000450EE  B982 0033                               3785          XGR   R3,R3
000450F2  5830 5024                     00000024  3786          L     R3,PLA             PL
000450F6  B982 0044                               3787          XGR   R4,R4
000450FA  5840 5020                     00000020  3788          L     R4,OPREPA
                                                   3789
000450FE  D50F 1000 3040      00000000  00000040  3790          CLC   0(16,R1),64(R3)
00045104  4770 5106                     00045126  3791          BNE   F361XX             OP1C Different, error
00045108  D50F 1010 3050      00000010  00000050  3792          CLC   16(16,R1),80(R3)
0004510E  4770 5106                     00045126  3793          BNE   F361XX             OP2C Different, error
00045112  D50F 1020 3060      00000020  00000060  3794          CLC   32(16,R1),96(R3)
00045118  4770 5106                     00045126  3795          BNE   F361XX             OP3C Different, error
0004511C  D50F 1030 3070      00000030  00000070  3796          CLC   48(16,R1),112(R3)
00045122  478B 0000                     00000000  3797          BE    0(R11)             OP4Csame, return
                                                   3798
                                00045126  00000001  3799 F361XX   EQU   *
00045126  45F0 8340                     00000540  3800          BAL   R15,FAILMSG
0004512A  07FB                                    3801          BR    R11
                                                   3802
0004512C  4700 5110                     00045130  3803 F361TSTCC BC    0,F361CCMSG      fail if unexpected condition code
                                00045130  00000001  3804 F361CCMSG EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00045130  45F0 80F2                      000002F2  3805           BAL   R15,CCMSG
00045134  47F0 50B2                      000450D2  3806           B     F361CKUR
                                                   3807
                                                   3808           TSTDATA
00045138                    00045138  00045400  3809+          ORG   PLOETST+(200*K)+(19*4*K)+K
                                                   3810           TSTLBL OPDST              Operand Destination
00045400                                           3811+OPDST57   DS    0C
00045400  00000000 00000000                        3812           DC    XL256'0'
00045408  00000000 00000000
00045410  00000000 00000000
00045418  00000000 00000000
00045420  00000000 00000000
00045428  00000000 00000000
00045430  00000000 00000000
00045438  00000000 00000000
00045440  00000000 00000000
00045448  00000000 00000000
00045450  00000000 00000000
00045458  00000000 00000000
00045460  00000000 00000000
00045468  00000000 00000000
00045470  00000000 00000000
00045478  00000000 00000000
00045480  00000000 00000000
00045488  00000000 00000000
00045490  00000000 00000000
00045498  00000000 00000000
000454A0  00000000 00000000
000454A8  00000000 00000000
000454B0  00000000 00000000
000454B8  00000000 00000000
000454C0  00000000 00000000
000454C8  00000000 00000000
000454D0  00000000 00000000
000454D8  00000000 00000000
000454E0  00000000 00000000
000454E8  00000000 00000000
000454F0  00000000 00000000
000454F8  00000000 00000000
                                                   3813
                                                   3814           TSTLBL OPSRC              Operand Source OP1-OP4
00045500                                           3815+OPSRC57   DS    0C
00045500  00000000 00000000                        3816           DC    XL256'0'
00045508  00000000 00000000
00045510  00000000 00000000
00045518  00000000 00000000
00045520  00000000 00000000
00045528  00000000 00000000
00045530  00000000 00000000
00045538  00000000 00000000
00045540  00000000 00000000
00045548  00000000 00000000
00045550  00000000 00000000
00045558  00000000 00000000
00045560  00000000 00000000
00045568  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00045570  00000000 00000000
00045578  00000000 00000000
00045580  00000000 00000000
00045588  00000000 00000000
00045590  00000000 00000000
00045598  00000000 00000000
000455A0  00000000 00000000
000455A8  00000000 00000000
000455B0  00000000 00000000
000455B8  00000000 00000000
000455C0  00000000 00000000
000455C8  00000000 00000000
000455D0  00000000 00000000
000455D8  00000000 00000000
000455E0  00000000 00000000
000455E8  00000000 00000000
000455F0  00000000 00000000
000455F8  00000000 00000000
                                          3817
                                          3818           TSTLBL PL                  Parameter List
00045600                                  3819+PL57      DS    0C
00045600  00000000 00000000               3820           DC    XL352'00'
00045608  00000000 00000000
00045610  00000000 00000000
00045618  00000000 00000000
00045620  00000000 00000000
00045628  00000000 00000000
00045630  00000000 00000000
00045638  00000000 00000000
00045640  00000000 00000000
00045648  00000000 00000000
00045650  00000000 00000000
00045658  00000000 00000000
00045660  00000000 00000000
00045668  00000000 00000000
00045670  00000000 00000000
00045678  00000000 00000000
00045680  00000000 00000000
00045688  00000000 00000000
00045690  00000000 00000000
00045698  00000000 00000000
000456A0  00000000 00000000
000456A8  00000000 00000000
000456B0  00000000 00000000
000456B8  00000000 00000000
000456C0  00000000 00000000
000456C8  00000000 00000000
000456D0  00000000 00000000
000456D8  00000000 00000000
000456E0  00000000 00000000
000456E8  00000000 00000000
000456F0  00000000 00000000
000456F8  00000000 00000000
00045700  00000000 00000000
00045708  00000000 00000000
00045710  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00045718  00000000 00000000
00045720  00000000 00000000
00045728  00000000 00000000
00045730  00000000 00000000
00045738  00000000 00000000
00045740  00000000 00000000
00045748  00000000 00000000
00045750  00000000 00000000
00045758  00000000 00000000
                                        3821
                                        3822          TSTLBL FMSG               fail message
00045760                                3823+FMSG57   DS    0C
00045760  86818993 858440D7             3824          DC    C'failed PLO-Extension '
00045768  D3D660C5 A7A38595
00045770  A2899695 40
00045775  D8A48184 99A49793             3825          DC    CL96'Quadruple Compare and Swap (QCSX: 36): 1st NE'
0004577D  8540C396 94978199
00045785  85408195 8440E2A6
0004578D  8197404D D8C3E2E7
00045795  7A40F3F6 5D7A40F1
0004579D  A2A340D5 C5404040
000457A5  40404040 40404040
000457AD  40404040 40404040
000457B5  40404040 40404040
000457BD  40404040 40404040
000457C5  40404040 40404040
000457CD  40404040 40404040
                                        3826
                                        3827          TSTEND
000457D6                                3828+         DS    0H
000457D6  C5958440 968640A3             3829+         DC    CL32'End of test: 57'
000457DE  85A2A37A 40F5F740
000457E6  40404040 40404040
000457EE  40404040 40404040
                                        3830
                                        3831 * ------------------------------------------------------------------
                                        3832 * Test Quadruple Compare and Swap (QCSX: 36) instruction : 2nd NE
                                        3833 * ------------------------------------------------------------------
                                        3834          TSTHDR 2,2,36             2nd Operand NOT Equal (cc=2, UR=2)
000457F6                  000457F6  00046000  3835+         ORG   PLOETST+(200*K)+(20*4*K)
00046000  E2E3C1D9 E3409686             3836+         DC    CL32'START of test: 58'
00046008  40A385A2 A37A40F5
00046010  F8404040 40404040
00046018  40404040 40404040
00046020                  00046020          3837+         USING *,R5                base for test data and test routine
00046020                  00000000          3838+         USING PLOETEST,R5
00046020  0004604C                        3839+T58      DC    A(CODE58)           address of test code
00046024  003A                            3840+         DC    H'58'               test number
00046026  00                              3841+         DC    X'00'
00046027  24                              3842+         DC    HL1'36'             Function
00046028  02                              3843+         DC    HL1'2'              expected unequal results (UR)
00046029  00                              3844+         DC    HL1'00'             found UR
0004602A  02                              3845+         DC    HL1'2'              cc
0004602B  0D                              3846+         DC    HL1'13'             not expected CC mask
0004602C  00                              3847+         DS    X                   extracted cc
00046030  00000000 00000000               3848+         DS    2F                  extract PSW after test (has CC)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00046038  00046760                                3849+          DC    A(FMSG58)           address of test fail message
0004603C  00046500                                3850+          DC    A(OPSRC58)          address of operand source
00046040  00046400                                3851+          DC    A(OPDST58)          address of operand destination
00046044  00046600                                3852+          DC    A(PL58)             address of parameter list
00046048  000011EB                                3853+          DC    A(RANDX+58)         address of random source data
                                                  3854
                                                  3855           TSTCODE
0004604C                                          3856+CODE58    DS    0F
                                                  3857 *                                   load random source
0004604C  B982 0022                               3858           XGR   R2,R2
00046050  5820 5028                    00000028   3859           L     R2,RANDXA
                                                  3860 *
                                                  3861 * Quadruple Compare and Swap (QCSX: 36)
                                                  3862 * Build Parambter List
                                                  3863 *
00046054  B982 0011                               3864           XGR   R1,R1
00046058  5810 501C                    0000001C   3865           L     R1,OPSRCA
                                                  3866
0004605C  B982 0033                               3867           XGR   R3,R3
00046060  5830 5024                    00000024   3868           L     R3,PLA
                                                  3869
00046064  B982 0044                               3870           XGR   R4,R4
00046068  5840 5020                    00000020   3871           L     R4,OPREPA
                                                  3872
0004606C  D23F 1000 2000      00000000  00000000  3873           MVC   0(64,R1),0(R2)      load Op1-Op4 data in source
00046072  D23F 3040 1000      00000040  00000000  3874           MVC   64(64,R3),0(R1)     copy to PL OP1-Op4 Compare
00046078  D23F 4000 2080      00000000  00000080  3875           MVC   0(64,R4),128(R2)    load initial Op1-Op4 data in ???
0004607E  D23F 30C0 4000      000000C0  00000000  3876           MVC   192(64,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                  3877 ****
                                                  3878 * force a failure by changing the 2nd operand
                                                  3879 ****
00046084  D207 1010 4000      00000010  00000000  3880           MVC   16(8,R1),0(R4)
                                                  3881 ****
                                                  3882
0004608A  E313 0000 0024                  00000000  3883           STG   R1,0(R3)            load OP1 address in PL
                                                  3884
00046090  4111 0010                       00000010  3885           LA    R1,16(R1)
00046094  E313 0008 0024                  00000008  3886           STG   R1,8(R3)            load OP2 address in PL
                                                  3887
0004609A  4111 0010                       00000010  3888           LA    R1,16(R1)
0004609E  E313 0010 0024                  00000010  3889           STG   R1,16(R3)           load OP3 address in PL
                                                  3890
000460A4  4111 0010                       00000010  3891           LA    R1,16(R1)
000460A8  E313 0018 0024                  00000018  3892           STG   R1,24(R3)           load OP4 address in PL
                                                  3893
000460AE  B982 0000                               3894           XGR   R0,R0
000460B2  4100 0024                       00000024  3895           LA    R0,36
000460B6  EE00 0000 3000      00000000  00000000  3896           PLO   0,0,0,0(R3)         Quadruple C/S (QCSX: 36)
                                                  3897
000460BC  B98D 0020                               3898           EPSW  R2,R0               extract psw
000460C0  5020 5010                       00000010  3899           ST    R2,CCPSW              to save CC
000460C4  E310 500B 0076                  0000000B  3900           LB    R1,CCMASK           (failure CC mask)
000460CA  8910 0004                       00000004  3901           SLL   R1,4                (shift to BC instr CC position)
000460CE  4410 510C                       0004612C  3902           EX    R1,F362TSTCC            fail if...
                                                  3903
                          000460D2        00000001  3904 F362CKUR  EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000460D2  D200 5009 3030      00000009  00000030  3905          MVC   URFOUND,48(R3)      Extract UR
000460D8  D500 5008 3030      00000008  00000030  3906          CLC   UR,48(R3)
000460DE  4780 50C6                     000460E6  3907          BE    F362CKRES
000460E2  45F0 8220                     00000420  3908          BAL   R15,URMSG
                                                  3909 *
                                                  3910 * compare results
                                                  3911 *
                                000460E6  00000001  3912 F362CKRES EQU   *
                                                  3913 *                                  reset adresses
000460E6  B982 0011                               3914          XGR   R1,R1
000460EA  5810 501C                     0000001C  3915          L     R1,OPSRCA           source
000460EE  B982 0033                               3916          XGR   R3,R3
000460F2  5830 5024                     00000024  3917          L     R3,PLA              PL
000460F6  B982 0044                               3918          XGR   R4,R4
000460FA  5840 5020                     00000020  3919          L     R4,OPREPA
                                                  3920
000460FE  D50F 1000 3040      00000000  00000040  3921          CLC   0(16,R1),64(R3)
00046104  4770 5106                     00046126  3922          BNE   F362XX              OP1C Different, error
00046108  D50F 1010 3050      00000010  00000050  3923          CLC   16(16,R1),80(R3)
0004610E  4770 5106                     00046126  3924          BNE   F362XX              OP2C Different, error
00046112  D50F 1020 3060      00000020  00000060  3925          CLC   32(16,R1),96(R3)
00046118  4770 5106                     00046126  3926          BNE   F362XX              OP3C Different, error
0004611C  D50F 1030 3070      00000030  00000070  3927          CLC   48(16,R1),112(R3)
00046122  478B 0000                     00000000  3928          BE    0(R11)              OP4Csame, return
                                                  3929
                                00046126  00000001  3930 F362XX   EQU   *
00046126  45F0 8340                     00000540  3931          BAL   R15,FAILMSG
0004612A  07FB                                    3932          BR    R11
                                                  3933
0004612C  4700 5110                     00046130  3934 F362TSTCC BC    0,F362CCMSG         fail if unexpected condition code
                                00046130  00000001  3935 F362CCMSG EQU   *
00046130  45F0 80F2                     000002F2  3936          BAL   R15,CCMSG
00046134  47F0 50B2                     000460D2  3937          B     F362CKUR
                                                  3938
                                                  3939          TSTDATA
00046138                      00046138  00046400  3940+         ORG   PLOETST+(200*K)+(20*4*K)+K
                                                  3941          TSTLBL OPDST              Operand Destination
00046400                                          3942+OPDST58  DS    0C
00046400  00000000 00000000               3943          DC    XL256'0'
00046408  00000000 00000000
00046410  00000000 00000000
00046418  00000000 00000000
00046420  00000000 00000000
00046428  00000000 00000000
00046430  00000000 00000000
00046438  00000000 00000000
00046440  00000000 00000000
00046448  00000000 00000000
00046450  00000000 00000000
00046458  00000000 00000000
00046460  00000000 00000000
00046468  00000000 00000000
00046470  00000000 00000000
00046478  00000000 00000000
00046480  00000000 00000000
00046488  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00046490  00000000 00000000
00046498  00000000 00000000
000464A0  00000000 00000000
000464A8  00000000 00000000
000464B0  00000000 00000000
000464B8  00000000 00000000
000464C0  00000000 00000000
000464C8  00000000 00000000
000464D0  00000000 00000000
000464D8  00000000 00000000
000464E0  00000000 00000000
000464E8  00000000 00000000
000464F0  00000000 00000000
000464F8  00000000 00000000
                                          3944
                                          3945          TSTLBL OPSRC              Operand Source OP1-OP4
00046500                                  3946+OPSRC58  DS     0C
00046500  00000000 00000000               3947          DC     XL256'0'
00046508  00000000 00000000
00046510  00000000 00000000
00046518  00000000 00000000
00046520  00000000 00000000
00046528  00000000 00000000
00046530  00000000 00000000
00046538  00000000 00000000
00046540  00000000 00000000
00046548  00000000 00000000
00046550  00000000 00000000
00046558  00000000 00000000
00046560  00000000 00000000
00046568  00000000 00000000
00046570  00000000 00000000
00046578  00000000 00000000
00046580  00000000 00000000
00046588  00000000 00000000
00046590  00000000 00000000
00046598  00000000 00000000
000465A0  00000000 00000000
000465A8  00000000 00000000
000465B0  00000000 00000000
000465B8  00000000 00000000
000465C0  00000000 00000000
000465C8  00000000 00000000
000465D0  00000000 00000000
000465D8  00000000 00000000
000465E0  00000000 00000000
000465E8  00000000 00000000
000465F0  00000000 00000000
000465F8  00000000 00000000
                                          3948
                                          3949          TSTLBL PL                 Parameter List
00046600                                  3950+PL58     DS     0C
00046600  00000000 00000000               3951          DC     XL352'00'
00046608  00000000 00000000
00046610  00000000 00000000
00046618  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00046620  00000000 00000000
00046628  00000000 00000000
00046630  00000000 00000000
00046638  00000000 00000000
00046640  00000000 00000000
00046648  00000000 00000000
00046650  00000000 00000000
00046658  00000000 00000000
00046660  00000000 00000000
00046668  00000000 00000000
00046670  00000000 00000000
00046678  00000000 00000000
00046680  00000000 00000000
00046688  00000000 00000000
00046690  00000000 00000000
00046698  00000000 00000000
000466A0  00000000 00000000
000466A8  00000000 00000000
000466B0  00000000 00000000
000466B8  00000000 00000000
000466C0  00000000 00000000
000466C8  00000000 00000000
000466D0  00000000 00000000
000466D8  00000000 00000000
000466E0  00000000 00000000
000466E8  00000000 00000000
000466F0  00000000 00000000
000466F8  00000000 00000000
00046700  00000000 00000000
00046708  00000000 00000000
00046710  00000000 00000000
00046718  00000000 00000000
00046720  00000000 00000000
00046728  00000000 00000000
00046730  00000000 00000000
00046738  00000000 00000000
00046740  00000000 00000000
00046748  00000000 00000000
00046750  00000000 00000000
00046758  00000000 00000000
                                          3952
                                          3953          TSTLBL FMSG               fail message
00046760                                  3954+FMSG58   DS    0C
00046760  86818993 858440D7               3955          DC    C'failed PLO-Extension '
00046768  D3D660C5 A7A38595
00046770  A2899695 40
00046775  D8A48184 99A49793               3956          DC    CL96'Quadruple Compare and Swap (QCSX: 36): 2nd NE'
0004677D  8540C396 94978199
00046785  85408195 8440E2A6
0004678D  8197404D D8C3E2E7
00046795  7A40F3F6 5D7A40F2
0004679D  958440D5 C5404040
000467A5  40404040 40404040
000467AD  40404040 40404040
000467B5  40404040 40404040
000467BD  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000467C5  40404040 40404040
000467CD  40404040 40404040
                                          3957
                                          3958          TSTEND
000467D6                                  3959+         DS    0H
000467D6  C5958440 968640A3               3960+         DC    CL32'End of test: 58'
000467DE  85A2A37A 40F5F840
000467E6  40404040 40404040
000467EE  40404040 40404040
                                          3961
                                          3962 * ------------------------------------------------------------------
                                          3963 * Test Quadruple Compare and Swap (QCSX: 36) instruction : 3rd NE
                                          3964 * ------------------------------------------------------------------
                                          3965          TSTHDR 2,3,36             3rd Operand NOT Equal (cc=2, UR=3)
000467F6                        000467F6  00047000  3966+         ORG   PLOETST+(200*K)+(21*4*K)
00047000  E2E3C1D9 E3409686               3967+         DC    CL32'START of test: 59'
00047008  40A385A2 A37A40F5
00047010  F9404040 40404040
00047018  40404040 40404040
00047020                        00047020            3968+         USING *,R5                base for test data and test routine
00047020                        00000000            3969+         USING PLOETEST,R5
00047020  0004704C                                  3970+T59      DC    A(CODE59)           address of test code
00047024  003B                                      3971+         DC    H'59'               test number
00047026  00                                        3972+         DC    X'00'
00047027  24                                        3973+         DC    HL1'36'             Function
00047028  03                                        3974+         DC    HL1'3'              expected unequal results (UR)
00047029  00                                        3975+         DC    HL1'00'             found UR
0004702A  02                                        3976+         DC    HL1'2'              cc
0004702B  0D                                        3977+         DC    HL1'13'             not expected CC mask
0004702C  00                                        3978+         DS    X                   extracted cc
00047030  00000000 00000000                         3979+         DS    2F                  extract PSW after test (has CC)
00047038  00047760                                  3980+         DC    A(FMSG59)           address of test fail message
0004703C  00047500                                  3981+         DC    A(OPSRC59)          address of operand source
00047040  00047400                                  3982+         DC    A(OPDST59)          address of operand destination
00047044  00047600                                  3983+         DC    A(PL59)             address of parameter list
00047048  000011EC                                  3984+         DC    A(RANDX+59)         address of random source data
                                                    3985
                                                    3986          TSTCODE
0004704C                                            3987+CODE59   DS    0F
                                                    3988 *                                  load random source
0004704C  B982 0022                                 3989          XGR   R2,R2
00047050  5820 5028                       00000028  3990          L     R2,RANDXA
                                                    3991 *
                                                    3992 * Quadruple Compare and Swap (QCSX: 36)
                                                    3993 * Build Parambter List
                                                    3994 *
00047054  B982 0011                                 3995          XGR   R1,R1
00047058  5810 501C                       0000001C  3996          L     R1,OPSRCA
                                                    3997
0004705C  B982 0033                                 3998          XGR   R3,R3
00047060  5830 5024                       00000024  3999          L     R3,PLA
                                                    4000
00047064  B982 0044                                 4001          XGR   R4,R4
00047068  5840 5020                       00000020  4002          L     R4,OPREPA
                                                    4003
0004706C  D23F 1000 2000        00000000  00000000  4004          MVC   0(64,R1),0(R2)      load Op1-Op4 data in source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00047072  D23F 3040 1000      00000040  00000000  4005          MVC   64(64,R3),0(R1)     copy to PL OP1-Op4 Compare
00047078  D23F 4000 2080      00000000  00000080  4006          MVC   0(64,R4),128(R2)    load initial Op1-Op4 data in ???
0004707E  D23F 30C0 4000      000000C0  00000000  4007          MVC   192(64,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                  4008 ****
                                                  4009 * force a failure by changing the 3nd operand
                                                  4010 ****
00047084  D207 1020 4000      00000020  00000000  4011          MVC   32(8,R1),0(R4)
                                                  4012 ****
                                                  4013
0004708A  E313 0000 0024                00000000  4014          STG   R1,0(R3)            load OP1 address in PL
                                                  4015
00047090  4111 0010                     00000010  4016          LA    R1,16(R1)
00047094  E313 0008 0024                00000008  4017          STG   R1,8(R3)            load OP2 address in PL
                                                  4018
0004709A  4111 0010                     00000010  4019          LA    R1,16(R1)
0004709E  E313 0010 0024                00000010  4020          STG   R1,16(R3)           load OP3 address in PL
                                                  4021
000470A4  4111 0010                     00000010  4022          LA    R1,16(R1)
000470A8  E313 0018 0024                00000018  4023          STG   R1,24(R3)           load OP4 address in PL
                                                  4024
000470AE  B982 0000                               4025          XGR   R0,R0
000470B2  4100 0024                     00000024  4026          LA    R0,36
000470B6  EE00 0000 3000      00000000  00000000  4027          PLO   0,0,0,0(R3)         Quadruple C/S (QCSX: 36)
                                                  4028
000470BC  B98D 0020                               4029          EPSW  R2,R0               extract psw
000470C0  5020 5010                     00000010  4030          ST    R2,CCPSW              to save CC
000470C4  E310 500B 0076                0000000B  4031          LB    R1,CCMASK           (failure CC mask)
000470CA  8910 0004                     00000004  4032          SLL   R1,4                (shift to BC instr CC position)
000470CE  4410 510C                     0004712C  4033          EX    R1,F363TSTCC            fail if...
                                                  4034
                                      000470D2    00000001  4035 F363CKUR  EQU   *
000470D2  D200 5009 3030      00000009  00000030  4036          MVC   URFOUND,48(R3)      Extract UR
000470D8  D500 5008 3030      00000008  00000030  4037          CLC   UR,48(R3)
000470DE  4780 50C6                     000470E6  4038          BE    F363CKRES
000470E2  45F0 8220                     00000420  4039          BAL   R15,URMSG
                                                  4040 *
                                                  4041 * compare results
                                                  4042 *
                                      000470E6    00000001  4043 F363CKRES EQU   *
                                                  4044 *                              reset adresses
000470E6  B982 0011                               4045          XGR   R1,R1
000470EA  5810 501C                     0000001C  4046          L     R1,OPSRCA           source
000470EE  B982 0033                               4047          XGR   R3,R3
000470F2  5830 5024                     00000024  4048          L     R3,PLA              PL
000470F6  B982 0044                               4049          XGR   R4,R4
000470FA  5840 5020                     00000020  4050          L     R4,OPREPA
                                                  4051
000470FE  D50F 1000 3040      00000000  00000040  4052          CLC   0(16,R1),64(R3)
00047104  4770 5106                     00047126  4053          BNE   F363XX              OP1C Different, error
00047108  D50F 1010 3050      00000010  00000050  4054          CLC   16(16,R1),80(R3)
0004710E  4770 5106                     00047126  4055          BNE   F363XX              OP2C Different, error
00047112  D50F 1020 3060      00000020  00000060  4056          CLC   32(16,R1),96(R3)
00047118  4770 5106                     00047126  4057          BNE   F363XX              OP3C Different, error
0004711C  D50F 1030 3070      00000030  00000070  4058          CLC   48(16,R1),112(R3)
00047122  478B 0000                     00000000  4059          BE    0(R11)              OP4Csame, return
                                                  4060
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                               00047126  00000001  4061 F363XX   EQU   *
00047126  45F0 8340                      00000540  4062          BAL   R15,FAILMSG
0004712A  07FB                           00000000  4063          BR    R11
                                                   4064
0004712C  4700 5110                      00047130  4065 F363TSTCC BC    0,F363CCMSG        fail if unexpected condition code
                               00047130  00000001  4066 F363CCMSG EQU   *
00047130  45F0 80F2                      000002F2  4067          BAL   R15,CCMSG
00047134  47F0 50B2                      000470D2  4068          B     F363CKUR
                                                   4069
                                                   4070          TSTDATA
00047138                       00047138  00047400  4071+         ORG   PLOETST+(200*K)+(21*4*K)+K
                                                   4072          TSTLBL OPDST              Operand Destination
00047400                                           4073+OPDST59  DS    0C
00047400  00000000 00000000                        4074          DC    XL256'0'
00047408  00000000 00000000
00047410  00000000 00000000
00047418  00000000 00000000
00047420  00000000 00000000
00047428  00000000 00000000
00047430  00000000 00000000
00047438  00000000 00000000
00047440  00000000 00000000
00047448  00000000 00000000
00047450  00000000 00000000
00047458  00000000 00000000
00047460  00000000 00000000
00047468  00000000 00000000
00047470  00000000 00000000
00047478  00000000 00000000
00047480  00000000 00000000
00047488  00000000 00000000
00047490  00000000 00000000
00047498  00000000 00000000
000474A0  00000000 00000000
000474A8  00000000 00000000
000474B0  00000000 00000000
000474B8  00000000 00000000
000474C0  00000000 00000000
000474C8  00000000 00000000
000474D0  00000000 00000000
000474D8  00000000 00000000
000474E0  00000000 00000000
000474E8  00000000 00000000
000474F0  00000000 00000000
000474F8  00000000 00000000
                                                   4075
                                                   4076          TSTLBL OPSRC              Operand Source OP1-OP4
00047500                                           4077+OPSRC59  DS    0C
00047500  00000000 00000000                        4078          DC    XL256'0'
00047508  00000000 00000000
00047510  00000000 00000000
00047518  00000000 00000000
00047520  00000000 00000000
00047528  00000000 00000000
00047530  00000000 00000000
00047538  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00047540  00000000 00000000
00047548  00000000 00000000
00047550  00000000 00000000
00047558  00000000 00000000
00047560  00000000 00000000
00047568  00000000 00000000
00047570  00000000 00000000
00047578  00000000 00000000
00047580  00000000 00000000
00047588  00000000 00000000
00047590  00000000 00000000
00047598  00000000 00000000
000475A0  00000000 00000000
000475A8  00000000 00000000
000475B0  00000000 00000000
000475B8  00000000 00000000
000475C0  00000000 00000000
000475C8  00000000 00000000
000475D0  00000000 00000000
000475D8  00000000 00000000
000475E0  00000000 00000000
000475E8  00000000 00000000
000475F0  00000000 00000000
000475F8  00000000 00000000
                                                4079
                                                4080          TSTLBL PL                 Parameter List
00047600                                        4081+PL59     DS    0C
00047600  00000000 00000000                     4082          DC    XL352'00'
00047608  00000000 00000000
00047610  00000000 00000000
00047618  00000000 00000000
00047620  00000000 00000000
00047628  00000000 00000000
00047630  00000000 00000000
00047638  00000000 00000000
00047640  00000000 00000000
00047648  00000000 00000000
00047650  00000000 00000000
00047658  00000000 00000000
00047660  00000000 00000000
00047668  00000000 00000000
00047670  00000000 00000000
00047678  00000000 00000000
00047680  00000000 00000000
00047688  00000000 00000000
00047690  00000000 00000000
00047698  00000000 00000000
000476A0  00000000 00000000
000476A8  00000000 00000000
000476B0  00000000 00000000
000476B8  00000000 00000000
000476C0  00000000 00000000
000476C8  00000000 00000000
000476D0  00000000 00000000
000476D8  00000000 00000000
000476E0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000476E8  00000000 00000000
000476F0  00000000 00000000
000476F8  00000000 00000000
00047700  00000000 00000000
00047708  00000000 00000000
00047710  00000000 00000000
00047718  00000000 00000000
00047720  00000000 00000000
00047728  00000000 00000000
00047730  00000000 00000000
00047738  00000000 00000000
00047740  00000000 00000000
00047748  00000000 00000000
00047750  00000000 00000000
00047758  00000000 00000000
                                        4083
                                        4084          TSTLBL FMSG               fail message
00047760                                4085+FMSG59   DS    0C
00047760  86818993 858440D7             4086          DC    C'failed PLO-Extension '
00047768  D3D660C5 A7A38595
00047770  A2899695 40
00047775  D8A48184 99A49793             4087          DC    CL96'Quadruple Compare and Swap (QCSX: 36): 3rd NE'
0004777D  8540C396 94978199
00047785  85408195 8440E2A6
0004778D  8197404D D8C3E2E7
00047795  7A40F3F6 5D7A40F3
0004779D  998440D5 C5404040
000477A5  40404040 40404040
000477AD  40404040 40404040
000477B5  40404040 40404040
000477BD  40404040 40404040
000477C5  40404040 40404040
000477CD  40404040 40404040
                                        4088
                                        4089          TSTEND
000477D6                                4090+         DS    0H
000477D6  C5958440 968640A3             4091+         DC    CL32'End of test: 59'
000477DE  85A2A37A 40F5F940
000477E6  40404040 40404040
000477EE  40404040 40404040
                                        4092
                                        4093 * ------------------------------------------------------------------
                                        4094 * Test Quadruple Compare and Swap (QCSX: 36) instruction : 4th NE
                                        4095 * ------------------------------------------------------------------
                                        4096          TSTHDR 2,4,36             4th Operand NOT Equal (cc=2, UR=4)
000477F6                  000477F6  00048000  4097+         ORG   PLOETST+(200*K)+(22*4*K)
00048000  E2E3C1D9 E3409686             4098+         DC    CL32'START of test: 60'
00048008  40A385A2 A37A40F6
00048010  F0404040 40404040
00048018  40404040 40404040
00048020                  00048020          4099+         USING *,R5                base for test data and test routine
00048020                  00000000          4100+         USING PLOETEST,R5
00048020  0004804C                        4101+T60      DC    A(CODE60)           address of test code
00048024  003C                            4102+         DC    H'60'               test number
00048026  00                              4103+         DC    X'00'
00048027  24                              4104+         DC    HL1'36'             Function
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00048028  04                                       4105+          DC    HL1'4'              expected unequal results (UR)
00048029  00                                       4106+          DC    HL1'00'             found UR
0004802A  02                                       4107+          DC    HL1'2'              cc
0004802B  0D                                       4108+          DC    HL1'13'             not expected CC mask
0004802C  00                                       4109+          DS    X                   extracted cc
00048030  00000000 00000000                        4110+          DS    2F                  extract PSW after test (has CC)
00048038  00048760                                 4111+          DC    A(FMSG60)           address of test fail message
0004803C  00048500                                 4112+          DC    A(OPSRC60)          address of operand source
00048040  00048400                                 4113+          DC    A(OPDST60)          address of operand destination
00048044  00048600                                 4114+          DC    A(PL60)             address of parameter list
00048048  000011ED                                 4115+          DC    A(RANDX+60)         address of random source data
                                                   4116
                                                   4117           TSTCODE
0004804C                                           4118+CODE60    DS    0F
                                                   4119 *                                   load random source
0004804C  B982 0022                                4120           XGR   R2,R2
00048050  5820 5028                    00000028    4121           L     R2,RANDXA
                                                   4122 *
                                                   4123 * Quadruple Compare and Swap (QCSX: 36)
                                                   4124 * Build Parambter List
                                                   4125 *
00048054  B982 0011                                4126           XGR   R1,R1
00048058  5810 501C                    0000001C    4127           L     R1,OPSRCA
                                                   4128
0004805C  B982 0033                                4129           XGR   R3,R3
00048060  5830 5024                    00000024    4130           L     R3,PLA
                                                   4131
00048064  B982 0044                                4132           XGR   R4,R4
00048068  5840 5020                    00000020    4133           L     R4,OPREPA
                                                   4134
0004806C  D23F 1000 2000       00000000  00000000  4135           MVC   0(64,R1),0(R2)      load Op1-Op4 data in source
00048072  D23F 3040 1000       00000040  00000000  4136           MVC   64(64,R3),0(R1)     copy to PL OP1-Op4 Compare
00048078  D23F 4000 2080       00000000  00000080  4137           MVC   0(64,R4),128(R2)    load initial Op1-Op4 data in ???
0004807E  D23F 30C0 4000       000000C0  00000000  4138           MVC   192(64,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                   4139 ****
                                                   4140 * force a failure by changing the 4th operand
                                                   4141 ****
00048084  D207 1030 4000       00000030  00000000  4142           MVC   48(8,R1),0(R4)
                                                   4143 ****
                                                   4144
0004808A  E313 0000 0024                00000000   4145           STG   R1,0(R3)            load OP1 address in PL
                                                   4146
00048090  4111 0010                     00000010   4147           LA    R1,16(R1)
00048094  E313 0008 0024                00000008   4148           STG   R1,8(R3)            load OP2 address in PL
                                                   4149
0004809A  4111 0010                     00000010   4150           LA    R1,16(R1)
0004809E  E313 0010 0024                00000010   4151           STG   R1,16(R3)           load OP3 address in PL
                                                   4152
000480A4  4111 0010                     00000010   4153           LA    R1,16(R1)
000480A8  E313 0018 0024                00000018   4154           STG   R1,24(R3)           load OP4 address in PL
                                                   4155
000480AE  B982 0000                                4156           XGR   R0,R0
000480B2  4100 0024                     00000024   4157           LA    R0,36
000480B6  EE00 0000 3000       00000000  00000000  4158           PLO   0,0,0,0(R3)         Quadruple C/S (QCSX: 36)
                                                   4159
000480BC  B98D 0020                                4160           EPSW  R2,R0               extract psw
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000480C0  5020 5010                     00000010  4161          ST    R2,CCPSW              to save CC
000480C4  E310 500B 0076                0000000B  4162          LB    R1,CCMASK             (failure CC mask)
000480CA  8910 0004                     00000004  4163          SLL   R1,4                  (shift to BC instr CC position)
000480CE  4410 510C                     0004812C  4164          EX    R1,F364TSTCC          fail if...
                                                  4165
                          000480D2  00000001  4166 F364CKUR  EQU   *
000480D2  D200 5009 3030  00000009  00000030  4167          MVC   URFOUND,48(R3)        Extract UR
000480D8  D500 5008 3030  00000008  00000030  4168          CLC   UR,48(R3)
000480DE  4780 50C6                     000480E6  4169          BE    F364CKRES
000480E2  45F0 8220                     00000420  4170          BAL   R15,URMSG
                                                  4171 *
                                                  4172 * compare results
                                                  4173 *
                          000480E6  00000001  4174 F364CKRES EQU   *
                                                  4175 *                              reset adresses
000480E6  B982 0011                             4176          XGR   R1,R1
000480EA  5810 501C                     0000001C  4177          L     R1,OPSRCA             source
000480EE  B982 0033                             4178          XGR   R3,R3
000480F2  5830 5024                     00000024  4179          L     R3,PLA                PL
000480F6  B982 0044                             4180          XGR   R4,R4
000480FA  5840 5020                     00000020  4181          L     R4,OPREPA
                                                  4182
000480FE  D50F 1000 3040  00000000  00000040  4183          CLC   0(16,R1),64(R3)
00048104  4770 5106                     00048126  4184          BNE   FXX437                OP1C Different, error
00048108  D50F 1010 3050  00000010  00000050  4185          CLC   16(16,R1),80(R3)
0004810E  4770 5106                     00048126  4186          BNE   FXX437                OP2C Different, error
00048112  D50F 1020 3060  00000020  00000060  4187          CLC   32(16,R1),96(R3)
00048118  4770 5106                     00048126  4188          BNE   FXX437                OP3C Different, error
0004811C  D50F 1030 3070  00000030  00000070  4189          CLC   48(16,R1),112(R3)
00048122  478B 0000                     00000000  4190          BE    0(R11)                OP4Csame, return
                                                  4191
                          00048126  00000001  4192 FXX437   EQU   *
00048126  45F0 8340                     00000540  4193          BAL   R15,FAILMSG
0004812A  07FB                                    4194          BR    R11
                                                  4195
0004812C  4700 5110                     00048130  4196 F364TSTCC BC    0,F364CCMSG         fail if unexpected condition code
                          00048130  00000001  4197 F364CCMSG EQU   *
00048130  45F0 80F2                     000002F2  4198          BAL   R15,CCMSG
00048134  47F0 50B2                     000480D2  4199          B     F364CKUR
                                                  4200
                                                  4201          TSTDATA
00048138                  00048138  00048400  4202+         ORG   PLOETST+(200*K)+(22*4*K)+K
                                                  4203          TSTLBL OPDST              Operand Destination
00048400                                          4204+OPDST60  DS    0C
00048400  00000000 00000000                     4205          DC    XL256'0'
00048408  00000000 00000000
00048410  00000000 00000000
00048418  00000000 00000000
00048420  00000000 00000000
00048428  00000000 00000000
00048430  00000000 00000000
00048438  00000000 00000000
00048440  00000000 00000000
00048448  00000000 00000000
00048450  00000000 00000000
00048458  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00048460  00000000 00000000
00048468  00000000 00000000
00048470  00000000 00000000
00048478  00000000 00000000
00048480  00000000 00000000
00048488  00000000 00000000
00048490  00000000 00000000
00048498  00000000 00000000
000484A0  00000000 00000000
000484A8  00000000 00000000
000484B0  00000000 00000000
000484B8  00000000 00000000
000484C0  00000000 00000000
000484C8  00000000 00000000
000484D0  00000000 00000000
000484D8  00000000 00000000
000484E0  00000000 00000000
000484E8  00000000 00000000
000484F0  00000000 00000000
000484F8  00000000 00000000
                                          4206
                                          4207           TSTLBL OPSRC              Operand Source OP1-OP4
00048500                                  4208+OPSRC60   DS    0C
00048500  00000000 00000000               4209           DC    XL256'0'
00048508  00000000 00000000
00048510  00000000 00000000
00048518  00000000 00000000
00048520  00000000 00000000
00048528  00000000 00000000
00048530  00000000 00000000
00048538  00000000 00000000
00048540  00000000 00000000
00048548  00000000 00000000
00048550  00000000 00000000
00048558  00000000 00000000
00048560  00000000 00000000
00048568  00000000 00000000
00048570  00000000 00000000
00048578  00000000 00000000
00048580  00000000 00000000
00048588  00000000 00000000
00048590  00000000 00000000
00048598  00000000 00000000
000485A0  00000000 00000000
000485A8  00000000 00000000
000485B0  00000000 00000000
000485B8  00000000 00000000
000485C0  00000000 00000000
000485C8  00000000 00000000
000485D0  00000000 00000000
000485D8  00000000 00000000
000485E0  00000000 00000000
000485E8  00000000 00000000
000485F0  00000000 00000000
000485F8  00000000 00000000
                                          4210
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   4211           TSTLBL PL                  Parameter List
00048600                                           4212+PL60      DS     0C
00048600  00000000 00000000                        4213           DC     XL352'00'
00048608  00000000 00000000
00048610  00000000 00000000
00048618  00000000 00000000
00048620  00000000 00000000
00048628  00000000 00000000
00048630  00000000 00000000
00048638  00000000 00000000
00048640  00000000 00000000
00048648  00000000 00000000
00048650  00000000 00000000
00048658  00000000 00000000
00048660  00000000 00000000
00048668  00000000 00000000
00048670  00000000 00000000
00048678  00000000 00000000
00048680  00000000 00000000
00048688  00000000 00000000
00048690  00000000 00000000
00048698  00000000 00000000
000486A0  00000000 00000000
000486A8  00000000 00000000
000486B0  00000000 00000000
000486B8  00000000 00000000
000486C0  00000000 00000000
000486C8  00000000 00000000
000486D0  00000000 00000000
000486D8  00000000 00000000
000486E0  00000000 00000000
000486E8  00000000 00000000
000486F0  00000000 00000000
000486F8  00000000 00000000
00048700  00000000 00000000
00048708  00000000 00000000
00048710  00000000 00000000
00048718  00000000 00000000
00048720  00000000 00000000
00048728  00000000 00000000
00048730  00000000 00000000
00048738  00000000 00000000
00048740  00000000 00000000
00048748  00000000 00000000
00048750  00000000 00000000
00048758  00000000 00000000
                                                   4214
                                                   4215           TSTLBL FMSG                fail message
00048760                                           4216+FMSG60    DS     0C
00048760  86818993 858440D7                        4217           DC     C'failed PLO-Extension '
00048768  D3D660C5 A7A38595
00048770  A2899695 40
00048775  D8A48184 99A49793                        4218           DC     CL96'Quadruple Compare and Swap (QCSX: 36): 4th NE'
0004877D  8540C396 94978199
00048785  85408195 8440E2A6
0004878D  8197404D D8C3E2E7
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00048795  7A40F3F6 5D7A40F4
0004879D  A38840D5 C5404040
000487A5  40404040 40404040
000487AD  40404040 40404040
000487B5  40404040 40404040
000487BD  40404040 40404040
000487C5  40404040 40404040
000487CD  40404040 40404040
                                       4219
                                       4220          TSTEND
000487D6                               4221+         DS    0H
000487D6  C5958440 968640A3            4222+         DC    CL32'End of test: 60'
000487DE  85A2A37A 40F6F040
000487E6  40404040 40404040
000487EE  40404040 40404040
                                       4223
                                       4224
                                       4225 * ------------------------------------------------------------------
                                       4226 * Test Quadruple Compare and Swap (QCSX: 36) instruction : All EQ
                                       4227 * ------------------------------------------------------------------
                                       4228          TSTHDR 0,99,36             All Equal (cc=0, UR= unchanged)
000487F6                  000487F6  00049000  4229+         ORG   PLOETST+(200*K)+(23*4*K)
00049000  E2E3C1D9 E3409686            4230+         DC    CL32'START of test: 61'
00049008  40A385A2 A37A40F6
00049010  F1404040 40404040
00049018  40404040 40404040
00049020                  00049020             4231+         USING *,R5                base for test data and test routine
00049020                  00000000             4232+         USING PLOETEST,R5
00049020  0004904C                             4233+T61      DC    A(CODE61)           address of test code
00049024  003D                                 4234+         DC    H'61'               test number
00049026  00                                   4235+         DC    X'00'
00049027  24                                   4236+         DC    HL1'36'             Function
00049028  63                                   4237+         DC    HL1'99'             expected unequal results (UR)
00049029  00                                   4238+         DC    HL1'00'             found UR
0004902A  00                                   4239+         DC    HL1'0'              cc
0004902B  07                                   4240+         DC    HL1'7'              not expected CC mask
0004902C  00                                   4241+         DS    X                   extracted cc
00049030  00000000 00000000                    4242+         DS    2F                  extract PSW after test (has CC)
00049038  00049760                             4243+         DC    A(FMSG61)           address of test fail message
0004903C  00049500                             4244+         DC    A(OPSRC61)          address of operand source
00049040  00049400                             4245+         DC    A(OPDST61)          address of operand destination
00049044  00049600                             4246+         DC    A(PL61)             address of parameter list
00049048  000011EE                             4247+         DC    A(RANDX+61)         address of random source data
                                               4248
                                               4249          TSTCODE
0004904C                                       4250+CODE61   DS    0F
                                               4251 *                                  load random source
0004904C  B982 0022                            4252          XGR   R2,R2
00049050  5820 5028                 00000028   4253          L     R2,RANDXA
                                               4254 *
                                               4255 * Quadruple Compare and Swap (QCSX: 36)
                                               4256 * Build Parambter List
                                               4257 *
00049054  B982 0011                            4258          XGR   R1,R1
00049058  5810 501C                 0000001C   4259          L     R1,OPSRCA
                                               4260
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004905C  B982 0033                            4261          XGR   R3,R3
00049060  5830 5024                  00000024  4262          L     R3,PLA
                                               4263
00049064  B982 0044                            4264          XGR   R4,R4
00049068  5840 5020                  00000020  4265          L     R4,OPREPA
                                               4266
0004906C  D23F 1000 2000      00000000  00000000  4267          MVC   0(64,R1),0(R2)      load Op1-Op4 data in source
00049072  D23F 3040 1000      00000040  00000000  4268          MVC   64(64,R3),0(R1)     copy to PL OP1-Op4 Compare
00049078  D23F 4000 2080      00000000  00000080  4269          MVC   0(64,R4),128(R2)    load initial Op1R-Op4R data
0004907E  D23F 30C0 4000      000000C0  00000000  4270          MVC   192(64,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                               4271 ****
                                               4272 * Do NOT force a failure by changing an operand
                                               4273 ****
                                               4274 *         MVC   96(8,R1),0(R4)
                                               4275 ****
                                               4276
00049084  E313 0000 0024                  00000000  4277          STG   R1,0(R3)            load OP1 address in PL
                                               4278
0004908A  4111 0010                  00000010  4279          LA    R1,16(R1)
0004908E  E313 0008 0024                  00000008  4280          STG   R1,8(R3)            load OP2 address in PL
                                               4281
00049094  4111 0010                  00000010  4282          LA    R1,16(R1)
00049098  E313 0010 0024                  00000010  4283          STG   R1,16(R3)           load OP3 address in PL
                                               4284
0004909E  4111 0010                  00000010  4285          LA    R1,16(R1)
000490A2  E313 0018 0024                  00000018  4286          STG   R1,24(R3)           load OP4 address in PL
                                               4287
000490A8  B982 0000                            4288          XGR   R0,R0
000490AC  4100 0024                  00000024  4289          LA    R0,36
000490B0  EE00 0000 3000      00000000  00000000  4290          PLO   0,0,0,0(R3)         Quadruple C/S (QCSX: 36)
                                               4291
000490B6  B98D 0020                            4292          EPSW  R2,R0               extract psw
000490BA  5020 5010                  00000010  4293          ST    R2,CCPSW              to save CC
000490BE  E310 500B 0076                  0000000B  4294          LB    R1,CCMASK           (failure CC mask)
000490C4  8910 0004                  00000004  4295          SLL   R1,4                (shift to BC instr CC position)
000490C8  4410 5106                  00049126  4296          EX    R1,F365TSTCC            fail if...
                                               4297
                                  000490CC  00000001  4298 F365CKUR  EQU   *
000490CC  D200 5009 3030      00000009  00000030  4299          MVC   URFOUND,48(R3)      Extract UR
000490D2  D500 5008 3030      00000008  00000030  4300          CLC   UR,48(R3)
000490D8  4780 50C0                  000490E0  4301          BE    F365CKRES
000490DC  45F0 8220                  00000420  4302          BAL   R15,URMSG
                                               4303 *
                                               4304 * compare results
                                               4305 *
                                  000490E0  00000001  4306 F365CKRES EQU   *
                                               4307 *                                    reset adresses
000490E0  B982 0011                            4308          XGR   R1,R1
000490E4  5810 501C                  0000001C  4309          L     R1,OPSRCA           source
000490E8  B982 0033                            4310          XGR   R3,R3
000490EC  5830 5024                  00000024  4311          L     R3,PLA              PL
000490F0  B982 0044                            4312          XGR   R4,R4
000490F4  5840 5020                  00000020  4313          L     R4,OPREPA
                                               4314
                                               4315 * All equal; replacement values copied to operand source
                                               4316
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000490F8  D50F 1000 3080      00000000  00000080  4317           CLC   0(16,R1),128(R3)
000490FE  4770 5100                     00049120  4318           BNE   FXX537                OP1R Different, error
00049102  D50F 1010 3090      00000010  00000090  4319           CLC   16(16,R1),144(R3)
00049108  4770 5100                     00049120  4320           BNE   FXX537                OP2R Different, error
0004910C  D50F 1020 30A0      00000020  000000A0  4321           CLC   32(16,R1),160(R3)
00049112  4770 5100                     00049120  4322           BNE   FXX537                OP3R Different, error
00049116  D50F 1030 30B0      00000030  000000B0  4323           CLC   48(16,R1),176(R3)
0004911C  478B 0000                     00000000  4324           BE    0(R11)                OP4R same, return
                                                  4325
                              00049120  00000001  4326 FXX537    EQU   *
00049120  45F0 8340                     00000540  4327           BAL   R15,FAILMSG
00049124  07FB                                    4328           BR    R11
                                                  4329
00049126  4700 510A                     0004912A  4330 F365TSTCC BC     0,F365CCMSG           fail if unexpected condition code
                              0004912A  00000001  4331 F365CCMSG EQU   *
0004912A  45F0 80F2                     000002F2  4332           BAL   R15,CCMSG
0004912E  47F0 50AC                     000490CC  4333           B     F365CKUR
                                                  4334
                                                  4335           TSTDATA
00049132                      00049132  00049400  4336+          ORG   PLOETST+(200*K)+(23*4*K)+K
                                                  4337           TSTLBL OPDST                Operand Destination
00049400                                          4338+OPDST61   DS    0C
00049400  00000000 00000000                       4339           DC    XL256'0'
00049408  00000000 00000000
00049410  00000000 00000000
00049418  00000000 00000000
00049420  00000000 00000000
00049428  00000000 00000000
00049430  00000000 00000000
00049438  00000000 00000000
00049440  00000000 00000000
00049448  00000000 00000000
00049450  00000000 00000000
00049458  00000000 00000000
00049460  00000000 00000000
00049468  00000000 00000000
00049470  00000000 00000000
00049478  00000000 00000000
00049480  00000000 00000000
00049488  00000000 00000000
00049490  00000000 00000000
00049498  00000000 00000000
000494A0  00000000 00000000
000494A8  00000000 00000000
000494B0  00000000 00000000
000494B8  00000000 00000000
000494C0  00000000 00000000
000494C8  00000000 00000000
000494D0  00000000 00000000
000494D8  00000000 00000000
000494E0  00000000 00000000
000494E8  00000000 00000000
000494F0  00000000 00000000
000494F8  00000000 00000000
                                                  4340
                                                  4341           TSTLBL OPSRC                Operand Source OP1-OP4
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00049500                                            4342+OPSRC61  DS    0C
00049500  00000000 00000000                         4343          DC    XL256'0'
00049508  00000000 00000000
00049510  00000000 00000000
00049518  00000000 00000000
00049520  00000000 00000000
00049528  00000000 00000000
00049530  00000000 00000000
00049538  00000000 00000000
00049540  00000000 00000000
00049548  00000000 00000000
00049550  00000000 00000000
00049558  00000000 00000000
00049560  00000000 00000000
00049568  00000000 00000000
00049570  00000000 00000000
00049578  00000000 00000000
00049580  00000000 00000000
00049588  00000000 00000000
00049590  00000000 00000000
00049598  00000000 00000000
000495A0  00000000 00000000
000495A8  00000000 00000000
000495B0  00000000 00000000
000495B8  00000000 00000000
000495C0  00000000 00000000
000495C8  00000000 00000000
000495D0  00000000 00000000
000495D8  00000000 00000000
000495E0  00000000 00000000
000495E8  00000000 00000000
000495F0  00000000 00000000
000495F8  00000000 00000000
                                                    4344
                                                    4345          TSTLBL PL                 Parameter List
00049600                                            4346+PL61     DS    0C
00049600  00000000 00000000                         4347          DC    XL352'00'
00049608  00000000 00000000
00049610  00000000 00000000
00049618  00000000 00000000
00049620  00000000 00000000
00049628  00000000 00000000
00049630  00000000 00000000
00049638  00000000 00000000
00049640  00000000 00000000
00049648  00000000 00000000
00049650  00000000 00000000
00049658  00000000 00000000
00049660  00000000 00000000
00049668  00000000 00000000
00049670  00000000 00000000
00049678  00000000 00000000
00049680  00000000 00000000
00049688  00000000 00000000
00049690  00000000 00000000
00049698  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000496A0  00000000 00000000
000496A8  00000000 00000000
000496B0  00000000 00000000
000496B8  00000000 00000000
000496C0  00000000 00000000
000496C8  00000000 00000000
000496D0  00000000 00000000
000496D8  00000000 00000000
000496E0  00000000 00000000
000496E8  00000000 00000000
000496F0  00000000 00000000
000496F8  00000000 00000000
00049700  00000000 00000000
00049708  00000000 00000000
00049710  00000000 00000000
00049718  00000000 00000000
00049720  00000000 00000000
00049728  00000000 00000000
00049730  00000000 00000000
00049738  00000000 00000000
00049740  00000000 00000000
00049748  00000000 00000000
00049750  00000000 00000000
00049758  00000000 00000000
                                                4348
                                                4349          TSTLBL FMSG               fail message
00049760                                        4350+FMSG61   DS    0C
00049760  86818993 858440D7                     4351          DC    C'failed PLO-Extension '
00049768  D3D660C5 A7A38595
00049770  A2899695 40
00049775  D8A48184 99A49793                     4352          DC    CL96'Quadruple Compare and Swap (QCSX: 36): All EQ'
0004977D  8540C396 94978199
00049785  85408195 8440E2A6
0004978D  8197404D D8C3E2E7
00049795  7A40F3F6 5D7A40C1
0004979D  939340C5 D8404040
000497A5  40404040 40404040
000497AD  40404040 40404040
000497B5  40404040 40404040
000497BD  40404040 40404040
000497C5  40404040 40404040
000497CD  40404040 40404040
                                                4353
                                                4354          TSTEND
000497D6                                        4355+         DS    0H
000497D6  C5958440 968640A3                     4356+         DC    CL32'End of test: 61'
000497DE  85A2A37A 40F6F140
000497E6  40404040 40404040
000497EE  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   4358 *=============================================================
                                                   4359 *       Quadruple Compare and Swap Tests
                                                   4360 *=============================================================
                                                   4361
                                                   4362 * -------------------------------------------------------------
                                                   4363 * Test Quadruple Compare and Swap (QCSG: 35) instruction : 1st NE
                                                   4364 * -------------------------------------------------------------
                                                   4365          TSTHDR 2,1,35             1st Operand NOT Equal (cc=2, UR=1)
000497F6                          000497F6  0004A000 4366+         ORG   PLOETST+(200*K)+(24*4*K)
0004A000  E2E3C1D9 E3409686                        4367+         DC    CL32'START of test: 62'
0004A008  40A385A2 A37A40F6
0004A010  F2404040 40404040
0004A018  40404040 40404040
0004A020                          0004A020         4368+         USING *,R5                base for test data and test routine
0004A020                          00000000         4369+         USING PLOETEST,R5
0004A020  0004A04C                                 4370+T62      DC    A(CODE62)           address of test code
0004A024  003E                                     4371+         DC    H'62'               test number
0004A026  00                                       4372+         DC    X'00'
0004A027  23                                       4373+         DC    HL1'35'             Function
0004A028  01                                       4374+         DC    HL1'1'              expected unequal results (UR)
0004A029  00                                       4375+         DC    HL1'00'             found UR
0004A02A  02                                       4376+         DC    HL1'2'              cc
0004A02B  0D                                       4377+         DC    HL1'13'             not expected CC mask
0004A02C  00                                       4378+         DS    X                   extracted cc
0004A030  00000000 00000000                        4379+         DS    2F                  extract PSW after test (has CC)
0004A038  0004A760                                 4380+         DC    A(FMSG62)           address of test fail message
0004A03C  0004A500                                 4381+         DC    A(OPSRC62)          address of operand source
0004A040  0004A400                                 4382+         DC    A(OPDST62)          address of operand destination
0004A044  0004A600                                 4383+         DC    A(PL62)             address of parameter list
0004A048  000011EF                                 4384+         DC    A(RANDX+62)         address of random source data
                                                   4385
                                                   4386          TSTCODE
0004A04C                                           4387+CODE62   DS    0F
                                                   4388 *                                  load random source
0004A04C  B982 0022                                4389          XGR   R2,R2
0004A050  5820 5028                    00000028    4390          L     R2,RANDXA
                                                   4391 *
                                                   4392 * Quadruple Compare and Swap (QCSG: 35)
                                                   4393 * Build Parambter List
                                                   4394 *
0004A054  B982 0011                                4395          XGR   R1,R1
0004A058  5810 501C                    0000001C    4396          L     R1,OPSRCA
                                                   4397
0004A05C  B982 0033                                4398          XGR   R3,R3
0004A060  5830 5024                    00000024    4399          L     R3,PLA
                                                   4400
0004A064  B982 0044                                4401          XGR   R4,R4
0004A068  5840 5020                    00000020    4402          L     R4,OPREPA
                                                   4403
0004A06C  D21F 1000 2000      00000000  00000000  4404          MVC   0(32,R1),0(R2)      load Op1-Op4 data in source
0004A072  D21F 3040 1000      00000040  00000000  4405          MVC   64(32,R3),0(R1)     copy to PL OP1-Op4 Compare
0004A078  D21F 4000 2080      00000000  00000080  4406          MVC   0(32,R4),128(R2)    load initial Op1-Op4 data in ???
0004A07E  D21F 3080 4000      00000080  00000000  4407          MVC   128(32,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                   4408 ****
                                                   4409 * force a failure by changing the first operand
                                                   4410 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004A084  D203 1000 4000      00000000  00000000  4411          MVC   0(4,R1),0(R4)
                                                  4412 ****
                                                  4413
0004A08A  E313 0000 0024                00000000  4414          STG   R1,0(R3)           load OP1 address in PL
                                                  4415
0004A090  4111 0008                     00000008  4416          LA    R1,8(R1)
0004A094  E313 0008 0024                00000008  4417          STG   R1,8(R3)           load OP2 address in PL
                                                  4418
0004A09A  4111 0008                     00000008  4419          LA    R1,8(R1)
0004A09E  E313 0010 0024                00000010  4420          STG   R1,16(R3)          load OP3 address in PL
                                                  4421
0004A0A4  4111 0008                     00000008  4422          LA    R1,8(R1)
0004A0A8  E313 0018 0024                00000018  4423          STG   R1,24(R3)          load OP4 address in PL
                                                  4424
0004A0AE  B982 0000                               4425          XGR   R0,R0
0004A0B2  4100 0023                     00000023  4426          LA    R0,35
0004A0B6  EE00 0000 3000      00000000  00000000  4427          PLO   0,0,0,0(R3)        Quadruple C/S (QCSG: 35)
                                                  4428
0004A0BC  B98D 0020                               4429          EPSW  R2,R0              extract psw
0004A0C0  5020 5010                     00000010  4430          ST    R2,CCPSW             to save CC
0004A0C4  E310 500B 0076                0000000B  4431          LB    R1,CCMASK          (failure CC mask)
0004A0CA  8910 0004                     00000004  4432          SLL   R1,4               (shift to BC instr CC position)
0004A0CE  4410 510C                     0004A12C  4433          EX    R1,F351TSTCC         fail if...
                                                  4434
                                      0004A0D2    00000001  4435 F351CKUR EQU   *
0004A0D2  D200 5009 3030      00000009  00000030  4436          MVC   URFOUND,48(R3)     Extract UR
0004A0D8  D500 5008 3030      00000008  00000030  4437          CLC   UR,48(R3)
0004A0DE  4780 50C6                     0004A0E6  4438          BE    F351CKRES
0004A0E2  45F0 8220                     00000420  4439          BAL   R15,URMSG
                                                  4440 *
                                                  4441 * compare results
                                                  4442 *
                                      0004A0E6    00000001  4443 F351CKRES EQU   *
                                                  4444 *                           reset adresses
0004A0E6  B982 0011                               4445          XGR   R1,R1
0004A0EA  5810 501C                     0000001C  4446          L     R1,OPSRCA          source
0004A0EE  B982 0033                               4447          XGR   R3,R3
0004A0F2  5830 5024                     00000024  4448          L     R3,PLA             PL
0004A0F6  B982 0044                               4449          XGR   R4,R4
0004A0FA  5840 5020                     00000020  4450          L     R4,OPREPA
                                                  4451
0004A0FE  D507 1000 3040      00000000  00000040  4452          CLC   0(8,R1),64(R3)
0004A104  4770 5106                     0004A126  4453          BNE   F351XX             OP1C Different, error
0004A108  D507 1008 3048      00000008  00000048  4454          CLC   8(8,R1),72(R3)
0004A10E  4770 5106                     0004A126  4455          BNE   F351XX             OP2C Different, error
0004A112  D507 1010 3050      00000010  00000050  4456          CLC   16(8,R1),80(R3)
0004A118  4770 5106                     0004A126  4457          BNE   F351XX             OP3C Different, error
0004A11C  D507 1018 3058      00000018  00000058  4458          CLC   24(8,R1),88(R3)
0004A122  478B 0000                     00000000  4459          BE    0(R11)             OP4Csame, return
                                                  4460
                                      0004A126    00000001  4461 F351XX   EQU   *
0004A126  45F0 8340                     00000540  4462          BAL   R15,FAILMSG
0004A12A  07FB                                    4463          BR    R11
                                                  4464
0004A12C  4700 5110                     0004A130  4465 F351TSTCC BC    0,F351CCMSG      fail if unexpected condition code
                                      0004A130    00000001  4466 F351CCMSG EQU   *
```

```
  LOC         OBJECT CODE        ADDR1     ADDR2     STMT
0004A130  45F0 80F2                        000002F2  4467           BAL   R15,CCMSG
0004A134  47F0 50B2                        0004A0D2  4468           B     F351CKUR
                                                     4469
                                                     4470           TSTDATA
0004A138                       0004A138    0004A400  4471+          ORG   PLOETST+(200*K)+(24*4*K)+K
                                                     4472           TSTLBL OPDST              Operand Destination
0004A400                                             4473+OPDST62   DS    0C
0004A400  00000000 00000000                          4474           DC    XL256'0'
0004A408  00000000 00000000
0004A410  00000000 00000000
0004A418  00000000 00000000
0004A420  00000000 00000000
0004A428  00000000 00000000
0004A430  00000000 00000000
0004A438  00000000 00000000
0004A440  00000000 00000000
0004A448  00000000 00000000
0004A450  00000000 00000000
0004A458  00000000 00000000
0004A460  00000000 00000000
0004A468  00000000 00000000
0004A470  00000000 00000000
0004A478  00000000 00000000
0004A480  00000000 00000000
0004A488  00000000 00000000
0004A490  00000000 00000000
0004A498  00000000 00000000
0004A4A0  00000000 00000000
0004A4A8  00000000 00000000
0004A4B0  00000000 00000000
0004A4B8  00000000 00000000
0004A4C0  00000000 00000000
0004A4C8  00000000 00000000
0004A4D0  00000000 00000000
0004A4D8  00000000 00000000
0004A4E0  00000000 00000000
0004A4E8  00000000 00000000
0004A4F0  00000000 00000000
0004A4F8  00000000 00000000
                                                     4475
                                                     4476           TSTLBL OPSRC              Operand Source OP1-OP4
0004A500                                             4477+OPSRC62   DS    0C
0004A500  00000000 00000000                          4478           DC    XL256'0'
0004A508  00000000 00000000
0004A510  00000000 00000000
0004A518  00000000 00000000
0004A520  00000000 00000000
0004A528  00000000 00000000
0004A530  00000000 00000000
0004A538  00000000 00000000
0004A540  00000000 00000000
0004A548  00000000 00000000
0004A550  00000000 00000000
0004A558  00000000 00000000
0004A560  00000000 00000000
0004A568  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004A570  00000000 00000000
0004A578  00000000 00000000
0004A580  00000000 00000000
0004A588  00000000 00000000
0004A590  00000000 00000000
0004A598  00000000 00000000
0004A5A0  00000000 00000000
0004A5A8  00000000 00000000
0004A5B0  00000000 00000000
0004A5B8  00000000 00000000
0004A5C0  00000000 00000000
0004A5C8  00000000 00000000
0004A5D0  00000000 00000000
0004A5D8  00000000 00000000
0004A5E0  00000000 00000000
0004A5E8  00000000 00000000
0004A5F0  00000000 00000000
0004A5F8  00000000 00000000
                                          4479
                                          4480           TSTLBL PL                 Parameter List
0004A600                                  4481+PL62      DS    0C
0004A600  00000000 00000000               4482           DC    XL352'00'
0004A608  00000000 00000000
0004A610  00000000 00000000
0004A618  00000000 00000000
0004A620  00000000 00000000
0004A628  00000000 00000000
0004A630  00000000 00000000
0004A638  00000000 00000000
0004A640  00000000 00000000
0004A648  00000000 00000000
0004A650  00000000 00000000
0004A658  00000000 00000000
0004A660  00000000 00000000
0004A668  00000000 00000000
0004A670  00000000 00000000
0004A678  00000000 00000000
0004A680  00000000 00000000
0004A688  00000000 00000000
0004A690  00000000 00000000
0004A698  00000000 00000000
0004A6A0  00000000 00000000
0004A6A8  00000000 00000000
0004A6B0  00000000 00000000
0004A6B8  00000000 00000000
0004A6C0  00000000 00000000
0004A6C8  00000000 00000000
0004A6D0  00000000 00000000
0004A6D8  00000000 00000000
0004A6E0  00000000 00000000
0004A6E8  00000000 00000000
0004A6F0  00000000 00000000
0004A6F8  00000000 00000000
0004A700  00000000 00000000
0004A708  00000000 00000000
0004A710  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004A718  00000000 00000000
0004A720  00000000 00000000
0004A728  00000000 00000000
0004A730  00000000 00000000
0004A738  00000000 00000000
0004A740  00000000 00000000
0004A748  00000000 00000000
0004A750  00000000 00000000
0004A758  00000000 00000000
                                        4483
                                        4484          TSTLBL FMSG               fail message
0004A760                                4485+FMSG62   DS    0C
0004A760  86818993 858440D7             4486          DC    C'failed PLO-Extension '
0004A768  D3D660C5 A7A38595
0004A770  A2899695 40
0004A775  D8A48184 99A49793             4487          DC    CL96'Quadruple Compare and Swap (QCSG: 35): 1st NE'
0004A77D  8540C396 94978199
0004A785  85408195 8440E2A6
0004A78D  8197404D D8C3E2C7
0004A795  7A40F3F5 5D7A40F1
0004A79D  A2A340D5 C5404040
0004A7A5  40404040 40404040
0004A7AD  40404040 40404040
0004A7B5  40404040 40404040
0004A7BD  40404040 40404040
0004A7C5  40404040 40404040
0004A7CD  40404040 40404040
                                        4488
                                        4489          TSTEND
0004A7D6                                4490+         DS    0H
0004A7D6  C5958440 968640A3             4491+         DC    CL32'End of test: 62'
0004A7DE  85A2A37A 40F6F240
0004A7E6  40404040 40404040
0004A7EE  40404040 40404040
                                        4492
                                        4493 * ------------------------------------------------------------------
                                        4494 * Test Quadruple Compare and Swap (QCSG: 35) instruction : 2nd NE
                                        4495 * ------------------------------------------------------------------
                                        4496          TSTHDR 2,2,35             2nd Operand NOT Equal (cc=2, UR=2)
0004A7F6                 0004A7F6  0004B000 4497+         ORG   PLOETST+(200*K)+(25*4*K)
0004B000  E2E3C1D9 E3409686             4498+         DC    CL32'START of test: 63'
0004B008  40A385A2 A37A40F6
0004B010  F3404040 40404040
0004B018  40404040 40404040
0004B020                 0004B020       4499+         USING *,R5                base for test data and test routine
0004B020                 00000000       4500+         USING PLOETEST,R5
0004B020  0004B04C                      4501+T63      DC    A(CODE63)           address of test code
0004B024  003F                          4502+         DC    H'63'               test number
0004B026  00                            4503+         DC    X'00'
0004B027  23                            4504+         DC    HL1'35'             Function
0004B028  02                            4505+         DC    HL1'2'              expected unequal results (UR)
0004B029  00                            4506+         DC    HL1'00'             found UR
0004B02A  02                            4507+         DC    HL1'2'              cc
0004B02B  0D                            4508+         DC    HL1'13'             not expected CC mask
0004B02C  00                            4509+         DS    X                   extracted cc
0004B030  00000000 00000000             4510+         DS    2F                  extract PSW after test (has CC)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004B038  0004B760                                4511+          DC    A(FMSG63)           address of test fail message
0004B03C  0004B500                                4512+          DC    A(OPSRC63)          address of operand source
0004B040  0004B400                                4513+          DC    A(OPDST63)          address of operand destination
0004B044  0004B600                                4514+          DC    A(PL63)             address of parameter list
0004B048  000011F0                                4515+          DC    A(RANDX+63)         address of random source data
                                                  4516
                                                  4517           TSTCODE
0004B04C                                          4518+CODE63    DS    0F
                                                  4519 *                                   load random source
0004B04C  B982 0022                               4520           XGR   R2,R2
0004B050  5820 5028                 00000028      4521           L     R2,RANDXA
                                                  4522 *
                                                  4523 * Quadruple Compare and Swap (QCSG: 35)
                                                  4524 * Build Parambter List
                                                  4525 *
0004B054  B982 0011                               4526           XGR   R1,R1
0004B058  5810 501C                 0000001C      4527           L     R1,OPSRCA
                                                  4528
0004B05C  B982 0033                               4529           XGR   R3,R3
0004B060  5830 5024                 00000024      4530           L     R3,PLA
                                                  4531
0004B064  B982 0044                               4532           XGR   R4,R4
0004B068  5840 5020                 00000020      4533           L     R4,OPREPA
                                                  4534
0004B06C  D21F 1000 2000  00000000  00000000      4535           MVC   0(32,R1),0(R2)      load Op1-Op4 data in source
0004B072  D21F 3040 1000  00000040  00000000      4536           MVC   64(32,R3),0(R1)     copy to PL OP1-Op4 Compare
0004B078  D21F 4000 2080  00000000  00000080      4537           MVC   0(32,R4),128(R2)    load initial Op1-Op4 data in ???
0004B07E  D21F 3080 4000  00000080  00000000      4538           MVC   128(32,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                  4539 ****
                                                  4540 * force a failure by changing the 2nd operand
                                                  4541 ****
0004B084  D203 1008 4000  00000008  00000000      4542           MVC   8(4,R1),0(R4)
                                                  4543 ****
                                                  4544
0004B08A  E313 0000 0024                00000000  4545           STG   R1,0(R3)            load OP1 address in PL
                                                  4546
0004B090  4111 0008                     00000008  4547           LA    R1,8(R1)
0004B094  E313 0008 0024                00000008  4548           STG   R1,8(R3)            load OP2 address in PL
                                                  4549
0004B09A  4111 0008                     00000008  4550           LA    R1,8(R1)
0004B09E  E313 0010 0024                00000010  4551           STG   R1,16(R3)           load OP3 address in PL
                                                  4552
0004B0A4  4111 0008                     00000008  4553           LA    R1,8(R1)
0004B0A8  E313 0018 0024                00000018  4554           STG   R1,24(R3)           load OP4 address in PL
                                                  4555
0004B0AE  B982 0000                               4556           XGR   R0,R0
0004B0B2  4100 0023                     00000023  4557           LA    R0,35
0004B0B6  EE00 0000 3000  00000000  00000000      4558           PLO   0,0,0,0(R3)         Quadruple C/S (QCSG: 35)
                                                  4559
0004B0BC  B98D 0020                               4560           EPSW  R2,R0               extract psw
0004B0C0  5020 5010                     00000010  4561           ST    R2,CCPSW              to save CC
0004B0C4  E310 500B 0076                0000000B  4562           LB    R1,CCMASK           (failure CC mask)
0004B0CA  8910 0004                     00000004  4563           SLL   R1,4                (shift to BC instr CC position)
0004B0CE  4410 510C                     0004B12C  4564           EX    R1,F352TSTCC            fail if...
                                                  4565
                          0004B0D2      00000001  4566 F352CKUR EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004B0D2  D200 5009 3030      00000009  00000030  4567          MVC   URFOUND,48(R3)      Extract UR
0004B0D8  D500 5008 3030      00000008  00000030  4568          CLC   UR,48(R3)
0004B0DE  4780 50C6                     0004B0E6  4569          BE    F352CKRES
0004B0E2  45F0 8220                     00000420  4570          BAL   R15,URMSG
                                                  4571 *
                                                  4572 * compare results
                                                  4573 *
                                00004B0E6 00000001  4574 F352CKRES EQU   *
                                                  4575 *                            reset adresses
0004B0E6  B982 0011                               4576          XGR   R1,R1
0004B0EA  5810 501C                     0000001C  4577          L     R1,OPSRCA           source
0004B0EE  B982 0033                               4578          XGR   R3,R3
0004B0F2  5830 5024                     00000024  4579          L     R3,PLA              PL
0004B0F6  B982 0044                               4580          XGR   R4,R4
0004B0FA  5840 5020                     00000020  4581          L     R4,OPREPA
                                                  4582
0004B0FE  D507 1000 3040      00000000  00000040  4583          CLC   0(8,R1),64(R3)
0004B104  4770 5106                     0004B126  4584          BNE   F352XX              OP1C Different, error
0004B108  D507 1008 3048      00000008  00000048  4585          CLC   8(8,R1),72(R3)
0004B10E  4770 5106                     0004B126  4586          BNE   F352XX              OP2C Different, error
0004B112  D507 1010 3050      00000010  00000050  4587          CLC   16(8,R1),80(R3)
0004B118  4770 5106                     0004B126  4588          BNE   F352XX              OP3C Different, error
0004B11C  D507 1018 3058      00000018  00000058  4589          CLC   24(8,R1),88(R3)
0004B122  478B 0000                     00000000  4590          BE    0(R11)              OP4Csame, return
                                                  4591
                                00004B126 00000001  4592 F352XX   EQU   *
0004B126  45F0 8340                     00000540  4593          BAL   R15,FAILMSG
0004B12A  07FB                                    4594          BR    R11
                                                  4595
0004B12C  4700 5110                     0004B130  4596 F352TSTCC BC    0,F352CCMSG         fail if unexpected condition code
                                00004B130 00000001  4597 F352CCMSG EQU   *
0004B130  45F0 80F2                     000002F2  4598          BAL   R15,CCMSG
0004B134  47F0 50B2                     0004B0D2  4599          B     F352CKUR
                                                  4600
                                                  4601          TSTDATA
0004B138                        0004B138  0004B400  4602+         ORG   PLOETST+(200*K)+(25*4*K)+K
                                                  4603          TSTLBL OPDST              Operand Destination
0004B400                                          4604+OPDST63  DS    0C
0004B400  00000000 00000000                       4605          DC    XL256'0'
0004B408  00000000 00000000
0004B410  00000000 00000000
0004B418  00000000 00000000
0004B420  00000000 00000000
0004B428  00000000 00000000
0004B430  00000000 00000000
0004B438  00000000 00000000
0004B440  00000000 00000000
0004B448  00000000 00000000
0004B450  00000000 00000000
0004B458  00000000 00000000
0004B460  00000000 00000000
0004B468  00000000 00000000
0004B470  00000000 00000000
0004B478  00000000 00000000
0004B480  00000000 00000000
0004B488  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004B490  00000000 00000000
0004B498  00000000 00000000
0004B4A0  00000000 00000000
0004B4A8  00000000 00000000
0004B4B0  00000000 00000000
0004B4B8  00000000 00000000
0004B4C0  00000000 00000000
0004B4C8  00000000 00000000
0004B4D0  00000000 00000000
0004B4D8  00000000 00000000
0004B4E0  00000000 00000000
0004B4E8  00000000 00000000
0004B4F0  00000000 00000000
0004B4F8  00000000 00000000
                                          4606
                                          4607          TSTLBL OPSRC              Operand Source OP1-OP4
0004B500                                  4608+OPSRC63  DS     0C
0004B500  00000000 00000000               4609          DC     XL256'0'
0004B508  00000000 00000000
0004B510  00000000 00000000
0004B518  00000000 00000000
0004B520  00000000 00000000
0004B528  00000000 00000000
0004B530  00000000 00000000
0004B538  00000000 00000000
0004B540  00000000 00000000
0004B548  00000000 00000000
0004B550  00000000 00000000
0004B558  00000000 00000000
0004B560  00000000 00000000
0004B568  00000000 00000000
0004B570  00000000 00000000
0004B578  00000000 00000000
0004B580  00000000 00000000
0004B588  00000000 00000000
0004B590  00000000 00000000
0004B598  00000000 00000000
0004B5A0  00000000 00000000
0004B5A8  00000000 00000000
0004B5B0  00000000 00000000
0004B5B8  00000000 00000000
0004B5C0  00000000 00000000
0004B5C8  00000000 00000000
0004B5D0  00000000 00000000
0004B5D8  00000000 00000000
0004B5E0  00000000 00000000
0004B5E8  00000000 00000000
0004B5F0  00000000 00000000
0004B5F8  00000000 00000000
                                          4610
                                          4611          TSTLBL PL                 Parameter List
0004B600                                  4612+PL63     DS     0C
0004B600  00000000 00000000               4613          DC     XL352'00'
0004B608  00000000 00000000
0004B610  00000000 00000000
0004B618  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004B620  00000000 00000000
0004B628  00000000 00000000
0004B630  00000000 00000000
0004B638  00000000 00000000
0004B640  00000000 00000000
0004B648  00000000 00000000
0004B650  00000000 00000000
0004B658  00000000 00000000
0004B660  00000000 00000000
0004B668  00000000 00000000
0004B670  00000000 00000000
0004B678  00000000 00000000
0004B680  00000000 00000000
0004B688  00000000 00000000
0004B690  00000000 00000000
0004B698  00000000 00000000
0004B6A0  00000000 00000000
0004B6A8  00000000 00000000
0004B6B0  00000000 00000000
0004B6B8  00000000 00000000
0004B6C0  00000000 00000000
0004B6C8  00000000 00000000
0004B6D0  00000000 00000000
0004B6D8  00000000 00000000
0004B6E0  00000000 00000000
0004B6E8  00000000 00000000
0004B6F0  00000000 00000000
0004B6F8  00000000 00000000
0004B700  00000000 00000000
0004B708  00000000 00000000
0004B710  00000000 00000000
0004B718  00000000 00000000
0004B720  00000000 00000000
0004B728  00000000 00000000
0004B730  00000000 00000000
0004B738  00000000 00000000
0004B740  00000000 00000000
0004B748  00000000 00000000
0004B750  00000000 00000000
0004B758  00000000 00000000
                                          4614
                                          4615          TSTLBL FMSG               fail message
0004B760                                  4616+FMSG63   DS    0C
0004B760  86818993 858440D7               4617          DC    C'failed PLO-Extension '
0004B768  D3D660C5 A7A38595
0004B770  A2899695 40
0004B775  D8A48184 99A49793               4618          DC    CL96'Quadruple Compare and Swap (QCSG: 35): 2nd NE'
0004B77D  8540C396 94978199
0004B785  85408195 8440E2A6
0004B78D  8197404D D8C3E2C7
0004B795  7A40F3F5 5D7A40F2
0004B79D  958440D5 C5404040
0004B7A5  40404040 40404040
0004B7AD  40404040 40404040
0004B7B5  40404040 40404040
0004B7BD  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004B7C5  40404040 40404040
0004B7CD  40404040 40404040
                                                   4619
                                                   4620          TSTEND
0004B7D6                                           4621+         DS    0H
0004B7D6  C5958440 968640A3                        4622+         DC    CL32'End of test: 63'
0004B7DE  85A2A37A 40F6F340
0004B7E6  40404040 40404040
0004B7EE  40404040 40404040
                                                   4623
                                                   4624 * ------------------------------------------------------------------
                                                   4625 * Test Quadruple Compare and Swap (QCSG: 35) instruction : 3rd NE
                                                   4626 * ------------------------------------------------------------------
                                                   4627          TSTHDR 2,3,35             3rd Operand NOT Equal (cc=2, UR=3)
0004B7F6                     0004B7F6  0004C000    4628+         ORG   PLOETST+(200*K)+(26*4*K)
0004C000  E2E3C1D9 E3409686                        4629+         DC    CL32'START of test: 64'
0004C008  40A385A2 A37A40F6
0004C010  F4404040 40404040
0004C018  40404040 40404040
0004C020                     0004C020              4630+         USING *,R5                base for test data and test routine
0004C020                     00000000              4631+         USING PLOETEST,R5
0004C020  0004C04C                                 4632+T64      DC    A(CODE64)           address of test code
0004C024  0040                                     4633+         DC    H'64'               test number
0004C026  00                                       4634+         DC    X'00'
0004C027  23                                       4635+         DC    HL1'35'             Function
0004C028  03                                       4636+         DC    HL1'3'              expected unequal results (UR)
0004C029  00                                       4637+         DC    HL1'00'             found UR
0004C02A  02                                       4638+         DC    HL1'2'              cc
0004C02B  0D                                       4639+         DC    HL1'13'             not expected CC mask
0004C02C  00                                       4640+         DS    X                   extracted cc
0004C030  00000000 00000000                        4641+         DS    2F                  extract PSW after test (has CC)
0004C038  0004C760                                 4642+         DC    A(FMSG64)           address of test fail message
0004C03C  0004C500                                 4643+         DC    A(OPSRC64)          address of operand source
0004C040  0004C400                                 4644+         DC    A(OPDST64)          address of operand destination
0004C044  0004C600                                 4645+         DC    A(PL64)             address of parameter list
0004C048  000011F1                                 4646+         DC    A(RANDX+64)         address of random source data
                                                   4647
                                                   4648          TSTCODE
0004C04C                                           4649+CODE64   DS    0F
                                                   4650 *                                  load random source
0004C04C  B982 0022                                4651          XGR   R2,R2
0004C050  5820 5028                  00000028      4652          L     R2,RANDXA
                                                   4653 *
                                                   4654 * Quadruple Compare and Swap (QCSG: 35)
                                                   4655 * Build Parambter List
                                                   4656 *
0004C054  B982 0011                                4657          XGR   R1,R1
0004C058  5810 501C                  0000001C      4658          L     R1,OPSRCA
                                                   4659
0004C05C  B982 0033                                4660          XGR   R3,R3
0004C060  5830 5024                  00000024      4661          L     R3,PLA
                                                   4662
0004C064  B982 0044                                4663          XGR   R4,R4
0004C068  5840 5020                  00000020      4664          L     R4,OPREPA
                                                   4665
0004C06C  D21F 1000 2000      00000000  00000000   4666          MVC   0(32,R1),0(R2)      load Op1-Op4 data in source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004C072  D21F 3040 1000      00000040  00000000  4667          MVC   64(32,R3),0(R1)     copy to PL OP1-Op4 Compare
0004C078  D21F 4000 2080      00000000  00000080  4668          MVC   0(32,R4),128(R2)    load initial Op1-Op4 data in ???
0004C07E  D21F 3080 4000      00000080  00000000  4669          MVC   128(32,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                  4670 ****
                                                  4671 * force a failure by changing the 3rd operand
                                                  4672 ****
0004C084  D203 1010 4000      00000010  00000000  4673          MVC   16(4,R1),0(R4)
                                                  4674 ****
                                                  4675
0004C08A  E313 0000 0024                00000000  4676          STG   R1,0(R3)            load OP1 address in PL
                                                  4677
0004C090  4111 0008                     00000008  4678          LA    R1,8(R1)
0004C094  E313 0008 0024                00000008  4679          STG   R1,8(R3)            load OP2 address in PL
                                                  4680
0004C09A  4111 0008                     00000008  4681          LA    R1,8(R1)
0004C09E  E313 0010 0024                00000010  4682          STG   R1,16(R3)           load OP3 address in PL
                                                  4683
0004C0A4  4111 0008                     00000008  4684          LA    R1,8(R1)
0004C0A8  E313 0018 0024                00000018  4685          STG   R1,24(R3)           load OP4 address in PL
                                                  4686
0004C0AE  B982 0000                               4687          XGR   R0,R0
0004C0B2  4100 0023                     00000023  4688          LA    R0,35
0004C0B6  EE00 0000 3000      00000000  00000000  4689          PLO   0,0,0,0(R3)         Quadruple C/S (QCSG: 35)
                                                  4690
0004C0BC  B98D 0020                               4691          EPSW  R2,R0               extract psw
0004C0C0  5020 5010                     00000010  4692          ST    R2,CCPSW              to save CC
0004C0C4  E310 500B 0076                0000000B  4693          LB    R1,CCMASK           (failure CC mask)
0004C0CA  8910 0004                     00000004  4694          SLL   R1,4                (shift to BC instr CC position)
0004C0CE  4410 510C                     0004C12C  4695          EX    R1,F353TSTCC            fail if...
                                                  4696
                                0004C0D2  00000001  4697 F353CKUR EQU   *
0004C0D2  D200 5009 3030      00000009  00000030  4698          MVC   URFOUND,48(R3)      Extract UR
0004C0D8  D500 5008 3030      00000008  00000030  4699          CLC   UR,48(R3)
0004C0DE  4780 50C6                     0004C0E6  4700          BE    F353CKRES
0004C0E2  45F0 8220                     00000420  4701          BAL   R15,URMSG
                                                  4702 *
                                                  4703 * compare results
                                                  4704 *
                                0004C0E6  00000001  4705 F353CKRES EQU   *
                                                  4706 *                                  reset adresses
0004C0E6  B982 0011                               4707          XGR   R1,R1
0004C0EA  5810 501C                     0000001C  4708          L     R1,OPSRCA           source
0004C0EE  B982 0033                               4709          XGR   R3,R3
0004C0F2  5830 5024                     00000024  4710          L     R3,PLA              PL
0004C0F6  B982 0044                               4711          XGR   R4,R4
0004C0FA  5840 5020                     00000020  4712          L     R4,OPREPA
                                                  4713
0004C0FE  D507 1000 3040      00000000  00000040  4714          CLC   0(8,R1),64(R3)
0004C104  4770 5106                     0004C126  4715          BNE   F353XX              OP1C Different, error
0004C108  D507 1008 3048      00000008  00000048  4716          CLC   8(8,R1),72(R3)
0004C10E  4770 5106                     0004C126  4717          BNE   F353XX              OP2C Different, error
0004C112  D507 1010 3050      00000010  00000050  4718          CLC   16(8,R1),80(R3)
0004C118  4770 5106                     0004C126  4719          BNE   F353XX              OP3C Different, error
0004C11C  D507 1018 3058      00000018  00000058  4720          CLC   24(8,R1),88(R3)
0004C122  478B 0000                     00000000  4721          BE    0(R11)              OP4Csame, return
                                                  4722
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                0004C126  00000001  4723 F353XX   EQU   *
0004C126  45F0 8340                       00000540  4724          BAL   R15,FAILMSG
0004C12A  07FB                            00000000  4725          BR    R11
                                                    4726
0004C12C  4700 5110                       0004C130  4727 F353TSTCC BC    0,F353CCMSG          fail if unexpected condition code
                                0004C130  00000001  4728 F353CCMSG EQU   *
0004C130  45F0 80F2                       000002F2  4729          BAL   R15,CCMSG
0004C134  47F0 50B2                       0004C0D2  4730          B     F353CKUR
                                                    4731
                                                    4732          TSTDATA
0004C138                        0004C138  0004C400  4733+         ORG   PLOETST+(200*K)+(26*4*K)+K
                                                    4734          TSTLBL OPDST              Operand Destination
0004C400                                            4735+OPDST64  DS    0C
0004C400  00000000 00000000                         4736          DC    XL256'0'
0004C408  00000000 00000000
0004C410  00000000 00000000
0004C418  00000000 00000000
0004C420  00000000 00000000
0004C428  00000000 00000000
0004C430  00000000 00000000
0004C438  00000000 00000000
0004C440  00000000 00000000
0004C448  00000000 00000000
0004C450  00000000 00000000
0004C458  00000000 00000000
0004C460  00000000 00000000
0004C468  00000000 00000000
0004C470  00000000 00000000
0004C478  00000000 00000000
0004C480  00000000 00000000
0004C488  00000000 00000000
0004C490  00000000 00000000
0004C498  00000000 00000000
0004C4A0  00000000 00000000
0004C4A8  00000000 00000000
0004C4B0  00000000 00000000
0004C4B8  00000000 00000000
0004C4C0  00000000 00000000
0004C4C8  00000000 00000000
0004C4D0  00000000 00000000
0004C4D8  00000000 00000000
0004C4E0  00000000 00000000
0004C4E8  00000000 00000000
0004C4F0  00000000 00000000
0004C4F8  00000000 00000000
                                                    4737
                                                    4738          TSTLBL OPSRC              Operand Source OP1-OP4
0004C500                                            4739+OPSRC64  DS    0C
0004C500  00000000 00000000                         4740          DC    XL256'0'
0004C508  00000000 00000000
0004C510  00000000 00000000
0004C518  00000000 00000000
0004C520  00000000 00000000
0004C528  00000000 00000000
0004C530  00000000 00000000
0004C538  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004C540  00000000 00000000
0004C548  00000000 00000000
0004C550  00000000 00000000
0004C558  00000000 00000000
0004C560  00000000 00000000
0004C568  00000000 00000000
0004C570  00000000 00000000
0004C578  00000000 00000000
0004C580  00000000 00000000
0004C588  00000000 00000000
0004C590  00000000 00000000
0004C598  00000000 00000000
0004C5A0  00000000 00000000
0004C5A8  00000000 00000000
0004C5B0  00000000 00000000
0004C5B8  00000000 00000000
0004C5C0  00000000 00000000
0004C5C8  00000000 00000000
0004C5D0  00000000 00000000
0004C5D8  00000000 00000000
0004C5E0  00000000 00000000
0004C5E8  00000000 00000000
0004C5F0  00000000 00000000
0004C5F8  00000000 00000000
                                                4741
                                                4742          TSTLBL PL                 Parameter List
0004C600                                        4743+PL64     DS    0C
0004C600  00000000 00000000                     4744          DC    XL352'00'
0004C608  00000000 00000000
0004C610  00000000 00000000
0004C618  00000000 00000000
0004C620  00000000 00000000
0004C628  00000000 00000000
0004C630  00000000 00000000
0004C638  00000000 00000000
0004C640  00000000 00000000
0004C648  00000000 00000000
0004C650  00000000 00000000
0004C658  00000000 00000000
0004C660  00000000 00000000
0004C668  00000000 00000000
0004C670  00000000 00000000
0004C678  00000000 00000000
0004C680  00000000 00000000
0004C688  00000000 00000000
0004C690  00000000 00000000
0004C698  00000000 00000000
0004C6A0  00000000 00000000
0004C6A8  00000000 00000000
0004C6B0  00000000 00000000
0004C6B8  00000000 00000000
0004C6C0  00000000 00000000
0004C6C8  00000000 00000000
0004C6D0  00000000 00000000
0004C6D8  00000000 00000000
0004C6E0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004C6E8  00000000 00000000
0004C6F0  00000000 00000000
0004C6F8  00000000 00000000
0004C700  00000000 00000000
0004C708  00000000 00000000
0004C710  00000000 00000000
0004C718  00000000 00000000
0004C720  00000000 00000000
0004C728  00000000 00000000
0004C730  00000000 00000000
0004C738  00000000 00000000
0004C740  00000000 00000000
0004C748  00000000 00000000
0004C750  00000000 00000000
0004C758  00000000 00000000
                                        4745
                                        4746          TSTLBL FMSG               fail message
0004C760                                4747+FMSG64   DS    0C
0004C760  86818993 858440D7             4748          DC    C'failed PLO-Extension '
0004C768  D3D660C5 A7A38595
0004C770  A2899695 40
0004C775  D8A48184 99A49793             4749          DC    CL96'Quadruple Compare and Swap (QCSG: 35): 4th NE'
0004C77D  8540C396 94978199
0004C785  85408195 8440E2A6
0004C78D  8197404D D8C3E2C7
0004C795  7A40F3F5 5D7A40F4
0004C79D  A38840D5 C5404040
0004C7A5  40404040 40404040
0004C7AD  40404040 40404040
0004C7B5  40404040 40404040
0004C7BD  40404040 40404040
0004C7C5  40404040 40404040
0004C7CD  40404040 40404040
                                        4750
                                        4751          TSTEND
0004C7D6                                4752+         DS    0H
0004C7D6  C5958440 968640A3             4753+         DC    CL32'End of test: 64'
0004C7DE  85A2A37A 40F6F440
0004C7E6  40404040 40404040
0004C7EE  40404040 40404040
                                        4754
                                        4755 * ------------------------------------------------------------------
                                        4756 * Test Quadruple Compare and Swap (QCSG: 35) instruction : 4th NE
                                        4757 * ------------------------------------------------------------------
                                        4758          TSTHDR 2,4,35             4th Operand NOT Equal (cc=2, UR=4)
0004C7F6                  0004C7F6  0004D000  4759+         ORG   PLOETST+(200*K)+(27*4*K)
0004D000  E2E3C1D9 E3409686             4760+         DC    CL32'START of test: 65'
0004D008  40A385A2 A37A40F6
0004D010  F5404040 40404040
0004D018  40404040 40404040
0004D020                  0004D020      4761+         USING *,R5                base for test data and test routine
0004D020                  00000000      4762+         USING PLOETEST,R5
0004D020  0004D04C                      4763+T65      DC    A(CODE65)           address of test code
0004D024  0041                          4764+         DC    H'65'               test number
0004D026  00                            4765+         DC    X'00'
0004D027  23                            4766+         DC    HL1'35'             Function
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004D028  04                                       4767+          DC    HL1'4'              expected unequal results (UR)
0004D029  00                                       4768+          DC    HL1'00'             found UR
0004D02A  02                                       4769+          DC    HL1'2'              cc
0004D02B  0D                                       4770+          DC    HL1'13'             not expected CC mask
0004D02C  00                                       4771+          DS    X                   extracted cc
0004D030  00000000 00000000                        4772+          DS    2F                  extract PSW after test (has CC)
0004D038  0004D760                                 4773+          DC    A(FMSG65)           address of test fail message
0004D03C  0004D500                                 4774+          DC    A(OPSRC65)          address of operand source
0004D040  0004D400                                 4775+          DC    A(OPDST65)          address of operand destination
0004D044  0004D600                                 4776+          DC    A(PL65)             address of parameter list
0004D048  000011F2                                 4777+          DC    A(RANDX+65)         address of random source data
                                                   4778
                                                   4779           TSTCODE
0004D04C                                           4780+CODE65    DS    0F
                                                   4781 *                                   load random source
0004D04C  B982 0022                                4782           XGR   R2,R2
0004D050  5820 5028                00000028        4783           L     R2,RANDXA
                                                   4784 *
                                                   4785 * Quadruple Compare and Swap (QCSG: 35)
                                                   4786 * Build Parambter List
                                                   4787 *
0004D054  B982 0011                                4788           XGR   R1,R1
0004D058  5810 501C                0000001C        4789           L     R1,OPSRCA
                                                   4790
0004D05C  B982 0033                                4791           XGR   R3,R3
0004D060  5830 5024                00000024        4792           L     R3,PLA
                                                   4793
0004D064  B982 0044                                4794           XGR   R4,R4
0004D068  5840 5020                00000020        4795           L     R4,OPREPA
                                                   4796
0004D06C  D21F 1000 2000  00000000  00000000        4797           MVC   0(32,R1),0(R2)      load Op1-Op4 data in source
0004D072  D21F 3040 1000  00000040  00000000        4798           MVC   64(32,R3),0(R1)     copy to PL OP1-Op4 Compare
0004D078  D21F 4000 2080  00000000  00000080        4799           MVC   0(32,R4),128(R2)    load initial Op1-Op4 data in ???
0004D07E  D21F 3080 4000  00000080  00000000        4800           MVC   128(32,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                   4801 ****
                                                   4802 * force a failure by changing the 3rd operand
                                                   4803 ****
0004D084  D203 1018 4000  00000018  00000000        4804           MVC   24(4,R1),0(R4)
                                                   4805 ****
                                                   4806
0004D08A  E313 0000 0024                00000000   4807           STG   R1,0(R3)            load OP1 address in PL
                                                   4808
0004D090  4111 0008                     00000008   4809           LA    R1,8(R1)
0004D094  E313 0008 0024                00000008   4810           STG   R1,8(R3)            load OP2 address in PL
                                                   4811
0004D09A  4111 0008                     00000008   4812           LA    R1,8(R1)
0004D09E  E313 0010 0024                00000010   4813           STG   R1,16(R3)           load OP3 address in PL
                                                   4814
0004D0A4  4111 0008                     00000008   4815           LA    R1,8(R1)
0004D0A8  E313 0018 0024                00000018   4816           STG   R1,24(R3)           load OP4 address in PL
                                                   4817
0004D0AE  B982 0000                                4818           XGR   R0,R0
0004D0B2  4100 0023                00000023        4819           LA    R0,35
0004D0B6  EE00 0000 3000  00000000  00000000        4820           PLO   0,0,0,0(R3)         Quadruple C/S (QCSG: 35)
                                                   4821
0004D0BC  B98D 0020                                4822           EPSW  R2,R0               extract psw
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004D0C0  5020 5010                     00000010  4823          ST    R2,CCPSW              to save CC
0004D0C4  E310 500B 0076                0000000B  4824          LB    R1,CCMASK            (failure CC mask)
0004D0CA  8910 0004                     00000004  4825          SLL   R1,4                  (shift to BC instr CC position)
0004D0CE  4410 510C                     0004D12C  4826          EX    R1,F354TSTCC           fail if...
                                                  4827
                          0004D0D2      00000001  4828 F354CKUR EQU   *
0004D0D2  D200 5009 3030  00000009      00000030  4829          MVC   URFOUND,48(R3)       Extract UR
0004D0D8  D500 5008 3030  00000008      00000030  4830          CLC   UR,48(R3)
0004D0DE  4780 50C6                     0004D0E6  4831          BE    F354CKRES
0004D0E2  45F0 8220                     00000420  4832          BAL   R15,URMSG
                                                  4833 *
                                                  4834 * compare results
                                                  4835 *
                          0004D0E6      00000001  4836 F354CKRES EQU   *
                                                  4837 *                                reset adresses
0004D0E6  B982 0011                               4838          XGR   R1,R1
0004D0EA  5810 501C                     0000001C  4839          L     R1,OPSRCA           source
0004D0EE  B982 0033                               4840          XGR   R3,R3
0004D0F2  5830 5024                     00000024  4841          L     R3,PLA              PL
0004D0F6  B982 0044                               4842          XGR   R4,R4
0004D0FA  5840 5020                     00000020  4843          L     R4,OPREPA
                                                  4844
0004D0FE  D507 1000 3040  00000000      00000040  4845          CLC   0(8,R1),64(R3)
0004D104  4770 5106                     0004D126  4846          BNE   F354XX              OP1C Different, error
0004D108  D507 1008 3048  00000008      00000048  4847          CLC   8(8,R1),72(R3)
0004D10E  4770 5106                     0004D126  4848          BNE   F354XX              OP2C Different, error
0004D112  D507 1010 3050  00000010      00000050  4849          CLC   16(8,R1),80(R3)
0004D118  4770 5106                     0004D126  4850          BNE   F354XX              OP3C Different, error
0004D11C  D507 1018 3058  00000018      00000058  4851          CLC   24(8,R1),88(R3)
0004D122  478B 0000                     00000000  4852          BE    0(R11)              OP4Csame, return
                                                  4853
                          0004D126      00000001  4854 F354XX   EQU   *
0004D126  45F0 8340                     00000540  4855          BAL   R15,FAILMSG
0004D12A  07FB                                    4856          BR    R11
                                                  4857
0004D12C  4700 5110                     0004D130  4858 F354TSTCC BC    0,F354CCMSG          fail if unexpected condition code
                          0004D130      00000001  4859 F354CCMSG EQU   *
0004D130  45F0 80F2                     000002F2  4860          BAL   R15,CCMSG
0004D134  47F0 50B2                     0004D0D2  4861          B     F354CKUR
                                                  4862
                                                  4863          TSTDATA
0004D138                  0004D138      0004D400  4864+         ORG   PLOETST+(200*K)+(27*4*K)+K
                                                  4865          TSTLBL OPDST              Operand Destination
0004D400                                          4866+OPDST65  DS    0C
0004D400  00000000 00000000                       4867          DC    XL256'0'
0004D408  00000000 00000000
0004D410  00000000 00000000
0004D418  00000000 00000000
0004D420  00000000 00000000
0004D428  00000000 00000000
0004D430  00000000 00000000
0004D438  00000000 00000000
0004D440  00000000 00000000
0004D448  00000000 00000000
0004D450  00000000 00000000
0004D458  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004D460  00000000 00000000
0004D468  00000000 00000000
0004D470  00000000 00000000
0004D478  00000000 00000000
0004D480  00000000 00000000
0004D488  00000000 00000000
0004D490  00000000 00000000
0004D498  00000000 00000000
0004D4A0  00000000 00000000
0004D4A8  00000000 00000000
0004D4B0  00000000 00000000
0004D4B8  00000000 00000000
0004D4C0  00000000 00000000
0004D4C8  00000000 00000000
0004D4D0  00000000 00000000
0004D4D8  00000000 00000000
0004D4E0  00000000 00000000
0004D4E8  00000000 00000000
0004D4F0  00000000 00000000
0004D4F8  00000000 00000000
                                        4868
                                        4869           TSTLBL OPSRC              Operand Source OP1-OP4
0004D500                                4870+OPSRC65   DS    0C
0004D500  00000000 00000000             4871           DC    XL256'0'
0004D508  00000000 00000000
0004D510  00000000 00000000
0004D518  00000000 00000000
0004D520  00000000 00000000
0004D528  00000000 00000000
0004D530  00000000 00000000
0004D538  00000000 00000000
0004D540  00000000 00000000
0004D548  00000000 00000000
0004D550  00000000 00000000
0004D558  00000000 00000000
0004D560  00000000 00000000
0004D568  00000000 00000000
0004D570  00000000 00000000
0004D578  00000000 00000000
0004D580  00000000 00000000
0004D588  00000000 00000000
0004D590  00000000 00000000
0004D598  00000000 00000000
0004D5A0  00000000 00000000
0004D5A8  00000000 00000000
0004D5B0  00000000 00000000
0004D5B8  00000000 00000000
0004D5C0  00000000 00000000
0004D5C8  00000000 00000000
0004D5D0  00000000 00000000
0004D5D8  00000000 00000000
0004D5E0  00000000 00000000
0004D5E8  00000000 00000000
0004D5F0  00000000 00000000
0004D5F8  00000000 00000000
                                        4872
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   4873           TSTLBL PL                  Parameter List
0004D600                                           4874+PL65      DS     0C
0004D600  00000000 00000000                        4875           DC     XL352'00'
0004D608  00000000 00000000
0004D610  00000000 00000000
0004D618  00000000 00000000
0004D620  00000000 00000000
0004D628  00000000 00000000
0004D630  00000000 00000000
0004D638  00000000 00000000
0004D640  00000000 00000000
0004D648  00000000 00000000
0004D650  00000000 00000000
0004D658  00000000 00000000
0004D660  00000000 00000000
0004D668  00000000 00000000
0004D670  00000000 00000000
0004D678  00000000 00000000
0004D680  00000000 00000000
0004D688  00000000 00000000
0004D690  00000000 00000000
0004D698  00000000 00000000
0004D6A0  00000000 00000000
0004D6A8  00000000 00000000
0004D6B0  00000000 00000000
0004D6B8  00000000 00000000
0004D6C0  00000000 00000000
0004D6C8  00000000 00000000
0004D6D0  00000000 00000000
0004D6D8  00000000 00000000
0004D6E0  00000000 00000000
0004D6E8  00000000 00000000
0004D6F0  00000000 00000000
0004D6F8  00000000 00000000
0004D700  00000000 00000000
0004D708  00000000 00000000
0004D710  00000000 00000000
0004D718  00000000 00000000
0004D720  00000000 00000000
0004D728  00000000 00000000
0004D730  00000000 00000000
0004D738  00000000 00000000
0004D740  00000000 00000000
0004D748  00000000 00000000
0004D750  00000000 00000000
0004D758  00000000 00000000
                                                   4876
                                                   4877           TSTLBL FMSG                fail message
0004D760                                           4878+FMSG65    DS     0C
0004D760  86818993 858440D7                        4879           DC     C'failed PLO-Extension '
0004D768  D3D660C5 A7A38595
0004D770  A2899695 40
0004D775  D8A48184 99A49793                        4880           DC     CL96'Quadruple Compare and Swap (QCSG: 35): 4th NE'
0004D77D  8540C396 94978199
0004D785  85408195 8440E2A6
0004D78D  8197404D D8C3E2C7
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004D795  7A40F3F5 5D7A40F4
0004D79D  A38840D5 C5404040
0004D7A5  40404040 40404040
0004D7AD  40404040 40404040
0004D7B5  40404040 40404040
0004D7BD  40404040 40404040
0004D7C5  40404040 40404040
0004D7CD  40404040 40404040
                                      4881
                                      4882          TSTEND
0004D7D6                              4883+         DS    0H
0004D7D6  C5958440 968640A3           4884+         DC    CL32'End of test: 65'
0004D7DE  85A2A37A 40F6F540
0004D7E6  40404040 40404040
0004D7EE  40404040 40404040
                                      4885
                                      4886 * -------------------------------------------------------------------
                                      4887 * Test Quadruple Compare and Swap (QCSG: 35) instruction : All EQ
                                      4888 * -------------------------------------------------------------------
                                      4889          TSTHDR 0,99,35              ALL EQ Equal (cc=0, UR=99
0004D7F6                    0004D7F6  0004E000  4890+         ORG   PLOETST+(200*K)+(28*4*K)
0004E000  E2E3C1D9 E3409686           4891+         DC    CL32'START of test: 66'
0004E008  40A385A2 A37A40F6
0004E010  F6404040 40404040
0004E018  40404040 40404040
0004E020                    0004E020  4892+         USING *,R5                  base for test data and test routine
0004E020                    00000000  4893+         USING PLOETEST,R5
0004E020  0004E04C                    4894+T66      DC    A(CODE66)             address of test code
0004E024  0042                        4895+         DC    H'66'                 test number
0004E026  00                          4896+         DC    X'00'
0004E027  23                          4897+         DC    HL1'35'               Function
0004E028  63                          4898+         DC    HL1'99'               expected unequal results (UR)
0004E029  00                          4899+         DC    HL1'00'               found UR
0004E02A  00                          4900+         DC    HL1'0'                cc
0004E02B  07                          4901+         DC    HL1'7'                not expected CC mask
0004E02C  00                          4902+         DS    X                     extracted cc
0004E030  00000000 00000000           4903+         DS    2F                    extract PSW after test (has CC)
0004E038  0004E760                    4904+         DC    A(FMSG66)             address of test fail message
0004E03C  0004E500                    4905+         DC    A(OPSRC66)            address of operand source
0004E040  0004E400                    4906+         DC    A(OPDST66)            address of operand destination
0004E044  0004E600                    4907+         DC    A(PL66)               address of parameter list
0004E048  000011F3                    4908+         DC    A(RANDX+66)           address of random source data
                                      4909
                                      4910          TSTCODE
0004E04C                              4911+CODE66   DS    0F
                                      4912 *                                    load random source
0004E04C  B982 0022                   4913          XGR   R2,R2
0004E050  5820 5028           00000028  4914          L     R2,RANDXA
                                      4915 *
                                      4916 * Quadruple Compare and Swap (QCSG: 35)
                                      4917 * Build Parambter List
                                      4918 *
0004E054  B982 0011                   4919          XGR   R1,R1
0004E058  5810 501C           0000001C  4920          L     R1,OPSRCA
                                      4921
0004E05C  B982 0033                   4922          XGR   R3,R3
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004E060  5830 5024                    00000024  4923          L     R3,PLA
                                                 4924
0004E064  B982 0044                              4925          XGR   R4,R4
0004E068  5840 5020                    00000020  4926          L     R4,OPREPA
                                                 4927
0004E06C  D21F 1000 2000       00000000  00000000  4928          MVC   0(32,R1),0(R2)      load Op1-Op4 data in source
0004E072  D21F 3040 1000       00000040  00000000  4929          MVC   64(32,R3),0(R1)     copy to PL OP1-Op4 Compare
0004E078  D21F 4000 2080       00000000  00000080  4930          MVC   0(32,R4),128(R2)    load initial Op1-Op4 data in ???
0004E07E  D21F 3080 4000       00000080  00000000  4931          MVC   128(32,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                 4932 ****
                                                 4933 * force a failure by changing the 3rd operand
                                                 4934 ****
                                                 4935 *        MVC   24(4,R1),0(R4)
                                                 4936 ****
                                                 4937
0004E084  E313 0000 0024                 00000000  4938          STG   R1,0(R3)            load OP1 address in PL
                                                 4939
0004E08A  4111 0008                    00000008  4940          LA    R1,8(R1)
0004E08E  E313 0008 0024               00000008  4941          STG   R1,8(R3)            load OP2 address in PL
                                                 4942
0004E094  4111 0008                    00000008  4943          LA    R1,8(R1)
0004E098  E313 0010 0024               00000010  4944          STG   R1,16(R3)           load OP3 address in PL
                                                 4945
0004E09E  4111 0008                    00000008  4946          LA    R1,8(R1)
0004E0A2  E313 0018 0024               00000018  4947          STG   R1,24(R3)           load OP4 address in PL
                                                 4948
0004E0A8  B982 0000                              4949          XGR   R0,R0
0004E0AC  4100 0023                    00000023  4950          LA    R0,35
0004E0B0  EE00 0000 3000       00000000  00000000  4951          PLO   0,0,0,0(R3)         Quadruple C/S (QCSG: 35)
                                                 4952
0004E0B6  B98D 0020                              4953          EPSW  R2,R0               extract psw
0004E0BA  5020 5010                    00000010  4954          ST    R2,CCPSW              to save CC
0004E0BE  E310 500B 0076               0000000B  4955          LB    R1,CCMASK           (failure CC mask)
0004E0C4  8910 0004                    00000004  4956          SLL   R1,4                (shift to BC instr CC position)
0004E0C8  4410 5106                    0004E126  4957          EX    R1,F355TSTCC            fail if...
                                                 4958
                                 0004E0CC  00000001  4959 F355CKUR EQU   *
0004E0CC  D200 5009 3030       00000009  00000030  4960          MVC   URFOUND,48(R3)      Extract UR
0004E0D2  D500 5008 3030       00000008  00000030  4961          CLC   UR,48(R3)
0004E0D8  4780 50C0                    0004E0E0  4962          BE    F355CKRES
0004E0DC  45F0 8220                    00000420  4963          BAL   R15,URMSG
                                                 4964 *
                                                 4965 * compare results
                                                 4966 *
                                 0004E0E0  00000001  4967 F355CKRES EQU   *
                                                 4968 *                    reset adresses
0004E0E0  B982 0011                              4969          XGR   R1,R1
0004E0E4  5810 501C                    0000001C  4970          L     R1,OPSRCA           source
0004E0E8  B982 0033                              4971          XGR   R3,R3
0004E0EC  5830 5024                    00000024  4972          L     R3,PLA              PL
0004E0F0  B982 0044                              4973          XGR   R4,R4
0004E0F4  5840 5020                    00000020  4974          L     R4,OPREPA
                                                 4975
                                                 4976 * All equal; replacement values copied to operand source
                                                 4977
0004E0F8  D507 1000 3060       00000000  00000060  4978          CLC   0(8,R1),96(R3)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT
0004E0FE  4770 5100                     0004E120  4979           BNE   F355XX              OP1R Different, error
0004E102  D507 1008 3068      00000008  00000068  4980           CLC   8(8,R1),104(R3)
0004E108  4770 5100                     0004E120  4981           BNE   F355XX              OP2R Different, error
0004E10C  D507 1010 3070      00000010  00000070  4982           CLC   16(8,R1),112(R3)
0004E112  4770 5100                     0004E120  4983           BNE   F355XX              OP3R Different, error
0004E116  D507 1018 3078      00000018  00000078  4984           CLC   24(8,R1),120(R3)
0004E11C  478B 0000                     00000000  4985           BE    0(R11)              OP4R same, return
                                                  4986
                              0004E120  00000001  4987 F355XX    EQU   *
0004E120  45F0 8340                     00000540  4988           BAL   R15,FAILMSG
0004E124  07FB                                    4989           BR    R11
                                                  4990
0004E126  4700 510A                     0004E12A  4991 F355TSTCC BC    0,F355CCMSG         fail if unexpected condition code
                              0004E12A  00000001  4992 F355CCMSG EQU    *
0004E12A  45F0 80F2                     000002F2  4993           BAL   R15,CCMSG
0004E12E  47F0 50AC                     0004E0CC  4994           B     F355CKUR
                                                  4995
                                                  4996           TSTDATA
0004E132                      0004E132  0004E400  4997+          ORG   PLOETST+(200*K)+(28*4*K)+K
                                                  4998           TSTLBL OPDST              Operand Destination
0004E400                                          4999+OPDST66   DS    0C
0004E400  00000000 00000000               00000000 5000          DC    XL256'0'
0004E408  00000000 00000000
0004E410  00000000 00000000
0004E418  00000000 00000000
0004E420  00000000 00000000
0004E428  00000000 00000000
0004E430  00000000 00000000
0004E438  00000000 00000000
0004E440  00000000 00000000
0004E448  00000000 00000000
0004E450  00000000 00000000
0004E458  00000000 00000000
0004E460  00000000 00000000
0004E468  00000000 00000000
0004E470  00000000 00000000
0004E478  00000000 00000000
0004E480  00000000 00000000
0004E488  00000000 00000000
0004E490  00000000 00000000
0004E498  00000000 00000000
0004E4A0  00000000 00000000
0004E4A8  00000000 00000000
0004E4B0  00000000 00000000
0004E4B8  00000000 00000000
0004E4C0  00000000 00000000
0004E4C8  00000000 00000000
0004E4D0  00000000 00000000
0004E4D8  00000000 00000000
0004E4E0  00000000 00000000
0004E4E8  00000000 00000000
0004E4F0  00000000 00000000
0004E4F8  00000000 00000000
                                                  5001
                                                  5002           TSTLBL OPSRC              Operand Source OP1-OP4
0004E500                                          5003+OPSRC66   DS    0C
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0004E500  00000000 00000000                        5004          DC    XL256'0'
0004E508  00000000 00000000
0004E510  00000000 00000000
0004E518  00000000 00000000
0004E520  00000000 00000000
0004E528  00000000 00000000
0004E530  00000000 00000000
0004E538  00000000 00000000
0004E540  00000000 00000000
0004E548  00000000 00000000
0004E550  00000000 00000000
0004E558  00000000 00000000
0004E560  00000000 00000000
0004E568  00000000 00000000
0004E570  00000000 00000000
0004E578  00000000 00000000
0004E580  00000000 00000000
0004E588  00000000 00000000
0004E590  00000000 00000000
0004E598  00000000 00000000
0004E5A0  00000000 00000000
0004E5A8  00000000 00000000
0004E5B0  00000000 00000000
0004E5B8  00000000 00000000
0004E5C0  00000000 00000000
0004E5C8  00000000 00000000
0004E5D0  00000000 00000000
0004E5D8  00000000 00000000
0004E5E0  00000000 00000000
0004E5E8  00000000 00000000
0004E5F0  00000000 00000000
0004E5F8  00000000 00000000
                                                   5005
                                                   5006          TSTLBL PL                 Parameter List
0004E600                                           5007+PL66     DS    0C
0004E600  00000000 00000000                        5008          DC    XL352'00'
0004E608  00000000 00000000
0004E610  00000000 00000000
0004E618  00000000 00000000
0004E620  00000000 00000000
0004E628  00000000 00000000
0004E630  00000000 00000000
0004E638  00000000 00000000
0004E640  00000000 00000000
0004E648  00000000 00000000
0004E650  00000000 00000000
0004E658  00000000 00000000
0004E660  00000000 00000000
0004E668  00000000 00000000
0004E670  00000000 00000000
0004E678  00000000 00000000
0004E680  00000000 00000000
0004E688  00000000 00000000
0004E690  00000000 00000000
0004E698  00000000 00000000
0004E6A0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004E6A8  00000000 00000000
0004E6B0  00000000 00000000
0004E6B8  00000000 00000000
0004E6C0  00000000 00000000
0004E6C8  00000000 00000000
0004E6D0  00000000 00000000
0004E6D8  00000000 00000000
0004E6E0  00000000 00000000
0004E6E8  00000000 00000000
0004E6F0  00000000 00000000
0004E6F8  00000000 00000000
0004E700  00000000 00000000
0004E708  00000000 00000000
0004E710  00000000 00000000
0004E718  00000000 00000000
0004E720  00000000 00000000
0004E728  00000000 00000000
0004E730  00000000 00000000
0004E738  00000000 00000000
0004E740  00000000 00000000
0004E748  00000000 00000000
0004E750  00000000 00000000
0004E758  00000000 00000000
                                          5009
                                          5010          TSTLBL FMSG               fail message
0004E760                                  5011+FMSG66   DS    0C
0004E760  86818993 858440D7               5012          DC    C'failed PLO-Extension '
0004E768  D3D660C5 A7A38595
0004E770  A2899695 40
0004E775  D8A48184 99A49793               5013          DC    CL96'Quadruple Compare and Swap (QCSG: 35): 3rd NE'
0004E77D  8540C396 94978199
0004E785  85408195 8440E2A6
0004E78D  8197404D D8C3E2C7
0004E795  7A40F3F5 5D7A40F3
0004E79D  998440D5 C5404040
0004E7A5  40404040 40404040
0004E7AD  40404040 40404040
0004E7B5  40404040 40404040
0004E7BD  40404040 40404040
0004E7C5  40404040 40404040
0004E7CD  40404040 40404040
                                          5014
                                          5015          TSTEND
0004E7D6                                  5016+         DS    0H
0004E7D6  C5958440 968640A3               5017+         DC    CL32'End of test: 66'
0004E7DE  85A2A37A 40F6F640
0004E7E6  40404040 40404040
0004E7EE  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5019 *=============================================================================
                                                    5020 *       Quadruple Compare and Swap Tests
                                                    5021 *=============================================================================
                                                    5022
                                                    5023 * -----------------------------------------------------------------------
                                                    5024 * Test Quadruple Compare and Swap (QCS: 34) instruction : 1st NE
                                                    5025 * -----------------------------------------------------------------------
                                                    5026          TSTHDR 2,1,34             1st Operand NOT Equal (cc=2, UR=1)
0004E7F6                          0004E7F6  0004F000  5027+         ORG   PLOETST+(200*K)+(29*4*K)
0004F000  E2E3C1D9 E3409686                         5028+         DC    CL32'START of test: 67'
0004F008  40A385A2 A37A40F6
0004F010  F7404040 40404040
0004F018  40404040 40404040
0004F020                          0004F020          5029+         USING *,R5                base for test data and test routine
0004F020                          00000000          5030+         USING PLOETEST,R5
0004F020  0004F04C                                  5031+T67      DC    A(CODE67)           address of test code
0004F024  0043                                      5032+         DC    H'67'               test number
0004F026  00                                        5033+         DC    X'00'
0004F027  22                                        5034+         DC    HL1'34'             Function
0004F028  01                                        5035+         DC    HL1'1'              expected unequal results (UR)
0004F029  00                                        5036+         DC    HL1'00'             found UR
0004F02A  02                                        5037+         DC    HL1'2'              cc
0004F02B  0D                                        5038+         DC    HL1'13'             not expected CC mask
0004F02C  00                                        5039+         DS    X                   extracted cc
0004F030  00000000 00000000                         5040+         DS    2F                  extract PSW after test (has CC)
0004F038  0004F760                                  5041+         DC    A(FMSG67)           address of test fail message
0004F03C  0004F500                                  5042+         DC    A(OPSRC67)          address of operand source
0004F040  0004F400                                  5043+         DC    A(OPDST67)          address of operand destination
0004F044  0004F600                                  5044+         DC    A(PL67)             address of parameter list
0004F048  000011F4                                  5045+         DC    A(RANDX+67)         address of random source data
                                                    5046
                                                    5047          TSTCODE
0004F04C                                            5048+CODE67   DS    0F
                                                    5049 *                                  load random source
0004F04C  B982 0022                                 5050          XGR   R2,R2
0004F050  5820 5028                     00000028    5051          L     R2,RANDXA
                                                    5052 *
                                                    5053 * Quadruple Compare and Swap (QCS: 34)
                                                    5054 * Build Parambter List
                                                    5055 *
0004F054  B982 0011                                 5056          XGR   R1,R1
0004F058  5810 501C                     0000001C    5057          L     R1,OPSRCA
                                                    5058
0004F05C  B982 0033                                 5059          XGR   R3,R3
0004F060  5830 5024                     00000024    5060          L     R3,PLA
                                                    5061
0004F064  B982 0044                                 5062          XGR   R4,R4
0004F068  5840 5020                     00000020    5063          L     R4,OPREPA
                                                    5064
0004F06C  D20F 1000 2000        00000000  00000000  5065          MVC   0(16,R1),0(R2)      load Op1-Op4 data in source
0004F072  D203 3040 1000        00000040  00000000  5066          MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
0004F078  D203 3048 1004        00000048  00000004  5067          MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
0004F07E  D203 3050 1008        00000050  00000008  5068          MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
0004F084  D203 3058 100C        00000058  0000000C  5069          MVC   88(4,R3),12(R1)     copy to PL OP4-Compare
                                                    5070
0004F08A  D21F 4000 2080        00000000  00000080  5071          MVC   0(32,R4),128(R2)    load initial Op1-Op4 data in ???
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004F090  D203 3060 4000      00000060  00000000  5072          MVC   96(4,R3),0(R4)      copy to PL OP1-Replacement
0004F096  D203 3068 4004      00000068  00000004  5073          MVC   104(4,R3),4(R4)     copy to PL OP2-Replacement
0004F09C  D203 3070 4008      00000070  00000008  5074          MVC   112(4,R3),8(R4)     copy to PL OP3-Replacement
0004F0A2  D203 3078 400C      00000078  0000000C  5075          MVC   120(4,R3),12(R4)    copy to PL OP4-Replacement
                                                  5076 ****
                                                  5077 * force a failure by changing the first operand
                                                  5078 ****
0004F0A8  D203 1000 4000      00000000  00000000  5079          MVC   0(4,R1),0(R4)
                                                  5080 ****
                                                  5081
0004F0AE  E313 0000 0024                00000000  5082          STG   R1,0(R3)            load OP1 address in PL
                                                  5083
0004F0B4  4111 0004                     00000004  5084          LA    R1,4(R1)
0004F0B8  E313 0008 0024                00000008  5085          STG   R1,8(R3)            load OP2 address in PL
                                                  5086
0004F0BE  4111 0004                     00000004  5087          LA    R1,4(R1)
0004F0C2  E313 0010 0024                00000010  5088          STG   R1,16(R3)           load OP3 address in PL
                                                  5089
0004F0C8  4111 0004                     00000004  5090          LA    R1,4(R1)
0004F0CC  E313 0018 0024                00000018  5091          STG   R1,24(R3)           load OP4 address in PL
                                                  5092
0004F0D2  B982 0000                               5093          XGR   R0,R0
0004F0D6  4100 0022                     00000022  5094          LA    R0,34
0004F0DA  EE00 0000 3000      00000000  00000000  5095          PLO   0,0,0,0(R3)         Quadruple C/S (QCS: 34)
                                                  5096
0004F0E0  B98D 0020                               5097          EPSW  R2,R0               extract psw
0004F0E4  5020 5010                     00000010  5098          ST    R2,CCPSW              to save CC
0004F0E8  E310 500B 0076                0000000B  5099          LB    R1,CCMASK           (failure CC mask)
0004F0EE  8910 0004                     00000004  5100          SLL   R1,4                (shift to BC instr CC position)
0004F0F2  4410 5130                     0004F150  5101          EX    R1,F341TSTCC            fail if...
                                                  5102
                                00F0F6  00000001  5103 F341CKUR EQU   *
0004F0F6  D200 5009 3030      00000009  00000030  5104          MVC   URFOUND,48(R3)      Extract UR
0004F0FC  D500 5008 3030      00000008  00000030  5105          CLC   UR,48(R3)
0004F102  4780 50EA                     0004F10A  5106          BE    F341CKRES
0004F106  45F0 8220                     00000420  5107          BAL   R15,URMSG
                                                  5108 *
                                                  5109 * compare results
                                                  5110 *
                                10F10A  00000001  5111 F341CKRES EQU   *
                                                  5112 *                              reset adresses
0004F10A  B982 0011                               5113          XGR   R1,R1
0004F10E  5810 501C                     0000001C  5114          L     R1,OPSRCA           source
0004F112  B982 0033                               5115          XGR   R3,R3
0004F116  5830 5024                     00000024  5116          L     R3,PLA              PL
0004F11A  B982 0044                               5117          XGR   R4,R4
0004F11E  5840 5020                     00000020  5118          L     R4,OPREPA
                                                  5119
0004F122  D503 1000 3040      00000000  00000040  5120          CLC   0(4,R1),64(R3)
0004F128  4770 512A                     0004F14A  5121          BNE   F341XX              OP1C Different, error
0004F12C  D503 1004 3048      00000004  00000048  5122          CLC   4(4,R1),72(R3)
0004F132  4770 512A                     0004F14A  5123          BNE   F341XX              OP2C Different, error
0004F136  D503 1008 3050      00000008  00000050  5124          CLC   8(4,R1),80(R3)
0004F13C  4770 512A                     0004F14A  5125          BNE   F341XX              OP3C Different, error
0004F140  D503 100C 3058      0000000C  00000058  5126          CLC   12(4,R1),88(R3)
0004F146  478B 0000                     00000000  5127          BE    0(R11)              OP4Csame, return
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5128
                               0004F14A  00000001  5129 F341XX   EQU   *
0004F14A  45F0 8340                      00000540  5130          BAL   R15,FAILMSG
0004F14E  07FB                                     5131          BR    R11
                                                   5132
0004F150  4700 5134                      0004F154  5133 F341TSTCC BC    0,F341CCMSG        fail if unexpected condition code
                               0004F154  00000001  5134 F341CCMSG EQU   *
0004F154  45F0 80F2                      000002F2  5135          BAL   R15,CCMSG
0004F158  47F0 50D6                      0004F0F6  5136          B     F341CKUR
                                                   5137
                                                   5138          TSTDATA
0004F15C                       0004F15C  0004F400  5139+         ORG   PLOETST+(200*K)+(29*4*K)+K
                                                   5140          TSTLBL OPDST              Operand Destination
0004F400                                           5141+OPDST67  DS    0C
0004F400  00000000 00000000                        5142          DC    XL256'0'
0004F408  00000000 00000000
0004F410  00000000 00000000
0004F418  00000000 00000000
0004F420  00000000 00000000
0004F428  00000000 00000000
0004F430  00000000 00000000
0004F438  00000000 00000000
0004F440  00000000 00000000
0004F448  00000000 00000000
0004F450  00000000 00000000
0004F458  00000000 00000000
0004F460  00000000 00000000
0004F468  00000000 00000000
0004F470  00000000 00000000
0004F478  00000000 00000000
0004F480  00000000 00000000
0004F488  00000000 00000000
0004F490  00000000 00000000
0004F498  00000000 00000000
0004F4A0  00000000 00000000
0004F4A8  00000000 00000000
0004F4B0  00000000 00000000
0004F4B8  00000000 00000000
0004F4C0  00000000 00000000
0004F4C8  00000000 00000000
0004F4D0  00000000 00000000
0004F4D8  00000000 00000000
0004F4E0  00000000 00000000
0004F4E8  00000000 00000000
0004F4F0  00000000 00000000
0004F4F8  00000000 00000000
                                                   5143
                                                   5144          TSTLBL OPSRC              Operand Source OP1-OP4
0004F500                                           5145+OPSRC67  DS    0C
0004F500  00000000 00000000                        5146          DC    XL256'0'
0004F508  00000000 00000000
0004F510  00000000 00000000
0004F518  00000000 00000000
0004F520  00000000 00000000
0004F528  00000000 00000000
0004F530  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004F538  00000000 00000000
0004F540  00000000 00000000
0004F548  00000000 00000000
0004F550  00000000 00000000
0004F558  00000000 00000000
0004F560  00000000 00000000
0004F568  00000000 00000000
0004F570  00000000 00000000
0004F578  00000000 00000000
0004F580  00000000 00000000
0004F588  00000000 00000000
0004F590  00000000 00000000
0004F598  00000000 00000000
0004F5A0  00000000 00000000
0004F5A8  00000000 00000000
0004F5B0  00000000 00000000
0004F5B8  00000000 00000000
0004F5C0  00000000 00000000
0004F5C8  00000000 00000000
0004F5D0  00000000 00000000
0004F5D8  00000000 00000000
0004F5E0  00000000 00000000
0004F5E8  00000000 00000000
0004F5F0  00000000 00000000
0004F5F8  00000000 00000000
                                                5147
                                                5148           TSTLBL PL                  Parameter List
0004F600                                        5149+PL67      DS    0C
0004F600  00000000 00000000                     5150           DC    XL352'00'
0004F608  00000000 00000000
0004F610  00000000 00000000
0004F618  00000000 00000000
0004F620  00000000 00000000
0004F628  00000000 00000000
0004F630  00000000 00000000
0004F638  00000000 00000000
0004F640  00000000 00000000
0004F648  00000000 00000000
0004F650  00000000 00000000
0004F658  00000000 00000000
0004F660  00000000 00000000
0004F668  00000000 00000000
0004F670  00000000 00000000
0004F678  00000000 00000000
0004F680  00000000 00000000
0004F688  00000000 00000000
0004F690  00000000 00000000
0004F698  00000000 00000000
0004F6A0  00000000 00000000
0004F6A8  00000000 00000000
0004F6B0  00000000 00000000
0004F6B8  00000000 00000000
0004F6C0  00000000 00000000
0004F6C8  00000000 00000000
0004F6D0  00000000 00000000
0004F6D8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004F6E0  00000000 00000000
0004F6E8  00000000 00000000
0004F6F0  00000000 00000000
0004F6F8  00000000 00000000
0004F700  00000000 00000000
0004F708  00000000 00000000
0004F710  00000000 00000000
0004F718  00000000 00000000
0004F720  00000000 00000000
0004F728  00000000 00000000
0004F730  00000000 00000000
0004F738  00000000 00000000
0004F740  00000000 00000000
0004F748  00000000 00000000
0004F750  00000000 00000000
0004F758  00000000 00000000
                                          5151
                                          5152          TSTLBL FMSG               fail message
0004F760                                  5153+FMSG67   DS    0C
0004F760  86818993 858440D7               5154          DC    C'failed PLO-Extension '
0004F768  D3D660C5 A7A38595
0004F770  A2899695 40
0004F775  D8A48184 99A49793               5155          DC    CL96'Quadruple Compare and Swap (QCS: 34): 1st NE'
0004F77D  8540C396 94978199
0004F785  85408195 8440E2A6
0004F78D  8197404D D8C3E27A
0004F795  40F3F45D 7A40F1A2
0004F79D  A340D5C5 40404040
0004F7A5  40404040 40404040
0004F7AD  40404040 40404040
0004F7B5  40404040 40404040
0004F7BD  40404040 40404040
0004F7C5  40404040 40404040
0004F7CD  40404040 40404040
                                          5156
                                          5157          TSTEND
0004F7D6                                  5158+         DS    0H
0004F7D6  C5958440 968640A3               5159+         DC    CL32'End of test: 67'
0004F7DE  85A2A37A 40F6F740
0004F7E6  40404040 40404040
0004F7EE  40404040 40404040
                                          5160
                                          5161 * ------------------------------------------------------------------
                                          5162 * Test Quadruple Compare and Swap (QCS: 34) instruction : 2nd NE
                                          5163 * ------------------------------------------------------------------
                                          5164          TSTHDR 2,2,34             2nd Operand NOT Equal (cc=2, UR=2)
0004F7F6                  0004F7F6  00050000  5165+         ORG   PLOETST+(200*K)+(30*4*K)
00050000  E2E3C1D9 E3409686               5166+         DC    CL32'START of test: 68'
00050008  40A385A2 A37A40F6
00050010  F8404040 40404040
00050018  40404040 40404040
00050020                  00050020          5167+         USING *,R5                base for test data and test routine
00050020                  00000000          5168+         USING PLOETEST,R5
00050020  0005004C                        5169+T68      DC    A(CODE68)           address of test code
00050024  0044                            5170+         DC    H'68'               test number
00050026  00                              5171+         DC    X'00'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00050027  22                                       5172+          DC    HL1'34'             Function
00050028  02                                       5173+          DC    HL1'2'              expected unequal results (UR)
00050029  00                                       5174+          DC    HL1'00'             found UR
0005002A  02                                       5175+          DC    HL1'2'              cc
0005002B  0D                                       5176+          DC    HL1'13'             not expected CC mask
0005002C  00                                       5177+          DS    X                   extracted cc
00050030  00000000 00000000                        5178+          DS    2F                  extract PSW after test (has CC)
00050038  00050760                                 5179+          DC    A(FMSG68)           address of test fail message
0005003C  00050500                                 5180+          DC    A(OPSRC68)          address of operand source
00050040  00050400                                 5181+          DC    A(OPDST68)          address of operand destination
00050044  00050600                                 5182+          DC    A(PL68)             address of parameter list
00050048  000011F5                                 5183+          DC    A(RANDX+68)         address of random source data
                                                   5184
                                                   5185           TSTCODE
0005004C                                           5186+CODE68    DS    0F
                                                   5187 *                                   load random source
0005004C  B982 0022                                5188           XGR   R2,R2
00050050  5820 5028                     00000028   5189           L     R2,RANDXA
                                                   5190 *
                                                   5191 * Quadruple Compare and Swap (QCS: 34)
                                                   5192 * Build Parambter List
                                                   5193 *
00050054  B982 0011                                5194           XGR   R1,R1
00050058  5810 501C                     0000001C   5195           L     R1,OPSRCA
                                                   5196
0005005C  B982 0033                                5197           XGR   R3,R3
00050060  5830 5024                     00000024   5198           L     R3,PLA
                                                   5199
00050064  B982 0044                                5200           XGR   R4,R4
00050068  5840 5020                     00000020   5201           L     R4,OPREPA
                                                   5202
0005006C  D20F 1000 2000       00000000  00000000  5203           MVC   0(16,R1),0(R2)      load Op1-Op4 data in source
00050072  D203 3040 1000       00000040  00000000  5204           MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00050078  D203 3048 1004       00000048  00000004  5205           MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
0005007E  D203 3050 1008       00000050  00000008  5206           MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
00050084  D203 3058 100C       00000058  0000000C  5207           MVC   88(4,R3),12(R1)     copy to PL OP4-Compare
                                                   5208
0005008A  D21F 4000 2080       00000000  00000080  5209           MVC   0(32,R4),128(R2)    load initial Op1-Op4 data in ???
00050090  D203 3060 4000       00000060  00000000  5210           MVC   96(4,R3),0(R4)      copy to PL OP1-Replacement
00050096  D203 3068 4004       00000068  00000004  5211           MVC   104(4,R3),4(R4)     copy to PL OP2-Replacement
0005009C  D203 3070 4008       00000070  00000008  5212           MVC   112(4,R3),8(R4)     copy to PL OP3-Replacement
000500A2  D203 3078 400C       00000078  0000000C  5213           MVC   120(4,R3),12(R4)    copy to PL OP4-Replacement
                                                   5214 ****
                                                   5215 * force a failure by changing the 2nd operand
                                                   5216 ****
000500A8  D203 1004 4000       00000004  00000000  5217           MVC   4(4,R1),0(R4)
                                                   5218 ****
                                                   5219
000500AE  E313 0000 0024                00000000   5220           STG   R1,0(R3)            load OP1 address in PL
                                                   5221
000500B4  4111 0004                     00000004   5222           LA    R1,4(R1)
000500B8  E313 0008 0024                00000008   5223           STG   R1,8(R3)            load OP2 address in PL
                                                   5224
000500BE  4111 0004                     00000004   5225           LA    R1,4(R1)
000500C2  E313 0010 0024                00000010   5226           STG   R1,16(R3)           load OP3 address in PL
                                                   5227
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000500C8  4111 0004                    00000004  5228          LA    R1,4(R1)
000500CC  E313 0018 0024               00000018  5229          STG   R1,24(R3)           load OP4 address in PL
                                                 5230
000500D2  B982 0000                              5231          XGR   R0,R0
000500D6  4100 0022                    00000022  5232          LA    R0,34
000500DA  EE00 0000 3000  00000000     00000000  5233          PLO   0,0,0,0(R3)         Quadruple C/S (QCS: 34)
                                                 5234
000500E0  B98D 0020                              5235          EPSW  R2,R0               extract psw
000500E4  5020 5010                    00000010  5236          ST    R2,CCPSW              to save CC
000500E8  E310 500B 0076               0000000B  5237          LB    R1,CCMASK           (failure CC mask)
000500EE  8910 0004                    00000004  5238          SLL   R1,4                (shift to BC instr CC position)
000500F2  4410 5130                    00050150  5239          EX    R1,F342TSTCC            fail if...
                                                 5240
                          000500F6     00000001  5241 F342CKUR EQU   *
000500F6  D200 5009 3030  00000009     00000030  5242          MVC   URFOUND,48(R3)      Extract UR
000500FC  D500 5008 3030  00000008     00000030  5243          CLC   UR,48(R3)
00050102  4780 50EA                    0005010A  5244          BE    F342CKRES
00050106  45F0 8220                    00000420  5245          BAL   R15,URMSG
                                                 5246 *
                                                 5247 * compare results
                                                 5248 *
                          0005010A     00000001  5249 F342CKRES EQU   *
                                                 5250 *                                  reset adresses
0005010A  B982 0011                              5251          XGR   R1,R1
0005010E  5810 501C                    0000001C  5252          L     R1,OPSRCA           source
00050112  B982 0033                              5253          XGR   R3,R3
00050116  5830 5024                    00000024  5254          L     R3,PLA              PL
0005011A  B982 0044                              5255          XGR   R4,R4
0005011E  5840 5020                    00000020  5256          L     R4,OPREPA
                                                 5257
00050122  D503 1000 3040  00000000     00000040  5258          CLC   0(4,R1),64(R3)
00050128  4770 512A                    0005014A  5259          BNE   F342XX              OP1C Different, error
0005012C  D503 1004 3048  00000004     00000048  5260          CLC   4(4,R1),72(R3)
00050132  4770 512A                    0005014A  5261          BNE   F342XX              OP2C Different, error
00050136  D503 1008 3050  00000008     00000050  5262          CLC   8(4,R1),80(R3)
0005013C  4770 512A                    0005014A  5263          BNE   F342XX              OP3C Different, error
00050140  D503 100C 3058  0000000C     00000058  5264          CLC   12(4,R1),88(R3)
00050146  478B 0000                    00000000  5265          BE    0(R11)              OP4Csame, return
                                                 5266
                          0005014A     00000001  5267 F342XX   EQU   *
0005014A  45F0 8340                    00000540  5268          BAL   R15,FAILMSG
0005014E  07FB                                   5269          BR    R11
                                                 5270
00050150  4700 5134                    00050154  5271 F342TSTCC BC    0,F342CCMSG         fail if unexpected condition code
                          00050154     00000001  5272 F342CCMSG EQU   *
00050154  45F0 80F2                    000002F2  5273          BAL   R15,CCMSG
00050158  47F0 50D6                    000500F6  5274          B     F342CKUR
                                                 5275
                                                 5276          TSTDATA
0005015C                  0005015C     00050400  5277+          ORG   PLOETST+(200*K)+(30*4*K)+K
                                                 5278          TSTLBL OPDST              Operand Destination
00050400                                         5279+OPDST68  DS    0C
00050400  00000000 00000000                      5280          DC    XL256'0'
00050408  00000000 00000000
00050410  00000000 00000000
00050418  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00050420  00000000 00000000
00050428  00000000 00000000
00050430  00000000 00000000
00050438  00000000 00000000
00050440  00000000 00000000
00050448  00000000 00000000
00050450  00000000 00000000
00050458  00000000 00000000
00050460  00000000 00000000
00050468  00000000 00000000
00050470  00000000 00000000
00050478  00000000 00000000
00050480  00000000 00000000
00050488  00000000 00000000
00050490  00000000 00000000
00050498  00000000 00000000
000504A0  00000000 00000000
000504A8  00000000 00000000
000504B0  00000000 00000000
000504B8  00000000 00000000
000504C0  00000000 00000000
000504C8  00000000 00000000
000504D0  00000000 00000000
000504D8  00000000 00000000
000504E0  00000000 00000000
000504E8  00000000 00000000
000504F0  00000000 00000000
000504F8  00000000 00000000
                                                  5281
                                                  5282           TSTLBL OPSRC              Operand Source OP1-OP4
00050500                                          5283+OPSRC68   DS    0C
00050500  00000000 00000000                       5284           DC    XL256'0'
00050508  00000000 00000000
00050510  00000000 00000000
00050518  00000000 00000000
00050520  00000000 00000000
00050528  00000000 00000000
00050530  00000000 00000000
00050538  00000000 00000000
00050540  00000000 00000000
00050548  00000000 00000000
00050550  00000000 00000000
00050558  00000000 00000000
00050560  00000000 00000000
00050568  00000000 00000000
00050570  00000000 00000000
00050578  00000000 00000000
00050580  00000000 00000000
00050588  00000000 00000000
00050590  00000000 00000000
00050598  00000000 00000000
000505A0  00000000 00000000
000505A8  00000000 00000000
000505B0  00000000 00000000
000505B8  00000000 00000000
000505C0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000505C8  00000000 00000000
000505D0  00000000 00000000
000505D8  00000000 00000000
000505E0  00000000 00000000
000505E8  00000000 00000000
000505F0  00000000 00000000
000505F8  00000000 00000000
                                          5285
                                          5286           TSTLBL PL                  Parameter List
00050600                                  5287+PL68      DS    0C
00050600  00000000 00000000               5288           DC    XL352'00'
00050608  00000000 00000000
00050610  00000000 00000000
00050618  00000000 00000000
00050620  00000000 00000000
00050628  00000000 00000000
00050630  00000000 00000000
00050638  00000000 00000000
00050640  00000000 00000000
00050648  00000000 00000000
00050650  00000000 00000000
00050658  00000000 00000000
00050660  00000000 00000000
00050668  00000000 00000000
00050670  00000000 00000000
00050678  00000000 00000000
00050680  00000000 00000000
00050688  00000000 00000000
00050690  00000000 00000000
00050698  00000000 00000000
000506A0  00000000 00000000
000506A8  00000000 00000000
000506B0  00000000 00000000
000506B8  00000000 00000000
000506C0  00000000 00000000
000506C8  00000000 00000000
000506D0  00000000 00000000
000506D8  00000000 00000000
000506E0  00000000 00000000
000506E8  00000000 00000000
000506F0  00000000 00000000
000506F8  00000000 00000000
00050700  00000000 00000000
00050708  00000000 00000000
00050710  00000000 00000000
00050718  00000000 00000000
00050720  00000000 00000000
00050728  00000000 00000000
00050730  00000000 00000000
00050738  00000000 00000000
00050740  00000000 00000000
00050748  00000000 00000000
00050750  00000000 00000000
00050758  00000000 00000000
                                          5289
                                          5290           TSTLBL FMSG                fail message
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00050760                                           5291+FMSG68   DS    0C
00050760  86818993 858440D7                        5292          DC    C'failed PLO-Extension '
00050768  D3D660C5 A7A38595
00050770  A2899695 40
00050775  D8A48184 99A49793                        5293          DC    CL96'Quadruple Compare and Swap (QCS: 34): 2nd NE'
0005077D  8540C396 94978199
00050785  85408195 8440E2A6
0005078D  8197404D D8C3E27A
00050795  40F3F45D 7A40F295
0005079D  8440D5C5 40404040
000507A5  40404040 40404040
000507AD  40404040 40404040
000507B5  40404040 40404040
000507BD  40404040 40404040
000507C5  40404040 40404040
000507CD  40404040 40404040
                                                   5294
                                                   5295          TSTEND
000507D6                                           5296+         DS    0H
000507D6  C5958440 968640A3                        5297+         DC    CL32'End of test: 68'
000507DE  85A2A37A 40F6F840
000507E6  40404040 40404040
000507EE  40404040 40404040
                                                   5298
                                                   5299 * ------------------------------------------------------------------
                                                   5300 * Test Quadruple Compare and Swap (QCS: 34) instruction : 3rd NE
                                                   5301 * ------------------------------------------------------------------
                                                   5302          TSTHDR 2,3,34             3rd Operand NOT Equal (cc=2, UR=3)
000507F6                        000507F6  00051000 5303+         ORG   PLOETST+(200*K)+(31*4*K)
00051000  E2E3C1D9 E3409686                        5304+         DC    CL32'START of test: 69'
00051008  40A385A2 A37A40F6
00051010  F9404040 40404040
00051018  40404040 40404040
00051020                        00051020          5305+         USING *,R5                base for test data and test routine
00051020                        00000000          5306+         USING PLOETEST,R5
00051020  0005104C                                 5307+T69      DC    A(CODE69)           address of test code
00051024  0045                                     5308+         DC    H'69'               test number
00051026  00                                       5309+         DC    X'00'
00051027  22                                       5310+         DC    HL1'34'             Function
00051028  03                                       5311+         DC    HL1'3'              expected unequal results (UR)
00051029  00                                       5312+         DC    HL1'00'             found UR
0005102A  02                                       5313+         DC    HL1'2'              cc
0005102B  0D                                       5314+         DC    HL1'13'             not expected CC mask
0005102C  00                                       5315+         DS    X                   extracted cc
00051030  00000000 00000000                        5316+         DS    2F                  extract PSW after test (has CC)
00051038  00051760                                 5317+         DC    A(FMSG69)           address of test fail message
0005103C  00051500                                 5318+         DC    A(OPSRC69)          address of operand source
00051040  00051400                                 5319+         DC    A(OPDST69)          address of operand destination
00051044  00051600                                 5320+         DC    A(PL69)             address of parameter list
00051048  000011F6                                 5321+         DC    A(RANDX+69)         address of random source data
                                                   5322
                                                   5323          TSTCODE
0005104C                                           5324+CODE69   DS    0F
                                                   5325 *                                  load random source
0005104C  B982 0022                                5326          XGR   R2,R2
00051050  5820 5028                     00000028   5327          L     R2,RANDXA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   5328 *
                                                   5329 * Quadruple Compare and Swap (QCS: 34)
                                                   5330 * Build Parambter List
                                                   5331 *
00051054  B982 0011                                5332          XGR   R1,R1
00051058  5810 501C                0000001C        5333          L     R1,OPSRCA
                                                   5334
0005105C  B982 0033                                5335          XGR   R3,R3
00051060  5830 5024                00000024        5336          L     R3,PLA
                                                   5337
00051064  B982 0044                                5338          XGR   R4,R4
00051068  5840 5020                00000020        5339          L     R4,OPREPA
                                                   5340
0005106C  D20F 1000 2000      00000000  00000000   5341          MVC   0(16,R1),0(R2)      load Op1-Op4 data in source
00051072  D203 3040 1000      00000040  00000000   5342          MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00051078  D203 3048 1004      00000048  00000004   5343          MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
0005107E  D203 3050 1008      00000050  00000008   5344          MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
00051084  D203 3058 100C      00000058  0000000C   5345          MVC   88(4,R3),12(R1)     copy to PL OP4-Compare
                                                   5346
0005108A  D21F 4000 2080      00000000  00000080   5347          MVC   0(32,R4),128(R2)    load initial Op1-Op4 data in ???
00051090  D203 3060 4000      00000060  00000000   5348          MVC   96(4,R3),0(R4)      copy to PL OP1-Replacement
00051096  D203 3068 4004      00000068  00000004   5349          MVC   104(4,R3),4(R4)     copy to PL OP2-Replacement
0005109C  D203 3070 4008      00000070  00000008   5350          MVC   112(4,R3),8(R4)     copy to PL OP3-Replacement
000510A2  D203 3078 400C      00000078  0000000C   5351          MVC   120(4,R3),12(R4)    copy to PL OP4-Replacement
                                                   5352 ****
                                                   5353 * force a failure by changing the 3rd operand
                                                   5354 ****
000510A8  D203 1008 4000      00000008  00000000   5355          MVC   8(4,R1),0(R4)
                                                   5356 ****
                                                   5357
000510AE  E313 0000 0024                00000000   5358          STG   R1,0(R3)            load OP1 address in PL
                                                   5359
000510B4  4111 0004                     00000004   5360          LA    R1,4(R1)
000510B8  E313 0008 0024                00000008   5361          STG   R1,8(R3)            load OP2 address in PL
                                                   5362
000510BE  4111 0004                     00000004   5363          LA    R1,4(R1)
000510C2  E313 0010 0024                00000010   5364          STG   R1,16(R3)           load OP3 address in PL
                                                   5365
000510C8  4111 0004                     00000004   5366          LA    R1,4(R1)
000510CC  E313 0018 0024                00000018   5367          STG   R1,24(R3)           load OP4 address in PL
                                                   5368
000510D2  B982 0000                                5369          XGR   R0,R0
000510D6  4100 0022                     00000022   5370          LA    R0,34
000510DA  EE00 0000 3000      00000000  00000000   5371          PLO   0,0,0,0(R3)         Quadruple C/S (QCS: 34)
                                                   5372
000510E0  B98D 0020                                5373          EPSW  R2,R0               extract psw
000510E4  5020 5010                     00000010   5374          ST    R2,CCPSW              to save CC
000510E8  E310 500B 0076                0000000B   5375          LB    R1,CCMASK           (failure CC mask)
000510EE  8910 0004                     00000004   5376          SLL   R1,4                (shift to BC instr CC position)
000510F2  4410 5130                     00051150   5377          EX    R1,F343TSTCC            fail if...
                                                   5378
                                000510F6  00000001   5379 F343CKUR EQU   *
000510F6  D200 5009 3030      00000009  00000030   5380          MVC   URFOUND,48(R3)      Extract UR
000510FC  D500 5008 3030      00000008  00000030   5381          CLC   UR,48(R3)
00051102  4780 50EA                     0005110A   5382          BE    F343CKRES
00051106  45F0 8220                     00000420   5383          BAL   R15,URMSG
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5384 *
                                                    5385 * compare results
                                                    5386 *
                               0005110A  00000001  5387 F343CKRES EQU   *
                                                    5388 *                          reset adresses
0005110A  B982 0011                                 5389          XGR   R1,R1
0005110E  5810 501C                 0000001C        5390          L     R1,OPSRCA           source
00051112  B982 0033                                 5391          XGR   R3,R3
00051116  5830 5024                 00000024        5392          L     R3,PLA              PL
0005111A  B982 0044                                 5393          XGR   R4,R4
0005111E  5840 5020                 00000020        5394          L     R4,OPREPA
                                                    5395
00051122  D503 1000 3040  00000000  00000040        5396          CLC   0(4,R1),64(R3)
00051128  4770 512A                 0005114A        5397          BNE   F343XX              OP1C Different, error
0005112C  D503 1004 3048  00000004  00000048        5398          CLC   4(4,R1),72(R3)
00051132  4770 512A                 0005114A        5399          BNE   F343XX              OP2C Different, error
00051136  D503 1008 3050  00000008  00000050        5400          CLC   8(4,R1),80(R3)
0005113C  4770 512A                 0005114A        5401          BNE   F343XX              OP3C Different, error
00051140  D503 100C 3058  0000000C  00000058        5402          CLC   12(4,R1),88(R3)
00051146  478B 0000                 00000000        5403          BE    0(R11)              OP4Csame, return
                                                    5404
                               0005114A  00000001  5405 F343XX   EQU   *
0005114A  45F0 8340                 00000540        5406          BAL   R15,FAILMSG
0005114E  07FB                                      5407          BR    R11
                                                    5408
00051150  4700 5134                 00051154        5409 F343TSTCC BC    0,F343CCMSG         fail if unexpected condition code
                               00051154  00000001  5410 F343CCMSG EQU   *
00051154  45F0 80F2                 000002F2        5411          BAL   R15,CCMSG
00051158  47F0 50D6                 000510F6        5412          B     F343CKUR
                                                    5413
                                                    5414          TSTDATA
0005115C                    0005115C  00051400      5415+         ORG   PLOETST+(200*K)+(31*4*K)+K
                                                    5416          TSTLBL OPDST              Operand Destination
00051400                                            5417+OPDST69  DS    0C
00051400  00000000 00000000                         5418          DC    XL256'0'
00051408  00000000 00000000
00051410  00000000 00000000
00051418  00000000 00000000
00051420  00000000 00000000
00051428  00000000 00000000
00051430  00000000 00000000
00051438  00000000 00000000
00051440  00000000 00000000
00051448  00000000 00000000
00051450  00000000 00000000
00051458  00000000 00000000
00051460  00000000 00000000
00051468  00000000 00000000
00051470  00000000 00000000
00051478  00000000 00000000
00051480  00000000 00000000
00051488  00000000 00000000
00051490  00000000 00000000
00051498  00000000 00000000
000514A0  00000000 00000000
000514A8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000514B0  00000000 00000000
000514B8  00000000 00000000
000514C0  00000000 00000000
000514C8  00000000 00000000
000514D0  00000000 00000000
000514D8  00000000 00000000
000514E0  00000000 00000000
000514E8  00000000 00000000
000514F0  00000000 00000000
000514F8  00000000 00000000
                                          5419
                                          5420           TSTLBL OPSRC              Operand Source OP1-OP4
00051500                                  5421+OPSRC69   DS    0C
00051500  00000000 00000000               5422           DC    XL256'0'
00051508  00000000 00000000
00051510  00000000 00000000
00051518  00000000 00000000
00051520  00000000 00000000
00051528  00000000 00000000
00051530  00000000 00000000
00051538  00000000 00000000
00051540  00000000 00000000
00051548  00000000 00000000
00051550  00000000 00000000
00051558  00000000 00000000
00051560  00000000 00000000
00051568  00000000 00000000
00051570  00000000 00000000
00051578  00000000 00000000
00051580  00000000 00000000
00051588  00000000 00000000
00051590  00000000 00000000
00051598  00000000 00000000
000515A0  00000000 00000000
000515A8  00000000 00000000
000515B0  00000000 00000000
000515B8  00000000 00000000
000515C0  00000000 00000000
000515C8  00000000 00000000
000515D0  00000000 00000000
000515D8  00000000 00000000
000515E0  00000000 00000000
000515E8  00000000 00000000
000515F0  00000000 00000000
000515F8  00000000 00000000
                                          5423
                                          5424           TSTLBL PL                 Parameter List
00051600                                  5425+PL69      DS    0C
00051600  00000000 00000000               5426           DC    XL352'00'
00051608  00000000 00000000
00051610  00000000 00000000
00051618  00000000 00000000
00051620  00000000 00000000
00051628  00000000 00000000
00051630  00000000 00000000
00051638  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00051640  00000000 00000000
00051648  00000000 00000000
00051650  00000000 00000000
00051658  00000000 00000000
00051660  00000000 00000000
00051668  00000000 00000000
00051670  00000000 00000000
00051678  00000000 00000000
00051680  00000000 00000000
00051688  00000000 00000000
00051690  00000000 00000000
00051698  00000000 00000000
000516A0  00000000 00000000
000516A8  00000000 00000000
000516B0  00000000 00000000
000516B8  00000000 00000000
000516C0  00000000 00000000
000516C8  00000000 00000000
000516D0  00000000 00000000
000516D8  00000000 00000000
000516E0  00000000 00000000
000516E8  00000000 00000000
000516F0  00000000 00000000
000516F8  00000000 00000000
00051700  00000000 00000000
00051708  00000000 00000000
00051710  00000000 00000000
00051718  00000000 00000000
00051720  00000000 00000000
00051728  00000000 00000000
00051730  00000000 00000000
00051738  00000000 00000000
00051740  00000000 00000000
00051748  00000000 00000000
00051750  00000000 00000000
00051758  00000000 00000000
                                      5427
                                      5428          TSTLBL FMSG               fail message
00051760                              5429+FMSG69   DS    0C
00051760  86818993 858440D7          5430          DC    C'failed PLO-Extension '
00051768  D3D660C5 A7A38595
00051770  A2899695 40
00051775  D8A48184 99A49793          5431          DC    CL96'Quadruple Compare and Swap (QCS: 34): 3rd NE'
0005177D  8540C396 94978199
00051785  85408195 8440E2A6
0005178D  8197404D D8C3E27A
00051795  40F3F45D 7A40F399
0005179D  8440D5C5 40404040
000517A5  40404040 40404040
000517AD  40404040 40404040
000517B5  40404040 40404040
000517BD  40404040 40404040
000517C5  40404040 40404040
000517CD  40404040 40404040
                                      5432
                                      5433          TSTEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000517D6                                          5434+         DS    0H
000517D6  C5958440 968640A3                       5435+         DC    CL32'End of test: 69'
000517DE  85A2A37A 40F6F940
000517E6  40404040 40404040
000517EE  40404040 40404040
                                                  5436
                                                  5437 * ------------------------------------------------------------------
                                                  5438 * Test Quadruple Compare and Swap (QCS: 34) instruction : 4th NE
                                                  5439 * ------------------------------------------------------------------
                                                  5440          TSTHDR 2,4,34             4th Operand NOT Equal (cc=2, UR=4)
000517F6                    000517F6  00052000    5441+         ORG   PLOETST+(200*K)+(32*4*K)
00052000  E2E3C1D9 E3409686                       5442+         DC    CL32'START of test: 70'
00052008  40A385A2 A37A40F7
00052010  F0404040 40404040
00052018  40404040 40404040
00052020                    00052020              5443+         USING *,R5                base for test data and test routine
00052020                    00000000              5444+         USING PLOETEST,R5
00052020  0005204C                                5445+T70      DC    A(CODE70)           address of test code
00052024  0046                                    5446+         DC    H'70'               test number
00052026  00                                      5447+         DC    X'00'
00052027  22                                      5448+         DC    HL1'34'             Function
00052028  04                                      5449+         DC    HL1'4'              expected unequal results (UR)
00052029  00                                      5450+         DC    HL1'00'             found UR
0005202A  02                                      5451+         DC    HL1'2'              cc
0005202B  0D                                      5452+         DC    HL1'13'             not expected CC mask
0005202C  00                                      5453+         DS    X                   extracted cc
00052030  00000000 00000000                       5454+         DS    2F                  extract PSW after test (has CC)
00052038  00052760                                5455+         DC    A(FMSG70)           address of test fail message
0005203C  00052500                                5456+         DC    A(OPSRC70)          address of operand source
00052040  00052400                                5457+         DC    A(OPDST70)          address of operand destination
00052044  00052600                                5458+         DC    A(PL70)             address of parameter list
00052048  000011F7                                5459+         DC    A(RANDX+70)         address of random source data
                                                  5460
                                                  5461          TSTCODE
0005204C                                          5462+CODE70   DS    0F
                                                  5463 *                                  load random source
0005204C  B982 0022                               5464          XGR   R2,R2
00052050  5820 5028                 00000028      5465          L     R2,RANDXA
                                                  5466 *
                                                  5467 * Quadruple Compare and Swap (QCS: 34)
                                                  5468 * Build Parambter List
                                                  5469 *
00052054  B982 0011                               5470          XGR   R1,R1
00052058  5810 501C                 0000001C      5471          L     R1,OPSRCA
                                                  5472
0005205C  B982 0033                               5473          XGR   R3,R3
00052060  5830 5024                 00000024      5474          L     R3,PLA
                                                  5475
00052064  B982 0044                               5476          XGR   R4,R4
00052068  5840 5020                 00000020      5477          L     R4,OPREPA
                                                  5478
0005206C  D20F 1000 2000      00000000  00000000  5479          MVC   0(16,R1),0(R2)      load Op1-Op4 data in source
00052072  D203 3040 1000      00000040  00000000  5480          MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00052078  D203 3048 1004      00000048  00000004  5481          MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
0005207E  D203 3050 1008      00000050  00000008  5482          MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
00052084  D203 3058 100C      00000058  0000000C  5483          MVC   88(4,R3),12(R1)     copy to PL OP4-Compare
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5484
0005208A  D21F 4000 2080      00000000  00000080  5485          MVC   0(32,R4),128(R2)    load initial Op1-Op4 data in ???
00052090  D203 3060 4000      00000060  00000000  5486          MVC   96(4,R3),0(R4)      copy to PL OP1-Replacement
00052096  D203 3068 4004      00000068  00000004  5487          MVC   104(4,R3),4(R4)     copy to PL OP2-Replacement
0005209C  D203 3070 4008      00000070  00000008  5488          MVC   112(4,R3),8(R4)     copy to PL OP3-Replacement
000520A2  D203 3078 400C      00000078  0000000C  5489          MVC   120(4,R3),12(R4)    copy to PL OP4-Replacement
                                                    5490 ****
                                                    5491 * force a failure by changing the 4th operand
                                                    5492 ****
000520A8  D203 100C 4000      0000000C  00000000  5493          MVC   12(4,R1),0(R4)
                                                    5494 ****
                                                    5495
000520AE  E313 0000 0024                00000000  5496          STG   R1,0(R3)            load OP1 address in PL
                                                    5497
000520B4  4111 0004                     00000004  5498          LA    R1,4(R1)
000520B8  E313 0008 0024                00000008  5499          STG   R1,8(R3)            load OP2 address in PL
                                                    5500
000520BE  4111 0004                     00000004  5501          LA    R1,4(R1)
000520C2  E313 0010 0024                00000010  5502          STG   R1,16(R3)           load OP3 address in PL
                                                    5503
000520C8  4111 0004                     00000004  5504          LA    R1,4(R1)
000520CC  E313 0018 0024                00000018  5505          STG   R1,24(R3)           load OP4 address in PL
                                                    5506
000520D2  B982 0000                               5507          XGR   R0,R0
000520D6  4100 0022                     00000022  5508          LA    R0,34
000520DA  EE00 0000 3000      00000000  00000000  5509          PLO   0,0,0,0(R3)         Quadruple C/S (QCS: 34)
                                                    5510
000520E0  B98D 0020                               5511          EPSW  R2,R0               extract psw
000520E4  5020 5010                     00000010  5512          ST    R2,CCPSW              to save CC
000520E8  E310 500B 0076                0000000B  5513          LB    R1,CCMASK           (failure CC mask)
000520EE  8910 0004                     00000004  5514          SLL   R1,4                (shift to BC instr CC position)
000520F2  4410 5130                     00052150  5515          EX    R1,F344TSTCC            fail if...
                                                    5516
                                000520F6  00000001  5517 F344CKUR EQU   *
000520F6  D200 5009 3030      00000009  00000030  5518          MVC   URFOUND,48(R3)      Extract UR
000520FC  D500 5008 3030      00000008  00000030  5519          CLC   UR,48(R3)
00052102  4780 50EA                     0005210A  5520          BE    F344CKRES
00052106  45F0 8220                     00000420  5521          BAL   R15,URMSG
                                                    5522 *
                                                    5523 * compare results
                                                    5524 *
                                0005210A  00000001  5525 F344CKRES EQU   *
                                                    5526 *                              reset adresses
0005210A  B982 0011                               5527          XGR   R1,R1
0005210E  5810 501C                     0000001C  5528          L     R1,OPSRCA           source
00052112  B982 0033                               5529          XGR   R3,R3
00052116  5830 5024                     00000024  5530          L     R3,PLA              PL
0005211A  B982 0044                               5531          XGR   R4,R4
0005211E  5840 5020                     00000020  5532          L     R4,OPREPA
                                                    5533
00052122  D503 1000 3040      00000000  00000040  5534          CLC   0(4,R1),64(R3)
00052128  4770 512A                     0005214A  5535          BNE   F344XX              OP1C Different, error
0005212C  D503 1004 3048      00000004  00000048  5536          CLC   4(4,R1),72(R3)
00052132  4770 512A                     0005214A  5537          BNE   F344XX              OP2C Different, error
00052136  D503 1008 3050      00000008  00000050  5538          CLC   8(4,R1),80(R3)
0005213C  4770 512A                     0005214A  5539          BNE   F344XX              OP3C Different, error
```

```
  LOC         OBJECT CODE        ADDR1     ADDR2     STMT
00052140  D503 100C 3058       0000000C  00000058  5540           CLC   12(4,R1),88(R3)
00052146  478B 0000                        00000000  5541           BE    0(R11)               OP4Csame, return
                                                     5542
                                0005214A  00000001  5543 F344XX    EQU   *
0005214A  45F0 8340                        00000540  5544           BAL   R15,FAILMSG
0005214E  07FB                             00000000  5545           BR    R11
                                                     5546
00052150  4700 5134                        00052154  5547 F344TSTCC BC     0,F344CCMSG          fail if unexpected condition code
                                00052154  00000001  5548 F344CCMSG EQU   *
00052154  45F0 80F2                        000002F2  5549           BAL   R15,CCMSG
00052158  47F0 50D6                        000520F6  5550           B     F344CKUR
                                                     5551
                                                     5552           TSTDATA
0005215C                        0005215C  00052400  5553+          ORG   PLOETST+(200*K)+(32*4*K)+K
                                                     5554           TSTLBL OPDST               Operand Destination
00052400                                             5555+OPDST70   DS    0C
00052400  00000000 00000000                          5556           DC    XL256'0'
00052408  00000000 00000000
00052410  00000000 00000000
00052418  00000000 00000000
00052420  00000000 00000000
00052428  00000000 00000000
00052430  00000000 00000000
00052438  00000000 00000000
00052440  00000000 00000000
00052448  00000000 00000000
00052450  00000000 00000000
00052458  00000000 00000000
00052460  00000000 00000000
00052468  00000000 00000000
00052470  00000000 00000000
00052478  00000000 00000000
00052480  00000000 00000000
00052488  00000000 00000000
00052490  00000000 00000000
00052498  00000000 00000000
000524A0  00000000 00000000
000524A8  00000000 00000000
000524B0  00000000 00000000
000524B8  00000000 00000000
000524C0  00000000 00000000
000524C8  00000000 00000000
000524D0  00000000 00000000
000524D8  00000000 00000000
000524E0  00000000 00000000
000524E8  00000000 00000000
000524F0  00000000 00000000
000524F8  00000000 00000000
                                                     5557
                                                     5558           TSTLBL OPSRC               Operand Source OP1-OP4
00052500                                             5559+OPSRC70   DS    0C
00052500  00000000 00000000                          5560           DC    XL256'0'
00052508  00000000 00000000
00052510  00000000 00000000
00052518  00000000 00000000
00052520  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00052528  00000000 00000000
00052530  00000000 00000000
00052538  00000000 00000000
00052540  00000000 00000000
00052548  00000000 00000000
00052550  00000000 00000000
00052558  00000000 00000000
00052560  00000000 00000000
00052568  00000000 00000000
00052570  00000000 00000000
00052578  00000000 00000000
00052580  00000000 00000000
00052588  00000000 00000000
00052590  00000000 00000000
00052598  00000000 00000000
000525A0  00000000 00000000
000525A8  00000000 00000000
000525B0  00000000 00000000
000525B8  00000000 00000000
000525C0  00000000 00000000
000525C8  00000000 00000000
000525D0  00000000 00000000
000525D8  00000000 00000000
000525E0  00000000 00000000
000525E8  00000000 00000000
000525F0  00000000 00000000
000525F8  00000000 00000000
                                                5561
                                                5562           TSTLBL PL                  Parameter List
00052600                                        5563+PL70      DS    0C
00052600  00000000 00000000                     5564           DC    XL352'00'
00052608  00000000 00000000
00052610  00000000 00000000
00052618  00000000 00000000
00052620  00000000 00000000
00052628  00000000 00000000
00052630  00000000 00000000
00052638  00000000 00000000
00052640  00000000 00000000
00052648  00000000 00000000
00052650  00000000 00000000
00052658  00000000 00000000
00052660  00000000 00000000
00052668  00000000 00000000
00052670  00000000 00000000
00052678  00000000 00000000
00052680  00000000 00000000
00052688  00000000 00000000
00052690  00000000 00000000
00052698  00000000 00000000
000526A0  00000000 00000000
000526A8  00000000 00000000
000526B0  00000000 00000000
000526B8  00000000 00000000
000526C0  00000000 00000000
000526C8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000526D0  00000000 00000000
000526D8  00000000 00000000
000526E0  00000000 00000000
000526E8  00000000 00000000
000526F0  00000000 00000000
000526F8  00000000 00000000
00052700  00000000 00000000
00052708  00000000 00000000
00052710  00000000 00000000
00052718  00000000 00000000
00052720  00000000 00000000
00052728  00000000 00000000
00052730  00000000 00000000
00052738  00000000 00000000
00052740  00000000 00000000
00052748  00000000 00000000
00052750  00000000 00000000
00052758  00000000 00000000
                                          5565
                                          5566           TSTLBL FMSG               fail message
00052760                                  5567+FMSG70    DS    0C
00052760  86818993 858440D7               5568           DC    C'failed PLO-Extension '
00052768  D3D660C5 A7A38595
00052770  A2899695 40
00052775  D8A48184 99A49793               5569           DC    CL96'Quadruple Compare and Swap (QCS: 34): 4th NE'
0005277D  8540C396 94978199
00052785  85408195 8440E2A6
0005278D  8197404D D8C3E27A
00052795  40F3F45D 7A40F4A3
0005279D  8840D5C5 40404040
000527A5  40404040 40404040
000527AD  40404040 40404040
000527B5  40404040 40404040
000527BD  40404040 40404040
000527C5  40404040 40404040
000527CD  40404040 40404040
                                       5570
                                       5571           TSTEND
000527D6                               5572+          DS    0H
000527D6  C5958440 968640A3            5573+          DC    CL32'End of test: 70'
000527DE  85A2A37A 40F7F040
000527E6  40404040 40404040
000527EE  40404040 40404040
                                       5574
                                       5575
                                       5576 * ------------------------------------------------------------------
                                       5577 * Test Quadruple Compare and Swap (QCS: 34) instruction : All EQ
                                       5578 * ------------------------------------------------------------------
                                       5579           TSTHDR 0,99,34             All EQ (cc=0, UR=99)
000527F6                 000527F6  00053000  5580+          ORG   PLOETST+(200*K)+(33*4*K)
00053000  E2E3C1D9 E3409686            5581+          DC    CL32'START of test: 71'
00053008  40A385A2 A37A40F7
00053010  F1404040 40404040
00053018  40404040 40404040
00053020                 00053020           5582+          USING *,R5                base for test data and test routine
00053020                 00000000           5583+          USING PLOETEST,R5
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00053020  0005304C                                5584+T71      DC    A(CODE71)           address of test code
00053024  0047                                    5585+         DC    H'71'               test number
00053026  00                                      5586+         DC    X'00'
00053027  22                                      5587+         DC    HL1'34'             Function
00053028  63                                      5588+         DC    HL1'99'             expected unequal results (UR)
00053029  00                                      5589+         DC    HL1'00'             found UR
0005302A  00                                      5590+         DC    HL1'0'              cc
0005302B  07                                      5591+         DC    HL1'7'              not expected CC mask
0005302C  00                                      5592+         DS    X                   extracted cc
00053030  00000000 00000000                       5593+         DS    2F                  extract PSW after test (has CC)
00053038  00053760                                5594+         DC    A(FMSG71)           address of test fail message
0005303C  00053500                                5595+         DC    A(OPSRC71)          address of operand source
00053040  00053400                                5596+         DC    A(OPDST71)          address of operand destination
00053044  00053600                                5597+         DC    A(PL71)             address of parameter list
00053048  000011F8                                5598+         DC    A(RANDX+71)         address of random source data
                                                  5599
                                                  5600          TSTCODE
0005304C                                          5601+CODE71   DS    0F
                                                  5602 *                                  load random source
0005304C  B982 0022                               5603          XGR   R2,R2
00053050  5820 5028                   00000028    5604          L     R2,RANDXA
                                                  5605 *
                                                  5606 * Quadruple Compare and Swap (QCS: 34)
                                                  5607 * Build Parambter List
                                                  5608 *
00053054  B982 0011                               5609          XGR   R1,R1
00053058  5810 501C                   0000001C    5610          L     R1,OPSRCA
                                                  5611
0005305C  B982 0033                               5612          XGR   R3,R3
00053060  5830 5024                   00000024    5613          L     R3,PLA
                                                  5614
00053064  B982 0044                               5615          XGR   R4,R4
00053068  5840 5020                   00000020    5616          L     R4,OPREPA
                                                  5617
0005306C  D20F 1000 2000      00000000  00000000  5618          MVC   0(16,R1),0(R2)      load Op1-Op4 data in source
00053072  D203 3040 1000      00000040  00000000  5619          MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00053078  D203 3048 1004      00000048  00000004  5620          MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
0005307E  D203 3050 1008      00000050  00000008  5621          MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
00053084  D203 3058 100C      00000058  0000000C  5622          MVC   88(4,R3),12(R1)     copy to PL OP4-Compare
                                                  5623
0005308A  D21F 4000 2080      00000000  00000080  5624          MVC   0(32,R4),128(R2)    load initial Op1-Op4 data in ???
00053090  D203 3060 4000      00000060  00000000  5625          MVC   96(4,R3),0(R4)      copy to PL OP1-Replacement
00053096  D203 3068 4004      00000068  00000004  5626          MVC   104(4,R3),4(R4)     copy to PL OP2-Replacement
0005309C  D203 3070 4008      00000070  00000008  5627          MVC   112(4,R3),8(R4)     copy to PL OP3-Replacement
000530A2  D203 3078 400C      00000078  0000000C  5628          MVC   120(4,R3),12(R4)    copy to PL OP4-Replacement
                                                  5629 ****
                                                  5630 * ALL EQ: do not force a failure by changing an operand
                                                  5631 ****
                                                  5632 *         MVC   12(4,R1),0(R4)
                                                  5633 ****
                                                  5634
000530A8  E313 0000 0024                  00000000 5635         STG   R1,0(R3)            load OP1 address in PL
                                                  5636
000530AE  4111 0004                       00000004 5637         LA    R1,4(R1)
000530B2  E313 0008 0024                  00000008 5638         STG   R1,8(R3)            load OP2 address in PL
                                                  5639
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000530B8  4111 0004                    00000004  5640          LA    R1,4(R1)
000530BC  E313 0010 0024               00000010  5641          STG   R1,16(R3)          load OP3 address in PL
                                                 5642
000530C2  4111 0004                    00000004  5643          LA    R1,4(R1)
000530C6  E313 0018 0024               00000018  5644          STG   R1,24(R3)          load OP4 address in PL
                                                 5645
000530CC  B982 0000                              5646          XGR   R0,R0
000530D0  4100 0022                    00000022  5647          LA    R0,34
000530D4  EE00 0000 3000  00000000     00000000  5648          PLO   0,0,0,0(R3)        Quadruple C/S (QCS: 34)
                                                 5649
000530DA  B98D 0020                              5650          EPSW  R2,R0              extract psw
000530DE  5020 5010                    00000010  5651          ST    R2,CCPSW             to save CC
000530E2  E310 500B 0076               0000000B  5652          LB    R1,CCMASK          (failure CC mask)
000530E8  8910 0004                    00000004  5653          SLL   R1,4               (shift to BC instr CC position)
000530EC  4410 5116                    00053136  5654          EX    R1,F345TSTCC         fail if...
                                                 5655 *
                                                 5656 * compare results
                                                 5657 *
                          000530F0     00000001  5658 F345CKRES EQU   *
                                                 5659 *                                reset adresses
000530F0  B982 0011                              5660          XGR   R1,R1
000530F4  5810 501C                    0000001C  5661          L     R1,OPSRCA          source
000530F8  B982 0033                              5662          XGR   R3,R3
000530FC  5830 5024                    00000024  5663          L     R3,PLA             PL
00053100  B982 0044                              5664          XGR   R4,R4
00053104  5840 5020                    00000020  5665          L     R4,OPREPA
                                                 5666
                                                 5667 * All equal; replacement values copied to operand source
                                                 5668
00053108  D503 1000 3060  00000000     00000060  5669          CLC   0(4,R1),96(R3)
0005310E  4770 5110                    00053130  5670          BNE   F345XX             OP1R Different, error
00053112  D503 1004 3068  00000004     00000068  5671          CLC   4(4,R1),104(R3)
00053118  4770 5110                    00053130  5672          BNE   F345XX             OP2R Different, error
0005311C  D503 1008 3070  00000008     00000070  5673          CLC   8(4,R1),112(R3)
00053122  4770 5110                    00053130  5674          BNE   F345XX             OP3R Different, error
00053126  D503 100C 3078  0000000C     00000078  5675          CLC   12(4,R1),120(R3)
0005312C  478B 0000                    00000000  5676          BE    0(R11)             OP4R same, return
                                                 5677
                          00053130     00000001  5678 F345XX   EQU   *
00053130  45F0 8340                    00000540  5679          BAL   R15,FAILMSG
00053134  07FB                                   5680          BR    R11
                                                 5681
00053136  4700 511A                    0005313A  5682 F345TSTCC BC    0,F345CCMSG        fail if unexpected condition code
                          0005313A     00000001  5683 F345CCMSG EQU   *
0005313A  45F0 80F2                    000002F2  5684          BAL   R15,CCMSG
0005313E  47F0 50D0                    000530F0  5685          B     F345CKRES
                                                 5686
                                                 5687          TSTDATA
00053142                  00053142     00053400  5688+         ORG   PLOETST+(200*K)+(33*4*K)+K
                                                 5689          TSTLBL OPDST             Operand Destination
00053400                                         5690+OPDST71  DS    0C
00053400  00000000 00000000                      5691          DC    XL256'0'
00053408  00000000 00000000
00053410  00000000 00000000
00053418  00000000 00000000
00053420  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00053428  00000000 00000000
00053430  00000000 00000000
00053438  00000000 00000000
00053440  00000000 00000000
00053448  00000000 00000000
00053450  00000000 00000000
00053458  00000000 00000000
00053460  00000000 00000000
00053468  00000000 00000000
00053470  00000000 00000000
00053478  00000000 00000000
00053480  00000000 00000000
00053488  00000000 00000000
00053490  00000000 00000000
00053498  00000000 00000000
000534A0  00000000 00000000
000534A8  00000000 00000000
000534B0  00000000 00000000
000534B8  00000000 00000000
000534C0  00000000 00000000
000534C8  00000000 00000000
000534D0  00000000 00000000
000534D8  00000000 00000000
000534E0  00000000 00000000
000534E8  00000000 00000000
000534F0  00000000 00000000
000534F8  00000000 00000000
                                                5692
                                                5693           TSTLBL OPSRC              Operand Source OP1-OP4
00053500                                        5694+OPSRC71   DS    0C
00053500  00000000 00000000                     5695           DC    XL256'0'
00053508  00000000 00000000
00053510  00000000 00000000
00053518  00000000 00000000
00053520  00000000 00000000
00053528  00000000 00000000
00053530  00000000 00000000
00053538  00000000 00000000
00053540  00000000 00000000
00053548  00000000 00000000
00053550  00000000 00000000
00053558  00000000 00000000
00053560  00000000 00000000
00053568  00000000 00000000
00053570  00000000 00000000
00053578  00000000 00000000
00053580  00000000 00000000
00053588  00000000 00000000
00053590  00000000 00000000
00053598  00000000 00000000
000535A0  00000000 00000000
000535A8  00000000 00000000
000535B0  00000000 00000000
000535B8  00000000 00000000
000535C0  00000000 00000000
000535C8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000535D0  00000000 00000000
000535D8  00000000 00000000
000535E0  00000000 00000000
000535E8  00000000 00000000
000535F0  00000000 00000000
000535F8  00000000 00000000
                                          5696
                                          5697           TSTLBL PL                  Parameter List
00053600                                  5698+PL71      DS    0C
00053600  00000000 00000000               5699           DC    XL352'00'
00053608  00000000 00000000
00053610  00000000 00000000
00053618  00000000 00000000
00053620  00000000 00000000
00053628  00000000 00000000
00053630  00000000 00000000
00053638  00000000 00000000
00053640  00000000 00000000
00053648  00000000 00000000
00053650  00000000 00000000
00053658  00000000 00000000
00053660  00000000 00000000
00053668  00000000 00000000
00053670  00000000 00000000
00053678  00000000 00000000
00053680  00000000 00000000
00053688  00000000 00000000
00053690  00000000 00000000
00053698  00000000 00000000
000536A0  00000000 00000000
000536A8  00000000 00000000
000536B0  00000000 00000000
000536B8  00000000 00000000
000536C0  00000000 00000000
000536C8  00000000 00000000
000536D0  00000000 00000000
000536D8  00000000 00000000
000536E0  00000000 00000000
000536E8  00000000 00000000
000536F0  00000000 00000000
000536F8  00000000 00000000
00053700  00000000 00000000
00053708  00000000 00000000
00053710  00000000 00000000
00053718  00000000 00000000
00053720  00000000 00000000
00053728  00000000 00000000
00053730  00000000 00000000
00053738  00000000 00000000
00053740  00000000 00000000
00053748  00000000 00000000
00053750  00000000 00000000
00053758  00000000 00000000
                                          5700
                                          5701           TSTLBL FMSG                fail message
00053760                                  5702+FMSG71    DS    0C
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00053760  86818993 858440D7                        5703          DC    C'failed PLO-Extension '
00053768  D3D660C5 A7A38595
00053770  A2899695 40
00053775  D8A48184 99A49793                        5704          DC    CL96'Quadruple Compare and Swap (QCS: 34): All EQ'
0005377D  8540C396 94978199
00053785  85408195 8440E2A6
0005378D  8197404D D8C3E27A
00053795  40F3F45D 7A40C193
0005379D  9340C5D8 40404040
000537A5  40404040 40404040
000537AD  40404040 40404040
000537B5  40404040 40404040
000537BD  40404040 40404040
000537C5  40404040 40404040
000537CD  40404040 40404040
                                                   5705
                                                   5706          TSTEND
000537D6                                           5707+         DS    0H
000537D6  C5958440 968640A3                        5708+         DC    CL32'End of test: 71'
000537DE  85A2A37A 40F7F140
000537E6  40404040 40404040
000537EE  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5710 *=============================================================
                                                    5711 *       Triple Compare and Swap Tests
                                                    5712 *=============================================================
                                                    5713
                                                    5714 * -------------------------------------------------------------
                                                    5715 * Test Triple Compare and Swap (TCSO: 33) instruction : 1st NE
                                                    5716 * -------------------------------------------------------------
                                                    5717          TSTHDR 2,1,33             1st Operand NOT Equal (cc=2, UR=1)
000537F6                          000537F6  00054000  5718+         ORG   PLOETST+(200*K)+(34*4*K)
00054000  E2E3C1D9 E3409686                         5719+         DC    CL32'START of test: 72'
00054008  40A385A2 A37A40F7
00054010  F2404040 40404040
00054018  40404040 40404040
00054020                          00054020          5720+         USING *,R5                base for test data and test routine
00054020                          00000000          5721+         USING PLOETEST,R5
00054020  0005404C                                  5722+T72      DC    A(CODE72)           address of test code
00054024  0048                                      5723+         DC    H'72'               test number
00054026  00                                        5724+         DC    X'00'
00054027  21                                        5725+         DC    HL1'33'             Function
00054028  01                                        5726+         DC    HL1'1'              expected unequal results (UR)
00054029  00                                        5727+         DC    HL1'00'             found UR
0005402A  02                                        5728+         DC    HL1'2'              cc
0005402B  0D                                        5729+         DC    HL1'13'             not expected CC mask
0005402C  00                                        5730+         DS    X                   extracted cc
00054030  00000000 00000000                         5731+         DS    2F                  extract PSW after test (has CC)
00054038  00054760                                  5732+         DC    A(FMSG72)           address of test fail message
0005403C  00054500                                  5733+         DC    A(OPSRC72)          address of operand source
00054040  00054400                                  5734+         DC    A(OPDST72)          address of operand destination
00054044  00054600                                  5735+         DC    A(PL72)             address of parameter list
00054048  000011F9                                  5736+         DC    A(RANDX+72)         address of random source data
                                                    5737
                                                    5738          TSTCODE
0005404C                                            5739+CODE72   DS    0F
                                                    5740 *                                  load random source
0005404C  B982 0022                                 5741          XGR   R2,R2
00054050  5820 5028                     00000028    5742          L     R2,RANDXA
                                                    5743 *
                                                    5744 * Triple Compare and Swap (TCSO: 33)
                                                    5745 * Build Parambter List
                                                    5746 *
00054054  B982 0011                                 5747          XGR   R1,R1
00054058  5810 501C                     0000001C    5748          L     R1,OPSRCA
                                                    5749
0005405C  B982 0033                                 5750          XGR   R3,R3
00054060  5830 5024                     00000024    5751          L     R3,PLA
                                                    5752
00054064  B982 0044                                 5753          XGR   R4,R4
00054068  5840 5020                     00000020    5754          L     R4,OPREPA
                                                    5755
0005406C  D25F 1000 2000        00000000  00000000  5756          MVC   0(96,R1),0(R2)      load Op1-Op3 data in source
00054072  D25F 3040 1000        00000040  00000000  5757          MVC   64(96,R3),0(R1)     copy to PL OP1-Op3 Compare
00054078  D25F 4000 2080        00000000  00000080  5758          MVC   0(96,R4),128(R2)    load initial Op1-Op3 data
0005407E  D25F 30A0 4000        000000A0  00000000  5759          MVC   160(96,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                    5760 ****
                                                    5761 * force a failure by changing the first operand
                                                    5762 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00054084  D207 1000 4000      00000000  00000000  5763           MVC   0(8,R1),0(R4)
                                                   5764 ****
                                                   5765
0005408A  E313 0000 0024                00000000  5766           STG   R1,0(R3)           load OP1 address in PL
                                                   5767
00054090  4111 0020                     00000020  5768           LA    R1,32(R1)
00054094  E313 0008 0024                00000008  5769           STG   R1,8(R3)           load OP2 address in PL
                                                   5770
0005409A  4111 0020                     00000020  5771           LA    R1,32(R1)
0005409E  E313 0010 0024                00000010  5772           STG   R1,16(R3)          load OP3 address in PL
                                                   5773
000540A4  B982 0000                               5774           XGR   R0,R0
000540A8  4100 0021                     00000021  5775           LA    R0,33
000540AC  EE00 0000 3000      00000000  00000000  5776           PLO   0,0,0,0(R3)        Triple C/S (TCSO: 33)
                                                   5777
000540B2  B98D 0020                               5778           EPSW  R2,R0              extract psw
000540B6  5020 5010                     00000010  5779           ST    R2,CCPSW             to save CC
000540BA  E310 500B 0076                0000000B  5780           LB    R1,CCMASK          (failure CC mask)
000540C0  8910 0004                     00000004  5781           SLL   R1,4               (shift to BC instr CC position)
000540C4  4410 50F8                     00054118  5782           EX    R1,F331TSTCC         fail if...
                                                   5783
                                000540C8  00000001  5784 F331CKUR EQU   *
000540C8  D200 5009 3030      00000009  00000030  5785           MVC   URFOUND,48(R3)     Extract UR
000540CE  D500 5008 3030      00000008  00000030  5786           CLC   UR,48(R3)
000540D4  4780 50BC                     000540DC  5787           BE    F331CKRES
000540D8  45F0 8220                     00000420  5788           BAL   R15,URMSG
                                                   5789 *
                                                   5790 * compare results
                                                   5791 *
                                000540DC  00000001  5792 F331CKRES EQU   *
                                                   5793 *                                reset adresses
000540DC  B982 0011                               5794           XGR   R1,R1
000540E0  5810 501C                     0000001C  5795           L     R1,OPSRCA          source
000540E4  B982 0033                               5796           XGR   R3,R3
000540E8  5830 5024                     00000024  5797           L     R3,PLA             PL
000540EC  B982 0044                               5798           XGR   R4,R4
000540F0  5840 5020                     00000020  5799           L     R4,OPREPA
                                                   5800
000540F4  D51F 1000 3040      00000000  00000040  5801           CLC   0(32,R1),64(R3)
000540FA  4770 50F2                     00054112  5802           BNE   F331XX             OP1C Different, error
000540FE  D51F 1020 3060      00000020  00000060  5803           CLC   32(32,R1),96(R3)
00054104  4770 50F2                     00054112  5804           BNE   F331XX             OP2C Different, error
00054108  D51F 1040 3080      00000040  00000080  5805           CLC   64(32,R1),128(R3)
0005410E  478B 0000                     00000000  5806           BE    0(R11)             OP3C same, return
                                                   5807
                                00054112  00000001  5808 F331XX   EQU   *
00054112  45F0 8340                     00000540  5809           BAL   R15,FAILMSG
00054116  07FB                                    5810           BR    R11
                                                   5811
00054118  4700 50FC                     0005411C  5812 F331TSTCC BC    0,F331CCMSG        fail if unexpected condition code
                                0005411C  00000001  5813 F331CCMSG EQU   *
0005411C  45F0 80F2                     000002F2  5814           BAL   R15,CCMSG
00054120  47F0 50A8                     000540C8  5815           B     F331CKUR
                                                   5816
                                                   5817           TSTDATA
00054124                        00054124  00054400  5818+          ORG   PLOETST+(200*K)+(34*4*K)+K
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

                                                     5819           TSTLBL OPDST              Operand Destination
00054400                                             5820+OPDST72   DS     0C
00054400  00000000 00000000                          5821           DC     XL256'0'
00054408  00000000 00000000
00054410  00000000 00000000
00054418  00000000 00000000
00054420  00000000 00000000
00054428  00000000 00000000
00054430  00000000 00000000
00054438  00000000 00000000
00054440  00000000 00000000
00054448  00000000 00000000
00054450  00000000 00000000
00054458  00000000 00000000
00054460  00000000 00000000
00054468  00000000 00000000
00054470  00000000 00000000
00054478  00000000 00000000
00054480  00000000 00000000
00054488  00000000 00000000
00054490  00000000 00000000
00054498  00000000 00000000
000544A0  00000000 00000000
000544A8  00000000 00000000
000544B0  00000000 00000000
000544B8  00000000 00000000
000544C0  00000000 00000000
000544C8  00000000 00000000
000544D0  00000000 00000000
000544D8  00000000 00000000
000544E0  00000000 00000000
000544E8  00000000 00000000
000544F0  00000000 00000000
000544F8  00000000 00000000
                                                     5822
                                                     5823           TSTLBL OPSRC              Operand Source OP1-OP4
00054500                                             5824+OPSRC72   DS     0C
00054500  00000000 00000000                          5825           DC     XL256'0'
00054508  00000000 00000000
00054510  00000000 00000000
00054518  00000000 00000000
00054520  00000000 00000000
00054528  00000000 00000000
00054530  00000000 00000000
00054538  00000000 00000000
00054540  00000000 00000000
00054548  00000000 00000000
00054550  00000000 00000000
00054558  00000000 00000000
00054560  00000000 00000000
00054568  00000000 00000000
00054570  00000000 00000000
00054578  00000000 00000000
00054580  00000000 00000000
00054588  00000000 00000000
00054590  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00054598  00000000 00000000
000545A0  00000000 00000000
000545A8  00000000 00000000
000545B0  00000000 00000000
000545B8  00000000 00000000
000545C0  00000000 00000000
000545C8  00000000 00000000
000545D0  00000000 00000000
000545D8  00000000 00000000
000545E0  00000000 00000000
000545E8  00000000 00000000
000545F0  00000000 00000000
000545F8  00000000 00000000
                                        5826
                                        5827           TSTLBL PL                  Parameter List
00054600                                5828+PL72      DS    0C
00054600  00000000 00000000             5829           DC    XL352'00'
00054608  00000000 00000000
00054610  00000000 00000000
00054618  00000000 00000000
00054620  00000000 00000000
00054628  00000000 00000000
00054630  00000000 00000000
00054638  00000000 00000000
00054640  00000000 00000000
00054648  00000000 00000000
00054650  00000000 00000000
00054658  00000000 00000000
00054660  00000000 00000000
00054668  00000000 00000000
00054670  00000000 00000000
00054678  00000000 00000000
00054680  00000000 00000000
00054688  00000000 00000000
00054690  00000000 00000000
00054698  00000000 00000000
000546A0  00000000 00000000
000546A8  00000000 00000000
000546B0  00000000 00000000
000546B8  00000000 00000000
000546C0  00000000 00000000
000546C8  00000000 00000000
000546D0  00000000 00000000
000546D8  00000000 00000000
000546E0  00000000 00000000
000546E8  00000000 00000000
000546F0  00000000 00000000
000546F8  00000000 00000000
00054700  00000000 00000000
00054708  00000000 00000000
00054710  00000000 00000000
00054718  00000000 00000000
00054720  00000000 00000000
00054728  00000000 00000000
00054730  00000000 00000000
00054738  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00054740  00000000 00000000
00054748  00000000 00000000
00054750  00000000 00000000
00054758  00000000 00000000
                                          5830
                                          5831           TSTLBL FMSG                fail message
00054760                                  5832+FMSG72    DS     0C
00054760  86818993 858440D7               5833           DC     C'failed PLO-Extension '
00054768  D3D660C5 A7A38595
00054770  A2899695 40
00054775  E3998997 938540C3               5834           DC     CL96'Triple Compare and Swap (TCSO: 33): 1st NE'
0005477D  96949781 99854081
00054785  958440E2 A6819740
0005478D  4DE3C3E2 D67A40F3
00054795  F35D7A40 F1A2A340
0005479D  D5C54040 40404040
000547A5  40404040 40404040
000547AD  40404040 40404040
000547B5  40404040 40404040
000547BD  40404040 40404040
000547C5  40404040 40404040
000547CD  40404040 40404040
                                          5835
                                          5836           TSTEND
000547D6                                  5837+          DS     0H
000547D6  C5958440 968640A3               5838+          DC     CL32'End of test: 72'
000547DE  85A2A37A 40F7F240
000547E6  40404040 40404040
000547EE  40404040 40404040
                                          5839
                                          5840 * ------------------------------------------------------------------
                                          5841 * Test Triple Compare and Swap (TCSO: 33) instruction : 2nd NE
                                          5842 * ------------------------------------------------------------------
                                          5843           TSTHDR 2,2,33              2nd Operand NOT Equal (cc=2, UR=2)
000547F6                      000547F6  00055000  5844+          ORG    PLOETST+(200*K)+(35*4*K)
00055000  E2E3C1D9 E3409686               5845+          DC     CL32'START of test: 73'
00055008  40A385A2 A37A40F7
00055010  F3404040 40404040
00055018  40404040 40404040
00055020                      00055020            5846+          USING *,R5                 base for test data and test routine
00055020                      00000000            5847+          USING PLOETEST,R5
00055020  0005504C                        5848+T73       DC     A(CODE73)          address of test code
00055024  0049                            5849+          DC     H'73'              test number
00055026  00                              5850+          DC     X'00'
00055027  21                              5851+          DC     HL1'33'            Function
00055028  02                              5852+          DC     HL1'2'             expected unequal results (UR)
00055029  00                              5853+          DC     HL1'00'            found UR
0005502A  02                              5854+          DC     HL1'2'             cc
0005502B  0D                              5855+          DC     HL1'13'            not expected CC mask
0005502C  00                              5856+          DS     X                  extracted cc
00055030  00000000 00000000               5857+          DS     2F                 extract PSW after test (has CC)
00055038  00055760                        5858+          DC     A(FMSG73)          address of test fail message
0005503C  00055500                        5859+          DC     A(OPSRC73)         address of operand source
00055040  00055400                        5860+          DC     A(OPDST73)         address of operand destination
00055044  00055600                        5861+          DC     A(PL73)            address of parameter list
00055048  000011FA                        5862+          DC     A(RANDX+73)        address of random source data
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5863
                                                    5864          TSTCODE
0005504C                                            5865+CODE73   DS    0F
                                                    5866 *                                load random source
0005504C  B982 0022                                 5867          XGR   R2,R2
00055050  5820 5028                   00000028      5868          L     R2,RANDXA
                                                    5869 *
                                                    5870 * Triple Compare and Swap (TCSO: 33)
                                                    5871 * Build Parambter List
                                                    5872 *
00055054  B982 0011                                 5873          XGR   R1,R1
00055058  5810 501C                   0000001C      5874          L     R1,OPSRCA
                                                    5875
0005505C  B982 0033                                 5876          XGR   R3,R3
00055060  5830 5024                   00000024      5877          L     R3,PLA
                                                    5878
00055064  B982 0044                                 5879          XGR   R4,R4
00055068  5840 5020                   00000020      5880          L     R4,OPREPA
                                                    5881
0005506C  D25F 1000 2000      00000000  00000000    5882          MVC   0(96,R1),0(R2)      load Op1-Op3 data in source
00055072  D25F 3040 1000      00000040  00000000    5883          MVC   64(96,R3),0(R1)     copy to PL OP1-Op3 Compare
00055078  D25F 4000 2080      00000000  00000080    5884          MVC   0(96,R4),128(R2)    load initial Op1-Op3 data
0005507E  D25F 30A0 4000      000000A0  00000000    5885          MVC   160(96,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                    5886 ****
                                                    5887 * force a failure by changing the 2nd operand
                                                    5888 ****
00055084  D207 1020 4000      00000020  00000000    5889          MVC   32(8,R1),0(R4)
                                                    5890 ****
                                                    5891
0005508A  E313 0000 0024                00000000    5892          STG   R1,0(R3)            load OP1 address in PL
                                                    5893
00055090  4111 0020                     00000020    5894          LA    R1,32(R1)
00055094  E313 0008 0024                00000008    5895          STG   R1,8(R3)            load OP2 address in PL
                                                    5896
0005509A  4111 0020                     00000020    5897          LA    R1,32(R1)
0005509E  E313 0010 0024                00000010    5898          STG   R1,16(R3)           load OP3 address in PL
                                                    5899
000550A4  B982 0000                                 5900          XGR   R0,R0
000550A8  4100 0021                     00000021    5901          LA    R0,33
000550AC  EE00 0000 3000      00000000  00000000    5902          PLO   0,0,0,0(R3)         Triple C/S (TCSO: 33)
                                                    5903
000550B2  B98D 0020                                 5904          EPSW  R2,R0               extract psw
000550B6  5020 5010                     00000010    5905          ST    R2,CCPSW              to save CC
000550BA  E310 500B 0076                0000000B    5906          LB    R1,CCMASK           (failure CC mask)
000550C0  8910 0004                     00000004    5907          SLL   R1,4                (shift to BC instr CC position)
000550C4  4410 50F8                     00055118    5908          EX    R1,F332TSTCC            fail if...
                                                    5909
                              000550C8  00000001    5910 F332CKUR EQU   *
000550C8  D200 5009 3030      00000009  00000030    5911          MVC   URFOUND,48(R3)      Extract UR
000550CE  D500 5008 3030      00000008  00000030    5912          CLC   UR,48(R3)
000550D4  4780 50BC                     000550DC    5913          BE    F332CKRES
000550D8  45F0 8220                     00000420    5914          BAL   R15,URMSG
                                                    5915 *
                                                    5916 * compare results
                                                    5917 *
                              000550DC  00000001    5918 F332CKRES EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5919 *                                  reset adresses
000550DC  B982 0011                                 5920          XGR   R1,R1
000550E0  5810 501C                       0000001C  5921          L     R1,OPSRCA           source
000550E4  B982 0033                                 5922          XGR   R3,R3
000550E8  5830 5024                       00000024  5923          L     R3,PLA              PL
000550EC  B982 0044                                 5924          XGR   R4,R4
000550F0  5840 5020                       00000020  5925          L     R4,OPREPA
                                                    5926
000550F4  D51F 1000 3040        00000000  00000040  5927          CLC   0(32,R1),64(R3)
000550FA  4770 50F2                       00055112  5928          BNE   F332XX              OP1C Different, error
000550FE  D51F 1020 3060        00000020  00000060  5929          CLC   32(32,R1),96(R3)
00055104  4770 50F2                       00055112  5930          BNE   F332XX              OP2C Different, error
00055108  D51F 1040 3080        00000040  00000080  5931          CLC   64(32,R1),128(R3)
0005510E  478B 0000                       00000000  5932          BE    0(R11)              OP3C same, return
                                                    5933
                                00055112  00000001  5934 F332XX   EQU   *
00055112  45F0 8340                       00000540  5935          BAL   R15,FAILMSG
00055116  07FB                                      5936          BR    R11
                                                    5937
00055118  4700 50FC                       0005511C  5938 F332TSTCC BC    0,F332CCMSG         fail if unexpected condition code
                                0005511C  00000001  5939 F332CCMSG EQU   *
0005511C  45F0 80F2                       000002F2  5940          BAL   R15,CCMSG
00055120  47F0 50A8                       000550C8  5941          B     F332CKUR
                                                    5942
                                                    5943          TSTDATA
00055124                        00055124  00055400  5944+         ORG   PLOETST+(200*K)+(35*4*K)+K
                                                    5945          TSTLBL OPDST              Operand Destination
00055400                                            5946+OPDST73  DS    0C
00055400  00000000 00000000                         5947          DC    XL256'0'
00055408  00000000 00000000
00055410  00000000 00000000
00055418  00000000 00000000
00055420  00000000 00000000
00055428  00000000 00000000
00055430  00000000 00000000
00055438  00000000 00000000
00055440  00000000 00000000
00055448  00000000 00000000
00055450  00000000 00000000
00055458  00000000 00000000
00055460  00000000 00000000
00055468  00000000 00000000
00055470  00000000 00000000
00055478  00000000 00000000
00055480  00000000 00000000
00055488  00000000 00000000
00055490  00000000 00000000
00055498  00000000 00000000
000554A0  00000000 00000000
000554A8  00000000 00000000
000554B0  00000000 00000000
000554B8  00000000 00000000
000554C0  00000000 00000000
000554C8  00000000 00000000
000554D0  00000000 00000000
000554D8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000554E0  00000000 00000000
000554E8  00000000 00000000
000554F0  00000000 00000000
000554F8  00000000 00000000
                                          5948
                                          5949           TSTLBL OPSRC              Operand Source OP1-OP4
00055500                                  5950+OPSRC73   DS    0C
00055500  00000000 00000000               5951           DC    XL256'0'
00055508  00000000 00000000
00055510  00000000 00000000
00055518  00000000 00000000
00055520  00000000 00000000
00055528  00000000 00000000
00055530  00000000 00000000
00055538  00000000 00000000
00055540  00000000 00000000
00055548  00000000 00000000
00055550  00000000 00000000
00055558  00000000 00000000
00055560  00000000 00000000
00055568  00000000 00000000
00055570  00000000 00000000
00055578  00000000 00000000
00055580  00000000 00000000
00055588  00000000 00000000
00055590  00000000 00000000
00055598  00000000 00000000
000555A0  00000000 00000000
000555A8  00000000 00000000
000555B0  00000000 00000000
000555B8  00000000 00000000
000555C0  00000000 00000000
000555C8  00000000 00000000
000555D0  00000000 00000000
000555D8  00000000 00000000
000555E0  00000000 00000000
000555E8  00000000 00000000
000555F0  00000000 00000000
000555F8  00000000 00000000
                                          5952
                                          5953           TSTLBL PL                 Parameter List
00055600                                  5954+PL73      DS    0C
00055600  00000000 00000000               5955           DC    XL352'00'
00055608  00000000 00000000
00055610  00000000 00000000
00055618  00000000 00000000
00055620  00000000 00000000
00055628  00000000 00000000
00055630  00000000 00000000
00055638  00000000 00000000
00055640  00000000 00000000
00055648  00000000 00000000
00055650  00000000 00000000
00055658  00000000 00000000
00055660  00000000 00000000
00055668  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00055670  00000000 00000000
00055678  00000000 00000000
00055680  00000000 00000000
00055688  00000000 00000000
00055690  00000000 00000000
00055698  00000000 00000000
000556A0  00000000 00000000
000556A8  00000000 00000000
000556B0  00000000 00000000
000556B8  00000000 00000000
000556C0  00000000 00000000
000556C8  00000000 00000000
000556D0  00000000 00000000
000556D8  00000000 00000000
000556E0  00000000 00000000
000556E8  00000000 00000000
000556F0  00000000 00000000
000556F8  00000000 00000000
00055700  00000000 00000000
00055708  00000000 00000000
00055710  00000000 00000000
00055718  00000000 00000000
00055720  00000000 00000000
00055728  00000000 00000000
00055730  00000000 00000000
00055738  00000000 00000000
00055740  00000000 00000000
00055748  00000000 00000000
00055750  00000000 00000000
00055758  00000000 00000000
                                          5956
                                          5957          TSTLBL FMSG               fail message
00055760                                  5958+FMSG73   DS    0C
00055760  86818993 858440D7              5959          DC    C'failed PLO-Extension '
00055768  D3D660C5 A7A38595
00055770  A2899695 40
00055775  E3998997 938540C3              5960          DC    CL96'Triple Compare and Swap (TCSO: 33): 2nd NE'
0005577D  96949781 99854081
00055785  958440E2 A6819740
0005578D  4DE3C3E2 D67A40F3
00055795  F35D7A40 F2958440
0005579D  D5C54040 40404040
000557A5  40404040 40404040
000557AD  40404040 40404040
000557B5  40404040 40404040
000557BD  40404040 40404040
000557C5  40404040 40404040
000557CD  40404040 40404040
                                          5961
                                          5962          TSTEND
000557D6                                  5963+         DS    0H
000557D6  C5958440 968640A3              5964+         DC    CL32'End of test: 73'
000557DE  85A2A37A 40F7F340
000557E6  40404040 40404040
000557EE  40404040 40404040
                                          5965
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   5966 * ------------------------------------------------------------------------
                                                   5967 * Test Triple Compare and Swap (TCSO: 33) instruction : 3rd NE
                                                   5968 * ------------------------------------------------------------------------
                                                   5969          TSTHDR 2,3,33             3rd Operand NOT Equal (cc=2, UR=3)
000557F6                        000557F6  00056000 5970+         ORG   PLOETST+(200*K)+(36*4*K)
00056000  E2E3C1D9 E3409686                        5971+         DC    CL32'START of test: 74'
00056008  40A385A2 A37A40F7
00056010  F4404040 40404040
00056018  40404040 40404040
00056020                        00056020          5972+         USING *,R5                base for test data and test routine
00056020                        00000000          5973+         USING PLOETEST,R5
00056020  0005604C                                5974+T74      DC    A(CODE74)           address of test code
00056024  004A                                    5975+         DC    H'74'               test number
00056026  00                                      5976+         DC    X'00'
00056027  21                                      5977+         DC    HL1'33'             Function
00056028  03                                      5978+         DC    HL1'3'              expected unequal results (UR)
00056029  00                                      5979+         DC    HL1'00'             found UR
0005602A  02                                      5980+         DC    HL1'2'              cc
0005602B  0D                                      5981+         DC    HL1'13'             not expected CC mask
0005602C  00                                      5982+         DS    X                   extracted cc
00056030  00000000 00000000                       5983+         DS    2F                  extract PSW after test (has CC)
00056038  00056760                                5984+         DC    A(FMSG74)           address of test fail message
0005603C  00056500                                5985+         DC    A(OPSRC74)          address of operand source
00056040  00056400                                5986+         DC    A(OPDST74)          address of operand destination
00056044  00056600                                5987+         DC    A(PL74)             address of parameter list
00056048  000011FB                                5988+         DC    A(RANDX+74)         address of random source data
                                                   5989
                                                   5990          TSTCODE
0005604C                                          5991+CODE74   DS    0F
                                                   5992 *                                  load random source
0005604C  B982 0022                               5993          XGR   R2,R2
00056050  5820 5028                     00000028  5994          L     R2,RANDXA
                                                   5995 *
                                                   5996 * Triple Compare and Swap (TCSO: 33)
                                                   5997 * Build Parambter List
                                                   5998 *
00056054  B982 0011                               5999          XGR   R1,R1
00056058  5810 501C                     0000001C  6000          L     R1,OPSRCA
                                                   6001
0005605C  B982 0033                               6002          XGR   R3,R3
00056060  5830 5024                     00000024  6003          L     R3,PLA
                                                   6004
00056064  B982 0044                               6005          XGR   R4,R4
00056068  5840 5020                     00000020  6006          L     R4,OPREPA
                                                   6007
0005606C  D25F 1000 2000        00000000  00000000 6008          MVC   0(96,R1),0(R2)      load Op1-Op3 data in source
00056072  D25F 3040 1000        00000040  00000000 6009          MVC   64(96,R3),0(R1)     copy to PL OP1-Op3 Compare
00056078  D25F 4000 2080        00000000  00000080 6010          MVC   0(96,R4),128(R2)    load initial Op1-Op3 data
0005607E  D25F 30A0 4000        000000A0  00000000 6011          MVC   160(96,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                   6012 ****
                                                   6013 * force a failure by changing the 3rd operand
                                                   6014 ****
00056084  D207 1040 4000        00000040  00000000 6015          MVC   64(8,R1),0(R4)
                                                   6016 ****
                                                   6017
0005608A  E313 0000 0024                  00000000 6018          STG   R1,0(R3)            load OP1 address in PL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6019
00056090  4111 0020                     00000020  6020          LA    R1,32(R1)
00056094  E313 0008 0024                 00000008  6021          STG   R1,8(R3)           load OP2 address in PL
                                                    6022
0005609A  4111 0020                     00000020  6023          LA    R1,32(R1)
0005609E  E313 0010 0024                 00000010  6024          STG   R1,16(R3)          load OP3 address in PL
                                                    6025
000560A4  B982 0000                               6026          XGR   R0,R0
000560A8  4100 0021                     00000021  6027          LA    R0,33
000560AC  EE00 0000 3000  00000000      00000000  6028          PLO   0,0,0,0(R3)        Triple C/S (TCSO: 33)
                                                    6029
000560B2  B98D 0020                               6030          EPSW  R2,R0              extract psw
000560B6  5020 5010                     00000010  6031          ST    R2,CCPSW             to save CC
000560BA  E310 500B 0076                0000000B  6032          LB    R1,CCMASK          (failure CC mask)
000560C0  8910 0004                     00000004  6033          SLL   R1,4               (shift to BC instr CC position)
000560C4  4410 50F8                     00056118  6034          EX    R1,F333TSTCC           fail if...
                                                    6035
                          000560C8      00000001  6036 F333CKUR EQU   *
000560C8  D200 5009 3030  00000009      00000030  6037          MVC   URFOUND,48(R3)     Extract UR
000560CE  D500 5008 3030  00000008      00000030  6038          CLC   UR,48(R3)
000560D4  4780 50BC                     000560DC  6039          BE    F333CKRES
000560D8  45F0 8220                     00000420  6040          BAL   R15,URMSG
                                                    6041 *
                                                    6042 * compare results
                                                    6043 *
                          000560DC      00000001  6044 F333CKRES EQU   *
                                                    6045 *                             reset adresses
000560DC  B982 0011                               6046          XGR   R1,R1
000560E0  5810 501C                     0000001C  6047          L     R1,OPSRCA          source
000560E4  B982 0033                               6048          XGR   R3,R3
000560E8  5830 5024                     00000024  6049          L     R3,PLA             PL
000560EC  B982 0044                               6050          XGR   R4,R4
000560F0  5840 5020                     00000020  6051          L     R4,OPREPA
                                                    6052
000560F4  D51F 1000 3040  00000000      00000040  6053          CLC   0(32,R1),64(R3)
000560FA  4770 50F2                     00056112  6054          BNE   F333XX             OP1C Different, error
000560FE  D51F 1020 3060  00000020      00000060  6055          CLC   32(32,R1),96(R3)
00056104  4770 50F2                     00056112  6056          BNE   F333XX             OP2C Different, error
00056108  D51F 1040 3080  00000040      00000080  6057          CLC   64(32,R1),128(R3)
0005610E  478B 0000                     00000000  6058          BE    0(R11)             OP3C same, return
                                                    6059
                          00056112      00000001  6060 F333XX   EQU   *
00056112  45F0 8340                     00000540  6061          BAL   R15,FAILMSG
00056116  07FB                                    6062          BR    R11
                                                    6063
00056118  4700 50FC                     0005611C  6064 F333TSTCC BC    0,F333CCMSG        fail if unexpected condition code
                          0005611C      00000001  6065 F333CCMSG EQU   *
0005611C  45F0 80F2                     000002F2  6066          BAL   R15,CCMSG
00056120  47F0 50A8                     000560C8  6067          B     F333CKUR
                                                    6068
                                                    6069          TSTDATA
00056124                  00056124      00056400  6070+         ORG   PLOETST+(200*K)+(36*4*K)+K
                                                    6071          TSTLBL OPDST             Operand Destination
00056400                                            6072+OPDST74  DS    0C
00056400  00000000 00000000                         6073          DC    XL256'0'
00056408  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00056410  00000000 00000000
00056418  00000000 00000000
00056420  00000000 00000000
00056428  00000000 00000000
00056430  00000000 00000000
00056438  00000000 00000000
00056440  00000000 00000000
00056448  00000000 00000000
00056450  00000000 00000000
00056458  00000000 00000000
00056460  00000000 00000000
00056468  00000000 00000000
00056470  00000000 00000000
00056478  00000000 00000000
00056480  00000000 00000000
00056488  00000000 00000000
00056490  00000000 00000000
00056498  00000000 00000000
000564A0  00000000 00000000
000564A8  00000000 00000000
000564B0  00000000 00000000
000564B8  00000000 00000000
000564C0  00000000 00000000
000564C8  00000000 00000000
000564D0  00000000 00000000
000564D8  00000000 00000000
000564E0  00000000 00000000
000564E8  00000000 00000000
000564F0  00000000 00000000
000564F8  00000000 00000000
                                          6074
                                          6075           TSTLBL OPSRC              Operand Source OP1-OP4
00056500                                  6076+OPSRC74   DS    0C
00056500  00000000 00000000               6077           DC    XL256'0'
00056508  00000000 00000000
00056510  00000000 00000000
00056518  00000000 00000000
00056520  00000000 00000000
00056528  00000000 00000000
00056530  00000000 00000000
00056538  00000000 00000000
00056540  00000000 00000000
00056548  00000000 00000000
00056550  00000000 00000000
00056558  00000000 00000000
00056560  00000000 00000000
00056568  00000000 00000000
00056570  00000000 00000000
00056578  00000000 00000000
00056580  00000000 00000000
00056588  00000000 00000000
00056590  00000000 00000000
00056598  00000000 00000000
000565A0  00000000 00000000
000565A8  00000000 00000000
000565B0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000565B8  00000000 00000000
000565C0  00000000 00000000
000565C8  00000000 00000000
000565D0  00000000 00000000
000565D8  00000000 00000000
000565E0  00000000 00000000
000565E8  00000000 00000000
000565F0  00000000 00000000
000565F8  00000000 00000000
                                        6078
                                        6079           TSTLBL PL                  Parameter List
00056600                                6080+PL74      DS    0C
00056600  00000000 00000000             6081           DC    XL352'00'
00056608  00000000 00000000
00056610  00000000 00000000
00056618  00000000 00000000
00056620  00000000 00000000
00056628  00000000 00000000
00056630  00000000 00000000
00056638  00000000 00000000
00056640  00000000 00000000
00056648  00000000 00000000
00056650  00000000 00000000
00056658  00000000 00000000
00056660  00000000 00000000
00056668  00000000 00000000
00056670  00000000 00000000
00056678  00000000 00000000
00056680  00000000 00000000
00056688  00000000 00000000
00056690  00000000 00000000
00056698  00000000 00000000
000566A0  00000000 00000000
000566A8  00000000 00000000
000566B0  00000000 00000000
000566B8  00000000 00000000
000566C0  00000000 00000000
000566C8  00000000 00000000
000566D0  00000000 00000000
000566D8  00000000 00000000
000566E0  00000000 00000000
000566E8  00000000 00000000
000566F0  00000000 00000000
000566F8  00000000 00000000
00056700  00000000 00000000
00056708  00000000 00000000
00056710  00000000 00000000
00056718  00000000 00000000
00056720  00000000 00000000
00056728  00000000 00000000
00056730  00000000 00000000
00056738  00000000 00000000
00056740  00000000 00000000
00056748  00000000 00000000
00056750  00000000 00000000
00056758  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6082
                                                    6083          TSTLBL FMSG               fail message
00056760                                            6084+FMSG74   DS    0C
00056760  86818993 858440D7                       6085          DC    C'failed PLO-Extension '
00056768  D3D660C5 A7A38595
00056770  A2899695 40
00056775  E3998997 938540C3                       6086          DC    CL96'Triple Compare and Swap (TCSO: 33): 3rd NE'
0005677D  96949781 99854081
00056785  958440E2 A6819740
0005678D  4DE3C3E2 D67A40F3
00056795  F35D7A40 F3998440
0005679D  D5C54040 40404040
000567A5  40404040 40404040
000567AD  40404040 40404040
000567B5  40404040 40404040
000567BD  40404040 40404040
000567C5  40404040 40404040
000567CD  40404040 40404040
                                                    6087
                                                    6088          TSTEND
000567D6                                            6089+         DS    0H
000567D6  C5958440 968640A3                       6090+         DC    CL32'End of test: 74'
000567DE  85A2A37A 40F7F440
000567E6  40404040 40404040
000567EE  40404040 40404040
                                                    6091
                                                    6092 * ------------------------------------------------------------------
                                                    6093 * Test Triple Compare and Swap (TCSO: 33) instruction : ALL EQ
                                                    6094 * ------------------------------------------------------------------
                                                    6095          TSTHDR 0,99,33             ALL EQ (cc=0, UR=99)
000567F6                      000567F6  00057000    6096+         ORG   PLOETST+(200*K)+(37*4*K)
00057000  E2E3C1D9 E3409686                       6097+         DC    CL32'START of test: 75'
00057008  40A385A2 A37A40F7
00057010  F5404040 40404040
00057018  40404040 40404040
00057020                      00057020            6098+         USING *,R5                base for test data and test routine
00057020                      00000000            6099+         USING PLOETEST,R5
00057020  0005704C                                6100+T75      DC    A(CODE75)           address of test code
00057024  004B                                    6101+         DC    H'75'               test number
00057026  00                                      6102+         DC    X'00'
00057027  21                                      6103+         DC    HL1'33'             Function
00057028  63                                      6104+         DC    HL1'99'             expected unequal results (UR)
00057029  00                                      6105+         DC    HL1'00'             found UR
0005702A  00                                      6106+         DC    HL1'0'              cc
0005702B  07                                      6107+         DC    HL1'7'              not expected CC mask
0005702C  00                                      6108+         DS    X                   extracted cc
00057030  00000000 00000000                       6109+         DS    2F                  extract PSW after test (has CC)
00057038  00057760                                6110+         DC    A(FMSG75)           address of test fail message
0005703C  00057500                                6111+         DC    A(OPSRC75)          address of operand source
00057040  00057400                                6112+         DC    A(OPDST75)          address of operand destination
00057044  00057600                                6113+         DC    A(PL75)             address of parameter list
00057048  000011FC                                6114+         DC    A(RANDX+75)         address of random source data
                                                    6115
                                                    6116          TSTCODE
0005704C                                            6117+CODE75   DS    0F
                                                    6118 *                                  load random source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0005704C  B982 0022                                6119          XGR   R2,R2
00057050  5820 5028                    00000028    6120          L     R2,RANDXA
                                                   6121 *
                                                   6122 * Triple Compare and Swap (TCSO: 33)
                                                   6123 * Build Parambter List
                                                   6124 *
00057054  B982 0011                                6125          XGR   R1,R1
00057058  5810 501C                    0000001C    6126          L     R1,OPSRCA
                                                   6127
0005705C  B982 0033                                6128          XGR   R3,R3
00057060  5830 5024                    00000024    6129          L     R3,PLA
                                                   6130
00057064  B982 0044                                6131          XGR   R4,R4
00057068  5840 5020                    00000020    6132          L     R4,OPREPA
                                                   6133
0005706C  D25F 1000 2000       00000000  00000000  6134          MVC   0(96,R1),0(R2)      load Op1-Op3 data in source
00057072  D25F 3040 1000       00000040  00000000  6135          MVC   64(96,R3),0(R1)     copy to PL OP1-Op3 Compare
00057078  D25F 4000 2080       00000000  00000080  6136          MVC   0(96,R4),128(R2)    load initial Op1-Op3 data
0005707E  D25F 30A0 4000       000000A0  00000000  6137          MVC   160(96,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                   6138 ****
                                                   6139 * Do NOTforce a failure by changing an operand
                                                   6140 ****
                                                   6141 *        MVC   64(8,R1),0(R4)
                                                   6142 ****
                                                   6143
00057084  E313 0000 0024                   00000000 6144          STG   R1,0(R3)            load OP1 address in PL
                                                   6145
0005708A  4111 0020                    00000020    6146          LA    R1,32(R1)
0005708E  E313 0008 0024               00000008    6147          STG   R1,8(R3)            load OP2 address in PL
                                                   6148
00057094  4111 0020                    00000020    6149          LA    R1,32(R1)
00057098  E313 0010 0024               00000010    6150          STG   R1,16(R3)           load OP3 address in PL
                                                   6151
0005709E  B982 0000                                6152          XGR   R0,R0
000570A2  4100 0021                    00000021    6153          LA    R0,33
000570A6  EE00 0000 3000       00000000  00000000  6154          PLO   0,0,0,0(R3)         Triple C/S (TCSO: 33)
                                                   6155
000570AC  B98D 0020                                6156          EPSW  R2,R0               extract psw
000570B0  5020 5010                    00000010    6157          ST    R2,CCPSW              to save CC
000570B4  E310 500B 0076               0000000B    6158          LB    R1,CCMASK           (failure CC mask)
000570BA  8910 0004                    00000004    6159          SLL   R1,4                (shift to BC instr CC position)
000570BE  4410 50DE                    000570FE    6160          EX    R1,F334TSTCC            fail if...
                                                   6161 *
                                                   6162 * compare results
                                                   6163 *
                               000570C2  00000001  6164 F334CKRES EQU   *
                                                   6165 *                                  reset adresses
000570C2  B982 0011                                6166          XGR   R1,R1
000570C6  5810 501C                    0000001C    6167          L     R1,OPSRCA           source
000570CA  B982 0033                                6168          XGR   R3,R3
000570CE  5830 5024                    00000024    6169          L     R3,PLA              PL
000570D2  B982 0044                                6170          XGR   R4,R4
000570D6  5840 5020                    00000020    6171          L     R4,OPREPA
                                                   6172
                                                   6173 * All equal; replacement values copied to operand source
                                                   6174
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000570DA  D51F 1000 30A0      00000000  000000A0  6175           CLC   0(32,R1),160(R3)
000570E0  4770 50D8                     000570F8  6176           BNE   F334XX               OP1R Different, error
000570E4  D51F 1020 30C0      00000020  000000C0  6177           CLC   32(32,R1),192(R3)
000570EA  4770 50D8                     000570F8  6178           BNE   F334XX               OP2R Different, error
000570EE  D51F 1040 30E0      00000040  000000E0  6179           CLC   64(32,R1),224(R3)
000570F4  478B 0000                     00000000  6180           BE    0(R11)
                                                  6181
                              000570F8  00000001  6182 F334XX    EQU   *
000570F8  45F0 8340                     00000540  6183           BAL   R15,FAILMSG
000570FC  07FB                                    6184           BR    R11
                                                  6185
000570FE  4700 50E2                     00057102  6186 F334TSTCC BC    0,F334CCMSG          fail if unexpected condition code
                              00057102  00000001  6187 F334CCMSG EQU   *
00057102  45F0 80F2                     000002F2  6188           BAL   R15,CCMSG
00057106  47F0 50A2                     000570C2  6189           B     F334CKRES
                                                  6190
                                                  6191           TSTDATA
0005710A                      0005710A  00057400  6192+          ORG   PLOETST+(200*K)+(37*4*K)+K
                                                  6193           TSTLBL OPDST               Operand Destination
00057400                                          6194+OPDST75   DS    0C
00057400  00000000 00000000                       6195           DC    XL256'0'
00057408  00000000 00000000
00057410  00000000 00000000
00057418  00000000 00000000
00057420  00000000 00000000
00057428  00000000 00000000
00057430  00000000 00000000
00057438  00000000 00000000
00057440  00000000 00000000
00057448  00000000 00000000
00057450  00000000 00000000
00057458  00000000 00000000
00057460  00000000 00000000
00057468  00000000 00000000
00057470  00000000 00000000
00057478  00000000 00000000
00057480  00000000 00000000
00057488  00000000 00000000
00057490  00000000 00000000
00057498  00000000 00000000
000574A0  00000000 00000000
000574A8  00000000 00000000
000574B0  00000000 00000000
000574B8  00000000 00000000
000574C0  00000000 00000000
000574C8  00000000 00000000
000574D0  00000000 00000000
000574D8  00000000 00000000
000574E0  00000000 00000000
000574E8  00000000 00000000
000574F0  00000000 00000000
000574F8  00000000 00000000
                                                  6196
                                                  6197           TSTLBL OPSRC               Operand Source OP1-OP4
00057500                                          6198+OPSRC75   DS    0C
00057500  00000000 00000000                       6199           DC    XL256'0'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00057508  00000000 00000000
00057510  00000000 00000000
00057518  00000000 00000000
00057520  00000000 00000000
00057528  00000000 00000000
00057530  00000000 00000000
00057538  00000000 00000000
00057540  00000000 00000000
00057548  00000000 00000000
00057550  00000000 00000000
00057558  00000000 00000000
00057560  00000000 00000000
00057568  00000000 00000000
00057570  00000000 00000000
00057578  00000000 00000000
00057580  00000000 00000000
00057588  00000000 00000000
00057590  00000000 00000000
00057598  00000000 00000000
000575A0  00000000 00000000
000575A8  00000000 00000000
000575B0  00000000 00000000
000575B8  00000000 00000000
000575C0  00000000 00000000
000575C8  00000000 00000000
000575D0  00000000 00000000
000575D8  00000000 00000000
000575E0  00000000 00000000
000575E8  00000000 00000000
000575F0  00000000 00000000
000575F8  00000000 00000000
                                                  6200
                                                  6201           TSTLBL PL                  Parameter List
00057600                                          6202+PL75      DS    0C
00057600  00000000 00000000                       6203           DC    XL352'00'
00057608  00000000 00000000
00057610  00000000 00000000
00057618  00000000 00000000
00057620  00000000 00000000
00057628  00000000 00000000
00057630  00000000 00000000
00057638  00000000 00000000
00057640  00000000 00000000
00057648  00000000 00000000
00057650  00000000 00000000
00057658  00000000 00000000
00057660  00000000 00000000
00057668  00000000 00000000
00057670  00000000 00000000
00057678  00000000 00000000
00057680  00000000 00000000
00057688  00000000 00000000
00057690  00000000 00000000
00057698  00000000 00000000
000576A0  00000000 00000000
000576A8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000576B0  00000000 00000000
000576B8  00000000 00000000
000576C0  00000000 00000000
000576C8  00000000 00000000
000576D0  00000000 00000000
000576D8  00000000 00000000
000576E0  00000000 00000000
000576E8  00000000 00000000
000576F0  00000000 00000000
000576F8  00000000 00000000
00057700  00000000 00000000
00057708  00000000 00000000
00057710  00000000 00000000
00057718  00000000 00000000
00057720  00000000 00000000
00057728  00000000 00000000
00057730  00000000 00000000
00057738  00000000 00000000
00057740  00000000 00000000
00057748  00000000 00000000
00057750  00000000 00000000
00057758  00000000 00000000
                                          6204
                                          6205          TSTLBL FMSG                fail message
00057760                                  6206+FMSG75   DS    0C
00057760  86818993 858440D7              6207          DC    C'failed PLO-Extension '
00057768  D3D660C5 A7A38595
00057770  A2899695 40
00057775  E3998997 938540C3              6208          DC    CL96'Triple Compare and Swap (TCSO: 33): 4th NE'
0005777D  96949781 99854081
00057785  958440E2 A6819740
0005778D  4DE3C3E2 D67A40F3
00057795  F35D7A40 F4A38840
0005779D  D5C54040 40404040
000577A5  40404040 40404040
000577AD  40404040 40404040
000577B5  40404040 40404040
000577BD  40404040 40404040
000577C5  40404040 40404040
000577CD  40404040 40404040
                                          6209
                                          6210          TSTEND
000577D6                                  6211+         DS    0H
000577D6  C5958440 968640A3              6212+         DC    CL32'End of test: 75'
000577DE  85A2A37A 40F7F540
000577E6  40404040 40404040
000577EE  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6214 *=====================================================================
                                                    6215 *       Triple Compare and Swap Tests
                                                    6216 *=====================================================================
                                                    6217
                                                    6218 * -------------------------------------------------------------------
                                                    6219 * Test Triple Compare and Swap (TCSX: 32) instruction : 1st NE
                                                    6220 * -------------------------------------------------------------------
                                                    6221          TSTHDR 2,1,32             1st Operand NOT Equal (cc=2, UR=1)
000577F6                          000577F6  00058000  6222+         ORG   PLOETST+(200*K)+(38*4*K)
00058000  E2E3C1D9 E3409686                         6223+         DC    CL32'START of test: 76'
00058008  40A385A2 A37A40F7
00058010  F6404040 40404040
00058018  40404040 40404040
00058020                          00058020          6224+         USING *,R5                base for test data and test routine
00058020                          00000000          6225+         USING PLOETEST,R5
00058020  0005804C                                  6226+T76      DC    A(CODE76)           address of test code
00058024  004C                                      6227+         DC    H'76'               test number
00058026  00                                        6228+         DC    X'00'
00058027  20                                        6229+         DC    HL1'32'             Function
00058028  01                                        6230+         DC    HL1'1'              expected unequal results (UR)
00058029  00                                        6231+         DC    HL1'00'             found UR
0005802A  02                                        6232+         DC    HL1'2'              cc
0005802B  0D                                        6233+         DC    HL1'13'             not expected CC mask
0005802C  00                                        6234+         DS    X                   extracted cc
00058030  00000000 00000000                         6235+         DS    2F                  extract PSW after test (has CC)
00058038  00058760                                  6236+         DC    A(FMSG76)           address of test fail message
0005803C  00058500                                  6237+         DC    A(OPSRC76)          address of operand source
00058040  00058400                                  6238+         DC    A(OPDST76)          address of operand destination
00058044  00058600                                  6239+         DC    A(PL76)             address of parameter list
00058048  000011FD                                  6240+         DC    A(RANDX+76)         address of random source data
                                                    6241
                                                    6242          TSTCODE
0005804C                                            6243+CODE76   DS    0F
                                                    6244 *                                  load random source
0005804C  B982 0022                                 6245          XGR   R2,R2
00058050  5820 5028                     00000028    6246          L     R2,RANDXA
                                                    6247 *
                                                    6248 * Triple Compare and Swap (TCSX: 32)
                                                    6249 * Build Parambter List
                                                    6250 *
00058054  B982 0011                                 6251          XGR   R1,R1
00058058  5810 501C                     0000001C    6252          L     R1,OPSRCA
                                                    6253
0005805C  B982 0033                                 6254          XGR   R3,R3
00058060  5830 5024                     00000024    6255          L     R3,PLA
                                                    6256
00058064  B982 0044                                 6257          XGR   R4,R4
00058068  5840 5020                     00000020    6258          L     R4,OPREPA
                                                    6259
0005806C  D22F 1000 2000        00000000  00000000  6260          MVC   0(48,R1),0(R2)      load Op1-Op3 data in source
00058072  D22F 3040 1000        00000040  00000000  6261          MVC   64(48,R3),0(R1)     copy to PL OP1-Op3 Compare
00058078  D22F 4000 2080        00000000  00000080  6262          MVC   0(48,R4),128(R2)    load initial Op1-Op3 data
0005807E  D22F 30A0 4000        000000A0  00000000  6263          MVC   160(48,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                    6264 ****
                                                    6265 * force a failure by changing the 1st operand
                                                    6266 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00058084  D203 1000 4000      00000000  00000000  6267           MVC   0(4,R1),0(R4)
                                                   6268 ****
                                                   6269
0005808A  E313 0000 0024                00000000  6270           STG   R1,0(R3)           load OP1 address in PL
                                                   6271
00058090  4111 0010                     00000010  6272           LA    R1,16(R1)
00058094  E313 0008 0024                00000008  6273           STG   R1,8(R3)           load OP2 address in PL
                                                   6274
0005809A  4111 0010                     00000010  6275           LA    R1,16(R1)
0005809E  E313 0010 0024                00000010  6276           STG   R1,16(R3)          load OP3 address in PL
                                                   6277
000580A4  B982 0000                               6278           XGR   R0,R0
000580A8  4100 0020                     00000020  6279           LA    R0,32
000580AC  EE00 0000 3000      00000000  00000000  6280           PLO   0,0,0,0(R3)        Triple C/S (TCSX: 32)
                                                   6281
000580B2  B98D 0020                               6282           EPSW  R2,R0              extract psw
000580B6  5020 5010                     00000010  6283           ST    R2,CCPSW             to save CC
000580BA  E310 500B 0076                0000000B  6284           LB    R1,CCMASK          (failure CC mask)
000580C0  8910 0004                     00000004  6285           SLL   R1,4               (shift to BC instr CC position)
000580C4  4410 50F8                     00058118  6286           EX    R1,F321TSTCC         fail if...
                                                   6287
                                000580C8  00000001  6288 F321CKUR EQU   *
000580C8  D200 5009 3030      00000009  00000030  6289           MVC   URFOUND,48(R3)     Extract UR
000580CE  D500 5008 3030      00000008  00000030  6290           CLC   UR,48(R3)
000580D4  4780 50BC                     000580DC  6291           BE    F321CKRES
000580D8  45F0 8220                     00000420  6292           BAL   R15,URMSG
                                                   6293 *
                                                   6294 * compare results
                                                   6295 *
                                000580DC  00000001  6296 F321CKRES EQU   *
                                                   6297 *                                reset adresses
000580DC  B982 0011                               6298           XGR   R1,R1
000580E0  5810 501C                     0000001C  6299           L     R1,OPSRCA          source
000580E4  B982 0033                               6300           XGR   R3,R3
000580E8  5830 5024                     00000024  6301           L     R3,PLA             PL
000580EC  B982 0044                               6302           XGR   R4,R4
000580F0  5840 5020                     00000020  6303           L     R4,OPREPA
                                                   6304
000580F4  D50F 1000 3040      00000000  00000040  6305           CLC   0(16,R1),64(R3)
000580FA  4770 50F2                     00058112  6306           BNE   F321XX             OP1C Different, error
000580FE  D50F 1010 3050      00000010  00000050  6307           CLC   16(16,R1),80(R3)
00058104  4770 50F2                     00058112  6308           BNE   F321XX             OP2C Different, error
00058108  D50F 1020 3060      00000020  00000060  6309           CLC   32(16,R1),96(R3)
0005810E  478B 0000                     00000000  6310           BE    0(R11)             OP3C same, return
                                                   6311
                                00058112  00000001  6312 F321XX   EQU   *
00058112  45F0 8340                     00000540  6313           BAL   R15,FAILMSG
00058116  07FB                                    6314           BR    R11
                                                   6315
00058118  4700 50FC                     0005811C  6316 F321TSTCC BC    0,F321CCMSG        fail if unexpected condition code
                                0005811C  00000001  6317 F321CCMSG EQU   *
0005811C  45F0 80F2                     000002F2  6318           BAL   R15,CCMSG
00058120  47F0 50A8                     000580C8  6319           B     F321CKUR
                                                   6320
                                                   6321           TSTDATA
00058124                        00058124  00058400  6322+          ORG   PLOETST+(200*K)+(38*4*K)+K
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   6323           TSTLBL OPDST              Operand Destination
00058400                                           6324+OPDST76   DS     0C
00058400  00000000 00000000                      6325           DC     XL256'0'
00058408  00000000 00000000
00058410  00000000 00000000
00058418  00000000 00000000
00058420  00000000 00000000
00058428  00000000 00000000
00058430  00000000 00000000
00058438  00000000 00000000
00058440  00000000 00000000
00058448  00000000 00000000
00058450  00000000 00000000
00058458  00000000 00000000
00058460  00000000 00000000
00058468  00000000 00000000
00058470  00000000 00000000
00058478  00000000 00000000
00058480  00000000 00000000
00058488  00000000 00000000
00058490  00000000 00000000
00058498  00000000 00000000
000584A0  00000000 00000000
000584A8  00000000 00000000
000584B0  00000000 00000000
000584B8  00000000 00000000
000584C0  00000000 00000000
000584C8  00000000 00000000
000584D0  00000000 00000000
000584D8  00000000 00000000
000584E0  00000000 00000000
000584E8  00000000 00000000
000584F0  00000000 00000000
000584F8  00000000 00000000
                                                   6326
                                                   6327           TSTLBL OPSRC              Operand Source OP1-OP4
00058500                                           6328+OPSRC76   DS     0C
00058500  00000000 00000000                      6329           DC     XL256'0'
00058508  00000000 00000000
00058510  00000000 00000000
00058518  00000000 00000000
00058520  00000000 00000000
00058528  00000000 00000000
00058530  00000000 00000000
00058538  00000000 00000000
00058540  00000000 00000000
00058548  00000000 00000000
00058550  00000000 00000000
00058558  00000000 00000000
00058560  00000000 00000000
00058568  00000000 00000000
00058570  00000000 00000000
00058578  00000000 00000000
00058580  00000000 00000000
00058588  00000000 00000000
00058590  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00058598  00000000 00000000
000585A0  00000000 00000000
000585A8  00000000 00000000
000585B0  00000000 00000000
000585B8  00000000 00000000
000585C0  00000000 00000000
000585C8  00000000 00000000
000585D0  00000000 00000000
000585D8  00000000 00000000
000585E0  00000000 00000000
000585E8  00000000 00000000
000585F0  00000000 00000000
000585F8  00000000 00000000
                                          6330
                                          6331           TSTLBL PL                  Parameter List
00058600                                  6332+PL76      DS    0C
00058600  00000000 00000000               6333           DC    XL352'00'
00058608  00000000 00000000
00058610  00000000 00000000
00058618  00000000 00000000
00058620  00000000 00000000
00058628  00000000 00000000
00058630  00000000 00000000
00058638  00000000 00000000
00058640  00000000 00000000
00058648  00000000 00000000
00058650  00000000 00000000
00058658  00000000 00000000
00058660  00000000 00000000
00058668  00000000 00000000
00058670  00000000 00000000
00058678  00000000 00000000
00058680  00000000 00000000
00058688  00000000 00000000
00058690  00000000 00000000
00058698  00000000 00000000
000586A0  00000000 00000000
000586A8  00000000 00000000
000586B0  00000000 00000000
000586B8  00000000 00000000
000586C0  00000000 00000000
000586C8  00000000 00000000
000586D0  00000000 00000000
000586D8  00000000 00000000
000586E0  00000000 00000000
000586E8  00000000 00000000
000586F0  00000000 00000000
000586F8  00000000 00000000
00058700  00000000 00000000
00058708  00000000 00000000
00058710  00000000 00000000
00058718  00000000 00000000
00058720  00000000 00000000
00058728  00000000 00000000
00058730  00000000 00000000
00058738  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00058740  00000000 00000000
00058748  00000000 00000000
00058750  00000000 00000000
00058758  00000000 00000000
                                          6334
                                          6335           TSTLBL FMSG                fail message
00058760                                  6336+FMSG76    DS     0C
00058760  86818993 858440D7               6337           DC     C'failed PLO-Extension '
00058768  D3D660C5 A7A38595
00058770  A2899695 40
00058775  E3998997 938540C3               6338           DC     CL96'Triple Compare and Swap (TCSX: 32): 1st NE'
0005877D  96949781 99854081
00058785  958440E2 A6819740
0005878D  4DE3C3E2 E77A40F3
00058795  F25D7A40 F1A2A340
0005879D  D5C54040 40404040
000587A5  40404040 40404040
000587AD  40404040 40404040
000587B5  40404040 40404040
000587BD  40404040 40404040
000587C5  40404040 40404040
000587CD  40404040 40404040
                                          6339
                                          6340           TSTEND
000587D6                                  6341+          DS     0H
000587D6  C5958440 968640A3               6342+          DC     CL32'End of test: 76'
000587DE  85A2A37A 40F7F640
000587E6  40404040 40404040
000587EE  40404040 40404040
                                          6343
                                          6344 * ------------------------------------------------------------------
                                          6345 * Test Triple Compare and Swap (TCSX: 32) instruction : 2nd NE
                                          6346 * ------------------------------------------------------------------
                                          6347           TSTHDR 2,2,32              2nd Operand NOT Equal (cc=2, UR=2)
000587F6                        000587F6  00059000  6348+          ORG    PLOETST+(200*K)+(39*4*K)
00059000  E2E3C1D9 E3409686               6349+          DC     CL32'START of test: 77'
00059008  40A385A2 A37A40F7
00059010  F7404040 40404040
00059018  40404040 40404040
00059020                        00059020           6350+          USING *,R5                 base for test data and test routine
00059020                        00000000           6351+          USING PLOETEST,R5
00059020  0005904C                        6352+T77       DC     A(CODE77)           address of test code
00059024  004D                            6353+          DC     H'77'               test number
00059026  00                              6354+          DC     X'00'
00059027  20                              6355+          DC     HL1'32'             Function
00059028  02                              6356+          DC     HL1'2'              expected unequal results (UR)
00059029  00                              6357+          DC     HL1'00'             found UR
0005902A  02                              6358+          DC     HL1'2'              cc
0005902B  0D                              6359+          DC     HL1'13'             not expected CC mask
0005902C  00                              6360+          DS     X                   extracted cc
00059030  00000000 00000000               6361+          DS     2F                  extract PSW after test (has CC)
00059038  00059760                        6362+          DC     A(FMSG77)           address of test fail message
0005903C  00059500                        6363+          DC     A(OPSRC77)          address of operand source
00059040  00059400                        6364+          DC     A(OPDST77)          address of operand destination
00059044  00059600                        6365+          DC     A(PL77)             address of parameter list
00059048  000011FE                        6366+          DC     A(RANDX+77)         address of random source data
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6367
                                                    6368          TSTCODE
0005904C                                            6369+CODE77   DS    0F
                                                    6370 *                                load random source
0005904C  B982 0022                                 6371          XGR   R2,R2
00059050  5820 5028                   00000028      6372          L     R2,RANDXA
                                                    6373 *
                                                    6374 * Triple Compare and Swap (TCSX: 32)
                                                    6375 * Build Parambter List
                                                    6376 *
00059054  B982 0011                                 6377          XGR   R1,R1
00059058  5810 501C                   0000001C      6378          L     R1,OPSRCA
                                                    6379
0005905C  B982 0033                                 6380          XGR   R3,R3
00059060  5830 5024                   00000024      6381          L     R3,PLA
                                                    6382
00059064  B982 0044                                 6383          XGR   R4,R4
00059068  5840 5020                   00000020      6384          L     R4,OPREPA
                                                    6385
0005906C  D22F 1000 2000      00000000  00000000    6386          MVC   0(48,R1),0(R2)      load Op1-Op3 data in source
00059072  D22F 3040 1000      00000040  00000000    6387          MVC   64(48,R3),0(R1)     copy to PL OP1-Op3 Compare
00059078  D22F 4000 2080      00000000  00000080    6388          MVC   0(48,R4),128(R2)    load initial Op1-Op3 data
0005907E  D22F 30A0 4000      000000A0  00000000    6389          MVC   160(48,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                    6390 ****
                                                    6391 * force a failure by changing the 2nd operand
                                                    6392 ****
00059084  D207 1010 4000      00000010  00000000    6393          MVC   16(8,R1),0(R4)
                                                    6394 ****
                                                    6395
0005908A  E313 0000 0024                00000000    6396          STG   R1,0(R3)            load OP1 address in PL
                                                    6397
00059090  4111 0010                     00000010    6398          LA    R1,16(R1)
00059094  E313 0008 0024                00000008    6399          STG   R1,8(R3)            load OP2 address in PL
                                                    6400
0005909A  4111 0010                     00000010    6401          LA    R1,16(R1)
0005909E  E313 0010 0024                00000010    6402          STG   R1,16(R3)           load OP3 address in PL
                                                    6403
000590A4  B982 0000                                 6404          XGR   R0,R0
000590A8  4100 0020                     00000020    6405          LA    R0,32
000590AC  EE00 0000 3000      00000000  00000000    6406          PLO   0,0,0,0(R3)         Triple C/S (TCSX: 32)
                                                    6407
000590B2  B98D 0020                                 6408          EPSW  R2,R0               extract psw
000590B6  5020 5010                     00000010    6409          ST    R2,CCPSW              to save CC
000590BA  E310 500B 0076                0000000B    6410          LB    R1,CCMASK           (failure CC mask)
000590C0  8910 0004                     00000004    6411          SLL   R1,4                (shift to BC instr CC position)
000590C4  4410 50F8                     00059118    6412          EX    R1,F322TSTCC            fail if...
                                                    6413
                              000590C8  00000001    6414 F322CKUR EQU   *
000590C8  D200 5009 3030      00000009  00000030    6415          MVC   URFOUND,48(R3)      Extract UR
000590CE  D500 5008 3030      00000008  00000030    6416          CLC   UR,48(R3)
000590D4  4780 50BC                     000590DC    6417          BE    F322CKRES
000590D8  45F0 8220                     00000420    6418          BAL   R15,URMSG
                                                    6419 *
                                                    6420 * compare results
                                                    6421 *
                              000590DC  00000001    6422 F322CKRES EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6423 *                                 reset adresses
000590DC  B982 0011                                 6424           XGR   R1,R1
000590E0  5810 501C                       0000001C  6425           L     R1,OPSRCA         source
000590E4  B982 0033                                 6426           XGR   R3,R3
000590E8  5830 5024                       00000024  6427           L     R3,PLA            PL
000590EC  B982 0044                                 6428           XGR   R4,R4
000590F0  5840 5020                       00000020  6429           L     R4,OPREPA
                                                    6430
000590F4  D50F 1000 3040  00000000        00000040  6431           CLC   0(16,R1),64(R3)
000590FA  4770 50F2                       00059112  6432           BNE   F322XX            OP1C Different, error
000590FE  D50F 1010 3050  00000010        00000050  6433           CLC   16(16,R1),80(R3)
00059104  4770 50F2                       00059112  6434           BNE   F322XX            OP2C Different, error
00059108  D50F 1020 3060  00000020        00000060  6435           CLC   32(16,R1),96(R3)
0005910E  478B 0000                       00000000  6436           BE    0(R11)            OP3C same, return
                                                    6437
                          00059112        00000001  6438 F322XX    EQU   *
00059112  45F0 8340                       00000540  6439           BAL   R15,FAILMSG
00059116  07FB                                      6440           BR    R11
                                                    6441
00059118  4700 50FC                       0005911C  6442 F322TSTCC BC    0,F322CCMSG       fail if unexpected condition code
                          0005911C        00000001  6443 F322CCMSG EQU   *
0005911C  45F0 80F2                       000002F2  6444           BAL   R15,CCMSG
00059120  47F0 50A8                       000590C8  6445           B     F322CKUR
                                                    6446
                                                    6447           TSTDATA
00059124                  00059124        00059400  6448+          ORG   PLOETST+(200*K)+(39*4*K)+K
                                                    6449           TSTLBL OPDST            Operand Destination
00059400                                            6450+OPDST77   DS    0C
00059400  00000000 00000000                         6451           DC    XL256'0'
00059408  00000000 00000000
00059410  00000000 00000000
00059418  00000000 00000000
00059420  00000000 00000000
00059428  00000000 00000000
00059430  00000000 00000000
00059438  00000000 00000000
00059440  00000000 00000000
00059448  00000000 00000000
00059450  00000000 00000000
00059458  00000000 00000000
00059460  00000000 00000000
00059468  00000000 00000000
00059470  00000000 00000000
00059478  00000000 00000000
00059480  00000000 00000000
00059488  00000000 00000000
00059490  00000000 00000000
00059498  00000000 00000000
000594A0  00000000 00000000
000594A8  00000000 00000000
000594B0  00000000 00000000
000594B8  00000000 00000000
000594C0  00000000 00000000
000594C8  00000000 00000000
000594D0  00000000 00000000
000594D8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000594E0  00000000 00000000
000594E8  00000000 00000000
000594F0  00000000 00000000
000594F8  00000000 00000000
                                        6452
                                        6453           TSTLBL OPSRC              Operand Source OP1-OP4
00059500                                6454+OPSRC77   DS    0C
00059500  00000000 00000000             6455           DC    XL256'0'
00059508  00000000 00000000
00059510  00000000 00000000
00059518  00000000 00000000
00059520  00000000 00000000
00059528  00000000 00000000
00059530  00000000 00000000
00059538  00000000 00000000
00059540  00000000 00000000
00059548  00000000 00000000
00059550  00000000 00000000
00059558  00000000 00000000
00059560  00000000 00000000
00059568  00000000 00000000
00059570  00000000 00000000
00059578  00000000 00000000
00059580  00000000 00000000
00059588  00000000 00000000
00059590  00000000 00000000
00059598  00000000 00000000
000595A0  00000000 00000000
000595A8  00000000 00000000
000595B0  00000000 00000000
000595B8  00000000 00000000
000595C0  00000000 00000000
000595C8  00000000 00000000
000595D0  00000000 00000000
000595D8  00000000 00000000
000595E0  00000000 00000000
000595E8  00000000 00000000
000595F0  00000000 00000000
000595F8  00000000 00000000
                                        6456
                                        6457           TSTLBL PL                 Parameter List
00059600                                6458+PL77      DS    0C
00059600  00000000 00000000             6459           DC    XL352'00'
00059608  00000000 00000000
00059610  00000000 00000000
00059618  00000000 00000000
00059620  00000000 00000000
00059628  00000000 00000000
00059630  00000000 00000000
00059638  00000000 00000000
00059640  00000000 00000000
00059648  00000000 00000000
00059650  00000000 00000000
00059658  00000000 00000000
00059660  00000000 00000000
00059668  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00059670  00000000 00000000
00059678  00000000 00000000
00059680  00000000 00000000
00059688  00000000 00000000
00059690  00000000 00000000
00059698  00000000 00000000
000596A0  00000000 00000000
000596A8  00000000 00000000
000596B0  00000000 00000000
000596B8  00000000 00000000
000596C0  00000000 00000000
000596C8  00000000 00000000
000596D0  00000000 00000000
000596D8  00000000 00000000
000596E0  00000000 00000000
000596E8  00000000 00000000
000596F0  00000000 00000000
000596F8  00000000 00000000
00059700  00000000 00000000
00059708  00000000 00000000
00059710  00000000 00000000
00059718  00000000 00000000
00059720  00000000 00000000
00059728  00000000 00000000
00059730  00000000 00000000
00059738  00000000 00000000
00059740  00000000 00000000
00059748  00000000 00000000
00059750  00000000 00000000
00059758  00000000 00000000
                                          6460
                                          6461           TSTLBL FMSG               fail message
00059760                                  6462+FMSG77    DS    0C
00059760  86818993 858440D7               6463           DC    C'failed PLO-Extension '
00059768  D3D660C5 A7A38595
00059770  A2899695 40
00059775  E3998997 938540C3               6464           DC    CL96'Triple Compare and Swap (TCSX: 32): 2nd NE'
0005977D  96949781 99854081
00059785  958440E2 A6819740
0005978D  4DE3C3E2 E77A40F3
00059795  F25D7A40 F2958440
0005979D  D5C54040 40404040
000597A5  40404040 40404040
000597AD  40404040 40404040
000597B5  40404040 40404040
000597BD  40404040 40404040
000597C5  40404040 40404040
000597CD  40404040 40404040
                                          6465
                                          6466           TSTEND
000597D6                                  6467+          DS    0H
000597D6  C5958440 968640A3               6468+          DC    CL32'End of test: 77'
000597DE  85A2A37A 40F7F740
000597E6  40404040 40404040
000597EE  40404040 40404040
                                          6469
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6470 * ------------------------------------------------------------------------
                                                    6471 * Test Triple Compare and Swap (TCSX: 32) instruction : 3nd NE
                                                    6472 * ------------------------------------------------------------------------
                                                    6473          TSTHDR 2,3,32             3nd Operand NOT Equal (cc=2, UR=3)
000597F6                      000597F6  0005A000    6474+         ORG   PLOETST+(200*K)+(40*4*K)
0005A000  E2E3C1D9 E3409686                         6475+         DC    CL32'START of test: 78'
0005A008  40A385A2 A37A40F7
0005A010  F8404040 40404040
0005A018  40404040 40404040
0005A020                      0005A020              6476+         USING *,R5                base for test data and test routine
0005A020                      00000000              6477+         USING PLOETEST,R5
0005A020  0005A04C                                  6478+T78      DC    A(CODE78)           address of test code
0005A024  004E                                      6479+         DC    H'78'               test number
0005A026  00                                        6480+         DC    X'00'
0005A027  20                                        6481+         DC    HL1'32'             Function
0005A028  03                                        6482+         DC    HL1'3'              expected unequal results (UR)
0005A029  00                                        6483+         DC    HL1'00'             found UR
0005A02A  02                                        6484+         DC    HL1'2'              cc
0005A02B  0D                                        6485+         DC    HL1'13'             not expected CC mask
0005A02C  00                                        6486+         DC    X                   extracted cc
0005A030  00000000 00000000                         6487+         DS    2F                  extract PSW after test (has CC)
0005A038  0005A760                                  6488+         DC    A(FMSG78)           address of test fail message
0005A03C  0005A500                                  6489+         DC    A(OPSRC78)          address of operand source
0005A040  0005A400                                  6490+         DC    A(OPDST78)          address of operand destination
0005A044  0005A600                                  6491+         DC    A(PL78)             address of parameter list
0005A048  000011FF                                  6492+         DC    A(RANDX+78)         address of random source data
                                                    6493
                                                    6494          TSTCODE
0005A04C                                            6495+CODE78   DS    0F
                                                    6496 *                                  load random source
0005A04C  B982 0022                                 6497          XGR   R2,R2
0005A050  5820 5028                   00000028      6498          L     R2,RANDXA
                                                    6499 *
                                                    6500 * Triple Compare and Swap (TCSX: 32)
                                                    6501 * Build Parambter List
                                                    6502 *
0005A054  B982 0011                                 6503          XGR   R1,R1
0005A058  5810 501C                   0000001C      6504          L     R1,OPSRCA
                                                    6505
0005A05C  B982 0033                                 6506          XGR   R3,R3
0005A060  5830 5024                   00000024      6507          L     R3,PLA
                                                    6508
0005A064  B982 0044                                 6509          XGR   R4,R4
0005A068  5840 5020                   00000020      6510          L     R4,OPREPA
                                                    6511
0005A06C  D22F 1000 2000      00000000  00000000    6512          MVC   0(48,R1),0(R2)      load Op1-Op3 data in source
0005A072  D22F 3040 1000      00000040  00000000    6513          MVC   64(48,R3),0(R1)     copy to PL OP1-Op3 Compare
0005A078  D22F 4000 2080      00000000  00000080    6514          MVC   0(48,R4),128(R2)    load initial Op1-Op3 data
0005A07E  D22F 30A0 4000      000000A0  00000000    6515          MVC   160(48,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                    6516 ****
                                                    6517 * force a failure by changing the 3nd operand
                                                    6518 ****
0005A084  D207 1020 4000      00000020  00000000    6519          MVC   32(8,R1),0(R4)
                                                    6520 ****
                                                    6521
0005A08A  E313 0000 0024                00000000    6522          STG   R1,0(R3)            load OP1 address in PL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6523
0005A090  4111 0010                     00000010  6524          LA    R1,16(R1)
0005A094  E313 0008 0024                00000008  6525          STG   R1,8(R3)           load OP2 address in PL
                                                    6526
0005A09A  4111 0010                     00000010  6527          LA    R1,16(R1)
0005A09E  E313 0010 0024                00000010  6528          STG   R1,16(R3)          load OP3 address in PL
                                                    6529
0005A0A4  B982 0000                               6530          XGR   R0,R0
0005A0A8  4100 0020                     00000020  6531          LA    R0,32
0005A0AC  EE00 0000 3000  00000000      00000000  6532          PLO   0,0,0,0(R3)        Triple C/S (TCSX: 32)
                                                    6533
0005A0B2  B98D 0020                               6534          EPSW  R2,R0              extract psw
0005A0B6  5020 5010                     00000010  6535          ST    R2,CCPSW             to save CC
0005A0BA  E310 500B 0076                0000000B  6536          LB    R1,CCMASK          (failure CC mask)
0005A0C0  8910 0004                     00000004  6537          SLL   R1,4               (shift to BC instr CC position)
0005A0C4  4410 50F8                     0005A118  6538          EX    R1,F323TSTCC           fail if...
                                                    6539
                          0005A0C8      00000001  6540 F323CKUR EQU   *
0005A0C8  D200 5009 3030  00000009      00000030  6541          MVC   URFOUND,48(R3)     Extract UR
0005A0CE  D500 5008 3030  00000008      00000030  6542          CLC   UR,48(R3)
0005A0D4  4780 50BC                     0005A0DC  6543          BE    F323CKRES
0005A0D8  45F0 8220                     00000420  6544          BAL   R15,URMSG
                                                    6545 *
                                                    6546 * compare results
                                                    6547 *
                          0005A0DC      00000001  6548 F323CKRES EQU   *
                                                    6549 *                             reset adresses
0005A0DC  B982 0011                               6550          XGR   R1,R1
0005A0E0  5810 501C                     0000001C  6551          L     R1,OPSRCA          source
0005A0E4  B982 0033                               6552          XGR   R3,R3
0005A0E8  5830 5024                     00000024  6553          L     R3,PLA             PL
0005A0EC  B982 0044                               6554          XGR   R4,R4
0005A0F0  5840 5020                     00000020  6555          L     R4,OPREPA
                                                    6556
0005A0F4  D50F 1000 3040  00000000      00000040  6557          CLC   0(16,R1),64(R3)
0005A0FA  4770 50F2                     0005A112  6558          BNE   F323XX             OP1C Different, error
0005A0FE  D50F 1010 3050  00000010      00000050  6559          CLC   16(16,R1),80(R3)
0005A104  4770 50F2                     0005A112  6560          BNE   F323XX             OP2C Different, error
0005A108  D50F 1020 3060  00000020      00000060  6561          CLC   32(16,R1),96(R3)
0005A10E  478B 0000                     00000000  6562          BE    0(R11)             OP3C same, return
                                                    6563
                          0005A112      00000001  6564 F323XX   EQU   *
0005A112  45F0 8340                     00000540  6565          BAL   R15,FAILMSG
0005A116  07FB                                    6566          BR    R11
                                                    6567
0005A118  4700 50FC                     0005A11C  6568 F323TSTCC BC    0,F323CCMSG        fail if unexpected condition code
                          0005A11C      00000001  6569 F323CCMSG EQU   *
0005A11C  45F0 80F2                     000002F2  6570          BAL   R15,CCMSG
0005A120  47F0 50A8                     0005A0C8  6571          B     F323CKUR
                                                    6572
                                                    6573          TSTDATA
0005A124                  0005A124      0005A400  6574+         ORG   PLOETST+(200*K)+(40*4*K)+K
                                                    6575          TSTLBL OPDST             Operand Destination
0005A400                                          6576+OPDST78  DS    0C
0005A400  00000000 00000000                       6577          DC    XL256'0'
0005A408  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0005A410  00000000 00000000
0005A418  00000000 00000000
0005A420  00000000 00000000
0005A428  00000000 00000000
0005A430  00000000 00000000
0005A438  00000000 00000000
0005A440  00000000 00000000
0005A448  00000000 00000000
0005A450  00000000 00000000
0005A458  00000000 00000000
0005A460  00000000 00000000
0005A468  00000000 00000000
0005A470  00000000 00000000
0005A478  00000000 00000000
0005A480  00000000 00000000
0005A488  00000000 00000000
0005A490  00000000 00000000
0005A498  00000000 00000000
0005A4A0  00000000 00000000
0005A4A8  00000000 00000000
0005A4B0  00000000 00000000
0005A4B8  00000000 00000000
0005A4C0  00000000 00000000
0005A4C8  00000000 00000000
0005A4D0  00000000 00000000
0005A4D8  00000000 00000000
0005A4E0  00000000 00000000
0005A4E8  00000000 00000000
0005A4F0  00000000 00000000
0005A4F8  00000000 00000000
                                          6578
                                          6579           TSTLBL OPSRC              Operand Source OP1-OP4
0005A500                                  6580+OPSRC78   DS    0C
0005A500  00000000 00000000               6581           DC    XL256'0'
0005A508  00000000 00000000
0005A510  00000000 00000000
0005A518  00000000 00000000
0005A520  00000000 00000000
0005A528  00000000 00000000
0005A530  00000000 00000000
0005A538  00000000 00000000
0005A540  00000000 00000000
0005A548  00000000 00000000
0005A550  00000000 00000000
0005A558  00000000 00000000
0005A560  00000000 00000000
0005A568  00000000 00000000
0005A570  00000000 00000000
0005A578  00000000 00000000
0005A580  00000000 00000000
0005A588  00000000 00000000
0005A590  00000000 00000000
0005A598  00000000 00000000
0005A5A0  00000000 00000000
0005A5A8  00000000 00000000
0005A5B0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0005A5B8  00000000 00000000
0005A5C0  00000000 00000000
0005A5C8  00000000 00000000
0005A5D0  00000000 00000000
0005A5D8  00000000 00000000
0005A5E0  00000000 00000000
0005A5E8  00000000 00000000
0005A5F0  00000000 00000000
0005A5F8  00000000 00000000
                                          6582
                                          6583           TSTLBL PL                  Parameter List
0005A600                                  6584+PL78      DS    0C
0005A600  00000000 00000000               6585           DC    XL352'00'
0005A608  00000000 00000000
0005A610  00000000 00000000
0005A618  00000000 00000000
0005A620  00000000 00000000
0005A628  00000000 00000000
0005A630  00000000 00000000
0005A638  00000000 00000000
0005A640  00000000 00000000
0005A648  00000000 00000000
0005A650  00000000 00000000
0005A658  00000000 00000000
0005A660  00000000 00000000
0005A668  00000000 00000000
0005A670  00000000 00000000
0005A678  00000000 00000000
0005A680  00000000 00000000
0005A688  00000000 00000000
0005A690  00000000 00000000
0005A698  00000000 00000000
0005A6A0  00000000 00000000
0005A6A8  00000000 00000000
0005A6B0  00000000 00000000
0005A6B8  00000000 00000000
0005A6C0  00000000 00000000
0005A6C8  00000000 00000000
0005A6D0  00000000 00000000
0005A6D8  00000000 00000000
0005A6E0  00000000 00000000
0005A6E8  00000000 00000000
0005A6F0  00000000 00000000
0005A6F8  00000000 00000000
0005A700  00000000 00000000
0005A708  00000000 00000000
0005A710  00000000 00000000
0005A718  00000000 00000000
0005A720  00000000 00000000
0005A728  00000000 00000000
0005A730  00000000 00000000
0005A738  00000000 00000000
0005A740  00000000 00000000
0005A748  00000000 00000000
0005A750  00000000 00000000
0005A758  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6586
                                                    6587          TSTLBL FMSG               fail message
0005A760                                            6588+FMSG78   DS    0C
0005A760  86818993 858440D7                         6589          DC    C'failed PLO-Extension '
0005A768  D3D660C5 A7A38595
0005A770  A2899695 40
0005A775  E3998997 938540C3                         6590          DC    CL96'Triple Compare and Swap (TCSX: 32): 3nd NE'
0005A77D  96949781 99854081
0005A785  958440E2 A6819740
0005A78D  4DE3C3E2 E77A40F3
0005A795  F25D7A40 F3958440
0005A79D  D5C54040 40404040
0005A7A5  40404040 40404040
0005A7AD  40404040 40404040
0005A7B5  40404040 40404040
0005A7BD  40404040 40404040
0005A7C5  40404040 40404040
0005A7CD  40404040 40404040
                                                    6591
                                                    6592          TSTEND
0005A7D6                                            6593+         DS    0H
0005A7D6  C5958440 968640A3                         6594+         DC    CL32'End of test: 78'
0005A7DE  85A2A37A 40F7F840
0005A7E6  40404040 40404040
0005A7EE  40404040 40404040
                                                    6595
                                                    6596 * -------------------------------------------------------------------
                                                    6597 * Test Triple Compare and Swap (TCSX: 32) instruction : All EQ
                                                    6598 * -------------------------------------------------------------------
                                                    6599          TSTHDR 0,99,32             All Equal (cc=0, UR=99)
0005A7F6                        0005A7F6  0005B000  6600+         ORG   PLOETST+(200*K)+(41*4*K)
0005B000  E2E3C1D9 E3409686                         6601+         DC    CL32'START of test: 79'
0005B008  40A385A2 A37A40F7
0005B010  F9404040 40404040
0005B018  40404040 40404040
0005B020                        0005B020            6602+         USING *,R5                base for test data and test routine
0005B020                        00000000            6603+         USING PLOETEST,R5
0005B020  0005B04C                                  6604+T79      DC    A(CODE79)           address of test code
0005B024  004F                                      6605+         DC    H'79'               test number
0005B026  00                                        6606+         DC    X'00'
0005B027  20                                        6607+         DC    HL1'32'             Function
0005B028  63                                        6608+         DC    HL1'99'             expected unequal results (UR)
0005B029  00                                        6609+         DC    HL1'00'             found UR
0005B02A  00                                        6610+         DC    HL1'0'              cc
0005B02B  07                                        6611+         DC    HL1'7'              not expected CC mask
0005B02C  00                                        6612+         DS    X                   extracted cc
0005B030  00000000 00000000                         6613+         DS    2F                  extract PSW after test (has CC)
0005B038  0005B760                                  6614+         DC    A(FMSG79)           address of test fail message
0005B03C  0005B500                                  6615+         DC    A(OPSRC79)          address of operand source
0005B040  0005B400                                  6616+         DC    A(OPDST79)          address of operand destination
0005B044  0005B600                                  6617+         DC    A(PL79)             address of parameter list
0005B048  00001200                                  6618+         DC    A(RANDX+79)         address of random source data
                                                    6619
                                                    6620          TSTCODE
0005B04C                                            6621+CODE79   DS    0F
                                                    6622 *                                  load random source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0005B04C  B982 0022                              6623          XGR   R2,R2
0005B050  5820 5028                   00000028   6624          L     R2,RANDXA
                                                 6625 *
                                                 6626 * Triple Compare and Swap (TCSX: 32)
                                                 6627 * Build Parambter List
                                                 6628 *
0005B054  B982 0011                              6629          XGR   R1,R1
0005B058  5810 501C                   0000001C   6630          L     R1,OPSRCA
                                                 6631
0005B05C  B982 0033                              6632          XGR   R3,R3
0005B060  5830 5024                   00000024   6633          L     R3,PLA
                                                 6634
0005B064  B982 0044                              6635          XGR   R4,R4
0005B068  5840 5020                   00000020   6636          L     R4,OPREPA
                                                 6637
0005B06C  D22F 1000 2000      00000000  00000000   6638          MVC   0(48,R1),0(R2)      load Op1-Op3 data in source
0005B072  D22F 3040 1000      00000040  00000000   6639          MVC   64(48,R3),0(R1)     copy to PL OP1-Op3 Compare
0005B078  D22F 4000 2080      00000000  00000080   6640          MVC   0(48,R4),128(R2)    load initial Op1-Op3 data
0005B07E  D22F 30A0 4000      000000A0  00000000   6641          MVC   160(48,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                 6642 ****
                                                 6643 * Do Not force a failure by changing the 3nd operand
                                                 6644 ****
                                                 6645 *         MVC   32(8,R1),0(R4)
                                                 6646 ****
                                                 6647
0005B084  E313 0000 0024                   00000000  6648          STG   R1,0(R3)            load OP1 address in PL
                                                 6649
0005B08A  4111 0010                   00000010   6650          LA    R1,16(R1)
0005B08E  E313 0008 0024                   00000008  6651          STG   R1,8(R3)            load OP2 address in PL
                                                 6652
0005B094  4111 0010                   00000010   6653          LA    R1,16(R1)
0005B098  E313 0010 0024                   00000010  6654          STG   R1,16(R3)           load OP3 address in PL
                                                 6655
0005B09E  B982 0000                              6656          XGR   R0,R0
0005B0A2  4100 0020                   00000020   6657          LA    R0,32
0005B0A6  EE00 0000 3000      00000000  00000000   6658          PLO   0,0,0,0(R3)         Triple C/S (TCSX: 32)
                                                 6659
0005B0AC  B98D 0020                              6660          EPSW  R2,R0               extract psw
0005B0B0  5020 5010                   00000010   6661          ST    R2,CCPSW              to save CC
0005B0B4  E310 500B 0076                   0000000B  6662          LB    R1,CCMASK           (failure CC mask)
0005B0BA  8910 0004                   00000004   6663          SLL   R1,4                (shift to BC instr CC position)
0005B0BE  4410 50DE                   0005B0FE   6664          EX    R1,F324TSTCC            fail if...
                                                 6665 *
                                                 6666 * compare results
                                                 6667 *
                              0005B0C2  00000001   6668 F324CKRES EQU   *
                                                 6669 *                                    reset adresses
0005B0C2  B982 0011                              6670          XGR   R1,R1
0005B0C6  5810 501C                   0000001C   6671          L     R1,OPSRCA           source
0005B0CA  B982 0033                              6672          XGR   R3,R3
0005B0CE  5830 5024                   00000024   6673          L     R3,PLA              PL
0005B0D2  B982 0044                              6674          XGR   R4,R4
0005B0D6  5840 5020                   00000020   6675          L     R4,OPREPA
                                                 6676
                                                 6677 * All equal; replacement values copied to operand source
                                                 6678
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0005B0DA  D50F 1000 3070      00000000  00000070  6679           CLC   0(16,R1),112(R3)
0005B0E0  4770 50D8                     0005B0F8  6680           BNE   F324XX               OP1R Different, error
0005B0E4  D50F 1010 3080      00000010  00000080  6681           CLC   16(16,R1),128(R3)
0005B0EA  4770 50D8                     0005B0F8  6682           BNE   F324XX               OP2R Different, error
0005B0EE  D50F 1020 3090      00000020  00000090  6683           CLC   32(16,R1),144(R3)
0005B0F4  478B 0000                     00000000  6684           BE    0(R11)
                                                  6685
                                        0005B0F8  00000001  6686 F324XX   EQU   *
0005B0F8  45F0 8340                     00000540  6687           BAL   R15,FAILMSG
0005B0FC  07FB                                    6688           BR    R11
                                                  6689
0005B0FE  4700 50E2                     0005B102  6690 F324TSTCC BC    0,F324CCMSG          fail if unexpected condition code
                                        0005B102  00000001  6691 F324CCMSG EQU   *
0005B102  45F0 80F2                     000002F2  6692           BAL   R15,CCMSG
0005B106  47F0 50A2                     0005B0C2  6693           B     F324CKRES
                                                  6694
                                                  6695           TSTDATA
0005B10A                        0005B10A  0005B400  6696+          ORG   PLOETST+(200*K)+(41*4*K)+K
                                                  6697           TSTLBL OPDST               Operand Destination
0005B400                                          6698+OPDST79   DS    0C
0005B400  00000000 00000000               0005B400  6699           DC    XL256'0'
0005B408  00000000 00000000
0005B410  00000000 00000000
0005B418  00000000 00000000
0005B420  00000000 00000000
0005B428  00000000 00000000
0005B430  00000000 00000000
0005B438  00000000 00000000
0005B440  00000000 00000000
0005B448  00000000 00000000
0005B450  00000000 00000000
0005B458  00000000 00000000
0005B460  00000000 00000000
0005B468  00000000 00000000
0005B470  00000000 00000000
0005B478  00000000 00000000
0005B480  00000000 00000000
0005B488  00000000 00000000
0005B490  00000000 00000000
0005B498  00000000 00000000
0005B4A0  00000000 00000000
0005B4A8  00000000 00000000
0005B4B0  00000000 00000000
0005B4B8  00000000 00000000
0005B4C0  00000000 00000000
0005B4C8  00000000 00000000
0005B4D0  00000000 00000000
0005B4D8  00000000 00000000
0005B4E0  00000000 00000000
0005B4E8  00000000 00000000
0005B4F0  00000000 00000000
0005B4F8  00000000 00000000
                                                  6700
                                                  6701           TSTLBL OPSRC               Operand Source OP1-OP4
0005B500                                          6702+OPSRC79   DS    0C
0005B500  00000000 00000000               0005B500  6703           DC    XL256'0'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0005B508  00000000 00000000
0005B510  00000000 00000000
0005B518  00000000 00000000
0005B520  00000000 00000000
0005B528  00000000 00000000
0005B530  00000000 00000000
0005B538  00000000 00000000
0005B540  00000000 00000000
0005B548  00000000 00000000
0005B550  00000000 00000000
0005B558  00000000 00000000
0005B560  00000000 00000000
0005B568  00000000 00000000
0005B570  00000000 00000000
0005B578  00000000 00000000
0005B580  00000000 00000000
0005B588  00000000 00000000
0005B590  00000000 00000000
0005B598  00000000 00000000
0005B5A0  00000000 00000000
0005B5A8  00000000 00000000
0005B5B0  00000000 00000000
0005B5B8  00000000 00000000
0005B5C0  00000000 00000000
0005B5C8  00000000 00000000
0005B5D0  00000000 00000000
0005B5D8  00000000 00000000
0005B5E0  00000000 00000000
0005B5E8  00000000 00000000
0005B5F0  00000000 00000000
0005B5F8  00000000 00000000
                                                  6704
                                                  6705           TSTLBL PL                  Parameter List
0005B600                                          6706+PL79      DS    0C
0005B600  00000000 00000000                       6707           DC    XL352'00'
0005B608  00000000 00000000
0005B610  00000000 00000000
0005B618  00000000 00000000
0005B620  00000000 00000000
0005B628  00000000 00000000
0005B630  00000000 00000000
0005B638  00000000 00000000
0005B640  00000000 00000000
0005B648  00000000 00000000
0005B650  00000000 00000000
0005B658  00000000 00000000
0005B660  00000000 00000000
0005B668  00000000 00000000
0005B670  00000000 00000000
0005B678  00000000 00000000
0005B680  00000000 00000000
0005B688  00000000 00000000
0005B690  00000000 00000000
0005B698  00000000 00000000
0005B6A0  00000000 00000000
0005B6A8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0005B6B0  00000000 00000000
0005B6B8  00000000 00000000
0005B6C0  00000000 00000000
0005B6C8  00000000 00000000
0005B6D0  00000000 00000000
0005B6D8  00000000 00000000
0005B6E0  00000000 00000000
0005B6E8  00000000 00000000
0005B6F0  00000000 00000000
0005B6F8  00000000 00000000
0005B700  00000000 00000000
0005B708  00000000 00000000
0005B710  00000000 00000000
0005B718  00000000 00000000
0005B720  00000000 00000000
0005B728  00000000 00000000
0005B730  00000000 00000000
0005B738  00000000 00000000
0005B740  00000000 00000000
0005B748  00000000 00000000
0005B750  00000000 00000000
0005B758  00000000 00000000
                                          6708
                                          6709          TSTLBL FMSG               fail message
0005B760                                  6710+FMSG79   DS    0C
0005B760  86818993 858440D7               6711          DC    C'failed PLO-Extension '
0005B768  D3D660C5 A7A38595
0005B770  A2899695 40
0005B775  E3998997 938540C3               6712          DC    CL96'Triple Compare and Swap (TCSX: 32): All EQ'
0005B77D  96949781 99854081
0005B785  958440E2 A6819740
0005B78D  4DE3C3E2 E77A40F3
0005B795  F25D7A40 C1939340
0005B79D  C5D84040 40404040
0005B7A5  40404040 40404040
0005B7AD  40404040 40404040
0005B7B5  40404040 40404040
0005B7BD  40404040 40404040
0005B7C5  40404040 40404040
0005B7CD  40404040 40404040
                                          6713
                                          6714          TSTEND
0005B7D6                                  6715+         DS    0H
0005B7D6  C5958440 968640A3               6716+         DC    CL32'End of test: 79'
0005B7DE  85A2A37A 40F7F940
0005B7E6  40404040 40404040
0005B7EE  40404040 40404040
                                          6717
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6719 *=============================================================================
                                                    6720 *       Triple Compare and Swap Tests
                                                    6721 *=============================================================================
                                                    6722
                                                    6723 * -----------------------------------------------------------------------
                                                    6724 * Test Triple Compare and Swap (TCSG: 31) instruction : 1st NE
                                                    6725 * -----------------------------------------------------------------------
                                                    6726          TSTHDR 2,1,31             1st Operand NOT Equal (cc=2, UR=1)
0005B7F6                          0005B7F6  0005C000  6727+         ORG   PLOETST+(200*K)+(42*4*K)
0005C000  E2E3C1D9 E3409686                         6728+         DC    CL32'START of test: 80'
0005C008  40A385A2 A37A40F8
0005C010  F0404040 40404040
0005C018  40404040 40404040
0005C020                          0005C020          6729+         USING *,R5                base for test data and test routine
0005C020                          00000000          6730+         USING PLOETEST,R5
0005C020  0005C04C                                  6731+T80      DC    A(CODE80)           address of test code
0005C024  0050                                      6732+         DC    H'80'               test number
0005C026  00                                        6733+         DC    X'00'
0005C027  1F                                        6734+         DC    HL1'31'             Function
0005C028  01                                        6735+         DC    HL1'1'              expected unequal results (UR)
0005C029  00                                        6736+         DC    HL1'00'             found UR
0005C02A  02                                        6737+         DC    HL1'2'              cc
0005C02B  0D                                        6738+         DC    HL1'13'             not expected CC mask
0005C02C  00                                        6739+         DS    X                   extracted cc
0005C030  00000000 00000000                         6740+         DS    2F                  extract PSW after test (has CC)
0005C038  0005C760                                  6741+         DC    A(FMSG80)           address of test fail message
0005C03C  0005C500                                  6742+         DC    A(OPSRC80)          address of operand source
0005C040  0005C400                                  6743+         DC    A(OPDST80)          address of operand destination
0005C044  0005C600                                  6744+         DC    A(PL80)             address of parameter list
0005C048  00001201                                  6745+         DC    A(RANDX+80)         address of random source data
                                                    6746
                                                    6747          TSTCODE
0005C04C                                            6748+CODE80   DS    0F
                                                    6749 *                                  load random source
0005C04C  B982 0022                                 6750          XGR   R2,R2
0005C050  5820 5028                     00000028    6751          L     R2,RANDXA
                                                    6752 *
                                                    6753 * Triple Compare and Swap (TCSG: 31)
                                                    6754 * Build Parambter List
                                                    6755 *
0005C054  B982 0011                                 6756          XGR   R1,R1
0005C058  5810 501C                     0000001C    6757          L     R1,OPSRCA
                                                    6758
0005C05C  B982 0033                                 6759          XGR   R3,R3
0005C060  5830 5024                     00000024    6760          L     R3,PLA
                                                    6761
0005C064  B982 0044                                 6762          XGR   R4,R4
0005C068  5840 5020                     00000020    6763          L     R4,OPREPA
                                                    6764
0005C06C  D217 1000 2000        00000000  00000000  6765          MVC   0(24,R1),0(R2)      load Op1-Op3 data in source
0005C072  D217 3040 1000        00000040  00000000  6766          MVC   64(24,R3),0(R1)     copy to PL OP1-Op3 Compare
0005C078  D217 4000 2080        00000000  00000080  6767          MVC   0(24,R4),128(R2)    load initial Op1-Op3 data
0005C07E  D217 3058 4000        00000058  00000000  6768          MVC   88(24,R3),0(R4)     copy to PL Op1-Op3 Replacement
                                                    6769 ****
                                                    6770 * force a failure by changing the 1st operand
                                                    6771 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0005C084  D203 1000 4000      00000000  00000000  6772           MVC   0(4,R1),0(R4)
                                                  6773 ****
                                                  6774
0005C08A  E313 0000 0024                00000000  6775           STG   R1,0(R3)           load OP1 address in PL
                                                  6776
0005C090  4111 0008                     00000008  6777           LA    R1,8(R1)
0005C094  E313 0008 0024                00000008  6778           STG   R1,8(R3)           load OP2 address in PL
                                                  6779
0005C09A  4111 0008                     00000008  6780           LA    R1,8(R1)
0005C09E  E313 0010 0024                00000010  6781           STG   R1,16(R3)          load OP3 address in PL
                                                  6782
0005C0A4  B982 0000                               6783           XGR   R0,R0
0005C0A8  4100 001F                     0000001F  6784           LA    R0,31
0005C0AC  EE00 0000 3000      00000000  00000000  6785           PLO   0,0,0,0(R3)        Triple C/S (TCSG: 31)
                                                  6786
0005C0B2  B98D 0020                               6787           EPSW  R2,R0              extract psw
0005C0B6  5020 5010                     00000010  6788           ST    R2,CCPSW             to save CC
0005C0BA  E310 500B 0076                0000000B  6789           LB    R1,CCMASK          (failure CC mask)
0005C0C0  8910 0004                     00000004  6790           SLL   R1,4               (shift to BC instr CC position)
0005C0C4  4410 50F8                     0005C118  6791           EX    R1,F311TSTCC         fail if...
                                                  6792
                                00005C0C8 00000001  6793 F311CKUR EQU   *
0005C0C8  D200 5009 3030      00000009  00000030  6794           MVC   URFOUND,48(R3)     Extract UR
0005C0CE  D500 5008 3030      00000008  00000030  6795           CLC   UR,48(R3)
0005C0D4  4780 50BC                     0005C0DC  6796           BE    F311CKRES
0005C0D8  45F0 8220                     00000420  6797           BAL   R15,URMSG
                                                  6798 *
                                                  6799 * compare results
                                                  6800 *
                                0005C0DC  00000001  6801 F311CKRES EQU   *
                                                  6802 *                                   reset adresses
0005C0DC  B982 0011                               6803           XGR   R1,R1
0005C0E0  5810 501C                     0000001C  6804           L     R1,OPSRCA          source
0005C0E4  B982 0033                               6805           XGR   R3,R3
0005C0E8  5830 5024                     00000024  6806           L     R3,PLA             PL
0005C0EC  B982 0044                               6807           XGR   R4,R4
0005C0F0  5840 5020                     00000020  6808           L     R4,OPREPA
                                                  6809
0005C0F4  D507 1000 3040      00000000  00000040  6810           CLC   0(8,R1),64(R3)
0005C0FA  4770 50F2                     0005C112  6811           BNE   F311XX             OP1C Different, error
0005C0FE  D507 1008 3048      00000008  00000048  6812           CLC   8(8,R1),72(R3)
0005C104  4770 50F2                     0005C112  6813           BNE   F311XX             OP2C Different, error
0005C108  D507 1010 3050      00000010  00000050  6814           CLC   16(8,R1),80(R3)
0005C10E  478B 0000                     00000000  6815           BE    0(R11)             OP3C same, return
                                                  6816
                                0005C112  00000001  6817 F311XX   EQU   *
0005C112  45F0 8340                     00000540  6818           BAL   R15,FAILMSG
0005C116  07FB                                    6819           BR    R11
                                                  6820
0005C118  4700 50FC                     0005C11C  6821 F311TSTCC BC    0,F311CCMSG        fail if unexpected condition code
                                0005C11C  00000001  6822 F311CCMSG EQU   *
0005C11C  45F0 80F2                     000002F2  6823           BAL   R15,CCMSG
0005C120  47F0 50A8                     0005C0C8  6824           B     F311CKUR
                                                  6825
                                                  6826           TSTDATA
0005C124                        0005C124  0005C400  6827+          ORG   PLOETST+(200*K)+(42*4*K)+K
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   6828           TSTLBL OPDST              Operand Destination
0005C400                                           6829+OPDST80   DS     0C
0005C400  00000000 00000000                        6830           DC     XL256'0'
0005C408  00000000 00000000
0005C410  00000000 00000000
0005C418  00000000 00000000
0005C420  00000000 00000000
0005C428  00000000 00000000
0005C430  00000000 00000000
0005C438  00000000 00000000
0005C440  00000000 00000000
0005C448  00000000 00000000
0005C450  00000000 00000000
0005C458  00000000 00000000
0005C460  00000000 00000000
0005C468  00000000 00000000
0005C470  00000000 00000000
0005C478  00000000 00000000
0005C480  00000000 00000000
0005C488  00000000 00000000
0005C490  00000000 00000000
0005C498  00000000 00000000
0005C4A0  00000000 00000000
0005C4A8  00000000 00000000
0005C4B0  00000000 00000000
0005C4B8  00000000 00000000
0005C4C0  00000000 00000000
0005C4C8  00000000 00000000
0005C4D0  00000000 00000000
0005C4D8  00000000 00000000
0005C4E0  00000000 00000000
0005C4E8  00000000 00000000
0005C4F0  00000000 00000000
0005C4F8  00000000 00000000
                                                   6831
                                                   6832           TSTLBL OPSRC              Operand Source OP1-OP4
0005C500                                           6833+OPSRC80   DS     0C
0005C500  00000000 00000000                        6834           DC     XL256'0'
0005C508  00000000 00000000
0005C510  00000000 00000000
0005C518  00000000 00000000
0005C520  00000000 00000000
0005C528  00000000 00000000
0005C530  00000000 00000000
0005C538  00000000 00000000
0005C540  00000000 00000000
0005C548  00000000 00000000
0005C550  00000000 00000000
0005C558  00000000 00000000
0005C560  00000000 00000000
0005C568  00000000 00000000
0005C570  00000000 00000000
0005C578  00000000 00000000
0005C580  00000000 00000000
0005C588  00000000 00000000
0005C590  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0005C598  00000000 00000000
0005C5A0  00000000 00000000
0005C5A8  00000000 00000000
0005C5B0  00000000 00000000
0005C5B8  00000000 00000000
0005C5C0  00000000 00000000
0005C5C8  00000000 00000000
0005C5D0  00000000 00000000
0005C5D8  00000000 00000000
0005C5E0  00000000 00000000
0005C5E8  00000000 00000000
0005C5F0  00000000 00000000
0005C5F8  00000000 00000000
                                          6835
                                          6836           TSTLBL PL                  Parameter List
0005C600                                  6837+PL80      DS    0C
0005C600  00000000 00000000               6838           DC    XL352'00'
0005C608  00000000 00000000
0005C610  00000000 00000000
0005C618  00000000 00000000
0005C620  00000000 00000000
0005C628  00000000 00000000
0005C630  00000000 00000000
0005C638  00000000 00000000
0005C640  00000000 00000000
0005C648  00000000 00000000
0005C650  00000000 00000000
0005C658  00000000 00000000
0005C660  00000000 00000000
0005C668  00000000 00000000
0005C670  00000000 00000000
0005C678  00000000 00000000
0005C680  00000000 00000000
0005C688  00000000 00000000
0005C690  00000000 00000000
0005C698  00000000 00000000
0005C6A0  00000000 00000000
0005C6A8  00000000 00000000
0005C6B0  00000000 00000000
0005C6B8  00000000 00000000
0005C6C0  00000000 00000000
0005C6C8  00000000 00000000
0005C6D0  00000000 00000000
0005C6D8  00000000 00000000
0005C6E0  00000000 00000000
0005C6E8  00000000 00000000
0005C6F0  00000000 00000000
0005C6F8  00000000 00000000
0005C700  00000000 00000000
0005C708  00000000 00000000
0005C710  00000000 00000000
0005C718  00000000 00000000
0005C720  00000000 00000000
0005C728  00000000 00000000
0005C730  00000000 00000000
0005C738  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0005C740  00000000 00000000
0005C748  00000000 00000000
0005C750  00000000 00000000
0005C758  00000000 00000000
                                          6839
                                          6840          TSTLBL FMSG                fail message
0005C760                                  6841+FMSG80   DS     0C
0005C760  86818993 858440D7               6842          DC     C'failed PLO-Extension '
0005C768  D3D660C5 A7A38595
0005C770  A2899695 40
0005C775  E3998997 938540C3               6843          DC     CL96'Triple Compare and Swap (TCSG: 31): 1st NE'
0005C77D  96949781 99854081
0005C785  958440E2 A6819740
0005C78D  4DE3C3E2 C77A40F3
0005C795  F15D7A40 F1A2A340
0005C79D  D5C54040 40404040
0005C7A5  40404040 40404040
0005C7AD  40404040 40404040
0005C7B5  40404040 40404040
0005C7BD  40404040 40404040
0005C7C5  40404040 40404040
0005C7CD  40404040 40404040
                                          6844
                                          6845          TSTEND
0005C7D6                                  6846+         DS     0H
0005C7D6  C5958440 968640A3               6847+         DC     CL32'End of test: 80'
0005C7DE  85A2A37A 40F8F040
0005C7E6  40404040 40404040
0005C7EE  40404040 40404040
                                          6848
                                          6849 * ------------------------------------------------------------------
                                          6850 * Test Triple Compare and Swap (TCSG: 31) instruction : 2nd NE
                                          6851 * ------------------------------------------------------------------
                                          6852          TSTHDR 2,2,31              2nd Operand NOT Equal (cc=2, UR=2)
0005C7F6                  0005C7F6  0005D000  6853+         ORG    PLOETST+(200*K)+(43*4*K)
0005D000  E2E3C1D9 E3409686               6854+         DC     CL32'START of test: 81'
0005D008  40A385A2 A37A40F8
0005D010  F1404040 40404040
0005D018  40404040 40404040
0005D020                  0005D020        6855+         USING  *,R5               base for test data and test routine
0005D020                  00000000        6856+         USING  PLOETEST,R5
0005D020  0005D04C                        6857+T81      DC     A(CODE81)          address of test code
0005D024  0051                            6858+         DC     H'81'              test number
0005D026  00                              6859+         DC     X'00'
0005D027  1F                              6860+         DC     HL1'31'            Function
0005D028  02                              6861+         DC     HL1'2'             expected unequal results (UR)
0005D029  00                              6862+         DC     HL1'00'            found UR
0005D02A  02                              6863+         DC     HL1'2'             cc
0005D02B  0D                              6864+         DC     HL1'13'            not expected CC mask
0005D02C  00                              6865+         DS     X                  extracted cc
0005D030  00000000 00000000               6866+         DS     2F                 extract PSW after test (has CC)
0005D038  0005D760                        6867+         DC     A(FMSG81)          address of test fail message
0005D03C  0005D500                        6868+         DC     A(OPSRC81)         address of operand source
0005D040  0005D400                        6869+         DC     A(OPDST81)         address of operand destination
0005D044  0005D600                        6870+         DC     A(PL81)            address of parameter list
0005D048  00001202                        6871+         DC     A(RANDX+81)        address of random source data
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6872
                                                    6873          TSTCODE
0005D04C                                            6874+CODE81   DS    0F
                                                    6875 *                                load random source
0005D04C  B982 0022                                 6876          XGR   R2,R2
0005D050  5820 5028                       00000028  6877          L     R2,RANDXA
                                                    6878 *
                                                    6879 * Triple Compare and Swap (TCSG: 31)
                                                    6880 * Build Parambter List
                                                    6881 *
0005D054  B982 0011                                 6882          XGR   R1,R1
0005D058  5810 501C                       0000001C  6883          L     R1,OPSRCA
                                                    6884
0005D05C  B982 0033                                 6885          XGR   R3,R3
0005D060  5830 5024                       00000024  6886          L     R3,PLA
                                                    6887
0005D064  B982 0044                                 6888          XGR   R4,R4
0005D068  5840 5020                       00000020  6889          L     R4,OPREPA
                                                    6890
0005D06C  D217 1000 2000      00000000    00000000  6891          MVC   0(24,R1),0(R2)      load Op1-Op3 data in source
0005D072  D217 3040 1000      00000040    00000000  6892          MVC   64(24,R3),0(R1)     copy to PL OP1-Op3 Compare
0005D078  D217 4000 2080      00000000    00000080  6893          MVC   0(24,R4),128(R2)    load initial Op1-Op3 data
0005D07E  D217 3058 4000      00000058    00000000  6894          MVC   88(24,R3),0(R4)     copy to PL Op1-Op3 Replacement
                                                    6895 ****
                                                    6896 * force a failure by changing the 2nd operand
                                                    6897 ****
0005D084  D203 1008 4000      00000008    00000000  6898          MVC   8(4,R1),0(R4)
                                                    6899 ****
                                                    6900
0005D08A  E313 0000 0024                  00000000  6901          STG   R1,0(R3)            load OP1 address in PL
                                                    6902
0005D090  4111 0008                       00000008  6903          LA    R1,8(R1)
0005D094  E313 0008 0024                  00000008  6904          STG   R1,8(R3)            load OP2 address in PL
                                                    6905
0005D09A  4111 0008                       00000008  6906          LA    R1,8(R1)
0005D09E  E313 0010 0024                  00000010  6907          STG   R1,16(R3)           load OP3 address in PL
                                                    6908
0005D0A4  B982 0000                                 6909          XGR   R0,R0
0005D0A8  4100 001F                       0000001F  6910          LA    R0,31
0005D0AC  EE00 0000 3000      00000000    00000000  6911          PLO   0,0,0,0(R3)         Triple C/S (TCSG: 31)
                                                    6912
0005D0B2  B98D 0020                                 6913          EPSW  R2,R0               extract psw
0005D0B6  5020 5010                       00000010  6914          ST    R2,CCPSW              to save CC
0005D0BA  E310 500B 0076                  0000000B  6915          LB    R1,CCMASK           (failure CC mask)
0005D0C0  8910 0004                       00000004  6916          SLL   R1,4                (shift to BC instr CC position)
0005D0C4  4410 50F8                       0005D118  6917          EX    R1,F312TSTCC            fail if...
                                                    6918
                              0005D0C8    00000001  6919 F312CKUR EQU   *
0005D0C8  D200 5009 3030      00000009    00000030  6920          MVC   URFOUND,48(R3)      Extract UR
0005D0CE  D500 5008 3030      00000008    00000030  6921          CLC   UR,48(R3)
0005D0D4  4780 50BC                       0005D0DC  6922          BE    F312CKRES
0005D0D8  45F0 8220                       00000420  6923          BAL   R15,URMSG
                                                    6924 *
                                                    6925 * compare results
                                                    6926 *
                              0005D0DC    00000001  6927 F312CKRES EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6928 *                                  reset adresses
0005D0DC  B982 0011                                 6929          XGR   R1,R1
0005D0E0  5810 501C                       0000001C  6930          L     R1,OPSRCA           source
0005D0E4  B982 0033                                 6931          XGR   R3,R3
0005D0E8  5830 5024                       00000024  6932          L     R3,PLA              PL
0005D0EC  B982 0044                                 6933          XGR   R4,R4
0005D0F0  5840 5020                       00000020  6934          L     R4,OPREPA
                                                    6935
0005D0F4  D507 1000 3040  00000000        00000040  6936          CLC   0(8,R1),64(R3)
0005D0FA  4770 50F2                       0005D112  6937          BNE   F312XX              OP1C Different, error
0005D0FE  D507 1008 3048  00000008        00000048  6938          CLC   8(8,R1),72(R3)
0005D104  4770 50F2                       0005D112  6939          BNE   F312XX              OP2C Different, error
0005D108  D507 1010 3050  00000010        00000050  6940          CLC   16(8,R1),80(R3)
0005D10E  478B 0000                       00000000  6941          BE    0(R11)              OP3C same, return
                                                    6942
                          0005D112        00000001  6943 F312XX   EQU   *
0005D112  45F0 8340                       00000540  6944          BAL   R15,FAILMSG
0005D116  07FB                                      6945          BR    R11
                                                    6946
0005D118  4700 50FC                       0005D11C  6947 F312TSTCC BC    0,F312CCMSG         fail if unexpected condition code
                          0005D11C        00000001  6948 F312CCMSG EQU   *
0005D11C  45F0 80F2                       000002F2  6949          BAL   R15,CCMSG
0005D120  47F0 50A8                       0005D0C8  6950          B     F312CKUR
                                                    6951
                                                    6952          TSTDATA
0005D124                  0005D124        0005D400  6953+         ORG   PLOETST+(200*K)+(43*4*K)+K
                                                    6954          TSTLBL OPDST              Operand Destination
0005D400                                            6955+OPDST81  DS    0C
0005D400  00000000 00000000                         6956          DC    XL256'0'
0005D408  00000000 00000000
0005D410  00000000 00000000
0005D418  00000000 00000000
0005D420  00000000 00000000
0005D428  00000000 00000000
0005D430  00000000 00000000
0005D438  00000000 00000000
0005D440  00000000 00000000
0005D448  00000000 00000000
0005D450  00000000 00000000
0005D458  00000000 00000000
0005D460  00000000 00000000
0005D468  00000000 00000000
0005D470  00000000 00000000
0005D478  00000000 00000000
0005D480  00000000 00000000
0005D488  00000000 00000000
0005D490  00000000 00000000
0005D498  00000000 00000000
0005D4A0  00000000 00000000
0005D4A8  00000000 00000000
0005D4B0  00000000 00000000
0005D4B8  00000000 00000000
0005D4C0  00000000 00000000
0005D4C8  00000000 00000000
0005D4D0  00000000 00000000
0005D4D8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0005D4E0  00000000 00000000
0005D4E8  00000000 00000000
0005D4F0  00000000 00000000
0005D4F8  00000000 00000000
                                        6957
                                        6958          TSTLBL OPSRC              Operand Source OP1-OP4
0005D500                                6959+OPSRC81  DS    0C
0005D500  00000000 00000000             6960          DC    XL256'0'
0005D508  00000000 00000000
0005D510  00000000 00000000
0005D518  00000000 00000000
0005D520  00000000 00000000
0005D528  00000000 00000000
0005D530  00000000 00000000
0005D538  00000000 00000000
0005D540  00000000 00000000
0005D548  00000000 00000000
0005D550  00000000 00000000
0005D558  00000000 00000000
0005D560  00000000 00000000
0005D568  00000000 00000000
0005D570  00000000 00000000
0005D578  00000000 00000000
0005D580  00000000 00000000
0005D588  00000000 00000000
0005D590  00000000 00000000
0005D598  00000000 00000000
0005D5A0  00000000 00000000
0005D5A8  00000000 00000000
0005D5B0  00000000 00000000
0005D5B8  00000000 00000000
0005D5C0  00000000 00000000
0005D5C8  00000000 00000000
0005D5D0  00000000 00000000
0005D5D8  00000000 00000000
0005D5E0  00000000 00000000
0005D5E8  00000000 00000000
0005D5F0  00000000 00000000
0005D5F8  00000000 00000000
                                        6961
                                        6962          TSTLBL PL                 Parameter List
0005D600                                6963+PL81     DS    0C
0005D600  00000000 00000000             6964          DC    XL352'00'
0005D608  00000000 00000000
0005D610  00000000 00000000
0005D618  00000000 00000000
0005D620  00000000 00000000
0005D628  00000000 00000000
0005D630  00000000 00000000
0005D638  00000000 00000000
0005D640  00000000 00000000
0005D648  00000000 00000000
0005D650  00000000 00000000
0005D658  00000000 00000000
0005D660  00000000 00000000
0005D668  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0005D670  00000000 00000000
0005D678  00000000 00000000
0005D680  00000000 00000000
0005D688  00000000 00000000
0005D690  00000000 00000000
0005D698  00000000 00000000
0005D6A0  00000000 00000000
0005D6A8  00000000 00000000
0005D6B0  00000000 00000000
0005D6B8  00000000 00000000
0005D6C0  00000000 00000000
0005D6C8  00000000 00000000
0005D6D0  00000000 00000000
0005D6D8  00000000 00000000
0005D6E0  00000000 00000000
0005D6E8  00000000 00000000
0005D6F0  00000000 00000000
0005D6F8  00000000 00000000
0005D700  00000000 00000000
0005D708  00000000 00000000
0005D710  00000000 00000000
0005D718  00000000 00000000
0005D720  00000000 00000000
0005D728  00000000 00000000
0005D730  00000000 00000000
0005D738  00000000 00000000
0005D740  00000000 00000000
0005D748  00000000 00000000
0005D750  00000000 00000000
0005D758  00000000 00000000
                                          6965
                                          6966           TSTLBL FMSG               fail message
0005D760                                  6967+FMSG81    DS    0C
0005D760  86818993 858440D7               6968           DC    C'failed PLO-Extension '
0005D768  D3D660C5 A7A38595
0005D770  A2899695 40
0005D775  E3998997 938540C3               6969           DC    CL96'Triple Compare and Swap (TCSG: 31): 2nd NE'
0005D77D  96949781 99854081
0005D785  958440E2 A6819740
0005D78D  4DE3C3E2 C77A40F3
0005D795  F15D7A40 F2958440
0005D79D  D5C54040 40404040
0005D7A5  40404040 40404040
0005D7AD  40404040 40404040
0005D7B5  40404040 40404040
0005D7BD  40404040 40404040
0005D7C5  40404040 40404040
0005D7CD  40404040 40404040
                                          6970
                                          6971           TSTEND
0005D7D6                                  6972+          DS    0H
0005D7D6  C5958440 968640A3               6973+          DC    CL32'End of test: 81'
0005D7DE  85A2A37A 40F8F140
0005D7E6  40404040 40404040
0005D7EE  40404040 40404040
                                          6974
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6975 * ------------------------------------------------------------------------
                                                    6976 * Test Triple Compare and Swap (TCSG: 31) instruction : 3rd NE
                                                    6977 * ------------------------------------------------------------------------
                                                    6978          TSTHDR 2,3,31             3rd Operand NOT Equal (cc=2, UR=3)
0005D7F6                      0005D7F6  0005E000    6979+         ORG    PLOETST+(200*K)+(44*4*K)
0005E000  E2E3C1D9 E3409686                         6980+         DC     CL32'START of test: 82'
0005E008  40A385A2 A37A40F8
0005E010  F2404040 40404040
0005E018  40404040 40404040
0005E020                      0005E020            6981+         USING *,R5                base for test data and test routine
0005E020                      00000000            6982+         USING PLOETEST,R5
0005E020  0005E04C                                6983+T82      DC     A(CODE82)          address of test code
0005E024  0052                                    6984+         DC     H'82'              test number
0005E026  00                                      6985+         DC     X'00'
0005E027  1F                                      6986+         DC     HL1'31'            Function
0005E028  03                                      6987+         DC     HL1'3'             expected unequal results (UR)
0005E029  00                                      6988+         DC     HL1'00'            found UR
0005E02A  02                                      6989+         DC     HL1'2'             cc
0005E02B  0D                                      6990+         DC     HL1'13'            not expected CC mask
0005E02C  00                                      6991+         DS     X                  extracted cc
0005E030  00000000 00000000                       6992+         DS     2F                 extract PSW after test (has CC)
0005E038  0005E760                                6993+         DC     A(FMSG82)          address of test fail message
0005E03C  0005E500                                6994+         DC     A(OPSRC82)         address of operand source
0005E040  0005E400                                6995+         DC     A(OPDST82)         address of operand destination
0005E044  0005E600                                6996+         DC     A(PL82)            address of parameter list
0005E048  00001203                                6997+         DC     A(RANDX+82)        address of random source data
                                                    6998
                                                    6999          TSTCODE
0005E04C                                          7000+CODE82   DS     0F
                                                    7001 *                                  load random source
0005E04C  B982 0022                               7002          XGR    R2,R2
0005E050  5820 5028                 00000028      7003          L      R2,RANDXA
                                                    7004 *
                                                    7005 * Triple Compare and Swap (TCSG: 31)
                                                    7006 * Build Parambter List
                                                    7007 *
0005E054  B982 0011                               7008          XGR    R1,R1
0005E058  5810 501C                 0000001C      7009          L      R1,OPSRCA
                                                    7010
0005E05C  B982 0033                               7011          XGR    R3,R3
0005E060  5830 5024                 00000024      7012          L      R3,PLA
                                                    7013
0005E064  B982 0044                               7014          XGR    R4,R4
0005E068  5840 5020                 00000020      7015          L      R4,OPREPA
                                                    7016
0005E06C  D217 1000 2000      00000000  00000000  7017          MVC    0(24,R1),0(R2)     load Op1-Op3 data in source
0005E072  D217 3040 1000      00000040  00000000  7018          MVC    64(24,R3),0(R1)    copy to PL OP1-Op3 Compare
0005E078  D217 4000 2080      00000000  00000080  7019          MVC    0(24,R4),128(R2)   load initial Op1-Op3 data
0005E07E  D217 3058 4000      00000058  00000000  7020          MVC    88(24,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                    7021 ****
                                                    7022 * force a failure by changing the 3rd operand
                                                    7023 ****
0005E084  D203 1010 4000      00000010  00000000  7024          MVC    16(4,R1),0(R4)
                                                    7025 ****
                                                    7026
0005E08A  E313 0000 0024                00000000  7027          STG    R1,0(R3)           load OP1 address in PL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    7028
0005E090  4111 0008                     00000008  7029          LA    R1,8(R1)
0005E094  E313 0008 0024                00000008  7030          STG   R1,8(R3)          load OP2 address in PL
                                                    7031
0005E09A  4111 0008                     00000008  7032          LA    R1,8(R1)
0005E09E  E313 0010 0024                00000010  7033          STG   R1,16(R3)         load OP3 address in PL
                                                    7034
0005E0A4  B982 0000                               7035          XGR   R0,R0
0005E0A8  4100 001F                     0000001F  7036          LA    R0,31
0005E0AC  EE00 0000 3000  00000000      00000000  7037          PLO   0,0,0,0(R3)       Triple C/S (TCSG: 31)
                                                    7038
0005E0B2  B98D 0020                               7039          EPSW  R2,R0             extract psw
0005E0B6  5020 5010                     00000010  7040          ST    R2,CCPSW            to save CC
0005E0BA  E310 500B 0076                0000000B  7041          LB    R1,CCMASK         (failure CC mask)
0005E0C0  8910 0004                     00000004  7042          SLL   R1,4              (shift to BC instr CC position)
0005E0C4  4410 50F8                     0005E118  7043          EX    R1,F313TSTCC        fail if...
                                                    7044
                          0005E0C8      00000001  7045 F313CKUR EQU   *
0005E0C8  D200 5009 3030  00000009      00000030  7046          MVC   URFOUND,48(R3)    Extract UR
0005E0CE  D500 5008 3030  00000008      00000030  7047          CLC   UR,48(R3)
0005E0D4  4780 50BC                     0005E0DC  7048          BE    F313CKRES
0005E0D8  45F0 8220                     00000420  7049          BAL   R15,URMSG
                                                    7050 *
                                                    7051 * compare results
                                                    7052 *
                          0005E0DC      00000001  7053 F313CKRES EQU   *
                                                    7054 *                          reset adresses
0005E0DC  B982 0011                               7055          XGR   R1,R1
0005E0E0  5810 501C                     0000001C  7056          L     R1,OPSRCA         source
0005E0E4  B982 0033                               7057          XGR   R3,R3
0005E0E8  5830 5024                     00000024  7058          L     R3,PLA            PL
0005E0EC  B982 0044                               7059          XGR   R4,R4
0005E0F0  5840 5020                     00000020  7060          L     R4,OPREPA
                                                    7061
0005E0F4  D507 1000 3040  00000000      00000040  7062          CLC   0(8,R1),64(R3)
0005E0FA  4770 50F2                     0005E112  7063          BNE   F313XX            OP1C Different, error
0005E0FE  D507 1008 3048  00000008      00000048  7064          CLC   8(8,R1),72(R3)
0005E104  4770 50F2                     0005E112  7065          BNE   F313XX            OP2C Different, error
0005E108  D507 1010 3050  00000010      00000050  7066          CLC   16(8,R1),80(R3)
0005E10E  478B 0000                     00000000  7067          BE    0(R11)            OP3C same, return
                                                    7068
                          0005E112      00000001  7069 F313XX   EQU   *
0005E112  45F0 8340                     00000540  7070          BAL   R15,FAILMSG
0005E116  07FB                                    7071          BR    R11
                                                    7072
0005E118  4700 50FC                     0005E11C  7073 F313TSTCC BC    0,F313CCMSG       fail if unexpected condition code
                          0005E11C      00000001  7074 F313CCMSG EQU   *
0005E11C  45F0 80F2                     000002F2  7075          BAL   R15,CCMSG
0005E120  47F0 50A8                     0005E0C8  7076          B     F313CKUR
                                                    7077
                                                    7078          TSTDATA
0005E124                  0005E124      0005E400  7079+         ORG   PLOETST+(200*K)+(44*4*K)+K
                                                    7080          TSTLBL OPDST            Operand Destination
0005E400                                          7081+OPDST82  DS    0C
0005E400  00000000 00000000                       7082          DC    XL256'0'
0005E408  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0005E410  00000000 00000000
0005E418  00000000 00000000
0005E420  00000000 00000000
0005E428  00000000 00000000
0005E430  00000000 00000000
0005E438  00000000 00000000
0005E440  00000000 00000000
0005E448  00000000 00000000
0005E450  00000000 00000000
0005E458  00000000 00000000
0005E460  00000000 00000000
0005E468  00000000 00000000
0005E470  00000000 00000000
0005E478  00000000 00000000
0005E480  00000000 00000000
0005E488  00000000 00000000
0005E490  00000000 00000000
0005E498  00000000 00000000
0005E4A0  00000000 00000000
0005E4A8  00000000 00000000
0005E4B0  00000000 00000000
0005E4B8  00000000 00000000
0005E4C0  00000000 00000000
0005E4C8  00000000 00000000
0005E4D0  00000000 00000000
0005E4D8  00000000 00000000
0005E4E0  00000000 00000000
0005E4E8  00000000 00000000
0005E4F0  00000000 00000000
0005E4F8  00000000 00000000
                                          7083
                                          7084           TSTLBL OPSRC              Operand Source OP1-OP4
0005E500                                  7085+OPSRC82   DS    0C
0005E500  00000000 00000000               7086           DC    XL256'0'
0005E508  00000000 00000000
0005E510  00000000 00000000
0005E518  00000000 00000000
0005E520  00000000 00000000
0005E528  00000000 00000000
0005E530  00000000 00000000
0005E538  00000000 00000000
0005E540  00000000 00000000
0005E548  00000000 00000000
0005E550  00000000 00000000
0005E558  00000000 00000000
0005E560  00000000 00000000
0005E568  00000000 00000000
0005E570  00000000 00000000
0005E578  00000000 00000000
0005E580  00000000 00000000
0005E588  00000000 00000000
0005E590  00000000 00000000
0005E598  00000000 00000000
0005E5A0  00000000 00000000
0005E5A8  00000000 00000000
0005E5B0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0005E5B8  00000000 00000000
0005E5C0  00000000 00000000
0005E5C8  00000000 00000000
0005E5D0  00000000 00000000
0005E5D8  00000000 00000000
0005E5E0  00000000 00000000
0005E5E8  00000000 00000000
0005E5F0  00000000 00000000
0005E5F8  00000000 00000000
                                          7087
                                          7088           TSTLBL PL                  Parameter List
0005E600                                  7089+PL82      DS    0C
0005E600  00000000 00000000               7090           DC    XL352'00'
0005E608  00000000 00000000
0005E610  00000000 00000000
0005E618  00000000 00000000
0005E620  00000000 00000000
0005E628  00000000 00000000
0005E630  00000000 00000000
0005E638  00000000 00000000
0005E640  00000000 00000000
0005E648  00000000 00000000
0005E650  00000000 00000000
0005E658  00000000 00000000
0005E660  00000000 00000000
0005E668  00000000 00000000
0005E670  00000000 00000000
0005E678  00000000 00000000
0005E680  00000000 00000000
0005E688  00000000 00000000
0005E690  00000000 00000000
0005E698  00000000 00000000
0005E6A0  00000000 00000000
0005E6A8  00000000 00000000
0005E6B0  00000000 00000000
0005E6B8  00000000 00000000
0005E6C0  00000000 00000000
0005E6C8  00000000 00000000
0005E6D0  00000000 00000000
0005E6D8  00000000 00000000
0005E6E0  00000000 00000000
0005E6E8  00000000 00000000
0005E6F0  00000000 00000000
0005E6F8  00000000 00000000
0005E700  00000000 00000000
0005E708  00000000 00000000
0005E710  00000000 00000000
0005E718  00000000 00000000
0005E720  00000000 00000000
0005E728  00000000 00000000
0005E730  00000000 00000000
0005E738  00000000 00000000
0005E740  00000000 00000000
0005E748  00000000 00000000
0005E750  00000000 00000000
0005E758  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    7091
                                                    7092          TSTLBL FMSG               fail message
0005E760                                            7093+FMSG82   DS    0C
0005E760  86818993 858440D7                         7094          DC    C'failed PLO-Extension '
0005E768  D3D660C5 A7A38595
0005E770  A2899695 40
0005E775  E3998997 938540C3                         7095          DC    CL96'Triple Compare and Swap (TCSG: 31): 3rd NE'
0005E77D  96949781 99854081
0005E785  958440E2 A6819740
0005E78D  4DE3C3E2 C77A40F3
0005E795  F15D7A40 F3998440
0005E79D  D5C54040 40404040
0005E7A5  40404040 40404040
0005E7AD  40404040 40404040
0005E7B5  40404040 40404040
0005E7BD  40404040 40404040
0005E7C5  40404040 40404040
0005E7CD  40404040 40404040
                                                    7096
                                                    7097          TSTEND
0005E7D6                                            7098+         DS    0H
0005E7D6  C5958440 968640A3                         7099+         DC    CL32'End of test: 82'
0005E7DE  85A2A37A 40F8F240
0005E7E6  40404040 40404040
0005E7EE  40404040 40404040
                                                    7100
                                                    7101 * ------------------------------------------------------------------
                                                    7102 * Test Triple Compare and Swap (TCSG: 31) instruction : All EQ
                                                    7103 * ------------------------------------------------------------------
                                                    7104          TSTHDR 0,99,31            All Operands Equal (cc=0, UR=99)
0005E7F6                        0005E7F6  0005F000  7105+         ORG   PLOETST+(200*K)+(45*4*K)
0005F000  E2E3C1D9 E3409686                         7106+         DC    CL32'START of test: 83'
0005F008  40A385A2 A37A40F8
0005F010  F3404040 40404040
0005F018  40404040 40404040
0005F020                        0005F020            7107+         USING *,R5                base for test data and test routine
0005F020                        00000000            7108+         USING PLOETEST,R5
0005F020  0005F04C                                  7109+T83      DC    A(CODE83)           address of test code
0005F024  0053                                      7110+         DC    H'83'               test number
0005F026  00                                        7111+         DC    X'00'
0005F027  1F                                        7112+         DC    HL1'31'             Function
0005F028  63                                        7113+         DC    HL1'99'             expected unequal results (UR)
0005F029  00                                        7114+         DC    HL1'00'             found UR
0005F02A  00                                        7115+         DC    HL1'0'              cc
0005F02B  07                                        7116+         DC    HL1'7'              not expected CC mask
0005F02C  00                                        7117+         DS    X                   extracted cc
0005F030  00000000 00000000                         7118+         DS    2F                  extract PSW after test (has CC)
0005F038  0005F760                                  7119+         DC    A(FMSG83)           address of test fail message
0005F03C  0005F500                                  7120+         DC    A(OPSRC83)          address of operand source
0005F040  0005F400                                  7121+         DC    A(OPDST83)          address of operand destination
0005F044  0005F600                                  7122+         DC    A(PL83)             address of parameter list
0005F048  00001204                                  7123+         DC    A(RANDX+83)         address of random source data
                                                    7124
                                                    7125          TSTCODE
0005F04C                                            7126+CODE83   DS    0F
                                                    7127 *                                  load random source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0005F04C  B982 0022                              7128          XGR   R2,R2
0005F050  5820 5028                   00000028   7129          L     R2,RANDXA
                                                 7130 *
                                                 7131 * Triple Compare and Swap (TCSG: 31)
                                                 7132 * Build Parambter List
                                                 7133 *
0005F054  B982 0011                              7134          XGR   R1,R1
0005F058  5810 501C                   0000001C   7135          L     R1,OPSRCA
                                                 7136
0005F05C  B982 0033                              7137          XGR   R3,R3
0005F060  5830 5024                   00000024   7138          L     R3,PLA
                                                 7139
0005F064  B982 0044                              7140          XGR   R4,R4
0005F068  5840 5020                   00000020   7141          L     R4,OPREPA
                                                 7142
0005F06C  D217 1000 2000      00000000  00000000   7143          MVC   0(24,R1),0(R2)      load Op1-Op3 data in source
0005F072  D217 3040 1000      00000040  00000000   7144          MVC   64(24,R3),0(R1)     copy to PL OP1-Op3 Compare
0005F078  D217 4000 2080      00000000  00000080   7145          MVC   0(24,R4),128(R2)    load initial Op1-Op3 data
0005F07E  D217 3058 4000      00000058  00000000   7146          MVC   88(24,R3),0(R4)     copy to PL Op1-Op3 Replacement
                                                 7147 ****
                                                 7148 * do not force a failure by changing the 3rd operand
                                                 7149 ****
                                                 7150 *         MVC   16(4,R1),0(R4)
                                                 7151 ****
                                                 7152
0005F084  E313 0000 0024                   00000000  7153          STG   R1,0(R3)            load OP1 address in PL
                                                 7154
0005F08A  4111 0008                   00000008   7155          LA    R1,8(R1)
0005F08E  E313 0008 0024                   00000008  7156          STG   R1,8(R3)            load OP2 address in PL
                                                 7157
0005F094  4111 0008                   00000008   7158          LA    R1,8(R1)
0005F098  E313 0010 0024                   00000010  7159          STG   R1,16(R3)           load OP3 address in PL
                                                 7160
0005F09E  B982 0000                              7161          XGR   R0,R0
0005F0A2  4100 001F                   0000001F   7162          LA    R0,31
0005F0A6  EE00 0000 3000      00000000  00000000   7163          PLO   0,0,0,0(R3)         Triple C/S (TCSG: 31)
                                                 7164
0005F0AC  B98D 0020                              7165          EPSW  R2,R0               extract psw
0005F0B0  5020 5010                   00000010   7166          ST    R2,CCPSW              to save CC
0005F0B4  E310 500B 0076                   0000000B  7167          LB    R1,CCMASK           (failure CC mask)
0005F0BA  8910 0004                   00000004   7168          SLL   R1,4                (shift to BC instr CC position)
0005F0BE  4410 50DE                   0005F0FE   7169          EX    R1,F314TSTCC            fail if...
                                                 7170 *
                                                 7171 * compare results
                                                 7172 *
                              0005F0C2  00000001   7173 F314CKRES EQU   *
                                                 7174 *                                    reset adresses
0005F0C2  B982 0011                              7175          XGR   R1,R1
0005F0C6  5810 501C                   0000001C   7176          L     R1,OPSRCA           source
0005F0CA  B982 0033                              7177          XGR   R3,R3
0005F0CE  5830 5024                   00000024   7178          L     R3,PLA              PL
0005F0D2  B982 0044                              7179          XGR   R4,R4
0005F0D6  5840 5020                   00000020   7180          L     R4,OPREPA
                                                 7181
                                                 7182 * All equal; replacement values copied to operand source
                                                 7183
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0005F0DA  D507 1000 3058      00000000  00000058  7184           CLC   0(8,R1),88(R3)
0005F0E0  4770 50D8                     0005F0F8  7185           BNE   F314XX               OP1R Different, error
0005F0E4  D507 1008 3060      00000008  00000060  7186           CLC   8(8,R1),96(R3)
0005F0EA  4770 50D8                     0005F0F8  7187           BNE   F314XX               OP2R Different, error
0005F0EE  D507 1010 3068      00000010  00000068  7188           CLC   16(8,R1),104(R3)
0005F0F4  478B 0000                     00000000  7189           BE    0(R11)
                                                  7190
                                        0005F0F8  00000001  7191 F314XX   EQU   *
0005F0F8  45F0 8340                     00000540  7192           BAL   R15,FAILMSG
0005F0FC  07FB                                    7193           BR    R11
                                                  7194
0005F0FE  4700 50E2                     0005F102  7195 F314TSTCC BC    0,F314CCMSG          fail if unexpected condition code
                                        0005F102  00000001  7196 F314CCMSG EQU  *
0005F102  45F0 80F2                     000002F2  7197           BAL   R15,CCMSG
0005F106  47F0 50A2                     0005F0C2  7198           B     F314CKRES
                                                  7199
                                                  7200           TSTDATA
0005F10A                        0005F10A  0005F400  7201+          ORG   PLOETST+(200*K)+(45*4*K)+K
                                                  7202           TSTLBL OPDST               Operand Destination
0005F400                                          7203+OPDST83    DS    0C
0005F400  00000000 00000000                       7204           DC    XL256'0'
0005F408  00000000 00000000
0005F410  00000000 00000000
0005F418  00000000 00000000
0005F420  00000000 00000000
0005F428  00000000 00000000
0005F430  00000000 00000000
0005F438  00000000 00000000
0005F440  00000000 00000000
0005F448  00000000 00000000
0005F450  00000000 00000000
0005F458  00000000 00000000
0005F460  00000000 00000000
0005F468  00000000 00000000
0005F470  00000000 00000000
0005F478  00000000 00000000
0005F480  00000000 00000000
0005F488  00000000 00000000
0005F490  00000000 00000000
0005F498  00000000 00000000
0005F4A0  00000000 00000000
0005F4A8  00000000 00000000
0005F4B0  00000000 00000000
0005F4B8  00000000 00000000
0005F4C0  00000000 00000000
0005F4C8  00000000 00000000
0005F4D0  00000000 00000000
0005F4D8  00000000 00000000
0005F4E0  00000000 00000000
0005F4E8  00000000 00000000
0005F4F0  00000000 00000000
0005F4F8  00000000 00000000
                                                  7205
                                                  7206           TSTLBL OPSRC               Operand Source OP1-OP4
0005F500                                          7207+OPSRC83    DS    0C
0005F500  00000000 00000000                       7208           DC    XL256'0'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0005F508  00000000 00000000
0005F510  00000000 00000000
0005F518  00000000 00000000
0005F520  00000000 00000000
0005F528  00000000 00000000
0005F530  00000000 00000000
0005F538  00000000 00000000
0005F540  00000000 00000000
0005F548  00000000 00000000
0005F550  00000000 00000000
0005F558  00000000 00000000
0005F560  00000000 00000000
0005F568  00000000 00000000
0005F570  00000000 00000000
0005F578  00000000 00000000
0005F580  00000000 00000000
0005F588  00000000 00000000
0005F590  00000000 00000000
0005F598  00000000 00000000
0005F5A0  00000000 00000000
0005F5A8  00000000 00000000
0005F5B0  00000000 00000000
0005F5B8  00000000 00000000
0005F5C0  00000000 00000000
0005F5C8  00000000 00000000
0005F5D0  00000000 00000000
0005F5D8  00000000 00000000
0005F5E0  00000000 00000000
0005F5E8  00000000 00000000
0005F5F0  00000000 00000000
0005F5F8  00000000 00000000
                                                  7209
                                                  7210           TSTLBL PL                  Parameter List
0005F600                                          7211+PL83      DS    0C
0005F600  00000000 00000000                       7212           DC    XL352'00'
0005F608  00000000 00000000
0005F610  00000000 00000000
0005F618  00000000 00000000
0005F620  00000000 00000000
0005F628  00000000 00000000
0005F630  00000000 00000000
0005F638  00000000 00000000
0005F640  00000000 00000000
0005F648  00000000 00000000
0005F650  00000000 00000000
0005F658  00000000 00000000
0005F660  00000000 00000000
0005F668  00000000 00000000
0005F670  00000000 00000000
0005F678  00000000 00000000
0005F680  00000000 00000000
0005F688  00000000 00000000
0005F690  00000000 00000000
0005F698  00000000 00000000
0005F6A0  00000000 00000000
0005F6A8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0005F6B0  00000000 00000000
0005F6B8  00000000 00000000
0005F6C0  00000000 00000000
0005F6C8  00000000 00000000
0005F6D0  00000000 00000000
0005F6D8  00000000 00000000
0005F6E0  00000000 00000000
0005F6E8  00000000 00000000
0005F6F0  00000000 00000000
0005F6F8  00000000 00000000
0005F700  00000000 00000000
0005F708  00000000 00000000
0005F710  00000000 00000000
0005F718  00000000 00000000
0005F720  00000000 00000000
0005F728  00000000 00000000
0005F730  00000000 00000000
0005F738  00000000 00000000
0005F740  00000000 00000000
0005F748  00000000 00000000
0005F750  00000000 00000000
0005F758  00000000 00000000
                                          7213
                                          7214          TSTLBL FMSG               fail message
0005F760                                  7215+FMSG83   DS    0C
0005F760  86818993 858440D7               7216          DC    C'failed PLO-Extension '
0005F768  D3D660C5 A7A38595
0005F770  A2899695 40
0005F775  E3998997 938540C3               7217          DC    CL96'Triple Compare and Swap (TCSG: 31): 3rd NE'
0005F77D  96949781 99854081
0005F785  958440E2 A6819740
0005F78D  4DE3C3E2 C77A40F3
0005F795  F15D7A40 F3998440
0005F79D  D5C54040 40404040
0005F7A5  40404040 40404040
0005F7AD  40404040 40404040
0005F7B5  40404040 40404040
0005F7BD  40404040 40404040
0005F7C5  40404040 40404040
0005F7CD  40404040 40404040
                                          7218
                                          7219          TSTEND
0005F7D6                                  7220+         DS    0H
0005F7D6  C5958440 968640A3               7221+         DC    CL32'End of test: 83'
0005F7DE  85A2A37A 40F8F340
0005F7E6  40404040 40404040
0005F7EE  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   7223 *=====================================================================
                                                   7224 *       Triple Compare and Swap Tests
                                                   7225 *=====================================================================
                                                   7226
                                                   7227 * -------------------------------------------------------------------
                                                   7228 * Test Triple Compare and Swap (TCS: 30) instruction : 1st NE
                                                   7229 * -------------------------------------------------------------------
                                                   7230          TSTHDR 2,1,30             1st Operand NOT Equal (cc=2, UR=1)
0005F7F6                          0005F7F6  00060000 7231+         ORG   PLOETST+(200*K)+(46*4*K)
00060000  E2E3C1D9 E3409686                        7232+         DC    CL32'START of test: 84'
00060008  40A385A2 A37A40F8
00060010  F4404040 40404040
00060018  40404040 40404040
00060020                          00060020         7233+         USING *,R5                base for test data and test routine
00060020                          00000000         7234+         USING PLOETEST,R5
00060020  0006004C                                 7235+T84      DC    A(CODE84)           address of test code
00060024  0054                                     7236+         DC    H'84'               test number
00060026  00                                       7237+         DC    X'00'
00060027  1E                                       7238+         DC    HL1'30'             Function
00060028  01                                       7239+         DC    HL1'1'              expected unequal results (UR)
00060029  00                                       7240+         DC    HL1'00'             found UR
0006002A  02                                       7241+         DC    HL1'2'              cc
0006002B  0D                                       7242+         DC    HL1'13'             not expected CC mask
0006002C  00                                       7243+         DS    X                   extracted cc
00060030  00000000 00000000                        7244+         DS    2F                  extract PSW after test (has CC)
00060038  00060760                                 7245+         DC    A(FMSG84)           address of test fail message
0006003C  00060500                                 7246+         DC    A(OPSRC84)          address of operand source
00060040  00060400                                 7247+         DC    A(OPDST84)          address of operand destination
00060044  00060600                                 7248+         DC    A(PL84)             address of parameter list
00060048  00001205                                 7249+         DC    A(RANDX+84)         address of random source data
                                                   7250
                                                   7251          TSTCODE
0006004C                                           7252+CODE84   DS    0F
                                                   7253 *                                  load random source
0006004C  B982 0022                                7254          XGR   R2,R2
00060050  5820 5028                     00000028   7255          L     R2,RANDXA
                                                   7256 *
                                                   7257 * Triple Compare and Swap (TCS: 30)
                                                   7258 * Build Parambter List
                                                   7259 *
00060054  B982 0011                                7260          XGR   R1,R1
00060058  5810 501C                     0000001C   7261          L     R1,OPSRCA
                                                   7262
0006005C  B982 0033                                7263          XGR   R3,R3
00060060  5830 5024                     00000024   7264          L     R3,PLA
                                                   7265
00060064  B982 0044                                7266          XGR   R4,R4
00060068  5840 5020                     00000020   7267          L     R4,OPREPA
                                                   7268
0006006C  D20B 1000 2000       00000000  00000000  7269          MVC   0(12,R1),0(R2)      load Op1-Op3 data in source
00060072  D203 3040 1000       00000040  00000000  7270          MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00060078  D203 3048 1004       00000048  00000004  7271          MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
0006007E  D203 3050 1008       00000050  00000008  7272          MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
                                                   7273
00060084  D20B 4000 2080       00000000  00000080  7274          MVC   0(12,R4),128(R2)    load initial Op1-Op3 data
0006008A  D203 3058 4000       00000058  00000000  7275          MVC   88(4,R3),0(R4)      copy to PL OP1-Replacement
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00060090  D203 3060 4004      00000060  00000004  7276          MVC   96(4,R3),4(R4)      copy to PL OP2-Replacement
00060096  D203 3068 4008      00000068  00000008  7277          MVC   104(4,R3),8(R4)     copy to PL OP3-Replacement
                                                  7278
                                                  7279 ****
                                                  7280 * force a failure by changing the 1st operand
                                                  7281 ****
0006009C  D203 1000 4000      00000000  00000000  7282          MVC   0(4,R1),0(R4)
                                                  7283 ****
                                                  7284
000600A2  E313 0000 0024                00000000  7285          STG   R1,0(R3)           load OP1 address in PL
                                                  7286
000600A8  4111 0004                     00000004  7287          LA    R1,4(R1)
000600AC  E313 0008 0024                00000008  7288          STG   R1,8(R3)           load OP2 address in PL
                                                  7289
000600B2  4111 0004                     00000004  7290          LA    R1,4(R1)
000600B6  E313 0010 0024                00000010  7291          STG   R1,16(R3)          load OP3 address in PL
                                                  7292
000600BC  B982 0000                               7293          XGR   R0,R0
000600C0  4100 001E                     0000001E  7294          LA    R0,30
000600C4  EE00 0000 3000      00000000  00000000  7295          PLO   0,0,0,0(R3)        Triple C/S (TCS: 30)
                                                  7296
000600CA  B98D 0020                               7297          EPSW  R2,R0              extract psw
000600CE  5020 5010                     00000010  7298          ST    R2,CCPSW             to save CC
000600D2  E310 500B 0076                0000000B  7299          LB    R1,CCMASK          (failure CC mask)
000600D8  8910 0004                     00000004  7300          SLL   R1,4               (shift to BC instr CC position)
000600DC  4410 5110                     00060130  7301          EX    R1,F301TSTCC         fail if...
                                                  7302
                          000600E0      00000001  7303 F301CKUR EQU   *
000600E0  D200 5009 3030      00000009  00000030  7304          MVC   URFOUND,48(R3)     Extract UR
000600E6  D500 5008 3030      00000008  00000030  7305          CLC   UR,48(R3)
000600EC  4780 50D4                     000600F4  7306          BE    F301CKRES
000600F0  45F0 8220                     00000420  7307          BAL   R15,URMSG
                                                  7308 *
                                                  7309 * compare results
                                                  7310 *
                          000600F4      00000001  7311 F301CKRES EQU   *
                                                  7312 *                               reset adresses
000600F4  B982 0011                               7313          XGR   R1,R1
000600F8  5810 501C                     0000001C  7314          L     R1,OPSRCA          source
000600FC  B982 0033                               7315          XGR   R3,R3
00060100  5830 5024                     00000024  7316          L     R3,PLA             PL
00060104  B982 0044                               7317          XGR   R4,R4
00060108  5840 5020                     00000020  7318          L     R4,OPREPA
                                                  7319
0006010C  D503 1000 3040      00000000  00000040  7320          CLC   0(4,R1),64(R3)
00060112  4770 510A                     0006012A  7321          BNE   F301XX             OP1C Different, error
00060116  D503 1004 3048      00000004  00000048  7322          CLC   4(4,R1),72(R3)
0006011C  4770 510A                     0006012A  7323          BNE   F301XX             OP2C Different, error
00060120  D503 1008 3050      00000008  00000050  7324          CLC   8(4,R1),80(R3)
00060126  478B 0000                     00000000  7325          BE    0(R11)             OP3C same, return
                                                  7326
                          0006012A      00000001  7327 F301XX   EQU   *
0006012A  45F0 8340                     00000540  7328          BAL   R15,FAILMSG
0006012E  07FB                                    7329          BR    R11
                                                  7330
00060130  4700 5114                     00060134  7331 F301TSTCC BC    0,F301CCMSG        fail if unexpected condition code
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                               00060134  00000001  7332 F301CCMSG EQU  *
00060134  45F0 80F2                      000002F2  7333           BAL   R15,CCMSG
00060138  47F0 50C0                      000600E0  7334           B     F301CKUR
                                                   7335
                                                   7336           TSTDATA
0006013C                       0006013C  00060400  7337+          ORG   PLOETST+(200*K)+(46*4*K)+K
                                                   7338           TSTLBL OPDST              Operand Destination
00060400                                           7339+OPDST84   DS    0C
00060400  00000000 00000000                        7340           DC    XL256'0'
00060408  00000000 00000000
00060410  00000000 00000000
00060418  00000000 00000000
00060420  00000000 00000000
00060428  00000000 00000000
00060430  00000000 00000000
00060438  00000000 00000000
00060440  00000000 00000000
00060448  00000000 00000000
00060450  00000000 00000000
00060458  00000000 00000000
00060460  00000000 00000000
00060468  00000000 00000000
00060470  00000000 00000000
00060478  00000000 00000000
00060480  00000000 00000000
00060488  00000000 00000000
00060490  00000000 00000000
00060498  00000000 00000000
000604A0  00000000 00000000
000604A8  00000000 00000000
000604B0  00000000 00000000
000604B8  00000000 00000000
000604C0  00000000 00000000
000604C8  00000000 00000000
000604D0  00000000 00000000
000604D8  00000000 00000000
000604E0  00000000 00000000
000604E8  00000000 00000000
000604F0  00000000 00000000
000604F8  00000000 00000000
                                                   7341
                                                   7342           TSTLBL OPSRC              Operand Source OP1-OP4
00060500                                           7343+OPSRC84   DS    0C
00060500  00000000 00000000                        7344           DC    XL256'0'
00060508  00000000 00000000
00060510  00000000 00000000
00060518  00000000 00000000
00060520  00000000 00000000
00060528  00000000 00000000
00060530  00000000 00000000
00060538  00000000 00000000
00060540  00000000 00000000
00060548  00000000 00000000
00060550  00000000 00000000
00060558  00000000 00000000
00060560  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00060568  00000000 00000000
00060570  00000000 00000000
00060578  00000000 00000000
00060580  00000000 00000000
00060588  00000000 00000000
00060590  00000000 00000000
00060598  00000000 00000000
000605A0  00000000 00000000
000605A8  00000000 00000000
000605B0  00000000 00000000
000605B8  00000000 00000000
000605C0  00000000 00000000
000605C8  00000000 00000000
000605D0  00000000 00000000
000605D8  00000000 00000000
000605E0  00000000 00000000
000605E8  00000000 00000000
000605F0  00000000 00000000
000605F8  00000000 00000000
                                          7345
                                          7346          TSTLBL PL                 Parameter List
00060600                                  7347+PL84     DS    0C
00060600  00000000 00000000               7348          DC    XL352'00'
00060608  00000000 00000000
00060610  00000000 00000000
00060618  00000000 00000000
00060620  00000000 00000000
00060628  00000000 00000000
00060630  00000000 00000000
00060638  00000000 00000000
00060640  00000000 00000000
00060648  00000000 00000000
00060650  00000000 00000000
00060658  00000000 00000000
00060660  00000000 00000000
00060668  00000000 00000000
00060670  00000000 00000000
00060678  00000000 00000000
00060680  00000000 00000000
00060688  00000000 00000000
00060690  00000000 00000000
00060698  00000000 00000000
000606A0  00000000 00000000
000606A8  00000000 00000000
000606B0  00000000 00000000
000606B8  00000000 00000000
000606C0  00000000 00000000
000606C8  00000000 00000000
000606D0  00000000 00000000
000606D8  00000000 00000000
000606E0  00000000 00000000
000606E8  00000000 00000000
000606F0  00000000 00000000
000606F8  00000000 00000000
00060700  00000000 00000000
00060708  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00060710  00000000 00000000
00060718  00000000 00000000
00060720  00000000 00000000
00060728  00000000 00000000
00060730  00000000 00000000
00060738  00000000 00000000
00060740  00000000 00000000
00060748  00000000 00000000
00060750  00000000 00000000
00060758  00000000 00000000
                                          7349
                                          7350           TSTLBL FMSG                fail message
00060760                                  7351+FMSG84    DS     0C
00060760  86818993 858440D7               7352           DC     C'failed PLO-Extension '
00060768  D3D660C5 A7A38595
00060770  A2899695 40
00060775  E3998997 938540C3               7353           DC     CL96'Triple Compare and Swap (TCS: 30): 1st NE'
0006077D  96949781 99854081
00060785  958440E2 A6819740
0006078D  4DE3C3E2 7A40F3F0
00060795  5D7A40F1 A2A340D5
0006079D  C5404040 40404040
000607A5  40404040 40404040
000607AD  40404040 40404040
000607B5  40404040 40404040
000607BD  40404040 40404040
000607C5  40404040 40404040
000607CD  40404040 40404040
                                          7354           TSTEND
000607D6                                  7355+          DS     0H
000607D6  C5958440 968640A3               7356+          DC     CL32'End of test: 84'
000607DE  85A2A37A 40F8F440
000607E6  40404040 40404040
000607EE  40404040 40404040
                                          7357
                                          7358 * ------------------------------------------------------------------
                                          7359 * Test Triple Compare and Swap (TCS: 30) instruction : 2nd NE
                                          7360 * ------------------------------------------------------------------
                                          7361           TSTHDR 2,2,30              2nd Operand NOT Equal (cc=2, UR=2)
000607F6                  000607F6  00061000  7362+          ORG    PLOETST+(200*K)+(47*4*K)
00061000  E2E3C1D9 E3409686               7363+          DC     CL32'START of test: 85'
00061008  40A385A2 A37A40F8
00061010  F5404040 40404040
00061018  40404040 40404040
00061020                  00061020          7364+          USING *,R5                 base for test data and test routine
00061020                  00000000          7365+          USING PLOETEST,R5
00061020  0006104C                        7366+T85       DC     A(CODE85)           address of test code
00061024  0055                            7367+          DC     H'85'               test number
00061026  00                              7368+          DC     X'00'
00061027  1E                              7369+          DC     HL1'30'             Function
00061028  02                              7370+          DC     HL1'2'              expected unequal results (UR)
00061029  00                              7371+          DC     HL1'00'             found UR
0006102A  02                              7372+          DC     HL1'2'              cc
0006102B  0D                              7373+          DC     HL1'13'             not expected CC mask
0006102C  00                              7374+          DS     X                   extracted cc
00061030  00000000 00000000               7375+          DS     2F                  extract PSW after test (has CC)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00061038  00061760                                7376+          DC    A(FMSG85)           address of test fail message
0006103C  00061500                                7377+          DC    A(OPSRC85)          address of operand source
00061040  00061400                                7378+          DC    A(OPDST85)          address of operand destination
00061044  00061600                                7379+          DC    A(PL85)             address of parameter list
00061048  00001206                                7380+          DC    A(RANDX+85)         address of random source data
                                                  7381
                                                  7382           TSTCODE
0006104C                                          7383+CODE85    DS    0F
                                                  7384 *                                   load random source
0006104C  B982 0022                               7385           XGR   R2,R2
00061050  5820 5028                   00000028    7386           L     R2,RANDXA
                                                  7387 *
                                                  7388 * Triple Compare and Swap (TCS: 30)
                                                  7389 * Build Parambter List
                                                  7390 *
00061054  B982 0011                               7391           XGR   R1,R1
00061058  5810 501C                   0000001C    7392           L     R1,OPSRCA
                                                  7393
0006105C  B982 0033                               7394           XGR   R3,R3
00061060  5830 5024                   00000024    7395           L     R3,PLA
                                                  7396
00061064  B982 0044                               7397           XGR   R4,R4
00061068  5840 5020                   00000020    7398           L     R4,OPREPA
                                                  7399
0006106C  D20B 1000 2000      00000000  00000000  7400           MVC   0(12,R1),0(R2)      load Op1-Op3 data in source
00061072  D203 3040 1000      00000040  00000000  7401           MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00061078  D203 3048 1004      00000048  00000004  7402           MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
0006107E  D203 3050 1008      00000050  00000008  7403           MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
                                                  7404
00061084  D20B 4000 2080      00000000  00000080  7405           MVC   0(12,R4),128(R2)    load initial Op1-Op3 data
0006108A  D203 3058 4000      00000058  00000000  7406           MVC   88(4,R3),0(R4)      copy to PL OP1-Replacement
00061090  D203 3060 4004      00000060  00000004  7407           MVC   96(4,R3),4(R4)      copy to PL OP2-Replacement
00061096  D203 3068 4008      00000068  00000008  7408           MVC   104(4,R3),8(R4)     copy to PL OP3-Replacement
                                                  7409
                                                  7410 ****
                                                  7411 * force a failure by changing the 2nd operand
                                                  7412 ****
0006109C  D203 1004 4000      00000004  00000000  7413           MVC   4(4,R1),0(R4)
                                                  7414 ****
                                                  7415
000610A2  E313 0000 0024                  00000000  7416         STG   R1,0(R3)            load OP1 address in PL
                                                  7417
000610A8  4111 0004                       00000004  7418         LA    R1,4(R1)
000610AC  E313 0008 0024                  00000008  7419         STG   R1,8(R3)            load OP2 address in PL
                                                  7420
000610B2  4111 0004                       00000004  7421         LA    R1,4(R1)
000610B6  E313 0010 0024                  00000010  7422         STG   R1,16(R3)           load OP3 address in PL
                                                  7423
000610BC  B982 0000                               7424           XGR   R0,R0
000610C0  4100 001E                       0000001E  7425         LA    R0,30
000610C4  EE00 0000 3000      00000000  00000000  7426           PLO   0,0,0,0(R3)         Triple C/S (TCS: 30)
                                                  7427
000610CA  B98D 0020                               7428           EPSW  R2,R0               extract psw
000610CE  5020 5010                       00000010  7429         ST    R2,CCPSW              to save CC
000610D2  E310 500B 0076                  0000000B  7430         LB    R1,CCMASK           (failure CC mask)
000610D8  8910 0004                       00000004  7431         SLL   R1,4                (shift to BC instr CC position)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000610DC  4410 5110                    00061130  7432          EX    R1,F302TSTCC            fail if...
                                                 7433
                                000610E0  00000001  7434 F302CKUR EQU   *
000610E0  D200 5009 3030        00000009  00000030  7435          MVC   URFOUND,48(R3)      Extract UR
000610E6  D500 5008 3030        00000008  00000030  7436          CLC   UR,48(R3)
000610EC  4780 50D4                    000610F4  7437          BE    F302CKRES
000610F0  45F0 8220                    00000420  7438          BAL   R15,URMSG
                                                 7439 *
                                                 7440 * compare results
                                                 7441 *
                                000610F4  00000001  7442 F302CKRES EQU   *
                                                 7443 *                                  reset adresses
000610F4  B982 0011                              7444          XGR   R1,R1
000610F8  5810 501C                    0000001C  7445          L     R1,OPSRCA           source
000610FC  B982 0033                              7446          XGR   R3,R3
00061100  5830 5024                    00000024  7447          L     R3,PLA              PL
00061104  B982 0044                              7448          XGR   R4,R4
00061108  5840 5020                    00000020  7449          L     R4,OPREPA
                                                 7450
0006110C  D503 1000 3040        00000000  00000040  7451          CLC   0(4,R1),64(R3)
00061112  4770 510A                    0006112A  7452          BNE   F302XX              OP1C Different, error
00061116  D503 1004 3048        00000004  00000048  7453          CLC   4(4,R1),72(R3)
0006111C  4770 510A                    0006112A  7454          BNE   F302XX              OP2C Different, error
00061120  D503 1008 3050        00000008  00000050  7455          CLC   8(4,R1),80(R3)
00061126  478B 0000                    00000000  7456          BE    0(R11)              OP3C same, return
                                                 7457
                                0006112A  00000001  7458 F302XX   EQU   *
0006112A  45F0 8340                    00000540  7459          BAL   R15,FAILMSG
0006112E  07FB                                   7460          BR    R11
                                                 7461
00061130  4700 5114                    00061134  7462 F302TSTCC BC    0,F302CCMSG         fail if unexpected condition code
                                00061134  00000001  7463 F302CCMSG EQU   *
00061134  45F0 80F2                    000002F2  7464          BAL   R15,CCMSG
00061138  47F0 50C0                    000610E0  7465          B     F302CKUR
                                                 7466
                                                 7467          TSTDATA
0006113C                        0006113C  00061400  7468+         ORG   PLOETST+(200*K)+(47*4*K)+K
                                                 7469          TSTLBL OPDST              Operand Destination
00061400                                         7470+OPDST85  DS    0C
00061400  00000000 00000000                      7471          DC    XL256'0'
00061408  00000000 00000000
00061410  00000000 00000000
00061418  00000000 00000000
00061420  00000000 00000000
00061428  00000000 00000000
00061430  00000000 00000000
00061438  00000000 00000000
00061440  00000000 00000000
00061448  00000000 00000000
00061450  00000000 00000000
00061458  00000000 00000000
00061460  00000000 00000000
00061468  00000000 00000000
00061470  00000000 00000000
00061478  00000000 00000000
00061480  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00061488  00000000 00000000
00061490  00000000 00000000
00061498  00000000 00000000
000614A0  00000000 00000000
000614A8  00000000 00000000
000614B0  00000000 00000000
000614B8  00000000 00000000
000614C0  00000000 00000000
000614C8  00000000 00000000
000614D0  00000000 00000000
000614D8  00000000 00000000
000614E0  00000000 00000000
000614E8  00000000 00000000
000614F0  00000000 00000000
000614F8  00000000 00000000
                                          7472
                                          7473           TSTLBL OPSRC              Operand Source OP1-OP4
00061500                                  7474+OPSRC85   DS    0C
00061500  00000000 00000000               7475           DC    XL256'0'
00061508  00000000 00000000
00061510  00000000 00000000
00061518  00000000 00000000
00061520  00000000 00000000
00061528  00000000 00000000
00061530  00000000 00000000
00061538  00000000 00000000
00061540  00000000 00000000
00061548  00000000 00000000
00061550  00000000 00000000
00061558  00000000 00000000
00061560  00000000 00000000
00061568  00000000 00000000
00061570  00000000 00000000
00061578  00000000 00000000
00061580  00000000 00000000
00061588  00000000 00000000
00061590  00000000 00000000
00061598  00000000 00000000
000615A0  00000000 00000000
000615A8  00000000 00000000
000615B0  00000000 00000000
000615B8  00000000 00000000
000615C0  00000000 00000000
000615C8  00000000 00000000
000615D0  00000000 00000000
000615D8  00000000 00000000
000615E0  00000000 00000000
000615E8  00000000 00000000
000615F0  00000000 00000000
000615F8  00000000 00000000
                                          7476
                                          7477           TSTLBL PL                 Parameter List
00061600                                  7478+PL85      DS    0C
00061600  00000000 00000000               7479           DC    XL352'00'
00061608  00000000 00000000
00061610  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00061618  00000000 00000000
00061620  00000000 00000000
00061628  00000000 00000000
00061630  00000000 00000000
00061638  00000000 00000000
00061640  00000000 00000000
00061648  00000000 00000000
00061650  00000000 00000000
00061658  00000000 00000000
00061660  00000000 00000000
00061668  00000000 00000000
00061670  00000000 00000000
00061678  00000000 00000000
00061680  00000000 00000000
00061688  00000000 00000000
00061690  00000000 00000000
00061698  00000000 00000000
000616A0  00000000 00000000
000616A8  00000000 00000000
000616B0  00000000 00000000
000616B8  00000000 00000000
000616C0  00000000 00000000
000616C8  00000000 00000000
000616D0  00000000 00000000
000616D8  00000000 00000000
000616E0  00000000 00000000
000616E8  00000000 00000000
000616F0  00000000 00000000
000616F8  00000000 00000000
00061700  00000000 00000000
00061708  00000000 00000000
00061710  00000000 00000000
00061718  00000000 00000000
00061720  00000000 00000000
00061728  00000000 00000000
00061730  00000000 00000000
00061738  00000000 00000000
00061740  00000000 00000000
00061748  00000000 00000000
00061750  00000000 00000000
00061758  00000000 00000000
                                      7480
                                      7481          TSTLBL FMSG               fail message
00061760                              7482+FMSG85   DS    0C
00061760  86818993 858440D7          7483          DC    C'failed PLO-Extension '
00061768  D3D660C5 A7A38595
00061770  A2899695 40
00061775  E3998997 938540C3          7484          DC    CL96'Triple Compare and Swap (TCS: 30): 2nd NE'
0006177D  96949781 99854081
00061785  958440E2 A6819740
0006178D  4DE3C3E2 7A40F3F0
00061795  5D7A40F2 958440D5
0006179D  C5404040 40404040
000617A5  40404040 40404040
000617AD  40404040 40404040
000617B5  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000617BD  40404040 40404040
000617C5  40404040 40404040
000617CD  40404040 40404040
                                        7485          TSTEND
000617D6                                7486+         DS    0H
000617D6  C5958440 968640A3             7487+         DC    CL32'End of test: 85'
000617DE  85A2A37A 40F8F540
000617E6  40404040 40404040
000617EE  40404040 40404040
                                        7488
                                        7489 * ------------------------------------------------------------------
                                        7490 * Test Triple Compare and Swap (TCS: 30) instruction : 3rd NE
                                        7491 * ------------------------------------------------------------------
                                        7492          TSTHDR 2,3,30             3rd Operand NOT Equal (cc=2, UR=3)
000617F6                        000617F6  00062000    7493+         ORG   PLOETST+(200*K)+(48*4*K)
00062000  E2E3C1D9 E3409686             7494+         DC    CL32'START of test: 86'
00062008  40A385A2 A37A40F8
00062010  F6404040 40404040
00062018  40404040 40404040
00062020                        00062020              7495+         USING *,R5                base for test data and test routine
00062020                        00000000              7496+         USING PLOETEST,R5
00062020  0006204C                                    7497+T86      DC    A(CODE86)           address of test code
00062024  0056                                        7498+         DC    H'86'               test number
00062026  00                                          7499+         DC    X'00'
00062027  1E                                          7500+         DC    HL1'30'             Function
00062028  03                                          7501+         DC    HL1'3'              expected unequal results (UR)
00062029  00                                          7502+         DC    HL1'00'             found UR
0006202A  02                                          7503+         DC    HL1'2'              cc
0006202B  0D                                          7504+         DC    HL1'13'             not expected CC mask
0006202C  00                                          7505+         DS    X                   extracted cc
00062030  00000000 00000000                           7506+         DS    2F                  extract PSW after test (has CC)
00062038  00062760                                    7507+         DC    A(FMSG86)           address of test fail message
0006203C  00062500                                    7508+         DC    A(OPSRC86)          address of operand source
00062040  00062400                                    7509+         DC    A(OPDST86)          address of operand destination
00062044  00062600                                    7510+         DC    A(PL86)             address of parameter list
00062048  00001207                                    7511+         DC    A(RANDX+86)         address of random source data
                                                      7512
                                                      7513          TSTCODE
0006204C                                              7514+CODE86   DS    0F
                                                      7515 *                                  load random source
0006204C  B982 0022                                   7516          XGR   R2,R2
00062050  5820 5028                     00000028      7517          L     R2,RANDXA
                                                      7518 *
                                                      7519 * Triple Compare and Swap (TCS: 30)
                                                      7520 * Build Parambter List
                                                      7521 *
00062054  B982 0011                                   7522          XGR   R1,R1
00062058  5810 501C                     0000001C      7523          L     R1,OPSRCA
                                                      7524
0006205C  B982 0033                                   7525          XGR   R3,R3
00062060  5830 5024                     00000024      7526          L     R3,PLA
                                                      7527
00062064  B982 0044                                   7528          XGR   R4,R4
00062068  5840 5020                     00000020      7529          L     R4,OPREPA
                                                      7530
0006206C  D20B 1000 2000        00000000  00000000    7531          MVC   0(12,R1),0(R2)      load Op1-Op3 data in source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00062072  D203 3040 1000      00000040  00000000  7532          MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00062078  D203 3048 1004      00000048  00000004  7533          MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
0006207E  D203 3050 1008      00000050  00000008  7534          MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
                                                  7535
00062084  D20B 4000 2080      00000000  00000080  7536          MVC   0(12,R4),128(R2)    load initial Op1-Op3 data
0006208A  D203 3058 4000      00000058  00000000  7537          MVC   88(4,R3),0(R4)      copy to PL OP1-Replacement
00062090  D203 3060 4004      00000060  00000004  7538          MVC   96(4,R3),4(R4)      copy to PL OP2-Replacement
00062096  D203 3068 4008      00000068  00000008  7539          MVC   104(4,R3),8(R4)     copy to PL OP3-Replacement
                                                  7540
                                                  7541 ****
                                                  7542 * force a failure by changing the 3rd operand
                                                  7543 ****
0006209C  D203 1008 4000      00000008  00000000  7544          MVC   8(4,R1),0(R4)
                                                  7545 ****
                                                  7546
000620A2  E313 0000 0024                00000000  7547          STG   R1,0(R3)            load OP1 address in PL
                                                  7548
000620A8  4111 0004                     00000004  7549          LA    R1,4(R1)
000620AC  E313 0008 0024                00000008  7550          STG   R1,8(R3)            load OP2 address in PL
                                                  7551
000620B2  4111 0004                     00000004  7552          LA    R1,4(R1)
000620B6  E313 0010 0024                00000010  7553          STG   R1,16(R3)           load OP3 address in PL
                                                  7554
000620BC  B982 0000                               7555          XGR   R0,R0
000620C0  4100 001E                     0000001E  7556          LA    R0,30
000620C4  EE00 0000 3000      00000000  00000000  7557          PLO   0,0,0,0(R3)         Triple C/S (TCS: 30)
                                                  7558
000620CA  B98D 0020                               7559          EPSW  R2,R0               extract psw
000620CE  5020 5010                     00000010  7560          ST    R2,CCPSW              to save CC
000620D2  E310 500B 0076                0000000B  7561          LB    R1,CCMASK           (failure CC mask)
000620D8  8910 0004                     00000004  7562          SLL   R1,4                (shift to BC instr CC position)
000620DC  4410 5110                     00062130  7563          EX    R1,F303TSTCC            fail if...
                                                  7564
                                000620E0  00000001  7565 F303CKUR EQU   *
000620E0  D200 5009 3030      00000009  00000030  7566          MVC   URFOUND,48(R3)      Extract UR
000620E6  D500 5008 3030      00000008  00000030  7567          CLC   UR,48(R3)
000620EC  4780 50D4                     000620F4  7568          BE    F303CKRES
000620F0  45F0 8220                     00000420  7569          BAL   R15,URMSG
                                                  7570 *
                                                  7571 * compare results
                                                  7572 *
                                000620F4  00000001  7573 F303CKRES EQU   *
                                                  7574 *                                reset adresses
000620F4  B982 0011                               7575          XGR   R1,R1
000620F8  5810 501C                     0000001C  7576          L     R1,OPSRCA           source
000620FC  B982 0033                               7577          XGR   R3,R3
00062100  5830 5024                     00000024  7578          L     R3,PLA              PL
00062104  B982 0044                               7579          XGR   R4,R4
00062108  5840 5020                     00000020  7580          L     R4,OPREPA
                                                  7581
0006210C  D503 1000 3040      00000000  00000040  7582          CLC   0(4,R1),64(R3)
00062112  4770 510A                     0006212A  7583          BNE   F303XX              OP1C Different, error
00062116  D503 1004 3048      00000004  00000048  7584          CLC   4(4,R1),72(R3)
0006211C  4770 510A                     0006212A  7585          BNE   F303XX              OP2C Different, error
00062120  D503 1008 3050      00000008  00000050  7586          CLC   8(4,R1),80(R3)
00062126  478B 0000                     00000000  7587          BE    0(R11)              OP3C same, return
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    7588
                               0006212A  00000001  7589 F303XX   EQU   *
0006212A  45F0 8340                      00000540  7590          BAL   R15,FAILMSG
0006212E  07FB                                     7591          BR    R11
                                                   7592
00062130  4700 5114                      00062134  7593 F303TSTCC BC   0,F303CCMSG         fail if unexpected condition code
                               00062134  00000001  7594 F303CCMSG EQU  *
00062134  45F0 80F2                      000002F2  7595          BAL   R15,CCMSG
00062138  47F0 50C0                      000620E0  7596          B     F303CKUR
                                                   7597
                                                   7598          TSTDATA
0006213C                       0006213C  00062400  7599+         ORG   PLOETST+(200*K)+(48*4*K)+K
                                                   7600          TSTLBL OPDST              Operand Destination
00062400                                           7601+OPDST86  DS    0C
00062400  00000000 00000000                        7602          DC    XL256'0'
00062408  00000000 00000000
00062410  00000000 00000000
00062418  00000000 00000000
00062420  00000000 00000000
00062428  00000000 00000000
00062430  00000000 00000000
00062438  00000000 00000000
00062440  00000000 00000000
00062448  00000000 00000000
00062450  00000000 00000000
00062458  00000000 00000000
00062460  00000000 00000000
00062468  00000000 00000000
00062470  00000000 00000000
00062478  00000000 00000000
00062480  00000000 00000000
00062488  00000000 00000000
00062490  00000000 00000000
00062498  00000000 00000000
000624A0  00000000 00000000
000624A8  00000000 00000000
000624B0  00000000 00000000
000624B8  00000000 00000000
000624C0  00000000 00000000
000624C8  00000000 00000000
000624D0  00000000 00000000
000624D8  00000000 00000000
000624E0  00000000 00000000
000624E8  00000000 00000000
000624F0  00000000 00000000
000624F8  00000000 00000000
                                                   7603
                                                   7604          TSTLBL OPSRC              Operand Source OP1-OP4
00062500                                           7605+OPSRC86  DS    0C
00062500  00000000 00000000                        7606          DC    XL256'0'
00062508  00000000 00000000
00062510  00000000 00000000
00062518  00000000 00000000
00062520  00000000 00000000
00062528  00000000 00000000
00062530  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00062538  00000000 00000000
00062540  00000000 00000000
00062548  00000000 00000000
00062550  00000000 00000000
00062558  00000000 00000000
00062560  00000000 00000000
00062568  00000000 00000000
00062570  00000000 00000000
00062578  00000000 00000000
00062580  00000000 00000000
00062588  00000000 00000000
00062590  00000000 00000000
00062598  00000000 00000000
000625A0  00000000 00000000
000625A8  00000000 00000000
000625B0  00000000 00000000
000625B8  00000000 00000000
000625C0  00000000 00000000
000625C8  00000000 00000000
000625D0  00000000 00000000
000625D8  00000000 00000000
000625E0  00000000 00000000
000625E8  00000000 00000000
000625F0  00000000 00000000
000625F8  00000000 00000000
                                          7607
                                          7608           TSTLBL PL                  Parameter List
00062600                                  7609+PL86      DS    0C
00062600  00000000 00000000               7610           DC    XL352'00'
00062608  00000000 00000000
00062610  00000000 00000000
00062618  00000000 00000000
00062620  00000000 00000000
00062628  00000000 00000000
00062630  00000000 00000000
00062638  00000000 00000000
00062640  00000000 00000000
00062648  00000000 00000000
00062650  00000000 00000000
00062658  00000000 00000000
00062660  00000000 00000000
00062668  00000000 00000000
00062670  00000000 00000000
00062678  00000000 00000000
00062680  00000000 00000000
00062688  00000000 00000000
00062690  00000000 00000000
00062698  00000000 00000000
000626A0  00000000 00000000
000626A8  00000000 00000000
000626B0  00000000 00000000
000626B8  00000000 00000000
000626C0  00000000 00000000
000626C8  00000000 00000000
000626D0  00000000 00000000
000626D8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000626E0  00000000 00000000
000626E8  00000000 00000000
000626F0  00000000 00000000
000626F8  00000000 00000000
00062700  00000000 00000000
00062708  00000000 00000000
00062710  00000000 00000000
00062718  00000000 00000000
00062720  00000000 00000000
00062728  00000000 00000000
00062730  00000000 00000000
00062738  00000000 00000000
00062740  00000000 00000000
00062748  00000000 00000000
00062750  00000000 00000000
00062758  00000000 00000000
                                          7611
                                          7612          TSTLBL FMSG               fail message
00062760                                  7613+FMSG86   DS    0C
00062760  86818993 858440D7               7614          DC    C'failed PLO-Extension '
00062768  D3D660C5 A7A38595
00062770  A2899695 40
00062775  E3998997 938540C3               7615          DC    CL96'Triple Compare and Swap (TCS: 30): 3rd NE'
0006277D  96949781 99854081
00062785  958440E2 A6819740
0006278D  4DE3C3E2 7A40F3F0
00062795  5D7A40F3 998440D5
0006279D  C5404040 40404040
000627A5  40404040 40404040
000627AD  40404040 40404040
000627B5  40404040 40404040
000627BD  40404040 40404040
000627C5  40404040 40404040
000627CD  40404040 40404040
                                          7616          TSTEND
000627D6                                  7617+         DS    0H
000627D6  C5958440 968640A3               7618+         DC    CL32'End of test: 86'
000627DE  85A2A37A 40F8F640
000627E6  40404040 40404040
000627EE  40404040 40404040
                                          7619
                                          7620 * ------------------------------------------------------------------
                                          7621 * Test Triple Compare and Swap (TCS: 30) instruction : All EQ
                                          7622 * ------------------------------------------------------------------
                                          7623          TSTHDR 0,99,30            All Operand Equal (cc=0, UR=99)
000627F6                  000627F6  00063000  7624+         ORG   PLOETST+(200*K)+(49*4*K)
00063000  E2E3C1D9 E3409686               7625+         DC    CL32'START of test: 87'
00063008  40A385A2 A37A40F8
00063010  F7404040 40404040
00063018  40404040 40404040
00063020                  00063020          7626+         USING *,R5                base for test data and test routine
00063020                  00000000          7627+         USING PLOETEST,R5
00063020  0006304C                        7628+T87      DC    A(CODE87)           address of test code
00063024  0057                            7629+         DC    H'87'               test number
00063026  00                              7630+         DC    X'00'
00063027  1E                              7631+         DC    HL1'30'             Function
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00063028  63                                      7632+          DC    HL1'99'             expected unequal results (UR)
00063029  00                                      7633+          DC    HL1'00'             found UR
0006302A  00                                      7634+          DC    HL1'0'              cc
0006302B  07                                      7635+          DC    HL1'7'              not expected CC mask
0006302C  00                                      7636+          DS    X                   extracted cc
00063030  00000000 00000000                       7637+          DS    2F                  extract PSW after test (has CC)
00063038  00063760                                7638+          DC    A(FMSG87)           address of test fail message
0006303C  00063500                                7639+          DC    A(OPSRC87)          address of operand source
00063040  00063400                                7640+          DC    A(OPDST87)          address of operand destination
00063044  00063600                                7641+          DC    A(PL87)             address of parameter list
00063048  00001208                                7642+          DC    A(RANDX+87)         address of random source data
                                                  7643
                                                  7644           TSTCODE
0006304C                                          7645+CODE87    DS    0F
                                                  7646 *                                   load random source
0006304C  B982 0022                               7647           XGR   R2,R2
00063050  5820 5028                     00000028  7648           L     R2,RANDXA
                                                  7649 *
                                                  7650 * Triple Compare and Swap (TCS: 30)
                                                  7651 * Build Parambter List
                                                  7652 *
00063054  B982 0011                               7653           XGR   R1,R1
00063058  5810 501C                     0000001C  7654           L     R1,OPSRCA
                                                  7655
0006305C  B982 0033                               7656           XGR   R3,R3
00063060  5830 5024                     00000024  7657           L     R3,PLA
                                                  7658
00063064  B982 0044                               7659           XGR   R4,R4
00063068  5840 5020                     00000020  7660           L     R4,OPREPA
                                                  7661
0006306C  D20B 1000 2000        00000000  00000000  7662           MVC   0(12,R1),0(R2)      load Op1-Op3 data in source
00063072  D203 3040 1000        00000040  00000000  7663           MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00063078  D203 3048 1004        00000048  00000004  7664           MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
0006307E  D203 3050 1008        00000050  00000008  7665           MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
                                                  7666
00063084  D20B 4000 2080        00000000  00000080  7667           MVC   0(12,R4),128(R2)    load initial Op1-Op3 data
0006308A  D203 3058 4000        00000058  00000000  7668           MVC   88(4,R3),0(R4)      copy to PL OP1-Replacement
00063090  D203 3060 4004        00000060  00000004  7669           MVC   96(4,R3),4(R4)      copy to PL OP2-Replacement
00063096  D203 3068 4008        00000068  00000008  7670           MVC   104(4,R3),8(R4)     copy to PL OP3-Replacement
                                                  7671
                                                  7672 ****
                                                  7673 * Do NOT force a failure by changing the 3rd operand
                                                  7674 ****
                                                  7675 *          MVC   8(4,R1),0(R4)
                                                  7676 ****
                                                  7677
0006309C  E313 0000 0024                00000000  7678           STG   R1,0(R3)            load OP1 address in PL
                                                  7679
000630A2  4111 0004                     00000004  7680           LA    R1,4(R1)
000630A6  E313 0008 0024                00000008  7681           STG   R1,8(R3)            load OP2 address in PL
                                                  7682
000630AC  4111 0004                     00000004  7683           LA    R1,4(R1)
000630B0  E313 0010 0024                00000010  7684           STG   R1,16(R3)           load OP3 address in PL
                                                  7685
000630B6  B982 0000                               7686           XGR   R0,R0
000630BA  4100 001E                     0000001E  7687           LA    R0,30
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000630BE  EE00 0000 3000      00000000  00000000  7688          PLO   0,0,0,0(R3)         Triple C/S (TCS: 30)
                                                  7689
000630C4  B98D 0020                               7690          EPSW  R2,R0               extract psw
000630C8  5020 5010                     00000010  7691          ST    R2,CCPSW              to save CC
000630CC  E310 500B 0076                0000000B  7692          LB    R1,CCMASK           (failure CC mask)
000630D2  8910 0004                     00000004  7693          SLL   R1,4                (shift to BC instr CC position)
000630D6  4410 50F6                     00063116  7694          EX    R1,F304TSTCC            fail if...
                                                  7695 *
                                                  7696 * compare results
                                                  7697 *
                          000630DA      00000001  7698 F304CKRES EQU   *
                                                  7699 *                                    reset adresses
000630DA  B982 0011                               7700          XGR   R1,R1
000630DE  5810 501C                     0000001C  7701          L     R1,OPSRCA           source
000630E2  B982 0033                               7702          XGR   R3,R3
000630E6  5830 5024                     00000024  7703          L     R3,PLA              PL
000630EA  B982 0044                               7704          XGR   R4,R4
000630EE  5840 5020                     00000020  7705          L     R4,OPREPA
                                                  7706
                                                  7707 * All equal; replacement values copied to operand source
                                                  7708
000630F2  D503 1000 3058      00000000  00000058  7709          CLC   0(4,R1),88(R3)
000630F8  4770 50F0                     00063110  7710          BNE   F304XX              OP1R Different, error
000630FC  D503 1004 3060      00000004  00000060  7711          CLC   4(4,R1),96(R3)
00063102  4770 50F0                     00063110  7712          BNE   F304XX              OP2R Different, error
00063106  D503 1008 3068      00000008  00000068  7713          CLC   8(4,R1),104(R3)
0006310C  478B 0000                     00000000  7714          BE    0(R11)              OP4R same, return
                                                  7715
                          00063110      00000001  7716 F304XX   EQU   *
00063110  45F0 8340                     00000540  7717          BAL   R15,FAILMSG
00063114  07FB                                    7718          BR    R11
                                                  7719
00063116  4700 50FA                     0006311A  7720 F304TSTCC BC    0,F304CCMSG         fail if unexpected condition code
                          0006311A      00000001  7721 F304CCMSG EQU  *
0006311A  45F0 80F2                     000002F2  7722          BAL   R15,CCMSG
0006311E  47F0 50BA                     000630DA  7723          B     F304CKRES
                                                  7724
                                                  7725          TSTDATA
00063122                  00063122      00063400  7726+         ORG   PLOETST+(200*K)+(49*4*K)+K
                                                  7727          TSTLBL OPDST              Operand Destination
00063400                                          7728+OPDST87  DS    0C
00063400  00000000 00000000                       7729          DC    XL256'0'
00063408  00000000 00000000
00063410  00000000 00000000
00063418  00000000 00000000
00063420  00000000 00000000
00063428  00000000 00000000
00063430  00000000 00000000
00063438  00000000 00000000
00063440  00000000 00000000
00063448  00000000 00000000
00063450  00000000 00000000
00063458  00000000 00000000
00063460  00000000 00000000
00063468  00000000 00000000
00063470  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00063478  00000000 00000000
00063480  00000000 00000000
00063488  00000000 00000000
00063490  00000000 00000000
00063498  00000000 00000000
000634A0  00000000 00000000
000634A8  00000000 00000000
000634B0  00000000 00000000
000634B8  00000000 00000000
000634C0  00000000 00000000
000634C8  00000000 00000000
000634D0  00000000 00000000
000634D8  00000000 00000000
000634E0  00000000 00000000
000634E8  00000000 00000000
000634F0  00000000 00000000
000634F8  00000000 00000000
                                          7730
                                          7731           TSTLBL OPSRC              Operand Source OP1-OP4
00063500                                  7732+OPSRC87   DS    0C
00063500  00000000 00000000               7733           DC    XL256'0'
00063508  00000000 00000000
00063510  00000000 00000000
00063518  00000000 00000000
00063520  00000000 00000000
00063528  00000000 00000000
00063530  00000000 00000000
00063538  00000000 00000000
00063540  00000000 00000000
00063548  00000000 00000000
00063550  00000000 00000000
00063558  00000000 00000000
00063560  00000000 00000000
00063568  00000000 00000000
00063570  00000000 00000000
00063578  00000000 00000000
00063580  00000000 00000000
00063588  00000000 00000000
00063590  00000000 00000000
00063598  00000000 00000000
000635A0  00000000 00000000
000635A8  00000000 00000000
000635B0  00000000 00000000
000635B8  00000000 00000000
000635C0  00000000 00000000
000635C8  00000000 00000000
000635D0  00000000 00000000
000635D8  00000000 00000000
000635E0  00000000 00000000
000635E8  00000000 00000000
000635F0  00000000 00000000
000635F8  00000000 00000000
                                          7734
                                          7735           TSTLBL PL                 Parameter List
00063600                                  7736+PL87      DS    0C
00063600  00000000 00000000               7737           DC    XL352'00'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00063608  00000000 00000000
00063610  00000000 00000000
00063618  00000000 00000000
00063620  00000000 00000000
00063628  00000000 00000000
00063630  00000000 00000000
00063638  00000000 00000000
00063640  00000000 00000000
00063648  00000000 00000000
00063650  00000000 00000000
00063658  00000000 00000000
00063660  00000000 00000000
00063668  00000000 00000000
00063670  00000000 00000000
00063678  00000000 00000000
00063680  00000000 00000000
00063688  00000000 00000000
00063690  00000000 00000000
00063698  00000000 00000000
000636A0  00000000 00000000
000636A8  00000000 00000000
000636B0  00000000 00000000
000636B8  00000000 00000000
000636C0  00000000 00000000
000636C8  00000000 00000000
000636D0  00000000 00000000
000636D8  00000000 00000000
000636E0  00000000 00000000
000636E8  00000000 00000000
000636F0  00000000 00000000
000636F8  00000000 00000000
00063700  00000000 00000000
00063708  00000000 00000000
00063710  00000000 00000000
00063718  00000000 00000000
00063720  00000000 00000000
00063728  00000000 00000000
00063730  00000000 00000000
00063738  00000000 00000000
00063740  00000000 00000000
00063748  00000000 00000000
00063750  00000000 00000000
00063758  00000000 00000000
                                        7738
                                        7739          TSTLBL FMSG               fail message
00063760                                7740+FMSG87   DS    0C
00063760  86818993 858440D7            7741          DC    C'failed PLO-Extension '
00063768  D3D660C5 A7A38595
00063770  A2899695 40
00063775  E3998997 938540C3            7742          DC    CL96'Triple Compare and Swap (TCS: 30): All EQ'
0006377D  96949781 99854081
00063785  958440E2 A6819740
0006378D  4DE3C3E2 7A40F3F0
00063795  5D7A40C1 939340C5
0006379D  D8404040 40404040
000637A5  40404040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000637AD  40404040 40404040
000637B5  40404040 40404040
000637BD  40404040 40404040
000637C5  40404040 40404040
000637CD  40404040 40404040
                                          7743          TSTEND
000637D6                                  7744+         DS    0H
000637D6  C5958440 968640A3               7745+         DC    CL32'End of test: 87'
000637DE  85A2A37A 40F8F740
000637E6  40404040 40404040
000637EE  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    7747 *=======================================================================
                                                    7748 *       Compare and Swap and Triple Store Tests
                                                    7749 *=======================================================================
                                                    7750
                                                    7751 * -----------------------------------------------------------------
                                                    7752 * Test Compare and Swap and Triple Store (CSTSTO: 29) instruction : NE
                                                    7753 * -----------------------------------------------------------------
                                                    7754          TSTHDR 1,99,29           1st Operand NOT Equal (cc=1, UR=99)
000637F6                          000637F6  00064000  7755+         ORG   PLOETST+(200*K)+(50*4*K)
00064000  E2E3C1D9 E3409686                         7756+         DC    CL32'START of test: 88'
00064008  40A385A2 A37A40F8
00064010  F8404040 40404040
00064018  40404040 40404040
00064020                          00064020          7757+         USING *,R5               base for test data and test routine
00064020                          00000000          7758+         USING PLOETEST,R5
00064020  0006404C                                  7759+T88      DC    A(CODE88)          address of test code
00064024  0058                                      7760+         DC    H'88'              test number
00064026  00                                        7761+         DC    X'00'
00064027  1D                                        7762+         DC    HL1'29'            Function
00064028  63                                        7763+         DC    HL1'99'            expected unequal results (UR)
00064029  00                                        7764+         DC    HL1'00'            found UR
0006402A  01                                        7765+         DC    HL1'1'             cc
0006402B  0B                                        7766+         DC    HL1'11'            not expected CC mask
0006402C  00                                        7767+         DS    X                  extracted cc
00064030  00000000 00000000                         7768+         DS    2F                 extract PSW after test (has CC)
00064038  00064760                                  7769+         DC    A(FMSG88)          address of test fail message
0006403C  00064500                                  7770+         DC    A(OPSRC88)         address of operand source
00064040  00064400                                  7771+         DC    A(OPDST88)         address of operand destination
00064044  00064600                                  7772+         DC    A(PL88)            address of parameter list
00064048  00001209                                  7773+         DC    A(RANDX+88)        address of random source data
                                                    7774
                                                    7775          TSTCODE
0006404C                                            7776+CODE88   DS    0F
                                                    7777 *                                 load random source
0006404C  B982 0022                                 7778          XGR   R2,R2
00064050  5820 5028                     00000028    7779          L     R2,RANDXA
                                                    7780 *
                                                    7781 * Compare and Swap and Triple Store (CSTSTO: 29)
                                                    7782 * Build Parambter List
                                                    7783 *
00064054  B982 0011                                 7784          XGR   R1,R1
00064058  5810 501C                     0000001C    7785          L     R1,OPSRCA          Op1-C, Op3, Op5, Op7, Op1-R
                                                    7786
0006405C  B982 0033                                 7787          XGR   R3,R3
00064060  5830 5024                     00000024    7788          L     R3,PLA
                                                    7789
00064064  B982 0044                                 7790          XGR   R4,R4
00064068  5840 5020                     00000020    7791          L     R4,OPREPA          OP2, Op4, Op6, Op8
                                                    7792
0006406C  D29F 1000 2000        00000000  00000000  7793          MVC   0(160,R1),0(R2)    load Op1-C, Op3, Op5, Op7, Op1-R
00064072  D21F 3000 1000        00000000  00000000  7794          MVC   0(32,R3),0(R1)     copy to PL OP1-C
00064078  D21F 3090 1020        00000090  00000020  7795          MVC   144(32,R3),32(R1)  copy to PL OP3
0006407E  D21F 30B0 1040        000000B0  00000040  7796          MVC   176(32,R3),64(R1)  copy to PL OP5
00064084  D21F 30D0 1060        000000D0  00000060  7797          MVC   208(32,R3),96(R1)  copy to PL OP7
0006408A  D21F 3020 1080        00000020  00000080  7798          MVC   32(32,R3),128(R1)  copy to PL OP1-R
                                                    7799
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00064090  D27F 4000 2080      00000000  00000080  7800          MVC   0(128,R4),128(R2)   load initial Op2, Op4, Op6, Op8
                                                   7801
                                                   7802 ****
                                                   7803 * force a failure by changing the Op1-c
                                                   7804 ****
00064096  D203 3000 2021      00000000  00000021  7805          MVC   0(4,R3),33(R2)
                                                   7806 ****
0006409C  B982 0000                               7807          XGR   R0,R0
000640A0  4104 0020                     00000020  7808          LA    R0,32(R4)
000640A4  E303 0048 0024                00000048  7809          STG   R0,72(R3)           load OP4 address in PL
                                                   7810
000640AA  4104 0040                     00000040  7811          LA    R0,64(R4)
000640AE  E303 0068 0024                00000068  7812          STG   R0,104(R3)          load OP6 address in PL
                                                   7813
000640B4  4104 0060                     00000060  7814          LA    R0,96(R4)
000640B8  E303 0088 0024                00000088  7815          STG   R0,136(R3)          load OP8 address in PL
                                                   7816
000640BE  B982 0000                               7817          XGR   R0,R0
000640C2  4100 001D                     0000001D  7818          LA    R0,29
000640C6  EE00 4000 3000      00000000  00000000  7819          PLO   0,0(R4),0,0(R3)     C/S Triple Store (CSTSTO: 29)
                                                   7820
000640CC  B98D 0020                               7821          EPSW  R2,R0               extract psw
000640D0  5020 5010                     00000010  7822          ST    R2,CCPSW              to save CC
000640D4  E310 500B 0076                0000000B  7823          LB    R1,CCMASK           (failure CC mask)
000640DA  8910 0004                     00000004  7824          SLL   R1,4                (shift to BC instr CC position)
000640DE  4410 50EA                     0006410A  7825          EX    R1,F291TSTCC            fail if...
                                                   7826 *
                                                   7827 * compare results
                                                   7828 *
                          000640E2  00000001  7829 F291CKRES EQU   *
                                                   7830 *                                reset adresses
000640E2  B982 0011                               7831          XGR   R1,R1
000640E6  5810 501C                     0000001C  7832          L     R1,OPSRCA           Op1, Op3, Op5, Op7
000640EA  B982 0033                               7833          XGR   R3,R3
000640EE  5830 5024                     00000024  7834          L     R3,PLA              PL
000640F2  B982 0044                               7835          XGR   R4,R4
000640F6  5840 5020                     00000020  7836          L     R4,OPREPA           OP2, Op4, Op6, Op8
                                                   7837
000640FA  D51F 4000 3000      00000000  00000000  7838          CLC   0(32,R4),0(R3)      Op2 == OP1C?
00064100  478B 0000                     00000000  7839          BE    0(R11)              same, return
                                                   7840
                          00064104  00000001  7841 F291XX   EQU   *
00064104  45F0 8340                     00000540  7842          BAL   R15,FAILMSG
00064108  07FB                                    7843          BR    R11
                                                   7844
0006410A  4700 50EE                     0006410E  7845 F291TSTCC BC    0,F291CCMSG         fail if unexpected condition code
                          0006410E  00000001  7846 F291CCMSG EQU   *
0006410E  45F0 80F2                     000002F2  7847          BAL   R15,CCMSG
00064112  47F0 50C2                     000640E2  7848          B     F291CKRES
                                                   7849
                                                   7850          TSTDATA
00064116                  00064116  00064400  7851+          ORG   PLOETST+(200*K)+(50*4*K)+K
                                                   7852          TSTLBL OPDST              OP2, Op4, Op6, Op8
00064400                                           7853+OPDST88  DS    0C
00064400  00000000 00000000                       7854          DC    XL256'0'
00064408  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00064410  00000000 00000000
00064418  00000000 00000000
00064420  00000000 00000000
00064428  00000000 00000000
00064430  00000000 00000000
00064438  00000000 00000000
00064440  00000000 00000000
00064448  00000000 00000000
00064450  00000000 00000000
00064458  00000000 00000000
00064460  00000000 00000000
00064468  00000000 00000000
00064470  00000000 00000000
00064478  00000000 00000000
00064480  00000000 00000000
00064488  00000000 00000000
00064490  00000000 00000000
00064498  00000000 00000000
000644A0  00000000 00000000
000644A8  00000000 00000000
000644B0  00000000 00000000
000644B8  00000000 00000000
000644C0  00000000 00000000
000644C8  00000000 00000000
000644D0  00000000 00000000
000644D8  00000000 00000000
000644E0  00000000 00000000
000644E8  00000000 00000000
000644F0  00000000 00000000
000644F8  00000000 00000000
                                                7855
                                                7856           TSTLBL OPSRC              Op1-C, Op3, Op5, Op7, Op1-R
00064500                                        7857+OPSRC88   DS     0C
00064500  00000000 00000000                     7858           DC     XL256'0'
00064508  00000000 00000000
00064510  00000000 00000000
00064518  00000000 00000000
00064520  00000000 00000000
00064528  00000000 00000000
00064530  00000000 00000000
00064538  00000000 00000000
00064540  00000000 00000000
00064548  00000000 00000000
00064550  00000000 00000000
00064558  00000000 00000000
00064560  00000000 00000000
00064568  00000000 00000000
00064570  00000000 00000000
00064578  00000000 00000000
00064580  00000000 00000000
00064588  00000000 00000000
00064590  00000000 00000000
00064598  00000000 00000000
000645A0  00000000 00000000
000645A8  00000000 00000000
000645B0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000645B8  00000000 00000000
000645C0  00000000 00000000
000645C8  00000000 00000000
000645D0  00000000 00000000
000645D8  00000000 00000000
000645E0  00000000 00000000
000645E8  00000000 00000000
000645F0  00000000 00000000
000645F8  00000000 00000000                          7859
                                                     7860          TSTLBL PL                  Parameter List
00064600                                             7861+PL88     DS    0C
00064600  00000000 00000000                          7862          DC    XL352'00'
00064608  00000000 00000000
00064610  00000000 00000000
00064618  00000000 00000000
00064620  00000000 00000000
00064628  00000000 00000000
00064630  00000000 00000000
00064638  00000000 00000000
00064640  00000000 00000000
00064648  00000000 00000000
00064650  00000000 00000000
00064658  00000000 00000000
00064660  00000000 00000000
00064668  00000000 00000000
00064670  00000000 00000000
00064678  00000000 00000000
00064680  00000000 00000000
00064688  00000000 00000000
00064690  00000000 00000000
00064698  00000000 00000000
000646A0  00000000 00000000
000646A8  00000000 00000000
000646B0  00000000 00000000
000646B8  00000000 00000000
000646C0  00000000 00000000
000646C8  00000000 00000000
000646D0  00000000 00000000
000646D8  00000000 00000000
000646E0  00000000 00000000
000646E8  00000000 00000000
000646F0  00000000 00000000
000646F8  00000000 00000000
00064700  00000000 00000000
00064708  00000000 00000000
00064710  00000000 00000000
00064718  00000000 00000000
00064720  00000000 00000000
00064728  00000000 00000000
00064730  00000000 00000000
00064738  00000000 00000000
00064740  00000000 00000000
00064748  00000000 00000000
00064750  00000000 00000000
00064758  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    7863
                                                    7864          TSTLBL FMSG               fail message
00064760                                            7865+FMSG88   DS    0C
00064760  86818993 858440D7                         7866          DC    C'failed PLO-Extension '
00064768  D3D660C5 A7A38595
00064770  A2899695 40
00064775  C3969497 81998540                         7867          DC    C'Compare and Swap and Triple Store (CSTSTO: 29) : NE'
0006477D  81958440 E2A68197
00064785  40819584 40E39989
0006478D  97938540 E2A39699
00064795  85404DC3 E2E3E2E3
0006479D  D67A40F2 F95D407A
000647A5  40D5C5
000647A8  40404040 40404040                         7868          DC    CL64' '
000647B0  40404040 40404040
000647B8  40404040 40404040
000647C0  40404040 40404040
000647C8  40404040 40404040
000647D0  40404040 40404040
000647D8  40404040 40404040
000647E0  40404040 40404040
                                                    7869          TSTEND
000647E8                                            7870+         DS    0H
000647E8  C5958440 968640A3                         7871+         DC    CL32'End of test: 88'
000647F0  85A2A37A 40F8F840
000647F8  40404040 40404040
00064800  40404040 40404040
                                                    7872
                                                    7873 * ------------------------------------------------------------------------
                                                    7874 * Test Compare and Swap and Triple Store (CSTSTO: 29) instruction : EQ
                                                    7875 * ------------------------------------------------------------------------
                                                    7876          TSTHDR 0,99,29            1st Operand NOT Equal (cc=1, UR=99)
00064808                            00064808  00065000  7877+         ORG   PLOETST+(200*K)+(51*4*K)
00065000  E2E3C1D9 E3409686                         7878+         DC    CL32'START of test: 89'
00065008  40A385A2 A37A40F8
00065010  F9404040 40404040
00065018  40404040 40404040
00065020                            00065020          7879+         USING *,R5                base for test data and test routine
00065020                            00000000          7880+         USING PLOETEST,R5
00065020  0006504C                                  7881+T89      DC    A(CODE89)           address of test code
00065024  0059                                      7882+         DC    H'89'               test number
00065026  00                                        7883+         DC    X'00'
00065027  1D                                        7884+         DC    HL1'29'             Function
00065028  63                                        7885+         DC    HL1'99'             expected unequal results (UR)
00065029  00                                        7886+         DC    HL1'00'             found UR
0006502A  00                                        7887+         DC    HL1'0'              cc
0006502B  07                                        7888+         DC    HL1'7'              not expected CC mask
0006502C  00                                        7889+         DS    X                   extracted cc
00065030  00000000 00000000                         7890+         DS    2F                  extract PSW after test (has CC)
00065038  00065760                                  7891+         DC    A(FMSG89)           address of test fail message
0006503C  00065500                                  7892+         DC    A(OPSRC89)          address of operand source
00065040  00065400                                  7893+         DC    A(OPDST89)          address of operand destination
00065044  00065600                                  7894+         DC    A(PL89)             address of parameter list
00065048  0000120A                                  7895+         DC    A(RANDX+89)         address of random source data
                                                    7896
                                                    7897          TSTCODE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0006504C                                          7898+CODE89   DS    0F
                                                  7899 *                                load random source
0006504C  B982 0022                               7900          XGR   R2,R2
00065050  5820 5028                   00000028    7901          L     R2,RANDXA
                                                  7902 *
                                                  7903 * Compare and Swap and Triple Store (CSTSTO: 29)
                                                  7904 * Build Parambter List
                                                  7905 *
00065054  B982 0011                               7906          XGR   R1,R1
00065058  5810 501C                   0000001C    7907          L     R1,OPSRCA           Op1-C, Op3, Op5, Op7, Op1-R
                                                  7908
0006505C  B982 0033                               7909          XGR   R3,R3
00065060  5830 5024                   00000024    7910          L     R3,PLA
                                                  7911
00065064  B982 0044                               7912          XGR   R4,R4
00065068  5840 5020                   00000020    7913          L     R4,OPREPA           OP2, Op4, Op6, Op8
                                                  7914
0006506C  D29F 1000 2000      00000000  00000000  7915          MVC   0(160,R1),0(R2)     load Op1-C, Op3, Op5, Op7, Op1-R
00065072  D21F 3000 1000      00000000  00000000  7916          MVC   0(32,R3),0(R1)      copy to PL OP1-C
00065078  D21F 3090 1020      00000090  00000020  7917          MVC   144(32,R3),32(R1)   copy to PL OP3
0006507E  D21F 30B0 1040      000000B0  00000040  7918          MVC   176(32,R3),64(R1)   copy to PL OP5
00065084  D21F 30D0 1060      000000D0  00000060  7919          MVC   208(32,R3),96(R1)   copy to PL OP7
0006508A  D21F 3020 1080      00000020  00000080  7920          MVC   32(32,R3),128(R1)   copy to PL OP1-R
                                                  7921
00065090  D27F 4000 2080      00000000  00000080  7922          MVC   0(128,R4),128(R2)   load initial Op2, Op4, Op6, Op8
                                                  7923
                                                  7924 ****
                                                  7925 * Force a EQ by changing the Op1-C == Op2
                                                  7926 ****
00065096  D21F 3000 4000      00000000  00000000  7927          MVC   0(32,R3),0(R4)
                                                  7928 ****
0006509C  B982 0000                               7929          XGR   R0,R0
000650A0  4104 0020                   00000020    7930          LA    R0,32(R4)
000650A4  E303 0048 0024              00000048    7931          STG   R0,72(R3)           load OP4 address in PL
                                                  7932
000650AA  4104 0040                   00000040    7933          LA    R0,64(R4)
000650AE  E303 0068 0024              00000068    7934          STG   R0,104(R3)          load OP6 address in PL
                                                  7935
000650B4  4104 0060                   00000060    7936          LA    R0,96(R4)
000650B8  E303 0088 0024              00000088    7937          STG   R0,136(R3)          load OP8 address in PL
                                                  7938
000650BE  B982 0000                               7939          XGR   R0,R0
000650C2  4100 001D                   0000001D    7940          LA    R0,29
000650C6  EE00 4000 3000      00000000  00000000  7941          PLO   0,0(R4),0,0(R3)     C/S Triple Store (CSTSTO: 29)
                                                  7942
000650CC  B98D 0020                               7943          EPSW  R2,R0               extract psw
000650D0  5020 5010                   00000010    7944          ST    R2,CCPSW              to save CC
000650D4  E310 500B 0076              0000000B    7945          LB    R1,CCMASK           (failure CC mask)
000650DA  8910 0004                   00000004    7946          SLL   R1,4                (shift to BC instr CC position)
000650DE  4410 5108                   00065128    7947          EX    R1,F292TSTCC            fail if...
                                                  7948 *
                                                  7949 * compare results
                                                  7950 *
                              000650E2  00000001  7951 F292CKRES EQU   *
                                                  7952 *                                reset adresses
000650E2  B982 0011                               7953          XGR   R1,R1
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000650E6  5810 501C                     0000001C  7954           L     R1,OPSRCA           Op1, Op3, Op5, Op7
000650EA  B982 0033                               7955           XGR   R3,R3
000650EE  5830 5024                     00000024  7956           L     R3,PLA              PL
000650F2  B982 0044                               7957           XGR   R4,R4
000650F6  5840 5020                     00000020  7958           L     R4,OPREPA           OP2, Op4, Op6, Op8
                                                  7959
000650FA  D51F 4020 3090      00000020  00000090  7960           CLC   32(32,R4),144(R3)   OP4 ==  PL Op3
00065100  4770 5102                     00065122  7961           BNE   F292XX                    different, error
00065104  D51F 4040 30B0      00000040  000000B0  7962           CLC   64(32,R4),176(R3)   OP6 ==  PL Op5
0006510A  4770 5102                     00065122  7963           BNE   F292XX                    different, error
0006510E  D51F 4060 30D0      00000060  000000D0  7964           CLC   96(32,R4),208(R3)   OP8 ==  PL Op7
00065114  4770 5102                     00065122  7965           BNE   F292XX                    different, error
00065118  D51F 4000 3020      00000000  00000020  7966           CLC   0(32,R4),32(R3)     OP2 ==  PL Op-R
0006511E  478B 0000                     00000000  7967           BE    0(R11)                    same, return
                                                  7968
                              00065122  00000001  7969 F292XX    EQU   *
00065122  45F0 8340                     00000540  7970           BAL   R15,FAILMSG
00065126  07FB                                    7971           BR    R11
                                                  7972
00065128  4700 510C                     0006512C  7973 F292TSTCC BC    0,F292CCMSG         fail if unexpected condition code
                              0006512C  00000001  7974 F292CCMSG EQU   *
0006512C  45F0 80F2                     000002F2  7975           BAL   R15,CCMSG
00065130  47F0 50C2                     000650E2  7976           B     F292CKRES
                                                  7977
                                                  7978           TSTDATA
00065134                      00065134  00065400  7979+          ORG   PLOETST+(200*K)+(51*4*K)+K
                                                  7980           TSTLBL OPDST              OP2, Op4, Op6, Op8
00065400                                          7981+OPDST89   DS    0C
00065400  00000000 00000000               7982           DC    XL256'0'
00065408  00000000 00000000
00065410  00000000 00000000
00065418  00000000 00000000
00065420  00000000 00000000
00065428  00000000 00000000
00065430  00000000 00000000
00065438  00000000 00000000
00065440  00000000 00000000
00065448  00000000 00000000
00065450  00000000 00000000
00065458  00000000 00000000
00065460  00000000 00000000
00065468  00000000 00000000
00065470  00000000 00000000
00065478  00000000 00000000
00065480  00000000 00000000
00065488  00000000 00000000
00065490  00000000 00000000
00065498  00000000 00000000
000654A0  00000000 00000000
000654A8  00000000 00000000
000654B0  00000000 00000000
000654B8  00000000 00000000
000654C0  00000000 00000000
000654C8  00000000 00000000
000654D0  00000000 00000000
000654D8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000654E0  00000000 00000000
000654E8  00000000 00000000
000654F0  00000000 00000000
000654F8  00000000 00000000
                                          7983
                                          7984           TSTLBL OPSRC              Op1-C, Op3, Op5, Op7, Op1-R
00065500                                  7985+OPSRC89   DS    0C
00065500  00000000 00000000               7986           DC    XL256'0'
00065508  00000000 00000000
00065510  00000000 00000000
00065518  00000000 00000000
00065520  00000000 00000000
00065528  00000000 00000000
00065530  00000000 00000000
00065538  00000000 00000000
00065540  00000000 00000000
00065548  00000000 00000000
00065550  00000000 00000000
00065558  00000000 00000000
00065560  00000000 00000000
00065568  00000000 00000000
00065570  00000000 00000000
00065578  00000000 00000000
00065580  00000000 00000000
00065588  00000000 00000000
00065590  00000000 00000000
00065598  00000000 00000000
000655A0  00000000 00000000
000655A8  00000000 00000000
000655B0  00000000 00000000
000655B8  00000000 00000000
000655C0  00000000 00000000
000655C8  00000000 00000000
000655D0  00000000 00000000
000655D8  00000000 00000000
000655E0  00000000 00000000
000655E8  00000000 00000000
000655F0  00000000 00000000
000655F8  00000000 00000000
                                          7987
                                          7988           TSTLBL PL                 Parameter List
00065600                                  7989+PL89      DS    0C
00065600  00000000 00000000               7990           DC    XL352'00'
00065608  00000000 00000000
00065610  00000000 00000000
00065618  00000000 00000000
00065620  00000000 00000000
00065628  00000000 00000000
00065630  00000000 00000000
00065638  00000000 00000000
00065640  00000000 00000000
00065648  00000000 00000000
00065650  00000000 00000000
00065658  00000000 00000000
00065660  00000000 00000000
00065668  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00065670  00000000 00000000
00065678  00000000 00000000
00065680  00000000 00000000
00065688  00000000 00000000
00065690  00000000 00000000
00065698  00000000 00000000
000656A0  00000000 00000000
000656A8  00000000 00000000
000656B0  00000000 00000000
000656B8  00000000 00000000
000656C0  00000000 00000000
000656C8  00000000 00000000
000656D0  00000000 00000000
000656D8  00000000 00000000
000656E0  00000000 00000000
000656E8  00000000 00000000
000656F0  00000000 00000000
000656F8  00000000 00000000
00065700  00000000 00000000
00065708  00000000 00000000
00065710  00000000 00000000
00065718  00000000 00000000
00065720  00000000 00000000
00065728  00000000 00000000
00065730  00000000 00000000
00065738  00000000 00000000
00065740  00000000 00000000
00065748  00000000 00000000
00065750  00000000 00000000
00065758  00000000 00000000
                                          7991
                                          7992          TSTLBL FMSG               fail message
00065760                                  7993+FMSG89   DS    0C
00065760  86818993 858440D7              7994          DC    C'failed PLO-Extension '
00065768  D3D660C5 A7A38595
00065770  A2899695 40
00065775  C3969497 81998540              7995          DC    C'Compare and Swap and Triple Store (CSTSTO: 29) : EQ'
0006577D  81958440 E2A68197
00065785  40819584 40E39989
0006578D  97938540 E2A39699
00065795  85404DC3 E2E3E2E3
0006579D  D67A40F2 F95D407A
000657A5  40C5D8
000657A8  40404040 40404040              7996          DC    CL64' '
000657B0  40404040 40404040
000657B8  40404040 40404040
000657C0  40404040 40404040
000657C8  40404040 40404040
000657D0  40404040 40404040
000657D8  40404040 40404040
000657E0  40404040 40404040
                                          7997          TSTEND
000657E8                                  7998+         DS    0H
000657E8  C5958440 968640A3              7999+         DC    CL32'End of test: 89'
000657F0  85A2A37A 40F8F940
000657F8  40404040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00065800  40404040 40404040                           8000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   8002 *=======================================================================
                                                   8003 *       Compare and Swap and Double Store Tests
                                                   8004 *=======================================================================
                                                   8005
                                                   8006 * ---------------------------------------------------------------------
                                                   8007 * Test Compare and Swap and Double Store (CSDSTO: 28) instruction : NE
                                                   8008 * ---------------------------------------------------------------------
                                                   8009          TSTHDR 1,99,28           1Op1-C != OP2 (cc=1, UR=99)
00065808                          00065808  00066000 8010+         ORG   PLOETST+(200*K)+(52*4*K)
00066000  E2E3C1D9 E3409686                        8011+         DC    CL32'START of test: 90'
00066008  40A385A2 A37A40F9
00066010  F0404040 40404040
00066018  40404040 40404040
00066020                          00066020         8012+         USING *,R5               base for test data and test routine
00066020                          00000000         8013+         USING PLOETEST,R5
00066020  0006604C                                 8014+T90      DC    A(CODE90)          address of test code
00066024  005A                                     8015+         DC    H'90'              test number
00066026  00                                       8016+         DC    X'00'
00066027  1C                                       8017+         DC    HL1'28'            Function
00066028  63                                       8018+         DC    HL1'99'            expected unequal results (UR)
00066029  00                                       8019+         DC    HL1'00'            found UR
0006602A  01                                       8020+         DC    HL1'1'             cc
0006602B  0B                                       8021+         DC    HL1'11'            not expected CC mask
0006602C  00                                       8022+         DS    X                  extracted cc
00066030  00000000 00000000                        8023+         DS    2F                 extract PSW after test (has CC)
00066038  00066760                                 8024+         DC    A(FMSG90)          address of test fail message
0006603C  00066500                                 8025+         DC    A(OPSRC90)         address of operand source
00066040  00066400                                 8026+         DC    A(OPDST90)         address of operand destination
00066044  00066600                                 8027+         DC    A(PL90)            address of parameter list
00066048  0000120B                                 8028+         DC    A(RANDX+90)        address of random source data
                                                   8029
                                                   8030          TSTCODE
0006604C                                           8031+CODE90   DS    0F
                                                   8032 *                                 load random source
0006604C  B982 0022                                8033          XGR   R2,R2
00066050  5820 5028                     00000028   8034          L     R2,RANDXA
                                                   8035 *
                                                   8036 * Compare and Swap and Double Store (CSDSTO: 28)
                                                   8037 * Build Parambter List
                                                   8038 *
00066054  B982 0011                                8039          XGR   R1,R1
00066058  5810 501C                     0000001C   8040          L     R1,OPSRCA          Op1-C, Op3, Op5, Op1-R
                                                   8041
0006605C  B982 0033                                8042          XGR   R3,R3
00066060  5830 5024                     00000024   8043          L     R3,PLA
                                                   8044
00066064  B982 0044                                8045          XGR   R4,R4
00066068  5840 5020                     00000020   8046          L     R4,OPREPA          OP2, Op4, Op6
                                                   8047
0006606C  D27F 1000 2000      00000000  00000000   8048          MVC   0(128,R1),0(R2)    load Op1-C, Op3, Op5, Op1-R
00066072  D21F 3000 1000      00000000  00000000   8049          MVC   0(32,R3),0(R1)     copy to PL OP1-C
00066078  D21F 3070 1020      00000070  00000020   8050          MVC   112(32,R3),32(R1)  copy to PL OP3
0006607E  D21F 3090 1040      00000090  00000040   8051          MVC   144(32,R3),64(R1)  copy to PL OP5
00066084  D21F 3020 1060      00000020  00000060   8052          MVC   32(32,R3),96(R1)   copy to PL OP1-R
                                                   8053
0006608A  D25F 4000 2080      00000000  00000080   8054          MVC   0(96,R4),128(R2)   load initial Op2, Op4, Op6
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8055
                                                    8056 ****
                                                    8057 * force a failure by changing the Op1-c
                                                    8058 ****
00066090  D203 3000 2021      00000000  00000021    8059          MVC   0(4,R3),33(R2)
                                                    8060 ****
00066096  B982 0000                               8061          XGR   R0,R0
0006609A  4104 0020                     00000020  8062          LA    R0,32(R4)
0006609E  E303 0048 0024                00000048  8063          STG   R0,72(R3)           load OP4 address in PL
                                                    8064
000660A4  4104 0040                     00000040  8065          LA    R0,64(R4)
000660A8  E303 0068 0024                00000068  8066          STG   R0,104(R3)          load OP6 address in PL
                                                    8067
000660AE  B982 0000                               8068          XGR   R0,R0
000660B2  4100 001C                     0000001C  8069          LA    R0,28
000660B6  EE00 4000 3000      00000000  00000000  8070          PLO   0,0(R4),0,0(R3)     C/S Double Store (CSDSTO: 28)
                                                    8071
000660BC  B98D 0020                               8072          EPSW  R2,R0               extract psw
000660C0  5020 5010                     00000010  8073          ST    R2,CCPSW              to save CC
000660C4  E310 500B 0076                0000000B  8074          LB    R1,CCMASK           (failure CC mask)
000660CA  8910 0004                     00000004  8075          SLL   R1,4                (shift to BC instr CC position)
000660CE  4410 50DA                     000660FA  8076          EX    R1,F281TSTCC          fail if...
                                                    8077 *
                                                    8078 * compare results
                                                    8079 *
                          000660D2  00000001        8080 F281CKRES EQU   *
                                                    8081 *                                reset adresses
000660D2  B982 0011                               8082          XGR   R1,R1
000660D6  5810 501C                     0000001C  8083          L     R1,OPSRCA           Op1, Op3, Op5, Op7
000660DA  B982 0033                               8084          XGR   R3,R3
000660DE  5830 5024                     00000024  8085          L     R3,PLA              PL
000660E2  B982 0044                               8086          XGR   R4,R4
000660E6  5840 5020                     00000020  8087          L     R4,OPREPA           OP2, Op4, Op6, Op8
                                                    8088
000660EA  D51F 4000 3000      00000000  00000000  8089          CLC   0(32,R4),0(R3)      Op2 == OP1C?
000660F0  478B 0000                     00000000  8090          BE    0(R11)              same, return
                                                    8091
                          000660F4  00000001        8092 F281XX   EQU   *
000660F4  45F0 8340                     00000540  8093          BAL   R15,FAILMSG
000660F8  07FB                                    8094          BR    R11
                                                    8095
000660FA  4700 50DE                     000660FE  8096 F281TSTCC BC   0,F281CCMSG          fail if unexpected condition code
                          000660FE  00000001        8097 F281CCMSG EQU  *
000660FE  45F0 80F2                     000002F2  8098          BAL   R15,CCMSG
00066102  47F0 50B2                     000660D2  8099          B     F281CKRES
                                                    8100
                                                    8101          TSTDATA
00066106                  00066106  00066400      8102+         ORG   PLOETST+(200*K)+(52*4*K)+K
                                                    8103          TSTLBL OPDST              OP2, Op4, Op6
00066400                                            8104+OPDST90  DS    0C
00066400  00000000 00000000                        8105          DC    XL256'0'
00066408  00000000 00000000
00066410  00000000 00000000
00066418  00000000 00000000
00066420  00000000 00000000
00066428  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00066430  00000000 00000000
00066438  00000000 00000000
00066440  00000000 00000000
00066448  00000000 00000000
00066450  00000000 00000000
00066458  00000000 00000000
00066460  00000000 00000000
00066468  00000000 00000000
00066470  00000000 00000000
00066478  00000000 00000000
00066480  00000000 00000000
00066488  00000000 00000000
00066490  00000000 00000000
00066498  00000000 00000000
000664A0  00000000 00000000
000664A8  00000000 00000000
000664B0  00000000 00000000
000664B8  00000000 00000000
000664C0  00000000 00000000
000664C8  00000000 00000000
000664D0  00000000 00000000
000664D8  00000000 00000000
000664E0  00000000 00000000
000664E8  00000000 00000000
000664F0  00000000 00000000
000664F8  00000000 00000000
                             8106
                             8107           TSTLBL OPSRC              Op1-C, Op3, Op5, Op1-R
00066500                     8108+OPSRC90   DS    0C
00066500  00000000 00000000  8109           DC    XL256'0'
00066508  00000000 00000000
00066510  00000000 00000000
00066518  00000000 00000000
00066520  00000000 00000000
00066528  00000000 00000000
00066530  00000000 00000000
00066538  00000000 00000000
00066540  00000000 00000000
00066548  00000000 00000000
00066550  00000000 00000000
00066558  00000000 00000000
00066560  00000000 00000000
00066568  00000000 00000000
00066570  00000000 00000000
00066578  00000000 00000000
00066580  00000000 00000000
00066588  00000000 00000000
00066590  00000000 00000000
00066598  00000000 00000000
000665A0  00000000 00000000
000665A8  00000000 00000000
000665B0  00000000 00000000
000665B8  00000000 00000000
000665C0  00000000 00000000
000665C8  00000000 00000000
000665D0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000665D8  00000000 00000000
000665E0  00000000 00000000
000665E8  00000000 00000000
000665F0  00000000 00000000
000665F8  00000000 00000000
                                          8110
                                          8111           TSTLBL PL                  Parameter List
00066600                                  8112+PL90      DS    0C
00066600  00000000 00000000               8113           DC    XL352'00'
00066608  00000000 00000000
00066610  00000000 00000000
00066618  00000000 00000000
00066620  00000000 00000000
00066628  00000000 00000000
00066630  00000000 00000000
00066638  00000000 00000000
00066640  00000000 00000000
00066648  00000000 00000000
00066650  00000000 00000000
00066658  00000000 00000000
00066660  00000000 00000000
00066668  00000000 00000000
00066670  00000000 00000000
00066678  00000000 00000000
00066680  00000000 00000000
00066688  00000000 00000000
00066690  00000000 00000000
00066698  00000000 00000000
000666A0  00000000 00000000
000666A8  00000000 00000000
000666B0  00000000 00000000
000666B8  00000000 00000000
000666C0  00000000 00000000
000666C8  00000000 00000000
000666D0  00000000 00000000
000666D8  00000000 00000000
000666E0  00000000 00000000
000666E8  00000000 00000000
000666F0  00000000 00000000
000666F8  00000000 00000000
00066700  00000000 00000000
00066708  00000000 00000000
00066710  00000000 00000000
00066718  00000000 00000000
00066720  00000000 00000000
00066728  00000000 00000000
00066730  00000000 00000000
00066738  00000000 00000000
00066740  00000000 00000000
00066748  00000000 00000000
00066750  00000000 00000000
00066758  00000000 00000000
                                          8114
                                          8115           TSTLBL FMSG                fail message
00066760                                  8116+FMSG90    DS    0C
00066760  86818993 858440D7               8117           DC    C'failed PLO-Extension '
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00066768  D3D660C5 A7A38595
00066770  A2899695 40
00066775  C3969497 81998540                        8118          DC    C'Compare and Swap and Double Store (CSDSTO: 29) : NE'
0006677D  81958440 E2A68197
00066785  40819584 40C496A4
0006678D  82938540 E2A39699
00066795  85404DC3 E2C4E2E3
0006679D  D67A40F2 F95D407A
000667A5  40D5C5
000667A8  40404040 40404040                        8119          DC    CL64' '
000667B0  40404040 40404040
000667B8  40404040 40404040
000667C0  40404040 40404040
000667C8  40404040 40404040
000667D0  40404040 40404040
000667D8  40404040 40404040
000667E0  40404040 40404040
                                                   8120          TSTEND
000667E8                                           8121+         DS    0H
000667E8  C5958440 968640A3                        8122+         DC    CL32'End of test: 90'
000667F0  85A2A37A 40F9F040
000667F8  40404040 40404040
00066800  40404040 40404040
                                                   8123
                                                   8124 * ------------------------------------------------------------------
                                                   8125 * Test Compare and Swap and Double Store (CSDSTO: 28) instruction : EQ
                                                   8126 * ------------------------------------------------------------------
                                                   8127          TSTHDR 0,99,28            Op1-C == OP2 (cc=0, UR=99)
00066808                        00066808  00067000 8128+         ORG   PLOETST+(200*K)+(53*4*K)
00067000  E2E3C1D9 E3409686                        8129+         DC    CL32'START of test: 91'
00067008  40A385A2 A37A40F9
00067010  F1404040 40404040
00067018  40404040 40404040
00067020                        00067020          8130+         USING *,R5                base for test data and test routine
00067020                        00000000          8131+         USING PLOETEST,R5
00067020  0006704C                                 8132+T91      DC    A(CODE91)           address of test code
00067024  005B                                     8133+         DC    H'91'               test number
00067026  00                                       8134+         DC    X'00'
00067027  1C                                       8135+         DC    HL1'28'             Function
00067028  63                                       8136+         DC    HL1'99'             expected unequal results (UR)
00067029  00                                       8137+         DC    HL1'00'             found UR
0006702A  00                                       8138+         DC    HL1'0'              cc
0006702B  07                                       8139+         DC    HL1'7'              not expected CC mask
0006702C  00                                       8140+         DS    X                   extracted cc
00067030  00000000 00000000                        8141+         DS    2F                  extract PSW after test (has CC)
00067038  00067760                                 8142+         DC    A(FMSG91)           address of test fail message
0006703C  00067500                                 8143+         DC    A(OPSRC91)          address of operand source
00067040  00067400                                 8144+         DC    A(OPDST91)          address of operand destination
00067044  00067600                                 8145+         DC    A(PL91)             address of parameter list
00067048  0000120C                                 8146+         DC    A(RANDX+91)         address of random source data
                                                   8147
                                                   8148          TSTCODE
0006704C                                           8149+CODE91   DS    0F
                                                   8150 *                                    load random source
0006704C  B982 0022                                8151          XGR   R2,R2
00067050  5820 5028                     00000028   8152          L     R2,RANDXA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8153 *
                                                    8154 * Compare and Swap and Double Store (CSDSTO: 28)
                                                    8155 * Build Parambter List
                                                    8156 *
00067054  B982 0011                                 8157          XGR   R1,R1
00067058  5810 501C                 0000001C        8158          L     R1,OPSRCA           Op1-C, Op3, Op5, Op1-R
                                                    8159
0006705C  B982 0033                                 8160          XGR   R3,R3
00067060  5830 5024                 00000024        8161          L     R3,PLA
                                                    8162
00067064  B982 0044                                 8163          XGR   R4,R4
00067068  5840 5020                 00000020        8164          L     R4,OPREPA           OP2, Op4, Op6, Op8
                                                    8165
0006706C  D27F 1000 2000  00000000  00000000        8166          MVC   0(128,R1),0(R2)     load Op1-C, Op3, Op5, Op1-R
00067072  D21F 3000 1000  00000000  00000000        8167          MVC   0(32,R3),0(R1)      copy to PL OP1-C
00067078  D21F 3070 1020  00000070  00000020        8168          MVC   112(32,R3),32(R1)   copy to PL OP3
0006707E  D21F 3090 1040  00000090  00000040        8169          MVC   144(32,R3),64(R1)   copy to PL OP5
00067084  D21F 3020 1060  00000020  00000060        8170          MVC   32(32,R3),96(R1)    copy to PL OP1-R
                                                    8171
0006708A  D25F 4000 2080  00000000  00000080        8172          MVC   0(96,R4),128(R2)    load initial Op2, Op4, Op6
                                                    8173
                                                    8174 ****
                                                    8175 * Force a EQ by changing the Op1-C == Op2
                                                    8176 ****
00067090  D21F 3000 4000  00000000  00000000        8177          MVC   0(32,R3),0(R4)
                                                    8178 ****
00067096  B982 0000                                 8179          XGR   R0,R0
0006709A  4104 0020                 00000020        8180          LA    R0,32(R4)
0006709E  E303 0048 0024            00000048        8181          STG   R0,72(R3)           load OP4 address in PL
                                                    8182
000670A4  4104 0040                 00000040        8183          LA    R0,64(R4)
000670A8  E303 0068 0024            00000068        8184          STG   R0,104(R3)          load OP6 address in PL
                                                    8185
000670AE  B982 0000                                 8186          XGR   R0,R0
000670B2  4100 001C                 0000001C        8187          LA    R0,28
000670B6  EE00 4000 3000  00000000  00000000        8188          PLO   0,0(R4),0,0(R3)     C/S Double Store (CSDSTO: 28)
                                                    8189
000670BC  B98D 0020                                 8190          EPSW  R2,R0               extract psw
000670C0  5020 5010                 00000010        8191          ST    R2,CCPSW              to save CC
000670C4  E310 500B 0076            0000000B        8192          LB    R1,CCMASK           (failure CC mask)
000670CA  8910 0004                 00000004        8193          SLL   R1,4                (shift to BC instr CC position)
000670CE  4410 50EE                 0006710E        8194          EX    R1,F282TSTCC            fail if...
                                                    8195 *
                                                    8196 * compare results
                                                    8197 *
                            000670D2  00000001      8198 F282CKRES EQU   *
                                                    8199 *                                  reset adresses
000670D2  B982 0011                                 8200          XGR   R1,R1
000670D6  5810 501C                 0000001C        8201          L     R1,OPSRCA           Op1, Op3, Op5, Op7
000670DA  B982 0033                                 8202          XGR   R3,R3
000670DE  5830 5024                 00000024        8203          L     R3,PLA              PL
000670E2  B982 0044                                 8204          XGR   R4,R4
000670E6  5840 5020                 00000020        8205          L     R4,OPREPA           OP2, Op4, Op6, Op8
                                                    8206
000670EA  D51F 4020 3070  00000020  00000070        8207          CLC   32(32,R4),112(R3)   OP4 ==  PL Op3
000670F0  4770 50E8                 00067108        8208          BNE   F282XX                   different, error
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000670F4  D51F 4040 3090      00000040  00000090  8209          CLC   64(32,R4),144(R3)   OP6 ==  PL Op5
000670FA  4770 50E8                     00067108  8210          BNE   F282XX                 different, error
000670FE  D51F 4000 3020      00000000  00000020  8211          CLC   0(32,R4),32(R3)     OP2 ==  PL Op-R
00067104  478B 0000                     00000000  8212          BE    0(R11)                same, return
                                                  8213
                              00067108  00000001  8214 F282XX   EQU   *
00067108  45F0 8340                     00000540  8215          BAL   R15,FAILMSG
0006710C  07FB                                    8216          BR    R11
                                                  8217
0006710E  4700 50F2                     00067112  8218 F282TSTCC BC    0,F282CCMSG         fail if unexpected condition code
                              00067112  00000001  8219 F282CCMSG EQU   *
00067112  45F0 80F2                     000002F2  8220          BAL   R15,CCMSG
00067116  47F0 50B2                     000670D2  8221          B     F282CKRES
                                                  8222
                                                  8223          TSTDATA
0006711A                      0006711A  00067400  8224+         ORG   PLOETST+(200*K)+(53*4*K)+K
                                                  8225          TSTLBL OPDST              OP2, Op4, Op6, Op8
00067400                                          8226+OPDST91  DS    0C
00067400  00000000 00000000                       8227          DC    XL256'0'
00067408  00000000 00000000
00067410  00000000 00000000
00067418  00000000 00000000
00067420  00000000 00000000
00067428  00000000 00000000
00067430  00000000 00000000
00067438  00000000 00000000
00067440  00000000 00000000
00067448  00000000 00000000
00067450  00000000 00000000
00067458  00000000 00000000
00067460  00000000 00000000
00067468  00000000 00000000
00067470  00000000 00000000
00067478  00000000 00000000
00067480  00000000 00000000
00067488  00000000 00000000
00067490  00000000 00000000
00067498  00000000 00000000
000674A0  00000000 00000000
000674A8  00000000 00000000
000674B0  00000000 00000000
000674B8  00000000 00000000
000674C0  00000000 00000000
000674C8  00000000 00000000
000674D0  00000000 00000000
000674D8  00000000 00000000
000674E0  00000000 00000000
000674E8  00000000 00000000
000674F0  00000000 00000000
000674F8  00000000 00000000
                                                  8228
                                                  8229          TSTLBL OPSRC              Op1-C, Op3, Op5, Op7, Op1-R
00067500                                          8230+OPSRC91  DS    0C
00067500  00000000 00000000                       8231          DC    XL256'0'
00067508  00000000 00000000
00067510  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00067518  00000000 00000000
00067520  00000000 00000000
00067528  00000000 00000000
00067530  00000000 00000000
00067538  00000000 00000000
00067540  00000000 00000000
00067548  00000000 00000000
00067550  00000000 00000000
00067558  00000000 00000000
00067560  00000000 00000000
00067568  00000000 00000000
00067570  00000000 00000000
00067578  00000000 00000000
00067580  00000000 00000000
00067588  00000000 00000000
00067590  00000000 00000000
00067598  00000000 00000000
000675A0  00000000 00000000
000675A8  00000000 00000000
000675B0  00000000 00000000
000675B8  00000000 00000000
000675C0  00000000 00000000
000675C8  00000000 00000000
000675D0  00000000 00000000
000675D8  00000000 00000000
000675E0  00000000 00000000
000675E8  00000000 00000000
000675F0  00000000 00000000
000675F8  00000000 00000000
                                        8232
                                        8233          TSTLBL PL                 Parameter List
00067600                                8234+PL91     DS    0C
00067600  00000000 00000000             8235          DC    XL352'00'
00067608  00000000 00000000
00067610  00000000 00000000
00067618  00000000 00000000
00067620  00000000 00000000
00067628  00000000 00000000
00067630  00000000 00000000
00067638  00000000 00000000
00067640  00000000 00000000
00067648  00000000 00000000
00067650  00000000 00000000
00067658  00000000 00000000
00067660  00000000 00000000
00067668  00000000 00000000
00067670  00000000 00000000
00067678  00000000 00000000
00067680  00000000 00000000
00067688  00000000 00000000
00067690  00000000 00000000
00067698  00000000 00000000
000676A0  00000000 00000000
000676A8  00000000 00000000
000676B0  00000000 00000000
000676B8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000676C0  00000000 00000000
000676C8  00000000 00000000
000676D0  00000000 00000000
000676D8  00000000 00000000
000676E0  00000000 00000000
000676E8  00000000 00000000
000676F0  00000000 00000000
000676F8  00000000 00000000
00067700  00000000 00000000
00067708  00000000 00000000
00067710  00000000 00000000
00067718  00000000 00000000
00067720  00000000 00000000
00067728  00000000 00000000
00067730  00000000 00000000
00067738  00000000 00000000
00067740  00000000 00000000
00067748  00000000 00000000
00067750  00000000 00000000
00067758  00000000 00000000
                                        8236
                                        8237          TSTLBL FMSG               fail message
00067760                                8238+FMSG91   DS    0C
00067760  86818993 858440D7             8239          DC    C'failed PLO-Extension '
00067768  D3D660C5 A7A38595
00067770  A2899695 40
00067775  C3969497 81998540             8240          DC    C'Compare and Swap and Double Store (CSDSTO: 28) : EQ'
0006777D  81958440 E2A68197
00067785  40819584 40C496A4
0006778D  82938540 E2A39699
00067795  85404DC3 E2C4E2E3
0006779D  D67A40F2 F85D407A
000677A5  40C5D8
000677A8  40404040 40404040             8241          DC    CL64' '
000677B0  40404040 40404040
000677B8  40404040 40404040
000677C0  40404040 40404040
000677C8  40404040 40404040
000677D0  40404040 40404040
000677D8  40404040 40404040
000677E0  40404040 40404040
                                        8242          TSTEND
000677E8                                8243+         DS    0H
000677E8  C5958440 968640A3             8244+         DC    CL32'End of test: 91'
000677F0  85A2A37A 40F9F140
000677F8  40404040 40404040
00067800  40404040 40404040
                                        8245
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   8247 *=====================================================================
                                                   8248 *       Compare and Swap and Store Tests
                                                   8249 *=====================================================================
                                                   8250
                                                   8251 * ---------------------------------------------------------------------
                                                   8252 * Test Compare and Swap and Store (CSSTO: 27) instruction : NE
                                                   8253 * ---------------------------------------------------------------------
                                                   8254          TSTHDR 1,99,26           Op1-C != OP2 (cc=1, UR=99)
00067808                          00067808  00068000 8255+         ORG   PLOETST+(200*K)+(54*4*K)
00068000  E2E3C1D9 E3409686                        8256+         DC    CL32'START of test: 92'
00068008  40A385A2 A37A40F9
00068010  F2404040 40404040
00068018  40404040 40404040
00068020                          00068020         8257+         USING *,R5               base for test data and test routine
00068020                          00000000         8258+         USING PLOETEST,R5
00068020  0006804C                                 8259+T92      DC    A(CODE92)          address of test code
00068024  005C                                     8260+         DC    H'92'              test number
00068026  00                                       8261+         DC    X'00'
00068027  1A                                       8262+         DC    HL1'26'            Function
00068028  63                                       8263+         DC    HL1'99'            expected unequal results (UR)
00068029  00                                       8264+         DC    HL1'00'            found UR
0006802A  01                                       8265+         DC    HL1'1'             cc
0006802B  0B                                       8266+         DC    HL1'11'            not expected CC mask
0006802C  00                                       8267+         DS    X                  extracted cc
00068030  00000000 00000000                        8268+         DS    2F                 extract PSW after test (has CC)
00068038  00068760                                 8269+         DC    A(FMSG92)          address of test fail message
0006803C  00068500                                 8270+         DC    A(OPSRC92)         address of operand source
00068040  00068400                                 8271+         DC    A(OPDST92)         address of operand destination
00068044  00068600                                 8272+         DC    A(PL92)            address of parameter list
00068048  0000120D                                 8273+         DC    A(RANDX+92)        address of random source data
                                                   8274
                                                   8275         TSTCODE
0006804C                                           8276+CODE92   DS    0F
                                                   8277 *                                 load random source
0006804C  B982 0022                                8278          XGR   R2,R2
00068050  5820 5028                     00000028   8279          L     R2,RANDXA
                                                   8280 *
                                                   8281 * Compare and Swap and Store (CSSTO: 27)
                                                   8282 * Build Parambter List
                                                   8283 *
00068054  B982 0011                                8284          XGR   R1,R1
00068058  5810 501C                     0000001C   8285          L     R1,OPSRCA          Op1-C, Op3, Op1-R
                                                   8286
0006805C  B982 0033                                8287          XGR   R3,R3
00068060  5830 5024                     00000024   8288          L     R3,PLA
                                                   8289
00068064  B982 0044                                8290          XGR   R4,R4
00068068  5840 5020                     00000020   8291          L     R4,OPREPA          OP2, Op4
                                                   8292
0006806C  D25F 1000 2000        00000000  00000000 8293          MVC   0(96,R1),0(R2)     load Op1-C, Op3, Op1-R
00068072  D21F 3000 1000        00000000  00000000 8294          MVC   0(32,R3),0(R1)     copy to PL OP1-C
00068078  D21F 3050 1020        00000050  00000020 8295          MVC   80(32,R3),32(R1)   copy to PL OP3
0006807E  D21F 3020 1060        00000020  00000060 8296          MVC   32(32,R3),96(R1)   copy to PL OP1-R
                                                   8297
00068084  D23F 4000 2080        00000000  00000080 8298          MVC   0(64,R4),128(R2)   load initial Op2, Op4
                                                   8299
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8300 ****
                                                    8301 * force a failure by changing the Op1-c
                                                    8302 ****
0006808A  D203 3000 2021      00000000  00000021  8303         MVC   0(4,R3),33(R2)
                                                    8304 ****
00068090  B982 0000                               8305         XGR   R0,R0
00068094  4104 0020                   00000020    8306         LA    R0,32(R4)
00068098  E303 0048 0024              00000048    8307         STG   R0,72(R3)          load OP4 address in PL
                                                    8308
0006809E  B982 0000                               8309         XGR   R0,R0
000680A2  4100 001B                   0000001B    8310         LA    R0,27
000680A6  EE00 4000 3000      00000000  00000000  8311         PLO   0,0(R4),0,0(R3)    C/S Store (CSSTO: 27)
                                                    8312
000680AC  B98D 0020                               8313         EPSW  R2,R0              extract psw
000680B0  5020 5010                   00000010    8314         ST    R2,CCPSW             to save CC
000680B4  E310 500B 0076              0000000B    8315         LB    R1,CCMASK          (failure CC mask)
000680BA  8910 0004                   00000004    8316         SLL   R1,4               (shift to BC instr CC position)
000680BE  4410 50CA                   000680EA    8317         EX    R1,F271TSTCC           fail if...
                                                    8318 *
                                                    8319 * compare results
                                                    8320 *
                              000680C2  00000001    8321 F271CKRES EQU   *
                                                    8322 *                           reset adresses
000680C2  B982 0011                               8323         XGR   R1,R1
000680C6  5810 501C                   0000001C    8324         L     R1,OPSRCA          Op1, Op3, Op1-R
000680CA  B982 0033                               8325         XGR   R3,R3
000680CE  5830 5024                   00000024    8326         L     R3,PLA             PL
000680D2  B982 0044                               8327         XGR   R4,R4
000680D6  5840 5020                   00000020    8328         L     R4,OPREPA          OP2, Op4
                                                    8329
000680DA  D51F 4000 3000      00000000  00000000  8330         CLC   0(32,R4),0(R3)     Op2 == OP1C?
000680E0  478B 0000                   00000000    8331         BE    0(R11)             same, return
                                                    8332
                              000680E4  00000001    8333 F271XX   EQU   *
000680E4  45F0 8340                   00000540    8334         BAL   R15,FAILMSG
000680E8  07FB                                    8335         BR    R11
                                                    8336
000680EA  4700 50CE                   000680EE    8337 F271TSTCC BC   0,F271CCMSG        fail if unexpected condition code
                              000680EE  00000001    8338 F271CCMSG EQU  *
000680EE  45F0 80F2                   000002F2    8339         BAL   R15,CCMSG
000680F2  47F0 50A2                   000680C2    8340         B     F271CKRES
                                                    8341
                                                    8342         TSTDATA
000680F6                      000680F6  00068400  8343+         ORG   PLOETST+(200*K)+(54*4*K)+K
                                                    8344         TSTLBL OPDST             OP2, Op4, Op6
00068400                                            8345+OPDST92 DS    0C
00068400  00000000 00000000                         8346         DC    XL256'0'
00068408  00000000 00000000
00068410  00000000 00000000
00068418  00000000 00000000
00068420  00000000 00000000
00068428  00000000 00000000
00068430  00000000 00000000
00068438  00000000 00000000
00068440  00000000 00000000
00068448  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00068450  00000000 00000000
00068458  00000000 00000000
00068460  00000000 00000000
00068468  00000000 00000000
00068470  00000000 00000000
00068478  00000000 00000000
00068480  00000000 00000000
00068488  00000000 00000000
00068490  00000000 00000000
00068498  00000000 00000000
000684A0  00000000 00000000
000684A8  00000000 00000000
000684B0  00000000 00000000
000684B8  00000000 00000000
000684C0  00000000 00000000
000684C8  00000000 00000000
000684D0  00000000 00000000
000684D8  00000000 00000000
000684E0  00000000 00000000
000684E8  00000000 00000000
000684F0  00000000 00000000
000684F8  00000000 00000000
                                          8347
                                          8348           TSTLBL OPSRC              Op1-C, Op3, Op5, Op1-R
00068500                                  8349+OPSRC92   DS    0C
00068500  00000000 00000000               8350           DC    XL256'0'
00068508  00000000 00000000
00068510  00000000 00000000
00068518  00000000 00000000
00068520  00000000 00000000
00068528  00000000 00000000
00068530  00000000 00000000
00068538  00000000 00000000
00068540  00000000 00000000
00068548  00000000 00000000
00068550  00000000 00000000
00068558  00000000 00000000
00068560  00000000 00000000
00068568  00000000 00000000
00068570  00000000 00000000
00068578  00000000 00000000
00068580  00000000 00000000
00068588  00000000 00000000
00068590  00000000 00000000
00068598  00000000 00000000
000685A0  00000000 00000000
000685A8  00000000 00000000
000685B0  00000000 00000000
000685B8  00000000 00000000
000685C0  00000000 00000000
000685C8  00000000 00000000
000685D0  00000000 00000000
000685D8  00000000 00000000
000685E0  00000000 00000000
000685E8  00000000 00000000
000685F0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000685F8  00000000 00000000                                8351
                                           8352          TSTLBL PL                  Parameter List
00068600                                   8353+PL92     DS    0C
00068600  00000000 00000000                8354          DC    XL352'00'
00068608  00000000 00000000
00068610  00000000 00000000
00068618  00000000 00000000
00068620  00000000 00000000
00068628  00000000 00000000
00068630  00000000 00000000
00068638  00000000 00000000
00068640  00000000 00000000
00068648  00000000 00000000
00068650  00000000 00000000
00068658  00000000 00000000
00068660  00000000 00000000
00068668  00000000 00000000
00068670  00000000 00000000
00068678  00000000 00000000
00068680  00000000 00000000
00068688  00000000 00000000
00068690  00000000 00000000
00068698  00000000 00000000
000686A0  00000000 00000000
000686A8  00000000 00000000
000686B0  00000000 00000000
000686B8  00000000 00000000
000686C0  00000000 00000000
000686C8  00000000 00000000
000686D0  00000000 00000000
000686D8  00000000 00000000
000686E0  00000000 00000000
000686E8  00000000 00000000
000686F0  00000000 00000000
000686F8  00000000 00000000
00068700  00000000 00000000
00068708  00000000 00000000
00068710  00000000 00000000
00068718  00000000 00000000
00068720  00000000 00000000
00068728  00000000 00000000
00068730  00000000 00000000
00068738  00000000 00000000
00068740  00000000 00000000
00068748  00000000 00000000
00068750  00000000 00000000
00068758  00000000 00000000
                                           8355
                                           8356          TSTLBL FMSG                fail message
00068760                                   8357+FMSG92   DS    0C
00068760  86818993 858440D7               8358          DC    C'failed PLO-Extension '
00068768  D3D660C5 A7A38595
00068770  A2899695 40
00068775  C3969497 81998540               8359          DC    C'Compare and Swap and Store (CSSTO: 28) : NE'
0006877D  81958440 E2A68197
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00068785  40819584 40E2A396
0006878D  9985404D C3E2E2E3
00068795  D67A40F2 F85D407A
0006879D  40D5C5
000687A0  40404040 40404040                        8360          DC    CL64' '
000687A8  40404040 40404040
000687B0  40404040 40404040
000687B8  40404040 40404040
000687C0  40404040 40404040
000687C8  40404040 40404040
000687D0  40404040 40404040
000687D8  40404040 40404040
                                                   8361          TSTEND
000687E0                                           8362+         DS    0H
000687E0  C5958440 968640A3                        8363+         DC    CL32'End of test: 92'
000687E8  85A2A37A 40F9F240
000687F0  40404040 40404040
000687F8  40404040 40404040
                                                   8364
                                                   8365 * ------------------------------------------------------------------
                                                   8366 * Test Compare and Swap and Store (CSSTO: 27) instruction : EQ
                                                   8367 * ------------------------------------------------------------------
                                                   8368          TSTHDR 0,99,27            Op1-C == OP2 (cc=0, UR=99)
00068800                        00068800  00069000 8369+         ORG   PLOETST+(200*K)+(55*4*K)
00069000  E2E3C1D9 E3409686                        8370+         DC    CL32'START of test: 93'
00069008  40A385A2 A37A40F9
00069010  F3404040 40404040
00069018  40404040 40404040
00069020                        00069020           8371+         USING *,R5                base for test data and test routine
00069020                        00000000           8372+         USING PLOETEST,R5
00069020  0006904C                                 8373+T93      DC    A(CODE93)           address of test code
00069024  005D                                     8374+         DC    H'93'               test number
00069026  00                                       8375+         DC    X'00'
00069027  1B                                       8376+         DC    HL1'27'             Function
00069028  63                                       8377+         DC    HL1'99'             expected unequal results (UR)
00069029  00                                       8378+         DC    HL1'00'             found UR
0006902A  00                                       8379+         DC    HL1'0'              cc
0006902B  07                                       8380+         DC    HL1'7'              not expected CC mask
0006902C  00                                       8381+         DS    X                   extracted cc
00069030  00000000 00000000                        8382+         DS    2F                  extract PSW after test (has CC)
00069038  00069760                                 8383+         DC    A(FMSG93)           address of test fail message
0006903C  00069500                                 8384+         DC    A(OPSRC93)          address of operand source
00069040  00069400                                 8385+         DC    A(OPDST93)          address of operand destination
00069044  00069600                                 8386+         DC    A(PL93)             address of parameter list
00069048  0000120E                                 8387+         DC    A(RANDX+93)         address of random source data
                                                   8388
                                                   8389          TSTCODE
0006904C                                           8390+CODE93   DS    0F
                                                   8391 *                                  load random source
0006904C  B982 0022                                8392          XGR   R2,R2
00069050  5820 5028                     00000028   8393          L     R2,RANDXA
                                                   8394 *
                                                   8395 * Compare and Swap and Store (CSSTO: 27)
                                                   8396 * Build Parambter List
                                                   8397 *
00069054  B982 0011                                8398          XGR   R1,R1
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00069058  5810 501C                     0000001C  8399           L     R1,OPSRCA           Op1-C, Op3, Op5, Op1-R
                                                   8400
0006905C  B982 0033                              8401           XGR   R3,R3
00069060  5830 5024                     00000024  8402           L     R3,PLA
                                                   8403
00069064  B982 0044                              8404           XGR   R4,R4
00069068  5840 5020                     00000020  8405           L     R4,OPREPA           OP2, Op4, Op6, Op8
                                                   8406
0006906C  D25F 1000 2000       00000000  00000000  8407           MVC   0(96,R1),0(R2)      load Op1-C, Op3, Op1-R
00069072  D21F 3000 1000       00000000  00000000  8408           MVC   0(32,R3),0(R1)      copy to PL OP1-C
00069078  D21F 3050 1020       00000050  00000020  8409           MVC   80(32,R3),32(R1)    copy to PL OP3
0006907E  D21F 3020 1060       00000020  00000060  8410           MVC   32(32,R3),96(R1)    copy to PL OP1-R
                                                   8411
00069084  D23F 4000 2080       00000000  00000080  8412           MVC   0(64,R4),128(R2)    load initial Op2, Op4
                                                   8413
                                                   8414 ****
                                                   8415 * Force a EQ by changing the Op1-C == Op2
                                                   8416 ****
0006908A  D21F 3000 4000       00000000  00000000  8417           MVC   0(32,R3),0(R4)
                                                   8418 ****
00069090  B982 0000                              8419           XGR   R0,R0
00069094  4104 0020                     00000020  8420           LA    R0,32(R4)
00069098  E303 0048 0024                00000048  8421           STG   R0,72(R3)           load OP4 address in PL
                                                   8422
0006909E  4104 0040                     00000040  8423           LA    R0,64(R4)
000690A2  E303 0068 0024                00000068  8424           STG   R0,104(R3)          load OP6 address in PL
                                                   8425
000690A8  B982 0000                              8426           XGR   R0,R0
000690AC  4100 001B                     0000001B  8427           LA    R0,27
000690B0  EE00 4000 3000       00000000  00000000  8428           PLO   0,0(R4),0,0(R3)     C/S Store (CSSTO: 27)
                                                   8429
000690B6  B98D 0020                              8430           EPSW  R2,R0               extract psw
000690BA  5020 5010                     00000010  8431           ST    R2,CCPSW              to save CC
000690BE  E310 500B 0076                0000000B  8432           LB    R1,CCMASK           (failure CC mask)
000690C4  8910 0004                     00000004  8433           SLL   R1,4                (shift to BC instr CC position)
000690C8  4410 50DE                     000690FE  8434           EX    R1,F272TSTCC            fail if...
                                                   8435 *
                                                   8436 * compare results
                                                   8437 *
                               000690CC  00000001  8438 F272CKRES EQU   *
                                                   8439 *                                  reset adresses
000690CC  B982 0011                              8440           XGR   R1,R1
000690D0  5810 501C                     0000001C  8441           L     R1,OPSRCA           Op1, Op3, Op1-R
000690D4  B982 0033                              8442           XGR   R3,R3
000690D8  5830 5024                     00000024  8443           L     R3,PLA              PL
000690DC  B982 0044                              8444           XGR   R4,R4
000690E0  5840 5020                     00000020  8445           L     R4,OPREPA           OP2, Op4, Op6
                                                   8446
000690E4  D51F 4020 3050       00000020  00000050  8447           CLC   32(32,R4),80(R3)    OP4 ==  PL Op3
000690EA  4770 50D8                     000690F8  8448           BNE   F272XX                   different, error
000690EE  D51F 4000 3020       00000000  00000020  8449           CLC   0(32,R4),32(R3)     OP2 ==  PL Op-R
000690F4  478B 0000                     00000000  8450           BE    0(R11)                   same, return
                                                   8451
                               000690F8  00000001  8452 F272XX   EQU   *
000690F8  45F0 8340                     00000540  8453           BAL   R15,FAILMSG
000690FC  07FB                                    8454           BR    R11
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8455
000690FE  4700 50E2                       00069102  8456 F272TSTCC BC   0,F272CCMSG         fail if unexpected condition code
                                 00069102 00000001  8457 F272CCMSG EQU  *
00069102  45F0 80F2                       000002F2  8458           BAL   R15,CCMSG
00069106  47F0 50AC                       000690CC  8459           B     F272CKRES
                                                    8460
                                                    8461           TSTDATA
0006910A                         0006910A 00069400  8462+          ORG   PLOETST+(200*K)+(55*4*K)+K
                                                    8463           TSTLBL OPDST              OP2, Op4, Op6, Op8
00069400                                            8464+OPDST93   DS    0C
00069400  00000000 00000000                         8465           DC    XL256'0'
00069408  00000000 00000000
00069410  00000000 00000000
00069418  00000000 00000000
00069420  00000000 00000000
00069428  00000000 00000000
00069430  00000000 00000000
00069438  00000000 00000000
00069440  00000000 00000000
00069448  00000000 00000000
00069450  00000000 00000000
00069458  00000000 00000000
00069460  00000000 00000000
00069468  00000000 00000000
00069470  00000000 00000000
00069478  00000000 00000000
00069480  00000000 00000000
00069488  00000000 00000000
00069490  00000000 00000000
00069498  00000000 00000000
000694A0  00000000 00000000
000694A8  00000000 00000000
000694B0  00000000 00000000
000694B8  00000000 00000000
000694C0  00000000 00000000
000694C8  00000000 00000000
000694D0  00000000 00000000
000694D8  00000000 00000000
000694E0  00000000 00000000
000694E8  00000000 00000000
000694F0  00000000 00000000
000694F8  00000000 00000000
                                                    8466
                                                    8467           TSTLBL OPSRC              Op1-C, Op3, Op5, Op7, Op1-R
00069500                                            8468+OPSRC93   DS    0C
00069500  00000000 00000000                         8469           DC    XL256'0'
00069508  00000000 00000000
00069510  00000000 00000000
00069518  00000000 00000000
00069520  00000000 00000000
00069528  00000000 00000000
00069530  00000000 00000000
00069538  00000000 00000000
00069540  00000000 00000000
00069548  00000000 00000000
00069550  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00069558  00000000 00000000
00069560  00000000 00000000
00069568  00000000 00000000
00069570  00000000 00000000
00069578  00000000 00000000
00069580  00000000 00000000
00069588  00000000 00000000
00069590  00000000 00000000
00069598  00000000 00000000
000695A0  00000000 00000000
000695A8  00000000 00000000
000695B0  00000000 00000000
000695B8  00000000 00000000
000695C0  00000000 00000000
000695C8  00000000 00000000
000695D0  00000000 00000000
000695D8  00000000 00000000
000695E0  00000000 00000000
000695E8  00000000 00000000
000695F0  00000000 00000000
000695F8  00000000 00000000
                                          8470
                                          8471           TSTLBL PL                 Parameter List
00069600                                  8472+PL93      DS    0C
00069600  00000000 00000000               8473           DC    XL352'00'
00069608  00000000 00000000
00069610  00000000 00000000
00069618  00000000 00000000
00069620  00000000 00000000
00069628  00000000 00000000
00069630  00000000 00000000
00069638  00000000 00000000
00069640  00000000 00000000
00069648  00000000 00000000
00069650  00000000 00000000
00069658  00000000 00000000
00069660  00000000 00000000
00069668  00000000 00000000
00069670  00000000 00000000
00069678  00000000 00000000
00069680  00000000 00000000
00069688  00000000 00000000
00069690  00000000 00000000
00069698  00000000 00000000
000696A0  00000000 00000000
000696A8  00000000 00000000
000696B0  00000000 00000000
000696B8  00000000 00000000
000696C0  00000000 00000000
000696C8  00000000 00000000
000696D0  00000000 00000000
000696D8  00000000 00000000
000696E0  00000000 00000000
000696E8  00000000 00000000
000696F0  00000000 00000000
000696F8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00069700  00000000 00000000
00069708  00000000 00000000
00069710  00000000 00000000
00069718  00000000 00000000
00069720  00000000 00000000
00069728  00000000 00000000
00069730  00000000 00000000
00069738  00000000 00000000
00069740  00000000 00000000
00069748  00000000 00000000
00069750  00000000 00000000
00069758  00000000 00000000
                                          8474
                                          8475           TSTLBL FMSG               fail message
00069760                                  8476+FMSG93    DS    0C
00069760  86818993 858440D7              8477           DC    C'failed PLO-Extension '
00069768  D3D660C5 A7A38595
00069770  A2899695 40
00069775  C3969497 81998540              8478           DC    C'Compare and Swap and Store (CSSTO: 27) : EQ'
0006977D  81958440 E2A68197
00069785  40819584 40E2A396
0006978D  9985404D C3E2E2E3
00069795  D67A40F2 F75D407A
0006979D  40C5D8
000697A0  40404040 40404040              8479           DC    CL64' '
000697A8  40404040 40404040
000697B0  40404040 40404040
000697B8  40404040 40404040
000697C0  40404040 40404040
000697C8  40404040 40404040
000697D0  40404040 40404040
000697D8  40404040 40404040
                                          8480           TSTEND
000697E0                                  8481+          DS    0H
000697E0  C5958440 968640A3              8482+          DC    CL32'End of test: 93'
000697E8  85A2A37A 40F9F340
000697F0  40404040 40404040
000697F8  40404040 40404040
                                          8483
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   8485 *=======================================================================
                                                   8486 *       Double Compare and Swap Tests
                                                   8487 *=======================================================================
                                                   8488
                                                   8489 * -----------------------------------------------------------------
                                                   8490 * Test Double Compare and Swap (DCSO: 26) instruction : NE Op1!=Op2
                                                   8491 * -----------------------------------------------------------------
                                                   8492          TSTHDR 1,99,26           Op1-C != OP2 (cc=1, UR=99)
00069800                          00069800  0006A000 8493+         ORG   PLOETST+(200*K)+(56*4*K)
0006A000  E2E3C1D9 E3409686                        8494+         DC    CL32'START of test: 94'
0006A008  40A385A2 A37A40F9
0006A010  F4404040 40404040
0006A018  40404040 40404040
0006A020                          0006A020         8495+         USING *,R5               base for test data and test routine
0006A020                          00000000         8496+         USING PLOETEST,R5
0006A020  0006A04C                                 8497+T94      DC    A(CODE94)          address of test code
0006A024  005E                                     8498+         DC    H'94'              test number
0006A026  00                                       8499+         DC    X'00'
0006A027  1A                                       8500+         DC    HL1'26'            Function
0006A028  63                                       8501+         DC    HL1'99'            expected unequal results (UR)
0006A029  00                                       8502+         DC    HL1'00'            found UR
0006A02A  01                                       8503+         DC    HL1'1'             cc
0006A02B  0B                                       8504+         DC    HL1'11'            not expected CC mask
0006A02C  00                                       8505+         DS    X                  extracted cc
0006A030  00000000 00000000                        8506+         DS    2F                 extract PSW after test (has CC)
0006A038  0006A760                                 8507+         DC    A(FMSG94)          address of test fail message
0006A03C  0006A500                                 8508+         DC    A(OPSRC94)         address of operand source
0006A040  0006A400                                 8509+         DC    A(OPDST94)         address of operand destination
0006A044  0006A600                                 8510+         DC    A(PL94)            address of parameter list
0006A048  0000120F                                 8511+         DC    A(RANDX+94)        address of random source data
                                                   8512
                                                   8513          TSTCODE
0006A04C                                           8514+CODE94   DS    0F
                                                   8515 *                                 load random source
0006A04C  B982 0022                                8516          XGR   R2,R2
0006A050  5820 5028                       00000028 8517          L     R2,RANDXA
                                                   8518 *
                                                   8519 * Double Compare and Swap (DCSO: 26)
                                                   8520 * Build Parambter List
                                                   8521 *
0006A054  B982 0011                                8522          XGR   R1,R1
0006A058  5810 501C                       0000001C 8523          L     R1,OPSRCA          Op1-C, Op3, Op1-R, Op3-R
                                                   8524
0006A05C  B982 0033                                8525          XGR   R3,R3
0006A060  5830 5024                       00000024 8526          L     R3,PLA
                                                   8527
0006A064  B982 0044                                8528          XGR   R4,R4
0006A068  5840 5020                       00000020 8529          L     R4,OPREPA          OP2, Op4
                                                   8530
0006A06C  D27F 1000 2000      00000000    00000000 8531          MVC   0(128,R1),0(R2)    load Op1-C, Op3, Op1-R, Op3-R
0006A072  D21F 3000 1000      00000000    00000000 8532          MVC   0(32,R3),0(R1)     copy to PL OP1-C
0006A078  D21F 3050 1020      00000050    00000020 8533          MVC   80(32,R3),32(R1)   copy to PL OP3
0006A07E  D21F 3020 1040      00000020    00000040 8534          MVC   32(32,R3),64(R1)   copy to PL OP1-R
0006A084  D21F 3070 1060      00000070    00000060 8535          MVC   112(32,R3),96(R1)  copy to PL OP3-R
                                                   8536
0006A08A  D23F 4000 2080      00000000    00000080 8537          MVC   0(64,R4),128(R2)   load initial Op2, Op4
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8538
                                                    8539 ****
                                                    8540 * force a failure by changing the Op1-c
                                                    8541 ****
0006A090  D203 3000 2021      00000000  00000021  8542         MVC   0(4,R3),33(R2)
                                                    8543 ****
0006A096  B982 0000                               8544         XGR   R0,R0
0006A09A  4104 0020                     00000020  8545         LA    R0,32(R4)
0006A09E  E303 0048 0024                00000048  8546         STG   R0,72(R3)          load OP4 address in PL
                                                    8547
0006A0A4  B982 0000                               8548         XGR   R0,R0
0006A0A8  4100 001A                     0000001A  8549         LA    R0,26
0006A0AC  EE00 4000 3000      00000000  00000000  8550         PLO   0,0(R4),0,0(R3)    Double C/S (DCSO: 26)
                                                    8551
0006A0B2  B98D 0020                               8552         EPSW  R2,R0              extract psw
0006A0B6  5020 5010                     00000010  8553         ST    R2,CCPSW             to save CC
0006A0BA  E310 500B 0076                0000000B  8554         LB    R1,CCMASK          (failure CC mask)
0006A0C0  8910 0004                     00000004  8555         SLL   R1,4               (shift to BC instr CC position)
0006A0C4  4410 50D0                     0006A0F0  8556         EX    R1,F261TSTCC           fail if...
                                                    8557 *
                                                    8558 * compare results
                                                    8559 *
                        0006A0C8  00000001  8560 F261CKRES EQU   *
                                                    8561 *                               reset adresses
0006A0C8  B982 0011                               8562         XGR   R1,R1
0006A0CC  5810 501C                     0000001C  8563         L     R1,OPSRCA          Op1-C, Op3, Op1-R, Op3-R
0006A0D0  B982 0033                               8564         XGR   R3,R3
0006A0D4  5830 5024                     00000024  8565         L     R3,PLA             PL
0006A0D8  B982 0044                               8566         XGR   R4,R4
0006A0DC  5840 5020                     00000020  8567         L     R4,OPREPA          OP2, Op4
                                                    8568
0006A0E0  D51F 4000 3000      00000000  00000000  8569         CLC   0(32,R4),0(R3)     Op2 == OP1C?
0006A0E6  478B 0000                     00000000  8570         BE    0(R11)             same, return
                                                    8571
                        0006A0EA  00000001  8572 F261XX   EQU   *
0006A0EA  45F0 8340                     00000540  8573         BAL   R15,FAILMSG
0006A0EE  07FB                                    8574         BR    R11
                                                    8575
0006A0F0  4700 50D4                     0006A0F4  8576 F261TSTCC BC    0,F261CCMSG        fail if unexpected condition code
                        0006A0F4  00000001  8577 F261CCMSG EQU  *
0006A0F4  45F0 80F2                     000002F2  8578         BAL   R15,CCMSG
0006A0F8  47F0 50A8                     0006A0C8  8579         B     F261CKRES
                                                    8580
                                                    8581         TSTDATA
0006A0FC                  0006A0FC  0006A400  8582+         ORG   PLOETST+(200*K)+(56*4*K)+K
                                                    8583         TSTLBL OPDST             OP2, Op4, Op6
0006A400                                          8584+OPDST94 DS    0C
0006A400  00000000 00000000                       8585         DC    XL256'0'
0006A408  00000000 00000000
0006A410  00000000 00000000
0006A418  00000000 00000000
0006A420  00000000 00000000
0006A428  00000000 00000000
0006A430  00000000 00000000
0006A438  00000000 00000000
0006A440  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0006A448  00000000 00000000
0006A450  00000000 00000000
0006A458  00000000 00000000
0006A460  00000000 00000000
0006A468  00000000 00000000
0006A470  00000000 00000000
0006A478  00000000 00000000
0006A480  00000000 00000000
0006A488  00000000 00000000
0006A490  00000000 00000000
0006A498  00000000 00000000
0006A4A0  00000000 00000000
0006A4A8  00000000 00000000
0006A4B0  00000000 00000000
0006A4B8  00000000 00000000
0006A4C0  00000000 00000000
0006A4C8  00000000 00000000
0006A4D0  00000000 00000000
0006A4D8  00000000 00000000
0006A4E0  00000000 00000000
0006A4E8  00000000 00000000
0006A4F0  00000000 00000000
0006A4F8  00000000 00000000
                                          8586
                                          8587          TSTLBL OPSRC              Op1-C, Op3, Op5, Op1-R
0006A500                                  8588+OPSRC94  DS    0C
0006A500  00000000 00000000               8589          DC    XL256'0'
0006A508  00000000 00000000
0006A510  00000000 00000000
0006A518  00000000 00000000
0006A520  00000000 00000000
0006A528  00000000 00000000
0006A530  00000000 00000000
0006A538  00000000 00000000
0006A540  00000000 00000000
0006A548  00000000 00000000
0006A550  00000000 00000000
0006A558  00000000 00000000
0006A560  00000000 00000000
0006A568  00000000 00000000
0006A570  00000000 00000000
0006A578  00000000 00000000
0006A580  00000000 00000000
0006A588  00000000 00000000
0006A590  00000000 00000000
0006A598  00000000 00000000
0006A5A0  00000000 00000000
0006A5A8  00000000 00000000
0006A5B0  00000000 00000000
0006A5B8  00000000 00000000
0006A5C0  00000000 00000000
0006A5C8  00000000 00000000
0006A5D0  00000000 00000000
0006A5D8  00000000 00000000
0006A5E0  00000000 00000000
0006A5E8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0006A5F0  00000000 00000000
0006A5F8  00000000 00000000
                                          8590
                                          8591           TSTLBL PL                  Parameter List
0006A600                                  8592+PL94      DS    0C
0006A600  00000000 00000000               8593           DC    XL352'00'
0006A608  00000000 00000000
0006A610  00000000 00000000
0006A618  00000000 00000000
0006A620  00000000 00000000
0006A628  00000000 00000000
0006A630  00000000 00000000
0006A638  00000000 00000000
0006A640  00000000 00000000
0006A648  00000000 00000000
0006A650  00000000 00000000
0006A658  00000000 00000000
0006A660  00000000 00000000
0006A668  00000000 00000000
0006A670  00000000 00000000
0006A678  00000000 00000000
0006A680  00000000 00000000
0006A688  00000000 00000000
0006A690  00000000 00000000
0006A698  00000000 00000000
0006A6A0  00000000 00000000
0006A6A8  00000000 00000000
0006A6B0  00000000 00000000
0006A6B8  00000000 00000000
0006A6C0  00000000 00000000
0006A6C8  00000000 00000000
0006A6D0  00000000 00000000
0006A6D8  00000000 00000000
0006A6E0  00000000 00000000
0006A6E8  00000000 00000000
0006A6F0  00000000 00000000
0006A6F8  00000000 00000000
0006A700  00000000 00000000
0006A708  00000000 00000000
0006A710  00000000 00000000
0006A718  00000000 00000000
0006A720  00000000 00000000
0006A728  00000000 00000000
0006A730  00000000 00000000
0006A738  00000000 00000000
0006A740  00000000 00000000
0006A748  00000000 00000000
0006A750  00000000 00000000
0006A758  00000000 00000000
                                          8594
                                          8595           TSTLBL FMSG                fail message
0006A760                                  8596+FMSG94    DS    0C
0006A760  86818993 858440D7               8597           DC    C'failed PLO-Extension '
0006A768  D3D660C5 A7A38595
0006A770  A2899695 40
0006A775  C496A482 938540C3               8598           DC    C'Double Compare and Swap (DCSO: 26)'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0006A77D  96949781 99854081
0006A785  958440E2 A6819740
0006A78D  4DC4C3E2 D67A40F2
0006A795  F65D
0006A797  407A40D5 C540D697                        8599          DC    C' : NE Op1!=Op2'
0006A79F  F15A7ED6 97F2
0006A7A5  40404040 40404040                        8600          DC    CL64' '
0006A7AD  40404040 40404040
0006A7B5  40404040 40404040
0006A7BD  40404040 40404040
0006A7C5  40404040 40404040
0006A7CD  40404040 40404040
0006A7D5  40404040 40404040
0006A7DD  40404040 40404040
                                                   8601          TSTEND
0006A7E6                                           8602+         DS    0H
0006A7E6  C5958440 968640A3                        8603+         DC    CL32'End of test: 94'
0006A7EE  85A2A37A 40F9F440
0006A7F6  40404040 40404040
0006A7FE  40404040 40404040
                                                   8604
                                                   8605 * ------------------------------------------------------------------
                                                   8606 * Test Double Compare and Swap (DCSO: 26) instruction : NE Op3!=Op4
                                                   8607 * ------------------------------------------------------------------
                                                   8608          TSTHDR 1,99,26            Op3 != OP4 (cc=2, UR=99)
0006A806                        0006A806  0006B000 8609+         ORG   PLOETST+(200*K)+(57*4*K)
0006B000  E2E3C1D9 E3409686                        8610+         DC    CL32'START of test: 95'
0006B008  40A385A2 A37A40F9
0006B010  F5404040 40404040
0006B018  40404040 40404040
0006B020                        0006B020           8611+         USING *,R5                base for test data and test routine
0006B020                        00000000           8612+         USING PLOETEST,R5
0006B020  0006B04C                                 8613+T95      DC    A(CODE95)           address of test code
0006B024  005F                                     8614+         DC    H'95'               test number
0006B026  00                                       8615+         DC    X'00'
0006B027  1A                                       8616+         DC    HL1'26'             Function
0006B028  63                                       8617+         DC    HL1'99'             expected unequal results (UR)
0006B029  00                                       8618+         DC    HL1'00'             found UR
0006B02A  01                                       8619+         DC    HL1'1'              cc
0006B02B  0B                                       8620+         DC    HL1'11'             not expected CC mask
0006B02C  00                                       8621+         DS    X                   extracted cc
0006B030  00000000 00000000                        8622+         DS    2F                  extract PSW after test (has CC)
0006B038  0006B760                                 8623+         DC    A(FMSG95)           address of test fail message
0006B03C  0006B500                                 8624+         DC    A(OPSRC95)          address of operand source
0006B040  0006B400                                 8625+         DC    A(OPDST95)          address of operand destination
0006B044  0006B600                                 8626+         DC    A(PL95)             address of parameter list
0006B048  00001210                                 8627+         DC    A(RANDX+95)         address of random source data
                                                   8628
                                                   8629          TSTCODE
0006B04C                                           8630+CODE95   DS    0F
                                                   8631 *                                  load random source
0006B04C  B982 0022                                8632          XGR   R2,R2
0006B050  5820 5028                     00000028   8633          L     R2,RANDXA
                                                   8634 *
                                                   8635 * Double Compare and Swap (DCSO: 26)
                                                   8636 * Build Parambter List
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8637 *
0006B054  B982 0011                                 8638          XGR   R1,R1
0006B058  5810 501C                 0000001C        8639          L     R1,OPSRCA           Op1-C, Op3, Op1-R, Op3-R
                                                    8640
0006B05C  B982 0033                                 8641          XGR   R3,R3
0006B060  5830 5024                 00000024        8642          L     R3,PLA
                                                    8643
0006B064  B982 0044                                 8644          XGR   R4,R4
0006B068  5840 5020                 00000020        8645          L     R4,OPREPA           OP2, Op4
                                                    8646
0006B06C  D27F 1000 2000      00000000  00000000    8647          MVC   0(128,R1),0(R2)     load Op1-C, Op3, Op1-R, Op3-R
0006B072  D21F 3000 1000      00000000  00000000    8648          MVC   0(32,R3),0(R1)      copy to PL OP1-C
0006B078  D21F 3050 1020      00000050  00000020    8649          MVC   80(32,R3),32(R1)    copy to PL OP3
0006B07E  D21F 3020 1040      00000020  00000040    8650          MVC   32(32,R3),64(R1)    copy to PL OP1-R
0006B084  D21F 3070 1060      00000070  00000060    8651          MVC   112(32,R3),96(R1)   copy to PL OP3-R
                                                    8652
0006B08A  D23F 4000 2080      00000000  00000080    8653          MVC   0(64,R4),128(R2)    load initial Op2, Op4
                                                    8654
                                                    8655 ****
                                                    8656 * force a failure by changing the Op3-C
                                                    8657 ****
0006B090  D203 3050 2021      00000050  00000021    8658          MVC   80(4,R3),33(R2)
                                                    8659 ****
0006B096  B982 0000                                 8660          XGR   R0,R0
0006B09A  4104 0020                 00000020        8661          LA    R0,32(R4)
0006B09E  E303 0048 0024            00000048        8662          STG   R0,72(R3)           load OP4 address in PL
                                                    8663
0006B0A4  B982 0000                                 8664          XGR   R0,R0
0006B0A8  4100 001A                 0000001A        8665          LA    R0,26
0006B0AC  EE00 4000 3000      00000000  00000000    8666          PLO   0,0(R4),0,0(R3)     Double C/S (DCSO: 26)
                                                    8667
0006B0B2  B98D 0020                                 8668          EPSW  R2,R0               extract psw
0006B0B6  5020 5010                 00000010        8669          ST    R2,CCPSW              to save CC
0006B0BA  E310 500B 0076            0000000B        8670          LB    R1,CCMASK           (failure CC mask)
0006B0C0  8910 0004                 00000004        8671          SLL   R1,4                (shift to BC instr CC position)
0006B0C4  4410 50DA                 0006B0FA        8672          EX    R1,F262TSTCC            fail if...
                                                    8673 *
                                                    8674 * compare results
                                                    8675 *
                            0006B0C8  00000001      8676 F262CKRES EQU   *
                                                    8677 *                                  reset adresses
0006B0C8  B982 0011                                 8678          XGR   R1,R1
0006B0CC  5810 501C                 0000001C        8679          L     R1,OPSRCA           Op1-C, Op3, Op1-R, Op3-R
0006B0D0  B982 0033                                 8680          XGR   R3,R3
0006B0D4  5830 5024                 00000024        8681          L     R3,PLA              PL
0006B0D8  B982 0044                                 8682          XGR   R4,R4
0006B0DC  5840 5020                 00000020        8683          L     R4,OPREPA           OP2, Op4
                                                    8684
0006B0E0  D51F 4000 3000      00000000  00000000    8685          CLC   0(32,R4),0(R3)      Op2 == OP1C?
0006B0E6  4770 50D4                 0006B0F4        8686          BNE   F262XX
0006B0EA  D51F 3050 4020      00000050  00000020    8687          CLC   80(32,R3),32(R4)    Op3-C == OP4
0006B0F0  477B 0000                 00000000        8688          BNE    0(R11)               NE, matches test Op3 != OP4
                                                    8689
                            0006B0F4  00000001      8690 F262XX   EQU   *
0006B0F4  45F0 8340                 00000540        8691          BAL   R15,FAILMSG
0006B0F8  07FB                                      8692          BR    R11
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8693
0006B0FA  4700 50DE                       0006B0FE  8694 F262TSTCC BC   0,F262CCMSG         fail if unexpected condition code
                                0006B0FE  00000001  8695 F262CCMSG EQU  *
0006B0FE  45F0 80F2                       000002F2  8696           BAL   R15,CCMSG
0006B102  47F0 50A8                       0006B0C8  8697           B     F262CKRES
                                                    8698
                                                    8699           TSTDATA
0006B106                        0006B106  0006B400  8700+          ORG   PLOETST+(200*K)+(57*4*K)+K
                                                    8701           TSTLBL OPDST              OP2, Op4, Op6
0006B400                                            8702+OPDST95   DS    0C
0006B400  00000000 00000000                         8703           DC    XL256'0'
0006B408  00000000 00000000
0006B410  00000000 00000000
0006B418  00000000 00000000
0006B420  00000000 00000000
0006B428  00000000 00000000
0006B430  00000000 00000000
0006B438  00000000 00000000
0006B440  00000000 00000000
0006B448  00000000 00000000
0006B450  00000000 00000000
0006B458  00000000 00000000
0006B460  00000000 00000000
0006B468  00000000 00000000
0006B470  00000000 00000000
0006B478  00000000 00000000
0006B480  00000000 00000000
0006B488  00000000 00000000
0006B490  00000000 00000000
0006B498  00000000 00000000
0006B4A0  00000000 00000000
0006B4A8  00000000 00000000
0006B4B0  00000000 00000000
0006B4B8  00000000 00000000
0006B4C0  00000000 00000000
0006B4C8  00000000 00000000
0006B4D0  00000000 00000000
0006B4D8  00000000 00000000
0006B4E0  00000000 00000000
0006B4E8  00000000 00000000
0006B4F0  00000000 00000000
0006B4F8  00000000 00000000
                                                    8704
                                                    8705           TSTLBL OPSRC              Op1-C, Op3, Op5, Op1-R
0006B500                                            8706+OPSRC95   DS    0C
0006B500  00000000 00000000                         8707           DC    XL256'0'
0006B508  00000000 00000000
0006B510  00000000 00000000
0006B518  00000000 00000000
0006B520  00000000 00000000
0006B528  00000000 00000000
0006B530  00000000 00000000
0006B538  00000000 00000000
0006B540  00000000 00000000
0006B548  00000000 00000000
0006B550  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0006B558  00000000 00000000
0006B560  00000000 00000000
0006B568  00000000 00000000
0006B570  00000000 00000000
0006B578  00000000 00000000
0006B580  00000000 00000000
0006B588  00000000 00000000
0006B590  00000000 00000000
0006B598  00000000 00000000
0006B5A0  00000000 00000000
0006B5A8  00000000 00000000
0006B5B0  00000000 00000000
0006B5B8  00000000 00000000
0006B5C0  00000000 00000000
0006B5C8  00000000 00000000
0006B5D0  00000000 00000000
0006B5D8  00000000 00000000
0006B5E0  00000000 00000000
0006B5E8  00000000 00000000
0006B5F0  00000000 00000000
0006B5F8  00000000 00000000
                                          8708
                                          8709           TSTLBL PL                 Parameter List
0006B600                                  8710+PL95      DS    0C
0006B600  00000000 00000000               8711           DC    XL352'00'
0006B608  00000000 00000000
0006B610  00000000 00000000
0006B618  00000000 00000000
0006B620  00000000 00000000
0006B628  00000000 00000000
0006B630  00000000 00000000
0006B638  00000000 00000000
0006B640  00000000 00000000
0006B648  00000000 00000000
0006B650  00000000 00000000
0006B658  00000000 00000000
0006B660  00000000 00000000
0006B668  00000000 00000000
0006B670  00000000 00000000
0006B678  00000000 00000000
0006B680  00000000 00000000
0006B688  00000000 00000000
0006B690  00000000 00000000
0006B698  00000000 00000000
0006B6A0  00000000 00000000
0006B6A8  00000000 00000000
0006B6B0  00000000 00000000
0006B6B8  00000000 00000000
0006B6C0  00000000 00000000
0006B6C8  00000000 00000000
0006B6D0  00000000 00000000
0006B6D8  00000000 00000000
0006B6E0  00000000 00000000
0006B6E8  00000000 00000000
0006B6F0  00000000 00000000
0006B6F8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0006B700  00000000 00000000
0006B708  00000000 00000000
0006B710  00000000 00000000
0006B718  00000000 00000000
0006B720  00000000 00000000
0006B728  00000000 00000000
0006B730  00000000 00000000
0006B738  00000000 00000000
0006B740  00000000 00000000
0006B748  00000000 00000000
0006B750  00000000 00000000
0006B758  00000000 00000000
                                          8712
                                          8713          TSTLBL FMSG               fail message
0006B760                                  8714+FMSG95   DS    0C
0006B760  86818993 858440D7               8715          DC    C'failed PLO-Extension '
0006B768  D3D660C5 A7A38595
0006B770  A2899695 40
0006B775  C496A482 938540C3               8716          DC    C'Double Compare and Swap (DCSO: 26)'
0006B77D  96949781 99854081
0006B785  958440E2 A6819740
0006B78D  4DC4C3E2 D67A40F2
0006B795  F65D
0006B797  407A40D6 97F3405A               8717          DC    C' : Op3 != OP4'
0006B79F  7E40D6D7 F4
0006B7A4  40404040 40404040               8718          DC    CL64' '
0006B7AC  40404040 40404040
0006B7B4  40404040 40404040
0006B7BC  40404040 40404040
0006B7C4  40404040 40404040
0006B7CC  40404040 40404040
0006B7D4  40404040 40404040
0006B7DC  40404040 40404040
                                          8719          TSTEND
0006B7E4                                  8720+         DS    0H
0006B7E4  C5958440 968640A3               8721+         DC    CL32'End of test: 95'
0006B7EC  85A2A37A 40F9F540
0006B7F4  40404040 40404040
0006B7FC  40404040 40404040
                                          8722
                                          8723 * ------------------------------------------------------------------
                                          8724 * Test Double Compare and Swap (DCSO: 26) instruction : EQ
                                          8725 * ------------------------------------------------------------------
                                          8726          TSTHDR 0,99,28            Op1-C == OP2 && Op3 == OP4
0006B804                        0006B804  0006C000  8727+         ORG   PLOETST+(200*K)+(58*4*K)
0006C000  E2E3C1D9 E3409686               8728+         DC    CL32'START of test: 96'
0006C008  40A385A2 A37A40F9
0006C010  F6404040 40404040
0006C018  40404040 40404040
0006C020                        0006C020  8729+         USING *,R5                base for test data and test routine
0006C020                        00000000  8730+         USING PLOETEST,R5
0006C020  0006C04C                        8731+T96      DC    A(CODE96)           address of test code
0006C024  0060                            8732+         DC    H'96'               test number
0006C026  00                              8733+         DC    X'00'
0006C027  1C                              8734+         DC    HL1'28'             Function
0006C028  63                              8735+         DC    HL1'99'             expected unequal results (UR)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0006C029  00                                       8736+          DC    HL1'00'             found UR
0006C02A  00                                       8737+          DC    HL1'0'              cc
0006C02B  07                                       8738+          DC    HL1'7'              not expected CC mask
0006C02C  00                                       8739+          DS    X                   extracted cc
0006C030  00000000 00000000                        8740+          DS    2F                  extract PSW after test (has CC)
0006C038  0006C760                                 8741+          DC    A(FMSG96)           address of test fail message
0006C03C  0006C500                                 8742+          DC    A(OPSRC96)          address of operand source
0006C040  0006C400                                 8743+          DC    A(OPDST96)          address of operand destination
0006C044  0006C600                                 8744+          DC    A(PL96)             address of parameter list
0006C048  00001211                                 8745+          DC    A(RANDX+96)         address of random source data
                                                   8746 *                                           (cc=0, UR=99)
                                                   8747           TSTCODE
0006C04C                                           8748+CODE96    DS    0F
                                                   8749 *                                   load random source
0006C04C  B982 0022                                8750           XGR   R2,R2
0006C050  5820 5028                    00000028    8751           L     R2,RANDXA
                                                   8752 *
                                                   8753 * Compare and Swap and Double Store (CSDSTO: 28)
                                                   8754 * Build Parambter List
                                                   8755 *
0006C054  B982 0011                                8756           XGR   R1,R1
0006C058  5810 501C                    0000001C    8757           L     R1,OPSRCA           Op1-C, Op3, Op1-R, Op3-R
                                                   8758
0006C05C  B982 0033                                8759           XGR   R3,R3
0006C060  5830 5024                    00000024    8760           L     R3,PLA
                                                   8761
0006C064  B982 0044                                8762           XGR   R4,R4
0006C068  5840 5020                    00000020    8763           L     R4,OPREPA           OP2, Op4
                                                   8764
0006C06C  D27F 1000 2000       00000000  00000000  8765           MVC   0(128,R1),0(R2)     load Op1-C, Op3, Op1-R, Op3-R
0006C072  D21F 3000 1000       00000000  00000000  8766           MVC   0(32,R3),0(R1)      copy to PL OP1-C
0006C078  D21F 3050 1020       00000050  00000020  8767           MVC   80(32,R3),32(R1)    copy to PL OP3
0006C07E  D21F 3020 1040       00000020  00000040  8768           MVC   32(32,R3),64(R1)    copy to PL OP1-R
0006C084  D21F 3070 1060       00000070  00000060  8769           MVC   112(32,R3),96(R1)   copy to PL OP3-R
                                                   8770
0006C08A  D23F 4000 2080       00000000  00000080  8771           MVC   0(64,R4),128(R2)    load initial Op2, Op4
                                                   8772
                                                   8773 ****
                                                   8774 * Force a EQ by changing Op1-C == OP2 && Op3 == OP4
                                                   8775 ****
0006C090  D21F 3000 4000       00000000  00000000  8776           MVC   0(32,R3),0(R4)      Op1-C <-- Op2
0006C096  D21F 3050 4020       00000050  00000020  8777           MVC   80(32,R3),32(R4)    Op3-C <-- Op4
                                                   8778 ****
0006C09C  B982 0000                                8779           XGR   R0,R0
0006C0A0  4104 0020                    00000020    8780           LA    R0,32(R4)
0006C0A4  E303 0048 0024               00000048    8781           STG   R0,72(R3)           load OP4 address in PL
                                                   8782
0006C0AA  B982 0000                                8783           XGR   R0,R0
0006C0AE  4100 001A                    0000001A    8784           LA    R0,26
0006C0B2  EE00 4000 3000       00000000  00000000  8785           PLO   0,0(R4),0,0(R3)     C/S Double Store (CSDSTO: 28)
                                                   8786
0006C0B8  B98D 0020                                8787           EPSW  R2,R0               extract psw
0006C0BC  5020 5010                    00000010    8788           ST    R2,CCPSW              to save CC
0006C0C0  E310 500B 0076               0000000B    8789           LB    R1,CCMASK           (failure CC mask)
0006C0C6  8910 0004                    00000004    8790           SLL   R1,4                (shift to BC instr CC position)
0006C0CA  4410 50E0                    0006C100    8791           EX    R1,F263TSTCC            fail if...
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8792 *
                                                    8793 * compare results
                                                    8794 *
                               0006C0CE  00000001   8795 F263CKRES EQU   *
                                                    8796 *                          reset adresses
0006C0CE  B982 0011                                 8797          XGR   R1,R1
0006C0D2  5810 501C                      0000001C   8798          L     R1,OPSRCA           Op1-C, Op3, Op1-R, Op3-R
0006C0D6  B982 0033                                 8799          XGR   R3,R3
0006C0DA  5830 5024                      00000024   8800          L     R3,PLA              PL
0006C0DE  B982 0044                                 8801          XGR   R4,R4
0006C0E2  5840 5020                      00000020   8802          L     R4,OPREPA           OP2, Op4
                                                    8803
0006C0E6  D51F 4020 3070      00000020   00000070   8804          CLC   32(32,R4),112(R3)   OP4 ==  PL Op3-R
0006C0EC  4770 50DA                      0006C0FA   8805          BNE   F263XX                   different, error
0006C0F0  D51F 4000 3020      00000000   00000020   8806          CLC   0(32,R4),32(R3)     OP2 ==  PL Op1-R
0006C0F6  478B 0000                      00000000   8807          BE    0(R11)                    same, return
                                                    8808
                               0006C0FA  00000001   8809 F263XX   EQU   *
0006C0FA  45F0 8340                      00000540   8810          BAL   R15,FAILMSG
0006C0FE  07FB                                      8811          BR    R11
                                                    8812
0006C100  4700 50E4                      0006C104   8813 F263TSTCC BC    0,F263CCMSG         fail if unexpected condition code
                               0006C104  00000001   8814 F263CCMSG EQU   *
0006C104  45F0 80F2                      000002F2   8815          BAL   R15,CCMSG
0006C108  47F0 50AE                      0006C0CE   8816          B     F263CKRES
                                                    8817
                                                    8818          TSTDATA
0006C10C                       0006C10C  0006C400   8819+         ORG   PLOETST+(200*K)+(58*4*K)+K
                                                    8820          TSTLBL OPDST              OP2, Op4
0006C400                                            8821+OPDST96  DS    0C
0006C400  00000000 00000000                         8822          DC    XL256'0'
0006C408  00000000 00000000
0006C410  00000000 00000000
0006C418  00000000 00000000
0006C420  00000000 00000000
0006C428  00000000 00000000
0006C430  00000000 00000000
0006C438  00000000 00000000
0006C440  00000000 00000000
0006C448  00000000 00000000
0006C450  00000000 00000000
0006C458  00000000 00000000
0006C460  00000000 00000000
0006C468  00000000 00000000
0006C470  00000000 00000000
0006C478  00000000 00000000
0006C480  00000000 00000000
0006C488  00000000 00000000
0006C490  00000000 00000000
0006C498  00000000 00000000
0006C4A0  00000000 00000000
0006C4A8  00000000 00000000
0006C4B0  00000000 00000000
0006C4B8  00000000 00000000
0006C4C0  00000000 00000000
0006C4C8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0006C4D0  00000000 00000000
0006C4D8  00000000 00000000
0006C4E0  00000000 00000000
0006C4E8  00000000 00000000
0006C4F0  00000000 00000000
0006C4F8  00000000 00000000
                                          8823
                                          8824           TSTLBL OPSRC              Op1-C, Op3, Op1-R, Op3-R
0006C500                                  8825+OPSRC96   DS    0C
0006C500  00000000 00000000               8826           DC    XL256'0'
0006C508  00000000 00000000
0006C510  00000000 00000000
0006C518  00000000 00000000
0006C520  00000000 00000000
0006C528  00000000 00000000
0006C530  00000000 00000000
0006C538  00000000 00000000
0006C540  00000000 00000000
0006C548  00000000 00000000
0006C550  00000000 00000000
0006C558  00000000 00000000
0006C560  00000000 00000000
0006C568  00000000 00000000
0006C570  00000000 00000000
0006C578  00000000 00000000
0006C580  00000000 00000000
0006C588  00000000 00000000
0006C590  00000000 00000000
0006C598  00000000 00000000
0006C5A0  00000000 00000000
0006C5A8  00000000 00000000
0006C5B0  00000000 00000000
0006C5B8  00000000 00000000
0006C5C0  00000000 00000000
0006C5C8  00000000 00000000
0006C5D0  00000000 00000000
0006C5D8  00000000 00000000
0006C5E0  00000000 00000000
0006C5E8  00000000 00000000
0006C5F0  00000000 00000000
0006C5F8  00000000 00000000
                                          8827
                                          8828           TSTLBL PL                 Parameter List
0006C600                                  8829+PL96      DS    0C
0006C600  00000000 00000000               8830           DC    XL352'00'
0006C608  00000000 00000000
0006C610  00000000 00000000
0006C618  00000000 00000000
0006C620  00000000 00000000
0006C628  00000000 00000000
0006C630  00000000 00000000
0006C638  00000000 00000000
0006C640  00000000 00000000
0006C648  00000000 00000000
0006C650  00000000 00000000
0006C658  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0006C660  00000000 00000000
0006C668  00000000 00000000
0006C670  00000000 00000000
0006C678  00000000 00000000
0006C680  00000000 00000000
0006C688  00000000 00000000
0006C690  00000000 00000000
0006C698  00000000 00000000
0006C6A0  00000000 00000000
0006C6A8  00000000 00000000
0006C6B0  00000000 00000000
0006C6B8  00000000 00000000
0006C6C0  00000000 00000000
0006C6C8  00000000 00000000
0006C6D0  00000000 00000000
0006C6D8  00000000 00000000
0006C6E0  00000000 00000000
0006C6E8  00000000 00000000
0006C6F0  00000000 00000000
0006C6F8  00000000 00000000
0006C700  00000000 00000000
0006C708  00000000 00000000
0006C710  00000000 00000000
0006C718  00000000 00000000
0006C720  00000000 00000000
0006C728  00000000 00000000
0006C730  00000000 00000000
0006C738  00000000 00000000
0006C740  00000000 00000000
0006C748  00000000 00000000
0006C750  00000000 00000000
0006C758  00000000 00000000
                                        8831
                                        8832          TSTLBL FMSG               fail message
0006C760                                8833+FMSG96   DS    0C
0006C760  86818993 858440D7            8834          DC    C'failed PLO-Extension '
0006C768  D3D660C5 A7A38595
0006C770  A2899695 40
0006C775  C496A482 938540C3            8835          DC    C'Double Compare and Swap (DCSO: 26)'
0006C77D  96949781 99854081
0006C785  958440E2 A6819740
0006C78D  4DC4C3E2 D67A40F2
0006C795  F65D
0006C797  407A40C5 D8                  8836          DC    C' : EQ'
0006C79C  40404040 40404040            8837          DC    CL64' '
0006C7A4  40404040 40404040
0006C7AC  40404040 40404040
0006C7B4  40404040 40404040
0006C7BC  40404040 40404040
0006C7C4  40404040 40404040
0006C7CC  40404040 40404040
0006C7D4  40404040 40404040
                                        8838          TSTEND
0006C7DC                                8839+         DS    0H
0006C7DC  C5958440 968640A3            8840+         DC    CL32'End of test: 96'
0006C7E4  85A2A37A 40F9F640
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0006C7EC  40404040 40404040
0006C7F4  40404040 40404040
                                                     8841
                                                     8842
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                  8844 *=====================================================================
                                                  8845 *       Compare and Swap Tests
                                                  8846 *=====================================================================
                                                  8847
                                                  8848 * -------------------------------------------------------------------
                                                  8849 * Test Compare and Swap (CSO: 25) instruction : NE Op1-C!=Op2
                                                  8850 * -------------------------------------------------------------------
                                                  8851          TSTHDR 1,99,26           Op1-C != OP2 (cc=1, UR=99)
0006C7FC                          0006C7FC  0006D000  8852+         ORG   PLOETST+(200*K)+(59*4*K)
0006D000  E2E3C1D9 E3409686                         8853+         DC    CL32'START of test: 97'
0006D008  40A385A2 A37A40F9
0006D010  F7404040 40404040
0006D018  40404040 40404040
0006D020                          0006D020          8854+         USING *,R5               base for test data and test routine
0006D020                          00000000          8855+         USING PLOETEST,R5
0006D020  0006D04C                                  8856+T97      DC    A(CODE97)          address of test code
0006D024  0061                                      8857+         DC    H'97'              test number
0006D026  00                                        8858+         DC    X'00'
0006D027  1A                                        8859+         DC    HL1'26'            Function
0006D028  63                                        8860+         DC    HL1'99'            expected unequal results (UR)
0006D029  00                                        8861+         DC    HL1'00'            found UR
0006D02A  01                                        8862+         DC    HL1'1'             cc
0006D02B  0B                                        8863+         DC    HL1'11'            not expected CC mask
0006D02C  00                                        8864+         DS    X                  extracted cc
0006D030  00000000 00000000                         8865+         DS    2F                 extract PSW after test (has CC)
0006D038  0006D760                                  8866+         DC    A(FMSG97)          address of test fail message
0006D03C  0006D500                                  8867+         DC    A(OPSRC97)         address of operand source
0006D040  0006D400                                  8868+         DC    A(OPDST97)         address of operand destination
0006D044  0006D600                                  8869+         DC    A(PL97)            address of parameter list
0006D048  00001212                                  8870+         DC    A(RANDX+97)        address of random source data
                                                  8871
                                                  8872          TSTCODE
0006D04C                                            8873+CODE97   DS    0F
                                                  8874 *                                   load random source
0006D04C  B982 0022                                 8875          XGR   R2,R2
0006D050  5820 5028                     00000028  8876          L     R2,RANDXA
                                                  8877 *
                                                  8878 * Compare and Swap (CSO: 25)
                                                  8879 * Build Parambter List
                                                  8880 *
0006D054  B982 0011                                 8881          XGR   R1,R1
0006D058  5810 501C                     0000001C  8882          L     R1,OPSRCA          Op1-C, Op1-R
                                                  8883
0006D05C  B982 0033                                 8884          XGR   R3,R3
0006D060  5830 5024                     00000024  8885          L     R3,PLA
                                                  8886
0006D064  B982 0044                                 8887          XGR   R4,R4
0006D068  5840 5020                     00000020  8888          L     R4,OPREPA          OP2
                                                  8889
0006D06C  D27F 1000 2000      00000000  00000000  8890          MVC   0(128,R1),0(R2)    load Op1-C, Op1-R
0006D072  D21F 3000 1000      00000000  00000000  8891          MVC   0(32,R3),0(R1)     copy to PL OP1-C
0006D078  D21F 3020 1020      00000020  00000020  8892          MVC   32(32,R3),32(R1)   copy to PL OP1-R
                                                  8893
0006D07E  D21F 4000 2080      00000000  00000080  8894          MVC   0(32,R4),128(R2)   load initial Op2
                                                  8895
                                                  8896 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   8897 * force a NE by changing the Op1-c
                                                   8898 ****
0006D084  D203 3000 2021      00000000  00000021   8899          MVC   0(4,R3),33(R2)
                                                   8900 ****
0006D08A  B982 0000                                8901          XGR   R0,R0
0006D08E  4100 0019                     00000019   8902          LA    R0,25
0006D092  EE00 4000 3000      00000000  00000000   8903          PLO   0,0(R4),0,0(R3)     C/S (CSO: 25)
                                                   8904
0006D098  B98D 0020                                8905          EPSW  R2,R0               extract psw
0006D09C  5020 5010                     00000010   8906          ST    R2,CCPSW              to save CC
0006D0A0  E310 500B 0076                0000000B   8907          LB    R1,CCMASK           (failure CC mask)
0006D0A6  8910 0004                     00000004   8908          SLL   R1,4                (shift to BC instr CC position)
0006D0AA  4410 50B6                     0006D0D6   8909          EX    R1,F251TSTCC            fail if...
                                                   8910 *
                                                   8911 * compare results
                                                   8912 *
                          0006D0AE      00000001   8913 F251CKRES EQU   *
                                                   8914 *                                  reset adresses
0006D0AE  B982 0011                                8915          XGR   R1,R1
0006D0B2  5810 501C                     0000001C   8916          L     R1,OPSRCA           Op1-C, Op3, Op1-R, Op3-R
0006D0B6  B982 0033                                8917          XGR   R3,R3
0006D0BA  5830 5024                     00000024   8918          L     R3,PLA              PL
0006D0BE  B982 0044                                8919          XGR   R4,R4
0006D0C2  5840 5020                     00000020   8920          L     R4,OPREPA           OP2, Op4
                                                   8921
0006D0C6  D51F 4000 3000      00000000  00000000   8922          CLC   0(32,R4),0(R3)      Op2 == OP1-C?
0006D0CC  478B 0000                     00000000   8923          BE    0(R11)              same, return
                                                   8924
                          0006D0D0      00000001   8925 F251XX   EQU   *
0006D0D0  45F0 8340                     00000540   8926          BAL   R15,FAILMSG
0006D0D4  07FB                                     8927          BR    R11
                                                   8928
0006D0D6  4700 50BA                     0006D0DA   8929 F251TSTCC BC    0,F251CCMSG         fail if unexpected condition code
                          0006D0DA      00000001   8930 F251CCMSG EQU   *
0006D0DA  45F0 80F2                     000002F2   8931          BAL   R15,CCMSG
0006D0DE  47F0 508E                     0006D0AE   8932          B     F251CKRES
                                                   8933
                                                   8934          TSTDATA
0006D0E2                  0006D0E2      0006D400   8935+         ORG   PLOETST+(200*K)+(59*4*K)+K
                                                   8936          TSTLBL OPDST              OP2, Op4, Op6
0006D400                                           8937+OPDST97  DS    0C
0006D400  00000000 00000000                        8938          DC    XL256'0'
0006D408  00000000 00000000
0006D410  00000000 00000000
0006D418  00000000 00000000
0006D420  00000000 00000000
0006D428  00000000 00000000
0006D430  00000000 00000000
0006D438  00000000 00000000
0006D440  00000000 00000000
0006D448  00000000 00000000
0006D450  00000000 00000000
0006D458  00000000 00000000
0006D460  00000000 00000000
0006D468  00000000 00000000
0006D470  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0006D478  00000000 00000000
0006D480  00000000 00000000
0006D488  00000000 00000000
0006D490  00000000 00000000
0006D498  00000000 00000000
0006D4A0  00000000 00000000
0006D4A8  00000000 00000000
0006D4B0  00000000 00000000
0006D4B8  00000000 00000000
0006D4C0  00000000 00000000
0006D4C8  00000000 00000000
0006D4D0  00000000 00000000
0006D4D8  00000000 00000000
0006D4E0  00000000 00000000
0006D4E8  00000000 00000000
0006D4F0  00000000 00000000
0006D4F8  00000000 00000000
                                        8939
                                        8940          TSTLBL OPSRC              Op1-C, Op3, Op5, Op1-R
0006D500                                8941+OPSRC97  DS    0C
0006D500  00000000 00000000             8942          DC    XL256'0'
0006D508  00000000 00000000
0006D510  00000000 00000000
0006D518  00000000 00000000
0006D520  00000000 00000000
0006D528  00000000 00000000
0006D530  00000000 00000000
0006D538  00000000 00000000
0006D540  00000000 00000000
0006D548  00000000 00000000
0006D550  00000000 00000000
0006D558  00000000 00000000
0006D560  00000000 00000000
0006D568  00000000 00000000
0006D570  00000000 00000000
0006D578  00000000 00000000
0006D580  00000000 00000000
0006D588  00000000 00000000
0006D590  00000000 00000000
0006D598  00000000 00000000
0006D5A0  00000000 00000000
0006D5A8  00000000 00000000
0006D5B0  00000000 00000000
0006D5B8  00000000 00000000
0006D5C0  00000000 00000000
0006D5C8  00000000 00000000
0006D5D0  00000000 00000000
0006D5D8  00000000 00000000
0006D5E0  00000000 00000000
0006D5E8  00000000 00000000
0006D5F0  00000000 00000000
0006D5F8  00000000 00000000
                                        8943
                                        8944          TSTLBL PL                 Parameter List
0006D600                                8945+PL97     DS    0C
0006D600  00000000 00000000             8946          DC    XL352'00'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0006D608  00000000 00000000
0006D610  00000000 00000000
0006D618  00000000 00000000
0006D620  00000000 00000000
0006D628  00000000 00000000
0006D630  00000000 00000000
0006D638  00000000 00000000
0006D640  00000000 00000000
0006D648  00000000 00000000
0006D650  00000000 00000000
0006D658  00000000 00000000
0006D660  00000000 00000000
0006D668  00000000 00000000
0006D670  00000000 00000000
0006D678  00000000 00000000
0006D680  00000000 00000000
0006D688  00000000 00000000
0006D690  00000000 00000000
0006D698  00000000 00000000
0006D6A0  00000000 00000000
0006D6A8  00000000 00000000
0006D6B0  00000000 00000000
0006D6B8  00000000 00000000
0006D6C0  00000000 00000000
0006D6C8  00000000 00000000
0006D6D0  00000000 00000000
0006D6D8  00000000 00000000
0006D6E0  00000000 00000000
0006D6E8  00000000 00000000
0006D6F0  00000000 00000000
0006D6F8  00000000 00000000
0006D700  00000000 00000000
0006D708  00000000 00000000
0006D710  00000000 00000000
0006D718  00000000 00000000
0006D720  00000000 00000000
0006D728  00000000 00000000
0006D730  00000000 00000000
0006D738  00000000 00000000
0006D740  00000000 00000000
0006D748  00000000 00000000
0006D750  00000000 00000000
0006D758  00000000 00000000
                                        8947
                                        8948          TSTLBL FMSG               fail message
0006D760                                8949+FMSG97   DS     0C
0006D760  86818993 858440D7             8950          DC     C'failed PLO-Extension '
0006D768  D3D660C5 A7A38595
0006D770  A2899695 40
0006D775  C3969497 81998540             8951          DC     C'Compare and Swap (CSO: 25)'
0006D77D  81958440 E2A68197
0006D785  404DC3E2 D67A40F2
0006D78D  F55D
0006D78F  407A40D5 C540D697             8952          DC     C' : NE Op1-C != Op2'
0006D797  F160C340 5A7E40D6
0006D79F  97F2
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0006D7A1  40404040 40404040                        8953          DC    CL64' '
0006D7A9  40404040 40404040
0006D7B1  40404040 40404040
0006D7B9  40404040 40404040
0006D7C1  40404040 40404040
0006D7C9  40404040 40404040
0006D7D1  40404040 40404040
0006D7D9  40404040 40404040
                                                   8954          TSTEND
0006D7E2                                           8955+         DS    0H
0006D7E2  C5958440 968640A3                        8956+         DC    CL32'End of test: 97'
0006D7EA  85A2A37A 40F9F740
0006D7F2  40404040 40404040
0006D7FA  40404040 40404040
                                                   8957
                                                   8958 * ------------------------------------------------------------------
                                                   8959 * Test Compare and Swap (DCSO: 25) instruction : EQ Op1-C == OP2
                                                   8960 * ------------------------------------------------------------------
                                                   8961          TSTHDR 0,99,25            EQ Op1-C == OP2 (cc=0, UR=99)
0006D802                          0006D802  0006E000 8962+         ORG   PLOETST+(200*K)+(60*4*K)
0006E000  E2E3C1D9 E3409686                        8963+         DC    CL32'START of test: 98'
0006E008  40A385A2 A37A40F9
0006E010  F8404040 40404040
0006E018  40404040 40404040
0006E020                          0006E020         8964+         USING *,R5                base for test data and test routine
0006E020                          00000000         8965+         USING PLOETEST,R5
0006E020  0006E04C                                 8966+T98      DC    A(CODE98)           address of test code
0006E024  0062                                     8967+         DC    H'98'               test number
0006E026  00                                       8968+         DC    X'00'
0006E027  19                                       8969+         DC    HL1'25'             Function
0006E028  63                                       8970+         DC    HL1'99'             expected unequal results (UR)
0006E029  00                                       8971+         DC    HL1'00'             found UR
0006E02A  00                                       8972+         DC    HL1'0'              cc
0006E02B  07                                       8973+         DC    HL1'7'              not expected CC mask
0006E02C  00                                       8974+         DS    X                   extracted cc
0006E030  00000000 00000000                        8975+         DS    2F                  extract PSW after test (has CC)
0006E038  0006E760                                 8976+         DC    A(FMSG98)           address of test fail message
0006E03C  0006E500                                 8977+         DC    A(OPSRC98)          address of operand source
0006E040  0006E400                                 8978+         DC    A(OPDST98)          address of operand destination
0006E044  0006E600                                 8979+         DC    A(PL98)             address of parameter list
0006E048  00001213                                 8980+         DC    A(RANDX+98)         address of random source data
                                                   8981
                                                   8982          TSTCODE
0006E04C                                           8983+CODE98   DS    0F
                                                   8984 *                                  load random source
0006E04C  B982 0022                                8985          XGR   R2,R2
0006E050  5820 5028                00000028        8986          L     R2,RANDXA
                                                   8987 *
                                                   8988 * Compare and Swap (DCSO: 25)
                                                   8989 * Build Parambter List
                                                   8990 *
0006E054  B982 0011                                8991          XGR   R1,R1
0006E058  5810 501C                0000001C        8992          L     R1,OPSRCA           Op1-C, Op3, Op1-R, Op3-R
                                                   8993
0006E05C  B982 0033                                8994          XGR   R3,R3
0006E060  5830 5024                00000024        8995          L     R3,PLA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8996
0006E064  B982 0044                                 8997          XGR   R4,R4
0006E068  5840 5020                   00000020      8998          L     R4,OPREPA           OP2, Op4
                                                    8999
0006E06C  D27F 1000 2000      00000000  00000000    9000          MVC   0(128,R1),0(R2)     load Op1-C, Op1-R
0006E072  D21F 3000 1000      00000000  00000000    9001          MVC   0(32,R3),0(R1)      copy to PL OP1-C
0006E078  D21F 3020 1020      00000020  00000020    9002          MVC   32(32,R3),32(R1)    copy to PL OP1-R
                                                    9003
0006E07E  D21F 4000 2080      00000000  00000080    9004          MVC   0(32,R4),128(R2)    load initial Op24
                                                    9005
                                                    9006 ****
                                                    9007 * force an EQ by changing the Op1-C
                                                    9008 ****
0006E084  D21F 3000 4000      00000000  00000000    9009          MVC   0(32,R3),0(R4)      Op1-C <-- Op2
                                                    9010 ****
0006E08A  B982 0000                                 9011          XGR   R0,R0
0006E08E  4100 0019                   00000019      9012          LA    R0,25
0006E092  EE00 4000 3000      00000000  00000000    9013          PLO   0,0(R4),0,0(R3)     C/S (CSO: 25)
                                                    9014
0006E098  B98D 0020                                 9015          EPSW  R2,R0               extract psw
0006E09C  5020 5010                   00000010      9016          ST    R2,CCPSW              to save CC
0006E0A0  E310 500B 0076              0000000B      9017          LB    R1,CCMASK           (failure CC mask)
0006E0A6  8910 0004                   00000004      9018          SLL   R1,4                (shift to BC instr CC position)
0006E0AA  4410 50B6                   0006E0D6      9019          EX    R1,F252TSTCC            fail if...
                                                    9020 *
                                                    9021 * compare results
                                                    9022 *
                              0006E0AE  00000001    9023 F252CKRES EQU   *
                                                    9024 *                                  reset adresses
0006E0AE  B982 0011                                 9025          XGR   R1,R1
0006E0B2  5810 501C                   0000001C      9026          L     R1,OPSRCA           Op1-C, Op3, Op1-R, Op3-R
0006E0B6  B982 0033                                 9027          XGR   R3,R3
0006E0BA  5830 5024                   00000024      9028          L     R3,PLA              PL
0006E0BE  B982 0044                                 9029          XGR   R4,R4
0006E0C2  5840 5020                   00000020      9030          L     R4,OPREPA           OP2, Op4
                                                    9031
0006E0C6  D51F 3020 4000      00000020  00000000    9032          CLC   32(32,R3),0(R4)     Op1-R == OP2
0006E0CC  478B 0000                   00000000      9033          BE    0(R11)              same, return
                                                    9034
                              0006E0D0  00000001    9035 F252XX   EQU   *
0006E0D0  45F0 8340                   00000540      9036          BAL   R15,FAILMSG
0006E0D4  07FB                                      9037          BR    R11
                                                    9038
0006E0D6  4700 50BA                   0006E0DA      9039 F252TSTCC BC    0,F252CCMSG         fail if unexpected condition code
                              0006E0DA  00000001    9040 F252CCMSG EQU   *
0006E0DA  45F0 80F2                   000002F2      9041          BAL   R15,CCMSG
0006E0DE  47F0 508E                   0006E0AE      9042          B     F252CKRES
                                                    9043
                                                    9044          TSTDATA
0006E0E2                      0006E0E2  0006E400    9045+         ORG   PLOETST+(200*K)+(60*4*K)+K
                                                    9046          TSTLBL OPDST              OP2, Op4, Op6
0006E400                                            9047+OPDST98  DS    0C
0006E400  00000000 00000000                         9048          DC    XL256'0'
0006E408  00000000 00000000
0006E410  00000000 00000000
0006E418  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0006E420  00000000 00000000
0006E428  00000000 00000000
0006E430  00000000 00000000
0006E438  00000000 00000000
0006E440  00000000 00000000
0006E448  00000000 00000000
0006E450  00000000 00000000
0006E458  00000000 00000000
0006E460  00000000 00000000
0006E468  00000000 00000000
0006E470  00000000 00000000
0006E478  00000000 00000000
0006E480  00000000 00000000
0006E488  00000000 00000000
0006E490  00000000 00000000
0006E498  00000000 00000000
0006E4A0  00000000 00000000
0006E4A8  00000000 00000000
0006E4B0  00000000 00000000
0006E4B8  00000000 00000000
0006E4C0  00000000 00000000
0006E4C8  00000000 00000000
0006E4D0  00000000 00000000
0006E4D8  00000000 00000000
0006E4E0  00000000 00000000
0006E4E8  00000000 00000000
0006E4F0  00000000 00000000
0006E4F8  00000000 00000000
                                                  9049
                                                  9050           TSTLBL OPSRC              Op1-C, Op3, Op5, Op1-R
0006E500                                          9051+OPSRC98   DS    0C
0006E500  00000000 00000000                       9052           DC    XL256'0'
0006E508  00000000 00000000
0006E510  00000000 00000000
0006E518  00000000 00000000
0006E520  00000000 00000000
0006E528  00000000 00000000
0006E530  00000000 00000000
0006E538  00000000 00000000
0006E540  00000000 00000000
0006E548  00000000 00000000
0006E550  00000000 00000000
0006E558  00000000 00000000
0006E560  00000000 00000000
0006E568  00000000 00000000
0006E570  00000000 00000000
0006E578  00000000 00000000
0006E580  00000000 00000000
0006E588  00000000 00000000
0006E590  00000000 00000000
0006E598  00000000 00000000
0006E5A0  00000000 00000000
0006E5A8  00000000 00000000
0006E5B0  00000000 00000000
0006E5B8  00000000 00000000
0006E5C0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0006E5C8  00000000 00000000
0006E5D0  00000000 00000000
0006E5D8  00000000 00000000
0006E5E0  00000000 00000000
0006E5E8  00000000 00000000
0006E5F0  00000000 00000000
0006E5F8  00000000 00000000
                                          9053
                                          9054           TSTLBL PL                 Parameter List
0006E600                                  9055+PL98      DS    0C
0006E600  00000000 00000000               9056           DC    XL352'00'
0006E608  00000000 00000000
0006E610  00000000 00000000
0006E618  00000000 00000000
0006E620  00000000 00000000
0006E628  00000000 00000000
0006E630  00000000 00000000
0006E638  00000000 00000000
0006E640  00000000 00000000
0006E648  00000000 00000000
0006E650  00000000 00000000
0006E658  00000000 00000000
0006E660  00000000 00000000
0006E668  00000000 00000000
0006E670  00000000 00000000
0006E678  00000000 00000000
0006E680  00000000 00000000
0006E688  00000000 00000000
0006E690  00000000 00000000
0006E698  00000000 00000000
0006E6A0  00000000 00000000
0006E6A8  00000000 00000000
0006E6B0  00000000 00000000
0006E6B8  00000000 00000000
0006E6C0  00000000 00000000
0006E6C8  00000000 00000000
0006E6D0  00000000 00000000
0006E6D8  00000000 00000000
0006E6E0  00000000 00000000
0006E6E8  00000000 00000000
0006E6F0  00000000 00000000
0006E6F8  00000000 00000000
0006E700  00000000 00000000
0006E708  00000000 00000000
0006E710  00000000 00000000
0006E718  00000000 00000000
0006E720  00000000 00000000
0006E728  00000000 00000000
0006E730  00000000 00000000
0006E738  00000000 00000000
0006E740  00000000 00000000
0006E748  00000000 00000000
0006E750  00000000 00000000
0006E758  00000000 00000000
                                          9057
                                          9058           TSTLBL FMSG               fail message
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0006E760                                          9059+FMSG98   DS    0C
0006E760  86818993 858440D7                       9060          DC    C'failed PLO-Extension '
0006E768  D3D660C5 A7A38595
0006E770  A2899695 40
0006E775  C3969497 81998540                       9061          DC    C'Compare and Swap (CSO: 25)'
0006E77D  81958440 E2A68197
0006E785  404DC3E2 D67A40F2
0006E78D  F55D
0006E78F  407AC5D8 40D697F1                       9062          DC    C' :EQ Op1-C == OP2'
0006E797  60C3407E 7E40D6D7
0006E79F  F2
0006E7A0  40404040 40404040                       9063          DC    CL64' '
0006E7A8  40404040 40404040
0006E7B0  40404040 40404040
0006E7B8  40404040 40404040
0006E7C0  40404040 40404040
0006E7C8  40404040 40404040
0006E7D0  40404040 40404040
0006E7D8  40404040 40404040
                                                  9064          TSTEND
0006E7E0                                          9065+         DS    0H
0006E7E0  C5958440 968640A3                       9066+         DC    CL32'End of test: 98'
0006E7E8  85A2A37A 40F9F840
0006E7F0  40404040 40404040
0006E7F8  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    9068 *=====================================================================
                                                    9069 *       Compare and Load Tests
                                                    9070 *=====================================================================
                                                    9071
                                                    9072 * -------------------------------------------------------------------
                                                    9073 * Test Compare and Load (CLO: 24) instruction : NE Op1-C!=Op2
                                                    9074 * -------------------------------------------------------------------
                                                    9075          TSTHDR 1,99,25           Op1-C != OP2 (cc=1, UR=99)
0006E800                          0006E800  0006F000 9076+         ORG   PLOETST+(200*K)+(61*4*K)
0006F000  E2E3C1D9 E3409686                         9077+         DC    CL32'START of test: 99'
0006F008  40A385A2 A37A40F9
0006F010  F9404040 40404040
0006F018  40404040 40404040
0006F020                          0006F020          9078+         USING *,R5               base for test data and test routine
0006F020                          00000000          9079+         USING PLOETEST,R5
0006F020  0006F04C                                  9080+T99      DC    A(CODE99)          address of test code
0006F024  0063                                      9081+         DC    H'99'              test number
0006F026  00                                        9082+         DC    X'00'
0006F027  19                                        9083+         DC    HL1'25'            Function
0006F028  63                                        9084+         DC    HL1'99'            expected unequal results (UR)
0006F029  00                                        9085+         DC    HL1'00'            found UR
0006F02A  01                                        9086+         DC    HL1'1'             cc
0006F02B  0B                                        9087+         DC    HL1'11'            not expected CC mask
0006F02C  00                                        9088+         DS    X                  extracted cc
0006F030  00000000 00000000                         9089+         DS    2F                 extract PSW after test (has CC)
0006F038  0006F760                                  9090+         DC    A(FMSG99)          address of test fail message
0006F03C  0006F500                                  9091+         DC    A(OPSRC99)         address of operand source
0006F040  0006F400                                  9092+         DC    A(OPDST99)         address of operand destination
0006F044  0006F600                                  9093+         DC    A(PL99)            address of parameter list
0006F048  00001214                                  9094+         DC    A(RANDX+99)        address of random source data
                                                    9095
                                                    9096          TSTCODE
0006F04C                                            9097+CODE99   DS    0F
                                                    9098 *                                 load random source
0006F04C  B982 0022                                 9099          XGR   R2,R2
0006F050  5820 5028                       00000028  9100          L     R2,RANDXA
                                                    9101 *
                                                    9102 * Compare and Load (CLO: 24)
                                                    9103 * Build Parambter List
                                                    9104 *
0006F054  B982 0011                                 9105          XGR   R1,R1
0006F058  5810 501C                       0000001C  9106          L     R1,OPSRCA          Op1-C, Op3
                                                    9107
0006F05C  B982 0033                                 9108          XGR   R3,R3
0006F060  5830 5024                       00000024  9109          L     R3,PLA
                                                    9110
0006F064  B982 0044                                 9111          XGR   R4,R4
0006F068  5840 5020                       00000020  9112          L     R4,OPREPA          OP2, Op4
                                                    9113
0006F06C  D27F 1000 2000      00000000    00000000  9114          MVC   0(128,R1),0(R2)    load Op1-C, Op3
0006F072  D21F 3000 1000      00000000    00000000  9115          MVC   0(32,R3),0(R1)     copy to PL Op1-C
0006F078  D21F 3020 1020      00000020    00000020  9116          MVC   32(32,R3),32(R1)   copy to PL Op3
                                                    9117
0006F07E  D23F 4000 2080      00000000    00000080  9118          MVC   0(64,R4),128(R2)   load initial Op2, Op4
                                                    9119
                                                    9120 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                          9121 * force a NE by changing the Op1-c
                                          9122 ****
0006F084  D203 3000 2021      00000000  00000021  9123          MVC   0(4,R3),33(R2)
                                          9124 ****
0006F08A  B982 0000                       9125          XGR   R0,R0
0006F08E  4104 0020                       9126          LA    R0,32(R4)
0006F092  E303 0048 0024                  9127          STG   R0,72(R3)           load OP4 address in PL
                                          9128
0006F098  B982 0000                       9129          XGR   R0,R0
0006F09C  4100 0018                       9130          LA    R0,24
0006F0A0  EE00 4000 3000      00000000  00000000  9131          PLO   0,0(R4),0,0(R3)     C/L (CLO: 24)
                                          9132
0006F0A6  B98D 0020                       9133          EPSW  R2,R0               extract psw
0006F0AA  5020 5010                       9134          ST    R2,CCPSW              to save CC
0006F0AE  E310 500B 0076                  0000000B  9135          LB    R1,CCMASK           (failure CC mask)
0006F0B4  8910 0004                       00000004  9136          SLL   R1,4                (shift to BC instr CC position)
0006F0B8  4410 50C4                       0006F0E4  9137          EX    R1,F241TSTCC            fail if...
                                          9138 *
                                          9139 * compare results
                                          9140 *
                          0006F0BC  00000001  9141 F241CKRES EQU   *
                                          9142 *                                    reset adresses
0006F0BC  B982 0011                       9143          XGR   R1,R1
0006F0C0  5810 501C                       0000001C  9144          L     R1,OPSRCA           Op1-C, Op3
0006F0C4  B982 0033                       9145          XGR   R3,R3
0006F0C8  5830 5024                       00000024  9146          L     R3,PLA              PL
0006F0CC  B982 0044                       9147          XGR   R4,R4
0006F0D0  5840 5020                       00000020  9148          L     R4,OPREPA           OP2, Op4
                                          9149
0006F0D4  D51F 4000 3000      00000000  00000000  9150          CLC   0(32,R4),0(R3)      Op2 == OP1-C?
0006F0DA  478B 0000                       00000000  9151          BE    0(R11)              same, return
                                          9152
                          0006F0DE  00000001  9153 F241XX   EQU   *
0006F0DE  45F0 8340                       00000540  9154          BAL   R15,FAILMSG
0006F0E2  07FB                            9155          BR    R11
                                          9156
0006F0E4  4700 50C8                       0006F0E8  9157 F241TSTCC BC   0,F241CCMSG         fail if unexpected condition code
                          0006F0E8  00000001  9158 F241CCMSG EQU  *
0006F0E8  45F0 80F2                       000002F2  9159          BAL   R15,CCMSG
0006F0EC  47F0 509C                       0006F0BC  9160          B     F241CKRES
                                          9161
                                          9162          TSTDATA
0006F0F0                  0006F0F0  0006F400  9163+          ORG   PLOETST+(200*K)+(61*4*K)+K
                                          9164          TSTLBL OPDST              OP2, Op4, Op6
0006F400                                  9165+OPDST99  DS    0C
0006F400  00000000 00000000               9166          DC    XL256'0'
0006F408  00000000 00000000
0006F410  00000000 00000000
0006F418  00000000 00000000
0006F420  00000000 00000000
0006F428  00000000 00000000
0006F430  00000000 00000000
0006F438  00000000 00000000
0006F440  00000000 00000000
0006F448  00000000 00000000
0006F450  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0006F458  00000000 00000000
0006F460  00000000 00000000
0006F468  00000000 00000000
0006F470  00000000 00000000
0006F478  00000000 00000000
0006F480  00000000 00000000
0006F488  00000000 00000000
0006F490  00000000 00000000
0006F498  00000000 00000000
0006F4A0  00000000 00000000
0006F4A8  00000000 00000000
0006F4B0  00000000 00000000
0006F4B8  00000000 00000000
0006F4C0  00000000 00000000
0006F4C8  00000000 00000000
0006F4D0  00000000 00000000
0006F4D8  00000000 00000000
0006F4E0  00000000 00000000
0006F4E8  00000000 00000000
0006F4F0  00000000 00000000
0006F4F8  00000000 00000000
                                        9167
                                        9168           TSTLBL OPSRC              Op1-C, Op3, Op5, Op1-R
0006F500                                9169+OPSRC99   DS    0C
0006F500  00000000 00000000             9170           DC    XL256'0'
0006F508  00000000 00000000
0006F510  00000000 00000000
0006F518  00000000 00000000
0006F520  00000000 00000000
0006F528  00000000 00000000
0006F530  00000000 00000000
0006F538  00000000 00000000
0006F540  00000000 00000000
0006F548  00000000 00000000
0006F550  00000000 00000000
0006F558  00000000 00000000
0006F560  00000000 00000000
0006F568  00000000 00000000
0006F570  00000000 00000000
0006F578  00000000 00000000
0006F580  00000000 00000000
0006F588  00000000 00000000
0006F590  00000000 00000000
0006F598  00000000 00000000
0006F5A0  00000000 00000000
0006F5A8  00000000 00000000
0006F5B0  00000000 00000000
0006F5B8  00000000 00000000
0006F5C0  00000000 00000000
0006F5C8  00000000 00000000
0006F5D0  00000000 00000000
0006F5D8  00000000 00000000
0006F5E0  00000000 00000000
0006F5E8  00000000 00000000
0006F5F0  00000000 00000000
0006F5F8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   9171
                                                   9172           TSTLBL PL                  Parameter List
0006F600                                           9173+PL99      DS    0C
0006F600  00000000 00000000                        9174           DC    XL352'00'
0006F608  00000000 00000000
0006F610  00000000 00000000
0006F618  00000000 00000000
0006F620  00000000 00000000
0006F628  00000000 00000000
0006F630  00000000 00000000
0006F638  00000000 00000000
0006F640  00000000 00000000
0006F648  00000000 00000000
0006F650  00000000 00000000
0006F658  00000000 00000000
0006F660  00000000 00000000
0006F668  00000000 00000000
0006F670  00000000 00000000
0006F678  00000000 00000000
0006F680  00000000 00000000
0006F688  00000000 00000000
0006F690  00000000 00000000
0006F698  00000000 00000000
0006F6A0  00000000 00000000
0006F6A8  00000000 00000000
0006F6B0  00000000 00000000
0006F6B8  00000000 00000000
0006F6C0  00000000 00000000
0006F6C8  00000000 00000000
0006F6D0  00000000 00000000
0006F6D8  00000000 00000000
0006F6E0  00000000 00000000
0006F6E8  00000000 00000000
0006F6F0  00000000 00000000
0006F6F8  00000000 00000000
0006F700  00000000 00000000
0006F708  00000000 00000000
0006F710  00000000 00000000
0006F718  00000000 00000000
0006F720  00000000 00000000
0006F728  00000000 00000000
0006F730  00000000 00000000
0006F738  00000000 00000000
0006F740  00000000 00000000
0006F748  00000000 00000000
0006F750  00000000 00000000
0006F758  00000000 00000000
                                                   9175
                                                   9176           TSTLBL FMSG                fail message
0006F760                                           9177+FMSG99    DS    0C
0006F760  86818993 858440D7                        9178           DC    C'failed PLO-Extension '
0006F768  D3D660C5 A7A38595
0006F770  A2899695 40
0006F775  C3969497 81998540                        9179           DC    C'Compare and Load (CLO: 24)'
0006F77D  81958440 D3968184
0006F785  404DC3D3 D67A40F2
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0006F78D  F45D
0006F78F  407A40D5 C540D697                        9180          DC    C' : NE Op1-C != Op2'
0006F797  F160C340 5A7E40D6
0006F79F  97F2
0006F7A1  40404040 40404040                        9181          DC    CL64' '
0006F7A9  40404040 40404040
0006F7B1  40404040 40404040
0006F7B9  40404040 40404040
0006F7C1  40404040 40404040
0006F7C9  40404040 40404040
0006F7D1  40404040 40404040
0006F7D9  40404040 40404040
                                                   9182          TSTEND
0006F7E2                                           9183+         DS    0H
0006F7E2  C5958440 968640A3                        9184+         DC    CL32'End of test: 99'
0006F7EA  85A2A37A 40F9F940
0006F7F2  40404040 40404040
0006F7FA  40404040 40404040
                                                   9185
                                                   9186 * ------------------------------------------------------------------
                                                   9187 * Test Compare and Load (CLO: 24) instruction : EQ Op1-C == OP2
                                                   9188 * ------------------------------------------------------------------
                                                   9189          TSTHDR 0,99,25            EQ Op1-C == OP2 (cc=0, UR=99)
0006F802                     0006F802  00070000    9190+         ORG   PLOETST+(200*K)+(62*4*K)
00070000  E2E3C1D9 E3409686                        9191+         DC    CL32'START of test: 100'
00070008  40A385A2 A37A40F1
00070010  F0F04040 40404040
00070018  40404040 40404040
00070020                     00070020              9192+         USING *,R5                base for test data and test routine
00070020                     00000000              9193+         USING PLOETEST,R5
00070020  0007004C                                 9194+T100     DC    A(CODE100)          address of test code
00070024  0064                                     9195+         DC    H'100'              test number
00070026  00                                       9196+         DC    X'00'
00070027  19                                       9197+         DC    HL1'25'             Function
00070028  63                                       9198+         DC    HL1'99'             expected unequal results (UR)
00070029  00                                       9199+         DC    HL1'00'             found UR
0007002A  00                                       9200+         DC    HL1'0'              cc
0007002B  07                                       9201+         DC    HL1'7'              not expected CC mask
0007002C  00                                       9202+         DS    X                   extracted cc
00070030  00000000 00000000                        9203+         DS    2F                  extract PSW after test (has CC)
00070038  00070760                                 9204+         DC    A(FMSG100)          address of test fail message
0007003C  00070500                                 9205+         DC    A(OPSRC100)         address of operand source
00070040  00070400                                 9206+         DC    A(OPDST100)         address of operand destination
00070044  00070600                                 9207+         DC    A(PL100)            address of parameter list
00070048  00001215                                 9208+         DC    A(RANDX+100)        address of random source data
                                                   9209
                                                   9210          TSTCODE
0007004C                                           9211+CODE100  DS    0F
                                                   9212 *                                  load random source
0007004C  B982 0022                                9213          XGR   R2,R2
00070050  5820 5028                  00000028      9214          L     R2,RANDXA
                                                   9215 *
                                                   9216 * Compare and Swap (DCSO: 25)
                                                   9217 * Build Parambter List
                                                   9218 *
00070054  B982 0011                                9219          XGR   R1,R1
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00070058  5810 501C                     0000001C  9220          L     R1,OPSRCA           Op1-C, Op3
                                                   9221
0007005C  B982 0033                               9222          XGR   R3,R3
00070060  5830 5024                     00000024  9223          L     R3,PLA
                                                   9224
00070064  B982 0044                               9225          XGR   R4,R4
00070068  5840 5020                     00000020  9226          L     R4,OPREPA           Op2, Op4
                                                   9227
0007006C  D27F 1000 2000  00000000 00000000 9228          MVC   0(128,R1),0(R2)     load Op1-C, Op3
00070072  D21F 3000 1000  00000000 00000000 9229          MVC   0(32,R3),0(R1)      copy to PL Op1-C
00070078  D21F 3020 1020  00000020 00000020 9230          MVC   32(32,R3),32(R1)    copy to PL Op3
                                                   9231
0007007E  D23F 4000 2080  00000000 00000080 9232          MVC   0(64,R4),128(R2)    load initial Op2, Op4
                                                   9233
                                                   9234 ****
                                                   9235 * force an EQ by changing the Op1-C
                                                   9236 ****
00070084  D21F 3000 4000  00000000 00000000 9237          MVC   0(32,R3),0(R4)      Op1-C <-- Op2
                                                   9238 ****
0007008A  B982 0000                               9239          XGR   R0,R0
0007008E  4104 0020                     00000020  9240          LA    R0,32(R4)
00070092  E303 0048 0024                00000048  9241          STG   R0,72(R3)           load OP4 address in PL
                                                   9242
00070098  B982 0000                               9243          XGR   R0,R0
0007009C  4100 0018                     00000018  9244          LA    R0,24
000700A0  EE00 4000 3000  00000000 00000000 9245          PLO   0,0(R4),0,0(R3)     C/S (CLO: 24)
                                                   9246
000700A6  B98D 0020                               9247          EPSW  R2,R0               extract psw
000700AA  5020 5010                     00000010  9248          ST    R2,CCPSW              to save CC
000700AE  E310 500B 0076                0000000B  9249          LB    R1,CCMASK           (failure CC mask)
000700B4  8910 0004                     00000004  9250          SLL   R1,4                (shift to BC instr CC position)
000700B8  4410 50C4                     000700E4  9251          EX    R1,F242TSTCC            fail if...
                                                   9252 *
                                                   9253 * compare results
                                                   9254 *
                          000700BC  00000001 9255 F242CKRES EQU   *
                                                   9256 *                            reset adresses
000700BC  B982 0011                               9257          XGR   R1,R1
000700C0  5810 501C                     0000001C  9258          L     R1,OPSRCA           Op1-C, Op3
000700C4  B982 0033                               9259          XGR   R3,R3
000700C8  5830 5024                     00000024  9260          L     R3,PLA              PL
000700CC  B982 0044                               9261          XGR   R4,R4
000700D0  5840 5020                     00000020  9262          L     R4,OPREPA           OP2, Op4
                                                   9263
000700D4  D51F 3020 4020  00000020 00000020 9264          CLC   32(32,R3),32(R4)    Op3 == OP4
000700DA  478B 0000                     00000000  9265          BE    0(R11)              same, return
                                                   9266
                          000700DE  00000001 9267 F242XX   EQU   *
000700DE  45F0 8340                     00000540  9268          BAL   R15,FAILMSG
000700E2  07FB                                    9269          BR    R11
                                                   9270
000700E4  4700 50C8                     000700E8  9271 F242TSTCC BC    0,F242CCMSG         fail if unexpected condition code
                          000700E8  00000001 9272 F242CCMSG EQU   *
000700E8  45F0 80F2                     000002F2  9273          BAL   R15,CCMSG
000700EC  47F0 509C                     000700BC  9274          B     F242CKRES
                                                   9275
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   9276           TSTDATA
000700F0                     000700F0  00070400  9277+          ORG   PLOETST+(200*K)+(62*4*K)+K
                                                   9278           TSTLBL OPDST              OP2, Op4, Op6
00070400                                           9279+OPDST100 DS    0C
00070400  00000000 00000000                        9280           DC    XL256'0'
00070408  00000000 00000000
00070410  00000000 00000000
00070418  00000000 00000000
00070420  00000000 00000000
00070428  00000000 00000000
00070430  00000000 00000000
00070438  00000000 00000000
00070440  00000000 00000000
00070448  00000000 00000000
00070450  00000000 00000000
00070458  00000000 00000000
00070460  00000000 00000000
00070468  00000000 00000000
00070470  00000000 00000000
00070478  00000000 00000000
00070480  00000000 00000000
00070488  00000000 00000000
00070490  00000000 00000000
00070498  00000000 00000000
000704A0  00000000 00000000
000704A8  00000000 00000000
000704B0  00000000 00000000
000704B8  00000000 00000000
000704C0  00000000 00000000
000704C8  00000000 00000000
000704D0  00000000 00000000
000704D8  00000000 00000000
000704E0  00000000 00000000
000704E8  00000000 00000000
000704F0  00000000 00000000
000704F8  00000000 00000000
                                                   9281
                                                   9282           TSTLBL OPSRC              Op1-C, Op3, Op5, Op1-R
00070500                                           9283+OPSRC100 DS    0C
00070500  00000000 00000000                        9284           DC    XL256'0'
00070508  00000000 00000000
00070510  00000000 00000000
00070518  00000000 00000000
00070520  00000000 00000000
00070528  00000000 00000000
00070530  00000000 00000000
00070538  00000000 00000000
00070540  00000000 00000000
00070548  00000000 00000000
00070550  00000000 00000000
00070558  00000000 00000000
00070560  00000000 00000000
00070568  00000000 00000000
00070570  00000000 00000000
00070578  00000000 00000000
00070580  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00070588  00000000 00000000
00070590  00000000 00000000
00070598  00000000 00000000
000705A0  00000000 00000000
000705A8  00000000 00000000
000705B0  00000000 00000000
000705B8  00000000 00000000
000705C0  00000000 00000000
000705C8  00000000 00000000
000705D0  00000000 00000000
000705D8  00000000 00000000
000705E0  00000000 00000000
000705E8  00000000 00000000
000705F0  00000000 00000000
000705F8  00000000 00000000
                                          9285
                                          9286          TSTLBL PL                 Parameter List
00070600                                  9287+PL100    DS    0C
00070600  00000000 00000000               9288          DC    XL352'00'
00070608  00000000 00000000
00070610  00000000 00000000
00070618  00000000 00000000
00070620  00000000 00000000
00070628  00000000 00000000
00070630  00000000 00000000
00070638  00000000 00000000
00070640  00000000 00000000
00070648  00000000 00000000
00070650  00000000 00000000
00070658  00000000 00000000
00070660  00000000 00000000
00070668  00000000 00000000
00070670  00000000 00000000
00070678  00000000 00000000
00070680  00000000 00000000
00070688  00000000 00000000
00070690  00000000 00000000
00070698  00000000 00000000
000706A0  00000000 00000000
000706A8  00000000 00000000
000706B0  00000000 00000000
000706B8  00000000 00000000
000706C0  00000000 00000000
000706C8  00000000 00000000
000706D0  00000000 00000000
000706D8  00000000 00000000
000706E0  00000000 00000000
000706E8  00000000 00000000
000706F0  00000000 00000000
000706F8  00000000 00000000
00070700  00000000 00000000
00070708  00000000 00000000
00070710  00000000 00000000
00070718  00000000 00000000
00070720  00000000 00000000
00070728  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00070730  00000000 00000000
00070738  00000000 00000000
00070740  00000000 00000000
00070748  00000000 00000000
00070750  00000000 00000000
00070758  00000000 00000000
                                        9289
                                        9290          TSTLBL FMSG               fail message
00070760                                9291+FMSG100  DS    0C
00070760  86818993 858440D7             9292          DC    C'failed PLO-Extension '
00070768  D3D660C5 A7A38595
00070770  A2899695 40
00070775  C3969497 81998540             9293          DC    C'Compare and Load (CLO: 24)'
0007077D  81958440 D3968184
00070785  404DC3D3 D67A40F2
0007078D  F45D
0007078F  407AC5D8 40D697F1             9294          DC    C' :EQ Op1-C == OP2'
00070797  60C3407E 7E40D6D7
0007079F  F2
000707A0  40404040 40404040             9295          DC    CL64' '
000707A8  40404040 40404040
000707B0  40404040 40404040
000707B8  40404040 40404040
000707C0  40404040 40404040
000707C8  40404040 40404040
000707D0  40404040 40404040
000707D8  40404040 40404040
                                        9296          TSTEND
000707E0                                9297+         DS    0H
000707E0  C5958440 968640A3             9298+         DC    CL32'End of test: 100'
000707E8  85A2A37A 40F1F0F0
000707F0  40404040 40404040
000707F8  40404040 40404040
                                        9299
                                        9300 ********************************************************************************
                                        9301 * table of pointers to individual tests
                                        9302 ********************************************************************************
                                        9303
00070800                                9304 PLOTESTS  DS    0F
                                        9305          PTTABLE
00070800                                9306+TTABLE   DS    0F
00070800  000015B4                      9307+         DC    A(T1)
00070804  000015D4                      9308+         DC    A(T2)
00070808  000015F4                      9309+         DC    A(T3)
0007080C  00001614                      9310+         DC    A(T4)
00070810  00001634                      9311+         DC    A(T5)
00070814  00001654                      9312+         DC    A(T6)
00070818  00001674                      9313+         DC    A(T7)
0007081C  00001694                      9314+         DC    A(T8)
00070820  000016B4                      9315+         DC    A(T9)
00070824  000016D4                      9316+         DC    A(T10)
00070828  000016F4                      9317+         DC    A(T11)
0007082C  00001714                      9318+         DC    A(T12)
00070830  00001734                      9319+         DC    A(T13)
00070834  00001754                      9320+         DC    A(T14)
00070838  00001774                      9321+         DC    A(T15)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0007083C  00001794                          9322+          DC    A(T16)
00070840  000017B4                          9323+          DC    A(T17)
00070844  000017D4                          9324+          DC    A(T18)
00070848  000017F4                          9325+          DC    A(T19)
0007084C  00001814                          9326+          DC    A(T20)
00070850  00001834                          9327+          DC    A(T21)
00070854  00001854                          9328+          DC    A(T22)
00070858  00001874                          9329+          DC    A(T23)
0007085C  00001894                          9330+          DC    A(T24)
00070860  000018B4                          9331+          DC    A(T25)
00070864  000018D4                          9332+          DC    A(T26)
00070868  000018F4                          9333+          DC    A(T27)
0007086C  00001914                          9334+          DC    A(T28)
00070870  00001934                          9335+          DC    A(T29)
00070874  00001954                          9336+          DC    A(T30)
00070878  00001974                          9337+          DC    A(T31)
0007087C  00001994                          9338+          DC    A(T32)
00070880  000019B4                          9339+          DC    A(T33)
00070884  000019D4                          9340+          DC    A(T34)
00070888  000019F4                          9341+          DC    A(T35)
0007088C  00001A14                          9342+          DC    A(T36)
00070890  00001A34                          9343+          DC    A(T37)
00070894  00001A54                          9344+          DC    A(T38)
00070898  00033020                          9345+          DC    A(T39)
0007089C  00034020                          9346+          DC    A(T40)
000708A0  00035020                          9347+          DC    A(T41)
000708A4  00036020                          9348+          DC    A(T42)
000708A8  00037020                          9349+          DC    A(T43)
000708AC  00038020                          9350+          DC    A(T44)
000708B0  00039020                          9351+          DC    A(T45)
000708B4  0003A020                          9352+          DC    A(T46)
000708B8  0003B020                          9353+          DC    A(T47)
000708BC  0003C020                          9354+          DC    A(T48)
000708C0  0003D020                          9355+          DC    A(T49)
000708C4  0003E020                          9356+          DC    A(T50)
000708C8  0003F020                          9357+          DC    A(T51)
000708CC  00040020                          9358+          DC    A(T52)
000708D0  00041020                          9359+          DC    A(T53)
000708D4  00042020                          9360+          DC    A(T54)
000708D8  00043020                          9361+          DC    A(T55)
000708DC  00044020                          9362+          DC    A(T56)
000708E0  00045020                          9363+          DC    A(T57)
000708E4  00046020                          9364+          DC    A(T58)
000708E8  00047020                          9365+          DC    A(T59)
000708EC  00048020                          9366+          DC    A(T60)
000708F0  00049020                          9367+          DC    A(T61)
000708F4  0004A020                          9368+          DC    A(T62)
000708F8  0004B020                          9369+          DC    A(T63)
000708FC  0004C020                          9370+          DC    A(T64)
00070900  0004D020                          9371+          DC    A(T65)
00070904  0004E020                          9372+          DC    A(T66)
00070908  0004F020                          9373+          DC    A(T67)
0007090C  00050020                          9374+          DC    A(T68)
00070910  00051020                          9375+          DC    A(T69)
00070914  00052020                          9376+          DC    A(T70)
00070918  00053020                          9377+          DC    A(T71)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0007091C  00054020                                9378+          DC    A(T72)
00070920  00055020                                9379+          DC    A(T73)
00070924  00056020                                9380+          DC    A(T74)
00070928  00057020                                9381+          DC    A(T75)
0007092C  00058020                                9382+          DC    A(T76)
00070930  00059020                                9383+          DC    A(T77)
00070934  0005A020                                9384+          DC    A(T78)
00070938  0005B020                                9385+          DC    A(T79)
0007093C  0005C020                                9386+          DC    A(T80)
00070940  0005D020                                9387+          DC    A(T81)
00070944  0005E020                                9388+          DC    A(T82)
00070948  0005F020                                9389+          DC    A(T83)
0007094C  00060020                                9390+          DC    A(T84)
00070950  00061020                                9391+          DC    A(T85)
00070954  00062020                                9392+          DC    A(T86)
00070958  00063020                                9393+          DC    A(T87)
0007095C  00064020                                9394+          DC    A(T88)
00070960  00065020                                9395+          DC    A(T89)
00070964  00066020                                9396+          DC    A(T90)
00070968  00067020                                9397+          DC    A(T91)
0007096C  00068020                                9398+          DC    A(T92)
00070970  00069020                                9399+          DC    A(T93)
00070974  0006A020                                9400+          DC    A(T94)
00070978  0006B020                                9401+          DC    A(T95)
0007097C  0006C020                                9402+          DC    A(T96)
00070980  0006D020                                9403+          DC    A(T97)
00070984  0006E020                                9404+          DC    A(T98)
00070988  0006F020                                9405+          DC    A(T99)
0007098C  00070020                                9406+          DC    A(T100)
                                                  9407+*
00070990  00000000                                9408+          DC    A(0)              END OF TABLE
00070994  00000000                                9409+          DC    A(0)
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                        9411 ********************************************************************************
                                        9412 *        Register equates
                                        9413 ********************************************************************************


                             00000000  00000001  9415 R0       EQU   0
                             00000001  00000001  9416 R1       EQU   1
                             00000002  00000001  9417 R2       EQU   2
                             00000003  00000001  9418 R3       EQU   3
                             00000004  00000001  9419 R4       EQU   4
                             00000005  00000001  9420 R5       EQU   5
                             00000006  00000001  9421 R6       EQU   6
                             00000007  00000001  9422 R7       EQU   7
                             00000008  00000001  9423 R8       EQU   8
                             00000009  00000001  9424 R9       EQU   9
                             0000000A  00000001  9425 R10      EQU   10
                             0000000B  00000001  9426 R11      EQU   11
                             0000000C  00000001  9427 R12      EQU   12
                             0000000D  00000001  9428 R13      EQU   13
                             0000000E  00000001  9429 R14      EQU   14
                             0000000F  00000001  9430 R15      EQU   15


                                        9432 ********************************************************************************
                                        9433 *        Register equates
                                        9434 ********************************************************************************


                             00000000  00000001  9436 V0       EQU   0
                             00000001  00000001  9437 V1       EQU   1
                             00000002  00000001  9438 V2       EQU   2
                             00000003  00000001  9439 V3       EQU   3
                             00000004  00000001  9440 V4       EQU   4
                             00000005  00000001  9441 V5       EQU   5
                             00000006  00000001  9442 V6       EQU   6
                             00000007  00000001  9443 V7       EQU   7
                             00000008  00000001  9444 V8       EQU   8
                             00000009  00000001  9445 V9       EQU   9
                             0000000A  00000001  9446 V10      EQU   10
                             0000000B  00000001  9447 V11      EQU   11
                             0000000C  00000001  9448 V12      EQU   12
                             0000000D  00000001  9449 V13      EQU   13
                             0000000E  00000001  9450 V14      EQU   14
                             0000000F  00000001  9451 V15      EQU   15
                             00000010  00000001  9452 V16      EQU   16
                             00000011  00000001  9453 V17      EQU   17
                             00000012  00000001  9454 V18      EQU   18
                             00000013  00000001  9455 V19      EQU   19
                             00000014  00000001  9456 V20      EQU   20
                             00000015  00000001  9457 V21      EQU   21
```

```
 LOC         OBJECT CODE       ADDR1     ADDR2     STMT
                             00000016  00000001  9458 V22       EQU   22
                             00000017  00000001  9459 V23       EQU   23
                             00000018  00000001  9460 V24       EQU   24
                             00000019  00000001  9461 V25       EQU   25
                             0000001A  00000001  9462 V26       EQU   26
                             0000001B  00000001  9463 V27       EQU   27
                             0000001C  00000001  9464 V28       EQU   28
                             0000001D  00000001  9465 V29       EQU   29
                             0000001E  00000001  9466 V30       EQU   30
                             0000001F  00000001  9467 V31       EQU   31
                                                 9468
                                                 9469           END
```

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
AFUN	U	000007	1	620	397
ATNUM	H	000004	2	618	390
ATSUB	A	000000	4	617	
BEGIN	I	000200	2	146	117 143 144
CC	U	00000A	1	599	228 257
CCFOUND	X	00000C	1	602	239 264
CCMASK	U	00000B	1	600	3107 3238 3369 3500 3632 3769 3900 4031 4162 4294 4431 4562 4693 4824 4955 5099 5237 5375 5513 5652 5780 5906 6032 6158 6284 6410 6536 6662 6789 6915 7041 7167 7299 7430 7561 7692 7823 7945 8074 8192 8315 8432 8554 8670 8789 8907 9017 9135 9249
CCMSG	U	0002F2	1	224	3143 3274 3405 3536 3670 3805 3936 4067 4198 4332 4467 4598 4729 4860 4993 5135 5273 5411 5549 5684 5814 5940 6066 6188 6318 6444 6570 6692 6823 6949 7075 7197 7333 7464 7595 7722 7847 7975 8098 8220 8339 8458 8578 8696 8815 8931 9041 9159 9273
CCPRTEXP	C	0010CE	1	535	261
CCPRTFUN	C	0010BD	2	532	254
CCPRTGOT	C	0010DE	1	538	268
CCPRTLINE	C	001081	16	528	540 273
CCPRTLNG	U	00005F	1	540	272
CCPRTNUM	C	001091	3	529	247
CCPSW	F	000010	4	603	236 3106 3237 3368 3499 3631 3768 3899 4030 4161 4293 4430 4561 4692 4823 4954 5098 5236 5374 5512 5651 5779 5905 6031 6157 6283 6409 6535 6661 6788 6914 7040 7166 7298 7429 7560 7691 7822 7944 8073 8191 8314 8431 8553 8669 8788 8906 9016 9134 9248
CODE1	F	0015BC	4	871	866
CODE10	F	0016DC	4	1051	1046
CODE100	F	07004C	4	9211	9194
CODE11	F	0016FC	4	1071	1066
CODE12	F	00171C	4	1091	1086
CODE13	F	00173C	4	1111	1106
CODE14	F	00175C	4	1131	1126
CODE15	F	00177C	4	1151	1146
CODE16	F	00179C	4	1171	1166
CODE17	F	0017BC	4	1191	1186
CODE18	F	0017DC	4	1211	1206
CODE19	F	0017FC	4	1231	1226
CODE2	F	0015DC	4	891	886
CODE20	F	00181C	4	1251	1246
CODE21	F	00183C	4	1271	1266
CODE22	F	00185C	4	1291	1286
CODE23	F	00187C	4	1311	1306
CODE24	F	00189C	4	1331	1326
CODE25	F	0018BC	4	1351	1346
CODE26	F	0018DC	4	1371	1366
CODE27	F	0018FC	4	1391	1386
CODE28	F	00191C	4	1411	1406
CODE29	F	00193C	4	1431	1426
CODE3	F	0015FC	4	911	906
CODE30	F	00195C	4	1451	1446
CODE31	F	00197C	4	1471	1466
CODE32	F	00199C	4	1491	1486
CODE33	F	0019BC	4	1511	1506
CODE34	F	0019DC	4	1531	1526
CODE35	F	0019FC	4	1551	1546
CODE36	F	001A1C	4	1571	1566
CODE37	F	001A3C	4	1591	1586

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
CODE38	F	001A5C	4	1611	1606
CODE39	F	03304C	4	1650	1633
CODE4	F	00161C	4	931	926
CODE40	F	03404C	4	1766	1749
CODE41	F	03504C	4	1882	1865
CODE42	F	03604C	4	1998	1981
CODE43	F	03704C	4	2118	2101
CODE44	F	03804C	4	2226	2209
CODE45	F	03904C	4	2334	2317
CODE46	F	03A04C	4	2442	2425
CODE47	F	03B04C	4	2554	2537
CODE48	F	03C04C	4	2654	2637
CODE49	F	03D04C	4	2764	2747
CODE5	F	00163C	4	951	946
CODE50	F	03E04C	4	2864	2847
CODE51	F	03F04C	4	2967	2950
CODE52	F	04004C	4	3063	3046
CODE53	F	04104C	4	3194	3177
CODE54	F	04204C	4	3325	3308
CODE55	F	04304C	4	3456	3439
CODE56	F	04404C	4	3588	3571
CODE57	F	04504C	4	3725	3708
CODE58	F	04604C	4	3856	3839
CODE59	F	04704C	4	3987	3970
CODE6	F	00165C	4	971	966
CODE60	F	04804C	4	4118	4101
CODE61	F	04904C	4	4250	4233
CODE62	F	04A04C	4	4387	4370
CODE63	F	04B04C	4	4518	4501
CODE64	F	04C04C	4	4649	4632
CODE65	F	04D04C	4	4780	4763
CODE66	F	04E04C	4	4911	4894
CODE67	F	04F04C	4	5048	5031
CODE68	F	05004C	4	5186	5169
CODE69	F	05104C	4	5324	5307
CODE7	F	00167C	4	991	986
CODE70	F	05204C	4	5462	5445
CODE71	F	05304C	4	5601	5584
CODE72	F	05404C	4	5739	5722
CODE73	F	05504C	4	5865	5848
CODE74	F	05604C	4	5991	5974
CODE75	F	05704C	4	6117	6100
CODE76	F	05804C	4	6243	6226
CODE77	F	05904C	4	6369	6352
CODE78	F	05A04C	4	6495	6478
CODE79	F	05B04C	4	6621	6604
CODE8	F	00169C	4	1011	1006
CODE80	F	05C04C	4	6748	6731
CODE81	F	05D04C	4	6874	6857
CODE82	F	05E04C	4	7000	6983
CODE83	F	05F04C	4	7126	7109
CODE84	F	06004C	4	7252	7235
CODE85	F	06104C	4	7383	7366
CODE86	F	06204C	4	7514	7497
CODE87	F	06304C	4	7645	7628
CODE88	F	06404C	4	7776	7759

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
CODE89	F	06504C	4	7898	7881
CODE9	F	0016BC	4	1031	1026
CODE90	F	06604C	4	8031	8014
CODE91	F	06704C	4	8149	8132
CODE92	F	06804C	4	8276	8259
CODE93	F	06904C	4	8390	8373
CODE94	F	06A04C	4	8514	8497
CODE95	F	06B04C	4	8630	8613
CODE96	F	06C04C	4	8748	8731
CODE97	F	06D04C	4	8873	8856
CODE98	F	06E04C	4	8983	8966
CODE99	F	06F04C	4	9097	9080
CTLR0	F	000864	4	486	153 154 155 156
DECNUM	C	0011A1	16	585	244 246 251 253 258 260 265 267 304 306 311 313 318 320
					325 327 354 356 391 393 398 400
EDIT	X	001175	18	580	245 252 259 266 305 312 319 326 355 392 399
ENDTEST	U	0002E4	1	213	200
EOJ	I	0007B8	4	468	188 216
EOJPSW	D	0007A8	8	466	468
F241CCMSG	U	06F0E8	1	9158	9157
F241CKRES	U	06F0BC	1	9141	9160
F241TSTCC	I	06F0E4	4	9157	9137
F241XX	U	06F0DE	1	9153	
F242CCMSG	U	0700E8	1	9272	9271
F242CKRES	U	0700BC	1	9255	9274
F242TSTCC	I	0700E4	4	9271	9251
F242XX	U	0700DE	1	9267	
F251CCMSG	U	06D0DA	1	8930	8929
F251CKRES	U	06D0AE	1	8913	8932
F251TSTCC	I	06D0D6	4	8929	8909
F251XX	U	06D0D0	1	8925	
F252CCMSG	U	06E0DA	1	9040	9039
F252CKRES	U	06E0AE	1	9023	9042
F252TSTCC	I	06E0D6	4	9039	9019
F252XX	U	06E0D0	1	9035	
F261CCMSG	U	06A0F4	1	8577	8576
F261CKRES	U	06A0C8	1	8560	8579
F261TSTCC	I	06A0F0	4	8576	8556
F261XX	U	06A0EA	1	8572	
F262CCMSG	U	06B0FE	1	8695	8694
F262CKRES	U	06B0C8	1	8676	8697
F262TSTCC	I	06B0FA	4	8694	8672
F262XX	U	06B0F4	1	8690	8686
F263CCMSG	U	06C104	1	8814	8813
F263CKRES	U	06C0CE	1	8795	8816
F263TSTCC	I	06C100	4	8813	8791
F263XX	U	06C0FA	1	8809	8805
F271CCMSG	U	0680EE	1	8338	8337
F271CKRES	U	0680C2	1	8321	8340
F271TSTCC	I	0680EA	4	8337	8317
F271XX	U	0680E4	1	8333	
F272CCMSG	U	069102	1	8457	8456
F272CKRES	U	0690CC	1	8438	8459
F272TSTCC	I	0690FE	4	8456	8434
F272XX	U	0690F8	1	8452	8448
F281CCMSG	U	0660FE	1	8097	8096

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES		
F281CKRES	U	0660D2	1	8080	8099		
F281TSTCC	I	0660FA	4	8096	8076		
F281XX	U	0660F4	1	8092			
F282CCMSG	U	067112	1	8219	8218		
F282CKRES	U	0670D2	1	8198	8221		
F282TSTCC	I	06710E	4	8218	8194		
F282XX	U	067108	1	8214	8208	8210	
F291CCMSG	U	06410E	1	7846	7845		
F291CKRES	U	0640E2	1	7829	7848		
F291TSTCC	I	06410A	4	7845	7825		
F291XX	U	064104	1	7841			
F292CCMSG	U	06512C	1	7974	7973		
F292CKRES	U	0650E2	1	7951	7976		
F292TSTCC	I	065128	4	7973	7947		
F292XX	U	065122	1	7969	7961	7963	7965
F301CCMSG	U	060134	1	7332	7331		
F301CKRES	U	0600F4	1	7311	7306		
F301CKUR	U	0600E0	1	7303	7334		
F301TSTCC	I	060130	4	7331	7301		
F301XX	U	06012A	1	7327	7321	7323	
F302CCMSG	U	061134	1	7463	7462		
F302CKRES	U	0610F4	1	7442	7437		
F302CKUR	U	0610E0	1	7434	7465		
F302TSTCC	I	061130	4	7462	7432		
F302XX	U	06112A	1	7458	7452	7454	
F303CCMSG	U	062134	1	7594	7593		
F303CKRES	U	0620F4	1	7573	7568		
F303CKUR	U	0620E0	1	7565	7596		
F303TSTCC	I	062130	4	7593	7563		
F303XX	U	06212A	1	7589	7583	7585	
F304CCMSG	U	06311A	1	7721	7720		
F304CKRES	U	0630DA	1	7698	7723		
F304TSTCC	I	063116	4	7720	7694		
F304XX	U	063110	1	7716	7710	7712	
F311CCMSG	U	05C11C	1	6822	6821		
F311CKRES	U	05C0DC	1	6801	6796		
F311CKUR	U	05C0C8	1	6793	6824		
F311TSTCC	I	05C118	4	6821	6791		
F311XX	U	05C112	1	6817	6811	6813	
F312CCMSG	U	05D11C	1	6948	6947		
F312CKRES	U	05D0DC	1	6927	6922		
F312CKUR	U	05D0C8	1	6919	6950		
F312TSTCC	I	05D118	4	6947	6917		
F312XX	U	05D112	1	6943	6937	6939	
F313CCMSG	U	05E11C	1	7074	7073		
F313CKRES	U	05E0DC	1	7053	7048		
F313CKUR	U	05E0C8	1	7045	7076		
F313TSTCC	I	05E118	4	7073	7043		
F313XX	U	05E112	1	7069	7063	7065	
F314CCMSG	U	05F102	1	7196	7195		
F314CKRES	U	05F0C2	1	7173	7198		
F314TSTCC	I	05F0FE	4	7195	7169		
F314XX	U	05F0F8	1	7191	7185	7187	
F321CCMSG	U	05811C	1	6317	6316		
F321CKRES	U	0580DC	1	6296	6291		
F321CKUR	U	0580C8	1	6288	6319		

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES		
F321TSTCC	I	058118	4	6316	6286		
F321XX	U	058112	1	6312	6306	6308	
F322CCMSG	U	05911C	1	6443	6442		
F322CKRES	U	0590DC	1	6422	6417		
F322CKUR	U	0590C8	1	6414	6445		
F322TSTCC	I	059118	4	6442	6412		
F322XX	U	059112	1	6438	6432	6434	
F323CCMSG	U	05A11C	1	6569	6568		
F323CKRES	U	05A0DC	1	6548	6543		
F323CKUR	U	05A0C8	1	6540	6571		
F323TSTCC	I	05A118	4	6568	6538		
F323XX	U	05A112	1	6564	6558	6560	
F324CCMSG	U	05B102	1	6691	6690		
F324CKRES	U	05B0C2	1	6668	6693		
F324TSTCC	I	05B0FE	4	6690	6664		
F324XX	U	05B0F8	1	6686	6680	6682	
F331CCMSG	U	05411C	1	5813	5812		
F331CKRES	U	0540DC	1	5792	5787		
F331CKUR	U	0540C8	1	5784	5815		
F331TSTCC	I	054118	4	5812	5782		
F331XX	U	054112	1	5808	5802	5804	
F332CCMSG	U	05511C	1	5939	5938		
F332CKRES	U	0550DC	1	5918	5913		
F332CKUR	U	0550C8	1	5910	5941		
F332TSTCC	I	055118	4	5938	5908		
F332XX	U	055112	1	5934	5928	5930	
F333CCMSG	U	05611C	1	6065	6064		
F333CKRES	U	0560DC	1	6044	6039		
F333CKUR	U	0560C8	1	6036	6067		
F333TSTCC	I	056118	4	6064	6034		
F333XX	U	056112	1	6060	6054	6056	
F334CCMSG	U	057102	1	6187	6186		
F334CKRES	U	0570C2	1	6164	6189		
F334TSTCC	I	0570FE	4	6186	6160		
F334XX	U	0570F8	1	6182	6176	6178	
F341CCMSG	U	04F154	1	5134	5133		
F341CKRES	U	04F10A	1	5111	5106		
F341CKUR	U	04F0F6	1	5103	5136		
F341TSTCC	I	04F150	4	5133	5101		
F341XX	U	04F14A	1	5129	5121	5123	5125
F342CCMSG	U	050154	1	5272	5271		
F342CKRES	U	05010A	1	5249	5244		
F342CKUR	U	0500F6	1	5241	5274		
F342TSTCC	I	050150	4	5271	5239		
F342XX	U	05014A	1	5267	5259	5261	5263
F343CCMSG	U	051154	1	5410	5409		
F343CKRES	U	05110A	1	5387	5382		
F343CKUR	U	0510F6	1	5379	5412		
F343TSTCC	I	051150	4	5409	5377		
F343XX	U	05114A	1	5405	5397	5399	5401
F344CCMSG	U	052154	1	5548	5547		
F344CKRES	U	05210A	1	5525	5520		
F344CKUR	U	0520F6	1	5517	5550		
F344TSTCC	I	052150	4	5547	5515		
F344XX	U	05214A	1	5543	5535	5537	5539
F345CCMSG	U	05313A	1	5683	5682		

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES		
F345CKRES	U	0530F0	1	5658	5685		
F345TSTCC	I	053136	4	5682	5654		
F345XX	U	053130	1	5678	5670	5672	5674
F351CCMSG	U	04A130	1	4466	4465		
F351CKRES	U	04A0E6	1	4443	4438		
F351CKUR	U	04A0D2	1	4435	4468		
F351TSTCC	I	04A12C	4	4465	4433		
F351XX	U	04A126	1	4461	4453	4455	4457
F352CCMSG	U	04B130	1	4597	4596		
F352CKRES	U	04B0E6	1	4574	4569		
F352CKUR	U	04B0D2	1	4566	4599		
F352TSTCC	I	04B12C	4	4596	4564		
F352XX	U	04B126	1	4592	4584	4586	4588
F353CCMSG	U	04C130	1	4728	4727		
F353CKRES	U	04C0E6	1	4705	4700		
F353CKUR	U	04C0D2	1	4697	4730		
F353TSTCC	I	04C12C	4	4727	4695		
F353XX	U	04C126	1	4723	4715	4717	4719
F354CCMSG	U	04D130	1	4859	4858		
F354CKRES	U	04D0E6	1	4836	4831		
F354CKUR	U	04D0D2	1	4828	4861		
F354TSTCC	I	04D12C	4	4858	4826		
F354XX	U	04D126	1	4854	4846	4848	4850
F355CCMSG	U	04E12A	1	4992	4991		
F355CKRES	U	04E0E0	1	4967	4962		
F355CKUR	U	04E0CC	1	4959	4994		
F355TSTCC	I	04E126	4	4991	4957		
F355XX	U	04E120	1	4987	4979	4981	4983
F361CCMSG	U	045130	1	3804	3803		
F361CKRES	U	0450E6	1	3781	3776		
F361CKUR	U	0450D2	1	3773	3806		
F361TSTCC	I	04512C	4	3803	3771		
F361XX	U	045126	1	3799	3791	3793	3795
F362CCMSG	U	046130	1	3935	3934		
F362CKRES	U	0460E6	1	3912	3907		
F362CKUR	U	0460D2	1	3904	3937		
F362TSTCC	I	04612C	4	3934	3902		
F362XX	U	046126	1	3930	3922	3924	3926
F363CCMSG	U	047130	1	4066	4065		
F363CKRES	U	0470E6	1	4043	4038		
F363CKUR	U	0470D2	1	4035	4068		
F363TSTCC	I	04712C	4	4065	4033		
F363XX	U	047126	1	4061	4053	4055	4057
F364CCMSG	U	048130	1	4197	4196		
F364CKRES	U	0480E6	1	4174	4169		
F364CKUR	U	0480D2	1	4166	4199		
F364TSTCC	I	04812C	4	4196	4164		
F365CCMSG	U	04912A	1	4331	4330		
F365CKRES	U	0490E0	1	4306	4301		
F365CKUR	U	0490CC	1	4298	4333		
F365TSTCC	I	049126	4	4330	4296		
F371CCMSG	U	040130	1	3142	3141		
F371CKRES	U	0400E6	1	3119	3114		
F371CKUR	U	0400D2	1	3111	3144		
F371TSTCC	I	04012C	4	3141	3109		
F372CCMSG	U	041130	1	3273	3272		

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES												
F372CKRES	U	0410E6	1	3250	3245												
F372CKUR	U	0410D2	1	3242	3275												
F372TSTCC	I	04112C	4	3272	3240												
F373CCMSG	U	042130	1	3404	3403												
F373CKRES	U	0420E6	1	3381	3376												
F373CKUR	U	0420D2	1	3373	3406												
F373TSTCC	I	04212C	4	3403	3371												
F373XX	U	042126	1	3399	3391	3393	3395										
F374CCMSG	U	043130	1	3535	3534												
F374CKRES	U	0430E6	1	3512	3507												
F374CKUR	U	0430D2	1	3504	3537												
F374TSTCC	I	04312C	4	3534	3502												
F374XX	U	043126	1	3530	3522	3524	3526										
F375CCMSG	U	04412A	1	3669	3668												
F375CKRES	U	0440E0	1	3644	3639												
F375CKUR	U	0440CC	1	3636	3671												
F375TSTCC	I	044126	4	3668	3634												
F375XX	U	044120	1	3664	3656	3658	3660										
F4053	U	03B0DA	1	2602	2598												
F4154	U	03A0F8	1	2498	2492	2494											
F4255	U	0390F8	1	2390	2384	2386											
F4352	U	03C0EE	1	2708	2704												
F4356	U	0380F8	1	2282	2276	2278											
F4457	U	0370F8	1	2174	2168	2170											
F4558	U	036116	1	2062	2054	2056	2058										
F4659	U	035116	1	1946	1938	1940	1942										
F4760	U	034116	1	1830	1822	1824	1826										
F4861	U	033116	1	1714	1706	1708	1710										
FAILED	F	001000	4	504	214	232	299	351	386								
FAILHLNG	U	000015	1	516													
FAILLINE	C	001008	16	513	516	519	368										
FAILLNG	U	000079	1	519	367												
FAILMSG	U	000540	1	349	1715	1831	1947	2063	2175	2283	2391	2499	2603	2709	2813	2911	3007	3138
					3269	3400	3531	3665	3800	3931	4062	4193	4327	4462	4593	4724	4855	4988
					5130	5268	5406	5544	5679	5809	5935	6061	6183	6313	6439	6565	6687	6818
					6944	7070	7192	7328	7459	7590	7717	7842	7970	8093	8215	8334	8453	8573
					8691	8810	8926	9036	9154	9268								
FAILPSW	D	0007C0	8	470	472												
FAILTEST	I	0007D0	4	472	217												
FAILTNUM	C	001018	3	514	357												
FAILTXT	C	00101D	100	518	359	362											
FAILTXTA	A	000018	4	605	361												
FB0001	F	000278	8	172	176	177	179										
FCCSAV	D	0003A0	8	282	275	279											
FLSAV	D	000590	8	377	370	374											
FMSG100	C	070760	1	9291	9204												
FMSG39	C	033600	1	1733	1643												
FMSG40	C	034600	1	1849	1759												
FMSG41	C	035600	1	1965	1875												
FMSG42	C	036600	1	2081	1991												
FMSG43	C	037600	1	2193	2111												
FMSG44	C	038600	1	2301	2219												
FMSG45	C	039600	1	2409	2327												
FMSG46	C	03A600	1	2517	2435												
FMSG47	C	03B600	1	2621	2547												
FMSG48	C	03C600	1	2727	2647												

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
FMSG49	C	03D600	1	2831	2757
FMSG50	C	03E600	1	2929	2857
FMSG51	C	03F600	1	3025	2960
FMSG52	C	040760	1	3161	3056
FMSG53	C	041760	1	3292	3187
FMSG54	C	042760	1	3423	3318
FMSG55	C	043760	1	3554	3449
FMSG56	C	044760	1	3688	3581
FMSG57	C	045760	1	3823	3718
FMSG58	C	046760	1	3954	3849
FMSG59	C	047760	1	4085	3980
FMSG60	C	048760	1	4216	4111
FMSG61	C	049760	1	4350	4243
FMSG62	C	04A760	1	4485	4380
FMSG63	C	04B760	1	4616	4511
FMSG64	C	04C760	1	4747	4642
FMSG65	C	04D760	1	4878	4773
FMSG66	C	04E760	1	5011	4904
FMSG67	C	04F760	1	5153	5041
FMSG68	C	050760	1	5291	5179
FMSG69	C	051760	1	5429	5317
FMSG70	C	052760	1	5567	5455
FMSG71	C	053760	1	5702	5594
FMSG72	C	054760	1	5832	5732
FMSG73	C	055760	1	5958	5858
FMSG74	C	056760	1	6084	5984
FMSG75	C	057760	1	6206	6110
FMSG76	C	058760	1	6336	6236
FMSG77	C	059760	1	6462	6362
FMSG78	C	05A760	1	6588	6488
FMSG79	C	05B760	1	6710	6614
FMSG80	C	05C760	1	6841	6741
FMSG81	C	05D760	1	6967	6867
FMSG82	C	05E760	1	7093	6993
FMSG83	C	05F760	1	7215	7119
FMSG84	C	060760	1	7351	7245
FMSG85	C	061760	1	7482	7376
FMSG86	C	062760	1	7613	7507
FMSG87	C	063760	1	7740	7638
FMSG88	C	064760	1	7865	7769
FMSG89	C	065760	1	7993	7891
FMSG90	C	066760	1	8116	8024
FMSG91	C	067760	1	8238	8142
FMSG92	C	068760	1	8357	8269
FMSG93	C	069760	1	8476	8383
FMSG94	C	06A760	1	8596	8507
FMSG95	C	06B760	1	8714	8623
FMSG96	C	06C760	1	8833	8741
FMSG97	C	06D760	1	8949	8866
FMSG98	C	06E760	1	9059	8976
FMSG99	C	06F760	1	9177	9090
FUN	U	000007	1	595	250 310
FXX137	U	040126	1	3137	3129 3131 3133
FXX237	U	041126	1	3268	3260 3262 3264
FXX437	U	048126	1	4192	4184 4186 4188
FXX50	U	03E0CC	1	2910	2906

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
FXX51	U	03D0D4	1	2812	2808
FXX537	U	049120	1	4326	4318 4320 4322
IMAGE	1	000000	461208	0	
K	U	000400	1	491	492 493 494 1629 1719 1745 1835 1861 1951 1977 2067 2097 2179 2205 2287 2313 2395 2421 2503 2533 2607 2633 2713 2743 2817 2843 2915 2946 3011 3042 3147 3173 3278 3304 3409 3435 3540 3567 3674 3704 3809 3835 3940 3966 4071 4097 4202 4229 4336 4366 4471 4497 4602 4628 4733 4759 4864 4890 4997 5027 5139 5165 5277 5303 5415 5441 5553 5580 5688 5718 5818 5844 5944 5970 6070 6096 6192 6222 6322 6348 6448 6474 6574 6600 6696 6727 6827 6853 6953 6979 7079 7105 7201 7231 7337 7362 7468 7493 7599 7624 7726 7755 7851 7877 7979 8010 8102 8128 8224 8255 8343 8369 8462 8493 8582 8609 8700 8727 8819 8852 8935 8962 9045 9076 9163 9190 9277
K64	U	010000	1	493	
MB	U	100000	1	494	
MSG	I	0006F0	4	426	187 277 337 372 412
MSGCMD	C	00073E	9	457	439 440
MSGMSG	C	000747	95	458	433 455 431
MSGMVC	I	000738	6	455	437
MSGOK	I	000706	2	435	432
MSGRET	I	000726	4	450	443 446
MSGSAVE	F	00072C	4	453	429 450
NACCSAV	D	000670	8	417	410 414
NAMSG	U	000610	1	384	876 896 916 936 956 976 996 1016 1036 1056 1076 1096 1116 1136 1156 1176 1196 1216 1236 1256 1276 1296 1316 1336 1356 1376 1396 1416 1436 1456 1476 1496 1516 1536 1556 1576 1596 1616
NAPRTFUN	C	001161	2	571	401
NAPRTLINE	C	00113F	16	568	574 408
NAPRTLNG	U	000036	1	574	407
NAPRTNUM	C	00114F	3	569	394
NEXTTST	U	0002CC	1	197	208
OPDST100	C	070400	1	9279	9206
OPDST39	C	033400	1	1721	1645
OPDST40	C	034400	1	1837	1761
OPDST41	C	035400	1	1953	1877
OPDST42	C	036400	1	2069	1993
OPDST43	C	037400	1	2181	2113
OPDST44	C	038400	1	2289	2221
OPDST45	C	039400	1	2397	2329
OPDST46	C	03A400	1	2505	2437
OPDST47	C	03B400	1	2609	2549
OPDST48	C	03C400	1	2715	2649
OPDST49	C	03D400	1	2819	2759
OPDST50	C	03E400	1	2917	2859
OPDST51	C	03F400	1	3013	2962
OPDST52	C	040400	1	3149	3058
OPDST53	C	041400	1	3280	3189
OPDST54	C	042400	1	3411	3320
OPDST55	C	043400	1	3542	3451
OPDST56	C	044400	1	3676	3583
OPDST57	C	045400	1	3811	3720
OPDST58	C	046400	1	3942	3851
OPDST59	C	047400	1	4073	3982
OPDST60	C	048400	1	4204	4113
OPDST61	C	049400	1	4338	4245
OPDST62	C	04A400	1	4473	4382

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
OPDST63	C	04B400	1	4604	4513
OPDST64	C	04C400	1	4735	4644
OPDST65	C	04D400	1	4866	4775
OPDST66	C	04E400	1	4999	4906
OPDST67	C	04F400	1	5141	5043
OPDST68	C	050400	1	5279	5181
OPDST69	C	051400	1	5417	5319
OPDST70	C	052400	1	5555	5457
OPDST71	C	053400	1	5690	5596
OPDST72	C	054400	1	5820	5734
OPDST73	C	055400	1	5946	5860
OPDST74	C	056400	1	6072	5986
OPDST75	C	057400	1	6194	6112
OPDST76	C	058400	1	6324	6238
OPDST77	C	059400	1	6450	6364
OPDST78	C	05A400	1	6576	6490
OPDST79	C	05B400	1	6698	6616
OPDST80	C	05C400	1	6829	6743
OPDST81	C	05D400	1	6955	6869
OPDST82	C	05E400	1	7081	6995
OPDST83	C	05F400	1	7203	7121
OPDST84	C	060400	1	7339	7247
OPDST85	C	061400	1	7470	7378
OPDST86	C	062400	1	7601	7509
OPDST87	C	063400	1	7728	7640
OPDST88	C	064400	1	7853	7771
OPDST89	C	065400	1	7981	7893
OPDST90	C	066400	1	8104	8026
OPDST91	C	067400	1	8226	8144
OPDST92	C	068400	1	8345	8271
OPDST93	C	069400	1	8464	8385
OPDST94	C	06A400	1	8584	8509
OPDST95	C	06B400	1	8702	8625
OPDST96	C	06C400	1	8821	8743
OPDST97	C	06D400	1	8937	8868
OPDST98	C	06E400	1	9047	8978
OPDST99	C	06F400	1	9165	9092
OPDSTA	A	000020	4	608	1680 1704 1796 1820 1912 1936 2028 2052 2145 2166 2253 2274 2361 2382
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					4926 4974 5063 5118 5201 5256 5339 5394 5477 5532 5616 5665 5754 5799
					5880 5925 6006 6051 6132 6171 6258 6303 6384 6429 6510 6555 6636 6675
					6763 6808 6889 6934 7015 7060 7141 7180 7267 7318 7398 7449 7529 7580
					7660 7705 7791 7836 7913 7958 8046 8087 8164 8205 8291 8328 8405 8445
					8529 8567 8645 8683 8763 8802 8888 8920 8998 9030 9112 9148 9226 9262
OPSRC100	C	070500	1	9283	9205
OPSRC39	C	033480	1	1725	1644
OPSRC40	C	034480	1	1841	1760
OPSRC41	C	035480	1	1957	1876
OPSRC42	C	036480	1	2073	1992
OPSRC43	C	037480	1	2185	2112
OPSRC44	C	038480	1	2293	2220
OPSRC45	C	039480	1	2401	2328
OPSRC46	C	03A480	1	2509	2436
OPSRC47	C	03B480	1	2613	2548

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
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OPSRC49	C	03D480	1	2823	2758
OPSRC50	C	03E480	1	2921	2858
OPSRC51	C	03F480	1	3017	2961
OPSRC52	C	040500	1	3153	3057
OPSRC53	C	041500	1	3284	3188
OPSRC54	C	042500	1	3415	3319
OPSRC55	C	043500	1	3546	3450
OPSRC56	C	044500	1	3680	3582
OPSRC57	C	045500	1	3815	3719
OPSRC58	C	046500	1	3946	3850
OPSRC59	C	047500	1	4077	3981
OPSRC60	C	048500	1	4208	4112
OPSRC61	C	049500	1	4342	4244
OPSRC62	C	04A500	1	4477	4381
OPSRC63	C	04B500	1	4608	4512
OPSRC64	C	04C500	1	4739	4643
OPSRC65	C	04D500	1	4870	4774
OPSRC66	C	04E500	1	5003	4905
OPSRC67	C	04F500	1	5145	5042
OPSRC68	C	050500	1	5283	5180
OPSRC69	C	051500	1	5421	5318
OPSRC70	C	052500	1	5559	5456
OPSRC71	C	053500	1	5694	5595
OPSRC72	C	054500	1	5824	5733
OPSRC73	C	055500	1	5950	5859
OPSRC74	C	056500	1	6076	5985
OPSRC75	C	057500	1	6198	6111
OPSRC76	C	058500	1	6328	6237
OPSRC77	C	059500	1	6454	6363
OPSRC78	C	05A500	1	6580	6489
OPSRC79	C	05B500	1	6702	6615
OPSRC80	C	05C500	1	6833	6742
OPSRC81	C	05D500	1	6959	6868
OPSRC82	C	05E500	1	7085	6994
OPSRC83	C	05F500	1	7207	7120
OPSRC84	C	060500	1	7343	7246
OPSRC85	C	061500	1	7474	7377
OPSRC86	C	062500	1	7605	7508
OPSRC87	C	063500	1	7732	7639
OPSRC88	C	064500	1	7857	7770
OPSRC89	C	065500	1	7985	7892
OPSRC90	C	066500	1	8108	8025
OPSRC91	C	067500	1	8230	8143
OPSRC92	C	068500	1	8349	8270
OPSRC93	C	069500	1	8468	8384
OPSRC94	C	06A500	1	8588	8508
OPSRC95	C	06B500	1	8706	8624
OPSRC96	C	06C500	1	8825	8742
OPSRC97	C	06D500	1	8941	8867
OPSRC98	C	06E500	1	9051	8977
OPSRC99	C	06F500	1	9169	9091
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SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
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					7785 7832 7907 7954 8040 8083 8158 8201 8285 8324 8399 8441 8523 8563
					8639 8679 8757 8798 8882 8916 8992 9026 9106 9144 9220 9258
PAGE	U	001000	1	492	
PL100	C	070600	1	9287	9207
PL39	C	033500	1	1729	1646
PL40	C	034500	1	1845	1762
PL41	C	035500	1	1961	1878
PL42	C	036500	1	2077	1994
PL43	C	037500	1	2189	2114
PL44	C	038500	1	2297	2222
PL45	C	039500	1	2405	2330
PL46	C	03A500	1	2513	2438
PL47	C	03B500	1	2617	2550
PL48	C	03C500	1	2723	2650
PL49	C	03D500	1	2827	2760
PL50	C	03E500	1	2925	2860
PL51	C	03F500	1	3021	2963
PL52	C	040600	1	3157	3059
PL53	C	041600	1	3288	3190
PL54	C	042600	1	3419	3321
PL55	C	043600	1	3550	3452
PL56	C	044600	1	3684	3584
PL57	C	045600	1	3819	3721
PL58	C	046600	1	3950	3852
PL59	C	047600	1	4081	3983
PL60	C	048600	1	4212	4114
PL61	C	049600	1	4346	4246
PL62	C	04A600	1	4481	4383
PL63	C	04B600	1	4612	4514
PL64	C	04C600	1	4743	4645
PL65	C	04D600	1	4874	4776
PL66	C	04E600	1	5007	4907
PL67	C	04F600	1	5149	5044
PL68	C	050600	1	5287	5182
PL69	C	051600	1	5425	5320
PL70	C	052600	1	5563	5458
PL71	C	053600	1	5698	5597
PL72	C	054600	1	5828	5735
PL73	C	055600	1	5954	5861
PL74	C	056600	1	6080	5987
PL75	C	057600	1	6202	6113
PL76	C	058600	1	6332	6239
PL77	C	059600	1	6458	6365
PL78	C	05A600	1	6584	6491
PL79	C	05B600	1	6706	6617
PL80	C	05C600	1	6837	6744
PL81	C	05D600	1	6963	6870
PL82	C	05E600	1	7089	6996
PL83	C	05F600	1	7211	7122
PL84	C	060600	1	7347	7248
PL85	C	061600	1	7478	7379
PL86	C	062600	1	7609	7510
PL87	C	063600	1	7736	7641

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
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PL89	C	065600	1	7989	7894
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PL91	C	067600	1	8234	8145
PL92	C	068600	1	8353	8272
PL93	C	069600	1	8472	8386
PL94	C	06A600	1	8592	8510
PL95	C	06B600	1	8710	8626
PL96	C	06C600	1	8829	8744
PL97	C	06D600	1	8945	8869
PL98	C	06E600	1	9055	8979
PL99	C	06F600	1	9173	9093
PLA	A	000024	4	609	1661 1683 1777 1799 1893 1915 2009 2031 2129 2148 2237 2256 2345 2364 2453 2472 2565 2581 2665 2684 2777 2791 2875 2889 2978 2991 3075 3124 3206 3255 3337 3386 3468 3517 3600 3649 3737 3786 3868 3917 3999 4048 4130 4179 4262 4311 4399 4448 4530 4579 4661 4710 4792 4841 4923 4972 5060 5116 5198 5254 5336 5392 5474 5530 5613 5663 5751 5797 5877 5923 6003 6049 6129 6169 6255 6301 6381 6427 6507 6553 6633 6673 6760 6806 6886 6932 7012 7058 7138 7178 7264 7316 7395 7447 7526 7578 7657 7703 7788 7834 7910 7956 8043 8085 8161 8203 8288 8326 8402 8443 8526 8565 8642 8681 8760 8800 8885 8918 8995 9028 9109 9146 9223 9260
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PLOETST	J	000000	461208	110	113 115 119 123 503 1629 1719 1745 1835 1861 1951 1977 2067 2097 2179 2205 2287 2313 2395 2421 2503 2533 2607 2633 2713 2743 2817 2843 2915 2946 3011 3042 3147 3173 3278 3304 3409 3435 3540 3567 3674 3704 3809 3835 3940 3966 4071 4097 4202 4229 4336 4366 4471 4497 4602 4628 4733 4759 4864 4890 4997 5027 5139 5165 5277 5303 5415 5441 5553 5580 5688 5718 5818 5844 5944 5970 6070 6096 6192 6222 6322 6348 6448 6474 6574 6600 6696 6727 6827 6853 6953 6979 7079 7105 7201 7231 7337 7362 7468 7493 7599 7624 7726 7755 7851 7877 7979 8010 8102 8128 8224 8255 8343 8369 8462 8493 8582 8609 8700 8727 8819 8852 8935 8962 9045 9076 9163 9190 9277 111
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SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES													
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SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
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R11	U	00000B	1	9426	204 205 875 877 895 897 915 917 935 937 955 957 975 977
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SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
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R13	U	00000D	1	9428	
R14	U	00000E	1	9429	
R15	U	00000F	1	9430	229 275 279 280 296 335 339 340 370 374 375 410 414 415
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					5135 5245 5268 5273 5383 5406 5411 5521 5544 5549 5679 5684 5788 5809
					5814 5914 5935 5940 6040 6061 6066 6183 6188 6292 6313 6318 6418 6439
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R3	U	000003	1	9418	1660 1661 1662 1665 1668 1671 1675 1682 1683 1684 1687 1690 1693 1697
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SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES													
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					3227	3230	3234	3243	3244	3254	3255	3259	3261	3263	3265	3336	3337	3343
					3345	3352	3355	3358	3361	3365	3374	3375	3385	3386	3390	3392	3394	3396
					3467	3468	3474	3476	3483	3486	3489	3492	3496	3505	3506	3516	3517	3521
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					3886	3889	3892	3896	3905	3906	3916	3917	3921	3923	3925	3927	3998	3999
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					6011	6018	6021	6024	6028	6037	6038	6048	6049	6053	6055	6057	6128	6129
					6135	6137	6144	6147	6150	6154	6168	6169	6175	6177	6179	6254	6255	6261
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					7702	7703	7709	7711	7713	7787	7788	7794	7795	7796	7797	7798	7805	7809
					7812	7815	7819	7833	7834	7838	7909	7910	7916	7917	7918	7919	7920	7927
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SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
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					9145 9146 9150 9222 9223 9229 9230 9237 9241 9245 9259 9260 9264
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					4133 4137 4138 4142 4180 4181 4264 4265 4269 4270 4312 4313 4401 4402
					4406 4407 4411 4449 4450 4532 4533 4537 4538 4542 4580 4581 4663 4664
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R5	U	000005	1	9420	198 199 202 275 279 335 339 370 374 388 403 410 414 864
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					964 965 978 984 985 998 1004 1005 1018 1024 1025 1038 1044 1045
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					1145 1158 1164 1165 1178 1184 1185 1198 1204 1205 1218 1224 1225 1238
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					7627 7757 7758 7879 7880 8012 8013 8130 8131 8257 8258 8371 8372 8495
					8496 8611 8612 8729 8730 8854 8855 8964 8965 9078 9079 9192 9193
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R7	U	000007	1	9422	
R8	U	000008	1	9423	143 146 147 148 150

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
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					6871 6997 7123 7249 7380 7511 7642 7773 7895 8028 8146 8273 8387 8511
					8627 8745 8870 8980 9094 9208
RANDXA	A	000028	4	610	1655 1771 1887 2003 2123 2231 2339 2447 2559 2659 2770 2869 2972 3066
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SKT0001	C	000222	26	166	169 186
SVOLDPSW	U	000140	0	113	
T1	A	0015B4	4	866	9307
T10	A	0016D4	4	1046	9316
T100	A	070020	4	9194	9406
T11	A	0016F4	4	1066	9317
T12	A	001714	4	1086	9318
T13	A	001734	4	1106	9319
T14	A	001754	4	1126	9320
T15	A	001774	4	1146	9321
T16	A	001794	4	1166	9322
T17	A	0017B4	4	1186	9323
T18	A	0017D4	4	1206	9324
T19	A	0017F4	4	1226	9325
T2	A	0015D4	4	886	9308
T20	A	001814	4	1246	9326
T21	A	001834	4	1266	9327
T22	A	001854	4	1286	9328
T23	A	001874	4	1306	9329
T24	A	001894	4	1326	9330
T25	A	0018B4	4	1346	9331
T26	A	0018D4	4	1366	9332
T27	A	0018F4	4	1386	9333
T28	A	001914	4	1406	9334
T29	A	001934	4	1426	9335
T3	A	0015F4	4	906	9309
T30	A	001954	4	1446	9336
T31	A	001974	4	1466	9337
T32	A	001994	4	1486	9338
T33	A	0019B4	4	1506	9339
T34	A	0019D4	4	1526	9340
T35	A	0019F4	4	1546	9341
T36	A	001A14	4	1566	9342
T37	A	001A34	4	1586	9343
T38	A	001A54	4	1606	9344
T39	A	033020	4	1633	9345
T4	A	001614	4	926	9310
T40	A	034020	4	1749	9346
T41	A	035020	4	1865	9347
T42	A	036020	4	1981	9348
T43	A	037020	4	2101	9349
T44	A	038020	4	2209	9350
T45	A	039020	4	2317	9351
T46	A	03A020	4	2425	9352

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
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T48	A	03C020	4	2637	9354
T49	A	03D020	4	2747	9355
T5	A	001634	4	946	9311
T50	A	03E020	4	2847	9356
T51	A	03F020	4	2950	9357
T52	A	040020	4	3046	9358
T53	A	041020	4	3177	9359
T54	A	042020	4	3308	9360
T55	A	043020	4	3439	9361
T56	A	044020	4	3571	9362
T57	A	045020	4	3708	9363
T58	A	046020	4	3839	9364
T59	A	047020	4	3970	9365
T6	A	001654	4	966	9312
T60	A	048020	4	4101	9366
T61	A	049020	4	4233	9367
T62	A	04A020	4	4370	9368
T63	A	04B020	4	4501	9369
T64	A	04C020	4	4632	9370
T65	A	04D020	4	4763	9371
T66	A	04E020	4	4894	9372
T67	A	04F020	4	5031	9373
T68	A	050020	4	5169	9374
T69	A	051020	4	5307	9375
T7	A	001674	4	986	9313
T70	A	052020	4	5445	9376
T71	A	053020	4	5584	9377
T72	A	054020	4	5722	9378
T73	A	055020	4	5848	9379
T74	A	056020	4	5974	9380
T75	A	057020	4	6100	9381
T76	A	058020	4	6226	9382
T77	A	059020	4	6352	9383
T78	A	05A020	4	6478	9384
T79	A	05B020	4	6604	9385
T8	A	001694	4	1006	9314
T80	A	05C020	4	6731	9386
T81	A	05D020	4	6857	9387
T82	A	05E020	4	6983	9388
T83	A	05F020	4	7109	9389
T84	A	060020	4	7235	9390
T85	A	061020	4	7366	9391
T86	A	062020	4	7497	9392
T87	A	063020	4	7628	9393
T88	A	064020	4	7759	9394
T89	A	065020	4	7881	9395
T9	A	0016B4	4	1026	9315
T90	A	066020	4	8014	9396
T91	A	067020	4	8132	9397
T92	A	068020	4	8259	9398
T93	A	069020	4	8373	9399
T94	A	06A020	4	8497	9400
T95	A	06B020	4	8613	9401
T96	A	06C020	4	8731	9402
T97	A	06D020	4	8856	9403

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
T98	A	06E020	4	8966	9404
T99	A	06F020	4	9080	9405
TESTING	F	001004	4	505	
TNUM	H	000004	2	593	243 303 353
TSUB	A	000000	4	592	204
TTABLE	F	070800	4	9306	
UR	U	000008	1	596	295 317 3113 3244 3375 3506 3638 3775 3906 4037 4168 4300 4437 4568 4699 4830 4961 5105 5243 5381 5519 5786 5912 6038 6290 6416 6542 6795 6921 7047 7305 7436 7567
URFOUND	U	000009	1	597	324 3112 3243 3374 3505 3637 3774 3905 4036 4167 4299 4436 4567 4698 4829 4960 5104 5242 5380 5518 5785 5911 6037 6289 6415 6541 6794 6920 7046 7304 7435 7566
URMSG	U	000420	1	291	3115 3246 3377 3508 3640 3777 3908 4039 4170 4302 4439 4570 4701 4832 4963 5107 5245 5383 5521 5788 5914 6040 6292 6418 6544 6797 6923 7049 7307 7438 7569
URPRTEXP	C	00112D	1	555	321
URPRTFUN	C	00111C	2	552	314
URPRTGOT	C	00113D	1	558	328
URPRTLINE	C	0010E0	16	548	560 333
URPRTLNG	U	00005F	1	560	332
URPRTNUM	C	0010F0	3	549	307
URSAV	D	0004C0	8	342	335 339
V0	U	000000	1	9436	
V1	U	000001	1	9437	
V10	U	00000A	1	9446	
V11	U	00000B	1	9447	
V12	U	00000C	1	9448	
V13	U	00000D	1	9449	
V14	U	00000E	1	9450	
V15	U	00000F	1	9451	
V16	U	000010	1	9452	
V17	U	000011	1	9453	
V18	U	000012	1	9454	
V19	U	000013	1	9455	
V2	U	000002	1	9438	
V20	U	000014	1	9456	
V21	U	000015	1	9457	
V22	U	000016	1	9458	
V23	U	000017	1	9459	
V24	U	000018	1	9460	
V25	U	000019	1	9461	
V26	U	00001A	1	9462	
V27	U	00001B	1	9463	
V28	U	00001C	1	9464	
V29	U	00001D	1	9465	
V3	U	000003	1	9439	
V30	U	00001E	1	9466	
V31	U	00001F	1	9467	
V4	U	000004	1	9440	
V5	U	000005	1	9441	
V6	U	000006	1	9442	
V7	U	000007	1	9443	
V8	U	000008	1	9444	
V9	U	000009	1	9445	
X0001	U	0002A0	1	175	163 176
XC0001	U	0002C8	1	189	181

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
=A(PLOTESTS)	A	0007D8	4	480	195
=AL2(L'MSGMSG)	R	0007E2	2	483	431
=CL128' '	C	0007E4	128	484	359
=F'1'	F	0007D4	4	479	180 231 298 350 385
=H'0'	H	0007E0	2	482	426
=XL4'3'	X	0007DC	4	481	238

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 MACRO    DEFN   REFERENCES

FCHECK     61    162
PTTABLE   764   9305
TSTCODE   685   1649   1765   1881   1997   2117   2225   2333   2441   2553   2653   2763   2863   2966   3062   3193   3324   3455
                3587   3724   3855   3986   4117   4249   4386   4517   4648   4779   4910   5047   5185   5323   5461   5600   5738
                5864   5990   6116   6242   6368   6494   6620   6747   6873   6999   7125   7251   7382   7513   7644   7775   7897
                8030   8148   8275   8389   8513   8629   8747   8872   8982   9096   9210
TSTDATA   697   1718   1834   1950   2066   2178   2286   2394   2502   2606   2712   2816   2914   3010   3146   3277   3408   3539
                3673   3808   3939   4070   4201   4335   4470   4601   4732   4863   4996   5138   5276   5414   5552   5687   5817
                5943   6069   6191   6321   6447   6573   6695   6826   6952   7078   7200   7336   7467   7598   7725   7850   7978
                8101   8223   8342   8461   8581   8699   8818   8934   9044   9162   9276
TSTEND    675   1737   1853   1969   2085   2197   2305   2413   2521   2625   2731   2835   2933   3029   3165   3296   3427   3558
                3692   3827   3958   4089   4220   4354   4489   4620   4751   4882   5015   5157   5295   5433   5571   5706   5836
                5962   6088   6210   6340   6466   6592   6714   6845   6971   7097   7219   7354   7485   7616   7743   7869   7997
                8120   8242   8361   8480   8601   8719   8838   8954   9064   9182   9296
TSTHDR    711   1628   1744   1860   1976   2096   2204   2312   2420   2532   2632   2742   2842   2945   3041   3172   3303   3434
                3566   3703   3834   3965   4096   4228   4365   4496   4627   4758   4889   5026   5164   5302   5440   5579   5717
                5843   5969   6095   6221   6347   6473   6599   6726   6852   6978   7104   7230   7361   7492   7623   7754   7876
                8009   8127   8254   8368   8492   8608   8726   8851   8961   9075   9189
TSTLBL    664   1720   1724   1728   1732   1836   1840   1844   1848   1952   1956   1960   1964   2068   2072   2076   2080   2180
                2184   2188   2192   2288   2292   2296   2300   2396   2400   2404   2408   2504   2508   2512   2516   2608   2612
                2616   2620   2714   2718   2722   2726   2818   2822   2826   2830   2916   2920   2924   2928   3012   3016   3020
                3024   3148   3152   3156   3160   3279   3283   3287   3291   3410   3414   3418   3422   3541   3545   3549   3553
                3675   3679   3683   3687   3810   3814   3818   3822   3941   3945   3949   3953   4072   4076   4080   4084   4203
                4207   4211   4215   4337   4341   4345   4349   4472   4476   4480   4484   4603   4607   4611   4615   4734   4738
                4742   4746   4865   4869   4873   4877   4998   5002   5006   5010   5140   5144   5148   5152   5278   5282   5286
                5290   5416   5420   5424   5428   5554   5558   5562   5566   5689   5693   5697   5701   5819   5823   5827   5831
                5945   5949   5953   5957   6071   6075   6079   6083   6193   6197   6201   6205   6323   6327   6331   6335   6449
                6453   6457   6461   6575   6579   6583   6587   6697   6701   6705   6709   6828   6832   6836   6840   6954   6958
                6962   6966   7080   7084   7088   7092   7202   7206   7210   7214   7338   7342   7346   7350   7469   7473   7477
                7481   7600   7604   7608   7612   7727   7731   7735   7739   7852   7856   7860   7864   7980   7984   7988   7992
                8103   8107   8111   8115   8225   8229   8233   8237   8344   8348   8352   8356   8463   8467   8471   8475   8583
                8587   8591   8595   8701   8705   8709   8713   8820   8824   8828   8832   8936   8940   8944   8948   9046   9050
                9054   9058   9164   9168   9172   9176   9278   9282   9286   9290
TSTPLO    633    860    880    900    920    940    960    980   1000   1020   1040   1060   1080   1100   1120   1140   1160   1180
                1200   1220   1240   1260   1280   1300   1320   1340   1360   1380   1400   1420   1440   1460   1480   1500   1520
                1540   1560   1580   1600
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   DESC      SYMBOL    SIZE       POS           ADDR

Entry: 0

Image       IMAGE     461208  00000-70997  00000-70997
  Region              461208  00000-70997  00000-70997
    CSECT   PLOETST   461208  00000-70997  00000-70997
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   STMT                     FILE NAME
1     /devstor/sharedvfp/tests/plo-01-extension.asm

** NO ERRORS FOUND **
```